The heat generator on the retained inverse-fibre cover

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Abstract

We transport the actual de Bruijn–Newman initial function through the retained cover $s = -r^2/2$. We give its connection, square, branch commutators, analytic heat realization, and the exact evolution of its arithmetic relation. We also compute the full Euclidean spatial diffusion of its actual scalar observable and its transported covector. The original inverse-fibre coordinates and all analytic-unit terms remain present. Complete proofs specify the relations between branch escape, arithmetic zeros, and the transported operators. The inverse-fibre construction alone does not exhibit a counterexample to the Riemann hypothesis or to the classical Euclidean Navier–Stokes problem. The source's exact fourth-order interacting electric-correlation coefficient is now propagated into this arithmetic heat flow through an explicitly invertible Fourier operator. Its full source proof, spectral derivative term, coupling scales, and spatial pullbacks are included. The complete arithmetic quotient continuation proves exact heat transport, closure and multiplicity-jet results in its stated topologies, and Mellin, principal-part and compact-test morphisms. The common-representative map identifies its heat dynamics with the original Fourier dynamics. The complete second Euclidean diffusion has an explicit jet residue. We prove that its divergence along the original escaping trajectory survives in the actual fourth coefficient at sufficiently negative Newman times, with a full error bound. The original three special points of the supplied period construction also give an exact real three-torus quotient. Their retained twists and cusp direction generate a three-dimensional forced fluid with all mixed interactions and localization terms calculated explicitly. For a supplied incompressible flow we also construct three profiles in the original polynomial target and recover its velocity from their arithmetic temporal defect by an explicit inverse integral and norm identity. The complete force-dependent acceleration, material diffusion and translated zero divisors are calculated.

The actual corrected Navier–Stokes source is also retained as an explicit construction input. Its exact axis vorticity gives an arithmetic endpoint derivative and a growing Mellin residue. An invertible radial map carries the complete corrected central angular momentum and all its fluxes into order-one entire functions with an equivalent entire evolution equation. Their exact positive value at the fixed spectral point $s = -5i/2$ grows at the source rate. The full solenoidal correspondence isolates the orthogonal swirl component carrying this signal. The original quotient map and every kernel remain explicit; its actual topology and relation-space invariance are not assumed.

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1 Original coordinates and the retained cover

The original source coordinate triple is (x, y, w) and its target is (a, b, c). Its polynomial is

$$\begin{aligned} F_1 &= (1+xy)^3 w + y^2(1+xy)(4+3xy), \\ F_2 &= y + 3x(1+xy)^2 w + 3xy^2(4+3xy), \\ F_3 &= 2x - 3x^2 y - x^3 w. \end{aligned} \tag{1}$$

All coordinate spaces here are complex unless a real locus is stated. The original $x \neq 0$ inverse chart is

$$\begin{aligned} P_{a,b,c}(r) &= cr^3 - 2r^2 + br - 2a, \qquad \alpha = \tfrac{1}{2} P_r(r), \\ (x, y, w) &= (\alpha^{-1}, r - \alpha, 5\alpha^2 - 3r\alpha - c\alpha^3), \qquad \alpha \neq 0. \end{aligned} \tag{2}$$

Here is the complete verification needed for this chart. At a point with $x \neq 0$, set $\alpha = 1/x$ and $r = y + 1/x$; solving $F_3 = c$ gives the third coordinate in (2). Substitution into the other original components gives

$$F_2 = 4r + 2\alpha - 3cr^2, \qquad F_1 = r^2 + r\alpha - cr^3.$$

The first equation equals b precisely when $2\alpha = P_r(r)$; after that substitution $2(F_1 - a) = P(r)$. This proves both directions and uniqueness of the root label. A multiple root gives no finite point in this chart. Directly, $F(0, y, w) = (w + 4y^2, y, 0)$, so $c = 0$ has the additional inverse point $(0, b, a - 4b^2)$ and $c \neq 0$ has none.

On $b = c = 0$ the root cover is $a = -r^2$, with $\alpha = -2r$. The arithmetic coordinate is the specific further map

$$a = 2s, \qquad s = -r^2/2. \tag{3}$$

Its unramified domain is $\mathbb{C}_r^* = \mathbb{C}_r \setminus \{0\}$ over $\mathbb{C}_s^*$. The point $r = 0$ is the ramification point of the completed cover and is excluded by $\alpha \neq 0$ from the original inverse chart. These are explicit restrictions of the original map.

The constant-coefficient heat evolution in the original root variable has the following exact map to the arithmetic base, retaining both sheets. For $V \subset \mathbb{C}_s^*$ and $\widetilde{V} = \{r : -r^2/2 \in V\}$ put $T_V : \mathcal{O}(V)^2 \to \mathcal{O}(\widetilde{V})$, $T_V(q_0, q_1)(r) = q_0(-r^2/2) + r q_1(-r^2/2)$. Its inverse is explicitly the even part and the odd part divided by r; their invariance under $r \mapsto -r$ makes them holomorphic

on V. Thus the formulas give an isomorphism in both directions. In the original basis $(1, r)$ put

$$R = \begin{pmatrix} 0 & -2s \\ 1 & 0 \end{pmatrix}, \qquad D = \partial_s + \begin{pmatrix} 0 & 0 \\ 0 & 1/(2s) \end{pmatrix}.$$

The identities $r^2 = -2s$ and $\partial_s r = -1/r = r/(2s)$ give $\partial_r T_V = T_V(-RD)$ by direct differentiation of both components. Also $DR - RD = -R^{-1}$, since differentiation of the factor r gives $-1/r$. Applying the first identity twice proves the full typed operator morphism

$$T_V^{-1}\partial_r^2 T_V = RDRD = R^2D^2 - D = -2sD^2 - D.$$

The arithmetic Newman generator has the corresponding pullback $T_V(-D^2/4)T_V^{-1}$, since D transports ∂_s. This proves the exact relationship between both generators, including the first derivative and both sheet components. The original polynomial has component vector $(4s - 2a, b - 2cs)$ under this map. Consequently its full root heat evolution is

$$P(\tau, r) = P_{a_0+2\tau, b_0+6c\tau, c}(r), \qquad P_\tau = 6cr - 4 = P_{rr}.$$

Here τ is forward heat time. The explicit reversal $\widetilde{P}(t, r) = P(-t, r)$ satisfies $\widetilde{P}_t = -\widetilde{P}_{rr}$. This is the backward sign convention; the arithmetic Newman generator below acts in its specified s coordinate. The arithmetic pullback below transports differentiation in $s = a/2$, which must also carry the derivative of r with respect to s. Both operators will therefore be displayed with their exact coordinates. In the displayed component vector, $(-2sD^2 - D)$ gives $(-4, 6c)$: the first component is $-\partial_s(4s - 2a) = -4$, while for the second $D(b - 2cs) = b/(2s) - 3c$ and $D^2(b - 2cs) = -b/(4s^2) - 3c/(2s)$. Their combination is exactly $6c$. These are precisely the parameter derivatives $a_\tau = 2$, $b_\tau = 6c$, $c_\tau = 0$ in the original polynomial, verifying the whole morphism on the original family.

For an open set $V \subset \mathbb{C}_s^*$, set $\widetilde{V} = \{r : -r^2/2 \in V\}$. Every holomorphic function on $\widetilde{V}$ has a unique decomposition

$$f(r) = q_0(s) + rq_1(s), \qquad q_0, q_1 \in \mathcal{O}(V). \tag{4}$$

Indeed its even part is $(f(r) + f(-r))/2$ and its odd coefficient is $(f(r) - f(-r))/(2r)$. Both are invariant under $r \mapsto -r$ and therefore define holomorphic functions of s on local inverse charts, agreeing on their overlaps. These formulas also prove uniqueness. They define an isomorphism $T_V : \mathcal{O}(V)^2 \to \mathcal{O}(\widetilde{V})$. The same decomposition holds over V containing zero: power series show that the odd numerator divided by r extends holomorphically and that each even power series is a power series in $s = -r^2/2$. The derivative operator below, however, is first defined only over $V \subset \mathbb{C}^*$.

2 Connection, square, and branch coupling

Write a section as the column $q = (q_0, q_1)^{\mathsf{T}}$ in the ordered basis $(1, r)$. On the unramified cover, $ds/dr = -r$, so differentiation in the original spectral coordinate is $\partial_s^{\text{lift}} = -r^{-1}\partial_r$. In particular $\partial_s r = -1/r = r/(2s)$. Consequently

$$D = \partial_s + A(s), \quad A(s) = \begin{pmatrix} 0 & 0 \\ 0 & 1/(2s) \end{pmatrix}, \qquad T_V D = \partial_s^{\text{lift}} T_V. \tag{5}$$

This is a connection: for $g \in \mathcal{O}(V)$, $D(gq) = g'q + gDq$. Its curvature in this one complex coordinate is zero; the nontrivial information is its ramification and monodromy.

Proposition 2.1 (The full second derivative and its commutators)**.** *On $\mathcal{O}(V)^2$ one has*

$$D^2 = \begin{pmatrix} \partial_s^2 & 0 \\ 0 & \partial_s^2 + s^{-1}\partial_s - 1/(4s^2) \end{pmatrix}, \tag{6}$$

$$R = \begin{pmatrix} 0 & -2s \\ 1 & 0 \end{pmatrix}, \quad R^2 = -2sI, \quad [D, R] = -R^{-1}, \tag{7}$$

$$[D^2, R] = -2R^{-1}D - R^{-3}. \tag{8}$$

Here R is multiplication by the original root r; commutators include differentiation of the matrix coefficients.

Proof. Expanding $(\partial_s + A)^2$ gives $\partial_s^2 + 2A\partial_s + A' + A^2$. Its lower diagonal constant is $-1/(2s^2) + 1/(4s^2) = -1/(4s^2)$, proving (6). Multiplication $r(q_0 + rq_1) = -2sq_1 + rq_0$ proves R and R^2. Matrix multiplication then gives

$$R' + AR - RA = \begin{pmatrix} 0 & -1 \\ 1/(2s) & 0 \end{pmatrix} = -R^{-1}.$$

Differentiating $RR^{-1} = I$ covariantly gives $[D, R^{-1}] = -R^{-1}[D, R]R^{-1} = R^{-3}$. Finally $[D^2, R] = D[D, R] + [D, R]D = -2R^{-1}D - R^{-3}$. This proves every term, including the zeroth-order branch term. □

The sheet involution is $J = \mathrm{diag}(1, -1)$; it commutes with D, D^2 and anticommutes with R. The trace is $(q_0, q_1) \mapsto 2q_0$, with kernel the odd summand, and its half has the section $q_0 \mapsto (q_0, 0)$. D preserves the two summands on the punctured base, whereas R exchanges them and couples the odd coefficient into $-2sq_1$. This describes exactly the information lost by tracing.

On a simply connected $V \subset \mathbb{C}^*$ choose a holomorphic root $r(s)$ satisfying $r(s)^2 = -2s$. The invertible matrix $G = \mathrm{diag}(1, r(s))$ obeys $D = G^{-1}\partial_s G$. Horizontal coefficients satisfy $q_0' = 0$ and $q_1' + q_1/(2s) = 0$, hence $(q_0, q_1) = (c_0, c_1/r(s))$. Continuation once around zero changes the sign of the latter coefficient. Thus the monodromy is exactly $\mathrm{diag}(1, -1)$; no single-valued global choice of $r(s)$ on $\mathbb{C}^*$ is implied by this local gauge formula.

3 The actual arithmetic heat function

We use the convention of [1, Section 1, equations (1)–(4)]:

$$\begin{aligned} \Phi(u) &= \sum_{n\geq 1} (2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u}) e^{-\pi n^2 e^{4u}}, \\ H_t(z) &= \int_0^\infty e^{tu^2} \Phi(u) \cos(zu)\, du, \\ Z_t(s) &= 8H_t(2s), \qquad Z_0(s) = \Xi(s) = \xi(1/2 + is), \\ \xi(\sigma) &= \tfrac{1}{2}\sigma(\sigma - 1)\pi^{-\sigma/2}\Gamma(\sigma/2)\zeta(\sigma). \end{aligned} \tag{9}$$

The Mellin and spectral coordinates are related by $\sigma = 1/2 + is$, with inverse $s = -i(\sigma - 1/2)$. They are not silently identified.

We rederive the integral identity and its analytic bounds. Set

$$h(x) = \frac{\pi}{2} x^2 (2\pi x^2 - 3) e^{-\pi x^2}, \quad S(\lambda) = \sum_{n\geq 1} h(n\lambda), \quad k(v) = e^{v/2} S(e^v).$$

For the Fourier convention $\widehat{h}(q) = \int_{\mathbb{R}} h(x)e^{-2\pi ixq}dx$, the Gaussian transform and its second and fourth derivatives give

$$\widehat{x^2e^{-\pi x^2}} = (1/(2\pi) - q^2)e^{-\pi q^2},$$
$$\widehat{x^4e^{-\pi x^2}} = (q^4 - 3q^2/\pi + 3/(4\pi^2))e^{-\pi q^2}.$$

Multiplying by $-3\pi/2$ and π^2 respectively gives $\widehat{h} = h$. Since $h(0) = 0$, Poisson summation gives $S(\lambda) = \lambda^{-1}S(1/\lambda)$ and $k(v) = k(-v)$. For $\lambda \geq 1$, $|S(\lambda)| \leq C\lambda^4 e^{-\pi\lambda^2}$, because the remaining series is bounded by a constant times $\sum_{n\geq 1} n^4 e^{-\pi(n^2-1)} < \infty$. The symmetry gives the corresponding bound as $\lambda \downarrow 0$. Each fixed number of v derivatives introduces only finitely many additional powers of ne^v. It follows, with constants depending on the derivative order, that

$$|k^{(j)}(v)| \leq C_j e^{C_j|v|}e^{-\pi e^{2|v|}}. \tag{10}$$

Termwise differentiation on compact sets is justified by the same summable bounds. On the negative half-line differentiate the proved evenness identity. This proves (10) on both ends.

For $\Re\sigma > 1$, absolute integrability permits summation inside the Mellin integral. The substitution $q = \pi x^2$ gives

$$\begin{aligned}\int_0^\infty h(x)x^{\sigma-1}dx &= \pi^{-\sigma/2}\left[\tfrac{1}{2}\Gamma(\sigma/2+2) - \tfrac{3}{4}\Gamma(\sigma/2+1)\right]\\ &= \frac{\sigma(\sigma-1)}{8}\pi^{-\sigma/2}\Gamma(\sigma/2),\\ \int_0^\infty S(\lambda)\lambda^{\sigma-1}d\lambda &= \xi(\sigma)/4.\end{aligned}$$

The two-ended bounds make the last integral entire in σ, so the equality extends by analytic continuation. Substituting $\lambda = e^v$ and $\sigma = 1/2 + is$ proves

$$Z_t(s) = 4\int_{\mathbb{R}} e^{tv^2/4}k(v)e^{isv}\,dv. \tag{11}$$

Indeed direct expansion also gives $\Phi(u) = 2k(2u)$; the substitution $v = 2u$ and the evenness of k carry the stated H_t integral to (11), retaining the factors $2, 4, 8$. The bound (10) dominates every fixed derivative uniformly on compact complex (t, s) sets. Thus Z is jointly entire and differentiation under its integral proves exactly

$$\partial_t Z_t = -\tfrac{1}{4}\partial_s^2 Z_t. \tag{12}$$

This supplies the actual initial datum and the actual time orientation.

4 An explicit domain for the heat evolution

The differential expression alone does not specify a heat evolution on every entire-function space. We now give a precise invariant domain containing the actual kernel.

Definition 4.1. Let $\mathscr{E}$ consist of $\phi \in C^\infty(\mathbb{R}, \mathbb{C})$ for which

$$p_{A,B,m,j}(\phi) = \int_{\mathbb{R}} |\phi^{(j)}(v)|(1+|v|)^m e^{Av^2+B|v|}\,dv < \infty$$

for every nonnegative integer A, B, m, j. Give it these seminorms. Define

$$T\phi(s) = 4\int_{\mathbb{R}} \phi(v)e^{isv}\,dv, \qquad \mathscr{A} = T\mathscr{E},$$

and equip $\mathscr{A}$ with the transported topology.

The map T is injective. To give the needed uniqueness argument, let $T\phi$ vanish on the real line and let $g_\epsilon(v) = (4\pi\epsilon)^{-1/2}e^{-v^2/(4\epsilon)}$. Its Gaussian integral representation is $g_\epsilon(v) = (2\pi)^{-1}\int_{\mathbb{R}} e^{-\epsilon s^2}e^{isv}ds$. Fubini is valid because $\phi \in L^1$ and the Gaussian is integrable; it expresses $\phi * g_\epsilon$ as a Gaussian-weighted integral of the vanishing Fourier transform. Thus $\phi * g_\epsilon = 0$. As $\epsilon \downarrow 0$, these convolutions converge to ϕ in L^1: translations are continuous in L^1, first for continuous compactly supported functions by uniform continuity, then for all L^1 functions by approximation; splitting the Gaussian integral into a small translation ball and its vanishing tail proves the claim. Consequently $\phi = 0$ almost everywhere and, being continuous, everywhere. This also justifies the transported topology without an unspecified choice of representative.

Proposition 4.2 (Heat evolution with its inverse)**.** *For every complex t, multiplication $E_t\phi(v) = e^{tv^2/4}\phi(v)$ is a continuous automorphism of $\mathscr{E}$. The operators $U_t = TE_tT^{-1}$ on $\mathscr{A}$ satisfy*

$$\begin{aligned} U_tU_u = U_{t+u}, \qquad U_t^{-1} = U_{-t}, \qquad \partial_t U_t = -\tfrac{1}{4}\partial_s^2 U_t, \\ U_t\Xi = Z_t, \qquad U_t\partial_s = \partial_s U_t, \qquad U_tM_sU_{-t} = M_s - \tfrac{t}{2}\partial_s. \end{aligned} \tag{13}$$

Here M_s is multiplication by the unchanged spectral coordinate.

Proof. Every v derivative of $e^{tv^2/4}\phi$ is a finite sum of $e^{tv^2/4}$ times a polynomial in v times a derivative of ϕ. Each target seminorm is bounded by finitely many source seminorms with larger A, m; on compact t sets these bounds are uniform. The inverse multiplier is $e^{-tv^2/4}$. Differentiation in t inserts $v^2/4$, while two s derivatives in the transform insert $-v^2$, proving the generator and the group identities. The kernel bound proves $k \in \mathscr{E}$, so $U_t\Xi = Z_t$ follows from (11).

Integration by parts gives $M_sT\phi = T(i\phi')$. For completeness the boundary terms vanish: each derivative of $e^{isv}\phi$ is integrable on a horizontal real-v line and the function itself is integrable, so its limits at both ends exist and are zero. The same argument applies after exponential weights. Thus multiplication by s preserves $\mathscr{A}$; differentiation does also, since $\partial_sT\phi = T(iv\phi)$. Finally

$$e^{tv^2/4}i\partial_v(e^{-tv^2/4}\phi) = i\phi' - \tfrac{t}{2}iv\phi.$$

Applying T proves the last identity in (13). All displayed operators have the domains just specified. □

5 Transport of the moving arithmetic relation

Put $B_t = M_s - (t/2)\partial_s$. For every polynomial p, the automorphism calculation proves

$$U_t(p(s)\Xi(s)) = p(B_t)Z_t(s). \tag{14}$$

The right side is the polynomial in the actual differential operator, with composition order retained. For example the first two nonconstant relations are

$$\begin{aligned} B_tZ_t &= sZ_t - \tfrac{t}{2}Z_t', \\ B_t^2Z_t &= (s^2 - t/2)Z_t - tsZ_t' + \tfrac{t^2}{4}Z_t''. \end{aligned}$$

These follow from $\partial_s(sq) = q + s\partial_s q$. In particular, an evolving polynomial multiple is not obtained by silently leaving multiplication by s fixed.

One can take a closure in the topology that has actually been defined: let $N_0 = \overline{\mathbb{C}[s]\Xi}^{\mathscr{A}}$ and $N_t = U_tN_0$. Continuity and invertibility give $N_t = \overline{\{p(B_t)Z_t : p \in \mathbb{C}[s]\}}^{\mathscr{A}}$. M_s preserves

N_0, since it is continuous and preserves its dense generating subspace; hence B_t preserves N_t. The map $[f] \mapsto [U_t f]$ is a well-defined continuous isomorphism $\mathscr{A}/N_0 \to \mathscr{A}/N_t$ with inverse induced by U_{-t} and intertwines the induced M_s with the induced B_t. The exact map from these explicit spaces to the original Connes quotient is constructed immediately below, with both original topologies retained.

5.1 A coordinate-level correspondence with the actual Connes quotient

We now construct the comparison with the actual object of [2, Section 3.6, Proposition 3.6]. Write $\mathcal{H} = \mathcal{H}_{\leq 1}$ for its ring of entire functions of Hadamard order at most one, retaining its stated Hadamard topology. Write

$$N_{\mathrm{C}} = \overline{\mathbb{C}[s]\Xi}^{\mathcal{H}}, \qquad Q_{\mathrm{C}} = \mathcal{H}/N_{\mathrm{C}}, \qquad Q_t = \mathscr{A}/N_t.$$

Here s is the same coordinate in both quotients. The closure defining N_{C} is precisely the closure of the generators in that proposition; the following calculation proves the equality of their algebraic spans, including the transform conventions. The comparison constructed below uses the actual topology on $\mathcal{H}$, through a subspace of a product; it does not substitute the topology of $\mathscr{A}$ for it.

The original summation, Mellin transform, and their coordinates. Let $\mathcal{G}_0$ be the vector space of the finite sums

$$f(x) = e^{-\pi x^2} \sum_{j=0}^{n} c_j \pi^j x^{2j}, \qquad c_0 = 0, \qquad \sum_{j=0}^{n} c_j \frac{(2j)!}{4^j j!} = 0. \tag{15}$$

The two constraints say exactly $f(0) = \widehat{f}(0) = 0$ for $\widehat{f}(y) = \int_{\mathbb{R}} f(x) e^{-2\pi i x y} dx$. Indeed the Gaussian moment is $\int_{\mathbb{R}} \pi^j x^{2j} e^{-\pi x^2} dx = (2j)!/(4^j j!)$. The original maps of [2, equations (3.3), (3.4), and Section 3.6] are

$$\begin{aligned} (\mathcal{E}_{\mathrm{C}} f)(\lambda) &= \lambda^{1/2} \sum_{n \geq 1} f(n\lambda), \qquad \lambda > 0, \\ (\mathcal{F}_\mu g)(s) &= \int_0^\infty g(\lambda) \lambda^{-is} \frac{d\lambda}{\lambda}. \end{aligned} \tag{16}$$

Both signs and the measure are retained. With $\lambda = e^v$, define

$$k_f(v) = e^{v/2} \sum_{n \geq 1} f(n e^v), \qquad \iota f(v) = \frac{1}{4} k_f(-v). \tag{17}$$

Then there is an exact typed map

$$\iota : \mathcal{G}_0 \longrightarrow \mathscr{E}, \qquad T\iota f(s) = \int_{\mathbb{R}} k_f(v) e^{-isv} dv = \mathcal{F}_\mu \mathcal{E}_{\mathrm{C}} f(s). \tag{18}$$

The reflection and the factor $1/4$ in (17) are necessary for the defined T, and have not been suppressed.

Here is the convergence proof. For $v \geq 0$, every derivative of k_f is bounded by $C_j e^{C_j v} e^{-\pi e^{2v}}$, by differentiating the finite polynomial times Gaussian and summing the resulting convergent series in n. The Fourier transform of a polynomial times this Gaussian is again a polynomial times the same Gaussian. Poisson summation and the two zero constraints give

$$\sum_{n \geq 1} f(n\lambda) = \lambda^{-1} \sum_{n \geq 1} \widehat{f}(n/\lambda), \qquad k_f(v) = k_{\widehat{f}}(-v).$$

The positive-half-line estimate for $\widehat{f}$ therefore proves the estimate on the negative half-line as well, for every derivative. Every defining seminorm of $\mathscr{E}$ is finite. These bounds justify the change of variable, reflection, and all entire continuations in (18).

Put $z = 1/2 - is$, whose inverse is $s = i(z - 1/2)$, and put

$$P_f(z) = \sum_{j=0}^{n} c_j (z/2)_j, \qquad (z/2)_j = \prod_{k=0}^{j-1} (z/2 + k), \qquad (z/2)_0 = 1.$$

For $\Re z > 1$, the absolutely convergent Mellin integral gives

$$\begin{aligned} \mathcal{F}_\mu \mathcal{E}_{\mathrm{C}} f(s) &= \zeta(z) \int_0^\infty f(x) x^{z-1} dx \\ &= \frac{1}{2} \pi^{-z/2} \Gamma(z/2) \zeta(z) P_f(z). \end{aligned} \tag{19}$$

The two constraints in (15) are respectively $P_f(0) = 0$ and $P_f(1) = 0$, since $(1/2)_j = (2j)!/(4^j j!)$. There is consequently a unique polynomial

$$q_f(s) = \left. \frac{P_f(z)}{z(z-1)} \right|_{z=1/2-is}, \qquad T\iota f(s) = q_f(s) \Xi(s). \tag{20}$$

For the last equality, the right side of (19) is $q_f(s)\xi(1/2 - is) = q_f(s)\Xi(-s)$, and the proved evenness $\Xi(-s) = \Xi(s)$ supplies the final sign conversion. Entire continuation then proves the identity on all $\mathbb{C}_s$.

This map is onto every polynomial multiple, with an explicit inverse. For $q \in \mathbb{C}[s]$, expand the unchanged polynomial

$$z(z-1) q\big(i(z - 1/2)\big) = \sum_{j=0}^{\deg q + 2} c_j (z/2)_j. \tag{21}$$

There is a unique such expansion because the jth basis polynomial has degree j and leading coefficient 2^{-j}. Evaluation at 0 and 1 proves the two required constraints, giving $f \in \mathcal{G}_0$. This proves both injectivity and surjectivity of $f \mapsto q_f$ and of $f \mapsto q_f \Xi$, since Ξ is not zero identically. Here is the full basis argument connecting to the source's Hermite presentation. Its functions are

$$h_n(x) = \frac{2^{1/4}}{\sqrt{2^n n!}} H_n(\sqrt{2\pi} x) e^{-\pi x^2}, \qquad H_n(y) = n! \sum_{j=0}^{\lfloor n/2 \rfloor} \frac{(-1)^j (2y)^{n-2j}}{j!(n-2j)!}.$$

Their even members are a triangular basis of all even polynomial Gaussians, since each has nonzero leading coefficient in its successive even degree. Put $L = \sqrt{\pi} x - (2\sqrt{\pi})^{-1} \partial_x$. Differentiating the displayed polynomial proves $h_n = L^n h_0 / \sqrt{n!}$. The Gaussian transform gives $\widehat{h}_0 = h_0$, while integration by parts gives $\mathcal{F}_{\mathbb{R}} L = -iL\mathcal{F}_{\mathbb{R}}$. Thus $\widehat{h}_{2n} = (-1)^n h_{2n}$, with both signs proved in the original Fourier convention. The finite polynomial formula also gives $h_{2n}(0) = (-1)^n 2^{1/4-n} \sqrt{(2n)!}/n! \neq 0$. Split an even polynomial Gaussian into the spans of the $h_{4\ell}$ and the $h_{4\ell+2}$. The two constraints $f(0) = \widehat{f}(0) = 0$ are exactly vanishing at zero of each of those two components. In the first span subtracting its constant-basis coordinate gives $h_{4\ell} - h_{4\ell}(0) h_0 / h_0(0)$; in the second it gives $-h_{4\ell+2} + h_{4\ell+2}(0) h_2 / h_2(0)$, with the displayed minus sign retained. These are precisely the source's ψ_ℓ^+, ψ_ℓ^-, for $\ell \geq 1$. They span the two constraint kernels by solving for their respective h_0, h_2 coefficients. Thus their union spans $\mathcal{G}_0$, and (20) proves the required equality of spans. For the particular original kernel $h(x) = \pi^2 x^4 e^{-\pi x^2} - (3\pi/2) x^2 e^{-\pi x^2}$, $P_h(z) = z(z-1)/4$, so $q_h = 1/4$, $\iota h = k/4$, and

$$\mathcal{F}_\mu \mathcal{E}_{\mathrm{C}} h = \Xi/4, \qquad T\iota(4h) = \Xi. \tag{22}$$

The factor 4 in the actual arithmetic function is thereby accounted for.

The original scaling differential operator is $S = -i(x\partial_x + 1/2) : \mathcal{G}_0 \to \mathcal{G}_0$. It preserves the two constraints: its value at zero is $-if(0)/2$, and integration by parts gives $\int Sf = (i/2)\int f$. Termwise differentiation proves $\mathcal{E}_{\mathrm{C}} Sf = -i\lambda\partial_\lambda \mathcal{E}_{\mathrm{C}} f$. Integration by parts, with the proved two-ended bounds, then gives

$$T\iota Sf = M_s T\iota f, \qquad \iota Sf = i\partial_v \iota f. \tag{23}$$

The second identity also follows from differentiating $k_f(-v)/4$; it records the sign produced by the reflection. Thus the comparison intertwines the actual scaling operator and multiplication by the same s.

The correspondence between the two completed quotients. Define a space of common representatives, with its topology specified by the displayed embedding into the actual product:

$$C = \mathscr{A} \cap \mathcal{H}, \qquad C \longrightarrow \mathscr{A} \times \mathcal{H}, \quad f \longmapsto (f, f). \tag{24}$$

Give C the subspace topology of this embedding and put

$$K = N_0 \cap N_{\mathrm{C}}, \qquad \mathscr{R} = C/K. \tag{25}$$

Both intersections mean equality of entire functions in the unchanged s coordinate. In particular $K \subset C$. Since the two quotient maps are continuous, the following is a continuous injective linear map, with its complete coordinate formula:

$$j : \mathscr{R} \longrightarrow Q_0 \times Q_{\mathrm{C}}, \qquad j([f]_K) = ([f]_{N_0}, [f]_{N_{\mathrm{C}}}). \tag{26}$$

Indeed the map on C is continuous by the product-induced topology; its kernel consists exactly of representatives belonging to both N_0 and N_{C}, which is K. The definition of the quotient topology therefore proves continuity and injectivity, without asserting that this injection is a topological embedding into the product.

Writing π_0, π_{C} for its two coordinate projections, their images and kernels are exactly

$$\begin{aligned} \operatorname{im}\pi_0 &= \{[f]_{N_0} : f \in C\}, & \ker\pi_0 &= (N_0 \cap \mathcal{H})/K, \\ \operatorname{im}\pi_{\mathrm{C}} &= \{[f]_{N_{\mathrm{C}}} : f \in C\}, & \ker\pi_{\mathrm{C}} &= (\mathscr{A} \cap N_{\mathrm{C}})/K. \end{aligned} \tag{27}$$

For example $\pi_0([f]_K) = 0$ means precisely $f \in N_0$, while $f \in C$ already supplies $f \in \mathcal{H}$; this proves the first kernel identity in both directions. The other identity follows by the same direct test in the second coordinate. Formula (27) gives an explicit representative rule for the correspondence even when either coordinate has several possible lifts.

The first image in (27) is dense in Q_0. To prove this, take $\phi \in \mathscr{E}$, and choose a fixed smooth cutoff χ equal to one on $[-1, 1]$ and zero outside $[-2, 2]$. Set $\phi_R(v) = \chi(v/R)\phi(v)$ for $R \geq 1$. Leibniz's formula bounds every seminorm of $\phi - \phi_R$ by a finite sum of weighted integrals of derivatives of ϕ over $|v| \geq R$, multiplied by bounded constants and factors R^{-k}. Those integrals tend to zero by their absolute integrability. Thus $\phi_R \to \phi$ in $\mathscr{E}$. Moreover

$$|T\phi_R(s)| \leq 4\|\phi_R\|_{L^1} e^{2R|\Im s|},$$

so $T\phi_R$ is of order at most one and belongs to C. Consequently C is dense in $\mathscr{A}$; taking images under the continuous surjective quotient map proves the assertion. All generators (20) also lie in C by their proved $\mathscr{E}$ representatives and their order at most one.

Finally this same correspondence transports exactly through every complex Newman time. For $t \in \mathbb{C}$, define

$$j_t : \mathscr{R} \longrightarrow Q_t \times Q_{\mathrm{C}}, \qquad j_t([f]_K) = ([U_t f]_{N_t}, [f]_{N_{\mathrm{C}}}). \tag{28}$$

It is continuous and injective: U_t is a continuous automorphism and $U_t f \in N_t$ is equivalent to $f \in N_0$. Its two kernels as coordinate projections are the same as in (27), and its first image is dense in Q_t. These assertions follow by applying the proved isomorphism $Q_0 \to Q_t$ to the first coordinate. The space C is stable under M_s: it is stable in $\mathscr{A}$ by integration by parts and in $\mathcal{H}$ because the latter is the specified topological ring. Each closed relation subspace is stable under M_s, by continuity and invariance of $\mathbb{C}[s]\Xi$. Thus $[f]_K \mapsto [sf]_K$ is defined, continuous, and obeys

$$j_t([sf]_K) = \big([B_t U_t f]_{N_t}, [sf]_{N_C}\big), \qquad B_t = M_s - (t/2)\partial_s. \tag{29}$$

This is the exact operator correspondence: the first coordinate has the heat-transported multiplication operator, while the second has the original Connes multiplication operator. It follows directly from $U_t M_s = B_t U_t$ and retains the full derivative term. No inclusion of all of $\mathscr{A}$ into $\mathcal{H}$, no equality of their closures, and no identification of their spectra is needed for the proved maps (26)–(29).

5.2 Complete local transport at the moving divisor

There is a second exact description, local at the moving divisor. Let $Z = Z_t(s)$ and $\mathcal{P} = \partial_t + \frac{1}{4}\partial_s^2$. For every holomorphic $g(t, s)$ the complete product calculation is

$$\mathcal{P}(Zg) = Z\mathcal{P}g + \tfrac{1}{2} Z_s g_s. \tag{30}$$

The term $\mathcal{P}Z$ is zero by (12); the mixed product term is retained. On the complement of $Z = 0$, set $b = Z_s/Z$. Direct differentiation of f/Z gives the exact conjugation

$$Z\mathcal{P}Z^{-1} = \partial_t + \tfrac{1}{4}\partial_s^2 - \tfrac{1}{2} b\partial_s + \tfrac{1}{2} b^2. \tag{31}$$

To check the potential term before canceling any expression, it is $\frac{1}{2}(Z_s/Z)^2 - \frac{1}{4} Z_{ss}/Z - Z_t/Z$; the last two terms cancel by the established heat equation. Thus the operator in (31) sends Zg exactly to $Z\mathcal{P}g$. Its coefficients are meromorphic at zeros of Z, but its restriction to the ideal of holomorphic multiples of Z has the displayed holomorphic value. Division by Z on this ideal is well-defined and unique by analytic continuation. There is no claim that division is holomorphic on arbitrary sections at a zero.

The identical calculation on the two-component module replaces ∂_s by D:

$$Z(\partial_t + \tfrac{1}{4} D^2) Z^{-1} = \partial_t + \tfrac{1}{4} D^2 - \tfrac{1}{2} bD + \tfrac{1}{2} b^2 I. \tag{32}$$

Indeed $D(Zq) = Z_s q + ZDq$, so the product expansion is exactly the scalar one, retaining $A(s)$ in D. This gives the full moving arithmetic relation on the retained cover without assuming a derivative acts on a frozen quotient.

6 The exact branch-sensitive heat defect

The actual pulled arithmetic solution is $q(t, s) = (Z_t(s), 0)$. Equation (6) proves $(\partial_t + \frac{1}{4} D^2)q = 0$. Its function on the cover is $f(t, r) = Z_t(-r^2/2)$. By differentiating $s = -r^2/2$ twice, the scalar generator is

$$L_r = -\frac{1}{4r^2}\partial_r^2 + \frac{1}{4r^3}\partial_r, \qquad \partial_t f = L_r f \quad (r \neq 0). \tag{33}$$

This retains the original spectral diffusion; it does not substitute the constant-coefficient r-heat equation of the fibre polynomial.

The branch observable rZ_t is the column $Rq = (0, Z_t)$. From the already proved commutator its complete defect is

$$(\partial_t + \tfrac{1}{4}D^2)(Rq) = -\tfrac{1}{2}R^{-1}Dq - \tfrac{1}{4}R^{-3}q = \begin{pmatrix} 0 \\ Z_t'/(4s) - Z_t/(16s^2) \end{pmatrix}. \tag{34}$$

Equivalently, as a function on the cover it is $-Z_t'/(2r) - Z_t/(4r^3)$. These terms are the exact dynamics of the retained odd observable, not a reason to discard that observable or declare it unrelated. For real t, $Z_t(0) > 0$ (proved from the integral in (45)), so its leading branch pole is nonzero. The original arithmetic solution itself remains entire at the branch point.

7 Keeping the other original fibre coefficients

The preceding branch at $s = 0$ belongs to the particular slice $b = c = 0$. We now retain the full source family, with b, c fixed and $a = 2s$. Its cover is defined by

$$cr^3 - 2r^2 + br - 4s = 0, \qquad s = \frac{cr^3 - 2r^2 + br}{4}. \tag{35}$$

On its original inverse chart $P_r = 3cr^2 - 4r + b = 2\alpha \neq 0$, implicit differentiation gives

$$\partial_s^{\text{lift}} = \frac{4}{P_r}\partial_r, \qquad L_{b,c} = -\frac{4}{P_r^2}\partial_r^2 + \frac{4P_{rr}}{P_r^3}\partial_r. \tag{36}$$

To check the second identity, apply the first operator twice: $16P_r^{-2}\partial_r^2 - 16P_{rr}P_r^{-3}\partial_r$, then multiply by $-1/4$. Thus the actual function $Z_t((cr^3-2r^2+br)/4)$ satisfies this exact transported equation on $P_r \neq 0$. The branch equations, with no coefficient removed, are

$$b = 4r_0 - 3cr_0^2, \qquad s_0 = \frac{r_0^2 - cr_0^3}{2}. \tag{37}$$

They follow by solving $P_r = 0$ and substituting in (35).

For the two-sheet subfamily $c = 0$, retain arbitrary $b \in \mathbb{C}$ and put $\Delta = b^2 - 32s$. The algebraic relation is $r^2 - (b/2)r + 2s = 0$. In the unchanged ordered basis $(1, r)$,

$$R_b = \begin{pmatrix} 0 & -2s \\ 1 & b/2 \end{pmatrix}, \qquad D_b = \partial_s + A_b, \quad A_b = \begin{pmatrix} 0 & 4b/\Delta \\ 0 & -16/\Delta \end{pmatrix}. \tag{38}$$

These formulas hold on $\Delta \neq 0$. Indeed $(b - 4r)^2 = \Delta$ in the algebra, hence $\partial_s r = 4/(b - 4r) = 4(b - 4r)/\Delta$. This gives both coefficient entries of A_b. Multiplication by r gives R_b. Direct differentiation and multiplication then prove

$$\begin{aligned} R_b^2 - \tfrac{b}{2}R_b + 2sI &= 0, \\ [D_b, R_b] &= 4(bI - 4R_b)/\Delta, \\ D_b^2 &= \partial_s^2 + 2A_b\partial_s + (16/\Delta)A_b. \end{aligned} \tag{39}$$

For the last equality, $A_b' = (32/\Delta)A_b$ and $A_b^2 = (-16/\Delta)A_b$. At $b = 0$ these identities recover (5)–(6) exactly.

The other root is $b/2 - r$. Thus the sheet involution and trace in this original basis are respectively

$$J_b = \begin{pmatrix} 1 & b/2 \\ 0 & -1 \end{pmatrix}, \qquad \operatorname{Tr}_b(q_0, q_1) = 2q_0 + (b/2)q_1.$$

$J_b^2 = I$, and J_b commutes with D_b because b is fixed and $A_b J_b = J_b A_b$. Also $\operatorname{Tr}_b D_b q = \partial_s \operatorname{Tr}_b q$, as the two added terms are $8bq_1/\Delta - 8bq_1/\Delta = 0$. The trace kernel is exactly the set of pairs $(-bq_1/4, q_1)$; the half-trace has the section $q \mapsto (q, 0)$. This explicitly exhibits the branch information and its coupling when the original linear coefficient is present.

The branch point is $r_0 = b/4$, $s_0 = b^2/32$. The exact coordinate change $\eta = r - b/4$, with inverse $r = \eta + b/4$, gives

$$s = s_0 - \eta^2/2, \qquad Z_t(s) = Z_t(s_0 - \eta^2/2). \tag{40}$$

The arithmetic function and its time origin have not changed. The displayed translation applies to the cover only. For any specified $s_0 \in \mathbb{C}$ there is a coefficient b with $b^2 = 32s_0$, so every base point occurs as a branch point of an original $c = 0$ fibre family. If $Z_t(s_0) = 0$ has multiplicity m, its factorization $Z_t(s) = (s - s_0)^m A(s)$, $A(s_0) \neq 0$, pulls back to $(-1/2)^m \eta^{2m} A(s_0 - \eta^2/2)$ and has multiplicity $2m$. This proves the exact contact with any actual arithmetic zero and keeps its analytic unit.

The original inverse point there has

$$\alpha = b/2 - 2r = -2\eta, \quad x = -1/(2\eta), \quad y = b/4 + 3\eta, \quad w = 26\eta^2 + (3b/2)\eta.$$

These follow directly from (2). At $\eta = 0$ its two chart branches escape, while the separate point $(0, b, 2s_0 - 4b^2)$ remains in the finite fibre. Thus the zero-free origin of the $b = 0$ arithmetic pullback does not remove the branch mechanism from the entire original coefficient family. Conversely, choosing a coefficient to place a branch at a specified base point does not prove that that point is an arithmetic zero, or determine whether a zero lies off the real s axis. The formulas state exactly what each operation retains.

8 The actual arithmetic zero divisor on the retained inverse cover

This section uses the arithmetic spectral coordinate s, the Newman time t, and the original inverse-root coordinate r. In particular, the letter s here is not the completed-zeta argument ρ. The complete coordinate dictionary is

$$a = 2s = -r^2, \qquad z = 2s = -r^2, \qquad \rho = \tfrac{1}{2} + is = \tfrac{1}{2} - \tfrac{i}{2}r^2. \tag{41}$$

The maps $s \mapsto a$ and $s \mapsto z$ have inverses $a \mapsto a/2$ and $z \mapsto z/2$; $s \mapsto \rho$ has inverse $\rho \mapsto -i(\rho - 1/2)$. The last root map is a two-sheet map, whose two local inverses on a simply connected open subset of $\mathbb{C}_s \setminus \{0\}$ are $r = \sqrt{-2s}$ and $r = -\sqrt{-2s}$. No single inverse on all of $\mathbb{C}_s \setminus \{0\}$ is asserted.

The primary analytic conventions below are those of Brad Rodgers and Terence Tao[1, Section 1, equations (1)–(4)] and D. H. J. Polymath[4, Section 3, Proposition 3.1]. The local calculations below prove their transport in the coordinates (41), including the analytic factors in zero motion.

8.1 The specified datum, its moments, and the branch point

Keep the exact kernel and constants

$$\begin{aligned} \Phi(u) &= \sum_{n=1}^{\infty} (2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u}) e^{-\pi n^2 e^{4u}}, \\ H_t(z) &= \int_0^{\infty} e^{tu^2} \Phi(u) \cos(zu)\, du, \\ Z(t, s) &= 8H_t(2s), \qquad f(t, r) = Z(t, -r^2/2) = 8H_t(-r^2). \end{aligned} \tag{42}$$

Thus $Z(0,s) = \Xi(s) = \xi(1/2 + is)$ by the cited Fourier identity, where $\xi(\rho) = \rho(\rho - 1)\pi^{-\rho/2}\Gamma(\rho/2)\zeta(\rho)/2$, with its entire continuation at the removable exceptional points. This identity fixes the initial datum; the inverse cubic below is never substituted for it.

For $u \geq 0$ each summand of Φ is positive, since its polynomial factor is

$$\pi n^2 e^{5u}(2\pi n^2 e^{4u} - 3) > 0.$$

For a completely explicit integrable majorant put $C_\Phi = \sum_{n\geq 1} n^4 e^{-\pi(n^2-1)} < \infty$. The finiteness follows by comparison with $\sum_{n\geq 1} n^4 e^{-\pi(n-1)}$. With $\lambda = \pi e^{4u} \geq \pi$, one has

$$0 < \Phi(u) \leq 2\pi^2 e^{9u} \sum_{n\geq 1} n^4 e^{-\lambda n^2} \leq 2\pi^2 C_\Phi e^{9u - \pi e^{4u}}.$$

On $|t| \leq T$, $|s| \leq S$, every derivative of the integrand in (42) is bounded by a constant times $u^N \exp(Tu^2 + (9+2S)u - \pi e^{4u})$, for an integer $N \geq 0$. This is integrable: $\pi e^{4u}/2$ eventually exceeds $Tu^2 + (9+2S)u + N\log(1+u)$, and $\exp(-\pi e^{4u}/2)$ is integrable. The same argument applies on compact r sets, with $2S$ replaced by a bound for $|r|^2$. Differentiation under the integral and locally uniform holomorphic integration therefore show that Z is entire on $\mathbb{C}_t \times \mathbb{C}_s$, and f is entire on $\mathbb{C}_t \times \mathbb{C}_r$. They give the exact equations

$$Z_t = -\tfrac{1}{4} Z_{ss}, \qquad f_t = \mathcal{L} f, \qquad \mathcal{L} = -\frac{1}{4r^2}\partial_r^2 + \frac{1}{4r^3}\partial_r \quad (r \neq 0). \tag{43}$$

Indeed $f_r = -rZ_s$ and $f_{rr} = -Z_s + r^2 Z_{ss}$, so the two terms in $\mathcal{L}f$ are $Z_s/(4r^2) - Z_{ss}/4$ and $-Z_s/(4r^2)$, respectively. Their sum is $-Z_{ss}/4$.

Define the actual moments, with their original measure,

$$M_k(t) = \int_0^\infty u^{2k} e^{tu^2} \Phi(u)\, du \quad (k = 0, 1, 2, \ldots).$$

The majorant just proved gives $M_k'(t) = M_{k+1}(t)$ and

$$\partial_s^{2k} Z(t,0) = 8(-1)^k 2^{2k} M_k(t), \qquad \partial_s^{2k+1} Z(t,0) = 0,$$
$$f(t,r) = 8\sum_{k=0}^{\infty} \frac{(-1)^k M_k(t)}{(2k)!} r^{4k}. \tag{44}$$

The series converges locally uniformly in both variables, by expansion of the cosine and the same compact-set majorant. In particular

$$\Xi(0) = 8M_0(0) = -\frac{\Gamma(1/4)\zeta(1/2)}{8\pi^{1/4}} > 0,$$
$$\Xi'(0) = 0, \qquad \Xi''(0) = -32 M_1(0) < 0,$$
$$f(t,r) = 8M_0(t) - 4M_1(t) r^4 + \tfrac{1}{3} M_2(t) r^8 + O(r^{12}).$$

The equality with $\Gamma(1/4)\zeta(1/2)$ follows by evaluating the specified completed-zeta identity, while both strict signs follow directly from the positive integrals. Neither sign requires a numerical evaluation of $\zeta(1/2)$.

For every real t and v there is the stronger exact assertion, also recorded in the first paragraph of Polymath's Section 3[4],

$$Z(t, iv) = 8\int_0^\infty e^{tu^2} \Phi(u) \cosh(2vu)\, du > 0. \tag{45}$$

In particular the actual arithmetic zero divisor avoids $s = 0$ for every real Newman time. The completed root cover has a branch point there, but no arithmetic zero reaches that branch point at any real time. Evenness of the cosine also proves, for all complex t, s, r,

$$Z(t,-s) = Z(t,s), \qquad f(t,-r) = f(t,r), \qquad f(t,ir) = f(t,r). \tag{46}$$

For real t complex conjugation gives $\overline{Z(t,s)} = Z(t,\overline{s})$ and $\overline{f(t,r)} = f(t,\overline{r})$, because the kernel is real.

8.2 Exact function maps and the removable and nonremovable terms

Write $\mathcal{O}(\mathbb{C})$ for entire functions with the topology of uniform convergence on compact sets. The pullback

$$\pi^* : \mathcal{O}(\mathbb{C}_s) \longrightarrow \mathcal{O}(\mathbb{C}_r), \qquad q \longmapsto q(-r^2/2)$$

is injective, since every $s \in \mathbb{C}$ has a root of $r^2 = -2s$. Its image is exactly the even entire functions. To prove the latter statement without a choice of square root, expand an even entire function as $g(r) = \sum_{n\geq 0} a_{2n} r^{2n}$ and define $q(s) = \sum_{n\geq 0} (-2)^n a_{2n} s^n$. This series converges for every s: choose $R > \sqrt{2|s|}$ and apply the Cauchy bound $|a_{2n}| \leq \max_{|r|=R} |g(r)| R^{-2n}$. It gives $q(-r^2/2) = g(r)$, and proves uniqueness by comparing coefficients. It also proves continuity of the inverse on compact sets. Every entire cover function has the unique decomposition

$$g(r) = q_0(-r^2/2) + r q_1(-r^2/2).$$

Here q_0 is recovered from $(g(r) + g(-r))/2$ and q_1 from $(g(r) - g(-r))/(2r)$, with the latter expression extended at $r = 0$ by its power series. The sheet involution $\iota g(r) = g(-r)$ has trace map $g \mapsto g + \iota g$ with kernel exactly the odd entire functions. These are maps of actual holomorphic functions, before any arithmetic quotient is formed.

For every integer $n \geq 0$ the operator in (43) gives

$$\mathcal{L}(r^n) = \frac{n(2-n)}{4} r^{n-4}. \tag{47}$$

Consequently, for $g(r) = \sum_{n\geq 0} a_n r^n$, its Laurent expansion is

$$\mathcal{L}g = \frac{a_1}{4r^3} - \frac{3a_3}{4r} + \sum_{n\geq 4} \frac{n(2-n)}{4} a_n r^{n-4}.$$

The terms $n = 0, 2$ are zero. Thus $\mathcal{L}g$ extends to an entire function exactly when $a_1 = a_3 = 0$. Its value at zero in that case is $-2a_4 = -g^{(4)}(0)/12$. In particular $\mathcal{L}f(t, 0) = 8M_1(t) = f_t(t, 0)$: the singular coordinate coefficients cancel on the actual arithmetic datum.

The largest subspace of $\mathcal{O}(\mathbb{C}_r)$ on which every iterate $\mathcal{L}^j$ remains entire is the even subspace. For even monomials successive applications of (47) reach r^0 or r^2, which are annihilated, without producing a pole. Equivalently, $\mathcal{L}^j \pi^* q = (-1/4)^j \pi^* q^{(2j)}$. Conversely, let a_n be the first nonzero odd coefficient of g. When $n = 4j + 1$, applying $\mathcal{L}^{j+1}$ produces a nonzero multiple of r^{-3} from this term; when $n = 4j + 3$ it produces a nonzero multiple of r^{-1}. Every factor $(n - 4\ell)(2 - n + 4\ell)/4$ is nonzero for these odd exponents. Distinct initial powers retain distinct exponents after the same number of applications, so no cancellation with a higher power is possible. Even powers cannot contribute a pole. This proves the assertion and maximality. It asserts stability of all differential iterates, not existence of a heat evolution for every even entire datum.

There is an equivalent spacetime statement. A function g holomorphic on a product of a time disk with a full disk about $r = 0$, satisfying $g_t = \mathcal{L}g$ for $r \neq 0$, is even in r. Indeed successive time differentiation gives $\mathcal{L}^j g = \partial_t^j g$ on the punctured disk, and the right side is holomorphic at zero for every j. The preceding first-odd-coefficient argument, which is local in r, applies at every time. The equation therefore descends there through the proved even-function inverse to $q_t = -q_{ss}/4$. On a connected punctured plane the kernel of $\mathcal{L}$ consists exactly of $A + Br^2$: multiplying the equation by $4r^3$ gives $r g_{rr} = g_r$, hence $(g_r/r)' = 0$ and then $g = A + Br^2$.

8.3 The original cubic, all finite fibres, and the zero divisor

Keep the original map in spatial coordinates (x, y, w):

$$F_1 = (1+xy)^3 w + y^2(1+xy)(4+3xy),$$
$$F_2 = y + 3x(1+xy)^2 w + 3xy^2(4+3xy),$$
$$F_3 = 2x - 3x^2 y - x^3 w.$$

Its retained polynomial is

$$P_{a,b,c}(r) = cr^3 - 2r^2 + br - 2a, \qquad \alpha = \tfrac{1}{2}P'_{a,b,c}(r).$$

For $x \neq 0$ the inverse chart is

$$x = \alpha^{-1}, \qquad y = r - \alpha, \qquad w = 5\alpha^2 - 3r\alpha - c\alpha^3, \qquad P_{a,b,c}(r) = 0, \quad P'_{a,b,c}(r) \neq 0. \tag{48}$$

Here is the needed exact verification. Set $r = y + 1/x$ and $\alpha = 1/x$. Solving $F_3 = c$ gives the displayed expression for w. Substitution in the other components gives $F_2 = 4r + 2\alpha - 3cr^2$ and $F_1 = r^2 + r\alpha - cr^3$. The equality $F_2 = b$ is $2\alpha = P'(r)$, and after its substitution $2(F_1 - a) = P(r)$. These identities prove both directions and recover r from the inverse image. In particular a multiple root $P' = 0$ has no finite point in this chart. On $x = 0$, direct substitution gives $F(0, y, w) = (w + 4y^2, y, 0)$; there is the additional point $(0, b, a - 4b^2)$ precisely when $c = 0$.

The arithmetic slice $a = 2s$, $b = c = 0$ now gives, with no omitted coefficient,

$$P_{2s,0,0}(r) = -2r^2 - 4s, \qquad \alpha = -2r.$$

Its root relation is exactly $s = -r^2/2$. For $r \neq 0$ the spatial inverse is the explicit map

$$\Gamma(r) = \left(-\frac{1}{2r},\ 3r,\ 26r^2\right), \qquad F(\Gamma(r)) = (-r^2, 0, 0). \tag{49}$$

The inverse root is $y + 1/x = 3r - 2r = r$. For $s \neq 0$, its two roots give two distinct finite points $\Gamma(r)$ and $\Gamma(-r)$, and the complete original fibre also has the point $(0, 0, 2s)$. For $s = 0$ the polynomial has its double root at zero, both inverse chart points are excluded by $\alpha = 0$, and the sole finite fibre point is $(0, 0, 0)$. The completed root point $r = 0$ is not silently assigned the undefined spatial coordinate $x = -1/(2r)$.

For real t, every zero s_* of the specified $Z(t, \cdot)$ is nonzero by (45). It therefore has exactly two zero lifts r_* and $-r_*$ on the retained cover, and these have the finite spatial representatives in (49). The actual scalar $Z(t, F_1(x, y, w)/2)$ vanishes on all three points of the original fibre above $(2s_*, 0, 0)$. Thus the odd sheet observable is retained by the cover, and the extra $x = 0$ fibre point is separately recorded.

Multiplicity is also transported exactly. At any fixed complex time, write $Z(t, s) = (s - s_*)^m A(s)$ at a zero, where $A(s_*) \neq 0$. For $r_* \neq 0$ with $r_*^2 = -2s_*$,

$$f(t, r) = \left[-\tfrac{1}{2}(r - r_*)(r + r_*)\right]^m A(-r^2/2).$$

The factor $[-(r + r_*)/2]^m A(-r^2/2)$ is a unit near r_*, so the multiplicity is m there and also at $-r_*$. The local coordinate $s - s_* = -(r - r_*)(r + r_*)/2$ has derivative $-r_* \neq 0$; its holomorphic inverse is the corresponding branch of $r = \sqrt{-2s}$. It gives isomorphisms of the local function rings and of their quotients by the respective zero ideals. At $s_* = 0$ the exact ideal is instead $(s^m)\mathcal{O}_{\mathbb{C}_r,0} = (r^{2m})$; the factor $(-1/2)^m A(-r^2/2)$ is a unit. Thus the multiplicity doubles there. The quotient at that point is the rank-two module

$$\mathbb{C}\{r\}/(r^{2m}) = \mathbb{C}\{s\}/(s^m) \ \oplus\ r\mathbb{C}\{s\}/(s^m), \qquad s = -r^2/2.$$

Separating even and odd coefficients proves existence and uniqueness of this decomposition; the trace has precisely its odd summand as kernel. These local divisor rings do not assert that differentiation acts on a frozen global arithmetic spectral quotient.

For $r = u + iv$ with $u, v \in \mathbb{R}$, the original arithmetic point is

$$s = \frac{v^2 - u^2}{2} - iuv, \qquad \rho = \frac{1}{2} + uv + i\frac{v^2 - u^2}{2}. \tag{50}$$

It lies on $\operatorname{Re}\rho = 1/2$ exactly when $uv = 0$, which is exactly the union of the real and imaginary r axes. For real t there are no cover zeros on $u = v$ or $u = -v$, since these lines map into the imaginary s axis of (45). In particular an imaginary root coordinate alone does not give an off-critical arithmetic zero; the exact arithmetic test in this coordinate is $uv \neq 0$ at a zero of $f(0, r)$.

8.4 Simple zero motion with every analytic-unit term

Consider a point of the actual zero locus with $Z_s(t_0, s_0) \neq 0$. The holomorphic implicit function theorem applies because Z is jointly entire and this derivative is nonzero. It gives a unique holomorphic zero $s = \sigma(t)$ in a product of small disks. Differentiating $Z(t, \sigma(t)) = 0$ and using (43) gives the exact velocity

$$\sigma'(t) = \frac{Z_{ss}(t, \sigma(t))}{4Z_s(t, \sigma(t))}. \tag{51}$$

More generally write a local factorization $Z = AQ$, where A is a nonvanishing holomorphic function of (t, s) and Q is the full local zero factor under consideration. The product rule gives

$$Q_t = -\tfrac{1}{4}Q_{ss} - \frac{A_s}{2A}Q_s - \left(\frac{A_t}{A} + \frac{A_{ss}}{4A}\right)Q. \tag{52}$$

All functions in this equality are evaluated on that product of disks; dividing by A is valid there. At a simple root it gives

$$\sigma' = \frac{1}{4}\frac{Q_{ss}}{Q_s} + \frac{1}{2}\frac{A_s}{A}. \tag{53}$$

If $Q = \prod_{j=1}^{m}(s - \sigma_j(t))$ has distinct roots at the time in question, factor off $s - \sigma_j$ and differentiate the remaining product. This proves

$$\frac{Q_{ss}(\sigma_j)}{Q_s(\sigma_j)} = 2\sum_{k \neq j}\frac{1}{\sigma_j - \sigma_k}, \qquad \sigma_j' = \frac{1}{2}\sum_{k \neq j}\frac{1}{\sigma_j - \sigma_k} + \frac{1}{2}\frac{A_s(t, \sigma_j)}{A(t, \sigma_j)}.$$

For a single local root $Q = s - \sigma$, its entire velocity is $A_s/(2A)$. Removing the unit would remove this actual velocity.

Away from $r = 0$, a lifted simple root obeys $r^2 = -2\sigma$, so differentiation and the computed spatial derivatives give

$$r' = -\frac{\sigma'}{r} = -\frac{1}{4r}\frac{Z_{ss}}{Z_s} = \frac{1}{4r^2}\frac{f_{rr}}{f_r} - \frac{1}{4r^3}. \tag{54}$$

All ratios are evaluated at the moving zero. For a factorization $f = BV$ on a cover disk, the complete product equation is

$$\begin{aligned} V_t = &-\frac{1}{4r^2}V_{rr} + \left(\frac{1}{4r^3} - \frac{B_r}{2r^2B}\right)V_r \\ &+ \left(-\frac{B_{rr}}{4r^2B} + \frac{B_r}{4r^3B} - \frac{B_t}{B}\right)V, \end{aligned} \tag{55}$$

and at a simple root

$$r' = \frac{1}{4r^2}\frac{V_{rr}}{V_r} + \frac{B_r}{2r^2B} - \frac{1}{4r^3}.$$

For the two complete sheet lifts of a single base zero, take $V = r^2 + 2\sigma(t)$ and $B(t,r) = -A(t,-r^2/2)/2$, where $Z = A(s-\sigma)$. At a root $V_{rr}/V_r = 1/r$. Its term $1/(4r^3)$ cancels the displayed $-1/(4r^3)$ exactly, while $B_r/B = -rA_s/A$. The result is $r' = -A_s/(2rA) = -\sigma'/r$. This computation retains both sheet roots and the analytic unit; it identifies the cancellation rather than deleting either term. The original spatial velocity is also explicit:

$$\frac{d}{dt}\Gamma(r(t)) = \left(\frac{r'}{2r^2},\ 3r',\ 52rr'\right), \qquad \frac{d}{dt}F(\Gamma(r(t))) = (2\sigma', 0, 0).$$

8.5 A double zero: exact pair equations and the unit drift

At an actual double zero (t_0, s_0), put $h = t - t_0$ and write

$$Z(t_0, s_0 + \delta) = a_2\delta^2 + a_3\delta^3 + O(\delta^4), \qquad a_2 \neq 0.$$

The local zero factor can be expressed as $Q = (s-M)^2 - D^2$, where M and D^2 are holomorphic in t, $M(t_0) = s_0$, and $D^2(t_0) = 0$. A construction not depending on the naming of the two roots is to take a small circle containing only the double zero and no boundary zero. The integrals

$$p_k(t) = \frac{1}{2\pi i}\int_{\text{circle}} s^k \frac{Z_s(t,s)}{Z(t,s)}\, ds \quad (k = 1, 2)$$

are holomorphic and equal the first two root power sums with multiplicity, by the residue computation for a local zero factor. Thus $M = p_1/2$ and $D^2 = p_2/2 - M^2$ are holomorphic. The two zeros, counted with multiplicity, are $M + D$ and $M - D$. The holomorphic unit can itself be constructed on a smaller concentric disk by the contour formula

$$A(t,s) = \frac{1}{2\pi i}\int_{\text{circle}} \frac{Z(t,\zeta)}{Q(t,\zeta)(\zeta - s)}\, d\zeta.$$

The denominator is nonzero on the contour for all the retained times, so the integral is jointly holomorphic. At each fixed time Z/Q extends holomorphically across its two zeros, counted with their multiplicities, by their one-variable Taylor factorizations. Cauchy's formula therefore identifies the displayed integral with Z/Q, proving $Z = AQ$. At $t = t_0$ it is $Z(t_0, s)/(s - s_0)^2$, nonzero at s_0, so A is nonvanishing after shrinking the disks.

Put $b(t,s) = A_s(t,s)/A(t,s)$. At every nearby time with $D \neq 0$ the proved simple-zero equation gives the exact pair dynamics

$$\begin{aligned}
(M+D)' &= \frac{1}{4D} + \tfrac{1}{2}b(t, M+D),\\
(M-D)' &= -\frac{1}{4D} + \tfrac{1}{2}b(t, M-D),\\
M' &= \tfrac{1}{4}\big[b(t,M+D) + b(t,M-D)\big],\\
(D^2)' &= \tfrac{1}{2} + \tfrac{D}{2}\big[b(t,M+D) - b(t,M-D)\big].
\end{aligned} \tag{56}$$

The sum and D times the difference are even holomorphic functions of D, hence holomorphic functions of D^2. These equations extend to $h = 0$, and there give

$$(D^2)'(t_0) = \tfrac{1}{2}, \qquad M'(t_0) = \frac{a_3}{2a_2} = \frac{Z_{sss}(t_0,s_0)}{6Z_{ss}(t_0,s_0)}.$$

Indeed $A(t_0, s_0) = a_2$ and $A_s(t_0, s_0) = a_3$. The full equation (52) still carries A_t and A_{ss}; their factors multiply Q and vanish only when evaluated at a zero. They have not been assumed to vanish on the neighborhood.

The following explicit expansion proves the pair's first unit term and its error order. The heat equation gives the terms of weighted degree at most three, with δ of weight one and h of weight two:

$$Z(t_0 + h, s_0 + \delta) = a_2(\delta^2 - h/2) + a_3(\delta^3 - 3h\delta/2) + O((|\delta| + |h|^{1/2})^4).$$

Set $h = \varepsilon^2$. Dividing by ε^2 after $\delta = \varepsilon y$ gives a holomorphic function of (ε, y) whose value at $\varepsilon = 0$ is $a_2(y^2 - 1/2)$. Its two simple roots are $y = \pm 1/\sqrt{2}$, so the holomorphic implicit function theorem gives two holomorphic y branches. Their next coefficient is found by substituting $\delta = \lambda\varepsilon + \mu\varepsilon^2$: $\lambda^2 = 1/2$ and $2a_2\lambda\mu + a_3(\lambda^3 - 3\lambda/2) = 0$. Hence $\mu = a_3/(2a_2)$ and

$$\sigma_\pm(t) = s_0 \pm \sqrt{h/2} + \frac{a_3}{2a_2}h + O(h^{3/2}). \tag{57}$$

This means convergent holomorphic expansions in ε, with a chosen $\varepsilon^2 = h$; reversing ε exchanges the root labels. It follows that, on either punctured time branch,

$$\sigma'_\pm = \pm\frac{1}{2\sqrt{2h}} + \frac{a_3}{2a_2} + O(h^{1/2}).$$

For real t_0, s_0 the coefficients are real. For small positive h both branches are real, by uniqueness and conjugation of their real ε power series. For small negative real h their imaginary parts are $\pm\sqrt{|h|/2} + O(|h|^{3/2})$, so they are a nonreal conjugate pair. This describes an actual double-zero event wherever it occurs; it does not insert such an event at an unspecified real arithmetic time.

At real time $s_0 \neq 0$ by (45). Choose either $r_0^2 = -2s_0$. The inverse branch has $dr/ds = -1/r_0$ and $d^2r/ds^2 = -1/r_0^3$. Taylor substitution in (57) yields the two roots near that particular sheet point,

$$r_\pm(t) = r_0 \mp \frac{1}{r_0}\sqrt{h/2} - \left(\frac{a_3}{2a_2r_0} + \frac{1}{4r_0^3}\right)h + O(h^{3/2}). \tag{58}$$

The other sheet consists exactly of their negatives. Both the arithmetic unit drift and the curvature of the original root map occur in its displayed coefficient of h.

8.6 Every multiplicity and its interaction with ramification

At an actual zero of finite multiplicity $m \geq 1$ write $Z(t_0, s_0 + \delta) = \sum_{k \geq m} a_k\delta^k$, $a_m \neq 0$. There is finite multiplicity because $Z(t_0, \cdot)$ is not the zero function. For real t_0 positivity at zero already proves this. For complex t_0 put $K(u) = e^{t_0u^2}\Phi(|u|)$ on $\mathbb{R}$. The proved decay shows that K is continuous, bounded, and integrable, and $K(0) = \Phi(0) > 0$. Its Fourier transform satisfies $\widehat{K}(\xi) = 2H_{t_0}(\xi)$ for real ξ, by evenness. If $Z(t_0, \cdot)$ were identically zero, this transform would vanish. For $\epsilon > 0$, the Gaussian integral gives

$$G_\epsilon(x) = \frac{e^{-x^2/(4\epsilon)}}{\sqrt{4\pi\epsilon}} = \frac{1}{2\pi}\int_{\mathbb{R}} e^{-\epsilon\xi^2}e^{ix\xi}\,d\xi.$$

Fubini is justified by $K \in L^1$ and the integrable Gaussian in ξ; it gives $(K * G_\epsilon)(0) = (2\pi)^{-1}\int \widehat{K}(\xi)e^{-\epsilon\xi^2}\,d\xi = 0$. On the other hand $(K * G_\epsilon)(0) \to K(0) > 0$ as $\epsilon \downarrow 0$: on $|u| < \delta$ continuity bounds $|K(u) - K(0)|$ arbitrarily small, while on its complement boundedness of K and the vanishing Gaussian tail bound the remaining integral. This contradiction proves the required nonidentity with the zero function for every complex time.

Repeated differentiation of the heat equation gives $\partial_t^\ell Z = (-1/4)^\ell\partial_s^{2\ell}Z$. The bivariate Taylor coefficient of $h^\ell\delta^j$ is therefore

$$\frac{(-1/4)^\ell}{\ell!}\frac{(j + 2\ell)!}{j!}a_{j+2\ell}.$$

Set $h = \varepsilon^2$, $\delta = \varepsilon y/\sqrt{2}$, with the inverse substitutions retained whenever $\varepsilon \neq 0$. The holomorphic Taylor series is divisible by ε^m; grouping the terms of degree m and $m+1$ gives

$$\frac{Z(t_0+\varepsilon^2, s_0+\varepsilon y/\sqrt{2})}{a_m(\varepsilon/\sqrt{2})^m} = \mathrm{He}_m(y) + \frac{a_{m+1}}{a_m}\frac{\varepsilon}{\sqrt{2}}\mathrm{He}_{m+1}(y) + O(\varepsilon^2),$$

$$\mathrm{He}_m(y) = \sum_{\ell=0}^{\lfloor m/2\rfloor} \frac{m!(-1)^\ell}{2^\ell \ell!(m-2\ell)!} y^{m-2\ell}. \tag{59}$$

The quotient extends holomorphically at $\varepsilon = 0$; its remainder is uniform on each bounded y set. This follows directly from the convergent bivariate Taylor series after substitution, so the error does not assume a finite polynomial initial datum.

For completeness the polynomial has m distinct real roots. Differentiating its displayed finite sum proves $\mathrm{He}_m' = m\,\mathrm{He}_{m-1}$ and $\mathrm{He}_{m+1} = y\,\mathrm{He}_m - m\,\mathrm{He}_{m-1}$. Starting with $\mathrm{He}_0 = 1$ and $\mathrm{He}_1 = y$, induction shows that the roots of two successive polynomials strictly interlace: at consecutive roots of He_m, the values $-m\,\mathrm{He}_{m-1}$ alternate sign, providing one root of He_{m+1} between each consecutive pair. At the largest root, $\mathrm{He}_{m-1} > 0$, so $\mathrm{He}_{m+1} < 0$ there and is positive sufficiently far to the right. At the smallest root, the sign is $(-1)^m$ and the far-left sign of its monic degree $m+1$ polynomial is $(-1)^{m+1}$, providing a further root to the left. There are thus $m+1$ distinct roots. The induction begins with the single root of He_1 and gives the claimed reality, simplicity, and interlacing. Let these roots be λ_j. At a root of He_m the recurrences also give

$$\frac{\mathrm{He}_{m+1}(\lambda_j)}{\mathrm{He}_m'(\lambda_j)} = -1.$$

Apply the holomorphic implicit function theorem to (59) at each $(0, \lambda_j)$. The resulting roots are

$$\sigma_j(t) = s_0 + \lambda_j\sqrt{h/2} + \frac{a_{m+1}}{2a_m}h + O(h^{3/2}). \tag{60}$$

They are all the nearby roots: a small fixed circle about s_0 has no boundary zero at t_0, and continuity followed by Rouché's theorem keeps its zero count at m; the m disjoint roots just constructed account for that count. Writing $Z(t_0, s) = (s-s_0)^m A_0(s)$ gives $a_{m+1}/(2a_m) = A_0'(s_0)/(2A_0(s_0))$. Thus the first common drift at every multiplicity is exactly the initial analytic-unit derivative. Higher coefficients remain determined by the full Taylor series and the exact unit equation (52).

For $s_0 \neq 0$, on either chosen sheet $r_0^2 = -2s_0$ the same inverse coordinate Taylor expansion proves

$$r_j(t) = r_0 - \frac{\lambda_j}{r_0}\sqrt{h/2} - \left(\frac{a_{m+1}}{2a_m r_0} + \frac{\lambda_j^2}{4r_0^3}\right)h + O(h^{3/2}).$$

At $s_0 = 0$, an event of the actual family is possible only at nonreal complex time, by (45). On this locus the exact evenness $Z(t,-s) = Z(t,s)$ forces $m = 2k$ and $a_{m+1} = 0$. Its Hermite roots are nonzero, since $\mathrm{He}_{2k}(0) = (-1)^k(2k)!/(2^k k!)$. Set $h = \eta^4$ and $\varepsilon = \eta^2$. Equation (60) now gives

$$\sigma_j(t_0+\eta^4) = \frac{\lambda_j}{\sqrt{2}}\eta^2 + O(\eta^6), \qquad r_{j,\pm}(\eta) = \pm\eta\sqrt{-\sqrt{2}\lambda_j} + O(\eta^5).$$

The square root of each nonzero displayed constant has two choices. To justify the expansion, $-2\sigma_j(t_0+\eta^4)/\eta^2$ extends holomorphically and has nonzero value $-\sqrt{2}\lambda_j$ at zero; either holomorphic square root of that unit gives exactly $r_{j,\pm} = \pm\eta\sqrt{-2\sigma_j/\eta^2}$. The $2m$ lifted roots account for the cover multiplicity $2m$ proved above. This is the precise fourth-root time ramification over an arithmetic multiple zero at the cover branch locus. It does not occur there at real Newman time, and it has no finite spatial inverse chart at its central point because $P' = 0$.

Verification status. The accompanying exact rational-polynomial checker verifies the chain rule coefficients, monomial pole coefficients, original spatial lift, analytic-unit product identities, and finite coefficients of the collision expansions. The convergence, positivity, local holomorphic factorizations, root counts, and complete multiplicity statements are proved in the text; the checker is not a substitute for those proofs. No off-critical zero of the distinguished arithmetic initial function is asserted by these calculations.

9 The actual arithmetic observable under spatial diffusion

Retain the original real spatial coordinates $q = (x, y, w)$ and the polynomial F in (1). The complex formulas below are their holomorphic extensions; a sum of derivative squares is a Euclidean squared norm only on the stated real locus. Define the exact observable

$$\sigma(q) = F_1(q)/2, \qquad \Psi(t, q) = Z_t(\sigma(q)).$$

The exact map of time and spatial coordinates is $\rho : \mathbb{C}_t \times \mathbb{C}_q^3 \to \mathbb{C}_t \times \mathbb{C}_s$, $\rho(t, q) = (t, \sigma(q))$, with pullback $\rho^* Z = \Psi$. For each differentiable time map $\vartheta : I_\tau \to \mathbb{C}_t$ its restriction is $\rho_\vartheta(\tau, q) = (\vartheta(\tau), \sigma(q))$. The chain rule gives $\partial_\tau \rho_\vartheta^* Z = \vartheta' \rho_\vartheta^* Z_t$ and $\partial_{q_i} \rho_\vartheta^* Z = (\partial_{q_i}\sigma)\rho_\vartheta^* Z_s$. These formulas specify the typed relation between Newman time and spatial time with its derivative factor. The already proved entire Z makes Ψ entire in $(t, q) \in \mathbb{C} \times \mathbb{C}^3$.

9.1 Original vector fields and the rank-one heat operator

Here is a direct determinant proof needed to define the vector fields. Use the already proved inverse-chart map

$$H(\alpha, r, c) = (\alpha^{-1}, r - \alpha, 5\alpha^2 - 3r\alpha - c\alpha^3), \quad \alpha \neq 0.$$

Its inverse on $x \neq 0$ is $(1/x, y + 1/x, F_3(q))$. The two derivative matrices, in the displayed coordinate order, are

$$DH = \begin{pmatrix} -\alpha^{-2} & 0 & 0 \\ -1 & 1 & 0 \\ -3c\alpha^2 + 10\alpha - 3r & -3\alpha & -\alpha^3 \end{pmatrix}, \quad D(F \circ H) = \begin{pmatrix} r & \alpha - 3cr^2 + 2r & -r^3 \\ 2 & 4 - 6cr & -3r^2 \\ 0 & 0 & 1 \end{pmatrix}.$$

The first determinant is α. Expansion along the last row of the second gives $r(4 - 6cr) - 2(\alpha - 3cr^2 + 2r) = -2\alpha$. The chain rule therefore proves $\det DF = -2$ on $x \neq 0$. Both sides are polynomials in (x, y, w), so their equality extends to $x = 0$ by continuity. Put $J = DF$, $W_i = J^{-1} e_i$, and

$$U = 2W_1 + 6F_3 W_2.$$

These are globally defined polynomial vector fields, since $J^{-1} = -(1/2)\operatorname{adj} J$. With $\epsilon_{123} = 1$ and cyclic indices, the adjugate formula is $W_i = -\frac{1}{2}\nabla F_{i+1} \times \nabla F_{i+2}$. The divergence of each cross product is zero: its differentiated components contract a symmetric Hessian with an antisymmetric ϵ tensor. Also $W_i F_j = \delta_{ij}$, by $JW_i = e_i$. The product rule now gives

$$\operatorname{div} U = 2 \operatorname{div} W_1 + 6F_3 \operatorname{div} W_2 + 6W_2 F_3 = 0, \qquad U\sigma = 1.$$

For every twice differentiable scalar $h(s)$, direct differentiation proves $U(h \circ \sigma) = h'(\sigma)$ and $U^2(h \circ \sigma) = h''(\sigma)$. Consequently the actual observable satisfies the exact operator identity

$$\partial_t \Psi + \tfrac{1}{4} U^2 \Psi = 0. \tag{61}$$

Here the full spatial operator is

$$U^2h = U^iU^j\partial_i\partial_j h + U^i(\partial_i U^j)\partial_j h = \partial_i(U^iU^j\partial_j h),$$

where repeated indices run over the three original coordinates. The last equality uses the proved divergence identity. Its real principal tensor is $UU^{\mathsf{T}}/4$, of rank one because $U\sigma = 1$ implies $U \neq 0$. On $C_c^\infty(\mathbb{R}^3)$, one integration by parts gives $-\int \overline{h}U^2h = \int |Uh|^2$; polynomial coefficients are bounded on the support and there are no boundary terms. The explicit reversal $\Phi(\tau, q) = \Psi(-\tau, q)$ obeys $\partial_\tau\Phi = U^2\Phi/4$. This proves an exact forward spatial heat realization on the specified observable, with every first-order coefficient retained.

9.2 Every Euclidean diffusion coefficient

Let $Q = 1 + xy$, an abbreviation for the original polynomial factor. Differentiation before restriction to a fibre gives

$$\begin{aligned} g_1 &:= \partial_x\sigma = \tfrac{3}{2}yQ^2w + \tfrac{1}{2}y^3(7 + 6xy),\\ g_2 &:= \partial_y\sigma = \tfrac{3}{2}xQ^2w + \tfrac{1}{2}(8y + 21xy^2 + 12x^2y^3),\\ g_3 &:= \partial_w\sigma = \tfrac{1}{2}Q^3. \end{aligned}$$

Define $N = g_1^2 + g_2^2 + g_3^2$ and

$$L = \Delta_q\sigma = 3wQ(x^2 + y^2) + 3y^4 + 18x^2y^2 + 21xy + 4.$$

For example the second derivatives in the Laplacian are $\sigma_{xx} = 3wy^2Q + 3y^4$, $\sigma_{yy} = 3wx^2Q + 18x^2y^2 + 21xy + 4$, and $\sigma_{ww} = 0$, proving every term in L. The chain rule in each coordinate, followed by summation, proves

$$\Delta_q\Psi = N(q)Z_{ss}(t, \sigma(q)) + L(q)Z_s(t, \sigma(q)). \tag{62}$$

No term is suppressed by the one-variable presentation of Z. For real q, $N = |\nabla\sigma|^2 > 0$: an invertible J cannot have its first row zero.

On the original finite inverse lift $\Gamma(r) = (-1/(2r), 3r, 26r^2)$, $r \neq 0$, direct substitution gives

$$\nabla\sigma(\Gamma(r)) = (9r^3/4, 3r/8, -1/16)^{\mathsf{T}}, \qquad D^2\sigma(\Gamma(r)) = \begin{pmatrix} -108r^4 & 3r^2/4 & 9r/8 \\ 3r^2/4 & 13/4 & -3/(16r) \\ 9r/8 & -3/(16r) & 0 \end{pmatrix}.$$

Thus, retaining $\sigma(\Gamma(r)) = -r^2/2$,

$$\begin{aligned} N(\Gamma(r)) &= \frac{1296r^6 + 36r^2 + 1}{256},\\ L(\Gamma(r)) &= \frac{13}{4} - 108r^4,\\ (\Delta_q\Psi)(t, \Gamma(r)) &= \frac{1296r^6 + 36r^2 + 1}{256}Z_{ss}(t, -r^2/2) + \left(\frac{13}{4} - 108r^4\right)Z_s(t, -r^2/2). \end{aligned} \tag{63}$$

These are identities of rational holomorphic functions for complex $r \neq 0$. Their Euclidean-norm interpretation uses real $r \neq 0$.

For a real differentiable time map $t = \theta(\tau)$ and the original viscosity $\nu > 0$, the complete advection–diffusion residual is

$$\begin{aligned} &(\partial_\tau + U \cdot \nabla_q - \nu\Delta_q)\Psi(\theta(\tau), q)\\ &\quad = \left(-\frac{\theta'(\tau)}{4} - \nu N(q)\right)Z_{ss}(\theta(\tau), \sigma(q)) + (1 - \nu L(q))Z_s(\theta(\tau), \sigma(q)). \end{aligned} \tag{64}$$

The three preceding exact identities prove this formula term by term. It is the direct map between the actual arithmetic heat solution and the scalar spatial advection–diffusion operator of this polynomial flow. It does not assert that its displayed residual vanishes.

9.3 The particular arithmetic covector along the escaping orbit

Local existence and uniqueness of the smooth polynomial ODE $\partial_\tau X_\tau = U(X_\tau)$ follow on a closed finite ball from the contraction map $Y \mapsto q + \int_0^\tau U(Y(v))\,dv$ for sufficiently short time; bounded DU supplies its Lipschitz constant. Continuing these solutions defines the maximal local flow on its actual open domains. Smooth dependence follows by differentiating the integral equation; uniqueness supplies the inverse local flow. The identities $UF_1 = 2$, $UF_2 = 6F_3$, and $UF_3 = 0$ integrate to

$$F(X_\tau(q)) = (F_1(q) + 2\tau, F_2(q) + 6\tau F_3(q), F_3(q)).$$

In particular $\sigma(X_\tau(q)) = \sigma(q) + \tau$. Write $A_\tau = D_q X_\tau$ and $K_\tau = A_\tau^{-1} A_\tau^{-\mathsf{T}}$. Differentiating this last scalar identity gives the full covector map

$$A_\tau^{\mathsf{T}} \nabla\sigma(X_\tau(q)) = \nabla\sigma(q), \qquad \nu \nabla\sigma(q)^{\mathsf{T}} K_\tau \nabla\sigma(q) = \nu N(X_\tau(q)).$$

Every matrix is evaluated at the same initial point q and time τ; no arbitrary covector has replaced this actual one.

At $q_0 = (1, -3/2, 13/2) = \Gamma(-1/2)$ put $z = \sqrt{1 - 8\tau} > 0$, with inverse $\tau = (1 - z^2)/8$. For $-\infty < \tau < 1/8$ the explicit trajectory is $X_\tau(q_0) = \Gamma(-z/2) = (z^{-1}, -3z/2, 13z^2/2)$. To verify the ODE, its F image is $(-1/4 + 2\tau, 0, 0)$; differentiation gives $J\gamma' = (2, 0, 0) = JU(\gamma)$, so invertibility of J gives $\gamma' = U(\gamma)$. Its value at $\tau = 0$ is q_0, and its escape as $\tau \uparrow 1/8$ precludes a finite continuous extension of the ODE solution. For the specific initial covector

$$\ell_\sigma = \nabla\sigma(q_0) = (-9/32, -3/16, -1/16)^{\mathsf{T}}$$

we have the entire transported vector and cost

$$A_\tau^{-\mathsf{T}} \ell_\sigma = (-9z^3/32, -3z/16, -1/16)^{\mathsf{T}},$$
$$\nu \ell_\sigma^{\mathsf{T}} K_\tau \ell_\sigma = \nu \frac{81z^6 + 36z^2 + 4}{1024} \longrightarrow \frac{\nu}{256} \quad (z \downarrow 0).$$

This computes the precise diffusive cost of the covector selected by the arithmetic observable, even at the spatial escape endpoint. The zero condition itself remains $Z_t(\sigma(q)) = 0$ with its full analytic divisor as proved above; the transported cost is not an assertion of such a zero or of a Euclidean Navier–Stokes counterexample.

10 The source fourth coefficient and its exact Newman heat packet

The result supplied in the linked task uses the coupling symbol ξ. Keep its original finite open cubic box, $L \geq 2$, with $N = 3(2L)(2L+1)^2$ links and $M = 3(2L)^2(2L+1)$ elementary faces. On product Haar probability measure on $SU(2)^N$, the actual Hamiltonian and scales are

$$H = \kappa \sum_e E_e + b \sum_p (2 - W_p), \qquad \kappa = \frac{2g^2}{a}, \quad b = \frac{1}{2g^2 a}, \quad \xi = \frac{b}{\kappa} = \frac{1}{4g^4}, \qquad a, g > 0.$$

Here a is the source lattice spacing, and the polynomial target also uses the letter $a = F_1$. Their exact typed relationship is as follows. The shared letters are retained through the following explicit product space and maps. Subscripts in this paragraph identify the original coordinate being evaluated: $a_{\mathrm{lat}} = a$ in the Hamiltonian, $b_{\mathrm{mag}} = b$ in its potential, and $(a_{\mathrm{pol}}, b_{\mathrm{pol}}, c_{\mathrm{pol}}) = (a, b, c)$ in the polynomial target. Put

$$\mathcal{P}_L = (0, \infty)_{a_{\mathrm{lat}}} \times (0, \infty)_g, \quad \mathcal{X}_L = \mathcal{P}_L \times \mathbb{C}^3_{x,y,w}, \quad \mathcal{Y}_L = \mathcal{P}_L \times \mathbb{C}^3_{a_{\mathrm{pol}}, b_{\mathrm{pol}}, c_{\mathrm{pol}}}.$$

The exact morphism is

$$\widehat{F}_L : \mathcal{X}_L \longrightarrow \mathcal{Y}_L, \qquad (a_{\mathrm{lat}}, g, q) \longmapsto (a_{\mathrm{lat}}, g, F(q)).$$

Its parameter projection is unchanged, its polynomial projection is the full F in (1), and its derivative in this coordinate order is $\mathrm{diag}(I_2, DF(q))$. These assertions follow by differentiating each displayed component. The arithmetic projection $\widehat{\sigma}(a_{\mathrm{lat}}, g, q) = F_1(q)/2$ therefore satisfies $\widehat{\sigma} = \frac{1}{2}\,\mathrm{pr}_{a_{\mathrm{pol}}}\widehat{F}_L$. In particular the lattice spacing is a retained parameter of the map, while the polynomial coordinate is evaluated by its explicit component; neither coordinate is set equal to a constant or erased.

The physical parameter chart and its globally defined positive-real inverse are

$$\begin{aligned} \chi : \mathcal{P}_L \longrightarrow (0,\infty)_\kappa \times (0,\infty)_\xi, &\qquad \chi(a_{\mathrm{lat}}, g) = (2g^2/a_{\mathrm{lat}}, 1/(4g^4)), \\ \chi^{-1}(\kappa,\xi) = (1/(\kappa\sqrt{\xi}), (4\xi)^{-1/4}), &\qquad b_{\mathrm{mag}} = \kappa\xi. \end{aligned}$$

All roots here are the positive real roots. Indeed $g^2 = 1/(2\sqrt{\xi})$, so $2g^2/a_{\mathrm{lat}} = \kappa$ and $1/(2g^2 a_{\mathrm{lat}}) = \kappa\xi$; substitution in the other direction returns the original a_{lat}, g. Thus this is an invertible parameter chart retaining both physical parameters. In particular a fixed-κ coupling path has $a_{\mathrm{lat}}(\xi) = 1/(\kappa\sqrt{\xi})$ and $g(\xi) = (4\xi)^{-1/4}$; it does not silently hold the lattice spacing fixed. The source's finite-box coefficient has the displayed dependence on these parameters and makes no continuum-limit assertion.

On the original generators the Pauli trace identity gives $-\mathrm{tr}(T_i T_j)/2 = \delta_{ij}/4$. Consequently the orthonormal frame for that metric is exactly $(2T_1, 2T_2, 2T_3)$ and its left derivatives are $(2X_1, 2X_2, 2X_3)$. Their sum of squares is $4\sum_i X_i^2$, proving $-\Delta^c = 4E$ and the original electric coefficient $\kappa/4$. This records the metric map with its factor, as well as the parameter map. The electric operator is $E_e = -\sum_{a'=1}^{3} X_{e,a'}^2$ for $T_{a'} = -i\sigma_{a'}/2$; the prime distinguishes the generator index from both meanings of a. The potential trace W_p and the original operator and form domains are specified and proved in Appendix A. Let ψ_ξ be the positive unit vacuum, $\mathcal{E}$ its energy,

$$A_\xi = (H - \mathcal{E})/\kappa, \qquad v_e = (E_e - \langle \psi_\xi, E_e \psi_\xi \rangle)\psi_\xi.$$

For two distinct links e, f sharing no elementary face, retain the original ordered incidence count

$$a_{ef} = \#\{(p,q) : e \in \partial p,\ f \in \partial q, |\partial p \cap \partial q| = 1\}.$$

This is a count, not a continuum limit or a change of physical scale. Appendix A proves, in the full original box,

$$\begin{aligned} \langle v_e, v_f \rangle &= \frac{127 a_{ef}}{219024}\xi^4 + O_L(\xi^6), \\ \langle v_e, (H - \mathcal{E}) v_f \rangle &= 0. \end{aligned} \tag{65}$$

It retains both shared-link spin channels, their exact norms, the disconnected subtraction, and the full vacuum and spectral-band Gram factors. The complete source proof is included, not replaced by these two consequences.

For the energy variable λ define the distribution

$$\mathfrak{d} = c_0\delta_3 + \beta\delta_3' + c_1\delta_{9/2} + c_2\delta_{13/2}, \qquad \begin{aligned} c_0 &= -\frac{59}{275184}, \quad \beta = \frac{1}{336}, \\ c_1 &= \frac{1}{1296}, \quad c_2 = \frac{3}{132496}. \end{aligned} \tag{66}$$

The convention is $\delta_3'(h) = -h'(3)$, so every sign is fixed. The proved source functional is, for bounded continuous h differentiable at 3,

$$\lim_{\xi \to 0} \xi^{-4} \langle v_e, h(A_\xi) v_f \rangle = a_{ef}\mathfrak{d}(h). \tag{67}$$

The variable λ is the dimensionless energy ω/κ; the physical measure is its pushforward under $\lambda \mapsto \kappa\lambda$. Here is its complete typed map, including the derivative distribution. For each original $\kappa > 0$, let

$$p_\kappa : \mathbb{R}_\lambda \longrightarrow \mathbb{R}_\omega, \quad p_\kappa(\lambda) = \kappa\lambda, \qquad p_\kappa^{-1}(\omega) = \omega/\kappa.$$

For any compactly supported distribution d define $(p_{\kappa*}d)(h) = d(h \circ p_\kappa)$ for smooth h. This gives an isomorphism between the four-dimensional spaces

$$\mathcal{V} = \operatorname{span}_{\mathbb{C}}\{\delta_3, \delta_3', \delta_{9/2}, \delta_{13/2}\},$$
$$\mathcal{V}_\kappa = \operatorname{span}_{\mathbb{C}}\{\delta_{3\kappa}, \delta_{3\kappa}', \delta_{9\kappa/2}, \delta_{13\kappa/2}\}.$$

In these ordered bases the matrix and its inverse are

$$B_\kappa = \operatorname{diag}(1, \kappa, 1, 1), \qquad B_\kappa^{-1} = \operatorname{diag}(1, \kappa^{-1}, 1, 1).$$

To prove every column, evaluate a delta on $h(\kappa\lambda)$, and use $\delta_3'(h(\kappa\lambda)) = -\kappa h'(3\kappa)$ for the second column. Thus the physical fourth coefficient is exactly

$$p_{\kappa*}\mathfrak{d} = c_0\delta_{3\kappa} + \frac{\kappa}{336}\delta_{3\kappa}' + c_1\delta_{9\kappa/2} + c_2\delta_{13\kappa/2}. \tag{68}$$

The derivative coefficient retains its physical factor κ.

For the actual self-adjoint excitation operator $A_\xi = (H - \mathcal{E})/\kappa$, its projection-valued measure E_{A_ξ} obeys, for every Borel set $B \subset \mathbb{R}$,

$$E_{H-\mathcal{E}}(B) = E_{A_\xi}(p_\kappa^{-1}B), \qquad \Sigma_{ef}^{\omega}(B) = \langle v_e, E_{A_\xi}(p_\kappa^{-1}B)v_f\rangle = (p_{\kappa*}\Sigma_{ef}^{\lambda})(B).$$

Indeed the spectral integral for κA_ξ inserts $\kappa\lambda$, so its indicator function is $\mathbf{1}_B(\kappa\lambda)$, proving these identities. Integration of bounded Borel tests first for simple functions and then by dominated convergence proves the entire pushforward identity on the actual measures. For absolute Hamiltonian energy the additional exact affine coordinate is $\varepsilon = \mathcal{E} + \kappa\lambda$ and the same proof uses the preimage of B under this affine map. On the unchanged operator domain the inverse reconstruction is $H = \kappa A_\xi + \mathcal{E}I$, where $\mathcal{E} = 2bM + \kappa e(\xi)$; the scalar $2bM$ is recovered as well.

Multiplication by ω on $\mathcal{V}_\kappa$ has matrix

$$Q_\omega = \begin{pmatrix} 3\kappa & -1 & 0 & 0 \\ 0 & 3\kappa & 0 & 0 \\ 0 & 0 & 9\kappa/2 & 0 \\ 0 & 0 & 0 & 13\kappa/2 \end{pmatrix}.$$

Testing $\omega\delta_{3\kappa}'$ gives $-h(3\kappa) - 3\kappa h'(3\kappa)$ and proves its off-diagonal entry. With Q in (69), direct multiplication gives $Q_\omega B_\kappa = \kappa B_\kappa Q$. The zeroth moment is therefore unchanged, and the first moment is multiplied by its original κ, including the derivative term. The source proves that the original matrix-valued spectral measure is positive semidefinite on every Borel set. An off-diagonal entry retains its signs, and its fourth Taylor coefficient retains δ_3'.

10.1 A four-dimensional spectral distribution and its heat map

Let $\mathcal{V}$ be the ordered span $(\delta_3, \delta_3', \delta_{9/2}, \delta_{13/2})$. Multiplication of a distribution by the unchanged energy coordinate acts in this basis by

$$Q = \begin{pmatrix} 3 & -1 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 9/2 & 0 \\ 0 & 0 & 0 & 13/2 \end{pmatrix}. \tag{69}$$

Indeed $(\lambda\delta_3')(h) = -3h'(3) - h(3)$, while the other three basis vectors are multiplied by their displayed energies. This proves the entire matrix, including its off-diagonal entry. The exact exponential of $tQ^2/4$ has its first block

$$e^{9t/4}\begin{pmatrix} 1 & -3t/2 \\ 0 & 1 \end{pmatrix}$$

and remaining entries $e^{81t/16}, e^{169t/16}$. To verify it, Q^2 has first block $9I + \left(\begin{smallmatrix} 0 & -6 \\ 0 & 0 \end{smallmatrix}\right)$; the second matrix has square zero, so its exponential series stops after its linear term. The two other entries are scalar exponentials.

Define the exact Fourier morphism, with its displayed domain and codomain,

$$\mathcal{F}_{\mathcal{V}} : \mathcal{V} \longrightarrow \operatorname{span}_{\mathbb{C}}\{e^{3is}, -ise^{3is}, e^{9is/2}, e^{13is/2}\}, \qquad \mathcal{F}_{\mathcal{V}}(d)(s) = d_\lambda(e^{is\lambda}).$$

Its basis images are $e^{3is}, -ise^{3is}, e^{9is/2}, e^{13is/2}$. It is injective. If a linear combination vanishes, divide it by e^{3is} and put $s = iy$, $y \to +\infty$. The remaining expression is a polynomial of degree at most one in y, plus constant multiples of $e^{-3y/2}$ and $e^{-7y/2}$. Its polynomial coefficients must vanish; multiplying the remaining identity by $e^{3y/2}$ then taking the same limit makes the third coefficient zero, and then the fourth is zero. Differentiation of the four displayed functions gives

$$\partial_s\mathcal{F}_{\mathcal{V}}(d) = i\mathcal{F}_{\mathcal{V}}(Qd), \qquad -\tfrac{1}{4}\partial_s^2\mathcal{F}_{\mathcal{V}}(d) = \mathcal{F}_{\mathcal{V}}(Q^2d/4).$$

Thus the source spectral coefficient has an exact heat intertwiner, not merely a shared dimension or notation.

Applying this map to $a_{ef}e^{tQ^2/4}\mathfrak{d}$ yields

$$\begin{aligned} J_t(s) = a_{ef}\Big[&\left(-\frac{59}{275184} - \frac{t}{224} - \frac{is}{336}\right) e^{9t/4+3is} \\ &+ \frac{1}{1296}e^{81t/16+9is/2} + \frac{3}{132496}e^{169t/16+13is/2}\Big]. \end{aligned} \tag{70}$$

It is entire in both variables and satisfies $\partial_t J = -\partial_s^2 J/4$. Equivalently this follows directly from $J_t = a_{ef}\mathfrak{d}(e^{t\lambda^2/4+is\lambda})$. For the actual original spectral measure, this test is bounded on the real energy axis whenever $\operatorname{Re} t < 0$, for every complex s, and also when $t = 0, s \in \mathbb{R}$. Completing the real square in $\operatorname{Re}(t)\lambda^2/4 - \operatorname{Im}(s)\lambda$ proves that bound. Consequently (67) gives

$$\lim_{\xi\to 0}\xi^{-4}\langle v_e, e^{tA_\xi^2/4+isA_\xi}v_f\rangle = J_t(s)$$

on precisely these guaranteed domains. The finite coefficient (70) has the stated entire continuation; this does not assume Gaussian exponential moments of the full unbounded interacting spectral measure at positive real t.

Physical correlation time $\theta \geq 0$ and Newman time t have the following exact typed dynamical morphisms. The following coordinate maps prove the relationship between the two time actions. Put

$$\mathcal{W} = \operatorname{span}_{\mathbb{C}}\{e^{3is}, -ise^{3is}, e^{9is/2}, e^{13is/2}\} \subset \mathcal{O}(\mathbb{C}_s).$$

The previously proved Fourier morphism is the isomorphism $\mathcal{F}_{\mathcal{V}} : \mathcal{V} \to \mathcal{W}$. For physical time $\theta \geq 0$, define

$$P_{\kappa,\theta} = e^{-\kappa\theta Q} : \mathcal{V} \to \mathcal{V}, \quad S_{\kappa,\theta} : \mathcal{W} \to \mathcal{W}, \quad (S_{\kappa,\theta}f)(s) = f(s + i\kappa\theta).$$

The first block of $P_{\kappa,\theta}$ is

$$e^{-3\kappa\theta}\begin{pmatrix} 1 & \kappa\theta \\ 0 & 1 \end{pmatrix},$$

and the other diagonal entries are $e^{-9\kappa\theta/2}, e^{-13\kappa\theta/2}$. The nilpotent off-diagonal matrix squares to zero, so expanding its exponential proves the entire block, including $\kappa\theta$. For every $d \in \mathcal{V}$,

$$\mathcal{F}_{\mathcal{V}} P_{\kappa,\theta} d(s) = d(e^{-\kappa\theta\lambda} e^{is\lambda}) = \mathcal{F}_{\mathcal{V}} d(s + i\kappa\theta) = S_{\kappa,\theta} \mathcal{F}_{\mathcal{V}} d(s).$$

This proves the intertwiner on every vector, not only on the selected coefficient. Its infinitesimal identity is $\mathcal{F}_{\mathcal{V}}(-\kappa Q) = i\kappa\partial_s \mathcal{F}_{\mathcal{V}}$.

For Newman time $t \in \mathbb{C}$, the other action is $N_t = e^{tQ^2/4} : \mathcal{V} \to \mathcal{V}$, which intertwines with $e^{-t\partial_s^2/4} : \mathcal{W} \to \mathcal{W}$ by the proved matrix derivative identity. Both actions commute, since Q commutes with Q^2. This proves the full two-parameter correspondence, with the original physical and Newman generators retained. Finally evaluation at $s = 0$ after the coordinate map

$$\iota_\kappa : [0, \infty)_\theta \longrightarrow \mathbb{C}_t \times \mathbb{C}_s, \qquad \iota_\kappa(\theta) = (0, i\kappa\theta)$$

gives the displayed correlation coefficient $\iota_\kappa^* J(\theta) = J_0(i\kappa\theta)$. Its derivative is $i\kappa J_s(0, i\kappa\theta)$ by the chain rule.

The unchanged dimensionful correlation coefficient is

$$a_{ef}\mathfrak{d}(e^{-\kappa\theta\lambda}) = a_{ef}\left[\left(c_0 + \frac{\kappa\theta}{336}\right)e^{-3\kappa\theta} + c_1 e^{-9\kappa\theta/2} + c_2 e^{-13\kappa\theta/2}\right].$$

It equals $J_0(i\kappa\theta)$ by substitution, giving the exact time dictionary rather than identifying the two evolutions. Its value at zero is $127a_{ef}/219024$, and its first physical-time derivative is zero because $3c_0 - \beta + (9/2)c_1 + (13/2)c_2 = 0$. These are exactly (65), with the physical factor κ retained in the derivative.

11 The original fourth coupling coefficient as an arithmetic heat automorphism

This section uses the complete finite-box Wilson calculation retained in the complete source proof in Appendix A. Its parameter is $\xi = b/\kappa = 1/(4g^4)$, whereas the completed Riemann function is $\xi(\rho)$ and the unchanged arithmetic heat function is $Z_t(s)$. These notations retain their original meanings. We construct the actual operator-valued arithmetic observable, prove its fourth coupling coefficient, and give the inverse of the resulting arithmetic map.

11.1 Original source operators and the complete spectral functional

Fix the original open box with vertices $\{-L, \dots, L\}^3$, integer $L \geq 2$, its $N = 3(2L)(2L+1)^2$ positively oriented physical edges, and its $M = 3(2L)^2(2L+1)$ contained elementary faces. On the gauge-invariant subspace of product Haar probability $L^2(SU(2)^N)$, retain

$$T_a = -i\sigma_a/2, \qquad E_e = -\sum_{a=1}^{3} X_{e,a}^2, \qquad H_0 = \sum_e E_e, \qquad S = \sum_p W_p,$$
$$H = \kappa H_0 + b\sum_p (2 - W_p) = \kappa(H_0 - \xi S) + 2bMI,$$
$$\kappa = \frac{2g^2}{a}, \quad b = \frac{1}{2g^2 a}, \quad \xi = \frac{1}{4g^4}, \quad a, g > 0. \tag{71}$$

Here $X_{e,a}$ is the original left derivative generated by T_a, and W_p is the original oriented fundamental Wilson trace. The metric identity is $-\Delta_e^c = 4E_e$, so the electric coefficient in

that metric is $\kappa/4$. Neither this factor nor the scalar $2bM$ is changed. Let ψ_ξ be the positive unit ground vector and write

$$\mathcal{E}(\xi) = 2bM + \kappa e(\xi), \qquad A_\xi = \frac{H - \mathcal{E}(\xi)}{\kappa} = H_0 - \xi S - e(\xi) I,$$
$$\gamma_e(\xi) = \langle \psi_\xi, E_e \psi_\xi \rangle, \qquad v_e(\xi) = (E_e - \gamma_e(\xi))\psi_\xi. \tag{72}$$

The inner product is conjugate-linear in its first entry. For real ξ near zero, A_ξ is self-adjoint on the physical H^2 domain, nonnegative, and has the original smooth ground vector in its kernel. The formula for A_ξ is the exact physical excitation energy divided by its original κ; the scalar in H cancels its identical scalar in $\mathcal{E}$ in this displayed difference.

For distinct edges e, f sharing no face, let a_{ef} count the ordered-by-containment pairs of adjacent faces containing e and f, respectively. The complete spectral coefficient proved in the source calculation is, for bounded continuous complex $h : \mathbb{R} \to \mathbb{C}$ differentiable at 3,

$$\lim_{\xi \to 0} \xi^{-4} \langle v_e(\xi), h(A_\xi) v_f(\xi) \rangle = a_{ef} \mathcal{D}(h), \qquad \mathcal{D}(h) = c_0 h(3) - \frac{h'(3)}{336} + c_1 h(9/2) + c_2 h(13/2), \tag{73}$$

where every coefficient is retained:

$$c_0 = -\frac{59}{275184}, \qquad c_1 = \frac{1}{1296}, \qquad c_2 = \frac{3}{132496}. \tag{74}$$

Equivalently, $\mathcal{D} = c_0 \delta_3 + \delta_3'/336 + c_1 \delta_{9/2} + c_2 \delta_{13/2}$, with $\delta_3'(h) = -h'(3)$. Thus the derivative term is part of the original spectral coefficient. The source proof obtains it from the actual first interacting spectral band and its full Gram isometry, together with both exterior spin channels; it is not replaced here by an atomic measure.

11.2 Bounded arithmetic functional calculus at every complex heat time

Lemma 11.1 (The actual arithmetic test and its operator integral)**.** *For every complex t, s, the function $h_{t,s}(q) = Z_t(s + q)$ is bounded and continuously differentiable on the whole real q-axis. On real ξ near zero, its bounded spectral calculus is exactly*

$$Z_t(s + A_\xi) = 4 \int_{\mathbb{R}} k(v) e^{tv^2/4 + isv} e^{ivA_\xi} \, dv. \tag{75}$$

The integral is the strong integral of bounded operators, defined on each Hilbert-space vector; its operator norm is bounded by

$$4 \int_{\mathbb{R}} |k(v)| e^{(\operatorname{Re} t) v^2/4 + |\operatorname{Im} s||v|} \, dv. \tag{76}$$

It is jointly entire in t, s, in the operator norm, and obeys $\partial_t Z_t(s + A_\xi) = -\frac{1}{4} \partial_s^2 Z_t(s + A_\xi)$.

Proof. Use the unchanged arithmetic identity (11):

$$h_{t,s}(q) = 4 \int_{\mathbb{R}} k(v) e^{tv^2/4 + isv} e^{iqv} \, dv.$$

For real q, the last factor has modulus one. The previously proved two-ended superexponential bound on k dominates the absolute integral in (76). Its product with $|v|^j$ is also integrable for every nonnegative integer j. This proves boundedness of every real q-derivative, uniformly in q, by differentiating the integral. On compact complex (t, s) sets the same dominator works after every specified t, s, q derivative.

For any Hilbert-space vector u, the function $v \mapsto e^{ivA_\xi} u$ is strongly continuous and has constant norm. Thus the right side of (75) is a Bochner integral on that vector and satisfies

the stated norm bound. For any two vectors, the corresponding spectral measure has finite total variation, bounded by the product of their norms. Fubini therefore interchanges this measure integral with the absolute v integral and gives its matrix element as that of $h_{t,s}(A_\xi)$. This proves the operator identity, rather than postulating an unbounded entire functional calculus.

Expand the scalar exponential about any fixed (t, s). On every smaller compact polydisc, its absolutely summed Taylor series is dominated by the same integral with larger fixed quadratic and linear weights. The integral norm inequality proves convergence of the resulting operator Taylor series in norm. This establishes joint entire dependence. Differentiation in t inserts $v^2/4$, whereas two s derivatives insert $-v^2$, which proves the heat identity with the original sign and factor. □

Theorem 11.2 (The actual arithmetic fourth coefficient)**.** *Define the entire arithmetic function*

$$\mathcal{K}_t(s) = c_0 Z_t(s+3) - \frac{Z_t'(s+3)}{336} + c_1 Z_t(s+9/2) + c_2 Z_t(s+13/2). \tag{77}$$

For every complex t, s, the actual original observable

$$G_\xi(t,s) = \langle v_e(\xi), Z_t(s + A_\xi) v_f(\xi) \rangle \tag{78}$$

satisfies

$$\lim_{\xi \to 0} \xi^{-4} G_\xi(t,s) = a_{ef} \mathcal{K}_t(s). \tag{79}$$

The limit retains the actual Riemann initial datum:

$$\mathcal{K}_0(s) = c_0 \Xi(s+3) - \frac{\Xi'(s+3)}{336} + c_1 \Xi(s+9/2) + c_2 \Xi(s+13/2), \tag{80}$$

$$\partial_t \mathcal{K}_t = -\tfrac{1}{4} \partial_s^2 \mathcal{K}_t, \qquad \partial_t G_\xi = -\tfrac{1}{4} \partial_s^2 G_\xi. \tag{81}$$

Proof. Lemma 11.1 verifies every test-function hypothesis of (73) for the actual $h(q) = Z_t(s+q)$. Its values and derivative at the four specified source locations give (77) and (79) literally. Every translated term and its retained derivative is jointly entire. Translation by each of the original constants and ∂_s commute with ∂_s^2, so (12) proves the first identity in (81); the operator identity from Lemma 11.1 proves the second one. Finally $Z_0 = \Xi$ gives (80) with no change of arithmetic initial function. □

11.3 Actual coupling analyticity and the sixth-order remainder

Proposition 11.3 (Uniform remainder on compact arithmetic parameter sets)**.** *For each original finite box and each compact $\mathcal{Q} \subset \mathbb{C}^2_{t,s}$, there are numbers $r > 0$ and $C_{\mathcal{Q},L,e,f} < \infty$ such that, for real $|\xi| \le r/2$,*

$$\sup_{(t,s) \in \mathcal{Q}} |G_\xi(t,s) - a_{ef} \xi^4 \mathcal{K}_t(s)| \le C_{\mathcal{Q},L,e,f} |\xi|^6. \tag{82}$$

The construction gives a jointly holomorphic extension in a complex coupling disc and all complex t, s. The radius and constant retain their dependence on the original finite box.

Proof. We first specify the analytic continuation of the vectors and bras. The full free ground eigenvalue is simple and the next full-space eigenvalue is $3/4$; the original perturbation has $\|S\| = 2M$. The resolvent on $|z| = 3/8$ has norm at most $8/3$, so the source Neumann construction supplies the analytic rank-one vacuum projection in a coupling disc contained in $|\xi| < 3/(16M)$. Write $\chi(\xi) = P(\xi)1$. Complex conjugation C on functions satisfies $CP(\xi)C = P(\bar{\xi})$, since both H_0 and S have real coefficients. Thus $C\chi(\xi) = \chi(\bar{\xi})$.

The scalar $\mathfrak{n}(\xi) = \langle \chi(\bar{\xi}), \chi(\xi)\rangle$ is holomorphic and equals one at zero. On a smaller symmetric disc, it is nonzero and has the holomorphic square root with value one at zero. Set $\psi(\xi) = \chi(\xi)/\sqrt{\mathfrak{n}(\xi)}$. For small real ξ this is the original positive unit vacuum: it is the real normalized eigenvector with positive overlap with the constant vector. For complex ξ, this formula fixes its analytic extension; no assertion that its Hilbert norm remains one off the real axis is needed. Its eigenvalue $e(\xi)$ is holomorphic and real on the real axis.

Analyticity of ψ holds in every fixed Sobolev norm. Indeed the exact equation

$$\psi(\xi) = (I + H_0)^{-1}[(1 + e(\xi) + \xi S)\psi(\xi)]$$

increases the Sobolev index by two, since multiplication by the original smooth S preserves that index and the elliptic inverse increases it by two. Starting with the Hilbert-space analytic projection proves the assertion inductively. Consequently

$$\gamma_e(\xi) = \langle \psi(\bar{\xi}), E_e\psi(\xi)\rangle, \qquad v_e(\xi) = (E_e - \gamma_e(\xi))\psi(\xi)$$

are holomorphic in every required Sobolev space. Their restrictions to real ξ are exactly (72). The conjugate-linear-first functional $u \mapsto \langle v_e(\bar{\xi}), u\rangle$ is holomorphic in ξ; it is the precise continuation of the original bra.

On the same physical domain put $A_\xi = H_0 - \xi S - e(\xi)I$, now for complex ξ. The bounded perturbation of the free unitary group gives e^{ivA_ξ} for every real v. Its Dyson series is obtained by inserting $-\xi S$ between the unchanged free groups. The term with n insertions has norm at most $(2M|\xi||v|)^n/n!$; the ordered integrals are strong integrals on each vector. The resulting bounded-operator series converges in norm, uniformly on compact coupling sets. The scalar shift commutes with every term and contributes $e^{-ive(\xi)}$. Hence

$$\|e^{ivA_\xi}\| \le \exp\{|v|(2M|\xi| + |\operatorname{Im} e(\xi)|)\}. \tag{83}$$

This proves norm-holomorphic dependence on ξ for fixed real v, with the entire original bounded potential retained.

Define the continuation of (78) by

$$\widetilde{G}(\xi, t, s) = 4\int_{\mathbb{R}} k(v)e^{tv^2/4+isv}\langle v_e(\bar{\xi}), e^{ivA_\xi}v_f(\xi)\rangle\, dv. \tag{84}$$

Choose a closed coupling disc $|\xi| \le r$ strictly inside the disc just constructed. Put

$$B_r = \sup_{|\xi|\le r} \|v_e(\bar{\xi})\|\, \|v_f(\xi)\|, \qquad E_r = \sup_{|\xi|\le r} |\operatorname{Im} e(\xi)|.$$

Both numbers are finite by continuity on that closed disc. On a compact arithmetic parameter set $\mathcal{Q}$, set $T_\mathcal{Q} = \max_\mathcal{Q} \operatorname{Re} t$ and $S_\mathcal{Q} = \max_\mathcal{Q} |\operatorname{Im} s|$. The absolute integral in (84) is uniformly bounded by the explicit finite number

$$M_{r,\mathcal{Q}} = 4B_r \int_{\mathbb{R}} |k(v)|e^{T_\mathcal{Q}v^2/4+(S_\mathcal{Q}+2Mr+E_r)|v|}\, dv. \tag{85}$$

The superexponential arithmetic kernel absorbs all these retained weights. On larger compact arithmetic sets it also absorbs the weights introduced by each parameter derivative. The locally uniformly convergent Dyson and scalar exponential series, or their Cauchy integral formulas dominated by this same bound, therefore prove joint holomorphy of $\widetilde{G}$. For real coupling it equals the original observable by Lemma 11.1.

The source central-link unitary is

$$(\mathcal{Z}F)(U) = F\big(((-1)^{\sum_{j<i} n_j}U_{(n,i)})_{(n,i)}\big).$$

It preserves Haar measure, commutes with every original E_e, and changes the sign of every Wilson trace. Consequently $\mathcal{Z}(H_0 - \xi S)\mathcal{Z} = H_0 + \xi S$. Uniqueness and positivity of the real vacuum imply $\psi_{-\xi} = \mathcal{Z}\psi_\xi$, $e(-\xi) = e(\xi)$, $v_e(-\xi) = \mathcal{Z}v_e(\xi)$, and $A_{-\xi} = \mathcal{Z}A_\xi\mathcal{Z}$. Thus $G_{-\xi}(t,s) = G_\xi(t,s)$ for real coupling. The identity theorem extends this equality to the complex disc. No time or arithmetic coordinate changes in this symmetry.

Write the coupling Taylor series as $\widetilde{G} = \sum_{n\geq 0} g_n(t,s)\xi^n$. The actual limit (79) on the real axis forces $g_0 = g_1 = g_2 = g_3 = 0$ and $g_4 = a_{ef}\mathcal{K}_t(s)$: otherwise division by ξ^4 could not have the proved finite limit. Evenness makes every odd coefficient vanish. The Cauchy bound on $|\xi| = r$ gives $\sup_{\mathcal{Q}} |g_n| \leq M_{r,\mathcal{Q}} r^{-n}$. For $|\xi| \leq r/2$, summing the remaining series yields

$$\sup_{\mathcal{Q}} |\widetilde{G} - a_{ef}\xi^4\mathcal{K}| \leq 2M_{r,\mathcal{Q}} r^{-6}|\xi|^6.$$

This proves (82), for example with the specified constant $2M_{r,\mathcal{Q}}r^{-6}$. All box dependence in M, N, r, B_r, E_r remains present. The estimate does not exchange a volume limit with a coupling derivative. □

11.4 An invertible arithmetic map on the full retained Fourier domain

Retain the exact spaces $\mathscr{E}, \mathscr{A}$ and transform T already defined above, including every seminorm

$$p_{A,B,m,j}(\phi) = \int_{\mathbb{R}} |\phi^{(j)}(v)|(1+|v|)^m e^{Av^2+B|v|}\, dv, \qquad T\phi(s) = 4\int_{\mathbb{R}} \phi(v)e^{isv}\, dv.$$

Define the source multiplier and arithmetic operator by

$$m(v) = c_0 e^{3iv} - \frac{iv}{336}e^{3iv} + c_1 e^{9iv/2} + c_2 e^{13iv/2}, \tag{86}$$

$$(\mathcal{T}_{\mathcal{D}} f)(s) = c_0 f(s+3) - \frac{f'(s+3)}{336} + c_1 f(s+9/2) + c_2 f(s+13/2). \tag{87}$$

Lemma 11.4 (A uniform exact lower bound for the original multiplier)**.** *For every real v,*

$$|m(v)| \geq \delta, \qquad \delta = \frac{575}{1752192} > 0. \tag{88}$$

In particular its real-axis reciprocal exists everywhere, with no branch choice or removed spectral point.

Proof. Factoring the nonzero phase gives the exact identity

$$e^{-3iv}m(v) = c_0 - \frac{iv}{336} + c_1 e^{3iv/2} + c_2 e^{7iv/2}.$$

For $|v| \leq 1/2$, the elementary inequality $\cos x \geq 1 - x^2/2$ yields

$$\begin{aligned}\operatorname{Re}(e^{-3iv}m(v)) &= c_0 + c_1\cos(3v/2) + c_2\cos(7v/2)\\ &\geq c_0 + \frac{23}{32}c_1 - \frac{17}{32}c_2 = \frac{575}{1752192}.\end{aligned}$$

For completeness, this cosine bound follows directly by integration: for $x \geq 0$, $1 - \cos x = \int_0^x \sin u\, du \leq \int_0^x u\, du$, using $\sin u \leq u$; the latter follows from $u - \sin u = \int_0^u (1 - \cos w)\, dw \geq 0$. Evenness gives the negative case. Since the real part is positive, its displayed lower bound also bounds the modulus.

For $|v| \geq 1/2$, the triangle inequality with the original linear term gives

$$\begin{aligned}|m(v)| &\geq \frac{|v|}{336} - (|c_0| + c_1 + c_2)\\ &\geq \frac{1}{672} - \frac{10825}{10732176} = \frac{10291}{21464352} > \frac{575}{1752192}.\end{aligned}$$

All coefficients in these estimates are the original rational coefficients. The two intervals cover the whole real axis and prove (88). □

Theorem 11.5 (Exact arithmetic automorphism and heat conjugacy)**.** *Multiplication by m is a continuous automorphism of $\mathscr{E}$. Its inverse is literal pointwise multiplication by $1/m$ on the unchanged real Fourier axis. The operator $\mathcal{T}_{\mathcal{D}}$ is a continuous automorphism of $\mathscr{A}$, with exact inverse*

$$\mathcal{T}_{\mathcal{D}}^{-1} f = T\left(\frac{T^{-1} f}{m}\right). \tag{89}$$

For every complex t,

$$\mathcal{T}_{\mathcal{D}} = T M_m T^{-1}, \qquad \mathcal{T}_{\mathcal{D}} U_t = U_t \mathcal{T}_{\mathcal{D}}, \qquad \mathcal{T}_{\mathcal{D}}^{-1} U_t = U_t \mathcal{T}_{\mathcal{D}}^{-1}, \tag{90}$$

$$\mathcal{K}_t = \mathcal{T}_{\mathcal{D}} Z_t, \qquad Z_t = \mathcal{T}_{\mathcal{D}}^{-1} \mathcal{K}_t. \tag{91}$$

Thus the actual arithmetic function is exactly recoverable from the propagated source coefficient, on this fully specified domain.

Proof. Every derivative of m has at most linear growth on the real axis. More explicitly, for $j \geq 0$ it satisfies $|m^{(j)}(v)| \leq C_j(1 + |v|)$, where one may take

$$C_j = |c_0| 3^j + c_1 (9/2)^j + c_2 (13/2)^j + \frac{3^j + \mathbf{1}_{j \geq 1} j 3^{j-1}}{336}.$$

The product rule gives the seminorm estimate

$$p_{A,B,m,j}(m\phi) \leq \sum_{\ell=0}^{j} \binom{j}{\ell} C_\ell \, p_{A,B,m+1,j-\ell}(\phi).$$

This proves continuity of M_m on the original $\mathscr{E}$.

Let $r(v) = 1/m(v)$, which is smooth by Lemma 11.4. The exact product identity $mr = 1$ and its derivatives give

$$r^{(j)} = -\frac{1}{m} \sum_{\ell=1}^{j} \binom{j}{\ell} m^{(\ell)} r^{(j-\ell)} \quad (j \geq 1), \qquad |r| \leq \delta^{-1}.$$

Set $Q_0 = \delta^{-1}$ and recursively define the finite positive constants

$$Q_j = \delta^{-1} \sum_{\ell=1}^{j} \binom{j}{\ell} C_\ell Q_{j-\ell}.$$

Induction proves $|r^{(j)}(v)| \leq Q_j (1 + |v|)^j$: the term with index $\ell \geq 1$ has exponent at most $1 + j - \ell \leq j$, and the denominator is bounded by δ^{-1}. Therefore

$$p_{A,B,m,j}(r\phi) \leq \sum_{\ell=0}^{j} \binom{j}{\ell} Q_\ell \, p_{A,B,m+\ell,j-\ell}(\phi).$$

This proves that reciprocal multiplication is continuous on exactly the same $\mathscr{E}$. The two pointwise compositions are the identity, proving the Fourier automorphism and its inverse.

Applying the retained transform to each term of $m\phi$ gives the three translations and the derivative in (87); the factor iv is exactly the Fourier multiplier of ∂_s. The injectivity of T already proved above makes its inverse on $\mathscr{A}$ unambiguous. This proves the first identity in (90) and (89). Multiplication by m and by $1/m$ each commutes pointwise with multiplication by $e^{tv^2/4}$. All three maps preserve $\mathscr{E}$, so applying T proves the two full-domain heat commutation identities for every complex t. Finally (77) is exactly $\mathcal{T}_{\mathcal{D}} Z_t$, and applying the proved inverse gives (91). □

The source factor acts by the typed scalar map $M_{a_{ef}} : \mathscr{A} \to \mathscr{A},\ f \mapsto a_{ef} f$. For $a_{ef} > 0$ its inverse is $g \mapsto g/a_{ef}$, by direct composition; for $a_{ef} = 0$ its kernel is all of $\mathscr{A}$ and its image is $\{0\}$. This retains the complete pair observation map $M_{a_{ef}} \mathcal{T}_{\mathcal{D}}$, including its zero case. When $a_{ef} > 0$, its division also recovers the unscaled $\mathcal{K}$ from the coefficient in (79). The original boxes contain such pairs. In particular, for $e = ((0,0,0),1)$ and $f = ((0,2,0),1)$, there is exactly one common edge-adjacency neighbor, $((0,1,0),1)$. To verify uniqueness directly, a direction-1 neighbor of e is its parallel translate by one step in direction 2 or 3; the corresponding lists for e and f meet only at the displayed midpoint edge. A direction-2 neighbor has its direction-2 base coordinate in $\{-1,0\}$ for e, and in $\{1,2\}$ for f, so these lists are disjoint. A direction-3 neighbor has direction-2 coordinate zero for e and two for f, also disjoint. All faces in the unique midpoint channel are contained in every original box with $L \geq 2$. The exact source channel bijection gives $a_{ef} = 1$, including these open boundaries.

Equations (84) and (91) provide the complete source-to-arithmetic heat correspondence. They assert an invertible map of the specified arithmetic function space. The coordinate-level map on every pointwise zero jet and analytic unit is proved next. The exact morphism on zero data uses the transported jet maps below. Fix $z \in \mathbb{C}$ and an integer $n \geq 1$. Define continuous linear maps

$$\begin{aligned} &\mathcal{J}_{z,n} : \mathscr{A} \longrightarrow \mathbb{C}^n, && \mathcal{J}_{z,n} f = (f(z), f'(z), \ldots, f^{(n-1)}(z)), \\ &\mathcal{L}_{z,n} : \mathscr{A} \longrightarrow \mathbb{C}^n, && \mathcal{L}_{z,n} g = \mathcal{J}_{z,n} \mathcal{T}_{\mathcal{D}}^{-1} g. \end{aligned}$$

For $g = T\phi$ their individual coordinate functionals are exactly

$$(\mathcal{L}_{z,n} g)_j = 4 \int_{\mathbb{R}} (iv)^j \frac{\phi(v)}{m(v)} e^{izv}\, dv, \qquad j = 0, \ldots, n-1. \tag{92}$$

This follows by differentiating the proved inverse integral; it retains the factor 4, the original coordinate z and every derivative factor $(iv)^j$. Absolute values are bounded by

$$4\delta^{-1} \int_{\mathbb{R}} |\phi(v)||v|^j e^{|\operatorname{Im} z||v|}\, dv,$$

where $\delta = 575/1752192$. One defining seminorm with an integer linear weight at least $|\operatorname{Im} z|$ dominates this integral. The same estimate on compact sets justifies every displayed derivative and proves continuity. It also proves continuity of $\mathcal{J}_{z,n}$ by omitting the factor $1/m$.

Both maps are onto. To prove this without assuming a jet-extension theorem on $\mathscr{A}$, set $u_z(s) = \Xi(s - z)$. This lies in $\mathscr{A}$, since its kernel is $k(v)e^{-izv}$ and all differentiated weighted seminorms are bounded by finitely many seminorms of k with a larger linear weight. Each $(s-z)^j u_z(s)$ also lies in $\mathscr{A}$ by the already proved invariance under multiplication by s. Its first n derivatives at z form a triangular matrix with diagonal entries $j!\Xi(0) \neq 0$, $j = 0, \ldots, n-1$. The nonzero value was proved by the positive kernel integral. Back substitution therefore constructs every specified jet exactly. This proves surjectivity of $\mathcal{J}_{z,n}$, and that of $\mathcal{L}_{z,n}$ follows by composition with the automorphism. The finite back-substitution formula is also a continuous linear right inverse from $\mathbb{C}^n$ into the displayed finite-dimensional span.

Put $\mathcal{M}_{z,n} = \ker \mathcal{J}_{z,n}$. The exact zero-jet correspondence is

$$\mathcal{T}_{\mathcal{D}} \mathcal{M}_{z,n} = \ker \mathcal{L}_{z,n}, \qquad \mathscr{A}/\mathcal{M}_{z,n} \longrightarrow \mathscr{A}/\ker \mathcal{L}_{z,n}, \quad [f] \longmapsto [\mathcal{T}_{\mathcal{D}} f].$$

The first equality follows in both directions by applying the inverse automorphism. It makes the quotient map well-defined and gives the inverse $[g] \mapsto [\mathcal{T}_{\mathcal{D}}^{-1} g]$; both are continuous by their quotient topologies. Both quotients have the displayed coordinates in $\mathbb{C}^n$, and the map is the identity in those coordinates because $\mathcal{L}_{z,n} \mathcal{T}_{\mathcal{D}} = \mathcal{J}_{z,n}$.

For the actual $\mathcal{K}_t = \mathcal{T}_{\mathcal{D}} Z_t$ this proves every jet identity $\mathcal{L}_{z,n} \mathcal{K}_t = \mathcal{J}_{z,n} Z_t$. Thus a zero of order exactly n of Z_t is expressed on the transformed function by vanishing of coordinates

$0, \dots, n-1$ of $\mathcal{L}_{z,n+1}\mathcal{K}_t$ and nonvanishing of coordinate n. At every such zero the complete analytic unit is recovered as the convergent series

$$\frac{Z_t(s)}{(s-z)^n} = \sum_{j=0}^{\infty} \frac{(\mathcal{L}_{z,n+j+1}\mathcal{K}_t)_{n+j}}{(n+j)!}(s-z)^j.$$

It is the Taylor expansion of the recovered entire function, so the identity and convergence follow from its holomorphy. No derivative, unit, coordinate or multiplicity is discarded by this morphism.

12 Propagation through the original inverse fibres and spatial flow

Use the exact per-channel arithmetic heat function $\mathcal{K}_t(s)$ defined by (77), with all source constants in (66). Retain the incidence count by setting $G_t(s) = a_{ef}\mathcal{K}_t(s)$. Define

$$g_t(r) = G_t(-r^2/2), \qquad \Psi^{(4)}(t,q) = G_t(\sigma(q)), \quad \sigma(q) = F_1(q)/2.$$

The underlying typed maps and their exact relation are

$$\pi : \mathbb{C}_r \to \mathbb{C}_s, \quad \pi(r) = -r^2/2, \qquad \sigma : \mathbb{C}_q^3 \to \mathbb{C}_s, \quad \sigma(q) = F_1(q)/2,$$

$$\Gamma : \mathbb{C}_r^* \to \mathbb{C}_q^3, \quad \Gamma(r) = (-1/(2r), 3r, 26r^2), \qquad \sigma \circ \Gamma = \pi|_{\mathbb{C}^*}.$$

The last equality follows by the full polynomial substitution $F(\Gamma(r)) = (-r^2, 0, 0)$ proved in (2). Their pullbacks are respectively $\pi^* : \mathcal{O}(\mathbb{C}_s) \to \mathcal{O}_{\text{even}}(\mathbb{C}_r)$ and $\sigma^* : \mathcal{O}(\mathbb{C}_s) \to \mathcal{O}(\mathbb{C}_q^3)$. The first is the isomorphism with inverse proved in Section 8.2. The second is injective with explicit left inverse η^*, where $\eta : \mathbb{C}_s \to \mathbb{C}_q^3$, $\eta(s) = (0, 0, 2s)$: indeed $F_1(0, 0, 2s) = 2s$, so $\sigma\eta = \mathrm{id}$ and $\eta^*\sigma^* = \mathrm{id}$. Composition of these polynomial or entire maps proves continuity for uniform convergence on compact sets. In particular $\Gamma^*\sigma^* = \pi^*$ on $\mathbb{C}^*$, with the entire extension on the right retained. These are maps on the full functions and all their derivatives, not only on zero sets. These are entire compositions of the actual transformed arithmetic function; they retain the incidence factor a_{ef}. Since the new transform commutes with the actual Newman heat evolution, the already proved coordinate chain rules apply to it verbatim and give

$$\begin{aligned} \partial_t g_t &= -\frac{1}{4r^2}\partial_r^2 g_t + \frac{1}{4r^3}\partial_r g_t \qquad (r \neq 0), \\ \partial_t \Psi^{(4)} + \tfrac{1}{4}U^2\Psi^{(4)} &= 0, \\ \Delta_q \Psi^{(4)} &= N(q)G_{ss}(t,\sigma(q)) + L(q)G_s(t,\sigma(q)). \end{aligned} \tag{93}$$

Every coefficient N, L, U is the original one proved in Section 9; no Euclidean diffusion coefficient has been replaced by the root-coordinate generator. For any retained time map $t = \theta(\tau)$, the full residual is explicitly

$$(\partial_\tau + U \cdot \nabla_q - \nu\Delta_q)G_{\theta(\tau)}(\sigma(q)) = \left(-\frac{\theta'}{4} - \nu N\right) G_{ss} + (1 - \nu L)G_s,$$

with the derivatives on the right evaluated at $(\theta(\tau), \sigma(q))$. The proof is the time chain rule and the three identities in (93), so the force-sized remainder is calculated rather than assumed absent.

The three spectral translations also have exact maps in the retained original root coordinates. For $\lambda \in \{3, 9/2, 13/2\}$ define the entire algebraic correspondence

$$\mathcal{C}_\lambda = \{(r, u) \in \mathbb{C}^2 : u^2 = r^2 - 2\lambda\}.$$

Its morphisms are the coordinate projections $p_\lambda : \mathcal{C}_\lambda \to \mathbb{C}_r$, $(r,u) \mapsto r$, and $q_\lambda : \mathcal{C}_\lambda \to \mathbb{C}_u$, $(r,u) \mapsto u$. Both are defined on the entire stated algebraic curve. With $\tau_\lambda : \mathbb{C}_s \to \mathbb{C}_s$, $s \mapsto s+\lambda$, the exact commuting identity is $\pi q_\lambda = \tau_\lambda \pi p_\lambda$, proved by its defining equation. Pullback of any entire function therefore gives $q_\lambda^* \pi^* = p_\lambda^* \pi^* \tau_\lambda^*$; the derivative channel is computed below in the same u coordinate. It satisfies $-u^2/2 = -r^2/2 + \lambda$. Consequently

$$Z_t(-r^2/2+\lambda) = f_t(u), \qquad Z_s(t,-r^2/2+3) = -\frac{\partial_u f_t(u)}{u} \quad ((r,u) \in \mathcal{C}_3,\ u \neq 0),$$

where $f_t(u) = Z_t(-u^2/2)$ is the original function, already proved even and entire. The second formula follows from $\partial_u f_t = -u Z_s(t,-u^2/2)$; thus its quotient has a unique entire continuation at $u = 0$ equal to $-Z_s(t,0)$. Neither formula depends on the choice of sign of u.

In these actual correspondences the full propagated function is

$$g_t(r) = a_{ef}\left[c_0 f_t(u_3) + \beta \frac{\partial_u f_t(u_3)}{u_3} + c_1 f_t(u_{9/2}) + c_2 f_t(u_{13/2})\right], \qquad u_\lambda^2 = r^2 - 2\lambda.$$

Each potentially branched expression descends to the displayed entire function of r, by the proven evenness and removable quotient. The finite inverse point for a nonzero u_λ is exactly $\Gamma(u_\lambda) = (-1/(2u_\lambda), 3u_\lambda, 26u_\lambda^2)$; its target is

$$F(\Gamma(u_\lambda)) = (-r^2 + 2\lambda, 0, 0).$$

At $u_\lambda = 0$ the completed root point still has no finite inverse in the $x \neq 0$ chart, as its unchanged first coordinate shows. The additional finite point with $x = 0$ is retained by the earlier complete fibre calculation. The three new branch values are precisely $r^2 = 6, 9, 13$; they arise from the source's three specified energies.

All of this also retains the full original cubic coefficient family. For fixed b, c and $s(r) = (cr^3 - 2r^2 + br)/4$, set $g_{t;b,c}(r) = G_t(s(r))$. The exact same chain rule gives

$$\partial_t g_{t;b,c} = -\frac{4}{P_r^2}\partial_r^2 g_{t;b,c} + \frac{4P_{rr}}{P_r^3}\partial_r g_{t;b,c} \quad (P_r = 3cr^2 - 4r + b \neq 0).$$

Thus the source fourth coefficient is propagated into the coefficient family as well as its $b = c = 0$ slice. This is a completed map of the source observable and its actual heat dynamics. For the per-channel function, recovery of the original Ξ uses the full-domain inverse automorphism proved in Section 11. For any actual pair with $a_{ef} > 0$, the recovered input is $Z_t = \mathcal{T}_{\mathcal{D}}^{-1}(G_t/a_{ef})$; the explicit separated pair in the source proof has $a_{ef} \geq 1$. At $a_{ef} = 0$ this fourth coefficient is exactly zero and carries no pair observation. A zero of a finite linear combination is not silently substituted for a zero of its recovered arithmetic input.

13 The second Euclidean diffusion of the actual arithmetic observable

The first diffusion coefficient in the preceding calculation has a finite limit along the escaping inverse-fibre trajectory. We compute the next Euclidean diffusion without dropping any derivative of that coefficient. All spatial derivatives in this section refer to the original Cartesian coordinates (x, y, w), with their Euclidean metric. Newman time t remains distinct from the fluid trajectory time τ.

Keep

$$Q = 1 + xy, \qquad \sigma(x,y,w) = \tfrac{1}{2}[Q^3 w + y^2 Q(4+3xy)], \qquad \Psi_t(x,y,w) = Z(t, \sigma(x,y,w)).$$

Here Z is the actual entire function defined by (42); its initial datum is the specified completed zeta function. Its analyticity and all differentiated integral bounds were proved above. In particular, for each real t,

$$Z_s(t,0) = Z_{sss}(t,0) = 0, \qquad Z_{ss}(t,0) = -32M_1(t) < 0.$$

The strict inequality retains the positive original moment $M_1(t) = \int_0^\infty u^2 e^{tu^2}\Phi(u)\,du$.

13.1 The full fourth-order chain rule

Write $g = \nabla\sigma$, $H = D^2\sigma$, $N = g \cdot g$, and $L = \Delta\sigma$, where $\Delta = \partial_x^2 + \partial_y^2 + \partial_w^2$. For any entire function h of one variable, ordinary differentiation gives

$$\Delta(h \circ \sigma) = Nh''(\sigma) + Lh'(\sigma), \tag{94}$$

$$\begin{aligned}\Delta^2(h \circ \sigma) = {} & N^2 h^{(4)}(\sigma) + (4g^{\mathsf{T}}Hg + 2NL)h^{(3)}(\sigma) \\ & + (2\|H\|_{\mathrm{F}}^2 + 4g \cdot \nabla L + L^2)h''(\sigma) + (\Delta L)h'(\sigma).\end{aligned} \tag{95}$$

Here $\|H\|_{\mathrm{F}}^2 = \sum_{i,j=1}^3 H_{ij}^2$ on the real locus. To prove every coefficient, differentiate the first formula by the product rule:

$$\begin{aligned}\Delta(Nh'') &= (\Delta N)h'' + 2(g \cdot \nabla N)h^{(3)} + NLh^{(3)} + N^2 h^{(4)}, \\ \Delta(Lh') &= (\Delta L)h' + 2(g \cdot \nabla L)h'' + L^2 h'' + LNh^{(3)}.\end{aligned}$$

In these lines every derivative of h is evaluated at σ. For the remaining two identities, write $N = \sum_i (\partial_i\sigma)^2$ and differentiate each summand:

$$\partial_j N = 2\sum_i (\partial_i\sigma)(\partial_j\partial_i\sigma), \qquad \Delta N = 2\sum_{i,j}(\partial_j\partial_i\sigma)^2 + 2\sum_i (\partial_i\sigma)\partial_i\Delta\sigma.$$

Thus $g \cdot \nabla N = 2g^{\mathsf{T}}Hg$ and $\Delta N = 2\|H\|_{\mathrm{F}}^2 + 2g \cdot \nabla L$. Substitution proves (95), including its second- and first-derivative terms.

13.2 All coefficients on the unchanged inverse trajectory

For $z > 0$ set

$$\gamma(z) = (z^{-1}, -3z/2, 13z^2/2), \qquad \tau = (1 - z^2)/8.$$

The inverse is $z = \sqrt{1 - 8\tau}$ on $\tau < 1/8$. This is precisely the original trajectory, parametrized by a positive coordinate; it has not been rescaled. Direct substitution gives $Q(\gamma) = -1/2$ and $\sigma(\gamma) = -z^2/8$. For clarity the original first derivatives and Laplacian before substitution are

$$\begin{aligned}g_1 &= \tfrac{3}{2}yQ^2 w + \tfrac{1}{2}y^3(7 + 6xy), \\ g_2 &= \tfrac{3}{2}xQ^2 w + \tfrac{1}{2}(8y + 21xy^2 + 12x^2y^3), \\ g_3 &= \tfrac{1}{2}Q^3, \\ L &= 3wQ(x^2 + y^2) + 3y^4 + 18x^2y^2 + 21xy + 4.\end{aligned}$$

Differentiating these polynomials, followed by the stated substitution, gives

$$g(\gamma) = \begin{pmatrix} -9z^3/32 \\ -3z/16 \\ -1/16 \end{pmatrix}, \qquad H(\gamma) = \begin{pmatrix} -27z^4/4 & 3z^2/16 & -9z/16 \\ 3z^2/16 & 13/4 & 3/(8z) \\ -9z/16 & 3/(8z) & 0 \end{pmatrix}.$$

The full contractions needed in (95) are

$$\begin{aligned}
N(\gamma) &= \frac{81z^6+36z^2+4}{1024}, & L(\gamma) &= \frac{13-27z^4}{4},\\
\|H(\gamma)\|_{\mathrm{F}}^2 &= \frac{729z^8}{16}+\frac{9z^4}{128}+\frac{81z^2}{128}+\frac{169}{16}+\frac{9}{32z^2},\\
(g\cdot\nabla L)(\gamma) &= \frac{9477z^8}{512}-\frac{405z^4}{64}+\frac{27z^2}{128}+\frac{81}{32}+\frac{3}{32z^2},\\
(g^{\mathsf{T}}Hg)(\gamma) &= -\frac{2187z^{10}}{4096}+\frac{81z^6}{4096}-\frac{81z^4}{4096}+\frac{117z^2}{1024}+\frac{9}{1024},\\
(\Delta L)(\gamma) &= -111z^2+\frac{36}{z^2}.
\end{aligned}$$

These are polynomial differentiation and finite matrix products, with both occurrences of each off-diagonal Hessian entry retained in its squared Frobenius norm. They also give a direct way to check each term in the following fully expanded result:

$$(\Delta^2\Psi_t)(\gamma(z)) = \sum_{j=1}^{4} C_j(z)\,\partial_s^j Z(t,-z^2/8), \tag{96}$$

where

$$\begin{aligned}
C_4(z) &= \left(\frac{81z^6+36z^2+4}{1024}\right)^2,\\
C_3(z) &= -\frac{6561z^{10}}{2048}+\frac{243z^6}{2048}-\frac{135z^4}{1024}+\frac{351z^2}{512}+\frac{31}{512},\\
C_2(z) &= \frac{26973z^8}{128}-\frac{4419z^4}{64}+\frac{135z^2}{64}+\frac{669}{16}+\frac{15}{16z^2},\\
C_1(z) &= -111z^2+\frac{36}{z^2}.
\end{aligned}$$

For example, the coefficient of z^{-2} in C_2 is $2(9/32)+4(3/32)=15/16$; the L^2 term has no negative power. Thus this coefficient includes the differentiated first-order coefficient as well as the Hessian square.

Proposition 13.1 (Two successive Euclidean diffusion limits)**.** *For every fixed real Newman time t,*

$$\lim_{z\downarrow 0}(\Delta\Psi_t)(\gamma(z)) = \frac{Z_{ss}(t,0)}{256} = -\frac{M_1(t)}{8}, \tag{97}$$

$$\lim_{z\downarrow 0} z^2(\Delta^2\Psi_t)(\gamma(z)) = \frac{15}{16}Z_{ss}(t,0) = -30M_1(t) < 0. \tag{98}$$

In particular $\Delta^2\Psi_t$ is unbounded on $\mathbb{R}^3$, although Ψ_t is real analytic and smooth at every finite spatial point.

Proof. At this fixed t, the entire even Taylor expansion of $Z(t,s)$ and its termwise derivatives give, with $d_j = \partial_s^j Z(t,0)$,

$$\begin{aligned}
Z_s(t,-z^2/8) &= -d_2z^2/8+O(z^6),\\
Z_{ss}(t,-z^2/8) &= d_2+O(z^4),\\
Z_{sss}(t,-z^2/8) &= -d_4z^2/8+O(z^6),\\
Z_{ssss}(t,-z^2/8) &= d_4+O(z^4).
\end{aligned}$$

The remainders follow from convergence on a fixed closed disk around zero; they are bounded there by the convergent Taylor tails. Equation (94), $N(\gamma) \to 1/256$, and $L(\gamma) \to 13/4$ prove (97). In (96), C_4 and C_3 are bounded as $z \downarrow 0$, $z^2 C_2 \to 15/16$, and $z^2 C_1 \to 36$. Multiplying that equation by z^2 and using the four Taylor estimates proves (98). The last term tends to zero because $Z_s(t, -z^2/8) \to 0$; its unmultiplied finite contribution is $-36d_2/8 + O(z^4)$ and has not been discarded from the exact formula. The moment identity and positivity give the displayed strict sign. Consequently, for sufficiently small $z > 0$ the bilaplacian is at most $-15M_1(t)z^{-2}$, proving unboundedness. Entire composition proves smoothness at each finite point. Finally $|\gamma(z)| \geq z^{-1} \to \infty$, so this sequence is explicitly at spatial infinity. □

13.3 The exact relation to the material diffusion operator

Retain the original divergence-free polynomial flow X_τ and its Jacobian A_τ from the material-generator construction. On each of its open flow domains, X_τ is a smooth diffeomorphism, $\det A_\tau = 1$, and $K_\tau = A_\tau^{-1} A_\tau^{-\mathsf{T}}$. The scalar operator proved there is $\mathcal{L}_\tau = \operatorname{div}(K_\tau \nabla)$. For every smooth physical scalar v on the image domain the full chain and cofactor identity give

$$\mathcal{L}_\tau(v \circ X_\tau) = (\Delta v) \circ X_\tau.$$

Apply this same identity a second time, now to the smooth scalar Δv. At this fixed τ it proves the exact intertwining

$$\mathcal{L}_\tau^2(v \circ X_\tau) = (\Delta^2 v) \circ X_\tau.$$

The square is operator composition; it differentiates every spatial coefficient of K_τ. It is not the square of a coefficient evaluated at one point.

The already established original flow identity $F_1(X_\tau(q)) = F_1(q) + 2\tau$ gives

$$\Psi_t(X_\tau(q)) = Z(t, \sigma(q) + \tau).$$

At $q_0 = (1, -3/2, 13/2)$ and $\tau < 1/8$ one has $X_\tau(q_0) = \gamma(\sqrt{1 - 8\tau})$. Therefore (97) and (98) give, without interchanging the two time variables,

$$\lim_{\tau \uparrow 1/8} \mathcal{L}_\tau[Z(t, \sigma + \tau)](q_0) = -M_1(t)/8,$$
$$\lim_{\tau \uparrow 1/8} (1 - 8\tau) \mathcal{L}_\tau^2[Z(t, \sigma + \tau)](q_0) = -30 M_1(t).$$

For the physical operator $\nu\Delta$ with the unchanged $\nu > 0$, the respective limits acquire the factors ν and ν^2. The scalar in brackets itself has an entire extension in τ. Its material coefficients nevertheless have the displayed second diffusion behavior at the end of this incomplete chart. These identities are an exact relation between the arithmetic observable and Euclidean viscosity. They do not give a singular velocity at a finite physical point, admissible initial velocity or forcing, or a counterexample to the stated Navier–Stokes alternatives.

13.4 The full material-coordinate intertwiner

Here is a complete coordinate proof of the material identity used above. At a fixed permitted real flow time τ, write $X = X_\tau$ on an open flow domain Ω_τ, and let $\Omega'_\tau = X(\Omega_\tau)$. The map

$$X^* : C^\infty(\Omega'_\tau) \longrightarrow C^\infty(\Omega_\tau), \qquad v \longmapsto v \circ X$$

is a linear homeomorphism for the usual compact derivative seminorms; its inverse is composition with the smooth inverse flow. The chain rule on compact sets proves both continuity assertions. Put $A_{ij} = \partial_{q_j} X_i$. Differentiating the flow equation gives $\partial_\tau A = (DU)(X)A$ and $A_0 = I$. On every interval of local invertibility, the determinant derivative is $\partial_\tau \det A =$

$(\operatorname{div} U)(X)\det A = 0$. The proved zero divergence and the initial value consequently give $\det A = 1$ throughout the flow domain.

For completeness the needed cofactor identity, in the unchanged index order, is

$$(A^{-1})_{ji} = \frac{1}{2}\epsilon_{iab}\epsilon_{jcd}(\partial_{q_c}X_a)(\partial_{q_d}X_b), \qquad \sum_j \partial_{q_j}(A^{-1})_{ji} = 0.$$

The first formula is the cofactor formula with $\det A = 1$. In the derivative of its first factor, the symmetric pair (j,c) contracts against ϵ_{jcd}; in the derivative of its second factor the symmetric pair (j,d) does so. Each term is zero, proving the second formula. Therefore every smooth vector field w on Ω'_τ obeys

$$\operatorname{div}_q(A^{-1}(w\circ X)) = \sum_{i,j,k}(A^{-1})_{ji}(\partial_{x_k}w_i)(X)A_{kj} = (\operatorname{div}_x w)\circ X.$$

The gradient chain rule gives $\nabla_q(v\circ X) = A^{\mathsf{T}}(\nabla_x v)\circ X$. With the full matrix $K_\tau = A^{-1}A^{-\mathsf{T}}$ it follows that

$$\mathcal{L}_\tau X^* = X^*\Delta_x, \qquad \mathcal{L}_\tau^2 X^* = X^*\Delta_x^2, \qquad \mathcal{L}_\tau = \operatorname{div}_q(K_\tau\nabla_q). \tag{99}$$

The second equality is the first one applied to $\Delta_x v$; all derivatives of K_τ are part of the composed operator. This proves the typed isomorphism of the differential operators, including every spatial coefficient, on the stated smooth domains.

13.5 The general entire-function residue and the actual fourth coefficient

Let $\mathcal{O}(\mathbb{C}_s)$ carry its compact-convergence topology. The original polynomial pullback σ^* has domain $\mathcal{O}(\mathbb{C}_s)$ and codomain $\mathcal{O}(\mathbb{C}_q^3)$, and is injective with the left inverse η^* already proved above. The four coefficients C_j in (96) are independent of the particular entire input. The complete chain rule therefore gives the following equality for every $h\in\mathcal{O}(\mathbb{C}_s)$:

$$(\Delta_q^2\sigma^*h)(\gamma(z)) = \sum_{j=1}^{4} C_j(z)h^{(j)}(-z^2/8), \tag{100}$$

$$\mathfrak{r}(h) := \lim_{z\downarrow 0} z^2(\Delta_q^2\sigma^*h)(\gamma(z)) = 36h'(0) + \frac{15}{16}h''(0). \tag{101}$$

To prove the limit without discarding a derivative, all four $h^{(j)}(-z^2/8)$ converge to $h^{(j)}(0)$; meanwhile $z^2C_4, z^2C_3 \to 0$, $z^2C_2 \to 15/16$ and $z^2C_1 \to 36$. These four separate limits prove the displayed equality. This is a continuous linear map $\mathfrak{r}:\mathcal{O}(\mathbb{C}_s)\to\mathbb{C}$: the Cauchy formula on any fixed circle about zero bounds both derivative evaluations by its compact supremum.

Its domain can also be restricted to the original $\mathscr{A}$ or the common space $C = \mathscr{A}\cap\mathcal{H}$. It is continuous there because weighted Fourier seminorms bound both derivatives. An explicit right inverse in those very spaces is

$$E:\mathbb{C}\longrightarrow C, \qquad E(a)(s) = \frac{8a}{15\Xi(0)}s^2\Xi(s), \qquad \mathfrak{r}E(a) = a. \tag{102}$$

Indeed $s^2\Xi\in C$, $E(a)'(0) = 0$ and $E(a)''(0) = 16a/15$. Its scalar-to-function continuity holds for the specified topological vector spaces. Thus the precise kernel is $\{h : 36h'(0) + (15/16)h''(0) = 0\}$, the image is all of $\mathbb{C}$, and $h\mapsto(h - E\mathfrak{r}(h), \mathfrak{r}(h))$ is a continuous linear isomorphism onto $\ker\mathfrak{r}\times\mathbb{C}$, with inverse $(b,a)\mapsto b + E(a)$, on each of these domains.

In particular retain $\mathcal{K}_t = \mathcal{T}_{\mathcal{D}} Z_t$ and the physical incidence factor a_{ef} from (77). Define $\mathcal{R}(t) = \mathfrak{r}(\mathcal{K}_t)$. The exact value is

$$\begin{aligned}\mathcal{R}(t) = c_0 \left(36 Z_t'(3) + \frac{15}{16} Z_t''(3)\right) - \frac{1}{336} \left(36 Z_t''(3) + \frac{15}{16} Z_t'''(3)\right) \\ + c_1 \left(36 Z_t'(9/2) + \frac{15}{16} Z_t''(9/2)\right) + c_2 \left(36 Z_t'(13/2) + \frac{15}{16} Z_t''(13/2)\right). \end{aligned} \tag{103}$$

Every derivative and shift remains present. The original evenness of Z_t supplies no deletion of the derivative at any of these nonzero arguments. Equations (100)–(103) prove

$$\lim_{z \downarrow 0} z^2 (\Delta_q^2 \Psi^{(4)})(t, \gamma(z)) = a_{ef} \mathcal{R}(t). \tag{104}$$

In the material coordinates the exact same number is

$$\lim_{\tau \uparrow 1/8} (1 - 8\tau) \mathcal{L}_\tau^2 [a_{ef} \mathcal{K}_t(\sigma + \tau)](q_0) = a_{ef} \mathcal{R}(t),$$

by (99) and the original inverse $z = \sqrt{1 - 8\tau}$, $\tau = (1 - z^2)/8$. For $(\nu \Delta_q)^2$ and $(\nu \mathcal{L}_\tau)^2$ the right sides acquire exactly ν^2, with the same original $\nu > 0$.

This calculation is also a full coordinate morphism through the invertible fourth-coefficient operator, on all of $\mathscr{A}$:

$$\mathfrak{r}\mathcal{T}_{\mathcal{D}} : \mathscr{A} \to \mathbb{C}, \qquad \ker(\mathfrak{r}\mathcal{T}_{\mathcal{D}}) = \mathcal{T}_{\mathcal{D}}^{-1}(\ker \mathfrak{r}), \qquad (\mathfrak{r}\mathcal{T}_{\mathcal{D}})(\mathcal{T}_{\mathcal{D}}^{-1} E) = \mathrm{id}_{\mathbb{C}}\,.$$

Its coordinate formula is (103) with Z_t replaced by its actual input h. The first equality of kernels is the definition tested in both directions, and the last equality uses the two proved inverses, so surjectivity and a continuous right inverse follow without an assertion about pointwise zero preservation.

13.6 A strictly nonzero transferred residue at negative Newman times

Set

$$d_0 = c_0 + c_1 + c_2 = \frac{127}{219024} > 0, \qquad d_1 = 3c_0 - \frac{1}{336} + \frac{9}{2} c_1 + \frac{13}{2} c_2 = 0.$$

These are the original zeroth and first energy moments of $\mathcal{D}$, including its derivative delta. For the unchanged Fourier multiplier $m(v)$ from Section 11, $m(0) = d_0$ and $m'(0) = i d_1 = 0$. Keep the original kernel k, with

$$k(0) = \sum_{n \geq 1} \frac{\pi}{2} n^2 (2\pi n^2 - 3) e^{-\pi n^2} > 0.$$

The sum converges by Gaussian decay and is strictly positive term by term. No constant kernel has replaced $k(v)$.

Proposition 13.2 (The full transferred asymptotic with a remainder bound)**.** *There is a finite explicitly defined constant C_* below such that, for every real $T \geq 1$,*

$$\mathcal{R}(-T) = -15 d_0 k(0) \sqrt{\pi}\, T^{-3/2} + \mathcal{E}(T), \qquad |\mathcal{E}(T)| \leq 96 C_* \sqrt{\pi}\, T^{-5/2}. \tag{105}$$

Consequently $\mathcal{R}(-T) < 0$ for all sufficiently large T. For every original pair with $a_{ef} > 0$, the actual transferred bilaplacian is unbounded along the original escaping trajectory at each such fixed Newman time.

Proof. Differentiation of the original Fourier integral, justified by its full weighted bounds, gives

$$\mathcal{R}(-T) = 4\int_{\mathbb{R}} e^{-Tv^2/4}k(v)\left(-\frac{15}{16}v^2 + 36iv\right)m(v)\,dv.$$

The original coefficients are real, so $m(-v) = \overline{m(v)}$. The kernel k is real and even. Write

$$F(v) = k(v)\,\mathrm{Re}\left[\left(-\frac{15}{16}v^2 + 36iv\right)m(v)\right], \qquad c = \frac{15}{16}d_0k(0) > 0.$$

Then F is smooth, real and even. Its Taylor coefficients through degree two are $F(0) = F'(0) = 0$ and $F''(0)/2 = -c$. Indeed the real quadratic coefficient from $36ivm(v)$ is $-36d_1v^2 = 0$; the other quadratic coefficient is $-(15/16)d_0v^2$. This retains and computes the derivative-channel contribution. Evenness also gives $F'''(0) = 0$.

All constants in the following definition are finite:

$$K_* := \sup_{v\in\mathbb{R}} |k(v)|, \quad A_* := |c_0| + c_1 + c_2, \quad B_* := 1/336,$$

$$C_* := \max\left\{\frac{1}{24}\max_{|v|\le 1}|F^{(4)}(v)|,\ K_*\left(\frac{15}{16}+36\right)(A_*+B_*) + c\right\}.$$

The proved two-ended decay of k makes K_* finite, and smoothness makes the displayed compact derivative maximum finite. For $|v| \le 1$, Taylor's theorem with its integral remainder gives $|F(v) + cv^2| \le C_*v^4$. For $|v| \ge 1$, use the complete bound $|m(v)| \le A_* + B_*|v|$ and

$$|F(v)| \le K_*\left(\frac{15}{16}v^2 + 36|v|\right)(A_* + B_*|v|).$$

Every power on the right is at most $|v|^4$ on that set. The second entry in the maximum consequently gives the same bound $|F(v)+cv^2| \le C_*v^4$ there. It is therefore a global remainder bound for the original integrand, not only a formal Taylor expansion. The imaginary integrand is odd and absolutely integrable, so its integral vanishes by the substitution $v \mapsto -v$.

The Gaussian integral and two derivatives with respect to its positive parameter give, in the original v coordinate,

$$\int_{\mathbb{R}} v^2e^{-Tv^2/4}\,dv = 4\sqrt{\pi}T^{-3/2}, \qquad \int_{\mathbb{R}} v^4e^{-Tv^2/4}\,dv = 24\sqrt{\pi}T^{-5/2}.$$

Differentiation is justified on each compact positive parameter interval by a polynomial times a fixed Gaussian. Integration of the global remainder bound gives $\mathcal{R}(-T) = -16c\sqrt{\pi}T^{-3/2} + \mathcal{E}(T)$ with the stated error. Substitution of c proves (105). For example every $T > \max\{1, 6C_*/c\}$ makes this value negative. Equation (104) then gives unboundedness, since $z^2 > 0$ and $a_{ef}\mathcal{R}(-T) < 0$. □

Every such observable is still entire at each finite spatial point. The exact coordinate map giving the unbounded sequence is $z \mapsto \gamma(z) = \Gamma(-z/2)$, with inverse $z = 1/x$ on its image and $|\gamma(z)| \ge z^{-1} \to \infty$. In material coordinates the same sequence is represented by (τ, q_0) with $\tau = (1 - z^2)/8$, and (99) proves the full operator correspondence there. The newly proved divergence is thus a property of these actual scalar observables and their complete diffusion operators along that image. It supplies no off-critical zero of Z_0 and no finite-point singular classical velocity.

14 The common heat space and the two moving quotient coordinates

The full continuation in Appendices B–I uses $\mathcal{H}_{\le 1} = \mathcal{H}_{\le 1}$ for the entire functions of order at most one. This is the same underlying function set denoted by $\mathcal{H}$ in Section 5.1. Its source topology is τ, its original relation is $N_{\mathrm{ar}} = N_{\mathrm{C}}$, and its original quotient is $Q_{\mathrm{ar}} = Q_{\mathrm{C}}$. These equalities identify the unchanged source objects. The symbol $\mathscr{E}$ elsewhere in this manuscript still denotes the weighted smooth Fourier-kernel space, and $\mathscr{A} = T\mathscr{E}$ retains its transported topology. In this section write $N_t^{\mathscr{A}}, Q_t^{\mathscr{A}}$ for the earlier Fourier relation and quotient, and $N_t^{\mathcal{H}}, Q_t^{\mathcal{H}}$ for those in (164). Thus every coordinate in the comparison has a specified source and target. The topology called κ in the continuation is its compact-convergence topology; the positive physical energy scale $\kappa = 2g^2/a$ remains the separate real parameter of Section 10.

Theorem 14.1 (Equality of the heat maps on the common representatives)**.** *On $C = \mathscr{A} \cap \mathcal{H}$, the two globally defined maps*

$$U_t^{\mathscr{A}} f = T(e^{tv^2/4} T^{-1} f), \qquad U_t^{\mathcal{H}} f = \sum_{k=0}^{\infty} \frac{(-t/4)^k}{k!} \partial_s^{2k} f$$

agree as entire functions, for every $t \in \mathbb{C}$. In particular C is invariant under each map and its inverse. Give $C_t = C$ the topology induced by

$$C_t \longrightarrow \mathscr{A} \times (\mathcal{H}, \tau_t), \qquad f \longmapsto (f, f).$$

Then $U_t : C_0 \longrightarrow C_t$ is a linear homeomorphism with inverse U_{-t}; the parameter in the inverse indicates the function map from C_t back to C_0.

Proof. Take $f = T\phi \in C$. For $|t| \le T_0$ and $|s| \le R_0$, the absolute value of the Fourier integrand is bounded by $4|\phi(v)| \exp(T_0 v^2/4 + R_0|v|)$. Its integral is finite by the defining seminorms of $\mathscr{E}$. The sum of the absolute values of the time-exponential terms has this same integrable bound. Absolute integration of the series therefore gives, at every (t, s),

$$\begin{aligned} U_t^{\mathscr{A}} f(s) &= 4 \sum_{k=0}^{\infty} \frac{(t/4)^k}{k!} \int_{\mathbb{R}} v^{2k} \phi(v) e^{isv}\, dv \\ &= \sum_{k=0}^{\infty} \frac{(-t/4)^k}{k!} \partial_s^{2k} f(s). \end{aligned}$$

The second identity uses $(iv)^{2k} = (-1)^k v^{2k}$, and the same weighted bounds justify every spatial derivative. The series on the right is precisely the convergent γ-series proved in Lemma C.1; convergence in γ implies convergence at every s. Hence the two maps agree with all constants and signs as displayed. Their common output is in both underlying spaces. Applying this conclusion at $-t$ proves invariance under the inverse.

The product map $U_t^{\mathscr{A}} \times U_t^{\mathcal{H}}$ is a homeomorphism from $\mathscr{A} \times (\mathcal{H}, \tau)$ to $\mathscr{A} \times (\mathcal{H}, \tau_t)$: the first assertion is the proved Fourier heat group, and the second is the exact definition of τ_t. The pointwise equality just proved maps the first diagonal onto the second diagonal. Restricting the product map and its continuous inverse proves the claimed homeomorphism. □

Theorem 14.2 (Simultaneous moving quotient correspondence)**.** *Put*

$$\mathscr{K}_t = N_t^{\mathscr{A}} \cap N_t^{\mathcal{H}}, \qquad \mathscr{R}_t = C_t / \mathscr{K}_t.$$

Then $\mathscr{K}_t = U_t \mathscr{K}_0$, and the following maps have the stated types:

$$\overline{U}_t : \mathscr{R}_0 \longrightarrow \mathscr{R}_t, \qquad [f]_{\mathscr{K}_0} \longmapsto [U_t f]_{\mathscr{K}_t}, \tag{106}$$

$$J_t : \mathscr{R}_t \longrightarrow Q_t^{\mathscr{A}} \times Q_t^{\mathcal{H}}, \qquad [g]_{\mathscr{K}_t} \longmapsto ([g]_{N_t^{\mathscr{A}}}, [g]_{N_t^{\mathcal{H}}}). \tag{107}$$

The first map is a homeomorphism and the second a continuous injection. Its two coordinate images and kernels are exactly

$$\operatorname{im}\pi_{\mathscr{A},t} = \{[g]_{N_t^{\mathscr{A}}} : g \in C\}, \qquad \ker\pi_{\mathscr{A},t} = (N_t^{\mathscr{A}} \cap \mathcal{H})/\mathscr{K}_t,$$
$$\operatorname{im}\pi_{\mathcal{H},t} = \{[g]_{N_t^{\mathcal{H}}} : g \in C\}, \qquad \ker\pi_{\mathcal{H},t} = (\mathscr{A} \cap N_t^{\mathcal{H}})/\mathscr{K}_t.$$

The first image is dense in $Q_t^{\mathscr{A}}$. For $B_t = M_s - (t/2)\partial_s$, the induced continuous operator on $\mathscr{R}_t$ satisfies the complete coordinate identity

$$J_t([B_t g]_{\mathscr{K}_t}) = \big([B_t g]_{N_t^{\mathscr{A}}}, [B_t g]_{N_t^{\mathcal{H}}}\big). \tag{108}$$

Proof. For $f \in \mathscr{K}_0$, both definitions of the moving relation put $U_t f$ in $\mathscr{K}_t$. Conversely, $g \in \mathscr{K}_t$ lies in both function spaces; Theorem 14.1 identifies both inverse images as the same $U_{-t}g$. Each moving-relation definition puts this inverse in its original relation, so it belongs to $\mathscr{K}_0$. This proves the equality, including both inclusions. The homeomorphism of C_0 and C_t consequently descends to the homeomorphism (106), with inverse $[g] \mapsto [U_{-t}g]$.

The two coordinate quotient maps are continuous on the product and therefore on its diagonal. Their joint kernel on C_t is exactly $\mathscr{K}_t$. The quotient universal property gives continuity and injectivity of J_t. Testing its first coordinate for zero means precisely $g \in N_t^{\mathscr{A}}$ in addition to $g \in C$; testing the second means $g \in N_t^{\mathcal{H}}$. This proves both kernel identities and the displayed image formulas, without an assumption about a topology on the image of the injection. Density follows from the cutoff proof in Section 5.1: C is dense in $\mathscr{A}$, and its image under the continuous surjective quotient map is dense in $Q_t^{\mathscr{A}}$.

At time zero, multiplication M_s is continuous on both spaces, preserves C, and preserves both closed original relations. It therefore preserves $\mathscr{K}_0$. Conjugation by the proved homeomorphism gives a continuous operator on C_t preserving $\mathscr{K}_t$. In both coordinates its entire-function formula is $U_t M_s U_{-t} = M_s - (t/2)\partial_s$. Passing to representatives proves (108) in both quotient coordinates. □

This also gives the exact map from the earlier fixed-second-coordinate presentation (28) to the present one. If $\mathcal{U}_t^{\mathcal{H}} : Q_{\mathrm{C}} \to Q_t^{\mathcal{H}}$ is the source heat homeomorphism, then

$$(\mathrm{id}_{Q_t^{\mathscr{A}}} \times \mathcal{U}_t^{\mathcal{H}}) j_t = J_t \overline{U}_t. \tag{109}$$

Both sides send $[f]_{\mathscr{K}_0}$ to $([U_t f]_{N_t^{\mathscr{A}}}, [U_t f]_{N_t^{\mathcal{H}}})$; this proves the identity with its domain, codomain and inverse coordinate change.

The actual fourth coefficient in the same function space. The transported coefficient retains its complete formula

$$\begin{aligned} K_t(s) = T_D Z_t(s) = &-\frac{59}{275184} Z_t(s+3) - \frac{1}{336} Z_t'(s+3) \\ &+ \frac{1}{1296} Z_t(s+9/2) + \frac{3}{132496} Z_t(s+13/2). \end{aligned}$$

The resulting map is explicitly

$$C \longrightarrow \mathscr{R}_t \xrightarrow{J_t} Q_t^{\mathscr{A}} \times Q_t^{\mathcal{H}}, \qquad f \longmapsto [f]_{\mathscr{K}_t} \longmapsto ([f]_{N_t^{\mathscr{A}}}, [f]_{N_t^{\mathcal{H}}}),$$

applied to the actual $f = K_t$. Its joint kernel is $\mathscr{K}_t$, as proved above. For this application K_t lies in C: $Z_t = U_t \Xi \in C$, and the four displayed operations preserve both underlying function spaces. On $\mathscr{A}$, a translation by h multiplies the kernel by e^{ihv} and differentiation multiplies it

by iv; every seminorm remains finite by Leibniz's rule and the weights already defining $\mathscr{E}$. On $\mathcal{H}$, the translation bound is, for $b = a/2^{p-1}$,

$$\|f(\cdot + h)\|_{p,a} \le e^{a|h|^p} \|f\|_{p,b},$$

because $|s+h|^p \le 2^{p-1}(|s|^p + |h|^p)$; differentiation has the bound (161). Thus $T_D : C \to C$ is a well-defined linear map on the common underlying representatives. The formula does not require that T_D preserve either quotient relation. Its exact inverse on $\mathscr{A}$, the entire jet quotients and the retained analytic-unit recovery were proved earlier; the present calculation places the actual coefficient in both moving quotient coordinates. The incidence factor remains $a_{ef}K_t$ in the original fourth-order covariance, so these same linear maps retain a_{ef}.

15 The original three special points and an exact real torus quotient

The input for this section is the three-point period and gluing data in the supplied unpublished manuscript *A compact complex threefold fibred by tori over the projective line, and the six-sphere*, its independent TeX reconstruction and its accompanying workbench proofs. The precise source loci are the reconstruction's Section 3 and `candidate_geometry.tex`, `exact_algebra.tex`, and `integral_leray.tex` in that workbench. We use their explicitly marked local period family and overlap formulas as source input. No identification of the global source manifold with a sphere is used in the following quotient calculation.

15.1 The retained base points, periods and real marking

The original base and its distinguished points are

$$B = \mathbb{P}^1_t, \qquad p_1 = 0, \quad p_2 = 1, \quad p_0 = \infty, \qquad t_c = 1/t, \qquad B^\circ = B \setminus \{p_0, p_1, p_2\}.$$

The finite orders are $m_1 = 3$ and $m_2 = 4$. On the marked orbifold cover $\pi : \mathfrak{h}_z \to B \setminus \{p_0\}$, put $g_0 = (g_1 g_2)^{-1}$. The chosen lifts z_j of p_j have coordinates

$$s_j(z) = \frac{z - z_j}{z - \overline{z_j}}, \qquad s_j(g_j z) = e^{-2\pi i/m_j} s_j(z), \qquad t_j \circ \pi = s_j^{m_j}.$$

The inverse of the displayed Cayley coordinate is $z = (z_j - s_j\overline{z_j})/(1 - s_j)$. The modular period is the separate map $\tau : \mathfrak{h}_z \to \mathfrak{h}_\tau$, with

$$j(\tau) = 1728(t \circ \pi), \qquad \tau(z_1) = \rho = e^{\pi i/3}, \qquad \tau(z_2) = i,$$

$$\mu(z_1) = \frac{2 - \rho}{3}, \qquad \mu(z_2) = \frac{1 - i}{2}.$$

Its exact local forms are $\tau - \rho = s_1 U_1(s_1)$ and $\tau - i = s_2^2 U_2(s_2)$ with $U_1(0)U_2(0) \neq 0$. These units and the local base-coordinate changes $t = t_1 V_1(t_1)$ and $t - 1 = t_2 V_2(t_2)$ remain part of the original data; $V_j(0) \neq 0$ does not assign them the value one.

The ordered source lattices and periods are

$$V = \mathbb{Z}\langle \gamma, u, w, \delta \rangle, \qquad \Lambda = V^* = \mathbb{Z}\langle \widehat{\gamma}, \widehat{u}, \widehat{w}, \widehat{\delta} \rangle,$$

$$\Pi(z) = [Z(z) \mid I_2], \qquad Z(z) = \begin{pmatrix} 6\mu(z) & \tau(z) \\ \beta(z) & \mu(z) \end{pmatrix}.$$

Vectors are columns in this ordered basis. The real map underlying the period map is exactly

$$\Pi(z)(a, b, c, d) = (6\mu a + \tau b + c,\ \beta a + \mu b + d). \tag{110}$$

The source admissible period parameter ensures

$$\operatorname{Im}\tau > 0, \qquad D = \operatorname{Im}\beta - \frac{6(\operatorname{Im}\mu)^2}{\operatorname{Im}\tau} < 0.$$

Thus the real determinant in the output order $(\operatorname{Re}\zeta_1, \operatorname{Im}\zeta_1, \operatorname{Re}\zeta_2, \operatorname{Im}\zeta_2)$ is $(\operatorname{Im}\tau)D \neq 0$. The admissible constant in β is retained; no special numerical value is selected here. For an ordinary lifted base open set U, the marking is the real analytic isomorphism

$$\mathcal{M}_U : U \times (\mathbb{R}^4/\mathbb{Z}^4) \longrightarrow \mathcal{T}|_U, \qquad (z, x + \mathbb{Z}^4) \longmapsto [z, \Pi(z)x]. \tag{111}$$

The inverse uses the real inverse of (110). This keeps the full source complex torus in the domain of the next map.

The original inverse-transpose monodromies on Λ are

$$A_1 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 6 & 0 & 1 & 0 \\ -6 & -1 & -1 & 0 \\ -2 & 1 & 0 & 1 \end{pmatrix}, \qquad A_2 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ -6 & 1 & 0 & 0 \\ 3 & 0 & 1 & 1 \end{pmatrix},$$
$$M_0 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{pmatrix}. \tag{112}$$

The source relations are $A_1^3 = A_2^4 = I_4$ and $A_1 A_2 M_0 = I_4$. The two original fixed twist vectors are

$$v_1 = (1, 2, -4, 0)^{\mathsf{T}}, \qquad v_2 = (-1, -3, 3, 0)^{\mathsf{T}}, \qquad A_j v_j = v_j. \tag{113}$$

In particular the fourth marked coordinate is present in every original matrix and twist before taking a quotient.

15.2 The exact circle quotient and its section defects

Let e_1, e_2, e_3, e_4 denote the four vectors of the original ordered real marking, and use $\bar{e}_1, \bar{e}_2, \bar{e}_3$ in the quotient. Define

$$P : \mathbb{R}^4 \longrightarrow \mathbb{R}^3, \quad P(a, b, c, d) = (a, b, c), \qquad \iota : \mathbb{R}^3 \longrightarrow \mathbb{R}^4, \quad \iota(a, b, c) = (a, b, c, 0).$$

Their matrices are $P = [I_3 \mid 0]$ and $\iota = [I_3 \mid 0]^{\mathsf{T}}$. Since $P\mathbb{Z}^4 = \mathbb{Z}^3$, they induce smooth homomorphisms

$$\overline{P} : \mathbb{T}^4 \longrightarrow \mathbb{T}^3, \quad [x] \longmapsto [Px], \qquad \bar{\iota} : \mathbb{T}^3 \longrightarrow \mathbb{T}^4, \quad [y] \longmapsto [\iota y], \qquad \mathbb{T}^n = \mathbb{R}^n/\mathbb{Z}^n. \tag{114}$$

Proposition 15.1 (Typed quotient, kernel and full monodromy intertwining)**.** *The maps in* (114) *give the split exact sequence of marked compact real Lie groups*

$$0 \longrightarrow \mathbb{R}e_4/\mathbb{Z}e_4 \longrightarrow \mathbb{T}^4 \xrightarrow{\overline{P}} \mathbb{T}^3 \longrightarrow 0, \qquad \overline{P}\,\bar{\iota} = \mathrm{id}_{\mathbb{T}^3}. \tag{115}$$

The quotient monodromies in the original first three coordinates are

$$\bar{A}_1 = \begin{pmatrix} 1 & 0 & 0 \\ 6 & 0 & 1 \\ -6 & -1 & -1 \end{pmatrix}, \qquad \bar{A}_2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ -6 & 1 & 0 \end{pmatrix}, \qquad \bar{M}_0 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}. \tag{116}$$

They satisfy

$$PA_j = \bar{A}_j P, \quad PM_0 = \bar{M}_0 P, \quad \bar{A}_1^3 = \bar{A}_2^4 = I_3, \quad \bar{A}_1\bar{A}_2\bar{M}_0 = I_3.$$

The section has the exact defects

$$\begin{aligned}(A_1\iota - \iota\bar{A}_1)(a,b,c) &= (-2a+b)e_4,\\ (A_2\iota - \iota\bar{A}_2)(a,b,c) &= (3a+c)e_4,\\ (M_0\iota - \iota\bar{M}_0)(a,b,c) &= -ae_4.\end{aligned} \tag{117}$$

Proof. For $[x] \in \ker \overline{P}$, the first three entries of x are integers. Subtracting that vector of $\mathbb{Z}^4$ leaves precisely de_4. Two such representatives give the same class exactly when their fourth coordinates differ by an integer. This proves the stated kernel. Surjectivity and the section identity follow by applying P to $\iota(a,b,c)$. The same equations show that $\mathbb{T}^4/(\mathbb{R}e_4/\mathbb{Z}e_4) \to \mathbb{T}^3$, $[[x]] \mapsto [Px]$, is bijective with inverse $[y] \mapsto [[\iota y]]$. Both maps are smooth: in the ordinary local covering coordinates they are the displayed linear projection and inclusion followed by the quotient.

Each matrix in (112) fixes e_4, and its first three rows have zero fourth entry. Taking those rows proves all three intertwiners and the matrices in (116) by literal multiplication. Applying P to the source product and power identities, and then composing with the surjection P, proves the corresponding identities in dimension three. The lower row of each full matrix applied to $(a,b,c,0)$ gives exactly (117). These defects are generally nonzero even as circle classes; thus the section is a right inverse of the marked quotient without being a monodromy-equivariant section. Indeed there is no real linear section equivariant even for M_0 alone. Any linear section has the form $\iota_L(y) = (y, Ly)$ for a row L. Equivariance would require $L(\bar{M}_0 - I_3) = (-1,0,0)$. Its left side is $(0, L_3, 0)$, contradicting the first coordinate of the required equality. □

On the source fibre, this gives the concrete map

$$p_z : \mathbb{C}^2/\Pi(z)\Lambda \longrightarrow \mathbb{T}^3, \qquad [\zeta] \longmapsto [P\Pi(z)^{-1}\zeta], \tag{118}$$

where the inverse is real linear. Its kernel is the original circle $\mathbb{R}\Pi(z)e_4/\mathbb{Z}\Pi(z)e_4$; here $\Pi(z)e_4 = (0,1)^{\mathsf{T}}$ exactly. The map $[y] \mapsto [\Pi(z)\iota y]$ is its marked right inverse. Equation (111) proves the smooth dependence of the quotient and section on the original base variable. This construction concerns a three-real-dimensional fibre quotient. Keeping the complex base produces a five-real-dimensional family; no identification of that total space with a three-dimensional physical domain is implicit.

15.3 The finite affine actions and the complete logarithmic overlaps

Put

$$c_1 = P(v_1/3) = \tfrac{1}{3}(1,2,-4)^{\mathsf{T}}, \qquad c_2 = P(v_2/4) = \tfrac{1}{4}(-1,-3,3)^{\mathsf{T}}. \tag{119}$$

For $j = 1,2$ the full marked map $L_j(x) = A_j x + v_j/m_j$ on $\mathbb{T}^4$ descends to

$$\bar{L}_j([y]) = [\bar{A}_j y + c_j], \qquad \overline{P}L_j = \bar{L}_j\overline{P}.$$

All matrices and translations are those obtained above. Since $\bar{A}_j c_j = c_j$, induction gives

$$\bar{L}_j^k([y]) = [\bar{A}_j^k y + k c_j]. \tag{120}$$

At $k = m_j$ the matrix is the identity and the translation is the integer vector Pv_j, so the map is the identity on $\mathbb{T}^3$. The first coordinate of every $\bar{A}_j^k y$ is y_1. For $0 < k < 3$, the first-coordinate translation of $\bar{L}_1^k$ is $k/3 \notin \mathbb{Z}$; for $0 < k < 4$, that of $\bar{L}_2^k$ is $-k/4 \notin \mathbb{Z}$. Thus every nonidentity

power is fixed-point free, and the two affine torus actions have exactly orders three and four. On $\Delta_j \times \mathbb{T}^3$ the corresponding maps are $(z,[y]) \mapsto (g_j z, \bar{L}_j[y])$, and the same argument proves freeness also over the fixed base point. The quotient fibre over p_j is therefore the smooth compact real quotient $\mathbb{T}^3/\langle \bar{L}_j \rangle$.

The overlap is a varying translation whose complete real expression must also be retained. Define

$$\begin{aligned} J_z &= \Pi(z)^{-1} i \Pi(z) : \mathbb{R}^4 \to \mathbb{R}^4, \qquad \alpha(z) = \frac{\log s_j(z)}{2\pi i}, \\ a_j(z) &= \operatorname{Re}\alpha(z)\, v_j + \operatorname{Im}\alpha(z)\, J_z v_j, \qquad \bar{a}_j(z) = P a_j(z). \end{aligned} \tag{121}$$

Here is an explicit coordinate formula for the full J_z, retaining all entries of the period matrix. Write $\mu_R = \operatorname{Re}\mu$, $\mu_I = \operatorname{Im}\mu$, and likewise for τ, β. For $x = (a,b,c,d)$ set

$$p = 6\mu a + \tau b + c, \qquad q = \beta a + \mu b + d, \qquad E = 6\mu_I^2 - \tau_I \beta_I = -\tau_I D \neq 0.$$

Then $J_z x = (a', b', c', d')$ is exactly

$$\begin{aligned} a' &= \frac{\mu_I \operatorname{Re} p - \tau_I \operatorname{Re} q}{E}, & b' &= \frac{-\beta_I \operatorname{Re} p + 6\mu_I \operatorname{Re} q}{E}, \\ c' &= -\operatorname{Im} p - 6\mu_R a' - \tau_R b', & d' &= -\operatorname{Im} q - \beta_R a' - \mu_R b'. \end{aligned} \tag{122}$$

Indeed the imaginary parts of $\Pi(z) J_z x = i(p,q)$ are the two linear equations for a', b' with determinant E; its real parts give c', d'. For the two actual twist inputs one uses, without changing their signs,

$$(p,q)_{v_1} = (6\mu + 2\tau - 4, \beta + 2\mu), \qquad (p,q)_{v_2} = (-6\mu - 3\tau + 3, -\beta - 3\mu).$$

Equations (121) and (122) consequently specify every coordinate of the descended translation.

The source period covariance gives $J_{g_j z} A_j = A_j J_z$. Clockwise continuation of the chosen logarithm changes α by $-1/m_j$, so

$$a_j(g_j z) = A_j a_j(z) - v_j/m_j, \qquad \bar{a}_j(g_j z) = \bar{A}_j \bar{a}_j(z) - c_j. \tag{123}$$

To verify the first identity directly, the real part of α acquires the displayed subtraction; its imaginary part stays fixed, and $A_j v_j = v_j$ gives $J_{g_j z} v_j = A_j J_z v_j$. Applying the proved intertwiner P gives the second identity.

On the punctured lifted family put

$$\bar{h}_j(z,[y]) = (z, [y + \bar{a}_j(z)]).$$

Changing $\log s_j$ by $2\pi i n$ changes $\bar{a}_j$ by the integer vector nPv_j, proving branch independence. Its inverse is translation by $-\bar{a}_j(z)$. Applying (123) to the three successive maps gives the exact conjugacy

$$\bar{h}_j\, (g_j, \bar{L}_j)\, \bar{h}_j^{-1} : (z,[y]) \longmapsto (g_j z, [\bar{A}_j y]). \tag{124}$$

The cancelled translation is $-\bar{A}_j \bar{a}_j(z) + c_j + \bar{a}_j(g_j z) = 0$. This proves descent of the complete logarithmic gluing to the real circle quotient, including its inverse and every period-dependent coefficient. Circle cosets are preserved because the full linear maps fix e_4 and fibre translations preserve cosets. No preservation of a chosen Euclidean or Hermitian metric is required or asserted.

There is a concrete reason to retain these overlap translations when combining the three source points. In a single uncorrected marked fibre, compose the two finite affine formulas and the linear cusp formula. Their linear product is $\bar{A}_1 \bar{A}_2 \bar{M}_0 = I_3$, but their translation is

$$c_1 + \bar{A}_1 c_2 = (1/12, -1/12, 1/6)^{\mathsf{T}} \notin \mathbb{Z}^3.$$

Thus those three uncorrected affine formulas are not the three meridians in the same middle-family marking. The exact marking change is (124); it identifies each finite formula with its linear middle-family representative, where the product relation holds. This computes the discrepancy and the maps correcting it.

15.4 The original cusp direction and three independent shear data

The cusp's complete period expression is

$$e^{2\pi i\tau} = t_c u_1(t_c), \quad u_1(0) = 1/1728, \quad h = (2\pi i)^{-1}\log u_1, \quad s = \tau - h, \quad b = \beta + \tau,$$

$$Z = sB_0 + C(t_c), \qquad B_0 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad C(t_c) = \begin{pmatrix} 6\mu(t_c) & h(t_c) \\ b(t_c) - h(t_c) & \mu(t_c) \end{pmatrix}. \tag{125}$$

The full holomorphic unit u_1 and all entries of C remain present. Define the original cusp sublattice and quotient with their ordered bases:

$$\Lambda_{\mathrm{tor}} = \mathbb{Z}\langle e_3, e_4\rangle, \qquad \bar{\Lambda} = \Lambda/\Lambda_{\mathrm{tor}} = \mathbb{Z}\langle [e_1], [e_2]\rangle.$$

On the original marked lattice,

$$(M_0 - I)(a, b, c, d) = (0, 0, b, -a), \qquad \ker(M_0 - I) = \mathrm{im}(M_0 - I) = \mathbb{Z}\langle e_3, e_4\rangle.$$

The quotient sends this cusp image lattice onto $\mathbb{Z}\bar{e}_3$, with kernel $\mathbb{Z}e_4$. In particular the unchanged source vector e_2 supplies

$$c_0 = P((M_0 - I)e_2) = \bar{e}_3 = (0, 0, 1)^{\mathsf{T}}. \tag{126}$$

The other primitive image $(M_0 - I)e_1 = -e_4$ is killed by this precise quotient. This records the lost cusp direction explicitly. Translation by the integer vector c_0 is the identity on $\mathbb{T}^3$; the same vector defines a nonzero tangent direction there. These two statements follow respectively from taking its class modulo $\mathbb{Z}^3$ and from the differential of the covering $\mathbb{R}^3 \to \mathbb{T}^3$.

The relation with the original two-dimensional cusp torus is therefore the following exact restriction. Denote its full marked torus by

$$\mathbb{T}^2_{\mathrm{tor}} = (\mathbb{R}e_3 \oplus \mathbb{R}e_4)/(\mathbb{Z}e_3 \oplus \mathbb{Z}e_4), \qquad K_4 = \mathbb{R}e_4/\mathbb{Z}e_4, \quad K_3 = \mathbb{R}\bar{e}_3/\mathbb{Z}\bar{e}_3.$$

In addition to (115), one has

$$\begin{aligned} 0 \longrightarrow K_4 \longrightarrow \mathbb{T}^2_{\mathrm{tor}} &\xrightarrow{[re_3+ue_4]\mapsto[r\bar{e}_3]} K_3 \longrightarrow 0, \\ 0 \longrightarrow \mathbb{R}e_4 \longrightarrow \mathbb{R}e_3 \oplus \mathbb{R}e_4 &\xrightarrow{(r,u)\mapsto r} \mathbb{R}\bar{e}_3 \longrightarrow 0. \end{aligned} \tag{127}$$

The vertical inclusions of the middle and final terms into $\mathbb{T}^4$ and $\mathbb{T}^3$ make a commuting diagram with (115): on every representative, $P(re_3 + ue_4) = r\bar{e}_3$. The same equality proves commutation of the differential sequences. Their kernels, surjections and sections $[r] \mapsto [re_3]$ follow by testing the displayed coordinates and their integer lattices. The original B_0 is also retained in this diagram: the map $\bar{\Lambda} \to \Lambda_{\mathrm{tor}}$ sends $(a, b) \mapsto be_3 - ae_4$, and composition with P is $(a, b) \mapsto b\bar{e}_3$. This proves the exact relationship between the two real cusp coordinates, their two-torus, and the one-dimensional image inside the three-dimensional marked quotient.

The cusp punctured-overlap map on the original complex covering is

$$(z, \zeta_1, \zeta_2) \longmapsto (e^{2\pi i\zeta_1}, e^{2\pi i\zeta_2}, e^{2\pi i s(z)}).$$

In these coordinates the e_4 circle acts by $(x_1, x_2, t_c) \mapsto (x_1, e^{2\pi ir}x_2, t_c)$, $r \in \mathbb{R}/\mathbb{Z}$. This follows by exponentiating $\Pi(z)e_4 = (0, 1)$. It commutes with the full toric multipliers

$$(x, t_c) \longmapsto \big(e^{2\pi i C(t_c)\bar{\lambda}} t_c^{B_0\bar{\lambda}} x, t_c\big), \qquad \bar{\lambda} \in \mathbb{Z}^2.$$

Thus the overlap identifies the same circle quotient on the punctured cusp model. The calculation makes no assertion that this circle action remains free on the singular central fibre; the defined three-dimensional marked fibre quotient is on the smooth torus locus.

Finally, in the original order (p_1, p_2, p_0) the three directions form

$$[c_1\ c_2\ c_0] = \begin{pmatrix} 1/3 & -1/4 & 0 \\ 2/3 & -3/4 & 0 \\ -4/3 & 3/4 & 1 \end{pmatrix}, \qquad \det[c_1\ c_2\ c_0] = -\frac{1}{12}. \tag{128}$$

The integer wave covectors in the same order are

$$k_1 = (4,0,1), \qquad k_2 = (3,0,1), \qquad k_0 = (0,1,0), \qquad \det \begin{pmatrix} k_1 \\ k_2 \\ k_0 \end{pmatrix} = -1. \tag{129}$$

Direct multiplication gives $k_j c_j = 0$ for all three labels: explicitly $(4-4)/3 = 0$, $(-3+3)/4 = 0$, and $0 = 0$. The first determinant is the upper-left minor $-1/4 + 1/6 = -1/12$; expansion along the last row of the second matrix gives $-(4-3) = -1$. Hence the source directions span the whole quotient tangent space and the wave covectors form an integral basis of its dual lattice. The direction map $(r_1, r_2, r_0) \mapsto r_1 c_1 + r_2 c_2 + r_0 c_0$ has the exact inverse matrix

$$[c_1\ c_2\ c_0]^{-1} = \begin{pmatrix} 9 & -3 & 0 \\ 8 & -4 & 0 \\ 6 & -1 & 1 \end{pmatrix}.$$

Multiplying this displayed matrix on either side by (128) gives I_3, proving the inverse in the original coordinates. For each j, each scalar amplitude a_j and each smooth one-periodic function F_j, the literal quotient field $y \mapsto a_j c_j F_j(k_j y)$ is periodic and has divergence $a_j (k_j c_j) F_j'(k_j y) = 0$. This final derivative identity uses the explicit quotient coordinates; constructing and checking its subsequent Navier–Stokes dynamics is a further calculation on these defined fields. These individual modes are fields on the marked $\mathbb{T}^3$. For clarity, their first two wave covectors obey $k_1 \bar{A}_1 = (-2,-1,-1)$ and $k_2 \bar{A}_2 = (-3,1,0)$, whereas $k_0 \bar{M}_0 = k_0$. Therefore no automatic descent of the individual finite-point modes to $\mathbb{T}^3/\langle \bar{L}_j \rangle$ is asserted. There is an explicit field map to that quotient: for a smooth one-periodic real function F, the finite orbit sum

$$\mathcal{R}_j F(y) = c_j \sum_{\ell=0}^{m_j - 1} F(k_j \bar{A}_j^{\ell} y)$$

is invariant as a vector field under $\bar{L}_j$. Indeed $\bar{A}_j c_j = c_j$ and $k_j \bar{A}_j^{\ell} c_j = k_j c_j = 0$; substitution of $\bar{A}_j y + c_j$ cycles the sum and gives $\mathcal{R}_j F(\bar{L}_j y) = \bar{A}_j \mathcal{R}_j F(y)$. Each summand has zero divergence by the same coordinate derivative calculation. The local quotient covering then gives a unique smooth field downstairs: at $[y]$ use its differential on $\mathcal{R}_j F(y)$; the displayed covariance proves independence of the lift. This supplies the exact field descent map while keeping its differential-operator and metric questions separate for the subsequent dynamics calculation.

15.5 Three source polarizations in the original Euclidean three-torus

Fix an ordinary lifted base point z and the real quotient map p_z in (118). The physical three-torus below is its marked target. Write t_B for the unchanged source base coordinate and t for fluid time; their product is $\mathbb{P}^1_{t_B} \times I_t$, with the original base recovered by projection to t_B. The fields below are defined on this regular marked target and use the data of all three special points as their labelled polarization inputs.

Use the three vectors from the preceding exact lattice quotient, in the retained order $(1,2,0)$:

$$c_1 = \tfrac{1}{3}(1,2,-4)^{\mathsf{T}}, \quad c_2 = \tfrac{1}{4}(-1,-3,3)^{\mathsf{T}}, \quad c_0 = (0,0,1)^{\mathsf{T}}.$$

The divisors $3, 4$ are the actual finite filling orders. Choose the following integral character rows on the quotient torus:

$$k_1 = (4, 0, 1), \qquad k_2 = (3, 0, 1), \qquad k_0 = (0, 1, 0).$$

They obey $k_j c_j = 0$. For $\nu > 0$ and any three real smooth amplitudes a_j on a time interval, define on the unit Euclidean torus $\mathbb{T}^3_x = \mathbb{R}^3/\mathbb{Z}^3$

$$u(t, x) = \sum_{j \in \{1,2,0\}} a_j(t) c_j \sin(2\pi k_j x). \tag{130}$$

This is an explicit linear map from the amplitude space to smooth vector fields. Its inverse on its image is

$$a_j(t) = \frac{2}{|c_j|^2} \int_{[0,1]^3} u(t, x) \cdot c_j \sin(2\pi k_j x)\, dx. \tag{131}$$

To prove it, the integral of a nonconstant integer exponential is zero by integrating a coordinate with nonzero frequency. Product-to-sum then gives zero for distinct displayed sine frequencies and $1/2$ for the square of each. This proves the inverse and also continuity in every compact-time smooth seminorm in both directions. Direct differentiation gives $\operatorname{div} u = 0$ and

$$\Delta_x u = -4\pi^2 \sum_j |k_j|^2 a_j c_j \sin(2\pi k_j x), \qquad (|k_1|^2, |k_2|^2, |k_0|^2) = (17, 10, 1).$$

The three phases are themselves a typed torus automorphism:

$$W : \mathbb{T}^3_x \longrightarrow \mathbb{T}^3_\theta, \qquad \theta = (4x_1 + x_3, 3x_1 + x_3, x_2),$$
$$W^{-1}(\theta) = (\theta_1 - \theta_2, \theta_0, -3\theta_1 + 4\theta_2).$$

Both matrices are integral and their composition is the identity; $\det W = -1$, so no finite covering kernel or factor in volume is introduced. The complete Euclidean metric transformation is

$$g_\theta = W^{-\mathsf{T}} W^{-1}, \qquad g_\theta^{-1} = W W^{\mathsf{T}} = G = \begin{pmatrix} 17 & 13 & 0 \\ 13 & 10 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \Delta_x = W^*(G^{ab} \partial_{\theta_a} \partial_{\theta_b})(W^*)^{-1}. \tag{132}$$

Indeed $\partial_{x_j} = \sum_a W_{aj} \partial_{\theta_a}$; applying this twice proves the last identity, including its mixed coefficient $26 \partial_{\theta_1} \partial_{\theta_2}$. For $v = W u \circ W^{-1}$, $p_\theta = p \circ W^{-1}$ and $f_\theta = W f \circ W^{-1}$, the full momentum equation becomes

$$\partial_t v + (v \cdot \nabla_\theta) v - \nu G^{ab} \partial_{\theta_a} \partial_{\theta_b} v + G \nabla_\theta p_\theta = f_\theta, \qquad \operatorname{div}_\theta v = 0.$$

The derivative chain rule proves convection and divergence; multiplying $\nabla_x p = W^{\mathsf{T}} \nabla_\theta p_\theta$ by W proves the pressure term. Thus the phase coordinates retain the original Euclidean viscosity rather than changing its matrix.

At every time when all a_j are nonzero, the field depends on all three spatial directions: if a constant real vector b obeys $(b \cdot \nabla_x) u = 0$, Fourier independence gives $k_j b = 0$ for all three j. Invertibility of W gives $b = 0$. Its exact kinetic energy on the unit cell is

$$\frac{1}{2} \int_{[0,1]^3} |u|^2\, dx = \frac{1}{4} \left(\frac{7}{3} a_1^2 + \frac{19}{16} a_2^2 + a_0^2 \right). \tag{133}$$

This follows from the same orthogonality and the unchanged lengths $|c_1|^2 = 7/3$, $|c_2|^2 = 19/16$, $|c_0|^2 = 1$.

15.6 Every mixed interaction, the pressure and the force

Use ordered pairs $(i,j) = (1,2),(1,0),(2,0)$ and define

$$k_{ij}^{\pm} = k_i \pm k_j, \qquad B_{ij}^{\pm} = (c_i \cdot k_j)c_j \pm (c_j \cdot k_i)c_i, \qquad P_\ell b = b - \ell^{\mathsf{T}} \frac{\ell b}{|\ell|^2}.$$

Here each k or ℓ is a row and each B or b a column. Every displayed cross frequency is nonzero. The six frequencies, in the same pair order and then $+,-$, are

$$(7,0,2),\ (1,0,0),\ (4,1,1),\ (4,-1,1),\ (3,1,1),\ (3,-1,1).$$

They are distinct up to sign and none is a base frequency. Differentiating (130) before collecting terms gives

$$N := (u \cdot \nabla)u = 2\pi \sum_{i \neq j} a_i a_j (c_i \cdot k_j) c_j \sin(2\pi k_i x)\cos(2\pi k_j x).$$

The self terms are zero by $k_j c_j = 0$. Applying both exact product-to-sum identities to each ordered pair gives

$$N = \pi \sum_{i<j} a_i a_j \sum_{\epsilon \in \{+,-\}} B_{ij}^{\epsilon} \sin(2\pi k_{ij}^{\epsilon} x). \tag{134}$$

The notation $i<j$ here means the three ordered pairs just fixed, not numerical ordering of their names.

Choose the following zero-mean, unit-periodic pressure and force:

$$p(t,x) = \frac{1}{2} \sum_{i<j} a_i a_j \sum_{\epsilon \in \{+,-\}} \frac{k_{ij}^{\epsilon} B_{ij}^{\epsilon}}{|k_{ij}^{\epsilon}|^2} \cos(2\pi k_{ij}^{\epsilon} x), \tag{135}$$

$$f(t,x) = \sum_j (a_j' + 4\pi^2 \nu |k_j|^2 a_j) c_j \sin(2\pi k_j x) + \pi \sum_{i<j} a_i a_j \sum_{\epsilon \in \{+,-\}} P_{k_{ij}^{\epsilon}} B_{ij}^{\epsilon} \sin(2\pi k_{ij}^{\epsilon} x). \tag{136}$$

The gradient of each cosine pressure term is minus the longitudinal part of the corresponding term in (134). Hence

$$\partial_t u + (u \cdot \nabla)u - \nu \Delta u + \nabla p = f, \qquad \operatorname{div} u = \operatorname{div} f = 0.$$

This proves every component of an exact forced Navier–Stokes solution, with the original positive viscosity and periodic pressure. It also proves the exact residual map $(a_1, a_2, a_0) \mapsto (u, p, f)$ and its inverse on the velocity coordinate (131).

The six transverse interaction columns are explicitly

$$P_{k_{12}^+} B_{12}^+ = (-7/318, 1/12, 49/636)^{\mathsf{T}}, \qquad P_{k_{12}^-} B_{12}^- = (0, 5/12, -7/12)^{\mathsf{T}},$$
$$P_{k_{10}^+} B_{10}^+ = (1, 16, -20)^{\mathsf{T}}/27, \qquad P_{k_{10}^-} B_{10}^- = (-17, -16, 52)^{\mathsf{T}}/27,$$
$$P_{k_{20}^+} B_{20}^+ = (7, -27, 6)^{\mathsf{T}}/44, \qquad P_{k_{20}^-} B_{20}^- = (29, 27, -60)^{\mathsf{T}}/44.$$

They follow by inserting the three original c_j into the displayed definitions; all are nonzero. For an independent component-level check of the non-gradient interaction,

$$B_{12}^+ = (0, 1/12, 1/12)^{\mathsf{T}}, \qquad (k_{12}^+)^{\mathsf{T}} \times B_{12}^+ = (-1/6, -7/12, 7/12)^{\mathsf{T}} \neq 0.$$

The coefficient of $\cos(2\pi(7x_1 + 2x_3))$ in $\operatorname{curl} N$ is exactly $2\pi^2 a_1 a_2$ times this vector. Distinct frequencies preclude cancellation with another pair. Thus the three-point image has actual transverse nonlinear forcing, with all pair interactions explicitly calculated.

The complete mean quadratic stress and its exact image cone are

$$\int_{[0,1]^3} u \otimes u \, dx = \frac{1}{2} C \operatorname{diag}(a_1^2, a_2^2, a_0^2) C^{\mathsf{T}}, \qquad C = (c_1, c_2, c_0), \quad \det C = -1/12. \tag{137}$$

Indeed all mixed sine products integrate to zero. A real symmetric matrix belongs to this image exactly when $C^{-1} R C^{-\mathsf{T}}$ is diagonal with nonnegative entries. Necessity is the displayed identity; for sufficiency take the three amplitudes to be either sign of the square roots of twice those entries. This proves both directions and all amplitude fibres, including the unique zero choice for a zero entry. It quantifies precisely the stress supplied by these three specific source channels.

15.7 The exact forcing class and what fixed modes permit

Define the original scalar function

$$b(t) = \begin{cases} \exp[-1/(t(1-t))], & 0 < t < 1, \\ 0, & t \notin (0,1). \end{cases}$$

Every derivative on $(0,1)$ is the same exponential times a rational function with poles only at $0, 1$. For every positive integer n, $y^n e^{-y} \to 0$ as $y \to \infty$, for example by comparison with the $(n+1)$st exponential-series term. Consequently each derivative tends to zero at both endpoints. Extending by zero proves $b \in C_c^\infty(\mathbb{R})$, with support $[0,1]$. Take $a_1 = a_2 = a_0 = b$ in (130)–(136). This gives an actual three-dimensional field starting from rest, a smooth periodic pressure, and a force compactly supported in time. All spatial and temporal derivatives of the force are bounded on its compact time support, and therefore satisfy every bound $C_{\alpha m K}(1+t)^{-K}$ on $t \geq 0$. Formula (133) proves a uniform finite kinetic-energy bound. The velocity and pressure are globally smooth, so this explicit admissible forced example itself is not a breakdown example.

For the entire fixed three-mode image there is an exact stronger calculation. Suppose an input amplitude curve on $[0,T)$ has a prescribed momentum force F, with any periodic pressure. Projecting the momentum equation onto the divergence-free base sine mode and using the six distinct cross frequencies gives

$$a_j'(t) + \lambda_j a_j(t) = b_j(t), \quad \lambda_j = 4\pi^2 \nu |k_j|^2 > 0, \qquad b_j(t) = \frac{2}{|c_j|^2} \int_{[0,1]^3} F(t,x) \cdot c_j \sin(2\pi k_j x) \, dx. \tag{138}$$

The pressure projection is zero by periodic integration by parts and $\operatorname{div}(c_j \sin(2\pi k_j x)) = 0$. Multiplication by $e^{\lambda_j t}$ and integration give the complete inverse formula

$$a_j(t) = e^{-\lambda_j t} a_j(0) + \int_0^t e^{-\lambda_j (t-s)} b_j(s) \, ds. \tag{139}$$

If F extends smoothly through a finite T, so do its integrals b_j and the explicit functions (139); differentiation under their finite integrals proves all time derivatives. The finite mode sum then extends smoothly as a velocity. In particular,

$$|a_j(t)| \leq |a_j(0)| + \lambda_j^{-1} \sup_{0 \leq s \leq T} |b_j(s)| \quad (0 \leq t < T).$$

This proves directly that bounded terminal forcing cannot generate velocity blowup within this fixed finite image. No choice of its periodic pressure changes the conclusion.

For zero force the exact classification is even more explicit: (139) gives $a_j(t) = a_j(0) e^{-\lambda_j t}$. Every cross-mode projection in (136) must also vanish. The six nonzero transverse columns

and distinct frequencies force $a_i a_j = 0$ for each of the three pairs. Hence at most one initial amplitude is nonzero. Conversely any one such decaying mode, with zero pressure, solves the unforced equations directly. This proves both directions of the unforced classification for the actual image. The full integer frequency group generated by the three source characters is nevertheless $\mathbb{Z}^3$, since W^{-1} is integral. The six calculated cross frequencies give the first required extra modes of a nonlinear extension; the group identity does not assert that every possible mode has already been dynamically constructed.

15.8 Compact spatial realization with every additional force term

Regard the periodic functions above as functions on $\mathbb{R}^3_x$ and fix any real $\chi \in C_c^\infty(\mathbb{R}^3)$ equal to one on a chosen open cube. The exact vector potential is

$$A(t,x) = \sum_j \frac{a_j(t)}{2\pi|k_j|^2}(k_j^{\mathsf{T}} \times c_j)\cos(2\pi k_j x), \qquad \operatorname{curl} A = u.$$

Indeed differentiating its cosine yields $-k_j^{\mathsf{T}} \times (k_j^{\mathsf{T}} \times c_j)/|k_j|^2 = c_j$ by $k_j c_j = 0$. Define

$$V = \operatorname{curl}(\chi A) = \chi u + d, \qquad d = \nabla\chi \times A, \qquad p_{\mathrm{loc}} = \chi p.$$

Then V is smooth, compactly supported and divergence-free. The complete force for this spatially localized field is

$$\begin{aligned} F_{\mathrm{loc}} = {} & \chi f + (\chi^2 - \chi)N + \partial_t d + \big(\chi(u\cdot\nabla\chi) + d\cdot\nabla\chi\big)u \\ & + \chi(u\cdot\nabla)d + \chi(d\cdot\nabla)u + (d\cdot\nabla)d \\ & - \nu\left(2\sum_l(\partial_l\chi)\partial_l u + (\Delta\chi)u + \Delta d\right) + p\nabla\chi. \end{aligned} \tag{140}$$

To verify every term, expand $\partial_t(\chi u + d)$, expand the two factors of $(\chi u + d)\cdot\nabla(\chi u + d)$, and use $\Delta(\chi u) = \chi\Delta u + 2\sum_l \chi_l u_l + (\Delta\chi)u$ and $\nabla(\chi p) = \chi\nabla p + p\nabla\chi$. The terms multiplied by χ form exactly χf; the remaining terms are precisely (140). Thus

$$\partial_t V + (V\cdot\nabla)V - \nu\Delta V + \nabla p_{\mathrm{loc}} = F_{\mathrm{loc}}.$$

On the cube where $\chi = 1$, the map recovers (u, p, f) exactly. It is a continuous map between the indicated smooth function spaces because each output is a finite sum of products and derivatives. For the specific compact-time amplitudes b, all three localized outputs have compact space-time support and $V(0) = 0$. Each force derivative therefore satisfies every whole-space bound $C_{\alpha m K}(1 + |x| + t)^{-K}$: multiply it by the displayed weight and take its finite maximum on the fixed compact support. The same argument proves the initial-data decay bounds and a uniform finite energy bound. This is an explicit whole-space counterpart with the complete forcing cost, again giving a smooth forced solution rather than asserting breakdown.

16 Exact arithmetic profiles driven by a supplied incompressible flow

16.1 Public source, retained input class and coordinates

The public manuscript *Finite Time Blowup for Navier–Stokes* [8, Theorem 1.1] claims a smooth forced velocity on $\mathbb{R}^3 \times [0,1)$, with one fixed compact spatial support, initial velocity zero, bounded kinetic energy and unbounded terminal supremum norm, for every $\nu > 0$. Its

complete proof and formal verification are not imported as independently certified results in this section. We construct and prove an operator correspondence for its stated smooth input class, which also contains the explicit localized fields in Section 15.8. The correspondence itself is established below without assuming any unproved endpoint. The first source statement is an attribution; the formulas and estimates that follow are our proofs.

For attribution, Buckmaster's primary statement [9, p. 1] identifies his collaboration with Levent Alpöge on smooth-forced Euler and credits Diego Córdoba and Luis Martínez-Zoroa for the underlying forced-blowup programme. The revised NS manuscript [8, pp. 2–3] now discusses that programme and the hypodissipative extension with Zheng explicitly. These statements identify the respective work; they do not establish a dependency on the three-point period construction used above.

Let $\mathcal{U}_{K,T,\nu}$ be the set of smooth real triples (u, p, f) on $[0, T) \times \mathbb{R}^3$ satisfying, component by component,

$$\partial_t u_i + \sum_{j=1}^{3} u_j \partial_j u_i = \nu \sum_{j=1}^{3} \partial_j^2 u_i - \partial_i p + f_i, \qquad \sum_i \partial_i u_i = 0, \qquad \operatorname{supp} u(t, \cdot) \subset K, \tag{141}$$

where $K \subset \mathbb{R}^3$ is compact, $0 < T < \infty$, and $\nu > 0$ is retained. We do not require compact support of p, f for the following map. The particular compactly supported inputs already constructed above satisfy these conditions on every finite interval. No assertion of existence of a singular member is part of the definition of this set.

In the original real polynomial coordinates $x = (x_1, x_2, x_3) = (x, y, w)$, retain every coefficient of F and put

$$S(x) = (F_1(x)/2, F_2(x), F_3(x))^{\mathsf{T}}, \quad J(x) = DS(x) = \operatorname{diag}(1/2, 1, 1) DF(x). \tag{142}$$

The earlier determinant proof gives $\det J = -1$, with the exact inverse

$$J^{-1} = (DF)^{-1} \operatorname{diag}(2, 1, 1) = [2W_1 \; W_2 \; W_3]. \tag{143}$$

Thus $S_1 = \sigma$ is precisely the earlier arithmetic coordinate. The other two coordinates retain the entire original target. No global injectivity of S or F is used; J^{-1} is a pointwise matrix inverse. Both J and J^{-1} have polynomial entries. The first-coordinate observation alone has the exact velocity kernel $\ker(d\sigma)_x = \operatorname{span}_{\mathbb{R}}\{W_2(x), W_3(x)\}$. Indeed $d\sigma(W_i) = \delta_{1i}/2$ and the W_i form a basis, by the full inverse matrix. The retained two additional target coordinates recover precisely those two directions in (143).

16.2 Flow, inverse and volume form on every preterminal interval

For an input of (141), define $X_t(q)$ by

$$\partial_t X_t(q) = u(t, X_t(q)), \qquad X_0(q) = q.$$

Here $q \in \mathbb{R}^3$ is the retained initial label and $x = X_t(q)$ the physical coordinate. This is a global smooth diffeomorphism for each $t < T$. To prove existence with its actual domain, fix $T_0 < T$. On $[0, T_0] \times K$, u and every spatial derivative are bounded by compactness; they vanish outside K. In particular u is globally Lipschitz in space, uniformly on this time interval. On a sufficiently short time interval, the map $Y \mapsto q + \int_0^t u(s, Y(s))\, ds$ is a contraction in the uniform norm on continuous curves, because its Lipschitz constant is at most the interval length times $\sup |Du|$. Successive contractions cover $[0, T_0]$; bounded u prevents escape in each such interval. The same argument backward in time gives the inverse flow. Differentiation of the integral equation gives successive linear integral equations for each label derivative,

proving their existence and continuity by the same local iteration, and proves smoothness by induction. Since T_0 is arbitrary this proves all the assertions on $[0, T)$.

Set $A = D_q X_t$. Differentiation gives

$$A' = Du(t, X_t)A, \qquad A(0) = I_3, \qquad (\det A)' = (\operatorname{div} u)(t, X_t) \det A = 0.$$

The determinant identity follows first near $t = 0$ by the product rule for the determinant and A^{-1}; it gives $\det A = 1$, which prevents loss of invertibility and continues that identity to every $t < T$. Thus $\det A = 1$ with its original positive orientation. Every point outside K has the constant trajectory, by uniqueness. If a trajectory starting in K ended outside K, the backward constant trajectory from its endpoint would contradict the inverse-flow uniqueness. Consequently

$$X_t(K) = K, \qquad X_t(q) = q \ (q \notin K), \qquad X_t(L) \subset K \cup L \quad \text{for every compact } L \subset \mathbb{R}^3. \tag{144}$$

The inverse equality follows in the same way. These conclusions require no bound on $\sup_{t<T} |u|$.

16.3 An invertible profile correspondence with a complex spectral coordinate

Write $\mathcal{B} = C^\infty(\mathbb{R}^3; \mathcal{O}(\mathbb{C}_\zeta))$, with seminorms given by uniform suprema of finitely many real q derivatives and complex ζ derivatives on compact products. Let

$$s_{j,t}(q) = S_j(X_t(q)), \qquad \mathcal{P}_t : \mathcal{O}(\mathbb{C})^3 \longrightarrow \mathcal{B}^3, \quad (h_j)_j \longmapsto \big(h_j(s_{j,t}(q) + \zeta)\big)_j.$$

Every $s_{j,t}$ is real. The domain variables of h_j are complex; the spatial labels remain real, since a general smooth NS velocity need not have a holomorphic extension in space. The map on the full product is explicitly

$$H_{j,t} : \mathbb{R}^3 \times \mathbb{C}_\zeta \longrightarrow \mathbb{R}^3 \times \mathbb{C}_s, \quad (q, \zeta) \mapsto (q, s_{j,t}(q) + \zeta), \qquad H_{j,t}^{-1}(q, s) = (q, s - s_{j,t}(q)). \tag{145}$$

These are smooth in the real variables and biholomorphic on every complex fibre. For any fixed label q_*, an exact left inverse of $\mathcal{P}_t$, and inverse on its image, is

$$(\mathcal{R}_t G)_j(s) = G_j(q_*, s - s_{j,t}(q_*)), \qquad \mathcal{R}_t \mathcal{P}_t = I. \tag{146}$$

Substitution proves both directions on the image. On each compact product the image argument stays in a compact subset of $\mathbb{C}$, and all the finitely many derivatives of $s_{j,t}$ involved are bounded. The chain and product rules prove continuity of $\mathcal{P}_t$; evaluation and translation prove continuity of $\mathcal{R}_t$. Thus the image carries a topology for which the stated maps are an actual topological isomorphism, namely its subspace topology in $\mathcal{B}^3$. The map retains the full original complex profile.

For comparison with the earlier spatial observable, let $e_0(q) = (q, 0)$. The precise restriction is

$$e_0^*(\mathcal{P}_t h)_1(q) = h_1(\sigma(X_t(q))) = X_t^*(\sigma^* h_1)(q).$$

The inverse in (146) uses the complex fibre; no holomorphic spatial flow is silently supplied to this real restriction.

16.4 Full first and second time generators, and Euclidean diffusion

Let $\vartheta : [0, T) \to \mathbb{R}$ be smooth. It is the chosen Newman-time map, independent of physical t; no value for it is inferred from NS. Take either $h_\theta = Z_\theta$ or the full four-term coefficient $h_\theta = K_\theta$ of Section 11, so that $\partial_\theta h_\theta = -\partial_s^2 h_\theta / 4$ with its original factor and sign. Set

$$\Psi_j(t, q, \zeta) = h_{\vartheta(t)}(s_{j,t}(q) + \zeta), \qquad v_j(t, q) = [\nabla S_j \cdot u](t, X_t(q)).$$

Then $\partial_t s_{j,t} = v_j$ and the chain rule proves the exact equation

$$\partial_t \Psi_j = -\frac{\vartheta'}{4}\partial_\zeta^2 \Psi_j + v_j \partial_\zeta \Psi_j. \tag{147}$$

The exact connection on the moving image of $\mathcal{P}_t$ is $\nabla_t = \partial_t - \mathrm{diag}(v_j)\partial_\zeta$: $\nabla_t(\mathcal{P}_t h_t) = \mathcal{P}_t(\partial_t h_t)$. This follows by the same chain rule for every smooth curve of entire profiles. It is a connection between the displayed moving images; multiplication by $v_j(q)$ is not being asserted to preserve a fixed image of the profile space. There is no ζ dependence in v_j. Differentiating this complete equation once more, including the time derivatives of its coefficients, gives

$$\partial_t^2 \Psi_j = \frac{(\vartheta')^2}{16}\partial_\zeta^4 \Psi_j - \frac{\vartheta' v_j}{2}\partial_\zeta^3 \Psi_j + \left(v_j^2 - \frac{\vartheta''}{4}\right)\partial_\zeta^2 \Psi_j + a_j \partial_\zeta \Psi_j, \tag{148}$$

where the acceleration coefficient is, with all NS terms retained,

$$a_j = \partial_t v_j = \left[\sum_i (\nu \Delta u_i - \partial_i p + f_i)\partial_i S_j + \sum_{i,l} u_i u_l \partial_i \partial_l S_j\right](t, X_t(q)). \tag{149}$$

Indeed apply $\partial_t + u \cdot \nabla_x$ to $\sum_i u_i \partial_i S_j$ and substitute (141); the derivative of $\partial_i S_j$ is exactly the displayed Hessian term. In (148), the cubic derivative occurs twice, each with coefficient $-\vartheta' v_j/4$, proving its coefficient $-\vartheta' v_j/2$.

The full physical Laplacian has an exact material counterpart. Define

$$G_t = A^{-1}A^{-\mathsf{T}}, \qquad \mathcal{L}_t = \sum_{a,b} \partial_{q_a}(G_t^{ab}\partial_{q_b}).$$

For every smooth scalar $b(x)$,

$$\mathcal{L}_t(X_t^* b) = X_t^*(\Delta_x b). \tag{150}$$

Here is a proof retaining the first-order terms in $\mathcal{L}_t$. The gradient chain rule gives $\nabla_q X_t^* b = A^{\mathsf{T}}(\nabla_x b) \circ X_t$. For a compactly supported smooth test ϕ, integration by parts and the change of variables $x = X_t(q)$, whose Jacobian is one, give

$$\int \phi\, \mathcal{L}_t X_t^* b\, dq = -\int \nabla_q \phi \cdot A^{-1}(\nabla_x b) \circ X_t\, dq = -\int \nabla_x(\phi \circ X_t^{-1}) \cdot \nabla_x b\, dx = \int \phi\, X_t^* \Delta_x b\, dq.$$

All functions are smooth, so the distributional equality is pointwise. Apply this at fixed ζ to the full profile to obtain

$$\mathcal{L}_t \Psi_j = (|\nabla S_j|^2 \circ X_t)\partial_\zeta^2 \Psi_j + (\Delta S_j \circ X_t)\partial_\zeta \Psi_j. \tag{151}$$

Consequently its complete advective viscous residual is

$$(\partial_t - \nu\mathcal{L}_t)\Psi_j = -\left(\frac{\vartheta'}{4} + \nu|\nabla S_j|^2 \circ X_t\right)\partial_\zeta^2 \Psi_j + (v_j - \nu \Delta S_j \circ X_t)\partial_\zeta \Psi_j.$$

These formulas identify the two differential operators through an explicit coordinate map and calculate their remaining coefficients; they do not delete the viscosity or equate two time variables.

16.5 Recovering the actual velocity from the arithmetic temporal defect

For real θ the functions Z_θ, K_θ and their spectral derivatives are Schwartz on the real s line, by the weighted Fourier space proved above. With the original real kernel k and multiplier m, their Fourier representations are

$$Z_\theta(s) = 4\int_{\mathbb{R}} k(\omega)e^{\theta\omega^2/4}e^{is\omega}\,d\omega, \qquad K_\theta(s) = 4\int_{\mathbb{R}} m(\omega)k(\omega)e^{\theta\omega^2/4}e^{is\omega}\,d\omega.$$

No source energy, derivative channel, or Fourier factor is changed. Define $d_h(\theta)^2 = \int_{\mathbb{R}} |h'_\theta(s)|^2\,ds$. Then

$$\begin{aligned} d_Z(\theta)^2 &= 32\pi\int_{\mathbb{R}} \omega^2 k(\omega)^2 e^{\theta\omega^2/2}\,d\omega, \\ d_K(\theta)^2 &= 32\pi\int_{\mathbb{R}} \omega^2 |m(\omega)|^2 k(\omega)^2 e^{\theta\omega^2/2}\,d\omega. \end{aligned} \tag{152}$$

For completeness, the factor follows without a Fourier convention change. For a Schwartz amplitude b, insert $e^{-\epsilon s^2}$ in the squared norm of $4\int b(\omega)e^{is\omega}d\omega$ and use Fubini. The result is $16\sqrt{\pi/\epsilon}\iint b(\omega)\overline{b(\eta)}e^{-(\omega-\eta)^2/(4\epsilon)}d\omega d\eta$. The Gaussian convolution, with mass 2π, tends to $2\pi b$; splitting at any fixed small distance and using continuity and the Gaussian tail proves this convergence under the integral. Thus the limit is $32\pi\int|b|^2$. On the original side dominated convergence applies because the transform is Schwartz. Use respectively $b = i\omega k e^{\theta\omega^2/4}$ and $b = i\omega m k e^{\theta\omega^2/4}$ to prove (152).

Both numbers are strictly positive. The previously proved $k(0) > 0$ and continuity give an interval of nonzero k containing nonzero ω. For the second integral the established full-real bound $|m| \geq 575/1752192$ applies. The weighted bounds also prove continuity of these integrals in θ by domination on each compact parameter interval. Hence for each compact interval $I \subset \mathbb{R}$ there are finite constants $0 < d_- \leq d_h(\theta) \leq d_+ < \infty$ on I.

Define the actual arithmetic temporal defect by

$$D_j(t,q,s) = \partial_t\Psi_j(t,q,s) + \frac{\vartheta'}{4}\partial_s^2\Psi_j(t,q,s), \qquad s \in \mathbb{R}. \tag{153}$$

Equation (147) gives $D_j = v_j h'_\vartheta(s + s_{j,t})$. Real translation preserves Lebesgue measure, so

$$\sum_j \|D_j(t,q,\cdot)\|^2_{L^2(\mathbb{R})} = d_h(\vartheta(t))^2|v(t,q)|^2. \tag{154}$$

Moreover the inverse is a literal integral followed by the original matrix inverse:

$$\begin{aligned} v_j &= \frac{\displaystyle\int_{\mathbb{R}} \overline{h'_\vartheta(s+s_{j,t})}D_j(t,q,s)\,ds}{d_h(\vartheta)^2}, \\ u(t,X_t(q)) &= 2W_1(X_t(q))v_1 + W_2(X_t(q))v_2 + W_3(X_t(q))v_3. \end{aligned} \tag{155}$$

Substitution proves both inverse identities. The correspondence retains $X_t(q)$ as part of both sides: $(X_t, u\circ X_t) \leftrightarrow (X_t, D)$. The data D alone are not claimed to select a globally unique inverse sheet of the polynomial map. At fixed (t,q), the formula identifies $\mathbb{R}^3$ with the three-dimensional real subspace of $L^2(\mathbb{R};\mathbb{C})^3$ spanned componentwise by the three displayed shifted profiles. It is not merely an analogy between their norms.

For a nonempty K, set $M = \max_K \|J\|_{\mathrm{op}}$ and $N = \max_K \|J^{-1}\|_{\mathrm{op}}$. They are finite and positive. If $\vartheta([0,T)) \subset I$, the exact map satisfies, for all $t < T$,

$$\frac{d_-}{N}\|u(t)\|_{L^\infty(\mathbb{R}^3)} \leq \sup_q \left(\sum_j \|D_j(t,q,\cdot)\|_2^2\right)^{1/2} \leq d_+ M\|u(t)\|_{L^\infty(\mathbb{R}^3)}. \tag{156}$$

Indeed $|Ju| \le M|u|$ and $|u| \le N|Ju|$ on K, while both u and v vanish off K; (144) preserves K and surjectivity preserves the supremum. Combine this with (154). This proves equivalence of unbounded terminal velocity and unbounded supremum of this specific arithmetic temporal-defect norm for every input in the defined class. It supplies an exact detection map even though no singular member is established here. At $u = 0$ all these defects vanish, as the same formulas require. The full space–spectral energy has an exact identity too. Since $v = 0$ off K, (154) is integrable in q and the volume-preserving change of variables gives

$$\sum_j \int_{\mathbb{R}^3} \int_{\mathbb{R}} |D_j(t,q,s)|^2 \, ds \, dq = d_h(\vartheta(t))^2 \int_{\mathbb{R}^3} |J(x)u(t,x)|^2 \, dx.$$

The same constants therefore bound its square root between $(d_-/N)\|u(t)\|_2$ and $d_+M\|u(t)\|_2$. In particular the map transfers a uniform kinetic-energy bound into a uniform joint $L^2(q,s)$ bound, while (156) detects concentration through the separate supremum in q. This conclusion follows from the displayed integrals without a spatial truncation of the defect. In particular $\vartheta(t) = 0$ for every physical time is an explicit allowed choice. Then the entire input stays equal to the actual Ξ (or K_0), and $D_j = \partial_t \Psi_j$; the constants in (156) use the single strictly positive value $d_h(0)$.

For an actual separated-link pair, the full coefficient is $a_{ef}K$. The defect is then $a_{ef}D^K$, and its squared norm acquires exactly a_{ef}^2; division in (155) is available for $a_{ef} \neq 0$. If $a_{ef} = 0$ this pair supplies the zero observation, not a spurious inverse. For a pair with $a_{ef} > 0$, recovery of Z from the profiles of K additionally uses the already proved full-domain inverse $\mathcal{T}_{\mathcal{D}}^{-1}$; its translation and derivative commutation hold because multiplication by m commutes with $e^{is_{j,t}\omega}$ and $i\omega$ in the unchanged Fourier domain.

16.6 Bounded profile values and the exact transported zero divisor

For compact label set L, compact spectral set $E \subset \mathbb{C}$, and bounded $\vartheta([0,T)) \subset I$, (144) puts every argument $s_{j,t}(q) + \zeta$ in the compact set $S_j(K \cup L) + E$. The jointly entire Z and K are bounded on that set times I. Therefore

$$\sup_{t<T,\ q\in L,\ \zeta\in E} |\Psi_j(t,q,\zeta)| < \infty.$$

The identical argument bounds every pure spectral derivative. It does not bound time or spatial-label derivatives; (156) specifies a quantity that can grow even while these values stay bounded.

Let $h_\theta(s)$ have a zero at z_* of multiplicity m and write its actual local factorization $h_\theta(s) = (s - z_*)^m A(s)$, $A(z_*) \neq 0$. The full inverse of (145) takes that zero to

$$\zeta_*(q) = z_* - s_{j,t}(q), \qquad \Psi_j = (\zeta - \zeta_*(q))^m A(s_{j,t}(q) + \zeta).$$

Thus it preserves multiplicity and the whole analytic unit on every complex fibre. Its complete jet formula is

$$\partial_\zeta^{m+n} \Psi_j(t,q,\zeta_*(q)) = \frac{(m+n)!}{n!} A^{(n)}(z_*), \qquad n \ge 0,$$

with all lower derivatives zero. This follows by multiplying the Taylor series of the unchanged A by $(\zeta - \zeta_*)^m$; no unit coefficient has been discarded. Since $s_{j,t}(q)$ is real, $\operatorname{Im}\zeta_*(q) = \operatorname{Im} z_*$. At Newman time $\vartheta = 0$ with $h = Z$, the full map to the completed-zeta argument is

$$\rho = \tfrac{1}{2} + i(s_{j,t}(q) + \zeta), \qquad \zeta = -i(\rho - \tfrac{1}{2}) - s_{j,t}(q), \qquad \operatorname{Re}\rho = \tfrac{1}{2} - \operatorname{Im}\zeta.$$

Consequently this NS-induced real translation preserves each zero's distance from the critical line exactly; the full arithmetic time evolution is still the specified Z_ϑ. For $h = K$ the same

coordinate proof transports the zeros of the four-term profile; these are converted to a statement about Z only by actually applying the proved inverse operator to the whole profile. The inverse does not identify individual zeros of two different entire functions. These explicit maps give both the velocity detection and the precise zero transport of the construction, without deducing an off-critical zero from growth of the temporal defect.

17 Provenance and reproducibility

The Newman convention and its public source are Rodgers–Tao, [1, Section 1, equations (1)–(4)]. The polynomial is Tao's displayed map [3, equation (1)]; its coordinate identities used here are reproved above, so no hidden local reference is required to check them. The actual topological Connes quotient is described in [2, Section 3.6, Proposition 3.6]. Section 5.1 proves the coordinate-level Mellin/Fourier map, the common-representative quotient correspondence with its complete kernels, and its heat and multiplication intertwiners, retaining both original topologies.

The readable proofs establish the analytic assertions. The scripts `scripts/check_heat_connection.py` and `scripts/zero_transport_check.py`, together with `scripts/check_euclidean_observable.py`, replay exact algebraic identities using rational polynomial arithmetic. Their JSON receipts state precisely which identities were checked. They are not Lean certificates or certificates of an off-critical zero. No Lean, Lake, or Elan run was used. The work was prepared with Codex and independent agent checks; the cited human literature remains the scholarly source. The fourth-order transfer is also checked by `scripts/check_xi4_transport.py`. The explicit physical, temporal, spatial and zero-jet morphisms are also checked by `scripts/check_xi4_typed_maps.py`. Appendix A contains the complete supplied spectral-coefficient proof; the original Markdown and its independent verification are preserved under `sources/xi4_source/`. The coupling notation there is ξ^4. The full arithmetic quotient continuation is preserved byte-for-byte under `sources/connes_continuation/`; its full proof is included in Appendices B–I. Its `checks/verify_exact.py` supplies 206 supplementary exact checks; the replay receipt is `checks/connes_continuation_replay.json`. Section 14 proves the common-space identification and its simultaneous moving quotient intertwiners. The complete incoming Cartesian bilaplacian proof is preserved in `sources/ns_bilaplacian/` and included in Section 13. Its fourth-coefficient continuation in Sections 13.4–13.6 proves the full material intertwiner, residue map and negative-time asymptotic; `scripts/check_xi4_bilaplacian.py` verifies the exact algebra. Section 15 retains the three actual marked special points of the supplied period construction and proves its real circle quotient, including the full logarithmic overlaps. Sections 15.5–15.8 give the resulting three-dimensional force, pressure, all mixed interactions, the exact finite-mode classification and every spatial localization term. The source provenance is retained under `sources/s6_three_points/`; the independent algebra replays are `scripts/check_s6_three_point_forcing.py` and `scripts/check_s6_three_point_fields.py`. The exact supplied-flow arithmetic correspondence is proved in Section 16; its source versions are identified in `sources/ns_public/provenance.json`, and its independent algebra checks are in `scripts/check_ns_arithmetic_flow.py`.

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A Complete source proof of the separated-link fourth coefficient

8 September 2026. Two links which share no elementary face nevertheless develop a nonzero connected electric covariance through two adjacent faces. This note calculates that coefficient in the actual full-box vacuum. The calculation keeps both intermediate spin channels and the disconnected expectation subtraction. It does not assume a volume-uniform remainder.

A.1 Original Hilbert space and the coefficient to be calculated

Fix $L \geq 2$, vertices $\{-L, \dots, L\}^3$, all positive contained links e, and all contained elementary faces p. Write their complete counts as $N = 3(2L)(2L+1)^2$ and $M = 3(2L)^2(2L+1)$. Link variables are $U_e \in SU(2)$, with the inverse on reverse traversal. The face trace is

$$W_{(n;i,j)} = \operatorname{tr}\{U_i(n)U_j(n+\mathbf{e}_i)U_i(n+\mathbf{e}_j)^{-1}U_j(n)^{-1}\}, \qquad i < j.$$

On product Haar probability space use $T_a = -i\sigma_a/2$, left derivatives $X_{e,a}$, $E_e = -\sum_a X_{e,a}^2$, and

$$H = \kappa \sum_e E_e + b \sum_p (2 - W_p), \quad \kappa = 2g^2/a, \quad b = 1/(2g^2 a), \quad \xi = b/\kappa = 1/(4g^4). \tag{S1}$$

The fundamental Casimir is $3/4$. In the original metric $-\operatorname{tr}(AB)/2$, the Laplacian is four times the sum-of-squares Laplacian in these generators, so the electric coefficient in that metric is $\kappa/4$. Neither this factor nor the Wilson scalar $2bM$ is changed.

The compact connected configuration manifold and bounded smooth potential give a self-adjoint compact-resolvent operator of domain H^2, form domain H^1, and unique smooth positive unit ground vector ψ_ξ. For precision, the modulus inequality supplies a nonnegative form minimizer, elliptic regularity and the maximum principle make it positive, and the identity $q_{H-\mathcal{E}}(\psi f) = \kappa \sum_{e,a} \int \psi^2 |X_{e,a} f|^2$ implies uniqueness by division by ψ. Gauge transformations preserve this vector and every link Casimir. All following vectors are thus physical.

Set $r_e = \#\{p : e \in \partial p\}$, $\gamma_e = \langle \psi, E_e \psi \rangle$, $v_e = (E_e - \gamma_e)\psi$, and

$$C_{ef} = \langle v_e, v_f \rangle = \langle \psi, E_e E_f \psi \rangle - \gamma_e \gamma_f. \tag{S2}$$

The operators commute for different links. Smoothness makes (S2) legitimate as an operator identity on the vacuum, not only a formal product of forms.

For distinct e,f that share no face, define the integer

$$a_{ef} = \#\{(p,q) : e \in \partial p,\ f \in \partial q, |\partial p \cap \partial q| = 1\}. \tag{S3}$$

The pair in (S3) is ordered by which face contains e and which contains f; there is no double counting because no face contains both links. All faces are those of the original open box. The result is

$$\boxed{C_{ef} = \frac{127}{219024}\, a_{ef} \xi^4 + O_L(\xi^6) \quad (e, f \text{ share no face}).} \tag{S4}$$

In particular the coefficient is positive whenever a two-face channel exists, even though the energy matrix between these links is exactly zero. The latter assertion is proved in Appendix A.4.

A.2 The full second vacuum coefficient, including both shared-link channels

Put $H_0 = \sum E_e$ and $S = \sum W_p$, so $H = \kappa(H_0 - \xi S) + 2bM$. On $|z| = 3/8$ in the full Hilbert space the resolvent of H_0 has norm at most $8/3$, since its first nonzero eigenvalue is $3/4$. Also $\|S\| = 2M$. Therefore the bounded perturbation Neumann series converges for $|\xi| < 3/(16M)$, gives the analytic rank-one bottom projection, and gives an analytic positive unit eigenvector for real ξ near zero. Elliptic regularity applied repeatedly to the eigen-equation upgrades this to analyticity in each fixed Sobolev space. Its radius and estimates may depend on L. This suffices for the finite derivative calculations below.

Product Haar integration gives $\int W_p = 0$, $\langle W_p, W_q \rangle = \delta_{pq}$, and $H_0 W_p = 3W_p$. Indeed a link in only one of two distinct face boundaries has odd central parity; for the same face integrate its Haar holonomy and use the fundamental character squared norm one. Comparing coefficients in the actual equation $(H_0 - \xi S)\psi = e(\xi)\psi$, including its unit-vector condition, gives

$$\psi = 1 + \xi A_1 + \xi^2 A_2 + O_{H^k,L}(\xi^3), \quad A_1 = S/3, \quad e(\xi) = -M\xi^2/3 + O_L(\xi^3),$$

$$H_0 A_2 = (S^2 - M)/3, \qquad \int A_2 = -M/18. \tag{S5}$$

The inverse in (S5) can be resolved entirely. For adjacent distinct faces p,q sharing g put

$$Z_{pq,0} = \int W_p W_q \, dU_g, \qquad Z_{pq,1} = W_p W_q - Z_{pq,0}.$$

After reversing one real trace if needed, their words are $\mathrm{tr}(U_g B)$, $\mathrm{tr}(U_g^{-1} D)$, with six distinct outer links. The fundamental Haar identity gives $Z_{pq,0} = \mathrm{tr}(BD)/2$. The shared-edge spin tensor $1/2 \otimes 1/2$ consists of spins zero and one, while all outer links have spin $1/2$. Thus

$$H_0 Z_{pq,0} = \tfrac{9}{2} Z_{pq,0}, \quad H_0 Z_{pq,1} = \tfrac{13}{2} Z_{pq,1}, \quad \|Z_{pq,0}\|^2 = \tfrac{1}{4}, \quad \|Z_{pq,1}\|^2 = \tfrac{3}{4}. \tag{S6}$$

The norms use orthogonal Haar projection on g and total product norm one. A repeated face has $W_p^2 - 1 = \chi_1(U_p)$ with free energy 8 and norm one. An edge-disjoint pair has energy 6 and norm one, even when the faces meet at a vertex. Consequently the full coefficient is

$$\boxed{\begin{aligned} A_2 = & -\frac{M}{18} + \sum_p \frac{W_p^2 - 1}{24} \\ & + \sum_{\{p,q\}\ \text{edge-disjoint}} \frac{W_p W_q}{9} \\ & + \sum_{\{p,q\}\ \text{adjacent}} \left(\frac{4}{27} Z_{pq,0} + \frac{4}{39} Z_{pq,1} \right). \end{aligned}} \tag{S7}$$

All distinct-pair sums in (S7) are unordered; the factor two in S^2 has already been included. No finite-face subsystem is substituted for H.

Different unordered pairs in (S7) have different odd-link parity supports. If their symmetric boundaries agreed, their symmetric face difference would be a nonempty closed mod-two collection of at most four squares. A face in that collection needs another face on each of its four boundary edges; two distinct elementary squares share at most one edge, so at least four other squares would be needed, a contradiction. Haar integration on a link where parity differs makes the cross term zero. Repeated-face terms are even everywhere, and different repeated-face terms have different joint link spins. Thus all the displayed components, with their two spin channels, are mutually orthogonal. This remains true after application of any link Casimir, which preserves those parity and spin subspaces.

A.3 Exact connected fourth coefficient

Suppose e,f share no face. Then $E_e E_f 1 = E_e E_f A_1 = 0$: each term in A_1 is a single face and depends on at most one of these links. Self-adjointness moves $E_e E_f$ to either of these vectors, so every term of degree less than four in $\langle \psi, E_e E_f \psi \rangle$ vanishes, and its degree-four term is exactly

$$\langle A_2, E_e E_f A_2 \rangle. \tag{S8}$$

In particular no third or fourth unknown coefficient of the vacuum is being omitted from (S8): its possible pairing with 1 or A_1 is identically zero for the stated reason. Similarly

$$\gamma_e = \xi^2 \langle A_1, E_e A_1 \rangle + O_L(\xi^3) = \frac{r_e}{12} \xi^2 + O_L(\xi^3). \tag{S9}$$

Only a pair p,q with $e \in p$, $f \in q$ contributes to (S8). Repeated faces contribute zero. There are exactly $r_e r_f$ such distinct unordered pairs when they are labeled as in (S3). For an edge-disjoint pair, both e and f carry fundamental spin, so its contribution is $(3/4)^2/9^2 = 1/144$. For an adjacent pair, neither e nor f can be its shared edge: otherwise the face containing the other link would contain both e and f. They are therefore both outer fundamental edges in each of the two channels (S6), and the contribution is

$$\left(\frac{3}{4}\right)^2 \left[\left(\frac{4}{27}\right)^2 \frac{1}{4} + \left(\frac{4}{39}\right)^2 \frac{3}{4} \right] = \frac{1}{324} + \frac{3}{676}. \tag{S10}$$

Equation (S9) subtracts $r_e r_f/144$ at degree four in (S2). Hence every edge-disjoint pair cancels its disconnected contribution exactly; every adjacent pair leaves

$$\frac{1}{324} + \frac{3}{676} - \frac{1}{144} = \frac{127}{219024} > 0. \tag{S11}$$

This proves the coefficient in (S4).

For its stated even error, the center transformation $U_{(n,i)} \mapsto (-1)^{\sum_{j<i} n_j} U_{(n,i)}$ changes every face trace sign, commutes with every E_e and preserves Haar measure. It intertwines $H_0 - \xi S$ with $H_0 + \xi S$ and hence the positive unit vacua at ξ and$-\xi$. Both terms in (S2) are consequently even analytic functions of ξ near zero. Their next possible order after four is six. This proves the error in (S4); it does not supply a volume-uniform analytic radius.

A.4 Exact support map and relation to the energy matrix

Define the edge adjacency matrix B by $B_{eg} = 1$ when e,g are distinct physical links belonging to a common elementary face, and zero otherwise. Two distinct links belong to at most one elementary square: perpendicular incident links determine its two plane directions and base, while parallel links in a square must be opposite sides and also determine that square. Thus, for the e,f considered above,

$$\boxed{a_{ef} = (B^2)_{ef}.} \tag{S12}$$

Indeed a pair p,q counted in (S3) has a unique shared edge g, which is a common neighbor of e,f. Conversely a common neighbor g determines the unique p containing e,g and q containing g,f; these are distinct because e,f share no face, and their unique shared edge is g. These two maps are inverse. The covariance coefficient thus has support at graph distance two, including boundary incidences, and no support beyond distance two at this order. It makes no all-orders finite-range claim about C.

There are actual such channels in every $L \geq 2$. Take $e = ((0,0,0),1)$ and $f = ((0,2,0),1)$. They share no face, but the faces in directions $1,2$ based at $(0,0,0)$ and $(0,1,0)$ share the edge $((0,1,0),1)$, giving $a_{ef} \geq 1$. All of their vertices are in the original box, including at L=2.

Put $\mathcal{N}_{ef} = \langle v_e, (H - \mathcal{E})v_f \rangle$. Commuting the link operators on the smooth real vacuum gives

$$\mathcal{N}_{ef} = \tfrac{1}{2}\langle \psi, [E_e, [H, E_f]]\psi \rangle. \tag{S13}$$

To check (S13), expand the double commutator, use $H\psi = \mathcal{E}\psi$, commute E_e,E_f, and use the real symmetry of their two matrix elements. Only potential faces containing f occur in $[H, E_f]$. If e shares no face with f, E_e commutes with every one of their multiplication and derivative terms. Thus

$$\boxed{\mathcal{N}_{ef} = 0 \text{ exactly, at every positive coupling, when e,f share no face.}} \tag{S14}$$

There is no contradiction between (S4) and (S14): a matrix-valued spectral measure can have nonzero zeroth moment and zero first moment off diagonal. Explicitly, for the spectral resolution of A=H−E on the physical space,

$$d\Sigma_{ef}(\omega) = \langle v_e, \mathbf{1}_{d\omega}(A) v_f \rangle, \quad C_{ef} = \int d\Sigma_{ef}, \quad \mathcal{N}_{ef} = \int \omega \, d\Sigma_{ef}. \tag{S15}$$

The measure matrix is positive semidefinite on every Borel set because its quadratic form is the squared norm of a projected linear combination of the v_e. An individual off-diagonal entry, however, need not be a nonnegative measure; (S15) retains its signs. Smoothness gives the finite moments used here. These identities are the exact relation between the nonlocal covariance contribution and the local energy matrix, not an assumption that one determines the other's support.

Finally $\xi^4 = 1/(256 g^{16})$. The physical coefficient in (S4) is therefore $127 a_{ef}/(56070144 g^{16})$; C has no extra energy factor, whereas $\mathcal{N}$ has one. The scalar energy shift remains in H and cancels only in H−E. The fixed-box coefficient is a concrete next input to the extensive denominator calculation; control of all higher orders and summability at fixed coupling is not inferred from its finite range at order four.

A.5 The complete fourth-order spectral functional and time correlation

The same calculation can resolve the energy dependence, rather than only its zeroth moment. Put $A_\xi = (H - \mathcal{E})/\kappa$, and write

$$F_{ef,\xi}(h) = \langle v_e, h(A_\xi) v_f \rangle.$$

For each fixed box, for distinct links sharing no face, and every bounded continuous real or complex function h which is differentiable at 3,

$$\boxed{\lim_{\xi \to 0} \xi^{-4} F_{ef,\xi}(h) = a_{ef} \Big[-\frac{59}{275184} h(3) - \frac{1}{336} h'(3) + \frac{1}{1296} h(9/2) + \frac{3}{132496} h(13/2) \Big].} \tag{S16}$$

Here the limit is a functional on the specified test functions. In particular the derivative term must not be omitted or represented as an ordinary positive spectral atom. Each original matrix-valued spectral measure is still the exact positive-semidefinite measure (S15).

We prove (S16), including the relation to the actual first interacting spectral subspace. Let P project onto the free physical energy-three space $\mathcal{P} = \operatorname{span}\{W_p\}$, and let

$$R_3 = (I - P)((H_0 - 3)|_{\mathcal{P}^\perp})^{-1}(I - P).$$

The inverse is on the physical space: its constant channel is -1/3. To verify isolation, a nonzero physical spin network has no degree-one support vertex. Its support contains a cycle of length at least four in the bipartite cubic graph, costing at least 3. With fewer than six nontrivial edges it must be one elementary square: a fifth edge would have a new degree-one endpoint, or be a forbidden same-bipartition chord of the four-cycle. The degree-two invariant contractions require equal spins on that square. Only spin 1/2 has energy below 9/2. A six-edge rectangular loop realizes energy 9/2. Thus the first physical free band is exactly P, with the next energy 9/2.

For sufficiently small real ξ, the contour $|z - 3| = 3/4$ defines the actual spectral projection P_ξ, and the exact isometry

$$V_\xi = P_\xi P (P P_\xi P|_{\mathcal{P}})^{-1/2} : \mathcal{P} \longrightarrow \operatorname{ran} P_\xi \tag{S17}$$

uses its full Gram factor. Indeed the free contour resolvent has norm at most 4/3. For $|\xi| \le 3/(64M)$, its Neumann factor is at most 1/8, and integrating the resolvent difference gives $\|P_\xi - P\| \le 1/7$. The displayed inverse square root therefore exists by its binomial series; idempotence and orthogonality of the real projection show $V_\xi^* V_\xi = I$. Its range is smooth, and the finite-rank resolvent formula and elliptic regularity justify all operator products and fixed Sobolev derivatives below. It is orthogonal to the actual vacuum for a smaller interval, for example $|\xi| \le 3/(128M)$, by the free gaps and the bounded perturbation estimate $2M|\xi|$.

Let Q be the unchanged open-face signless adjacency matrix, with diagonal the actual face degrees and off-diagonal 1 for shared-edge faces. Direct virtual-channel calculation gives

$$V_\xi^* A_\xi V_\xi = 3I + \xi^2 F + O_L(\xi^4), \qquad F = \frac{7}{15} I - \frac{1}{21} Q. \tag{S18}$$

Here is the coefficient calculation rather than an assumption of (S18). Differentiating (S17) gives $V_0' = R_3 S P$. The second matrix coefficient before vacuum subtraction is $-PSR_3SP$. The vacuum channel gives +1/3 to every entry; a repeated-face channel gives -1/5 to its diagonal; an edge-disjoint pair gives -1/3 to its two diagonals and mutual entries; an adjacent pair gives $-(1/4)/(9/2 - 3) - (3/4)/(13/2 - 3) = -8/21$ to that same four-entry pattern.

Thus the diagonal is $7/15 - M/3 - d_p/21$, and the off-diagonal is -1/21 exactly for adjacent faces. The actual vacuum coefficient -M/3 from (S5) gives (S18). The central cochain used in Appendix A.3 acts as -I on P; it makes the matrix in (S18) even and accounts for the fourth-order remainder.

The electric vector can be treated with arbitrary fixed real link weights w, retaining its linear dependence on all of them. Define $\Gamma_w = \sum w_e E_e$, $a_p(w) = \sum_{e \in p} w_e/4$, and $v(w) = (\Gamma_w - \langle \Gamma_w \rangle)\psi$. Put

$$y_\xi(w) = V_\xi^* v(w), \qquad \rho_\xi(w) = (I - P_\xi)v(w).$$

These are exact orthogonal components, both orthogonal to the actual vacuum. Coefficient comparison using (S7) and $V_0' = R_3 S P$ gives

$$y_\xi(w) = \xi x(w) + \xi^3 z(w) + O_L(\xi^5), \qquad x(w) = \sum_p a_p(w) W_p,$$

$$\rho_\xi(w) = \xi^2 r(w) + O_{H^k,L}(\xi^3), \tag{S19}$$

where, for $s = a_p(w) + a_q(w)$ and g the shared edge,

$$r(w) = \sum_p \frac{2a_p(w)}{15}(W_p^2 - 1) + \sum_{\{p,q\}\ \text{adjacent}} \left[-\frac{2(s + w_g)}{9} Z_{pq,0} + \frac{2(3s + 7w_g)}{273} Z_{pq,1} \right]. \tag{S20}$$

For completeness, Γ_w has eigenvalues $8a_p$ on the repeated face, $3s$ on an edge-disjoint pair, and $3s - 3w_g/2, 3s + w_g/2$ on the adjacent zero and one channels. The shared edge is counted twice as fundamental in 3s; replacing that by its actual spin accounts for precisely the last two corrections. In $\Gamma_w A_2$, the respective coefficients are $a_p/3, s/3, (4s - 2w_g)/9, (12s + 2w_g)/39$. Subtract the full dressing $R_3 S x(w)$, whose coefficients are $a_p/5, s/3, 2s/3, 2s/7$. The two constant coefficients are both $-\sum_p a_p/3$ and also cancel. This proves (S20) with no omitted channel. The centered projection identity $V_\xi^* \psi_\xi = 0$ is exact. The same central cochain makes $y_\xi(w)$ odd, proving the first expansion (S19).

For a single link e use the original weight $w_g^{(e)} = \delta_{eg}$, and denote the resulting x,z,r by $x_e, z_e, r_e^{\mathrm{vec}}$. This superscript distinguishes the vector from the incidence count r_e. Then $x_e = \sum_{p \ni e} W_p/4$. If e,f share no face, $\langle x_e, x_f \rangle = 0$, and (S18) gives exactly

$$\langle x_e, F x_f \rangle = -a_{ef}/336. \tag{S21}$$

Each ordered adjacent pair contributes $-1/(4 \cdot 21 \cdot 4)$. In the cross spectral measure of $r_e^{\mathrm{vec}}, r_f^{\mathrm{vec}}$, only the pairs counted by (S3) occur. Repeated-face cross terms vanish. For every contributing pair, e and f are outer links of different faces, $s = 1/4$ for each weight, and $w_g = 0$. Thus their zero-channel coefficients in (S20) are both -1/18, and their one-channel coefficients are both 1/182. Orthogonality and (S6) give the complete free cross measure

$$\langle r_e^{\mathrm{vec}}, h(H_0) r_f^{\mathrm{vec}} \rangle = a_{ef} \left[\frac{1}{1296} h(9/2) + \frac{3}{132496} h(13/2) \right]. \tag{S22}$$

No edge-disjoint channel survives in (S20).

As $A_\xi - H_0 = -\xi S - e(\xi) I$ tends to zero in operator norm, its resolvents converge in norm. For bounded continuous h, this implies convergence of the cross expectations on ρ_ξ/ξ^2 to (S22): first use continuous functions vanishing at infinity and norm resolvent convergence, then approximate h on a compact interval and control the discarded tail by the uniformly bounded second spectral moments. Those moments follow from the H^2 convergence in (S19). Polarization gives the cross statement from the four corresponding diagonal statements. Hence (S22) is the entire fourth-order out-of-band contribution.

The fourth coefficient of the band mass is obtained without knowing the third vacuum coefficient: subtract (S22) at h=1 from (S4). It is

$$\langle x_e, z_f\rangle + \langle z_e, x_f\rangle = a_{ef}\left(\frac{127}{219024} - \frac{1}{1296} - \frac{3}{132496}\right) = -\frac{59}{275184}a_{ef}. \tag{S23}$$

Differentiability of h at 3 and spectral calculus of the finite matrix (S18) give $h(V_\xi^* A_\xi V_\xi) = h(3)I + \xi^2 h'(3)F + o_L(\xi^2)$. Combine this with (S19), (S21), (S23), and the exact orthogonal spectral decomposition into the band and its complement. This proves (S16).

In particular the complete dimensionful ground-state time correlation, with physical $\kappa > 0$ and physical time $t \geq 0$ fixed, is

$$\boxed{\begin{aligned}\langle v_e, e^{-t(H-\mathcal{E})}v_f\rangle = a_{ef}\xi^4\Big[&-\frac{59}{275184}e^{-3\kappa t} + \frac{\kappa t}{336}e^{-3\kappa t} \\ &+ \frac{1}{1296}e^{-9\kappa t/2} + \frac{3}{132496}e^{-13\kappa t/2}\Big] + O_{L,t,\kappa}(\xi^6).\end{aligned}} \tag{S24}$$

Here $h(q) = e^{-\kappa t q}$ is used on the nonnegative excitation spectrum, with any bounded continuous extension below zero, so it belongs to the test class of (S16). The bounded potential perturbation gives $e^{-\kappa t K_\xi}$ its norm-convergent Duhamel series: each order is bounded by $(2M|\xi|\kappa t)^n/n!$. The excitation semigroup retains the additional analytic scalar $e^{\kappa t e(\xi)}$. The vacuum and vectors are analytic in fixed Sobolev norms, and the central cochain makes the whole expression even. These facts justify the even remainder in (S24), not only a pointwise little-o limit. For varying physical κ, the same proof uses the explicit parameter $\tau = \kappa t$ and keeps its dependence; no estimate uniform in τ or L is implied.

At t=0 the bracket is 127/219024. Its derivative at t=0 is exactly zero, because

$$3\left(-\frac{59}{275184}\right) - \frac{1}{336} + \frac{9}{2\cdot 1296} + \frac{39}{2\cdot 132496} = 0. \tag{S25}$$

This agrees with the exact identity (S14). It exhibits which separated-link spectral contributions cancel in the first moment: the actual band-mass correction, the actual band-energy derivative, and both higher spin channels. The term proportional to t is essential to that cancellation. Equations (S16)–(S25) describe the fixed-box fourth coefficient of the original interacting correlations; they do not exchange a volume limit with the coupling expansion or assert a continuum spectrum.

A.6 Primary context

The original Hamiltonian framework is J. Kogut and L. Susskind, *Hamiltonian formulation of Wilson's lattice gauge theories*, Physical Review D **11** (1975), 395–408, `https://doi.org/10.1103/PhysRevD.11.395`. Full conventions appear in (S1). For historical linked-cluster calculations in the same spatial dimension, see A. C. Irving, T. E. Preece and C. J. Hamer, *Cluster expansion approach to non-abelian lattice gauge theory in (3+1)D (I). SU(2)*, Nuclear Physics B **270** (1986), 536–552, `https://doi.org/10.1016/0550-3213(86)90567-5`. This note derives (S4) directly, with no historical novelty assertion or imported coefficient.

B The primary arithmetic object and the topology actually specified

The starting point is Connes–Consani–Moscovici, *Zeta zeros and prolate wave operators: Semilocal adelic operators*, Section 3.6, proof-PDF pages 17–19, especially Lemma 3.3, equations

(3.16)–(3.22), and Proposition 3.6 [5]. The preceding arithmetic map is

$$\mathcal{E}f(x) = x^{1/2}\sum_{n\geq 1} f(nx), \qquad \mathcal{F}_\mu g(s) = \int_0^\infty g(x)x^{-is}\,\frac{dx}{x}.$$

The input consists of even Schwartz functions with $f(0) = \widehat{f}(0) = 0$. The paper's two Hermite families are

$$\psi_\ell^+ = h_{4\ell} - \frac{h_{4\ell}(0)}{h_0(0)}h_0, \qquad \psi_\ell^- = -h_{4\ell+2} + \frac{h_{4\ell+2}(0)}{h_2(0)}h_2 \quad (\ell \geq 1).$$

These signs and the minus sign in the Mellin exponent are retained. The completed function is

$$\Xi(s) = \xi(\tfrac{1}{2} + is), \qquad \xi(z) = \tfrac{1}{2}z(z-1)\pi^{-z/2}\Gamma(z/2)\zeta(z), \qquad \Xi(s) = 8H_0(2s). \tag{157}$$

The paper proves

$$\mathcal{F}_\mu\mathcal{E}\psi_\ell^\pm = R_\ell^\pm\Xi, \qquad P_\ell^\pm = -\tfrac{1}{2}(\tfrac{1}{4} + s^2)R_\ell^\pm. \tag{158}$$

It also states immediately before Proposition 3.6 that these give all polynomial multiples of Ξ. Here is the precise algebra behind the last assertion. The top summands of (3.19) and (3.20) are nonzero; their degrees are 2ℓ and $2\ell+1$. Division by $-\frac{1}{2}(\frac{1}{4}+s^2)$ gives degrees $2\ell-2$ and $2\ell-1$, respectively. The even and odd parities remain. Thus the ordered family $R_1^+, R_1^-, R_2^+, R_2^-, \ldots$ has one polynomial of each degree, with a nonzero leading coefficient. Subtracting the appropriate multiple of the polynomial of largest degree, and iterating down to degree zero, expresses every polynomial as a finite linear combination. Linear independence follows by the largest-degree coefficient of a finite relation. Consequently

$$N_{\mathrm{ar}} = \overline{\mathrm{span}\{\mathcal{F}_\mu\mathcal{E}\psi_\ell^\pm\}}^{\,\tau} = \overline{\mathbb{C}[s]\Xi}^{\,\tau}, \qquad Q_{\mathrm{ar}} = (\mathcal{H}_{\leq 1}, \tau)/N_{\mathrm{ar}}. \tag{159}$$

Here τ denotes precisely the source's topology. No equality of this closed space with a principal algebraic ideal is being asserted. Multiplication $Mf(s) = sf(s)$ is continuous in the topological ring; it preserves the dense generating subspace, and hence N_{ar}. Its induced operator S is the operator of Proposition 3.6. We use the source's spectral assertion as an explicitly imported theorem:

$$\mathrm{Spec}(S) = \{\lambda \in \mathbb{C} : \zeta(\tfrac{1}{2} + i\lambda) = 0 \text{ at a nontrivial zero}\}. \tag{160}$$

Spectrum here means failure of a continuous two-sided inverse. No proof of the entire source spectral theorem is claimed as a new result of this manuscript.

There is a material source limitation. Page 19 calls $\mathcal{H}_{\leq 1}$ the Hadamard topological ring of entire functions of order at most one; it does not state seminorms or a convergence criterion. The cited Connes–Marcolli Proposition 2.24, printed pages 379–381, proves the polynomial-image assertion and explicitly discusses different function-space realizations [6]. It does not supply that missing specification. We therefore retain τ in every source quotient. Below, all assertions depending on another topology name and define that topology. The comparison with the source is an exact quotient diagram, rather than an identification by terminology. The bounded edition audit gives the same limitation in the primary arXiv v1 source, lines 683–704 (Proposition 3.9), and in v2, `mainc2m24fine.tex`, lines 668–689 (Proposition 3.6): each states the order class, polynomial trace image, and spectral set, without defining its topology or algebraic multiplicities. These exact edition locators supplement the author-proof locator in [5]; no claim of byte identity with a publisher's final subscription text is required. Following the book's cited Sobolev construction leads to explicit weighted Hilbert spaces in its Chapter 2, Section 9, and in Connes's original spectral realization [10]. Section I.1 retains those spaces and proves their exact common-test comparison with Q_{ar}, including the original generator and the allowed zero multiplicities. The inspected construction does not specify the topology on all of $\mathcal{H}_{\leq 1}$ or identify it with those Hilbert topologies.

C A globally defined heat map on the original underlying functions

For an entire function f, let $M_f(r) = \max_{|s|\le r} |f(s)|$ and define its order by

$$\limsup_{r\to\infty} \frac{\log\log\max(e, M_f(r))}{\log r}.$$

The zero function is included. For $1 < p < 2$, $a > 0$ set

$$\|f\|_{p,a} = \sup_{s\in\mathbb{C}} |f(s)|e^{-a|s|^p}.$$

The underlying set $\mathcal{H}_{\le 1}$ is exactly the set where all these seminorms are finite. Indeed order at most one gives, for $1 < q < p$, $M_f(r) \le \exp(r^q)$ for sufficiently large r; the expression $r^q - ar^p$ is bounded above. Conversely finiteness gives order at most p for every $p > 1$. Write γ for this explicitly defined growth topology. It is enough to take $p = 1 + 1/j$, $j \ge 2$, and $a = 1/k$, $k \ge 1$: given $p > 1, a > 0$, select one of these $p_j < p$ and use the boundedness of $|s|^{p_j}/k - a|s|^p$. This proves equivalence of the countable family with all displayed seminorms. A sequence Cauchy in these seminorms converges uniformly on compact sets to an entire function. Taking pointwise limits in each weighted Cauchy inequality gives convergence in that seminorm and finiteness of its norm. Thus $(\mathcal{H}_{\le 1}, \gamma)$ is a complete metrizable locally convex space.

Lemma C.1 (Derivative bounds and the heat group)**.** *Put $D = \partial_s$. Differentiation and multiplication by s map the original underlying set $\mathcal{H}_{\le 1}$ into itself. For $m \ge 1$,*

$$\|D^m f\|_{p,a} \le m! \left(\frac{epa}{m}\right)^{m/p} \|f\|_{p,a/2^{p-1}}. \tag{161}$$

For every $t \in \mathbb{C}$, the series

$$U_t f = \sum_{k=0}^{\infty} \frac{(-t/4)^k}{k!} D^{2k} f \tag{162}$$

converges in γ, locally uniformly in t; U_t is a continuous automorphism of $(\mathcal{H}_{\le 1}, \gamma)$ and $U_t U_u = U_{t+u}$, $U_t^{-1} = U_{-t}$.

Proof. Cauchy's circle formula on a circle of radius R about s gives

$$|D^m f(s)| \le m! R^{-m} \max_{|z-s|=R} |f(z)|.$$

For $b = a/2^{p-1}$ the inequality $|s+z|^p \le 2^{p-1}(|s|^p + |z|^p)$ bounds the last expression, after multiplication by $e^{-a|s|^p}$, by $m!R^{-m}e^{aR^p}\|f\|_{p,b}$. Choosing $R = (m/(ap))^{1/p}$ proves (161). For multiplication use

$$\|Mf\|_{p,a} \le \sup_{r\ge 0} re^{-ar^p/2} \|f\|_{p,a/2}.$$

The kth heat summand is at most

$$\frac{(|t|/4)^k}{k!}(2k)! \left(\frac{epa}{2k}\right)^{2k/p} \|f\|_{p,b}.$$

Since $(2k)!/k! \le (2k)^k$, its scalar factor is at most $[C_{p,a}|t|k^{1-2/p}]^k$, whose kth root tends to zero. This proves summability, continuity, and locally uniform convergence of every time-differentiated series by the same estimate with the additional polynomial factors in k. Commutation with D follows from continuity of D. In the double composition series group terms with $n = k + \ell$. The sum of the absolute scalar coefficients at $D^{2n}f$ is $((|t| + |u|)/4)^n/n!$; the preceding estimate justifies regrouping. The binomial theorem gives U_{t+u}. The value at zero is the identity, proving the inverse. □

Lemma C.2 (Polynomial approximation)**.** *The Taylor polynomials of any $f \in \mathcal{H}_{\le 1}$ converge to f in γ. Multiplication of two elements is continuous in γ.*

Proof. If $f = \sum c_n s^n$, Cauchy's coefficient bound with parameter $b < a$ gives, for $n \ge 1$,

$$|c_n| \le \|f\|_{p,b}(epb/n)^{n/p}.$$

Also $\|s^n\|_{p,a} = (n/(epa))^{n/p}$. Therefore $\|c_n s^n\|_{p,a} \le \|f\|_{p,b}(b/a)^{n/p}$. The tail is bounded by a convergent geometric series. Finally $\|fg\|_{p,a} \le \|f\|_{p,a/2}\|g\|_{p,a/2}$. □

Retain the actual real heat family, rather than substitute a polynomial:

$$\Phi(u) = \sum_{n\ge 1}(2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u})e^{-\pi n^2 e^{4u}},$$
$$Z_t(s) = 8H_t(2s) = 8\int_0^\infty e^{tu^2}\Phi(u)\cos(2su)\,du, \qquad \partial_t Z_t = -\tfrac{1}{4}D^2 Z_t, Z_0 = \Xi. \tag{163}$$

These are the Rodgers–Tao conventions, Section 1 [7]. For clarity the analytic exchanges in this transport can be proved directly. Since $n^2 e^{4u} \ge (n^2 + e^{4u})/2$, the absolutely convergent sums $\sum n^j e^{-\pi n^2/2}$ give constants $C, c > 0$ such that $|\Phi(u)| \le Ce^{9u}e^{-ce^{4u}}$ for $u \ge 0$. For $|t| \le T$ and $|s| \le R$, every derivative of the integrand is bounded by a polynomial in u times $C\exp(Tu^2 + (2R+9)u - ce^{4u})$, which is integrable. The exponential series in t can therefore be integrated absolutely. Because $D^{2k}\cos(2su) = (-1)^k(2u)^{2k}\cos(2su)$, it gives exactly $Z_t = U_t\Xi$ with the coefficient $(-t/4)^k/k!$. In particular Z_t lies in the original order-at-most-one set for every complex t, by Lemma C.1; it is never identically zero because U_t is invertible and $\Xi(0) \ne 0$.

D Transport of the actual closed arithmetic image

Define τ_t on the same underlying vector space $\mathcal{H}_{\le 1}$ by declaring $U_t : (\mathcal{H}_{\le 1}, \tau) \longrightarrow (\mathcal{H}_{\le 1}, \tau_t)$ a homeomorphism. This definition specifies every open set: V is τ_t-open exactly when $U_{-t}V$ is τ-open. It uses the globally defined bijection just proved and does not require unproved continuity in τ. Set

$$N_t = U_t N_{\mathrm{ar}} = \overline{\mathrm{span}\{U_t(R_\ell^{\pm}\Xi)\}}^{\tau_t}, \qquad Q_t = (\mathcal{H}_{\le 1}, \tau_t)/N_t. \tag{164}$$

The closure equality follows because homeomorphisms carry closure to closure. In particular N_t is closed. The quotient map

$$\mathcal{U}_t : Q_{\mathrm{ar}} \longrightarrow Q_t, \qquad [f] \longmapsto [U_t f] \tag{165}$$

is well defined, bijective, and continuous, with continuous inverse $[g] \mapsto [U_{-t}g]$. Indeed changing f by $n \in N_{\mathrm{ar}}$ changes $U_t f$ by $U_t n \in N_t$, and the inverse gives necessity; the quotient universal property proves both continuity statements.

Theorem D.1 (Exact arithmetic heat intertwiner)**.** *On the entire-function set,*

$$B_t = U_t M U_{-t} = M - \tfrac{t}{2}D. \tag{166}$$

This operator preserves N_t, is continuous in τ_t, and induces S_t on Q_t. It satisfies $S_t\mathcal{U}_t = \mathcal{U}_t S$ and

$$\mathrm{Spec}(S_t) = \mathrm{Spec}(S) = \{\lambda : \zeta(\tfrac{1}{2} + i\lambda) = 0 \text{ } \textit{nontrivially}\}. \tag{167}$$

The time-dependent trace functions are exactly $U_t(R_\ell^{\pm}\Xi) = R_\ell^{\pm}(B_t)Z_t$.

Proof. The product rule gives $D^{2k}Mf = MD^{2k}f + 2kD^{2k-1}f$. Insert this identity in the convergent heat series to obtain $U_tM = (M - (t/2)D)U_t$, proving (166). If $n \in N_{\mathrm{ar}}$, then $B_tU_tn = U_tMn \in N_t$. Continuity is conjugation of the continuous M by the defining homeomorphism. The induced identity follows on representatives. If T is a continuous inverse of $S - \lambda$, then $\mathcal{U}_t T \mathcal{U}_t^{-1}$ is a continuous inverse of $S_t - \lambda$. Conjugating in the opposite direction proves the converse, and hence the spectrum equality using (160). Apply the polynomial identity $U_tp(M) = p(B_t)U_t$ to Ξ for the last assertion. □

The heat map also transports the full multiplication, with every cross term retained. For $f, g \in \mathcal{H}_{\le 1}$ define

$$f \star_t g = U_t((U_{-t}f)(U_{-t}g)) = \sum_{n=0}^{\infty} \frac{(-t/2)^n}{n!}(D^n f)(D^n g). \tag{168}$$

We prove the series identity, including convergence. For $F_t = U_tf$, $G_t = U_tg$, the series

$$J_t = \sum_{n=0}^{\infty} \frac{(-t/2)^n}{n!}(D^n F_t)(D^n G_t)$$

converges in γ, uniformly on compact time sets: applying (161) with output norm $(p, a/2)$ to both factors bounds its nth term by

$$K\left[\frac{T}{2}\left(\frac{epa}{2}\right)^{2/p} n^{1-2/p}\right]^n \qquad (|t| \le T),$$

where K bounds the product of the two input norms $(p, a/2^p)$, and is finite by the heat estimate. The root test applies. Cauchy's formula in time permits termwise time derivatives; continuity of D permits spatial derivatives. Differentiating the coefficient contributes $-\frac{1}{2}\sum(-t/2)^n(D^{n+1}F_t)(D^{n+1}G_t)/n!$. Differentiating the factors contributes the two pure second-derivative terms with coefficient $-1/4$. The complete product rule therefore gives $\partial_t J_t = -D^2J_t/4$ and $J_0 = fg$. Its entire time Taylor coefficients are uniquely $(-1/4)^kD^{2k}(fg)/k!$, so $J_t = U_t(fg)$. Replacing the inputs by $U_{-t}f, U_{-t}g$ proves (168). Conjugation of ordinary multiplication proves associativity, commutativity, unit 1, and continuity for τ_t, even when continuity of the series in the source topology is unknown. The identity $s \star_t f = sf - tf'/2$ recovers exactly B_t. No ideal property of N_{ar} beyond its stated polynomial invariance is required for this identity or for the quotient in Theorem D.1.

The transported actual trace image has no common point-evaluation zero at any nonzero time. Suppose λ were such a common zero. It would annihilate each $U_t(s^n\Xi)$. For every $a \in \mathbb{C}$, the Taylor series of e^{as} converges in γ by Lemma C.2. Multiplication, U_t, and evaluation are continuous in γ, so

$$0 = \sum_{n=0}^{\infty} \frac{a^n}{n!} U_t(s^n\Xi)(\lambda) = e^{a\lambda - ta^2/4} Z_t(\lambda - ta/2).$$

The last equality follows from the just-proved product formula and the entire Taylor formula for Z_t. At $t \neq 0$ varying a forces Z_t to vanish everywhere, a contradiction. This argument uses only the polynomial generators contained in N_t; it assumes no source-topology convergence of the exponential series. Thus the source arithmetic spectrum in (167) is carried by the explicitly transported operator and relation, even though the transported relation has no common ordinary evaluation zero.

The first two nonconstant polynomial generators make the extra terms visible without changing any polynomial coefficient:

$$\begin{aligned} U_t(s\Xi) &= sZ_t - \tfrac{t}{2}Z_t', \\ U_t(s^2\Xi) &= s^2Z_t - tsZ_t' - \tfrac{t}{2}Z_t + \tfrac{t^2}{4}Z_t''. \end{aligned} \tag{169}$$

For $L = \partial_t + \frac{1}{4}D^2$, the complete product rule is

$$L(AP) = A(P_t + \tfrac{1}{4}P_{ss}) + (A_t + \tfrac{1}{4}A_{ss})P + \tfrac{1}{2}A_sP_s. \tag{170}$$

Thus $L(R(s)Z_t) = \frac{1}{4}R''Z_t + \frac{1}{2}R'Z_t'$. The family $R(B_t)Z_t$ solves the homogeneous heat equation, whereas the displayed expression is the full defect of replacing it by $R(s)Z_t$. For a local factorization $Z_t = AP$ with $A \neq 0$, the same formula retains the analytic unit A and its derivatives. At a simple zero $\lambda(t)$, differentiating $Z_t(\lambda(t)) = 0$ gives

$$\lambda'(t) = \frac{1}{4}\frac{Z_{ss}}{Z_s}(t, \lambda(t)) = \frac{1}{4}\frac{P_{ss}}{P_s}(t, \lambda(t)) + \frac{1}{2}\frac{A_s}{A}(t, \lambda(t)). \tag{171}$$

This statement is on the open locus $Z_s \neq 0$, on which the analytic implicit-function theorem supplies the local curve; it claims no chosen zero is simple. At a multiple zero the division is undefined, and the full equation (170) remains the equation to use.

E Derivative domains, multivalued operators, and jet extensions

Every derivative has the full underlying domain $\mathcal{H}_{\leq 1}$ by Lemma C.1. Continuity in the unnamed source topology is not assumed. The following gives exact source-quotient operators and their domains without imposing it. For $k \geq 1$ put

$$K_k = \{n \in N_{\mathrm{ar}} : D^k n \in N_{\mathrm{ar}}\}, \qquad W_k = (D^k N_{\mathrm{ar}} + N_{\mathrm{ar}})/N_{\mathrm{ar}} \subseteq Q_{\mathrm{ar}}.$$

The derivative correspondence is the actual linear relation

$$\mathfrak{D}_k = \{([f], [D^k f]) : f \in \mathcal{H}_{\leq 1}\} \subseteq Q_{\mathrm{ar}} \times Q_{\mathrm{ar}}. \tag{172}$$

For fixed $[f]$, its output fibre is precisely $[D^k f] + W_k$: all representatives are $f + n$ with $n \in N_{\mathrm{ar}}$, and linearity gives all and only the stated outputs. The kernel of its presenting map $f \mapsto ([f], [D^k f])$ is exactly K_k. The relation has domain all of Q_{ar}; its multivalued part at zero is exactly W_k. Thus it is the graph of an operator exactly when $D^k N_{\mathrm{ar}} \subseteq N_{\mathrm{ar}}$. This is a calculated equivalence, with an explicit obstruction space, rather than an assumption of invariance.

There is no nonempty domain of quotient classes on which the same representative formula is unambiguous if $W_k \neq 0$ and every representative of a class is admitted: addition of the same offending n gives two different outputs for every class. A chosen linear lift is a different map. More precisely any vector subspace $A \subseteq \mathcal{H}_{\leq 1}$ supplies a single-valued map on its quotient image $q(A)$ exactly when $D^k(A \cap N_{\mathrm{ar}}) \subseteq N_{\mathrm{ar}}$; the identical difference-of-representatives argument proves both directions. The first heat generator is $-D^2/4$, so its ambiguity space is $-W_2/4$, with the coefficient and sign unchanged.

E.1 Exact algebraic spectral defects of the source relation

These calculations keep the actual N_{ar} and require only the already proved $MN_{\mathrm{ar}} \subseteq N_{\mathrm{ar}}$. For every $\lambda \in \mathbb{C}$, division by $x = s - \lambda$ gives the algebraic identities

$$\begin{aligned} \operatorname{coker}_{\mathrm{alg}}(S - \lambda) &= \mathcal{H}_{\leq 1}/(N_{\mathrm{ar}} + x\mathcal{H}_{\leq 1}), \\ \ker(S - \lambda) &\simeq (N_{\mathrm{ar}} \cap x\mathcal{H}_{\leq 1})/(xN_{\mathrm{ar}}), \quad [n] \longmapsto [n/x]. \end{aligned} \tag{173}$$

Division belongs to the original order class: if $f(\lambda) = 0$, the filled quotient is entire, is bounded on a fixed disk, and outside $|s| \geq 2|\lambda| + 1$ its modulus is at most $2|f(s)|$. Thus $x\mathcal{H}_{\leq 1}$ is exactly

the kernel of evaluation at λ. The cokernel identity follows from the image of the induced multiplication. For the second identity, changing n by xn_0 changes n/x by $n_0 \in N_{\mathrm{ar}}$; its image is in the stated kernel. Conversely any kernel representative f has $xf \in N_{\mathrm{ar}} \cap x\mathcal{H}_{\le 1}$. Finally $[n/x] = 0$ exactly when $n \in xN_{\mathrm{ar}}$. This proves every asserted kernel and image.

The cokernel has only two possibilities. When every $n \in N_{\mathrm{ar}}$ vanishes at λ, it is one-dimensional, with its exact identification given by evaluation. Otherwise retain any $h_\lambda \in N_{\mathrm{ar}}$ with $h_\lambda(\lambda) \neq 0$. Then

$$\begin{aligned} A_\lambda f(s) &= \frac{f(s) - f(\lambda)h_\lambda(s)/h_\lambda(\lambda)}{s-\lambda}, \\ f &= \frac{f(\lambda)}{h_\lambda(\lambda)} h_\lambda + (s-\lambda)A_\lambda f \end{aligned} \tag{174}$$

proves surjectivity of $S - \lambda$, and its exact remaining kernel is

$$\ker(S-\lambda) \simeq N_{\mathrm{ar}}/\big((s-\lambda)N_{\mathrm{ar}} \oplus \mathbb{C}h_\lambda\big), \qquad n \longmapsto [A_\lambda n]. \tag{175}$$

The map is onto because a kernel class $[g]$ is the image of $n = (s-\lambda)g \in N_{\mathrm{ar}}$. Its kernel consists exactly of $n = (s-\lambda)n_0 + ch_\lambda$, $n_0 \in N_{\mathrm{ar}}$, by (174). The sum is direct by evaluation. In this second case algebraic invertibility is therefore equivalent to the exact identity $N_{\mathrm{ar}} = (s-\lambda)N_{\mathrm{ar}} \oplus \mathbb{C}h_\lambda$. On that locus the representative formula $[f] \mapsto [A_\lambda f]$ is independent of representatives, since $A_\lambda N_{\mathrm{ar}} \subseteq N_{\mathrm{ar}}$; multiplication in (174) proves the right inverse, and $A_\lambda((s-\lambda)f) = f$ proves the left inverse. These are algebraic statements; continuity of this inverse in τ has not been inserted.

The first derivative and the heat generator have exactly the same frozen invariance question for the source subspace:

$$DN_{\mathrm{ar}} \subseteq N_{\mathrm{ar}} \quad \Longleftrightarrow \quad D^2 N_{\mathrm{ar}} \subseteq N_{\mathrm{ar}} \quad \Longleftrightarrow \quad -\tfrac{1}{4}D^2 N_{\mathrm{ar}} \subseteq N_{\mathrm{ar}}. \tag{176}$$

Indeed the product rule is $D^2(sn) - sD^2 n = 2Dn$. Since $sn \in N_{\mathrm{ar}}$, second-derivative invariance puts both terms on the left in N_{ar}, proving first-derivative invariance. Iteration proves the reverse implication; the retained nonzero scalar $-1/4$ proves the last equivalence. If these equivalent inclusions hold, all $D^j\Xi$ belong to N_{ar}. At each λ at least one is nonzero: otherwise the Taylor series would make Ξ identically zero. Thus every $S - \lambda$ is then algebraically surjective by (174). Its prescribed topological spectral points would be represented entirely by the kernel (175) or a noncontinuous algebraic inverse. This computes the precise alternatives without assigning either one to the source's stated spectral set.

There is also a full multiplicity consequence of algebraic derivative descent. Writing $d[f] = [Df]$ on its exact descent locus, the product rule gives $[d, S] = I$. For $0 \neq v \in \ker(S-\lambda)$,

$$(S-\lambda)d^k v = -kd^{k-1}v, \qquad (S-\lambda)^k d^k v = (-1)^k k! v \quad (k \ge 0), \tag{177}$$

with zero right side in the first identity at $k = 0$. To prove induction, commute the next d using $(S-\lambda)d = d(S-\lambda) - I$; the coefficient $-k$ becomes $-(k+1)$. Iteration proves the second identity. A finite linear dependence among $v, dv, \dots, d^n v$, with largest nonzero coefficient at index j, becomes that coefficient times $(-1)^j j! v$ after applying $(S-\lambda)^j$, a contradiction. Thus every such eigenvector generates an infinite independent generalized eigenvector sequence. No finite source multiplicity assertion has been assumed.

Finally even a single forward nonzero heat inclusion has a calculated consequence. Fix $t \neq 0$ and $c = t/2$, and define

$$\begin{aligned} A_0 = U_t, \qquad A_{k+1} = MA_k - A_k M, \qquad A_k = P_k(D)U_t, \\ P_0(X) = 1, \qquad P_{k+1}(X) = cXP_k(X) - P_k'(X). \end{aligned} \tag{178}$$

The identity follows by induction from $[M, U_t] = cDU_t$ and $[M, D^j] = -jD^{j-1}$: commuting M through $P_k(D)U_t$ contributes $-P_k'(D)U_t + cP_k(D)DU_t$. The degree of P_k is exactly k, with leading coefficient $c^k \neq 0$. Under $U_t N_{\mathrm{ar}} \subseteq N_{\mathrm{ar}}$, induction with $MN_{\mathrm{ar}} \subseteq N_{\mathrm{ar}}$ gives $A_k N_{\mathrm{ar}} \subseteq N_{\mathrm{ar}}$; subtracting the lower-degree terms of P_k then gives exactly

$$D^k U_t N_{\mathrm{ar}} \subseteq N_{\mathrm{ar}} \quad (k \geq 0). \tag{179}$$

In particular every derivative of the nonzero $Z_t = U_t \Xi$ lies in N_{ar}. At every point one derivative is nonzero by the entire Taylor theorem, so every $S - \lambda$ is algebraically surjective in this case too. This proof uses only the one forward inclusion. For the stronger equality $U_t N_{\mathrm{ar}} = N_{\mathrm{ar}}$, it also proves $DN_{\mathrm{ar}} \subseteq N_{\mathrm{ar}}$, by writing each $n \in N_{\mathrm{ar}}$ as $U_t m$ in (179). None of these exact implications presumes the inclusion or equality for the actual source closure. The explicit obstruction spaces above remain the calculated source question.

Even when W_k is not closed, the smallest source-closed enlargement of the target relation subspace is explicit:

$$[f] \longmapsto [D^k f] : Q_{\mathrm{ar}} \longrightarrow \mathcal{H}_{\leq 1} / \overline{N_{\mathrm{ar}} + D^k N_{\mathrm{ar}}}^{\tau}. \tag{180}$$

It is well defined algebraically. Equip its source with the quotient of the graph topology $f \mapsto (f, D^k f)$ in $(\mathcal{H}_{\leq 1}, \tau)^2$ to obtain a continuous map. This specifies a topology change explicitly, without claiming it is the original one. The lost part of the original target Q_{ar} is exactly $\overline{N_{\mathrm{ar}} + D^k N_{\mathrm{ar}}}^{\tau} / N_{\mathrm{ar}}$.

Here is an extension which retains correlated derivatives rather than quotienting them away. For $j \geq 1$ define the bijection

$$T_j(f_0, \ldots, f_j) = (f_0, f_1 - Df_0, \ldots, f_j - D^j f_0).$$

Its inverse is $(f_0, g_1, \ldots, g_j) \mapsto (f_0, g_1 + Df_0, \ldots, g_j + D^j f_0)$. Give $V_j = \mathcal{H}_{\leq 1}^{j+1}$ the topology transported by T_j from $(\mathcal{H}_{\leq 1}, \tau) \times (\mathcal{H}_{\leq 1}, \gamma)^j$. The residual factors are explicitly given the proved Hausdorff growth topology; the source factor keeps its original topology. For $J_j f = (f, Df, \ldots, D^j f)$, $T_j(J_j N_{\mathrm{ar}}) = N_{\mathrm{ar}} \oplus 0^j$, so $J_j N_{\mathrm{ar}}$ is closed. Define

$$X_j = V_j / J_j N_{\mathrm{ar}}. \tag{181}$$

Then the map

$$[f_0, \ldots, f_j] \longmapsto ([f_0], f_1 - Df_0, \ldots, f_j - D^j f_0)$$

is a homeomorphism $X_j \simeq Q_{\mathrm{ar}} \oplus (\mathcal{H}_{\leq 1}, \gamma)^j$; independence, the inverse, and continuity follow directly from T_j and the quotient definition. In particular there is the split exact sequence

$$0 \longrightarrow \mathcal{H}_{\leq 1}^j \longrightarrow X_j \xrightarrow{[f_0, \ldots, f_j] \mapsto [f_0]} Q_{\mathrm{ar}} \longrightarrow 0, \qquad [f] \longmapsto [J_j f]. \tag{182}$$

Its relation on a source generator retains every product term:

$$D^k(R\Xi) = \sum_{h=0}^{k} \binom{k}{h} R^{(h)} \Xi^{(k-h)} \quad (0 \leq k \leq j). \tag{183}$$

The binomial formula follows by induction from the ordinary product rule, including the two edge terms at each step.

The original spectral coordinate acts on V_j by

$$(\mathcal{M}_j f)_k = s f_k + k f_{k-1} \quad (0 \leq k \leq j), \qquad f_{-1} = 0.$$

The identity $D^k(sf) = sD^k f + kD^{k-1} f$ proves $\mathcal{M}_j J_j f = J_j(sf)$; it therefore preserves the exact relation subspace. In the T_j coordinates it acts by S on Q_{ar} and by $g_k \mapsto s g_k + k g_{k-1}$ on the residual terms ($g_0 = 0$). This also proves continuity in the stated topology. Notice that (182) adds a free residual module: the spectrum of the entire extension is not claimed to equal the source spectrum. The arithmetic quotient and its split inclusion are explicitly preserved.

F A fully computed analytic realization and its exact comparison

We now compute the closures and the failure of descent in specified topologies. This does not identify them with τ. Let κ be the topology of uniform convergence on compact sets on the same set $\mathcal{H}_{\le 1}$, and define

$$N_\gamma = \overline{\mathbb{C}[s]\Xi}^{\,\gamma}, \qquad N_\kappa = \overline{\mathbb{C}[s]\Xi}^{\,\kappa}.$$

Evaluations of all derivatives are continuous in both topologies by Cauchy's formula. The growth topology is finer than κ, so $N_\gamma \subseteq N_\kappa$. If λ is a zero of Ξ with its actual multiplicity m_λ, every element of either closed subspace has vanishing derivatives of orders $0, \dots, m_\lambda - 1$ at λ.

Lemma F.1 (Entire division with its growth and topology retained)**.** *Let $g \in \mathcal{H}_{\le 1}$ with $g(0) \neq 0$. If $f \in \mathcal{H}_{\le 1}$ and $h = f/g$ extends to an entire function, then $h \in \mathcal{H}_{\le 1}$. For $1 < p < 2$, $a > 0$, put $b = a/(6 \cdot 2^p)$. There is the linear division estimate*

$$\|f/g\|_{p,a} \le \left(\frac{\max(1, \|g\|_{p,b})}{|g(0)|}\right)^3 \|f\|_{p,b}. \tag{184}$$

Multiplication by g is a topological linear isomorphism from $(\mathcal{H}_{\le 1}, \gamma)$ onto the closed subspace of functions divisible by g, with its subspace topology. Compact-topology division is also continuous.

Proof. Choose $R > 0$ with no zero of g on $|s| = R$. Factor its finitely many zeros in the disk, retaining multiplicities. The residual is holomorphic and zero-free there, so its log modulus is harmonic. For $|z| < R$ the boundary mean of $\log|Re^{i\theta} - z|$ equals $\log R$: expand $\log(1 - (z/R)e^{-i\theta})$ in its uniformly convergent series and take real parts. The mean-value formula for the residual gives the exact Jensen identity

$$\frac{1}{2\pi}\int_0^{2\pi} \log|g(Re^{i\theta})|\, d\theta = \log|g(0)| + \sum_{|z|<R} m_z \log(R/|z|) \ge \log|g(0)|.$$

There is no zero at zero. With $\log^+ x = \max(0, \log x)$ and $\log^- x = \max(0, -\log x)$, the inequality $\log^+|h| \le \log^+|f| + \log^-|g|$ gives

$$\frac{1}{2\pi}\int_0^{2\pi} \log^+|h(Re^{i\theta})|\, d\theta \le \log^+ M_f(R) + \log^+ M_g(R) - \log|g(0)|. \tag{185}$$

The right side is nonnegative since $M_g(R) \ge |g(0)|$. The harmonic Poisson extension of the continuous nonnegative boundary function $\log^+|h|$ majorizes $\log|h|$ by the maximum principle and is nonnegative; hence it majorizes $\log^+|h|$ in the disk. For $|s| \le r < R$ the Poisson kernel is at most $(R + r)/(R - r)$. Take $R \downarrow 2r$ through circles avoiding zeros. Continuity of the two maximum functions gives, for every $r > 0$,

$$\log^+ M_h(r) \le 3\big(\log^+ M_f(2r) + \log^+ M_g(2r) - \log|g(0)|\big). \tag{186}$$

No pointwise lower bound for g on a large circle has been assumed. For $f \neq 0$, put $F = \|f\|_{p,b} > 0$ and apply this to $f/F, h/F$. Then $\log^+ M_{f/F}(2r) \le b(2r)^p$, and $\log^+ M_g(2r) \le \log\max(1, \|g\|_{p,b}) + b(2r)^p$. The coefficient of r^p is $6b2^p = a$, giving (184); continuity fills in $r = 0$. The zero numerator is immediate. All p, a are allowed, so h belongs to the original order class and division is continuous. Multiplication is continuous by Lemma C.2. The requisite vanishing jets at every zero are closed conditions; local Taylor division identifies them with entire divisibility. The estimate proves this closed set is exactly $g\mathcal{H}_{\le 1}$. For the compact topology choose $R > r$ with no boundary zero and $\delta = \min_{|s|=R}|g(s)| > 0$. The maximum principle for the filled entire quotient gives $M_h(r) \le \delta^{-1} M_f(R)$, proving compact continuity too. □

For general nonzero $g \in \mathcal{H}_{\le 1}$, choose and retain a point c with $g(c) \neq 0$ and use disks centered at c. The functions and spectral coordinate remain unchanged. The bound $|s+c|^p \le 2^{p-1}(|s|^p + |c|^p)$ proves continuity of translations by c and $-c$ in γ; compact containment proves it in κ. Thus all conclusions hold with the stated translated estimates. The original center $c = 0$ already applies to Ξ.

Proposition F.2 (The computed closures are the same subspace)**.** *With their definitions unchanged,*

$$\begin{aligned} N_\gamma = N_\kappa &= \Xi\mathcal{H}_{\le 1} \\ &= \{f \in \mathcal{H}_{\le 1} : D^h f(\lambda) = 0 \text{ for every } \Xi(\lambda) = 0,\ 0 \le h < m_\lambda\}. \end{aligned} \tag{187}$$

This is equality of subsets of the original ring, not an identification of γ, κ, or τ.

Proof. Jet continuity puts both closures inside the displayed vanishing set and gives $N_\gamma \subseteq N_\kappa$. Local Taylor division makes $h = f/\Xi$ entire for any f in that set; the division lemma then puts h in $\mathcal{H}_{\le 1}$. Its Taylor polynomials p_n converge in γ, and multiplication gives $p_n\Xi \to f$ there. Thus the vanishing set is contained in N_γ, proving all equalities. The algebraic-ideal equality is the conclusion of this argument; it was not substituted for an unevaluated closure. □

Theorem F.3 (Compact closure, multiplicities, and coordinate spectrum)**.** *The compact closure is exactly*

$$\begin{aligned} N_\kappa &= \{f \in \mathcal{H}_{\le 1} : f/\Xi \text{ extends to an entire function}\} \\ &= \{f \in \mathcal{H}_{\le 1} : D^h f(\lambda) = 0 \ (h < m_\lambda),\ \Xi(\lambda) = 0\}. \end{aligned} \tag{188}$$

On $Q_\kappa = (\mathcal{H}_{\le 1}, \kappa)/N_\kappa$, multiplication by s has spectrum exactly the zeros of Ξ in the original s coordinate. For a zero λ of multiplicity m, there is a continuous surjection

$$Q_\kappa \longrightarrow \mathbb{C}[\epsilon]/(\epsilon^m), \qquad [f] \longmapsto \sum_{h=0}^{m-1} \frac{D^h f(\lambda)}{h!}\epsilon^h,$$

intertwining multiplication by s with $\lambda + \epsilon$.

Proof. The vanishing conditions are closed and hold on every $p\Xi$, so the closure is contained in the last displayed set. Local Taylor division at each isolated zero identifies that set with entire divisibility by Ξ. For such an f, put $g = f/\Xi \in \mathcal{O}(\mathbb{C})$. Its Taylor polynomials p_n converge uniformly on every compact; $p_n\Xi \to f$ in κ. Every $p_n\Xi$ belongs to the original order class even though no growth assertion about g is needed. This proves (188) with the closure retained. The local-jet map is independent of representatives; the polynomials $1, (s-\lambda), \dots, (s-\lambda)^{m-1}$ prove surjectivity. Leibniz's rule gives the stated action, and the finite jet evaluations are continuous.

If $\Xi(\lambda) \neq 0$, define

$$T_\lambda f(s) = \frac{f(s) - f(\lambda)\Xi(s)/\Xi(\lambda)}{s - \lambda},$$

filling in the removable value. Division by a linear factor preserves the order bound: outside a disk about λ the denominator is bounded below, and inside a disk the filled entire function is bounded. It is continuous in κ by Cauchy's formula on a larger disk applied to the numerator. For $n \in N_\kappa$ the quotient $T_\lambda n/\Xi = (n/\Xi - (n/\Xi)(\lambda))/(s-\lambda)$ is entire; thus it descends. Direct multiplication proves $(S_\kappa - \lambda)[T_\lambda f] = [f]$. Applying T_λ to $(s-\lambda)f$ gives f exactly, proving the other inverse identity. If $\Xi(\lambda) = 0$, evaluation at λ descends to a nonzero continuous functional which annihilates the image of $S_\kappa - \lambda$. Hence that operator is not surjective. This proves the spectrum claim. □

The same coordinate spectrum and the same finite-jet surjections hold for $Q_\gamma = (\mathcal{H}_{\le 1}, \gamma)/\Xi\mathcal{H}_{\le 1}$. Indeed the numerator in $T_\lambda f$ depends continuously on f in γ. Division by $s - \lambda$, on its entire-divisibility domain, is continuous by Lemma F.1 with a nonzero center retained when needed. The already proved two inverse identities therefore give a continuous resolvent in γ as well. The evaluation obstruction at each zero is continuous in that topology too. This proves the claim, without transferring a resolvent through an unproved topology identification.

The ambient topologies κ and γ are in fact distinct. The original functions $f_n(s) = \exp(ns - n^2)$ tend to zero uniformly on each compact set. On the positive real axis the maximum of $nx - x^{3/2}$ is $4n^3/27$, attained at $x = 4n^2/9$. Since $\operatorname{Re} s \le |s|$, that axis attains the full weighted supremum, giving the exact value

$$\|f_n\|_{3/2,1} = \exp(4n^3/27 - n^2) \longrightarrow \infty.$$

Thus equality of the closed arithmetic relation subspaces and of their spectral sets does not identify the ambient topologies. The identity induces a continuous linear bijection $Q_\gamma \to Q_\kappa$; the next theorem proves that its inverse is not continuous.

F.1 The actual difference between the two analytic quotient topologies

Retain the original entire-function set $\mathcal{H}_{\le 1} = \mathcal{H}_{\le 1}$, the original coordinate s, and $\Xi(s) = \xi(\frac{1}{2} + is)$ from (157). Let Λ be the set of its distinct zeros and let $m_\lambda = \operatorname{ord}_\lambda \Xi \ge 1$ be each actual multiplicity. Put

$$I = \Xi\mathcal{H}_{\le 1}, \qquad Q_\kappa = (\mathcal{H}_{\le 1}, \kappa)/I, \qquad Q_\gamma = (\mathcal{H}_{\le 1}, \gamma)/I, \qquad \mathcal{P}_\Xi = \prod_{\lambda \in \Lambda} \mathbb{C}^{m_\lambda}.$$

The product has its ordinary product topology. Coordinates in $\mathbb{C}^{m_\lambda}$ are the actual derivatives of orders $0, \dots, m_\lambda - 1$, with no factorial rescaling. Define

$$J_\Xi f = (f^{(j)}(\lambda))_{\lambda \in \Lambda,\ 0 \le j < m_\lambda}, \qquad \mathcal{V}_\Xi = J_\Xi(\mathcal{H}_{\le 1}) \subset \mathcal{P}_\Xi. \tag{189}$$

Every point of $\mathcal{V}_\Xi$ retains the entire input's arithmetic jets. The restriction to this image is essential.

Theorem F.4 (Jet topology and its completion)**.** *The map induced by* (189) *is a topological linear isomorphism*

$$\overline{J}_\Xi : Q_\kappa \xrightarrow{\cong} \mathcal{V}_\Xi$$

onto the subspace topology from $\mathcal{P}_\Xi$. The image is dense and proper. Consequently Q_κ is incomplete, and its Hausdorff completion is canonically $\mathcal{P}_\Xi$ via this map. In contrast Q_γ is a Fréchet space and is complete. The identity on the same underlying quotient vector space induces a continuous linear bijection

$$\iota : Q_\gamma \longrightarrow Q_\kappa \tag{190}$$

whose inverse is not continuous.

Proof. We first construct each finite interpolation map with its constants. For a finite nonempty set $F \subset \Lambda$, write $d_F = \sum_{\lambda \in F} m_\lambda$ and set

$$B_{\lambda,F}(s) = \prod_{\substack{\nu \in F \\ \nu \ne \lambda}} (s - \nu)^{m_\nu}.$$

The empty product is 1. Since $B_{\lambda,F}(\lambda) \neq 0$, the germ $(s-\lambda)^j/(j!B_{\lambda,F}(s))$ is holomorphic at λ. Denote its Taylor polynomial through degree $m_\lambda - 1$, in powers of the unchanged difference $s - \lambda$, by $T_{\lambda,m_\lambda-1}$ of that germ. Put

$$L_{\lambda,j,F}(s) = B_{\lambda,F}(s) T_{\lambda,m_\lambda-1}\left(\frac{(s-\lambda)^j}{j!B_{\lambda,F}(s)}\right), \qquad 0 \le j < m_\lambda. \tag{191}$$

This is a polynomial of degree at most $d_F - 1$. For $\nu \in F \setminus \{\lambda\}$ it vanishes to order at least m_ν at ν. At λ its difference from $(s-\lambda)^j/j!$ vanishes to order at least m_λ. Therefore, for all $\nu \in F$ and $0 \le k < m_\nu$,

$$L^{(k)}_{\lambda,j,F}(\nu) = \begin{cases} 1, & (\nu,k) = (\lambda,j), \\ 0, & \text{otherwise.} \end{cases}$$

It follows that the linear map to polynomials

$$H_F(v)(s) = \sum_{\lambda \in F} \sum_{j=0}^{m_\lambda - 1} v_{\lambda,j} L_{\lambda,j,F}(s) \tag{192}$$

realizes every prescribed finite array of actual derivatives $v = (v_{\lambda,j})$. For each $R > 0$ define

$$M_f(R) = \max_{|s| \le R} |f(s)|, \qquad C_{R,F} = \sum_{\lambda \in F} \sum_{j=0}^{m_\lambda - 1} M_{L_{\lambda,j,F}}(R).$$

The triangle inequality gives the exact finite bound

$$M_{H_F(v)}(R) \le C_{R,F} \max_{\lambda \in F,\, j < m_\lambda} |v_{\lambda,j}|. \tag{193}$$

When F is empty set $H_F = 0$, $C_{R,F} = 0$ and the maximum over that empty set to 0.

The kernel of J_Ξ is precisely I. Indeed $J_\Xi f = 0$ means that f/Ξ has a removable singularity at each zero of Ξ by its local Taylor expansions. Away from those zeros it is already holomorphic. Thus it extends to an entire function, which belongs to the original order class $\mathcal{H}_{\le 1}$ by Lemma F.1; consequently $f \in \Xi \mathcal{H}_{\le 1}$. The converse follows by the product rule at each zero. This proves that $\overline{J}_\Xi$ is a well-defined linear bijection onto $\mathcal{V}_\Xi$, with inverse sending the jet array of any $f \in \mathcal{H}_{\le 1}$ to its class $[f]$. The kernel calculation proves that this inverse is independent of the choice of such f.

The quotient topology on Q_κ is generated by

$$q_R([f]) = \inf_{h \in \mathcal{H}_{\le 1}} M_{f - \Xi h}(R), \qquad R > 0. \tag{194}$$

For completeness, these are seminorms because they are infima of the representatives' seminorms modulo a linear subspace. The identity $\{q_R < \epsilon\} = \pi_\kappa(\{M_f(R) < \epsilon\})$ holds for every $\epsilon > 0$, and the quotient map π_κ is open since the inverse image of the image of an open set U is the union $U + I$ of its translates. The increasing disk seminorms generate the compact-open topology; the displayed identity therefore proves the assertion about the quotient topology. Integer $R \ge 1$ suffice.

Fix $R > 0$. Choose $R' > R$ such that the circle $|s| = R'$ contains no zero of Ξ, and put $F = \{\lambda \in \Lambda : |\lambda| < R'\}$, a finite set. For $f \in \mathcal{H}_{\le 1}$ form $H = H_F((f^{(j)}(\lambda))_{\lambda \in F, j < m_\lambda})$. The function $(f - H)/\Xi$ is holomorphic throughout $|s| < R'$ by the exact finite interpolation just proved. Its Taylor polynomials $p_n(s)$ at $s = 0$ consequently converge uniformly on $|s| \le R$ to $(f - H)/\Xi$. In particular $p_n \in \mathcal{H}_{\le 1}$, and

$$M_{f - \Xi p_n}(R) \le M_H(R) + M_\Xi(R) M_{(f-H)/\Xi - p_n}(R).$$

Taking the infimum over admissible representatives and then the limit gives

$$q_R([f]) \le M_H(R) \le C_{R,F} \max_{\lambda\in F,\, j<m_\lambda} |f^{(j)}(\lambda)|. \tag{195}$$

No entire extension or growth bound for $(f-H)/\Xi$ outside this disk was used. In particular there is no additional division assumption hidden in the approximation. If F is empty, this proof gives $q_R = 0$, which is compatible with the other quotient seminorms separating points.

Conversely, fix a finite set $F \subset \Lambda$, and choose $R > \max_{\lambda\in F} |\lambda|$ and $0 < \delta < R - \max_{\lambda\in F} |\lambda|$. Cauchy's formula on the radius-δ circle about each λ gives, for every $h \in \mathcal{H}_{\le 1}$,

$$|f^{(j)}(\lambda)| = |(f-\Xi h)^{(j)}(\lambda)| \le j!\delta^{-j} M_{f-\Xi h}(R), \qquad \lambda \in F, \quad j < m_\lambda.$$

Here $0! = 1$, so this bound includes the zeroth derivative without altering its value. Taking infima gives

$$\max_{\lambda\in F,\, j<m_\lambda} |f^{(j)}(\lambda)| \le \left(\max_{\lambda\in F,\, j<m_\lambda} j!\delta^{-j} \right) q_R([f]). \tag{196}$$

Equations (195) and (196) prove continuity in both directions, with the exact stated topologies.

We next verify that the zero set needed for incompleteness is indeed infinite and has an enumeration escaping every compact set. The source Hadamard product for the original even Ξ, retaining its nonzero constant $\Xi(0)$, implies that finitely many zeros would make Ξ a polynomial. This is impossible: at the original coordinate $s = -i(2n - \frac{1}{2})$, $n \ge 1$, the defining completion gives exactly

$$\Xi(-i(2n-\tfrac{1}{2})) = \xi(2n) = n(2n-1)(n-1)!\,\pi^{-n}\zeta(2n). \tag{197}$$

For integer $n \ge 2$, $\zeta(2n) > 1$ by its positive Dirichlet series, and the last $\lfloor n/2 \rfloor$ factors of $(n-1)!$ are at least $n/2$. Thus the logarithm of the right hand side is at least

$$\log(n(2n-1)) + \lfloor n/2 \rfloor \log(n/2) - n\log\pi,$$

which exceeds $d\log(2n)$ for every fixed d and all sufficiently large n. Polynomial values of degree d on these points are bounded by a constant times $(2n)^d$, a contradiction. Since Ξ is nonzero and entire, its zeros are isolated, each multiplicity is finite and each closed disk contains finitely many distinct zeros. We may therefore enumerate all distinct zeros as $\Lambda = \{\lambda_1, \lambda_2, \ldots\}$ with $|\lambda_n| \to \infty$, retaining each m_{λ_n}. This argument uses neither reality nor simplicity of any zero.

Every projection of $\mathcal{V}_\Xi$ onto finitely many full zero-jet blocks is onto, by (192). Such finite coordinate cylinders form a base of the product topology. Consequently $\mathcal{V}_\Xi$ is dense in $\mathcal{P}_\Xi$. It is proper: define the particular array $v \in \mathcal{P}_\Xi$ by

$$v_{\lambda_n,0} = \exp(|\lambda_n|^2), \qquad v_{\lambda_n,j} = 0 \quad (1 \le j < m_{\lambda_n}). \tag{198}$$

For each $f \in \mathcal{H}_{\le 1}$ there is a finite constant $A_f > 0$ with $M_f(r) \le A_f \exp(r^{3/2})$ for every $r \ge 0$; this is exactly the order-at-most-one bound, extended over a bounded interval by increasing A_f. Were $J_\Xi f = v$, we would have

$$\exp(|\lambda_n|^2) \le A_f \exp(|\lambda_n|^{3/2}) \quad \text{for all } n,$$

which contradicts $|\lambda_n| \to \infty$.

The incompleteness can be exhibited by an actual sequence of polynomial classes. Set $F_n = \{\lambda_1, \ldots, \lambda_n\}$, $H_n = H_{F_n}(v|_{F_n})$, and $x_n = [H_n] \in Q_\kappa$. For each fixed zero-jet block, $J_\Xi H_n$ agrees with v from some index onward; hence $J_\Xi H_n \to v$ in the product. Equivalently, for each fixed R, the finite set in (195) lies in every sufficiently large F_n, so that inequality gives

$$q_R(x_n - x_k) = 0 \qquad (n, k \text{ sufficiently large}).$$

Thus (x_n) is Cauchy in Q_κ. If it converged to a class $[f]$, continuity of $\overline{J}_\Xi$ and uniqueness of limits in the product would give $J_\Xi f = v$, already disproved. For example the product is metrized by the complete translation-invariant metric

$$d(v,w) = \sum_{n=1}^{\infty} 2^{-n} \min\Big(1, \max_{0\le j<m_{\lambda_n}} |v_{\lambda_n,j} - w_{\lambda_n,j}|\Big).$$

A Cauchy sequence for this metric is Cauchy in each finite coordinate block, so has a coordinatewise limit; controlling a finite initial sum and the geometric tail proves convergence in the metric. This proves product completeness directly. The isomorphism $\overline{J}_\Xi$, being linear and continuous with a continuous linear inverse, preserves the additive uniformities. Its dense embedding in this complete product therefore gives the claimed Hausdorff completion.

Finally, we prove the completeness assertion for the stronger quotient explicitly. The growth seminorms are

$$\|f\|_{p,a} = \sup_{s\in\mathbb{C}} |f(s)| e^{-a|s|^p}, \qquad 1 < p < 2, \quad a > 0.$$

The countable family $p = 1 + 1/j$, $a = 1/k$ for integers $j \ge 2$ and $k \ge 1$ defines the same topology: for arbitrary $1 < p < 2, a > 0$ choose $p_j < p$, and the finite maximum of $r^{p_j} - ar^p$ controls $\|f\|_{p,a}$ by a constant times $\|f\|_{p_j,1}$. Each countable seminorm belongs to the original family, proving both comparisons. Enumerate this countable family as $(u_\ell)_{\ell\ge1}$ and put $P_n = \max_{1\le\ell\le n} u_\ell$. If (f_n) is Cauchy in all these seminorms, it is Cauchy uniformly on each compact set, and has an entire limit f. In the weighted Cauchy inequality for any fixed seminorm, passage to the pointwise limit in the second index yields the same uniform weighted bound for $f_n - f$. Thus f has all the finite growth seminorms, lies in the original $\mathcal{H}_{\le1}$, and $f_n \to f$ in γ. This proves completeness of $(\mathcal{H}_{\le1}, \gamma)$.

The ideal I is closed: the proved kernel description is an intersection of kernels of derivative evaluations, and Cauchy's estimate shows that each of these evaluations is continuous for γ. The quotient is therefore Hausdorff. Its topology has the countable increasing seminorm family

$$\widetilde{P}_n([f]) = \inf_{h\in I} P_n(f+h).$$

Let (y_n) be a Cauchy sequence in this quotient. Choose a strictly increasing subsequence of indices n_k so far out that

$$\widetilde{P}_k(y_{n_{k+1}} - y_{n_k}) < 2^{-k-1} \quad (k \ge 1).$$

This is possible by the Cauchy property and by taking n_k past the threshold for the kth seminorm before selecting the next index. Choose a representative $d_k \in \mathcal{H}_{\le1}$ of each displayed difference with $P_k(d_k) < 2^{-k}$. For any fixed j, the tail $k \ge j$ satisfies $P_j(d_k) \le P_k(d_k) < 2^{-k}$. Consequently the partial sums of $\sum_{k\ge1} d_k$ are Cauchy in γ and converge in $\mathcal{H}_{\le1}$. Adding a representative of y_{n_1} gives a function whose quotient class is the limit of (y_{n_k}). The full Cauchy sequence (y_n) then has the same limit: for each quotient neighbourhood choose a smaller additive neighbourhood whose sum with itself is contained in it, and combine the Cauchy tail with a convergent subsequence term. The countable seminorm topology is metrizable, so this proves quotient completeness and the Fréchet assertion.

The identity $(\mathcal{H}_{\le1}, \gamma) \to (\mathcal{H}_{\le1}, \kappa)$ is continuous: for every R, p, a, one has $M_f(R) \le e^{aR^p} \|f\|_{p,a}$. It preserves the identical algebraic subspace I, so the quotient universal property proves continuity and bijectivity of (190). If its inverse were continuous, it would be a continuous linear map, hence uniformly continuous for the additive uniformities. The Cauchy sequence (x_n) constructed above would then be Cauchy in the complete Q_γ, would converge there, and by continuity of ι would converge in Q_κ, a contradiction. This proves that its inverse is not continuous. No invariance of metric completeness under arbitrary nonlinear homeomorphisms has been used. □

The original coordinate action remains visible after completion. For $v = J_\Xi f$, the exact product rule gives

$$\big(J_\Xi(sf)\big)_{\lambda,j} = \lambda v_{\lambda,j} + j v_{\lambda,j-1}, \qquad 0 \le j < m_\lambda, \tag{199}$$

where the second term is 0 when $j = 0$. Each output coordinate depends on finitely many input coordinates, so this extends to a continuous linear operator on the entire product completion. The factors j are retained because these coordinates are actual derivatives. The completion adds precisely the nonrealizable jet arrays, including (198); it does not add an entire function of order at most one representing such an array. Neither analytic quotient topology has been identified with the source topology τ.

F.2 Exact derivative domains inside the analytic relation

Let $I = \Xi\mathcal{H}_{\le 1} = N_\gamma = N_\kappa$. For $k \ge 1$ the exact representative domain in the relation space is

$$\begin{aligned} I \cap D^{-k} I = \{\Xi h : h \in \mathcal{H}_{\le 1}, \quad & [z^n](A_\lambda(\lambda + z)h(\lambda + z)) = 0 \\ & \text{for } \max(0, k - m_\lambda) \le n < k, \text{ at every zero } \lambda\}. \end{aligned} \tag{200}$$

where the original local unit is $A_\lambda(s) = \Xi(s)/(s-\lambda)^{m_\lambda}$. To prove this, write $\Xi h = z^m \sum_{n\ge 0} b_n z^n$ at λ, with $m = m_\lambda$ and $b_n = [z^n](A_\lambda h)$. Its kth derivative is

$$\sum_{m+n\ge k} \frac{(m+n)!}{(m+n-k)!} b_n z^{m+n-k}.$$

Every factorial coefficient is nonzero. Vanishing to order at least m is equivalent to $b_n = 0$ exactly for $\max(0, k-m) \le n < k$, as stated. The derivative has the original order bound, so the vanishing condition is equivalent to membership in I by (187). This proves both directions globally, including all multiplicities and local unit coefficients.

For $k = 1$, this is simply $h(\lambda) = 0$ at every distinct zero, without changing or discarding its original multiplicity in Ξ. The derivative ambiguity has the exact presentation

$$\mathcal{H}_{\le 1}/\{h : h(\lambda) = 0 \text{ for every } \Xi(\lambda) = 0\} \xrightarrow{\cong} (DI + I)/I, \qquad [h] \longmapsto [\Xi' h]. \tag{201}$$

Indeed $D(\Xi h) = \Xi h' + \Xi' h$, and the first term lies in I. At each zero of multiplicity m, Ξ' has exactly order $m-1$, so its product with h is divisible by Ξ exactly when $h(\lambda) = 0$ for every zero. This proves the precise kernel, image, and inverse on the image in (201).

For the heat generator ($k = 2$), at any zero of multiplicity at least two the condition is $h(\lambda) = h'(\lambda) = 0$. At a simple zero it is instead

$$A_\lambda(\lambda) h'(\lambda) + A'_\lambda(\lambda) h(\lambda) = 0. \tag{202}$$

It does not require $h(\lambda) = 0$. These follow from the index range in (200): the simple zero has only the coefficient b_1 constrained. In particular the second-derivative relation domain need not be a multiplication module: at a simple zero, a local Ξh obeying (202) and with $h(\lambda) \ne 0$ has $D^2(s\Xi h)(\lambda) = 2(\Xi h)'(\lambda) \ne 0$. This is a local statement with its explicit witness condition; it does not assume the existence of a global simple zeta zero.

There is also an exact calculation of every finite projection of the ambiguity $W_k = (D^k I + I)/I$. In the factorial jet coordinates of any finite set F of distinct zeros, its image is precisely

$$\prod_{\lambda\in F} \epsilon^{\max(m_\lambda - k, 0)} \mathbb{C}[\epsilon]/(\epsilon^{m_\lambda}). \tag{203}$$

The local dimension is $\min(k, m_\lambda)$. The differentiated series above proves containment. For surjectivity, prescribe the allowed output coefficient y_j of z^j, where $\max(m-k, 0) \le j < m$.

Set $b_{j+k-m} = j!y_j/(j+k)!$ and give the other coefficients $b_0, \dots, b_{k-1}$ arbitrary values, say zero. Finite Taylor division by the nonzero unit A_λ then specifies the first k Taylor coefficients of h at λ. These finitely many specifications can be achieved by one polynomial: put $M_\lambda(s) = \prod_{\nu \in F, \nu \neq \lambda} (s-\nu)^k$; take the Taylor polynomial of degree $k-1$ of the prescribed local polynomial divided by M_λ, and sum the resulting M_λ multiples over F. At its own node each summand agrees with the prescribed polynomial modulo $(s-\lambda)^k$, and at the other nodes it vanishes to that order. This constructs the required $h \in \mathbb{C}[s] \subset \mathcal{H}_{\leq 1}$ and proves (203). No claim about arbitrary infinite jet interpolation has replaced this exact finite image.

Theorem F.5 (Direct differential and heat obstructions)**.** *Neither N_γ nor N_κ is invariant under D or D^2. For either frozen quotient, neither representative formula $[f']$ nor $[-f''/4]$ defines an operator on any nonempty class domain admitting all representatives. For every $t \neq 0$, neither $U_t N_\gamma \subseteq N_\gamma$ nor $U_t N_\kappa \subseteq N_\kappa$ holds. Ordinary multiplication by s preserves neither $U_t N_\gamma$ nor $U_t N_\kappa$.*

Proof. Choose any zero λ of the actual Ξ and let $m \geq 1$ be its multiplicity. The existence of its zeros is part of the classical arithmetic datum in the primary source; no realness or simplicity is used. Write $\Xi = (s-\lambda)^m A$ with $A(\lambda) \neq 0$. Then $\Xi' = (s-\lambda)^{m-1}(mA + (s-\lambda)A')$ does not vanish to order m. Since $\Xi \in N_\gamma \subseteq N_\kappa$, both derivative invariance assertions fail. If $m \geq 2$, Ξ'' has leading coefficient $m(m-1)A(\lambda)$ at order $m-2$, proving the second-derivative failure. If $m = 1$, use the actual polynomial multiple $n = (s-\lambda)\Xi$ instead: $n''(\lambda) = 2\Xi'(\lambda) \neq 0$. This retains the product-rule cross term and covers all multiplicities. The nonempty-domain assertion follows from the explicit fibre computation in (172).

For $a \in \mathbb{C}$, $e^{as}\Xi \in N_\gamma$ by polynomial approximation of e^{as} and continuous multiplication, and hence belongs to N_κ. The conjugation $D(e^{as}f) = e^{as}(D+a)f$ and the absolutely convergent heat and translation series give

$$U_t(e^{as}\Xi)(s) = e^{as - ta^2/4} Z_t(s - ta/2). \tag{204}$$

For completeness, the translation series for f is its Taylor series about s and sums to $f(s-ta/2)$; its local uniform convergence and the derivative estimates justify commutation of $e^{-(ta/2)D}$ with U_t. If $U_t N_\bullet \subseteq N_\bullet$ for either choice, evaluate (204) at λ. The exponential never vanishes, so $Z_t(\lambda - ta/2) = 0$ for every a. When $t \neq 0$ these arguments fill $\mathbb{C}$; this contradicts $Z_t \neq 0$. Finally if M preserved $U_t N_\bullet$, conjugation would imply $(M + (t/2)D)N_\bullet \subseteq N_\bullet$. Since M already preserves $N_\bullet$, this forces $DN_\bullet \subseteq N_\bullet$, contrary to the first part. □

The analytic moving-divisor quotient, which does carry the zeros of the actual Z_t under ordinary s, is

$$I_t = \overline{\mathbb{C}[s]Z_t}^{\,\kappa} = \{f \in \mathcal{H}_{\leq 1} : f/Z_t \text{ is entire}\}, \qquad A_t = (\mathcal{H}_{\leq 1}, \kappa)/I_t. \tag{205}$$

The division lemma and polynomial approximation further prove

$$I_t = Z_t \mathcal{H}_{\leq 1} = \overline{\mathbb{C}[s]Z_t}^{\,\gamma} \quad (t \in \mathbb{C}).$$

For a complex t with $Z_t(0) = 0$, use the retained nonzero center c in the division lemma; Z_t is not identically zero. The analytically transported relation has the different exact principal presentation for the proved transported multiplication:

$$U_t(\Xi \mathcal{H}_{\leq 1}) = Z_t \star_t \mathcal{H}_{\leq 1}. \tag{206}$$

Indeed $U_t(\Xi h) = Z_t \star_t U_t h$, and U_t is onto. The map $h \mapsto Z_t \star_t h$ is a continuous linear isomorphism of $(\mathcal{H}_{\leq 1}, \gamma)$ onto this closed relation; its inverse on that relation is $n \mapsto U_t((U_{-t}n)/\Xi)$, continuous by the division and heat lemmas. Thus the two multiplications and their exact

relation spaces are connected by a proved isomorphism with a proved inverse. The proof of Theorem F.3 applies verbatim with the nonzero entire order-at-most-one function Z_t: it uses only local Taylor division and compact polynomial approximation. Consequently $\mathrm{Spec}(M \text{ on } A_t) = \{\lambda : Z_t(\lambda) = 0\}$, with the full local multiplicity jets, and with the explicit inverse $[f] \mapsto [(f - f(\lambda)Z_t/Z_t(\lambda))/(s-\lambda)]$ off that set. Equation (171) is the evolution of this original-coordinate spectrum on its simple-zero locus. This spectrum can move; (167) is the spectrum of the different, exactly specified transported operator B_t.

We now give the promised comparison morphisms with every kernel retained. Put $J = N_{\mathrm{ar}} \cap N_\kappa$ and $C = (\mathcal{H}_{\le 1}, \tau \vee \kappa)/J$, where $\tau \vee \kappa$ is the topology generated by both topologies. Since each summand is closed in its own topology, J is closed in their join. The identity maps give continuous surjections

$$Q_{\mathrm{ar}} \longleftarrow C \longrightarrow Q_\kappa, \qquad \ker(C \to Q_{\mathrm{ar}}) = N_{\mathrm{ar}}/J, \quad \ker(C \to Q_\kappa) = N_\kappa/J. \tag{207}$$

Each assertion follows by taking the same representative f and testing membership in the respective relation subspace; the quotient universal property proves continuity. An algebraic common quotient is $\mathcal{H}_{\le 1}/(N_{\mathrm{ar}} + N_\kappa)$ with the quotient topology from the join; its Hausdorff version uses $\overline{N_{\mathrm{ar}} + N_\kappa}^{\tau \vee \kappa}$. The map from C to this further quotient is continuous. Continuity of maps to it from each of the two separately topologized quotients is not asserted. No closedness of the sum is presumed. This describes the precise information lost in that further quotient.

At time t replace the two subspaces by N_t and I_t and the join by $\tau_t \vee \kappa$. The same construction gives continuous surjections from their intersection quotient to the genuine transported arithmetic quotient Q_t and to A_t, with kernels $N_t/(N_t \cap I_t)$ and $I_t/(N_t \cap I_t)$, respectively. The two generator systems are linked by the explicit formula $R(B_t)Z_t$ versus $R(s)Z_t$ and the defects (169)–(170). These are exact morphisms and closure formulas for the actual source space. Determining that either kernel vanishes for τ is not established by the inspected primary loci. Accordingly the source-frozen statements $DN_{\mathrm{ar}} \subseteq N_{\mathrm{ar}}$ and $D^2 N_{\mathrm{ar}} \subseteq N_{\mathrm{ar}}$ have not been decided solely from that unnamed topology. The unqualified noninvariance results above have been proved in the two explicitly specified analytic realizations.

G Arithmetic traces and their exact Mellin obstructions

G.1 The full Schwartz trace and arithmetic principal parts

Retain the source domain

$$\mathcal{S}_0^{\mathrm{ev}} = \{\psi \in \mathcal{S}(\mathbb{R}) : \psi(-x) = \psi(x),\ \psi(0) = \widehat{\psi}(0) = 0\}, \qquad \widehat{\psi}(y) = \int_{\mathbb{R}} \psi(x) e^{-2\pi i x y}\, dx.$$

Its full trace map takes values in entire functions:

$$\begin{aligned} F_\psi(s) &= \mathcal{F}_\mu \mathcal{E} \psi(s), \\ F_\psi(i(z - \tfrac{1}{2})) &= \zeta(z) \mathcal{M}\psi(z), \quad \mathcal{M}\psi(z) = \int_0^\infty \psi(x) x^z \, \frac{dx}{x}. \end{aligned} \tag{208}$$

The Mellin integral is holomorphic for $\mathrm{Re}\, z > -2$, and the factorization holds there with its removable value at $z = 1$. This statement does not impose an unproved order bound on the entire F_ψ for arbitrary Schwartz inputs.

Here is a direct proof with all analytic domains retained. For fixed $x > 0$, periodize ψ by the smooth function

$$y \longmapsto \sum_{k \in \mathbb{Z}} \psi(x(k + y)).$$

Its Fourier coefficients are $x^{-1}\widehat{\psi}(\ell/x)$: integrate on $y \in [0,1]$, substitute $v = x(k+y)$, and sum the adjacent intervals, justified by Schwartz decay. These coefficients decrease faster than every power of ℓ, so their Fourier series converges with all derivatives and at $y = 0$ gives the Poisson identity. Evenness and the two exact zero conditions yield

$$\sum_{n\geq 1} \psi(nx) = x^{-1} \sum_{n\geq 1} \widehat{\psi}(n/x), \qquad \mathcal{E}\psi(x) = \mathcal{E}\widehat{\psi}(1/x).$$

For $x \geq 1$, a Schwartz bound of arbitrarily high order gives $|\sum \psi(nx)| \leq C_A x^{-A} \sum n^{-A}$ for $A > 1$. The same estimate after termwise differentiation follows by choosing the Schwartz exponent larger than the derivative order plus one. The Poisson identity supplies the corresponding bounds at zero. Thus $\mathcal{E}\psi(e^u)$ decreases faster than $e^{-A|u|}$ for every $A > 0$ at both ends. Its Fourier integral and each derivative in s are dominated locally uniformly in s by an integrable function. This proves that F_ψ is entire.

Since ψ is even and $\psi(0) = 0$, its Taylor formula gives $\psi(x) = O(x^2)$ at zero. Together with Schwartz decay at infinity this proves the stated holomorphy of $\mathcal{M}\psi$ by dominated differentiation, including all powers of $\log x$. For $\sigma = \operatorname{Re} z > 1$,

$$\sum_{n\geq 1} \int_0^\infty |\psi(nx)| x^\sigma \, \frac{dx}{x} = \Big(\sum_{n\geq 1} n^{-\sigma}\Big) \int_0^\infty |\psi(y)| y^\sigma \, \frac{dy}{y} < \infty.$$

Absolute interchange and the original exponent x^{-is} now give (208) on this half-plane. Moreover $\mathcal{M}\psi(1) = \int_0^\infty \psi(x)\, dx = \widehat{\psi}(0)/2 = 0$. This cancels the simple pole of ζ at 1, so the right side is holomorphic for $\operatorname{Re} z > -2$. The identity theorem extends the equality to that entire half-plane, proving the assertion.

Let $\mathcal{Z}$ denote the set of distinct nontrivial zeros ρ of ζ, with their original multiplicities m_ρ. The evenness of Ξ, immediate from the retained cosine integral, gives the exact reflected-coordinate identity

$$s = i(z - \tfrac{1}{2}), \qquad z = \tfrac{1}{2} - is, \qquad \Xi(i(z-\tfrac{1}{2})) = \xi(1-z) = \xi(z). \tag{209}$$

At each $\rho \in \mathcal{Z}$ the completion factor $z(z-1)\pi^{-z/2}\Gamma(z/2)/2$ is a nonvanishing holomorphic unit. Hence $\lambda_\rho = i(\rho - \frac{1}{2})$ is a zero of Ξ of exactly multiplicity m_ρ, and these are all its zeros. No reflection of the retained spectral operator s has been performed: (209) is the exact Mellin parameter dictionary for that same operator. The factorization (208) proves

$$\begin{gathered} D^j F_\psi(\lambda_\rho) = 0 \quad (\rho \in \mathcal{Z},\ 0 \leq j < m_\rho), \\ F_\psi \in \mathcal{H}_{\leq 1} \implies F_\psi \in \Xi\mathcal{H}_{\leq 1}. \end{gathered} \tag{210}$$

The first assertion follows because $\mathcal{M}\psi$ is holomorphic at every ρ and $s - \lambda_\rho = i(z-\rho)$; the second then follows from the proved division theorem. It is an exact intersection assertion about the full Schwartz trace, without enlarging the order-at-most-one domain or closing it in τ.

For each ρ retain the actual local unit $a_\rho(z) = \zeta(z)/(z-\rho)^{m_\rho}$ and the space

$$\mathcal{P}_\rho^- = \left\{ \sum_{j=1}^{m_\rho} c_{-j}(z-\rho)^{-j} : c_{-j} \in \mathbb{C} \right\}.$$

The principal-part map on the original entire-function set is

$$\mathfrak{P} : \mathcal{H}_{\leq 1} \longrightarrow \prod_{\rho\in\mathcal{Z}} \mathcal{P}_\rho^-, \qquad F \longmapsto \left(\operatorname{PP}_{z=\rho} \frac{F(i(z-\frac{1}{2}))}{\zeta(z)} \right)_\rho. \tag{211}$$

The kernel of this map is exactly $\Xi\mathcal{H}_{\le 1}$. It induces an injective linear map from the underlying vector space of each Q_γ, Q_κ onto the same specified image. Every finite projection is surjective. The map from Q_κ to its image, with the product subspace topology, is a topological linear isomorphism; the product of all principal-part spaces is its completion. From Q_γ it is a continuous bijection onto that image whose inverse is not continuous.

To prove each claim while retaining the analytic units, write $a_\rho(\rho+w)^{-1} = \sum_{\ell\ge 0}\alpha_{\rho,\ell}w^\ell$. The exact Laurent coefficient at degree $-m_\rho+q$, $0\le q<m_\rho$, of the ρ component is

$$c_{-m_\rho+q} = \sum_{j=0}^{q} \frac{i^j}{j!}F^{(j)}(\lambda_\rho)\alpha_{\rho,q-j}. \tag{212}$$

This follows by multiplying the two convergent local Taylor series and the retained factor w^{-m_ρ}. Its finite matrix is triangular with nonzero diagonal entries $i^j/(j!a_\rho(\rho))$. Explicitly, given a prescribed principal part $P_\rho(w)$, take the coefficients b_j through degree $m_\rho-1$ of $a_\rho(\rho+w)w^{m_\rho}P_\rho(w)$; the required original derivatives are $j!i^{-j}b_j$. Multiplying back by a_ρ^{-1} recovers the prescribed negative powers, since the discarded error is divisible by w^{m_ρ}. This proves a finite-dimensional linear isomorphism at every zero, including its inverse and every factorial and power of i.

The vanishing of all principal parts is therefore equivalent to the vanishing of all the original multiplicity jets of F. Theorem F.2 identifies that kernel exactly with $\Xi\mathcal{H}_{\le 1}$. At any finite set of zeros, the required derivatives can be realized by the explicit Hermite lift (192); this proves surjectivity of every finite principal-part projection. Each block map and its inverse are continuous because their finite matrices have fixed complex coefficients. Their product is therefore a homeomorphism of the two product spaces: the inverse image of a cylinder fixing finitely many coordinates is again controlled by those finitely many blocks. Applying this product isomorphism to Theorem F.4 proves all stated topology, completion, and comparison assertions for $\mathfrak{P}$.

This map also carries the original spectral action exactly. For $P_\rho = \mathfrak{P}_\rho(F)$ one has

$$\mathfrak{P}_\rho(sF) = \mathrm{PP}_{w=0}((\lambda_\rho+iw)P_\rho(w)). \tag{213}$$

Indeed the full Laurent quotient for sF is the quotient for F multiplied by $i(z-\frac{1}{2}) = \lambda_\rho + iw$. Its holomorphic residual contributes no negative power after this multiplication. Thus the finite block is the retained λ_ρ plus the explicitly specified nilpotent operator $P\mapsto \mathrm{PP}(iwP)$, whose nilpotency index is m_ρ: iteration on w^{-m_ρ} gives a nonzero multiple of w^{-1} at step $m_\rho-1$ and zero at step m_ρ.

For $I=\Xi\mathcal{H}_{\le 1}$, the derivative ambiguity $W_k=(D^kI+I)/I$ has, under $\mathfrak{P}$, the exact finite image

$$\prod_{\rho\in F}\left\{\sum_{j=1}^{\min(k,m_\rho)} c_{-j}(z-\rho)^{-j} : c_{-j}\in\mathbb{C}\right\} \quad (F\subset\mathcal{Z}\text{ finite}). \tag{214}$$

The finite image in (203) consists exactly of numerator jets divisible by $w^{\max(m_\rho-k,0)}$ after the invertible substitution $s-\lambda_\rho = iw$. Multiplication by the nonvanishing unit a_ρ^{-1} preserves this divisibility and is invertible on the truncated Taylor ring. Division by w^{m_ρ} therefore yields exactly all negative powers of order at most $\min(k,m_\rho)$, proving both containment and surjectivity in the displayed finite image. In particular the complete local-unit expression, rather than only a count of vanishing orders, transports the derivative defect to the arithmetic Mellin poles.

G.2 The exact Schwartz–Mellin image

Put $\theta = x\partial_x$. For real $a \le b$, retain the finite polynomial

$$P_{a,b}(z) = \prod_{\substack{k\ge 1\\ a-1\le -2k\le b+1}} (z+2k),$$

with empty product equal to 1. Define $\mathfrak{M}_0^{\mathrm{ev}}$ to be the vector space of meromorphic functions M on $\mathbb{C}$ whose only possible poles are simple poles at $-2k$, $k \ge 1$, which satisfy $M(1) = 0$, and for which all the seminorms

$$q_{a,b,N}(M) = \sup_{\substack{a\le\sigma\le b\\ v\in\mathbb{R}}} (1+|v|)^N \big|P_{a,b}(\sigma+iv)M(\sigma+iv)\big|, \qquad N = 0,1,2,\dots, \tag{215}$$

are finite. At a pole of M, the product in this formula has its removable value. The margin 1 in $P_{a,b}$ is retained in this definition. It ensures that all poles in a neighborhood of the closed strip have been removed. These seminorm conditions are equivalent to rapid decrease of M on every finite vertical strip for $|\operatorname{Im} z| \ge 1$: on that part of the strip each factor $|z+2k|$ is bounded above by a constant times $1+|v|$ and is at least $|v|$; on the remaining compact rectangle, $P_{a,b}M$ is holomorphic in a neighborhood and is bounded. In particular no bound uniform in the pole index k has been inserted into the definition. One can equivalently use the residue coordinates together with the explicit away-from-poles strip seminorms

$$r_{a,b,N,\delta}(M) = \sup_{\substack{a\le\operatorname{Re} z\le b\\ \operatorname{dist}(z,\{-2,-4,\dots\})\ge\delta}} (1+|\operatorname{Im} z|)^N |M(z)|, \qquad 0<\delta\le\tfrac12.$$

Here is the equivalence as a statement about topologies, not only about membership. On the domain of this last supremum every factor of $P_{a,b}$ has modulus at least δ, so $r_{a,b,N,\delta} \le \delta^{-\deg P_{a,b}} q_{a,b,N}$. Residues are controlled by q as shown explicitly below. Conversely, outside the radius-δ disks about the poles, $|P_{a,b}(z)| \le C_{a,b}(1+|\operatorname{Im} z|)^{\deg P_{a,b}}$ on the strip, so the corresponding part of q is bounded by an r seminorm with this degree added to its exponent. For a pole whose radius-δ disk meets the strip, that pole is among the factors of $P_{a,b}$. The function $P_{a,b}M$ is holomorphic throughout the disk; its boundary has distance at least δ from every pole, since distinct pole locations are separated by 2. That boundary lies in the enlarged strip $a-1 \le \operatorname{Re} z \le b+1$. The maximum-modulus principle bounds the product inside the disk by its boundary values, and $(1+|\operatorname{Im} z|)^N \le (1+\delta)^N$ there. An r seminorm on that enlarged strip therefore bounds the remaining part of q. There are finitely many relevant disks. These estimates prove equality of the two locally convex topologies; separate residue coordinates may be included without changing them.

The map

$$\mathcal{M} : \mathcal{S}_0^{\mathrm{ev}} \longrightarrow \mathfrak{M}_0^{\mathrm{ev}}, \qquad (\mathcal{M}\psi)(z) = \int_0^\infty \psi(x) x^z \,\frac{dx}{x} \quad (\operatorname{Re} z > -2), \tag{216}$$

is a linear bijection. Its inverse, for every fixed real $c > 1$, is

$$\psi(x) = \frac{1}{2\pi}\int_{\mathbb{R}} M(c+iv) x^{-c-iv}\,dv \quad (x>0), \qquad \psi(0)=0, \qquad \psi(-x)=\psi(x). \tag{217}$$

For every $k \ge 1$ its precise residue is

$$\operatorname*{Res}_{z=-2k} M(z) = \frac{\psi^{(2k)}(0)}{(2k)!}. \tag{218}$$

Both maps are continuous when the source has its ordinary Schwartz seminorms and the target has (215).

Here is a proof including the strip and reconstruction estimates. Fix a smooth function χ on $[0,\infty)$ which equals 1 on $[0,1]$ and 0 on $[2,\infty)$. Such a function is obtained, for example, by setting $\eta(y) = 0$ for $y \le 0$ and $\eta(y) = e^{-1/y}$ for $y > 0$, and taking $\chi(x) = \eta(2-x)/(\eta(2-x)+\eta(x-1))$. Its denominator is positive at every $x \ge 0$. Let $a_k = \psi^{(2k)}(0)/(2k)!$. For an integer $K \ge 1$ write

$$r_K(x) = \psi(x) - \chi(x) \sum_{k=1}^{K} a_k x^{2k}.$$

Evenness and $\psi(0) = 0$ show that r_K vanishes to order at least $2K+2$ at zero. Taylor's integral remainder, applied also after differentiating, proves

$$|\theta^J r_K(x)| \le C_{J,K}(\psi) x^{2K+2} \quad (0 < x \le 1), \qquad J \ge 0,$$

where $C_{J,K}(\psi)$ is bounded by a constant times finitely many Schwartz seminorms of ψ. To see explicitly that all J are covered, expand $\theta^J = \sum_{\ell=0}^{J} S(J,\ell) x^\ell \partial_x^\ell$, where $S(0,0) = 1$ and $S(J+1,\ell) = \ell S(J,\ell) + S(J,\ell-1)$. For $\ell \le 2K+2$, Taylor's remainder gives

$$|\partial_x^\ell r_K(x)| \le \frac{x^{2K+2-\ell}}{(2K+2-\ell)!} \sup_{0 \le y \le 1} |\psi^{(2K+2)}(y)|.$$

For $\ell > 2K+2$, the differentiated polynomial vanishes on this interval and $x^\ell \le x^{2K+2}$, which gives the claimed estimate using $\sup_{[0,1]} |\psi^{(\ell)}|$. At infinity $r_K = \psi$, so each $\theta^J r_K$ decreases faster than every inverse power.

The cutoff Mellin function

$$C(w) = \int_0^\infty \chi(x) x^w \, \frac{dx}{x} \quad (\operatorname{Re} w > 0)$$

has the meromorphic continuation

$$C(w) = -\frac{L(w)}{w}, \qquad L(w) = \int_1^2 \chi'(x) x^w \, dx. \tag{219}$$

The entire function L satisfies $L(0) = \chi(2) - \chi(1) = -1$, so the residue of C at 0 is 1. Repeated integration by parts with θ in the expression $L(w) = \int_1^2 (x\chi'(x)) x^w \, dx/x$ gives, on every finite vertical strip, rapid decrease for $|\operatorname{Im} w| \ge 1$. Indeed $x\chi'(x)$ is smooth with support in $[1,2]$ and all boundary derivatives vanish, and for every integer $J \ge 0$

$$w^J L(w) = (-1)^J \int_1^2 \theta^J (x\chi'(x)) x^w \, \frac{dx}{x}.$$

Thus C has the same rapid decrease away from its sole pole.

For any strip $a \le \operatorname{Re} z \le b$, choose K with $a + 2K + 2 > 0$. On the larger half-plane $\operatorname{Re} z > -2K-2$, the formula

$$M(z) = \int_0^\infty r_K(x) x^z \, \frac{dx}{x} + \sum_{k=1}^{K} a_k C(z+2k) \tag{220}$$

is meromorphic and agrees with the original Mellin integral where $\operatorname{Re} z > -2$. The expressions for successive K therefore agree by the identity theorem and define one meromorphic function on $\mathbb{C}$. Formula (219) proves that its only possible poles and residues are exactly (218). For every $J \ge 0$, integration by parts gives

$$z^J \int_0^\infty r_K(x) x^z \, \frac{dx}{x} = (-1)^J \int_0^\infty \theta^J r_K(x) x^z \, \frac{dx}{x}.$$

All boundary terms vanish: at zero they are bounded by a constant times x^{a+2K+2}, and at infinity the Schwartz exponent can be chosen larger than $b+J+1$. The absolute value of the final integral, uniformly in this strip, is at most

$$B_{K,J;a,b}(\psi) = \int_0^1 |\theta^J r_K(x)| x^{a-1}\,dx + \int_1^\infty |\theta^J r_K(x)| x^{b-1}\,dx < \infty. \tag{221}$$

For $|\operatorname{Im} z| \geq 1$, divide by $|z|^J \geq |\operatorname{Im} z|^J$. Taking J arbitrarily large proves all the required strip estimates, including the polynomial $P_{a,b}$. At bounded imaginary part, multiply (220) by $P_{a,b}$; every pole meeting the strip cancels and the resulting expression is bounded on the compact rectangle. The constants in these bounds involve only finitely many Schwartz seminorms: the first integral in (221) is at most $C_{J,K}(\psi)/(a+2K+2)$; on $[1,2]$ the cutoff has fixed bounded derivatives; on $[2,\infty)$ the expansion of θ^J above uses finitely many bounds $(1+x)^A|\psi^{(\ell)}(x)|$ with $A > b+J+1$. This proves continuity of the forward map. Finally $M(1) = \int_0^\infty \psi(x)\,dx = \widehat{\psi}(0)/2 = 0$, retaining the factor 2 from evenness.

Conversely, let $M \in \mathfrak{M}_0^{\mathrm{ev}}$ and define (217). The integral and its ordinary x derivatives converge uniformly on each compact subinterval of $(0,\infty)$. More precisely, with the rising factorial $(z)_0 = 1$ and $(z)_j = z(z+1)\cdots(z+j-1)$ for $j \geq 1$, they are

$$\psi^{(j)}(x) = \frac{(-1)^j}{2\pi} \int_{\mathbb{R}} (c+iv)_j M(c+iv) x^{-c-iv-j}\,dv. \tag{222}$$

The factor $(c+iv)_j$ grows at most polynomially in v, so a strip estimate with exponent greater than $j+1$ supplies an integrable dominating function. This proves smoothness for $x > 0$.

The inverse integral is independent of the positive choice of c. In fact integrate $M(z)x^{-z}$ around the rectangle between two positive vertical lines and heights $\pm T$. It has no poles there. On each horizontal side, its absolute value is at most a constant, depending on the fixed x and the bounded real interval, times $(1+T)^{-N}$ for arbitrary N. Thus the horizontal contributions tend to zero, and the two upward vertical integrals agree. In particular for every real $b > 0$ the differentiated formula holds with c replaced by b. If

$$I_{d,j}(M) = \frac{1}{2\pi} \int_{\mathbb{R}} |(d+iv)_j M(d+iv)|\,dv \quad \big(d \notin \{-2,-4,-6,\ldots\}\big), \tag{223}$$

then $I_{d,j}(M)$ is finite and

$$|\psi^{(j)}(x)| \leq I_{b,j}(M) x^{-b-j} \quad (x > 0,\ b > 0). \tag{224}$$

This gives every required decay estimate at infinity.

For the behavior at zero fix $K \geq 1$ and take the actual left vertical line $d = -2K-1$. The rectangle between d and c has precisely the possible poles $-2, -4, \ldots, -2K$. Traverse its right edge upward and its left edge downward; this is the positive orientation. The residue theorem and the same horizontal estimates give

$$\begin{aligned} \psi(x) &= \sum_{k=1}^{K} a_k x^{2k} + R_K(x), & a_k &= \operatorname*{Res}_{z=-2k} M(z), \\ R_K(x) &= \frac{1}{2\pi} \int_{\mathbb{R}} M(-2K-1+iv) x^{2K+1-iv}\,dv, \\ |R_K^{(j)}(x)| &\leq I_{-2K-1,j}(M) x^{2K+1-j}. \end{aligned} \tag{225}$$

The sign of each residue is positive. The last estimate follows by differentiating the remainder integral with the exact factor $(-1)^j(-2K-1+iv)_j$; this factor is included in (223). For any given nonnegative integer j, choose K so that $2K+1 > j$. Formula (225) then proves that all

derivatives through order j have limits at zero given by the displayed even polynomial. These limits define a C^j extension: inductively, if $\psi^{(\ell)}$ and $\psi^{(\ell+1)}$ have these limits, integrate $\psi^{(\ell+1)}$ from a positive ε to x and let $\varepsilon \downarrow 0$; the resulting integral identity proves differentiability at zero with the asserted derivative. Because j was arbitrary, the extension is smooth, has value zero, all its odd derivatives vanish at zero, and its $2k$th derivative is $(2k)!a_k$. Reflecting it evenly across zero preserves smoothness: on the negative half-line its jth derivative is $(-1)^j\psi^{(j)}(-x)$, and the one-sided values coincide because the odd jets vanish. Together with (224), this proves $\psi \in \mathcal{S}(\mathbb{R})$ and the residue assertion.

For completeness the inversion just used also proves that the Mellin transform of this reconstructed function is the prescribed M. On a line $c > 1$ put $h(u) = e^{cu}\psi(e^u)$ and $m(v) = M(c+iv)$. The formula says $h(u) = (2\pi)^{-1}\int m(v)e^{-ivu}\,dv$. Here m is continuous and integrable by the strip estimates, and h is integrable by the bounds at both ends just proved. For $\varepsilon > 0$, absolute Fubini and the Gaussian integral give

$$\int_{\mathbb{R}} h(u)e^{ivu}e^{-\varepsilon u^2/2}\,du = \int_{\mathbb{R}} m(w)\frac{e^{-(v-w)^2/(2\varepsilon)}}{\sqrt{2\pi\varepsilon}}\,dw.$$

The right side tends to $m(v)$: split the integral at $|v-w| = \delta$, use continuity on the inner interval, and use the Gaussian tail and boundedness of m on its complement. The left side tends by dominated convergence to $\int h(u)e^{ivu}\,du$. Substitution $x = e^u$ proves $\mathcal{M}\psi(c+iv) = M(c+iv)$. Both sides are holomorphic on $\operatorname{Re} z > -2$, so the identity theorem proves their equality there and hence equality of their meromorphic continuations. In particular $\widehat{\psi}(0) = 2\mathcal{M}\psi(1) = 2M(1) = 0$. Reconstruction is injective because its output has Mellin transform equal to its input. The forward map is injective as well, with the following direct uniqueness proof. If the Mellin transform of $\delta \in \mathcal{S}_0^{\mathrm{ev}}$ vanishes, the continuous integrable function $g(u) = e^{cu}\delta(e^u)$ has Fourier transform $\int g(u)e^{ivu}\,du = 0$. Put $G_\varepsilon(u) = (2\pi\varepsilon)^{-1/2}e^{-u^2/(2\varepsilon)}$. Absolute Fubini applied to the Gaussian integral gives

$$(g * G_\varepsilon)(u) = \frac{1}{2\pi}\int_{\mathbb{R}}\left(\int_{\mathbb{R}} g(w)e^{ivw}\,dw\right)e^{-\varepsilon v^2/2}e^{-ivu}\,dv = 0.$$

As $\varepsilon \downarrow 0$, the Gaussian convolution tends to $g(u)$ at every u by continuity, the unit total mass of G_ε, and its vanishing tail mass. Thus $g = 0$, $\delta = 0$ on the positive half-line, and evenness gives $\delta = 0$ everywhere. This proves that the two displayed maps are mutual inverses, not only that reconstructed inputs have the prescribed transform.

The inverse continuity can be read directly from the estimates. For nonnegative integers A, j, choose $b = A+1$ and $K \geq 1$ with $2K+1 > j$. On $|x| \geq 1$ use (224); on $|x| \leq 1$ use (225). They give

$$\sup_{x\in\mathbb{R}}(1+|x|)^A|\psi^{(j)}(x)| \leq 2^A\left(I_{b,j}(M) + I_{-2K-1,j}(M) + \sum_{\substack{1\leq k\leq K\\ 2k\geq j}}\frac{(2k)!}{(2k-j)!}|a_k|\right). \tag{226}$$

Each line integral on the right is bounded by a constant times a strip seminorm with exponent greater than $j+1$ on a strip containing its line and no pole on the line. Every one of the finitely many residues is controlled by a strip seminorm too: if $-2k$ belongs to $[a,b]$, then

$$|a_k| = \frac{|(P_{a,b}M)(-2k)|}{\left|\displaystyle\prod_{\substack{\ell\geq 1,\ \ell\neq k\\ a-1\leq -2\ell\leq b+1}}(2\ell-2k)\right|}.$$

Its denominator is a finite product of nonzero numbers. This proves continuity without a uniform assumption on the entire residue sequence and completes the characterization.

Apply this result to the exact arithmetic trace map already proved in (208). For $F \in \mathcal{H}_{\le 1}$ define the meromorphic function

$$M_F(z) = \frac{F(i(z-\frac{1}{2}))}{\zeta(z)}.$$

Its definition uses meromorphic division, including the removable values determined by any canceled zeros or pole. Then the precise intersection of the full source trace image with its original order-at-most-one entire-function set is

$$\begin{aligned} \mathcal{A}_{\mathrm{Sch}} &:= \{F_\psi : \psi \in \mathcal{S}_0^{\mathrm{ev}}\} \cap \mathcal{H}_{\le 1} \\ &= \{F \in \mathcal{H}_{\le 1} : M_F \in \mathfrak{M}_0^{\mathrm{ev}}\} \\ &= \left\{\Xi h : h \in \mathcal{H}_{\le 1}, \quad A(z)h(i(z-\tfrac{1}{2})) \in \mathfrak{M}_0^{\mathrm{ev}}\right\}, \\ A(z) &= \tfrac{1}{2}z(z-1)\pi^{-z/2}\Gamma(z/2). \end{aligned} \tag{227}$$

For the first equality after the definition, a trace has the factorization and Mellin properties already proved. Conversely, if $M_F \in \mathfrak{M}_0^{\mathrm{ev}}$, reconstruct the unique $\psi \in \mathcal{S}_0^{\mathrm{ev}}$. In $\operatorname{Re} z > 1$ the trace factorization gives $F_\psi(i(z-\frac{1}{2})) = \zeta(z)M_F(z) = F(i(z-\frac{1}{2}))$. Both F_ψ and F are entire; their equality on this open set proves equality everywhere. This argument retains the original s coordinate and does not assert that arbitrary Schwartz traces belong to $\mathcal{H}_{\le 1}$.

For the last equality in (227), the previous nontrivial-zero jet calculation and the proved entire division theorem give $F = \Xi h$ with $h \in \mathcal{H}_{\le 1}$ for every trace in this intersection. Conversely the retained identity $\Xi(i(z-\frac{1}{2})) = \xi(1-z) = \xi(z)$ gives exactly

$$M_{\Xi h}(z) = \tfrac{1}{2}z(z-1)\pi^{-z/2}\Gamma(z/2)h(i(z-\tfrac{1}{2})),$$

including its meromorphic continuation and all its local units. The factor A has only simple poles at $-2k$, $k \ge 1$: the gamma poles at $z = 0$ and the factor z cancel, the remaining gamma poles occur at the displayed negative even integers, and the factor $z-1$ gives $A(1) = 0$. Thus for an arbitrary $h \in \mathcal{H}_{\le 1}$ the pole locations, orders and the equality $M_{\Xi h}(1) = 0$ are automatic; the exact additional requirement is the family of strip bounds in (215), with the actual factor A retained. No topology on the unnamed source closure is identified by this characterization.

This criterion also retains every Taylor coefficient of its arithmetic preimage. The trivial zero of ζ at $-2k$ is simple, as follows directly from the retained completion. The gamma recurrence gives a simple pole of $\Gamma(z/2)$ there with nonzero residue $2(-1)^k/k!$, while the other factors in $A(z)$ are nonzero there. On the other hand $\xi(-2k) = \xi(1+2k) > 0$: all factors in the completion at $1+2k$ are positive, including the convergent positive Dirichlet series for $\zeta(1+2k)$. Thus $\xi = A\zeta$ forces a zero of exactly order one at $-2k$. Meromorphic division now gives

$$\begin{aligned} \frac{\psi^{(2k)}(0)}{(2k)!} &= \frac{F(i(-2k-\frac{1}{2}))}{\zeta'(-2k)} \\ &= \frac{2k(2k+1)(-1)^k\pi^k}{k!}h(-i(2k+\tfrac{1}{2})) \qquad (F = \Xi h \in \mathcal{A}_{\mathrm{Sch}}). \end{aligned} \tag{228}$$

In the first line there is no extra factor i: the residue is taken in the actual coordinate z and the denominator has leading term $\zeta'(-2k)(z+2k)$. For the second line the gamma recurrence gives $\operatorname{Res}_{z=-2k}\Gamma(z/2) = 2(-1)^k/k!$. Indeed $\Gamma(w) = \Gamma(w+k+1)/[w(w+1)\cdots(w+k)]$, whose numerator is 1 at $w = -k$ and whose denominator has derivative $(-1)^k k!$ there; substitution $w = z/2$ supplies the factor 2. Multiplying this exact residue by $(-2k)(-2k-1)\pi^k/2$ and the

value of the entire h gives the asserted coefficient. Thus the inclusion criterion also reconstructs the full even jet at the origin.

The inclusion $\mathcal{A}_{\mathrm{Sch}} \hookrightarrow \Xi\mathcal{H}_{\le 1}$ is the actual identity on entire functions, with exact image specified by (227). It contains $\mathbb{C}[s]\Xi$ because the retained finite Hermite trace families span precisely that polynomial image. Consequently its closures in each of the explicitly proved analytic topologies satisfy

$$\overline{\mathcal{A}_{\mathrm{Sch}}}^{\gamma} = \overline{\mathcal{A}_{\mathrm{Sch}}}^{\kappa} = \Xi\mathcal{H}_{\le 1}. \tag{229}$$

Indeed the left image lies between $\mathbb{C}[s]\Xi$ and the closed subspace $\Xi\mathcal{H}_{\le 1}$, and the closures of the smaller polynomial subspace were proved to equal that subspace. This constructs the exact inclusion, its image criterion and both analytic closures; it inserts no equality involving the unnamed topology τ.

G.3 An exact exponential orbit and its Schwartz boundary

For each unchanged complex parameter $a = u + iv$ define

$$F_a(s) = \Xi(s)e^{as}, \qquad C(z) = \tfrac{1}{2}z(z-1)\pi^{-z/2}\Gamma(z/2).$$

Every F_a belongs to $\Xi\mathcal{H}_{\le 1} = N_\gamma = N_\kappa$. Its membership in the actual unclosed Schwartz trace image has the exact boundary

$$F_a \in \mathcal{F}_\mu\mathcal{E}(\mathcal{S}_0^{\mathrm{ev}}) \quad \Longleftrightarrow \quad |\operatorname{Re} a| < \pi/4. \tag{230}$$

On this open strip its unique original Schwartz input is

$$\psi_a(x) = e^{-ia/2}\big(4\pi^2 e^{-4ia}x^4 - 6\pi e^{-2ia}x^2\big)e^{-\pi e^{-2ia}x^2} \quad (x \in \mathbb{R}). \tag{231}$$

All coefficients and the factor $e^{-ia/2}$ are retained. We prove both directions, including the two boundary lines.

Write $\Theta = x\partial_x$ and $G(x) = e^{-\pi x^2}$. Direct differentiation gives

$$\psi_0 = \Theta(\Theta+1)G = (4\pi^2x^4 - 6\pi x^2)e^{-\pi x^2}.$$

For $\operatorname{Re} z > 0$, substitution $y = \pi x^2$ in the Euler gamma integral gives $\mathcal{M}G(z) = \frac{1}{2}\pi^{-z/2}\Gamma(z/2)$. Integration by parts gives $\mathcal{M}(\Theta f)(z) = -z\mathcal{M}f(z)$ when the boundary terms vanish. They do for G and the needed derivatives on this half-plane. Consequently $\mathcal{M}\psi_0 = C(z)$, with the exact multiplier $(-z)(1-z) = z(z-1)$.

Suppose $|u| < \pi/4$ and put $\beta = e^{-2ia}$. Then $\operatorname{Re}\beta = e^{2v}\cos(2u) > 0$, and its logarithm in the right half-plane is exactly $-2ia$. The Gaussian integral with coefficient β satisfies

$$\int_0^\infty e^{-\pi\beta x^2}x^z\,\frac{dx}{x} = \tfrac{1}{2}\pi^{-z/2}e^{iaz}\Gamma(z/2) \quad (\operatorname{Re} z > 0).$$

Indeed both sides are holomorphic in the connected right half-plane of β, locally dominated there by a positive Gaussian; they agree for positive real β by a real substitution, and the identity theorem proves equality everywhere in that half-plane. Applying $\Theta(\Theta+1)$ and retaining $e^{-ia/2}$ gives

$$\mathcal{M}\psi_a(z) = C(z)e^{ia(z-1/2)}. \tag{232}$$

The function in (231) is even, Schwartz, and zero at the origin, since its Gaussian has positive real coefficient and its polynomial starts at degree two. At $z = 1$, (232) gives $\widehat{\psi_a}(0) = 2\mathcal{M}\psi_a(1) = 0$. Thus it belongs to the exact source domain. The proved trace factorization gives

$$F_{\psi_a}(i(z-\tfrac{1}{2})) = \zeta(z)C(z)e^{ia(z-1/2)} = \Xi(i(z-\tfrac{1}{2}))e^{ai(z-1/2)} \quad (\operatorname{Re} z > 1).$$

Both sides are entire in $s = i(z - \frac{1}{2})$, so they agree everywhere. This proves sufficiency.

We next derive the exact gamma-modulus formula used for necessity. For real y, the absolutely convergent two Euler integrals, followed by $r = x_1 + x_2$, $q = x_1/(x_1 + x_2)$, give

$$\begin{aligned}|\Gamma(1+iy)|^2 &= \int_0^\infty \int_0^\infty e^{-x_1-x_2}(x_1/x_2)^{iy}\,dx_1\,dx_2 \\ &= \int_0^1 (q/(1-q))^{iy}\,dq = \int_{\mathbb{R}} e^{iyw}\frac{e^w}{(1+e^w)^2}\,dw.\end{aligned}$$

The middle equality uses $\int_0^\infty re^{-r}\,dr = 1$, and the last substitution is $w = \log(q/(1-q))$. For $y > 0$, integrate $f_y(w) = e^{iyw}e^w/(1+e^w)^2$ on the positively oriented rectangle with vertices $-R, R, R+2\pi i, -R+2\pi i$. The integrals on the vertical sides tend to zero, because the rational exponential factor is $O(e^{-R})$ there, uniformly in the height. The top horizontal integral is $-e^{-2\pi y}$ times the bottom integral. The only pole in the strip is at $w = \pi i$. Writing $w = \pi i + \eta$ gives

$$\frac{e^w}{(1+e^w)^2} = -\frac{1}{4\sinh^2(\eta/2)} = -\eta^{-2} + O(1),$$

so the residue of f_y is $-iye^{-\pi y}$. The residue theorem therefore gives $(1 - e^{-2\pi y})\int_{\mathbb{R}} f_y(w)\,dw = 2\pi y e^{-\pi y}$. By conjugation for negative y and continuity at zero, we have proved for every real y

$$|\Gamma(1+iy)|^2 = \frac{\pi y}{\sinh(\pi y)}, \tag{233}$$

with value 1 at $y = 0$. This proof uses no asymptotic remainder.

If a Schwartz input had trace F_a, its Mellin transform on $\operatorname{Re} z > 1$ would have to be the quotient $C(z)e^{ia(z-1/2)}$, by the factorization and the nonvanishing of ζ on that half-plane. On the fixed line $z = 2 - 2iy$, its exact squared modulus is

$$\left|C(2-2iy)e^{ia(3/2-2iy)}\right|^2 = e^{-3v+4uy}\,\frac{(1+y^2)(1+4y^2)y}{\pi\sinh(\pi y)}. \tag{234}$$

To check every factor, $|z|^2 = 4(1+y^2)$, $|z-1|^2 = 1+4y^2$, $|\pi^{-z/2}|^2 = \pi^{-2}$, and (233) supplies the remaining gamma factor. The real part of $ia(3/2 - 2iy)$ is $2uy - 3v/2$. At $y = 0$ the right side has its removable value e^{-3v}/π^2.

For $y > 0$, $\sinh(\pi y) < e^{\pi y}/2$. If $u \geq \pi/4$, the right side of (234) is therefore greater than

$$\frac{2e^{-3v}}{\pi}(1+y^2)(1+4y^2)ye^{(4u-\pi)y},$$

which tends to infinity, including $u = \pi/4$. If $u \leq -\pi/4$, put $y = -Y$ and let $Y \to +\infty$; the same inequality with $-u$ proves divergence. But for every Schwartz input the defining integral gives the uniform bound

$$|\mathcal{M}\psi(2-2iy)| \leq \int_0^\infty |\psi(x)|x\,dx < \infty \qquad (y \in \mathbb{R}).$$

This contradiction proves necessity and the exact open boundary.

Uniqueness of the input is also exact. If two source inputs have the same trace, the Mellin transform of their difference vanishes on $\operatorname{Re} z = 2$, by the same factorization. Under $x = e^w$ this is the Fourier transform, with the retained sign change of frequency, of the integrable continuous function $e^{2w}(\psi_1 - \psi_2)(e^w)$. Fourier inversion gives zero on the positive half-line, and evenness gives zero on the negative half-line and at zero. Thus (231) is the unique original input.

On the strip $|\operatorname{Re} a| < \pi/4$ this is a holomorphic family in the Schwartz topology. Indeed on a compact subset of the strip, $\operatorname{Re}(e^{-2ia})$ has a positive lower bound. Every fixed derivative

in a and in x is a polynomial in x with locally bounded coefficients times this same Gaussian. It is bounded in each Schwartz seminorm by an integrable Gaussian majorant; the Cauchy integral formula in a, with the same seminorm bounds, proves complex differentiability in that topology. Direct differentiation, with no change of the input coordinate, gives the exact source-generator equation

$$\partial_a \psi_a = -i(\Theta + \tfrac{1}{2})\psi_a, \qquad \mathcal{F}_\mu \mathcal{E}\big(-i(\Theta + \tfrac{1}{2})\psi_a\big) = sF_a(s). \tag{235}$$

The second equality also follows from $\mathcal{M}(-i(\Theta+\frac{1}{2})\psi) = i(z-\frac{1}{2})\mathcal{M}\psi$ and the same factorization. This operator is defined on every $\mathcal{S}_0^{\mathrm{ev}}$ input and preserves that space. Multiplication by x and differentiation preserve Schwartz functions, and $x\psi'$ is even when ψ is even. Its value at zero is zero. Finally integration by parts on the whole real line gives $\int x\psi'(x)\,dx = -\int \psi(x)\,dx$, with vanishing boundary terms. Hence $\int -i(\Theta + \frac{1}{2})\psi\,dx = (i/2)\int \psi\,dx = 0$. This supplies the exact morphism from the original scaling generator to multiplication by the retained spectral coordinate. In particular the inclusion of the actual Schwartz trace intersection into $\Xi\mathcal{H}_{\le 1}$ is strict: any real $a = \pi/4$ already gives an element of the latter with no Schwartz preimage. Neither this explicit separating element nor this exact generator map identifies the source closure in the unnamed topology τ.

G.4 An exact arithmetic inverse for every heated Hermite trace

Let ψ be any finite linear combination of the original $\psi_\ell^{\pm}$, retaining all its coefficients, and write $F_0 = \mathcal{F}_\mu \mathcal{E}\psi = p\Xi$. For $x > 0$ and $t \in \mathbb{C}$ define

$$\begin{aligned} q(x) &= \sum_{n\ge 1} \psi(nx), & g_t(x) &= e^{t(\log x)^2/4}\mathcal{E}\psi(x), \\ q_t(x) &= x^{-1/2} g_t(x) = e^{t(\log x)^2/4} q(x), & \psi_t(x) &= \sum_{n\ge 1} \mu(n) q_t(nx). \end{aligned} \tag{236}$$

Here $\mu(1) = 1$, $\mu(n) = 0$ when a prime square divides n, and $\mu(n) = (-1)^j$ for a product of j distinct primes. For the inverse ψ_t, the symbol $\mathcal{E}\psi_t$ means the absolutely convergent trace-sum extension $x^{1/2}\sum_{m\ge 1}\psi_t(mx)$ on the positive half-line. Its domain here consists of smooth functions with rapid decrease at infinity; no Schwartz extension through zero is required. The transform and inverse assertions, with their domains, are

$$\mathcal{E}\psi_t = g_t, \qquad \mathcal{F}_\mu g_t = U_t(p\Xi) =: F_t, \qquad \int_0^\infty \psi_t(x) x^z \,\frac{dx}{x} = \frac{F_t(i(z-\frac{1}{2}))}{\zeta(z)} \quad (\operatorname{Re} z > 1). \tag{237}$$

The series for ψ_t converges locally with all derivatives on $(0,\infty)$, and ψ_t and all its derivatives decrease faster than every inverse power at infinity. No behavior at zero has been assumed for this inverse.

We prove all analytic assertions in these formulas. A finite Hermite combination and its Fourier transform are polynomials times the original Gaussian $e^{-\pi x^2}$. Thus at infinity, for every derivative order j, their derivatives are bounded by $C_j(1+x)^{d_j}e^{-\pi x^2}$. For $x \ge 1$, termwise derivatives of $\sum \psi(nx)$ are bounded by a sum of $C_j n^j (1+nx)^{d_j} e^{-\pi n^2 x^2}$; splitting the Gaussian into two equal exponential factors bounds that sum by $C_j'(1+x)^{d_j}e^{-\pi x^2/2}$. This also justifies the differentiation. Both zero conditions give, by the Poisson identity,

$$\mathcal{E}\psi(x) = \mathcal{E}\widehat{\psi}(1/x).$$

Consequently, in $u = \log x$, the functions $\mathcal{E}\psi(e^u)$ and all their u derivatives have bounds of the form $C\exp(C'|u| - ce^{2|u|})$ on the two ends. Multiplication by $e^{tu^2/4}$, and any finite number of its derivatives, preserves integrability against $e^{A|u|}$ for every $A > 0$, locally uniformly for t in

compact subsets of $\mathbb{C}$. It follows that $\mathcal{F}_\mu g_t$ is entire in both t, s. Since $D_s^{2k} e^{-isu} = (-1)^k u^{2k} e^{-isu}$, absolute termwise integration of the time exponential gives exactly

$$\int_{\mathbb{R}} e^{tu^2/4} \mathcal{E}\psi(e^u) e^{-isu}\, du = \sum_{k\geq 0} \frac{(-t/4)^k}{k!} D_s^{2k}(p(s)\Xi(s)) = F_t(s).$$

The heat lemma identifies the same series in γ; hence $F_t \in \mathcal{H}_{\leq 1}$ with the original coordinate and coefficient.

The proved estimates imply that q_t and all its x derivatives decrease faster than every power at infinity, and that q_t decreases faster than every power at zero. On a compact interval $x \in [\delta, L] \subset (0, \infty)$, choose $A > j + 1$ to bound the jth derivative of the nth inverse summand by $Cn^j (nx)^{-A} \leq C\delta^{-A} n^{j-A}$. The finitely many arguments below one are harmless. This proves local convergence with derivatives. On $x \geq 1$ the same bound, with arbitrarily large A, proves rapid decrease of each derivative. For a fixed $x > 0$ the double sum used below is absolutely bounded by

$$\sum_{m,n\geq 1} |q_t(mnx)| = \sum_{k\geq 1} d(k) |q_t(kx)| < \infty,$$

where $d(k) \leq k$ and rapid decrease permits an exponent greater than two. Unique prime factorization gives $\sum_{n|k} \mu(n) = 1$ for $k = 1$ and $\prod_{p|k}(1 - 1) = 0$ for $k > 1$. Therefore

$$\sum_{m\geq 1} \psi_t(mx) = \sum_{k\geq 1} \Big(\sum_{n|k} \mu(n)\Big) q_t(kx) = q_t(x).$$

Multiplication by the retained factor $x^{1/2}$ proves the first identity of (237). It also proves uniqueness of this inverse among functions decreasing faster than every power at infinity: for any such v with $\sum_m v(mx) = q_t(x)$, the same absolutely convergent double sum applied to $\sum_n \mu(n) \sum_m v(mnx)$ equals $v(x)$.

For $\sigma = \operatorname{Re} z > 1$, absolute integration gives

$$\sum_{n\geq 1} |\mu(n)| \int_0^\infty |q_t(nx)| x^\sigma \frac{dx}{x} \leq \Big(\sum_{n\geq 1} n^{-\sigma}\Big) \int_0^\infty |q_t(y)| y^\sigma \frac{dy}{y} < \infty.$$

The divisor identity also gives $(\sum \mu(n) n^{-z}) \zeta(z) = 1$ by absolute multiplication of the two Dirichlet series in this half-plane. Finally

$$\int_0^\infty q_t(x) x^z \frac{dx}{x} = \int_0^\infty g_t(x) x^{z-1/2} \frac{dx}{x} = F_t(i(z - \tfrac{1}{2})),$$

because $-i\, i(z - \frac{1}{2}) = z - \frac{1}{2}$. These equalities prove the last identity of (237), including its absolute domain.

At a nontrivial zero ρ of ζ of multiplicity m, put $\lambda = i(\rho - \frac{1}{2})$ and retain $\zeta(z) = (z - \rho)^m a_\rho(z)$ with $a_\rho(\rho) \neq 0$. The retained cosine integral makes Ξ even, so its definition gives

$$\Xi(i(z - \tfrac{1}{2})) = \xi(1 - z) = \xi(z).$$

At a nontrivial zero, the factor $z(z - 1)\pi^{-z/2}\Gamma(z/2)/2$ is a nonvanishing holomorphic unit. Thus this reflected coordinate sends the zero at ρ to the zero at λ with exactly the same multiplicity. The meromorphic continuation of the last quotient has pole order

$$\max\{m - \operatorname{ord}_\lambda F_t, 0\}. \tag{238}$$

For the zero seed $F_t = 0$, the quotient is zero and its pole order is zero; the displayed formula uses $\operatorname{ord}_\lambda 0 = +\infty$. Indeed $s - \lambda = i(z - \rho)$, so its numerator has exactly the original order

$\operatorname{ord}_\lambda F_t$, with its leading coefficient multiplied by the nonzero power of i. The unit a_ρ is retained in every principal coefficient.

For every fixed $t \neq 0$ and every zero λ of Ξ, there is a polynomial p for which $F_t(\lambda) \neq 0$: otherwise all $U_t(s^n\Xi)$ would vanish there, contrary to the no-common-zero proof in Section D. The exact polynomial-image calculation supplies a finite original Hermite seed for this p. For that seed the unique inverse ψ_t in (236) is unbounded near zero. To prove this last assertion, boundedness near zero together with its proved rapid decrease at infinity would make its Mellin integral holomorphic in $\operatorname{Re} z > 0$: on compact subsets, derivatives in z are dominated by powers of $|\log x| x^{\sigma-1}$ near zero with $\sigma > 0$. Its equality in $\operatorname{Re} z > 1$ would then continue as a holomorphic function through every nontrivial ρ, contradicting the pole of order m just proved at $\rho = \frac{1}{2} - i\lambda$. Thus for each nonzero heat time some actual finite Hermite trace has an explicit arithmetic inverse which fails even boundedness at the origin. This proves a concrete obstruction to Schwartz descent with a unique inverse and its poles; it does not replace the unresolved closure in τ by a different closure.

G.5 Global meromorphic arithmetic heat transport

The original coordinate dictionary permits an exact heat action on the entire arithmetic Mellin quotient, including its poles. Define

$$\begin{aligned} (\mathscr{J}F)(z) &= \frac{F(i(z-\frac{1}{2}))}{\zeta(z)}, \qquad & \mathfrak{R}_\zeta &= \mathscr{J}(\mathcal{H}_{\le 1}), \\ H_M(z) &= \zeta(z)M(z), \qquad & (\mathscr{J}^{-1}M)(s) &= \zeta(\tfrac{1}{2}-is)M(\tfrac{1}{2}-is). \end{aligned} \tag{239}$$

All divisions and products in this display are meromorphic, with their removable values retained. Thus $\mathfrak{R}_\zeta$ consists exactly of meromorphic M for which H_M extends to an entire function of order at most one. To verify the order statement in both directions, the affine coordinate maps are $z \mapsto i(z-\frac{1}{2})$ and $s \mapsto \frac{1}{2} - is$. Their moduli are bounded by $|z| + \frac{1}{2}$ and $|s| + \frac{1}{2}$, respectively. The inequality $(r + \frac{1}{2})^p \le 2^{p-1}(r^p + 2^{-p})$ for every $p > 1$ then transfers each of the defining order-class growth bounds in both directions. The displayed maps are inverse because $\frac{1}{2} - i\,i(z-\frac{1}{2}) = z$ and $i(\frac{1}{2} - is - \frac{1}{2}) = s$.

Give $\mathfrak{R}_\zeta$ the transported growth topology $\gamma_\zeta = \mathscr{J}_*\gamma$, namely the seminorms

$$\|M\|_{\zeta;p,a} = \sup_{s\in\mathbb{C}} |\zeta(\tfrac{1}{2}-is)M(\tfrac{1}{2}-is)| e^{-a|s|^p}, \qquad 1 < p < 2, \quad a > 0. \tag{240}$$

The canceled product is used at every apparent singularity. Consequently $\mathscr{J} : (\mathcal{H}_{\le 1}, \gamma) \to (\mathfrak{R}_\zeta, \gamma_\zeta)$ is a linear homeomorphism; completeness and the locally convex structure follow directly from those of $(\mathcal{H}_{\le 1}, \gamma)$. No topology has been assigned to $\mathfrak{R}_\zeta$ merely by referring to unspecified convergence of meromorphic functions.

Every $M \in \mathfrak{R}_\zeta$ is holomorphic away from zeros of ζ. At a zero of multiplicity m its pole order is at most m. At $z = 1$, where ζ has its simple pole, M is holomorphic and vanishes. These assertions follow by dividing the entire H_M by the actual local Laurent expansion of ζ. They concern all complex zeros, including the simple trivial zeros, and do not discard the local units.

Put

$$\nabla_\zeta M = M' + \frac{\zeta'}{\zeta}M = \frac{H_M'}{\zeta}, \qquad \mathcal{L}_\zeta M = \frac{1}{4}\nabla_\zeta^2 M.$$

These are globally defined meromorphic operators on $\mathfrak{R}_\zeta$; the right expression supplies every

removable value. With $D = \partial_s$, the exact conjugacies are

$$\begin{aligned} \mathscr{J} D \mathscr{J}^{-1} &= -i\nabla_\zeta, \\ \mathcal{L}_\zeta M &= \frac{1}{4\zeta}(\zeta M)'' = \frac{1}{4}\left(M'' + 2\frac{\zeta'}{\zeta}M' + \frac{\zeta''}{\zeta}M\right), \\ \mathscr{J}(-D^2/4)\mathscr{J}^{-1} &= \mathcal{L}_\zeta. \end{aligned} \tag{241}$$

Indeed, $H_M(z) = F(i(z-\frac{1}{2}))$ gives $H_M'(z) = iF'(i(z-\frac{1}{2}))$ and $H_M''(z) = -F''(i(z-\frac{1}{2}))$. Division by ζ proves the first and last equalities. Expanding $(\zeta M)'' = \zeta M'' + 2\zeta' M' + \zeta'' M$ proves every coefficient in the middle one. In particular the coefficient ζ''/ζ includes both the derivative and the square of ζ'/ζ; neither term has been omitted.

For every original complex time t, define

$$\begin{aligned} \mathcal{V}_t M &= \mathscr{J} U_t \mathscr{J}^{-1} M = \frac{H_{M,t}}{\zeta}, \\ H_{M,t}(z) &= \exp\left(\frac{t}{4}\partial_z^2\right) H_M(z) = \sum_{k=0}^{\infty} \frac{t^k}{4^k k!} H_M^{(2k)}(z). \end{aligned} \tag{242}$$

The last series is a definition by the proved entire-function heat calculus, not an operation on unrestricted meromorphic functions. More explicitly, its kth term equals $(-t/4)^k F^{(2k)}(i(z-\frac{1}{2}))/k!$. It therefore has exactly the growth-topology convergence already proved for $U_t F$, after the continuous affine change of argument. In particular the numerator series converges locally uniformly in (t, z); after division by ζ, it converges in every seminorm (240), locally uniformly in complex t. Each $\mathcal{V}_t$ is a continuous automorphism of $(\mathfrak{R}_\zeta, \gamma_\zeta)$, with

$$\mathcal{V}_t \mathcal{V}_v = \mathcal{V}_{t+v}, \qquad \mathcal{V}_t^{-1} = \mathcal{V}_{-t}, \qquad \partial_t \mathcal{V}_t M = \mathcal{L}_\zeta \mathcal{V}_t M.$$

These follow by conjugating the proved heat group and its differentiated series. Equivalently $\mathcal{V}_t = \exp(t\mathcal{L}_\zeta)$ in this precisely specified topology. Near a zero of ζ, local uniform convergence of the numerators also gives convergence of each Laurent coefficient and of the holomorphic remainders after the finite principal parts are removed.

The original multiplication coordinate is the actual function $i(z-\frac{1}{2})$, because $\mathscr{J}(sF) = i(z-\frac{1}{2})\mathscr{J}F$. Thus the transported operator $B_t = s - (t/2)D$ becomes

$$\mathcal{C}_t M := \mathscr{J} B_t \mathscr{J}^{-1} M = i(z-\tfrac{1}{2})M + \frac{it}{2}\nabla_\zeta M = \mathcal{V}_t\big(i(z-\tfrac{1}{2})\big)\mathcal{V}_{-t} M. \tag{243}$$

In the last expression the function in parentheses denotes multiplication. The equality follows by conjugating $B_t = U_t s U_{-t}$ and proves the retained coordinate and sign. It does not replace the original spectral value by z.

This also has an exact relation to the actual source quotient. On the same underlying set $\mathfrak{R}_\zeta$ let $\sigma_t = \mathscr{J}_* \tau_t$ and $\mathfrak{N}_t = \mathscr{J} N_t$. The already proved closedness of N_t in τ_t gives closedness of $\mathfrak{N}_t$ in σ_t. The map

$$[F]_{N_t} \longmapsto [\mathscr{J} F]_{\mathfrak{N}_t}$$

is a homeomorphism of the corresponding quotient spaces, because it and its inverse are induced by the defining homeomorphism $\mathscr{J}$ of the ambient spaces. Equation (243) intertwines the quotient spectral operators. For every scalar λ, it therefore also intertwines $B_t - \lambda$ and $\mathcal{C}_t - \lambda$, their kernels and images, and their continuous inverses whenever those exist. Their spectra are the same original arithmetic s-spectrum. This statement uses σ_t, while the entire-time heat estimates above use γ_ζ; it inserts no equality between those topologies.

G.6 Local pole cancellation with every arithmetic unit retained

Fix an actual nontrivial zero ρ of ζ, of its full multiplicity $m = m_\rho$, and use its unchanged local coordinate $w = z - \rho$. Write

$$\zeta(\rho + w) = w^m a_\rho(\rho + w), \qquad \frac{1}{a_\rho(\rho + w)} = \sum_{j\ge 0} c_j w^j, \qquad H_M(\rho + w) = \sum_{n\ge 0} b_n w^n. \tag{244}$$

Here $a_\rho(\rho) \neq 0$ and $c_0 = 1/a_\rho(\rho)$. All series converge on a sufficiently small disk. The principal coefficients of M and its heat generator are exactly

$$\begin{aligned} [w^{-m+q}]M &= \sum_{n=0}^{q} c_{q-n} b_n, \\ [w^{-m+q}]\mathcal{L}_\zeta M &= \frac{1}{4}\sum_{n=0}^{q} c_{q-n}(n+2)(n+1)b_{n+2}, \quad 0 \le q < m. \end{aligned} \tag{245}$$

In particular neither M nor its generator has a pole of order greater than m, and its potential order-m generator coefficient is $H_M''(\rho)/(4a_\rho(\rho)) = b_2/(2a_\rho(\rho))$. If that coefficient vanishes, the remaining coefficients in the display determine the full lower pole order without suppressing the unit.

There is a direct cancellation calculation for the apparently higher poles in (241). Put $v(w) = H_M(\rho + w)/a_\rho(\rho + w)$, so $M = w^{-m}v$, and temporarily write $a = a_\rho(\rho + w)$ with primes denoting the same original z derivative. Then

$$\begin{aligned} M' &= w^{-m}v' - mw^{-m-1}v, \\ M'' &= w^{-m}v'' - 2mw^{-m-1}v' + m(m+1)w^{-m-2}v, \\ \frac{\zeta'}{\zeta} &= \frac{m}{w} + \frac{a'}{a}, \\ \frac{\zeta''}{\zeta} &= \frac{m(m-1)}{w^2} + \frac{2m}{w}\frac{a'}{a} + \frac{a''}{a}. \end{aligned}$$

In $M'' + 2(\zeta'/\zeta)M' + (\zeta''/\zeta)M$, the coefficient multiplying $w^{-m-2}v$ is $m(m+1) - 2m^2 + m(m-1) = 0$. The coefficients multiplying $w^{-m-1}v'$ are $-2m + 2m = 0$, and those multiplying $w^{-m-1}(a'/a)v$ are $-2m + 2m = 0$. The complete remaining expression is

$$w^{-m}\left(v'' + 2\frac{a'}{a}v' + \frac{a''}{a}v\right) = \frac{w^{-m}}{a}(av)'' = \frac{H_M''}{\zeta}.$$

This proves the cancellation of both possible extra pole orders with every cross term and local unit present. Expanding the final numerator and $1/a$ proves (245).

The same calculation holds at every time. Set

$$b_n(t) = \frac{H_{M,t}^{(n)}(\rho)}{n!}, \qquad \lambda_\rho = i(\rho - \tfrac{1}{2}).$$

The globally convergent numerator heat series gives the exact entire functions of t

$$\begin{aligned} b_n(t) &= \sum_{k=0}^{\infty} \frac{t^k (n+2k)!}{4^k k! n!} b_{n+2k}(0) = \frac{i^n}{n!}(D^n U_t F)(\lambda_\rho), \\ \partial_t b_n(t) &= \frac{(n+2)(n+1)}{4} b_{n+2}(t), \qquad\qquad F = \mathscr{J}^{-1} M. \end{aligned} \tag{246}$$

Differentiation and evaluation are continuous on the entire growth space by the proved derivative estimates. They therefore justify termwise differentiation and evaluation of this series, including local uniform convergence in t. If $\ell_q(t) = [w^{-m+q}]\mathcal{V}_t M$, then

$$\ell_q(t) = \sum_{n=0}^{q} c_{q-n} b_n(t), \qquad \partial_t \ell_q(t) = \frac{1}{4} \sum_{n=0}^{q} c_{q-n}(n+2)(n+1) b_{n+2}(t), \quad 0 \le q < m. \tag{247}$$

Thus the evolution of the arithmetic principal parts is a calculation from the full entire numerator. It is not a closed system on only the finitely many coefficients ℓ_q; the displayed higher numerator jets are part of the exact map.

For an initial M holomorphic at ρ, write $M(\rho + w) = M_0 + M_1 w + O(w^2)$. The formula for the generator then gives the complete possible principal part

$$\mathrm{PP}_\rho(\mathcal{L}_\zeta M) = \frac{m(m-1)M_0}{4w^2} + \frac{m(m+1)M_1 + 2m(a_\rho'/a_\rho)(\rho)M_0}{4w}. \tag{248}$$

For $m = 1$ the order-two term is zero. For $m \ge 2$ this principal part vanishes exactly when $M_0 = M_1 = 0$. For $m = 1$ it vanishes exactly when $(a_\rho M)'(\rho) = 0$. These equivalences follow from the nonzero constants m, $m-1$ in the first case and $a_\rho(\rho) \neq 0$ in both cases. They describe the exact local derivative obstruction, including multiple zeros, rather than assuming their multiplicity is one.

G.7 The calculated domain for heat to retain an original Schwartz input

Let $\mathfrak{M}_0^{\mathrm{ev}}$, $P_{a,b}$, and $q_{a,b,N}$ be the explicitly proved meromorphic Mellin image and strip seminorms in (215). Its intersection

$$\mathfrak{A}_\zeta = \mathfrak{R}_\zeta \cap \mathfrak{M}_0^{\mathrm{ev}} = \mathscr{J}\,\mathcal{A}_{\mathrm{Sch}}$$

is the exact original Schwartz trace domain in this presentation, by (227). For each unchanged complex time t, its complete domain of retained Schwartz inputs is the following explicitly calculated subspace:

$$\begin{aligned} \mathfrak{D}_t = \Big\{ M \in \mathfrak{A}_\zeta : \quad & \sum_{k=0}^{\infty} \frac{t^k (n+2k)!}{4^k k! n!} b^\rho_{n+2k}(0) = 0 \quad (\rho \in \mathcal{Z},\ 0 \le n < m_\rho), \\ & \sup_{\substack{a \le \sigma \le b \\ v \in \mathbb{R}}} (1+|v|)^N \left| P_{a,b}(\sigma+iv) \frac{H_{M,t}(\sigma+iv)}{\zeta(\sigma+iv)} \right| < \infty \\ & \qquad\qquad \text{for every real } a \le b \text{ and every integer } N \ge 0 \Big\}. \end{aligned} \tag{249}$$

Here $\mathcal{Z}$ is the actual set of nontrivial zeros, $b_n^\rho(0) = H_M^{(n)}(\rho)/n!$, and $H_{M,t}$ is the globally convergent series in (242). The removable values in the strip expression are the values forced by the preceding zero-jet equalities and by the factors $P_{a,b}$. No unknown initial input, omitted pole cancellation, or assumed decay estimate occurs in this definition.

The exact characterization and reconstruction are

$$\begin{aligned} \mathfrak{D}_t &= \mathfrak{A}_\zeta \cap \mathcal{V}_{-t}\mathfrak{A}_\zeta, \\ M \in \mathfrak{D}_t \quad &\Longleftrightarrow \quad U_t \mathscr{J}^{-1} M \in \mathcal{A}_{\mathrm{Sch}}, \\ \psi_t(x) &= \frac{1}{2\pi} \int_{\mathbb{R}} \frac{H_{M,t}(c+iv)}{\zeta(c+iv)} x^{-c-iv}\, dv \quad (x > 0,\ c > 1), \\ \psi_t(0) &= 0, \qquad \psi_t(-x) = \psi_t(x). \end{aligned} \tag{250}$$

In the equivalence M is taken in its declared initial domain $\mathfrak{A}_\zeta$; the reconstruction formula is for $M \in \mathfrak{D}_t$. We now prove that the conditions in (249) are both necessary and sufficient, rather than leaving them as proof obligations.

The first family of equalities says exactly $H_{M,t}^{(n)}(\rho) = 0$ for $0 \le n < m_\rho$, by (246). Division by $(z-\rho)^{m_\rho} a_\rho(z)$ therefore shows that it is equivalent to holomorphy of $\mathcal{V}_t M$ at every nontrivial zero. At every trivial zero $-2k$, the zero of ζ is simple, so $H_{M,t}/\zeta$ automatically has at most a simple pole there. At $z = 1$ it automatically is holomorphic and zero, because the numerator is entire and $1/\zeta$ has a simple zero. At all remaining points it is holomorphic. Thus the first family is equivalent to all pole and value conditions in the definition of $\mathfrak{M}_0^{\mathrm{ev}}$. The second family is exactly that definition's complete finite-strip decay requirement, with its original polynomial and margin. This proves

$$M \in \mathfrak{D}_t \quad \Longleftrightarrow \quad M \in \mathfrak{A}_\zeta \text{ and } \mathcal{V}_t M \in \mathfrak{M}_0^{\mathrm{ev}}.$$

Since $\mathcal{V}_t M$ already belongs to $\mathfrak{R}_\zeta$, the last membership is precisely membership in $\mathfrak{A}_\zeta$. The group inverse proves the first identity in (250), and the proved exact trace-image characterization proves the second. The inverse Mellin theorem then proves that the displayed integral converges with the required differentiated estimates, is independent of $c > 1$, gives the unique original even Schwartz input with both zero conditions, and satisfies

$$\mathcal{F}_\mu \mathcal{E} \psi_t = \mathscr{J}^{-1} \mathcal{V}_t M = U_t \mathscr{J}^{-1} M.$$

These conclusions use the already proved inverse theorem with exactly its hypotheses verified above.

All Taylor data at the original input origin are also explicit: for $M \in \mathfrak{D}_t$ and every $k \ge 1$,

$$\frac{\psi_t^{(2k)}(0)}{(2k)!} = \operatorname*{Res}_{z=-2k} \frac{H_{M,t}(z)}{\zeta(z)} = \frac{H_{M,t}(-2k)}{\zeta'(-2k)}. \tag{251}$$

The first equality is the proved Mellin residue theorem for the reconstructed input. The second divides by the actual leading term $\zeta'(-2k)(z+2k)$; in particular it introduces no factor i into the residue coordinate z. The right side is defined as an entire function of t for every initial $M \in \mathfrak{R}_\zeta$, even at times outside $\mathfrak{D}_t$ when it cannot be interpreted as a Schwartz input's Taylor coefficient.

At $t = 0$ the definition gives all of $\mathfrak{A}_\zeta$. For every $t \neq 0$ it is a proper subspace. Indeed fix any nontrivial zero ρ and its retained spectral coordinate $\lambda_\rho = i(\rho - \frac{1}{2})$. The proved absence of a common evaluation zero among $U_t(\mathbb{C}[s]\Xi)$ gives a polynomial p for which $U_t(p\Xi)(\lambda_\rho) \neq 0$. The actual finite Hermite trace theorem places $p\Xi$ in $\mathcal{A}_{\mathrm{Sch}}$. For $M = \mathscr{J}(p\Xi)$ the computed numerator at time t is nonzero at ρ, so its quotient by ζ has a pole of the full order m_ρ there. This violates the first family in (249) and proves $M \notin \mathfrak{D}_t$. The argument is an obstruction in the actual unclosed Schwartz trace map and its calculated meromorphic heat domain; it makes no identification of the unnamed source closure.

G.8 The exact persistent heat core and all possible return times

Retain the original functions, coordinate, and analytic relation $I = \Xi\mathcal{H}_{\le 1} = N_\gamma = N_\kappa$. The obstruction can be computed for an entire orbit, not only for a chosen generator or one time. Define

$$\begin{aligned} I_{\mathrm{even}}^\infty &= \{f \in \mathcal{H}_{\le 1} : \ D^{2k} f \in I \text{ for every integer } k \ge 0\}, \\ I_{\mathrm{heat}}^\infty &= \{f \in \mathcal{H}_{\le 1} : \ U_t f \in I \text{ for every } t \in \mathbb{C}\}. \end{aligned} \tag{252}$$

Then the complete calculation is

$$I_{\mathrm{even}}^\infty = I_{\mathrm{heat}}^\infty = \{0\}. \tag{253}$$

More strongly, for every set $T \subset \mathbb{C}$ with a finite accumulation point,

$$\bigcap_{t \in T} U_{-t} I = \{0\}. \tag{254}$$

We give the complete proof, including the arithmetic obstruction to an exceptional periodic function.

First suppose $f \in I^{\infty}_{\text{even}}$. At every actual zero λ of Ξ of multiplicity m_λ, its Taylor coefficients satisfy

$$D^{2k+j} f(\lambda) = 0 \qquad (k \geq 0,\ 0 \leq j < m_\lambda).$$

If any $m_\lambda \geq 2$, the choices $j = 0, 1$ cover all nonnegative derivative orders. The entire Taylor series at that unchanged λ is zero, so $f = 0$. It remains to prove the result when all the actual zeros are simple; this is the other case of the proof, not a simplicity hypothesis. For each of these zeros, the vanishing of every even derivative and the entire Taylor expansion give the exact reflection identity

$$f(2\lambda - s) = -f(s) \qquad (s \in \mathbb{C}). \tag{255}$$

In particular f is odd about each actual zero without changing the spectral coordinate. Fix one zero λ_0. Composing the reflections at λ_0 and at λ gives

$$f\big(s + 2(\lambda - \lambda_0)\big) = f(s) \qquad (s \in \mathbb{C},\ \Xi(\lambda) = 0). \tag{256}$$

We prove the needed period assertion explicitly. Let f be a nonconstant entire function and put

$$P(f) = \{p \in \mathbb{C} :\ f(s + p) = f(s) \text{ for all } s \in \mathbb{C}\}.$$

It is an additive subgroup, closed because pointwise limits of periods remain periods by continuity of f. It has no sequence of nonzero periods tending to zero: such a sequence p_n would give $f'(s) = \lim_n (f(s + p_n) - f(s))/p_n = 0$ at every s. Thus some disk about zero contains no other period. Two periods p, q linearly independent over $\mathbb{R}$ are also impossible. Every complex number has the form $ap + bq$ with real a, b. Subtracting the integer parts of a, b puts it in the compact parallelogram $\{xp + yq : 0 \leq x, y \leq 1\}$ without changing the value of f. The resulting global bound and the Cauchy estimate $|f'(s)| \leq \sup_{\mathbb{C}} |f|/R$ for every $R > 0$ imply that f is constant, a contradiction.

If $P(f)$ contains a nonzero period p, all its periods therefore lie on the real line $\mathbb{R}p$. The set $L = \{a \in \mathbb{R} : ap \in P(f)\}$ is a closed additive subgroup of $\mathbb{R}$, contains 1, and has a positive gap about zero. Let $d = \inf(L \cap (0, \infty))$. This number is positive; a sequence approaching the infimum and closedness show $d \in L$. For any positive $a \in L$, subtract $\lfloor a/d \rfloor d$. The remainder lies in $L \cap [0, d)$ and must be zero by minimality. Negative elements are handled by changing sign. Hence $L = d\mathbb{Z}$ and

$$P(f) = \omega\mathbb{Z}, \qquad \omega = dp \neq 0.$$

This proves the full assertion about the period group used here.

Return to (256). A constant satisfying (255) is zero. If $f \neq 0$, it is therefore nonconstant. The set of distinct zeros of Ξ is infinite, as proved above from its retained product and completion, so it contains a zero different from λ_0. The period group consequently contains a nonzero period and is $\omega\mathbb{Z}$ by the preceding argument. Every actual zero must then lie in the single arithmetic progression

$$\lambda \in \lambda_0 + \tfrac{1}{2}\omega\mathbb{Z}. \tag{257}$$

The number of points in this progression in the disk $|s| \leq r$ is at most $1 + 4r/|\omega|$: their integer indices lie in an interval of length at most $4r/|\omega|$, by projecting onto the line of the progression. This bound does not assume that λ_0 or ω is real.

Here is a direct contradiction with the actual completed zeta function, using no zero-counting asymptotic. Retain its exact even Hadamard product from the source,

$$\Xi(s) = \Xi(0) \prod_{\alpha \in \Lambda_+} (1 - s^2/\alpha^2),$$

where one member of each pair $\{\lambda, -\lambda\}$ is retained with its original multiplicity. In the case currently under consideration every multiplicity is one. Set $N_+(u) = \#\{\alpha \in \Lambda_+ : |\alpha| \le u\}$. There is a number $r_* > 0$ such that $N_+(u) = 0$ for $u < r_*$, since $\Xi(0) \neq 0$ and its zeros are discrete. The progression bound implies

$$N_+(u) \le Au \quad (u \ge r_*), \qquad A = \frac{1}{r_*} + \frac{4}{|\omega|}.$$

For $r > 0$, the elementary identity

$$\log(1 + r^2/v^2) = \int_v^\infty \frac{2r^2}{u(u^2 + r^2)}\, du \qquad (v > 0)$$

and integration of nonnegative summands give

$$\begin{aligned} \sum_{\alpha \in \Lambda_+} \log(1 + r^2/|\alpha|^2) &= \int_0^\infty \frac{2r^2 N_+(u)}{u(u^2 + r^2)}\, du \\ &\le 2Ar^2 \int_0^\infty \frac{du}{u^2 + r^2} = \pi A r. \end{aligned}$$

The source product, first for its finite partial products and then by its locally uniform convergence, therefore gives

$$|\Xi(s)| \le |\Xi(0)| e^{\pi A |s|} \qquad (s \in \mathbb{C}). \tag{258}$$

No lower bound on an individual product factor was used. But at the original points $s = -i(2n - \frac{1}{2})$, $n \ge 2$, the unchanged completion is

$$\Xi(-i(2n - \tfrac{1}{2})) = \xi(2n) = n(2n-1)\pi^{-n}(n-1)!\zeta(2n).$$

Every factor is positive and $\zeta(2n) > 1$ from its positive Dirichlet series. For $n \ge 4$, let $b_n = \lfloor n/2 \rfloor$. The last b_n factors of $(n-1)!$ are at least b_n, so

$$\log \left| \Xi(-i(2n - \tfrac{1}{2})) \right| \ge b_n \log b_n - n \log \pi.$$

Dividing by $2n - \frac{1}{2}$ makes the right side tend to $+\infty$, in contradiction with (258). This excludes every nonzero f in the remaining case and proves $I^\infty_{\text{even}} = \{0\}$.

For the time statements, for every actual zero and every one of its required derivative orders, define

$$h_{\lambda,j}(t) = (D^j U_t f)(\lambda), \qquad 0 \le j < m_\lambda.$$

These are entire functions of the original complex time by the proved heat series and derivative estimates. Its differentiated coefficients at zero are exactly

$$h^{(k)}_{\lambda,j}(0) = (-1/4)^k D^{2k+j} f(\lambda). \tag{259}$$

If $U_t f \in I$ for all t in a set with a finite accumulation point, every $h_{\lambda,j}$ vanishes there, and the identity theorem makes every one of them identically zero. Equation (259) then puts f in I^∞_{even} and proves (254). Taking all complex times proves $I^\infty_{\text{heat}} = \{0\}$. Conversely, membership

of $I^{\infty}_{\mathrm{even}}$ would make all Taylor coefficients of each $h_{\lambda,j}$ zero and hence put every $U_t f$ in I, by the already proved entire division criterion. This also verifies directly the equality of the two cores, with both directions and the original coefficient retained.

In particular for each nonzero $f \in \mathcal{H}_{\le 1}$ the exact return set

$$\mathcal{T}_I(f) = \{t \in \mathbb{C} : \ U_t f \in I\} = \bigcap_{\lambda,j} \{t : h_{\lambda,j}(t) = 0\} \tag{260}$$

is closed and has no finite accumulation point. Thus it is a discrete set, possibly empty. More explicitly, at least one $h_{\lambda,j}$ is a nonzero entire function, since otherwise the core calculation would give $f = 0$; its isolated zero set already contains $\mathcal{T}_I(f)$. Every compact disk meets this return set finitely, so the set is countable by exhaustion by the integer-radius disks. Since the exact unclosed Schwartz trace intersection $\mathcal{A}_{\mathrm{Sch}}$ is a subspace of I,

$$\mathcal{T}_{\mathrm{Sch}}(f) := \{t \in \mathbb{C} : \ U_t f \in \mathcal{A}_{\mathrm{Sch}}\} \subseteq \mathcal{T}_I(f) \qquad (f \neq 0) \tag{261}$$

also has no finite accumulation point and is closed and discrete. The closedness follows because a subset of a set with no finite accumulation point is closed in $\mathbb{C}$. Its exact membership, including all strip conditions and reconstruction of the original input, was computed in Section G.7. Consequently no nonzero original Schwartz trace follows a heat orbit through original Schwartz traces on any nontrivial time interval, even though its entire heat evolution exists for all complex times.

There is also an exact invariant-subspace conclusion. If W is a linear subspace of I with $D^2 W \subseteq W$, then $D^{2k} f \in I$ for every $f \in W$ and every $k \ge 0$. The proved core calculation forces $W = \{0\}$. The same conclusion holds for a subspace of I invariant under all U_t. These conclusions do not assume that W is closed or that its topology has any particular form. They apply to the analytic relation I and to its unclosed Schwartz trace subspace, and do not identify either with the actual source closure N_{ar} in the still unnamed τ.

G.9 The original four-term coefficient and its arithmetic return times

Retain the companion mechanism's actual coefficient operator on the unchanged entire variable s:

$$\begin{aligned} (T_D f)(s) = & -\frac{59}{275184} f(s+3) - \frac{1}{336} f'(s+3) \\ & + \frac{1}{1296} f(s+\tfrac{9}{2}) + \frac{3}{132496} f(s+\tfrac{13}{2}), \\ K_t(s) = T_D Z_t(s), & \qquad Z_t = U_t \Xi = 8 H_t(2s). \end{aligned} \tag{262}$$

Here T_D denotes this given four-term operator; it is not replaced by a differential truncation. We prove directly from its complete formula that $K_0 \neq 0$, that $K_t = U_t K_0$, and that its arithmetic return times have the precise restrictions just established.

Translations by each of the three original shifts act continuously on $(\mathcal{H}_{\le 1}, \gamma)$. For a general fixed shift h, the unchanged growth estimate is

$$\|f(\cdot + h)\|_{p,a} \le e^{a|h|^p} \|f\|_{p,a/2^{p-1}}, \qquad 1 < p < 2, \quad a > 0,$$

because $|s+h|^p \le 2^{p-1}(|s|^p + |h|^p)$. Differentiation is continuous by Lemma C.1. Thus every displayed term, and T_D itself, maps $\mathcal{H}_{\le 1}$ continuously to $\mathcal{H}_{\le 1}$. Differentiation commutes with each fixed translation. Applying these continuous maps to the convergent series for U_t therefore proves term by term

$$T_D U_t = U_t T_D, \qquad K_t = U_t K_0. \tag{263}$$

In particular this calculation includes the derivative coefficient $-1/336$ at the shift 3, as well as the three function-value terms.

Nonvanishing of this particular coefficient is not assumed. Use the retained positive kernel Φ from (163), with its original formula

$$\Phi(u) = \sum_{n\geq 1} (2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u}) e^{-\pi n^2 e^{4u}} \qquad (u \geq 0).$$

The substitution $v = 2u$ in the original cosine integral gives the exact entire Fourier representation

$$\begin{aligned} Z_t(s) &= 2\int_{\mathbb{R}} \Phi(|v|/2) e^{tv^2/4} e^{isv}\, dv, \\ K_t(s) &= 2\int_{\mathbb{R}} \Phi(|v|/2) m(v) e^{tv^2/4} e^{isv}\, dv, \\ m(v) &= \left(-\frac{59}{275184} - \frac{iv}{336}\right) e^{3iv} + \frac{e^{(9/2)iv}}{1296} + \frac{3e^{(13/2)iv}}{132496}. \end{aligned} \tag{264}$$

The coefficient 2 is required: first $8\,du = 4\,dv$ on the positive half-line, then the even extension of the cosine integral halves that coefficient. A shift by h multiplies the kernel by e^{ihv}, and the derivative multiplies it by iv for this original positive Fourier exponent. These two facts prove the second identity from the first, with the precise multiplier in the third. Every interchange and derivative is justified by the previously proved bound $|\Phi(u)| \leq Ce^{9u}e^{-ce^{4u}}$ on $u \geq 0$. On a compact set of complex t, s, this double exponential dominates the factors $e^{(\operatorname{Re} t)v^2/4}$, $e^{-\operatorname{Im}(s)v}$, and every polynomial factor from $m(v)$ and the derivatives. Thus the integrals converge absolutely and locally uniformly with all needed derivatives.

The multiplier has the exact value

$$m(0) = -\frac{59}{275184} + \frac{1}{1296} + \frac{3}{132496} = \frac{127}{219024} > 0. \tag{265}$$

Also

$$\Phi(0) = \sum_{n\geq 1} \pi n^2 (2\pi n^2 - 3) e^{-\pi n^2} > 0;$$

every summand is positive and the series is absolutely convergent. The continuous integrable function $q(v) = 2\Phi(|v|/2)m(v)$ therefore has $q(0) = 2\Phi(0)\,127/219024 > 0$. If K_0 were identically zero, its Fourier transform in (264) would vanish at every real s. The Gaussian Fourier uniqueness argument already proved in Section G.2 applies to this continuous integrable q: absolute Fubini makes every Gaussian convolution $q * G_\varepsilon$ zero, while these convolutions tend to q pointwise. This would force $q(0) = 0$, a contradiction. Hence $K_0 \neq 0$. The invertibility of U_t in (263) also proves $K_t \neq 0$ for every original complex time t.

The previous theorem now applies to this exact coefficient, not to an arbitrary replacement. In particular the sets

$$\begin{aligned} \mathcal{T}_I(K_0) &= \{t :\ T_D(8H_t(2\,\cdot)) \in \Xi\mathcal{H}_{\leq 1}\}, \\ \mathcal{T}_{\mathrm{Sch}}(K_0) &= \{t :\ T_D(8H_t(2\,\cdot)) \in \mathcal{A}_{\mathrm{Sch}}\} \subseteq \mathcal{T}_I(K_0) \end{aligned} \tag{266}$$

are closed discrete subsets of $\mathbb{C}$. Their complete individual membership conditions are still the original multiplicity jets and, for the second set, the finite-strip Mellin conditions proved above. At any time outside $\mathcal{T}_I(K_0)$ there is an actual nontrivial zero ρ of multiplicity m_ρ at which

$$\frac{K_t(i(z-\frac{1}{2}))}{\zeta(z)}$$

has a pole of positive order $m_\rho - \operatorname{ord}_{i(\rho-1/2)} K_t$. Indeed nonmembership in I is exactly failure of at least one required zero jet, by the proved division theorem; local division by $(z-\rho)^{m_\rho} a_\rho(z)$

gives this pole order with the unchanged coordinate and unit. The full principal coefficients, including their time evolution, are those of Section G.6 with the actual entire numerator $K_t(i(z-\frac{1}{2}))$ inserted.

Finally retain the original incidence factor a_{ef} in $a_{ef}K_t$. For $a_{ef}\neq 0$, scalar multiplication preserves both subspaces I and $\mathcal{A}_{\mathrm{Sch}}$ and has inverse multiplication by a_{ef}^{-1}, so it leaves both return sets unchanged and multiplies every principal coefficient by a_{ef}. For $a_{ef}=0$, the coefficient is identically zero and belongs to both spaces at every time. This proves both incidence cases without suppressing that factor. These are exact arithmetic relations for the mechanism coefficient. They do not equate I or $\mathcal{A}_{\mathrm{Sch}}$ with N_{ar} in the unnamed source topology.

H The inverse-root cover, the branch locus, and arithmetic fibres

The retained polynomial inverse-fibre slice is $P_{a,0,0}(r)=-2r^2-2a$. Its root equation is $a=-r^2$. The arithmetic coordinate is $a=2s$, so

$$s=-r^2/2. \tag{267}$$

The finite free extension of the actual quotient is

$$\widehat{Q}=\mathbb{C}[r]\otimes_{\mathbb{C}[s]}Q_{\mathrm{ar}}=Q_{\mathrm{ar}}\oplus rQ_{\mathrm{ar}},\qquad \mathcal{R}=\begin{pmatrix}0&-2S\\1&0\end{pmatrix},\qquad -\tfrac{1}{2}\mathcal{R}^2=\begin{pmatrix}S&0\\0&S\end{pmatrix}.$$

The even–odd decomposition of a polynomial in r proves freeness with basis $(1,r)$ and uniqueness coefficient by coefficient. Multiplying q_0+rq_1 by r proves the displayed matrices; the finite direct-sum topology makes every induced map continuous. The involution is $(q_0,q_1)\mapsto(q_0,-q_1)$; the trace is $\mathrm{Tr}(q_0+rq_1)=2q_0$, with kernel rQ_{ar}. It is $\mathbb{C}[s]$-linear. It does not provide a $\mathbb{C}[r]$-module quotient with this kernel unless $S=0$: multiplying rq by r gives $-2Sq$ in the even summand. This proves the exact limitation, rather than suppressing the odd component. The same construction at time t uses S_t and is conjugated by $\mathrm{diag}(\mathcal{U}_t,\mathcal{U}_t)$.

The original s-spectrum is unchanged by the extension: an inverse of $S-\lambda$ duplicates to a diagonal inverse, and conversely projection of an inverse of the doubled matrix to its first component, following inclusion of that component, gives an inverse of $S-\lambda$. Both directions are continuous. Moreover

$$\mathrm{Spec}(\mathcal{R})=\{\rho\in\mathbb{C}:-\rho^2/2\in\mathrm{Spec}(S)\}. \tag{268}$$

To prove this without an unsupported spectral-mapping citation, solve $(\mathcal{R}-\rho)(u,v)=(a,b)$. The second row gives $u=b+\rho v$ and the first becomes $-2(S+\rho^2/2)v=a+\rho b$. A continuous inverse of $S+\rho^2/2$ therefore gives a continuous inverse matrix. Conversely the right-hand side $(a,0)$ and the second projection of an inverse matrix supply an inverse of $-2(S+\rho^2/2)$, with both identities checked by the same two rows.

Before any quotient by Ξ, let $\mathcal{O}_s$ be the sheaf of holomorphic functions in s and put $B=\mathcal{O}_s[r]/(r^2+2s)$, identified with $\mathcal{O}_s\oplus r\mathcal{O}_s$. On the punctured base $s\neq 0$, implicit differentiation gives $dr/ds=-1/r=r/(2s)$. Thus the actual lifted derivative is

$$\nabla_s\begin{pmatrix}f_0\\f_1\end{pmatrix}=\begin{pmatrix}f_0'\\f_1'+f_1/(2s)\end{pmatrix},\quad R=\begin{pmatrix}0&-2s\\1&0\end{pmatrix},\quad [\nabla_s,R]=-R^{-1}. \tag{269}$$

Indeed with $A=\mathrm{diag}(0,1/(2s))$, $R'+AR-RA=\left(\begin{smallmatrix}0&-1\\1/(2s)&0\end{smallmatrix}\right)=-R^{-1}$. A second application gives

$$\nabla_s^2=\mathrm{diag}\left(D^2,D^2+s^{-1}D-\frac{1}{4s^2}\right). \tag{270}$$

Equivalently, on a function of r the derivative is $-r^{-1}\partial_r$ and the heat generator is $-\frac{1}{4}r^{-2}\partial_r^2 + \frac{1}{4}r^{-3}\partial_r$. Squaring $-r^{-1}\partial_r$ by the product rule proves this with both signs preserved.

The branch extension is exact. The meromorphic connection sends B into $s^{-1}B$ and the connection one-form has a simple pole with residue $\mathrm{diag}(0, 1/2)$. Its maximal domain for holomorphic output at $s = 0$ is

$$\mathrm{Dom}_{\mathrm{hol}}\,\nabla_s = \mathcal{O}_s \oplus rs\mathcal{O}_s. \tag{271}$$

In fact the odd coefficient is $f_1' + f_1/(2s)$; its only possible pole has coefficient $f_1(0)/2$, proving necessity and sufficiency. For the square, write $f_1 = \sum_{n\geq 0} a_n s^n$. Its image has coefficient $(n^2 - 1/4)a_n s^{n-2}$. The negative powers vanish exactly when $a_0 = a_1 = 0$, so

$$\mathrm{Dom}_{\mathrm{hol}}\,\nabla_s^2 = \mathcal{O}_s \oplus rs^2\mathcal{O}_s.$$

For k iterates the odd coefficient is $\prod_{h=0}^{k-1}(n + 1/2 - h)a_n s^{n-k}$, whose prefactor is never zero for integral n. Thus its domain is $\mathcal{O}_s \oplus rs^k\mathcal{O}_s$. No derivation $B \to B$ extending ∂_s exists at the branch: derivation of $r^2 + 2s = 0$ would give $2r\delta(r) + 2 = 0$, forcing r to be a unit although it vanishes at the branch point. The logarithmic derivative $s\nabla_s$ does extend to all of B, with matrix $sD + \mathrm{diag}(0, 1/2)$.

These are sheaf and representative calculations; they do not assert that ∇_s descends through the frozen arithmetic relation. For a holomorphic local multiplier $R_0(s)$ the relation vector $(R_0\Xi, 0)$ acquires $(R_0'\Xi + R_0\Xi', 0)$, and the odd relation $(0, R_0\Xi)$ acquires $(0, R_0'\Xi + R_0\Xi' + R_0\Xi/(2s))$. The second iterate includes the full $2R_0'\Xi'$ term, as well as the odd $s^{-1}D - 1/(4s^2)$ terms of (270). Jet extension (181), or the moving closed quotient (164), retains these relations exactly.

Finally, the branch point is not an arithmetic spectral point at $t = 0$. The source gives $\Xi(0) \neq 0$, so its Proposition 3.6 makes S continuously invertible; then $\mathcal{R}$ is invertible by (268). In the analytic divisor realization the stronger local statement follows from Ξ being a holomorphic unit on a disk about zero: the quotient stalk is zero there. For real t, $Z_t(0) > 0$ directly from the retained kernel:

$$2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u} = \pi n^2 e^{5u}(2\pi n^2 e^{4u} - 3) > 0 \quad (u \geq 0),$$

and the positive integral converges by the bound already proved. Thus the actual real heat divisor never meets the cover branch point $s = 0$. On a simple zero curve its retained lift satisfies $r^2 = -2\lambda$, $r' = -\lambda'/r$; together with (171) these are the original-coordinate and cover equations, including the unit contribution. A nonreal r alone does not specify whether $\lambda = -r^2/2$ is nonreal.

I Exact ingress from compact Weil tests

The accompanying retained Weil-test calculation supplies the concrete space $\mathcal{E} = C_c^\infty(\mathbb{R}; \mathbb{C})$ with coordinate operator $T = -i\partial_u$ and Fourier transform

$$(\mathscr{F}f)(s) = \int_{\mathbb{R}} f(u)e^{-ius}\,du.$$

Here we give the complete heat ingress and its inverse domains; none of the following claims requires identifying the source quotient topology with a test-space topology. Integration by parts gives $\mathscr{F}(Tf) = s\mathscr{F}f$ with no endpoint term. If $\mathrm{supp}\, f \subset [-L, L]$, then

$$|\mathscr{F}f(s)| \leq \|f\|_1 e^{L|\operatorname{Im} s|}, \qquad \|\mathscr{F}f\|_{p,a} \leq \|f\|_1 \sup_{r\geq 0} e^{Lr - ar^p} < \infty.$$

Thus the Fourier transform maps the actual test space into the original entire-function set, continuously in γ on every fixed-support smooth-function space and hence in its usual inductive-limit topology.

Define, for all complex t,

$$(G_t f)(u) = e^{tu^2/4} f(u), \qquad U_t \mathscr{F} f = \mathscr{F} G_t f, \qquad T_t = G_t T G_{-t} = T + \frac{it}{2} u. \tag{272}$$

The multiplication G_t is a continuous automorphism on each fixed-support smooth-function space, with inverse G_{-t}. Indeed each derivative of the smooth multiplier is bounded on the fixed compact support, and Leibniz's finite sum bounds every output derivative seminorm by finitely many input ones. The same argument for the inverse proves the assertion. The exponential series and every u-derivative converge uniformly on a fixed compact set, locally uniformly in t; consequently the family is entire in time in that test topology. In the Fourier integral, $D^{2k}\mathscr{F} f = \mathscr{F}((-1)^k u^{2k} f)$. The compact support permits absolute interchange with the exponential series, proving the middle identity with the exact factor $1/4$. Finally differentiation of $e^{-tu^2/4} f$ gives the displayed T_t, and

$$\mathscr{F}(T_t f) = B_t \mathscr{F} f, \qquad D\mathscr{F} f = \mathscr{F}(-iuf).$$

Both equations follow directly from the integral, including their signs. The test-space version of the retained cover is therefore

$$\mathcal{R}_t^{\mathrm{test}} = \begin{pmatrix} 0 & -2T_t \\ 1 & 0 \end{pmatrix}, \qquad (\mathcal{R}_t^{\mathrm{test}})^2 = -2\,\mathrm{diag}(T_t, T_t).$$

Componentwise Fourier transformation intertwines this matrix with the representative matrix using B_t, and then with the actual quotient matrix using S_t.

To retain every possible arithmetic relation in this ingress, put

$$K = \{f \in \mathcal{E} : \mathscr{F} f \in N_{\mathrm{ar}}\}, \qquad K_t = \{f \in \mathcal{E} : \mathscr{F} f \in N_t\}. \tag{273}$$

Then $K_t = G_t K$: by (272), $\mathscr{F} f \in U_t N_{\mathrm{ar}}$ exactly when $\mathscr{F}(G_{-t} f) \in N_{\mathrm{ar}}$. Thus there are injective linear maps onto their explicitly specified images

$$\mathcal{E}/K \longrightarrow Q_{\mathrm{ar}}, \quad [f] \longmapsto [\mathscr{F} f], \qquad \mathcal{E}/K_t \longrightarrow Q_t, \quad [f] \longmapsto [\mathscr{F} f],$$

and the square formed by G_t and $\mathcal{U}_t$ commutes. The kernel statements follow exactly from (273); no claim that $K = 0$ in the unnamed source topology is inserted. To make the ingress continuous without that identification, give $\mathcal{E}$ the join η of its usual test topology with the inverse-image topology of τ under $\mathscr{F}$. Then K is closed and the first quotient map is continuous. Give the time-t test space the transported topology $(G_t)_* \eta$; the second map and the commutative square are continuous by (272). This specifies the exact topologies rather than claiming an unproved topological embedding of a quotient image equipped with its subspace topology.

There is a fully explicit inverse-domain calculation in the same test coordinates. For $\lambda \in \mathbb{C}$, $k \geq 1$, the image is

$$(T_t - \lambda)^k \mathcal{E} = \left\{ g \in \mathcal{E} : \int_{\mathbb{R}} v^j e^{-tv^2/4 - i\lambda v} g(v)\, dv = 0 \quad (0 \leq j < k) \right\}. \tag{274}$$

On that precise image the inverse is

$$[(T_t - \lambda)^{-k} g](u) = \frac{i^k e^{tu^2/4 + i\lambda u}}{(k-1)!} \int_{-\infty}^{u} (u - v)^{k-1} e^{-tv^2/4 - i\lambda v} g(v)\, dv. \tag{275}$$

To prove both directions, write $f = e^{tu^2/4+i\lambda u}h$. Direct differentiation gives $(T_t - \lambda)f = e^{tu^2/4+i\lambda u}(-ih')$. Iteration reduces the inverse equation to a kth primitive, giving (275) because $(-i)^k i^k = 1$. For g supported in $[a, b]$, that formula vanishes below a. Above b, expansion of $(u - v)^{k-1}$ shows it vanishes identically exactly when all k moments in (274) vanish; the binomial coefficients and the nonzero exponential do not change that equivalence. Its smoothness follows by repeated differentiation of the integral; smoothness of g with compact support gives matching derivatives at both endpoints. Necessity for any compact solution also follows by integrating $h^{(k)}$ against $1, v, \dots, v^{k-1}$ and integrating by parts k times. Uniqueness follows since a compactly supported solution of $h^{(k)} = 0$ is a polynomial of degree at most $k - 1$ which vanishes on a ray, and hence is zero. This proves the exact image, inverse, kernel zero, and preservation of the original support interval.

In particular the inverse of $\mathcal{R}_t^{\mathrm{test}}$ is

$$(g_0, g_1) \longmapsto (g_1, -\tfrac{1}{2}T_t^{-1}g_0), \qquad \mathrm{Dom}(\mathcal{R}_t^{\mathrm{test}})^{-1} = \left\{(g_0, g_1) : \int e^{-tv^2/4}g_0(v)\,dv = 0\right\}.$$

Substitution into its two rows proves both inverse identities. Thus the branch-locus obstruction in the concrete test realization is a retained Gaussian-weighted zero-moment condition. The exact arithmetic quotient instead has invertible S_t and root operator, as proved above. Their ingress map and its kernels explain precisely how these statements coexist; one operator's inverse domain is not silently assigned to the other realization.

I.1 The source Sobolev cokernel and an exact common test-space map

The Sobolev construction cited immediately after Connes–Marcolli Proposition 2.24 is specified in Chapter 2, Section 9. It is a different, explicitly defined quotient, and supplies a useful exact comparison. We retain its weight, its generator, and its limited set of detected jets. The primary locators are Connes–Marcolli, printed pp.407–408 and 416–420 [6], and Connes, Section III, pp.10–15 and Appendix I, pp.62–67 of the author PDF [10]. Here and below $\delta > 1$.

For the rational field, choose the splitting $C_{\mathbb{Q}} = C_{\mathbb{Q},1} \times \mathbb{R}_+^*$ used in the source, give the compact factor Haar mass one, and put $x = e^u$ on the second factor. Its trivial-character component is the Hilbert space

$$B_\delta = L^2\big(\mathbb{R}, (1 + u^2)^{\delta/2}\,du\big), \qquad \|f\|_\delta^2 = \int_{\mathbb{R}} |f(u)|^2(1 + u^2)^{\delta/2}\,du.$$

The source adelic map is

$$E_{\mathrm{ad}}\psi(g) = |g|^{1/2} \sum_{q\in\mathbb{Q}^*} \psi(qg), \quad \psi \in \mathcal{S}(\mathbb{A}_{\mathbb{Q}})_0 = \{\psi : \psi(0) = 0,\ \int \psi\,dx = 0\}.$$

Let P_1 denote averaging over $C_{\mathbb{Q},1}$, and define

$$R_\delta = \overline{P_1 E_{\mathrm{ad}}(\mathcal{S}(\mathbb{A}_{\mathbb{Q}})_0)}^{B_\delta}, \qquad H_{\delta,1} = B_\delta/R_\delta.$$

These are exactly the trivial-character relation space and cokernel of the source. Indeed the source completion uses the norm of its image, so the extended isometry has closed range equal to the closure of the uncompleted image. Compact averaging commutes with that equivariant closed range. If b is a limit of image vectors and satisfies $P_1 b = b$, averaging an approximating sequence gives a sequence in the averaged image converging to b. This proves that the displayed closure is the trivial-character part of the full relation space, without replacing any source relation by an orthogonal complement as a definition.

We import explicitly the following result of the source proof: under the bilinear pairing $\langle f,\eta\rangle = \int f(u)\eta(u)\,du$, the annihilator of R_δ in $B_{-\delta} = L^2(\mathbb{R},(1+u^2)^{-\delta/2}\,du)$ is the closed linear span of

$$u^a e^{itu}, \qquad \zeta(\tfrac{1}{2}+it)=0, \quad t\in\mathbb{R}, \quad 0\le a<m_{-t}, \quad a<\frac{\delta-1}{2}. \tag{276}$$

Here m_{-t} is the original multiplicity of the zero $-t$ of Ξ, equal to that of $\frac{1}{2}+it$ by the retained reflected Mellin coordinate. The source proves the finite combinations after a compact Fourier cutoff and proves that the cutoff vectors are dense by uniformly bounded approximate units. Thus the closed span, not only its finite-dimensional projections, is the imported statement. No such annihilator statement is imported for the unnamed topology τ of the entire-function ring.

The precise cutoff can also be checked directly from the defined norm:

$$\|u^a e^{itu}\|_{-\delta}^2 = \int_{\mathbb{R}} |u|^{2a}(1+u^2)^{-\delta/2}\,du<\infty \quad\Longleftrightarrow\quad 2a-\delta<-1.$$

Near zero the integrand is locally integrable for every integer $a\ge 0$; for $|u|\ge 1$ it is bounded above and below by positive constants times $|u|^{2a-\delta}$. Integration of this power proves the strict inequality and the logarithmic divergence at equality. Hence set

$$q_\delta = \left\lceil \frac{\delta-1}{2}\right\rceil, \qquad n_{\delta,\lambda}=\min(m_\lambda,q_\delta). \tag{277}$$

There are exactly $n_{\delta,\lambda}$ allowed degrees at a real zero λ. The original Connes Theorem 1, Section III, p.13, states the equivalent multiplicity bound $n<(1+\delta)/2$, $n\le m_\lambda$. The inspected Connes–Marcolli printed p.408 instead displays $n<1+\delta/2$ in Theorem 2.47; its proof, pp.419–420, equations (2.444) and (2.448), gives precisely the stricter bound proved above. This records the primary discrepancy rather than silently identifying the two printed formulas. No assumption that an actual zeta zero has multiplicity greater than one is used.

Let $\mathcal{E}=C_c^\infty(\mathbb{R};\mathbb{C})$ retain its usual fixed-support smooth-function inductive-limit topology, and let $\mathscr{F}f(s)=\int f(u)e^{-ius}\,du$ as above. Define

$$K_{\mathrm{full}} = \{f\in\mathcal{E}:\ D^a\mathscr{F}f(\lambda)=0 \text{ for every zero } \lambda \text{ of } \Xi,\ 0\le a<m_\lambda\}, \tag{278}$$

$$K_\delta = \{f\in\mathcal{E}:\ D^a\mathscr{F}f(\lambda)=0 \text{ for every real zero } \lambda \text{ of } \Xi,\ 0\le a<n_{\delta,\lambda}\}. \tag{279}$$

The already proved entire division theorem identifies $K_{\mathrm{full}}=\{f:\mathscr{F}f\in\Xi\mathcal{H}_{\le 1}\}$ exactly.

Theorem I.1 (Exact dense test ingress into the source Sobolev cokernel)**.** *The map $j_\delta:\mathcal{E}\to H_{\delta,1}$ sending a test function to its Hilbert quotient class is continuous, has dense image, and has kernel exactly K_δ. It induces the continuous map*

$$J_\delta:\mathcal{E}/K_{\mathrm{full}}\longrightarrow H_{\delta,1}, \qquad [f]\longmapsto[f], \qquad \ker J_\delta = K_\delta/K_{\mathrm{full}}. \tag{280}$$

Its image is dense. These statements neither assert surjectivity nor identify the topology of its image with a quotient topology. For the source translation generator D_1, this ingress retains the original spectral coordinate through

$$j_\delta(Tf)=iD_1 j_\delta(f), \qquad T=-i\partial_u, \qquad \mathscr{F}(Tf)=s\mathscr{F}f. \tag{281}$$

Proof. The inclusion $\mathcal{E}\to B_\delta$ is continuous on every fixed-support space: on $[-L,L]$,

$$\|f\|_\delta^2 \le \left(\int_{-L}^{L}(1+u^2)^{\delta/2}\,du\right)\sup|f|^2.$$

The inductive-limit universal property and the continuous Hilbert quotient map give continuity of j_δ. Smooth compactly supported functions are dense in B_δ: truncate an arbitrary f outside $[-L, L]$, whose squared norm tail tends to zero by integrability; on the enlarged interval $[-L-1, L+1]$, the weight is bounded above and below by positive constants, and convolution with a smooth compactly supported approximate identity approximates the truncation in ordinary L^2 and therefore in the weighted norm. These convolutions have compact support and are smooth. Consequently their quotient images are dense.

The continuous dual of B_δ is exactly $B_{-\delta}$ under the displayed bilinear pairing. To check this, multiply by $(1+u^2)^{\delta/4}$ to identify B_δ isometrically with ordinary L^2; the Riesz representation there, with complex conjugation absorbed in the representing function, gives exactly the inverse-weight dual and its norm. Since R_δ is closed, a vector belongs to it exactly when every member of its annihilator vanishes on that vector. One direction is the definition. For the other, a vector outside a closed Hilbert subspace has a nonzero orthogonal component, and its inner product defines a continuous separating functional. By the explicitly imported closed-span description (276), it is enough to test the listed log-monomial characters. But

$$D^a \mathscr{F} f(-t) = (-i)^a \int u^a e^{itu} f(u)\, du, \qquad \int u^a e^{itu} f(u)\, du = i^a D^a \mathscr{F} f(-t).$$

Every factor i^a is nonzero. The coordinate $\lambda = -t$ and the proved count (277) now give $\ker j_\delta = K_\delta$. Every equation defining K_δ appears among the equations defining K_{full}, proving $K_{\mathrm{full}} \subseteq K_\delta$. Thus (280) is well defined and continuous by the quotient universal property; changing the representative changes its image by zero exactly for a member of K_δ. This proves the kernel and the dense-image assertion.

Finally the source left translation is $(V(e^\varepsilon)f)(u) = f(u-\varepsilon)$, so its infinitesimal generator on $\mathcal{E}$ is $-\partial_u$. The difference quotient converges in B_δ because the supports remain in a fixed compact set and the smooth-function difference quotient converges uniformly there. Passing to the quotient proves that $j_\delta(f)$ belongs to the domain of D_1 and $D_1 j_\delta(f) = j_\delta(-f')$. Multiplication by i gives the first equation of (281). Integration by parts in the compactly supported Fourier integral gives the second. In particular the source eigenparameter it becomes $-t$ under iD_1, exactly the same s coordinate used in the jet equations; no sign change is absorbed into the source operator. □

For completeness the unspecified source topology also has an exact comparison, requiring no unproved inclusion of its kernel in K_δ. Retain

$$K_\tau = \{f \in \mathcal{E} : \mathscr{F} f \in N_{\mathrm{ar}}\}, \qquad L_{\tau,\delta} = K_\tau \cap K_\delta.$$

Give $\mathcal{E}$ the join of its usual test topology, the inverse-image topology $\mathscr{F}^{-1}\tau$, and the seminorm $\|\cdot\|_\delta$. The last seminorm is already continuous in the test topology, but is displayed to specify every target map. Then K_τ, K_δ and $L_{\tau,\delta}$ are closed, and there are continuous maps

$$Q_{\mathrm{ar}} \longleftarrow \mathcal{E}/L_{\tau,\delta} \longrightarrow H_{\delta,1}, \qquad [f] \longmapsto [\mathscr{F} f], \quad [f] \longmapsto [f]. \tag{282}$$

Their respective kernels are exactly $K_\tau/L_{\tau,\delta}$ and $K_\delta/L_{\tau,\delta}$; the right image is dense. Indeed each composite on $\mathcal{E}$ is continuous by definition, vanishes on the intersection, and has the asserted kernel before quotienting. The quotient universal property proves the assertions, and density follows from the same set of compact tests as above. The map to the product of the two targets is injective because its kernel is precisely the intersection already divided out. No topological embedding of that image is inferred from injectivity. This is a proved relation to the actual source quotient while its named topology remains retained. The comparison also intertwines the coordinate operators: T preserves K_τ because $\mathscr{F}(Tf) = s\mathscr{F} f$ and $MN_{\mathrm{ar}} \subset N_{\mathrm{ar}}$; it preserves K_δ because

$$D^a(sF)(\lambda) = \lambda D^a F(\lambda) + a D^{a-1} F(\lambda),$$

where the second term is zero for $a = 0$ and otherwise uses an already imposed lower jet. Thus T descends to the common span, whose left map intertwines it with S and whose right map intertwines it with iD_1 on the proved domain of the ingress.

There is also a literal compatibility of the original finite Hermite traces with the adelic map. For the source's even ψ, put $\psi_{\mathrm{ad}} = \psi \otimes \mathbf{1}_{\widehat{\mathbb{Z}}}$. It lies in $\mathcal{S}(\mathbb{A}_{\mathbb{Q}})_0$ because its value and additive integral are the two original zero constraints. On the idele represented by $(x, 1, 1, \ldots)$, a rational number belongs to every $\mathbb{Z}_p$ exactly when it is an integer: write it in lowest terms and use any prime dividing its denominator for the reverse implication. Therefore

$$E_{\mathrm{ad}}\psi_{\mathrm{ad}}(x) = x^{1/2} \sum_{n \in \mathbb{Z}\setminus\{0\}} \psi(nx) = 2\mathcal{E}\psi(x).$$

Every norm-one rational idele class has a representative consisting of a real sign and finite units. Indeed, for an idele g put $q = \prod_p p^{v_p(g_p)} \in \mathbb{Q}_{>0}$, a finite product. Then $g_p/q \in \mathbb{Z}_p^*$ for every prime, while the norm-one identity gives $|g_\infty| = q$. Dividing by the diagonal rational q gives the asserted representative. The function is unchanged by the finite units; evenness removes the real sign. It therefore lies in the entire trivial-character component. The Fourier transform in $u = \log x$ is thus exactly twice the original source trace, retaining the factor 2. In particular the finite Hermite generators yield $2R_\ell^{\pm}(s)\Xi(s)$ in this explicit realization. Their logarithmic representatives lie in every B_δ by their proved rapid decay. This proves a concrete arithmetic compatibility as well as the quotient comparison; it does not identify a Sobolev topology with the topology on all of $\mathcal{H}_{\le 1}$.

J The retained spatial bilaplacian and the actual source quotient

The original polynomial spatial map and its inverse-fibre trajectory are

$$Q(x, y, w) = 1 + xy, \qquad \sigma(x, y, w) = \tfrac{1}{2}\{Q^3 w + y^2 Q(4 + 3xy)\},$$
$$\gamma(z) = (z^{-1}, -3z/2, 13z^2/2), \quad z > 0.$$

Here the Euclidean coordinates are exactly (x, y, w), and $\Delta = \partial_x^2 + \partial_y^2 + \partial_w^2$. The symbol $\gamma(z)$ in this section denotes this spatial curve; the topology γ on $\mathcal{H}_{\le 1}$ retains its earlier definition. Substitution gives $Q(\gamma(z)) = -1/2$ and $\sigma(\gamma(z)) = -z^2/8$. The pullback $\sigma^* : \mathcal{O}(\mathbb{C}) \longrightarrow \mathcal{O}(\mathbb{C}^3)$, $h \longmapsto h \circ \sigma$, is a continuous injective algebra homomorphism for compact convergence. Indeed, $\eta(s) = (0, 0, 2s)$ satisfies $\sigma \circ \eta(s) = s$, so $\eta^*\sigma^* = 1$. A compact set has compact image under each polynomial map, proving continuity of both pullbacks by their defining supremum seminorms. This also proves that the inverse on the image is continuous. Restriction to $\mathcal{H}_{\le 1}$ with its growth topology is continuous because that topology dominates compact convergence.

J.1 Every retained coefficient and the residue functional

Write $g = \nabla\sigma$, $H = D^2\sigma$, $N = g \cdot g$, and $L = \Delta\sigma$. For every entire h,

$$\begin{aligned}
\Delta(h \circ \sigma) &= Nh''(\sigma) + Lh'(\sigma),\\
\Delta^2(h \circ \sigma) &= N^2 h^{(4)}(\sigma) + (4g^{\mathsf{T}} H g + 2NL)h^{(3)}(\sigma)\\
&\quad + (2\|H\|_{\mathrm{F}}^2 + 4g \cdot \nabla L + L^2)h''(\sigma) + (\Delta L)h'(\sigma).
\end{aligned} \tag{283}$$

These formulas use the bilinear polynomial contractions; on the real locus $\|H\|_{\mathrm{F}}^2 = \sum_{i,j} H_{ij}^2$ is the usual squared Frobenius norm. To prove the second identity, the two product rules give

$$\Delta(Nh'') = (\Delta N)h'' + (2g \cdot \nabla N + NL)h^{(3)} + N^2 h^{(4)},$$
$$\Delta(Lh') = (\Delta L)h' + (2g \cdot \nabla L + L^2)h'' + LNh^{(3)}.$$

All derivatives of h are evaluated at σ. Differentiating $N = \sum_i (\partial_i \sigma)^2$ gives $\nabla N = 2Hg$ and $\Delta N = 2\sum_{i,j} H_{ij}^2 + 2g \cdot \nabla L$. Substitution proves every term in (283).

The original derivatives before restriction are

$$\begin{aligned} g_1 &= \tfrac{3}{2} y Q^2 w + \tfrac{1}{2} y^3 (7 + 6xy), \\ g_2 &= \tfrac{3}{2} x Q^2 w + \tfrac{1}{2} (8y + 21xy^2 + 12x^2 y^3), \\ g_3 &= \tfrac{1}{2} Q^3, \\ L &= 3wQ(x^2 + y^2) + 3y^4 + 18x^2 y^2 + 21xy + 4. \end{aligned}$$

Differentiating these polynomials and substituting the unchanged curve yields

$$g(\gamma(z)) = \begin{pmatrix} -9z^3/32 \\ -3z/16 \\ -1/16 \end{pmatrix}, \qquad H(\gamma(z)) = \begin{pmatrix} -27z^4/4 & 3z^2/16 & -9z/16 \\ 3z^2/16 & 13/4 & 3/(8z) \\ -9z/16 & 3/(8z) & 0 \end{pmatrix}.$$

The complete contractions are

$$\begin{aligned} N(\gamma(z)) &= (81z^6 + 36z^2 + 4)/1024, \\ L(\gamma(z)) &= (13 - 27z^4)/4, \\ \|H(\gamma(z))\|_{\mathrm{F}}^2 &= 729z^8/16 + 9z^4/128 + 81z^2/128 + 169/16 + 9/(32z^2), \\ (g \cdot \nabla L)(\gamma(z)) &= 9477z^8/512 - 405z^4/64 + 27z^2/128 + 81/32 + 3/(32z^2), \\ (g^{\mathsf{T}} H g)(\gamma(z)) &= -2187z^{10}/4096 + 81z^6/4096 - 81z^4/4096 + 117z^2/1024 + 9/1024, \\ (\Delta L)(\gamma(z)) &= -111z^2 + 36/z^2. \end{aligned}$$

Consequently the exact restricted operator is

$$(\Delta^2 \sigma^* h)(\gamma(z)) = \sum_{j=1}^{4} C_j(z) h^{(j)}(-z^2/8), \tag{284}$$

where

$$\begin{aligned} C_4(z) &= \big((81z^6 + 36z^2 + 4)/1024\big)^2, \\ C_3(z) &= -6561z^{10}/2048 + 243z^6/2048 - 135z^4/1024 + 351z^2/512 + 31/512, \\ C_2(z) &= 26973z^8/128 - 4419z^4/64 + 135z^2/64 + 669/16 + 15/(16z^2), \\ C_1(z) &= -111z^2 + 36/z^2. \end{aligned}$$

In particular both off-diagonal Hessian occurrences and all derivatives of L have been retained.

For every entire h, continuity of its first four derivatives at zero and these four Laurent polynomials prove the limit

$$\mathfrak{r}(h) := \lim_{z \downarrow 0} z^2 (\Delta^2 \sigma^* h)(\gamma(z)) = 36h'(0) + \frac{15}{16} h''(0). \tag{285}$$

Indeed $z^2 C_4, z^2 C_3$ tend to zero, whereas $z^2 C_2$ and $z^2 C_1$ tend to $15/16$ and 36. No evenness assumption on h is needed. For any $R > 0$, Cauchy's integral formula gives

$$|\mathfrak{r}(h)| \le \left(\frac{36}{R} + \frac{15}{8R^2} \right) \sup_{|s| \le R} |h(s)|.$$

Thus $\mathfrak{r}$ is continuous for compact convergence and for the growth topology. This estimate asserts no continuity in the unnamed source topology τ.

For the actual coefficient $K_t = T_{\mathrm{D}} Z_t$ of Section G.9, differentiation of its four original terms gives the exact value

$$\begin{aligned}\mathfrak{r}(K_t) = &-\frac{59}{275184}\left(36 Z_t'(3) + \frac{15}{16} Z_t''(3)\right)\\ &- \frac{1}{336}\left(36 Z_t''(3) + \frac{15}{16} Z_t'''(3)\right)\\ &+ \frac{1}{1296}\left(36 Z_t'(9/2) + \frac{15}{16} Z_t''(9/2)\right)\\ &+ \frac{3}{132496}\left(36 Z_t'(13/2) + \frac{15}{16} Z_t''(13/2)\right).\end{aligned} \tag{286}$$

For the unchanged incidence factor a_{ef} and physical diffusion $\nu\Delta$, its scaled second diffusion is $\nu^2 a_{ef}\mathfrak{r}(K_t)$ by linearity and operator composition. The limit probes spatial infinity: $|\gamma(z)| \geq z^{-1}$. Every $\sigma^* h$ is smooth at every finite real point by its entire composition. These assertions specify the domain and meaning of the observable exactly.

J.2 The full ambiguity on the actual source quotient

Set $E = \mathcal{H}_{\leq 1}$, $N = N_{\mathrm{ar}}$, and $Q_{\mathrm{ar}} = (E, \tau)/N$. The original even function Ξ has $\Xi(0) > 0$, as proved from its retained positive kernel. Define

$$e(s) = \frac{8}{15\Xi(0)} s^2 \Xi(s), \qquad K = \ker \mathfrak{r}, \qquad Pf = f - e\,\mathfrak{r}(f). \tag{287}$$

Then $e \in \mathbb{C}[s]\Xi \subset N$, $e'(0) = 0$, and $e''(0) = 16/15$, so $\mathfrak{r}(e) = 1$. Consequently

$$\mathfrak{r}(N) = \mathbb{C}, \qquad \{([f]_N, \mathfrak{r}(f)) : f \in E\} = Q_{\mathrm{ar}} \times \mathbb{C}. \tag{288}$$

Here is the exact representative for every pair on the right. Given any representative f of q and any $a \in \mathbb{C}$, put $f_a = f + (a - \mathfrak{r}(f))e$. Then $[f_a]_N = q$ and $\mathfrak{r}(f_a) = a$. In particular the zero class itself has representatives ae with every residue a. No nonempty set of source quotient classes admits this scalar as a value independent of all its representatives.

The strongest corresponding quotient map is explicit. Algebraically $P^2 = P$, $\operatorname{im} P = K$, $\ker P = \mathbb{C}e$, and

$$N_0 := N \cap K = P(N), \qquad P^{-1}(N_0) = N.$$

The first equality for N_0 follows because $e \in N$ and P fixes K. If $Pf \in N$, the identity $f = Pf + e\mathfrak{r}(f)$ proves $f \in N$, establishing the inverse-image equality. Give K the quotient topology $\tau_P = P_*\tau$: a subset $V \subset K$ is open exactly when $P^{-1}(V)$ is open in (E, τ). This is the topology of $(E, \tau)/\mathbb{C}e$ under its displayed algebraic isomorphism with K. In particular it is a vector topology. Since N is closed, the same inverse-image equality proves that N_0 is closed in (K, τ_P). There is a homeomorphism

$$(K, \tau_P)/N_0 \xrightarrow{\;[k] \mapsto [k]_N\;} (E, \tau)/N. \tag{289}$$

For completeness, let $q_N : E \to E/N$ and $q_0 : K \to K/N_0$ denote the quotient maps. The proposed map A satisfies $Aq_0 P = q_N$. Since $q_0 P$ is a composition of quotient maps and hence is quotient, this identity proves continuity of A. The inverse is induced by $q_0 P$, because its kernel is N; the quotient property of q_N proves its continuity. These maps are inverse since $Pf - f \in N$ and $Pk = k$. This proves the assertion without using τ-continuity of $\mathfrak{r}$ or of P as an endomorphism of (E, τ).

There is also a proved comparison using the inherited topology: the identity $(K, \tau|_K) \to (K, \tau_P)$ is continuous. Indeed for V open in τ_P, $V = K \cap P^{-1}(V)$, which is inherited-open. It

induces a continuous bijection $(K,\tau|_K)/N_0 \to Q_{\mathrm{ar}}$. No equality of these two topologies on K is asserted. For the growth topology the stronger result is available: $\mathfrak{r}$ and P are continuous by the Cauchy bound, so $f \mapsto (Pf, \mathfrak{r}(f))$ and $(k,a) \mapsto k + ae$ are inverse continuous linear maps

$$(E,\gamma) \simeq (K,\gamma|_K) \oplus \mathbb{C}. \tag{290}$$

The continuity of $a \mapsto ae$ follows from each defining seminorm. This proves the full split topology at precisely the topology where its hypotheses have been established.

J.3 Every finite jet at the original coordinate origin

The residue calculation is a particular finite-jet phenomenon. For $d \geq 0$ retain the derivative coordinates $J_d f = (f(0), f'(0), \dots, f^{(d)}(0))$. Write the convergent germ $1/\Xi(s) = \sum_{n\geq 0} b_n s^n$; its coefficients are given without any change of Ξ by

$$b_0 = \Xi(0)^{-1}, \qquad b_n = -\Xi(0)^{-1} \sum_{k=1}^{n} \frac{\Xi^{(k)}(0)}{k!}\, b_{n-k}.$$

For $a = (a_0, \dots, a_d)$ define the entire polynomial trace

$$E_d a(s) = \Xi(s) \sum_{n=0}^{d} \left(\sum_{j=0}^{n} \frac{a_j}{j!} b_{n-j} \right) s^n. \tag{291}$$

Multiplying its truncated Taylor series by that of Ξ gives $E_d a(s) = \sum_{j=0}^{d} a_j s^j/j! + O(s^{d+1})$, because the defining reciprocal convolution has coefficient one at zero and zero at each positive degree. Therefore $J_d E_d = 1$ and $\operatorname{im} E_d \subset \mathbb{C}[s]\Xi \subset N$. This proves $J_d(N) = \mathbb{C}^{d+1}$, with its complete finite-dimensional section, rather than only the scalar image. For any $q = [f]_N$ and any jet a, the representative $f + E_d(a - J_d f)$ realizes (q,a).

Put $P_d = 1 - E_d J_d$, $K_d = \ker J_d$, and $N_d = N \cap K_d$. The same identities, now with $\ker P_d = \operatorname{im} E_d$, prove $P_d(N) = N_d$ and $P_d^{-1}(N_d) = N$. Thus $(K_d, (P_d)_*\tau)/N_d \simeq Q_{\mathrm{ar}}$ by the two quotient-map arguments just given. Cauchy's bounds $|f^{(j)}(0)| \leq j! R^{-j} \sup_{|s|\leq R} |f(s)|$ prove the continuous growth-topology decomposition $(E,\gamma) \simeq (K_d, \gamma|_{K_d}) \oplus \mathbb{C}^{d+1}$. All factorials are retained in the derivative coordinates and in the section.

J.4 Heat transport of the retained residue and its quotient

The scalar observable also has an exact heat transport. Define

$$\mathfrak{r}_t = \mathfrak{r} U_{-t}, \qquad e_t = U_t e, \qquad P_t = 1 - e_t \mathfrak{r}_t = U_t P U_{-t}, \qquad K_t^{\mathrm{res}} = \ker \mathfrak{r}_t.$$

These definitions retain $U_t = \exp(-tD^2/4)$ and hence give the convergent formula

$$\mathfrak{r}_t(f) = 36 \sum_{k=0}^{\infty} \frac{(t/4)^k}{k!} f^{(2k+1)}(0) + \frac{15}{16} \sum_{k=0}^{\infty} \frac{(t/4)^k}{k!} f^{(2k+2)}(0). \tag{292}$$

The earlier heat-series proof gives convergence in the growth topology, locally uniformly in t; applying the two continuous derivative evaluations gives the displayed convergence and its value. In particular $\mathfrak{r}_t U_t = \mathfrak{r}$, $\mathfrak{r}_t(e_t) = 1$, and $e_t \in N_t$. Also $U_t K = K_t^{\mathrm{res}}$ and $U_t N_0 = N_t \cap K_t^{\mathrm{res}}$ by these identities and invertibility of U_t.

Give K_t^{res} the topology $(P_t)_*\tau_t$. It equals $(U_t)_*\tau_P$: for $V \subset K_t^{\mathrm{res}}$, the identity $P_t U_t = U_t P$ gives $U_{-t}(P_t^{-1}(V)) = P^{-1}(U_{-t}V)$, which is exactly the equivalence of the two defining open-set conditions. Thus U_t gives a homeomorphism between the residue-zero quotient in (289) and

$$(K_t^{\mathrm{res}}, (P_t)_*\tau_t)/(N_t \cap K_t^{\mathrm{res}}).$$

The latter is homeomorphic to $(E,\tau_t)/N_t$ by $[k]\mapsto[k]_{N_t}$, with inverse $[f]\mapsto[P_tf]$; the two quotient universal properties apply exactly as above. The two homeomorphisms commute with the already proved heat homeomorphism of the actual source quotient because both send $[f]$ to $[U_tf]$. Finally, for each $q=[f]_{N_t}$ and $a\in\mathbb{C}$, the unchanged class has the representative $f+e_t(a-\mathfrak{r}_t(f))$ with transported residue a. This computes the full transported scalar relation without identifying $\mathfrak{r}_t$ with the fixed physical functional $\mathfrak{r}$.

J.5 Every one-point finite jet of the moving actual relation

The complete finite-jet statement persists on the actual transported source relation, at every point once the time is nonzero. Throughout this subsection the derivative coordinate is the original s, the time may be complex, and $N_t=U_tN_{\mathrm{ar}}$ retains its proved closure topology τ_t.

Proposition J.1 (Full finite jets on the moving source relation)**.** *For every $t\in\mathbb{C}\setminus\{0\}$, $\lambda\in\mathbb{C}$, and integer $d\geq 0$, put*

$$J_{\lambda,d}f=(D^jf(\lambda))_{j=0}^{d}.$$

Then

$$J_{\lambda,d}(U_t(\mathbb{C}[s]\Xi))=J_{\lambda,d}(N_t)=\mathbb{C}^{d+1}. \tag{293}$$

There is a continuous linear right inverse of this jet map from the usual finite-dimensional space into $(\mathcal{H}_{\leq 1},\tau_t)$, whose image consists of finite linear combinations of the actual transported polynomial traces.

Proof. Suppose a complex linear functional

$$Lf=\sum_{j=0}^{d}c_jD^jf(\lambda)$$

annihilates $U_t(\mathbb{C}[s]\Xi)$. We first justify the full exponential generating calculation in the established growth topology, without invoking convergence in τ or τ_t. For every $1<p<2$, $a>0$, and $n\geq 1$, direct maximization gives

$$\|s^n\Xi\|_{p,a}\leq\|\Xi\|_{p,a/2}\left(\frac{2n}{ape}\right)^{n/p}. \tag{294}$$

Indeed the maximum of $r^n\exp(-ar^p/2)$ occurs at $r^p=2n/(ap)$. The $n=0$ term is controlled directly by $\|\Xi\|_{p,a}$. For every $A>0$, the sum of the right side times $A^n/n!$ converges: the elementary bound $n!\geq(n/\mathrm{e})^n$ makes its n-th root tend to zero. Each fixed derivative of the parameter power series adds only a polynomial factor in n, together with a fixed shift of its power. The same root estimate proves locally uniform convergence of all those differentiated series in every growth seminorm. Thus

$$\alpha\longmapsto e^{\alpha s}\Xi(s)=\sum_{n=0}^{\infty}\frac{\alpha^n}{n!}s^n\Xi(s)$$

is entire with values in $(\mathcal{H}_{\leq 1},\gamma)$, with the displayed parameter derivatives. The proved continuity of U_t in γ, and Cauchy's estimate for every derivative evaluation in L, justify applying LU_t to every series. All its Taylor coefficients vanish by the assumed annihilation. Hence $LU_t(e^{\alpha s}\Xi)=0$ for every $\alpha\in\mathbb{C}$.

The exact heat and exponential identity, retaining its original sign and coefficient, is

$$U_t(e^{\alpha s}\Xi)(s)=e^{\alpha s-t\alpha^2/4}Z_t(s-t\alpha/2).$$

It also follows directly by applying the convergent heat series to $D(e^{\alpha s}f) = e^{\alpha s}(D+\alpha)f$ and expanding the commuting operators D and the scalar α. Leibniz's rule therefore gives

$$0 = e^{\alpha\lambda - t\alpha^2/4} \sum_{j=0}^{d} c_j \sum_{k=0}^{j} \binom{j}{k} \alpha^{j-k} Z_t^{(k)}(\lambda - t\alpha/2). \tag{295}$$

The exponential is nonzero. Set $w = \lambda - t\alpha/2$, $b = 2/t$, and $x = w - \lambda$. Since $t \neq 0$, w ranges over all of $\mathbb{C}$, and $\alpha = -bx$. Substitution in (295) gives

$$\sum_{j=0}^{d} c_j \mathcal{N}_j Z_t = 0, \qquad \mathcal{N}_j = \sum_{k=0}^{j} \binom{j}{k} (-bx)^{j-k} D_w^k. \tag{296}$$

The polynomial factors in this formula multiply after the indicated derivatives are taken. They do not commute with D_w; in particular this formula is not an uncorrected power of $D_w - bx$.

Here is the exact correction at every order. Define

$$A = D_w - bx, \qquad H_j(X;b) = j! \sum_{\ell=0}^{\lfloor j/2 \rfloor} \frac{(b/2)^\ell X^{j-2\ell}}{\ell!(j-2\ell)!}. \tag{297}$$

Then $\mathcal{N}_j = H_j(A;b)$. To prove this coefficient identity on any entire f, use

$$G(u,w) = e^{-bu(w-\lambda)} f(w+u) = \sum_{j \geq 0} \frac{u^j}{j!} \mathcal{N}_j f(w).$$

Differentiating the left side in both variables gives $\partial_u G = (A + bu)G$. Comparing its Taylor coefficients yields

$$\mathcal{N}_0 = 1, \quad \mathcal{N}_1 = A, \quad \mathcal{N}_{j+1} = A\mathcal{N}_j + jb\mathcal{N}_{j-1} \quad (j \geq 1).$$

The explicitly displayed polynomials have exponential generating series $\exp(uX + bu^2/2)$ and hence satisfy the same recurrence and initial values. Induction proves the claimed identity. For example, $\mathcal{N}_2 = A^2 + b$ and $\mathcal{N}_3 = A^3 + 3bA$, including the terms created by noncommutation. This argument compares finite-order differential operators coefficient by coefficient and makes no claim that multiplication by an exponential quadratic preserves $\mathcal{H}_{\leq 1}$.

In the coordinate w, retain the proved conjugated source operator $B_t = w - tD_w/2 = U_t M U_{-t}$. Then

$$A = -\frac{2}{t}(B_t - \lambda), \qquad U_{-t} A U_t = -\frac{2}{t}(M - \lambda).$$

All the finite-order operators in (296) act continuously in γ. Applying U_{-t} to that identity and using $Z_t = U_t \Xi$ therefore gives the exact entire identity

$$\mathcal{P}(w)\Xi(w) = 0, \qquad \mathcal{P}(w) = \sum_{j=0}^{d} c_j H_j\left(-\frac{2}{t}(w-\lambda); \frac{2}{t}\right). \tag{298}$$

If c_J is the last nonzero coefficient, $H_J(X;b)$ is monic of degree J, while every preceding polynomial has smaller degree. Thus $\mathcal{P}$ has leading coefficient $c_J(-2/t)^J \neq 0$. But $\Xi(0) \neq 0$, so (298) forces $\mathcal{P}$ to vanish on a disk about zero and hence to be the zero polynomial, a contradiction. The annihilating functional is therefore zero.

A proper linear subspace of $\mathbb{C}^{d+1}$ has a nonzero annihilating functional: extend a basis of the subspace to a basis of the whole space and take a coordinate functional on one of the added basis vectors. The result just proved therefore gives the first equality in (293). Since every transported polynomial trace belongs to N_t, it gives the second equality too.

Here is a finite construction of the promised section. Enumerate $h_n = U_t(s^n\Xi)$, $n = 0, 1, \ldots$. Choose n_0 to be the first index whose jet column is nonzero, and successively choose n_ℓ to be the first index whose jet column is outside the span of the previously selected columns. Full rank, already proved, makes this selection terminate at $d+1$ columns. Define

$$G_{j\ell} = D^j h_{n_\ell}(\lambda) \quad (0 \le j, \ell \le d), \qquad \mathcal{S}_{t,\lambda,d}(v) = \sum_{\ell=0}^{d} h_{n_\ell}(G^{-1}v)_\ell. \tag{299}$$

The determinant of G is nonzero by its construction; equivalently $G^{-1} = \operatorname{adj}(G)/\det G$. Direct multiplication gives $J_{\lambda,d}\mathcal{S}_{t,\lambda,d} = 1$. Its image is a finite-dimensional subspace of $U_t(\mathbb{C}[s]\Xi)$. Each scalar coefficient in the finite sum is a continuous linear function of v, and scalar multiplication and addition are continuous in the source vector topology τ_t. This proves continuity of the section into precisely $(\mathcal{H}_{\le 1}, \tau_t)$, as claimed. □

There is an exact topological quotient for these representatives, without requiring the jet map itself to be τ_t-continuous. For the fixed $t \neq 0$, λ, and d above, put

$$K_{\lambda,d} = \ker J_{\lambda,d}, \qquad P_{t,\lambda,d} = 1 - \mathcal{S}_{t,\lambda,d}J_{\lambda,d}, \qquad N_{t,\lambda,d} = N_t \cap K_{\lambda,d},$$
$$\theta_{t,\lambda,d} = (P_{t,\lambda,d})_*\tau_t.$$

Here the last topology is the quotient topology on $K_{\lambda,d}$; a set is open exactly when its inverse image under $P_{t,\lambda,d}$ is τ_t-open. Since the section has image in N_t and is a right inverse, direct substitution gives

$$P_{t,\lambda,d}^2 = P_{t,\lambda,d}, \qquad P_{t,\lambda,d}(N_t) = N_{t,\lambda,d}, \qquad P_{t,\lambda,d}^{-1}(N_{t,\lambda,d}) = N_t.$$

The last equality proves that $N_{t,\lambda,d}$ is closed in the stated quotient topology. There is a homeomorphism

$$(K_{\lambda,d}, \theta_{t,\lambda,d})/N_{t,\lambda,d} \xrightarrow{[k]\mapsto[k]_{N_t}} Q_t, \qquad [f]_{N_t} \longmapsto [P_{t,\lambda,d}f] \tag{300}$$

as its inverse. Well-definedness and both inverse identities follow from the three displayed relations and $f - P_{t,\lambda,d}f \in N_t$. The forward map is continuous because its composition with the quotient map $P_{t,\lambda,d}$ and the further quotient by $N_{t,\lambda,d}$ is the original continuous quotient $\mathcal{H}_{\le 1} \to Q_t$. The reverse map is continuous because it is induced by that continuous composite quotient map out of $(\mathcal{H}_{\le 1}, \tau_t)$ with kernel N_t. As before, restriction of $P_{t,\lambda,d}$ to its range is the identity, proving the continuous identity map from the inherited-topology range to the quotient-topology range. This induces a continuous bijection of their quotients; inverse continuity for the inherited topology has not been inserted.

Finally apply the proposition with $\lambda = 0$ and $d = 2$. The explicit choice

$$\varepsilon_t = \mathcal{S}_{t,0,2}(0, 0, 16/15) \quad (t \neq 0), \qquad \varepsilon_0 = \frac{8s^2\Xi(s)}{15\Xi(0)} \tag{301}$$

belongs to N_t and satisfies $\mathfrak{r}(\varepsilon_t) = 36\varepsilon_t'(0) + (15/16)\varepsilon_t''(0) = 1$. At $t = 0$ this is the previously computed polynomial trace; at every nonzero time the section fixes the two required derivatives to 0 and 16/15. Therefore, at every complex time including zero,

$$\mathfrak{r}(N_t) = \mathbb{C}, \qquad \{([f]_{N_t}, \mathfrak{r}(f)) : f \in \mathcal{H}_{\le 1}\} = Q_t \times \mathbb{C}. \tag{302}$$

For each pair $([f]_{N_t}, a)$ the exact representative $f + (a - \mathfrak{r}(f))\varepsilon_t$ proves surjectivity onto that pair. Put $P_t^{\mathfrak{r}} = 1 - \varepsilon_t\mathfrak{r}$ and retain $K = \ker\mathfrak{r}$. The identities

$$P_t^{\mathfrak{r}}(N_t) = N_t \cap K, \qquad (P_t^{\mathfrak{r}})^{-1}(N_t \cap K) = N_t$$

give, by the same two quotient-map arguments, the homeomorphism

$$(K, (P_t^{\mathrm{r}})_* \tau_t)/(N_t \cap K) \cong Q_t, \qquad [k] \mapsto [k]_{N_t}, \quad [f]_{N_t} \mapsto [P_t^{\mathrm{r}} f]. \tag{303}$$

This retains the actual transported source topology and the exact physical residue at each fixed time. No regular dependence of the selected finite pivot columns on t is needed or asserted. In particular, the full scalar ambiguity holds without a statement that the original frozen relation N_{ar} is invariant under heat, and without identifying that relation with $\Xi\mathcal{H}_{\leq 1}$.

K The Cartan construction and an exact antiunitary reversal

The supplied modular–Cartan manuscript [11] contains both a sector conjugation preserving a logarithmic generator and a standard-form involution reversing the left-minus-right generator. The following construction retains these two equations explicitly. It constructs the negative radial direction, its domains, and its maps to the arithmetic heat space and moving source quotients.

K.1 The generator equation determines the reflected parameter

Let $\mathscr{H}$ be a complex Hilbert space, K a self-adjoint operator, and E_K its projection-valued spectral measure. Define

$$\begin{aligned} T(\zeta) &= e^{-i\zeta K}, \qquad \zeta = a + ib \in \mathbb{C}, \\ \operatorname{Dom} T(\zeta) &= \left\{ f \in \mathscr{H} : \int_{\mathbb{R}} e^{2b\lambda}\, d\langle E_K(\lambda) f, f\rangle < \infty \right\}. \end{aligned} \tag{304}$$

This is a closed densely defined normal operator for every b, including every negative b. Density follows because the spectral cutoffs $E_K([-n,n])f$ belong to this domain and converge to f. Closedness follows from the spectral multiplication representation.

Proposition K.1. *Let J be an antiunitary involution and suppose the equality of self-adjoint operators $JKJ = \epsilon K$ holds, where $\epsilon \in \{1, -1\}$; thus it includes equality of domains. Then*

$$JT(\zeta)J = T(-\epsilon\overline{\zeta}), \qquad Je^{zK}J = e^{\epsilon\overline{z}K}. \tag{305}$$

Both equalities include the transported operator domains. In particular, with $\Delta = e^{-K}$, the Tomita relation $J\Delta J = \Delta^{-1}$ is equivalent to $JKJ = -K$ and gives

$$J\Delta^{i(a+ib)}J = \Delta^{i(a-ib)}, \qquad Je^{bK}J = e^{-bK} \quad (b \in \mathbb{R}). \tag{306}$$

Proof. The uniqueness of the spectral resolution gives $JE_K(B)J = E_{\epsilon K}(B)$ for every Borel set B. For a simple scalar function ϕ, antilinearity gives $J\phi(K)J = \overline{\phi}(\epsilon K)$. Uniform approximation proves this for bounded Borel functions. For an arbitrary finite-valued Borel function, apply the bounded identity to $\phi\, \mathbf{1}_{\{|\phi| \leq n\}}$ and pass to the spectral integral. The squared-integrability condition for $|\phi|$ is transported by J, proving equality of the full domains as well as of the values. Taking $\phi(\lambda) = e^{-i\zeta\lambda}$ and $e^{z\lambda}$ proves (305). Applying the same real-valued functional calculus to $-\log$ proves the equivalence involving Δ, whose kernel is zero. Substitution of $\epsilon = -1$ proves (306). □

The Cartan logarithm in the supplied construction is

$$\mathbf{L}_\ell(T) = \frac{1}{2\ell} \log(T^*T), \qquad \ell > 0.$$

For the operator in (304), multiplication of the spectral functions, including the product domain, gives $T(a+ib)^*T(a+ib) = e^{2bK}$. Indeed the product domain is precisely the set on which the squared multiplier $e^{2b\lambda}$ is square-integrable; its integrability also implies that of $e^{b\lambda}$ by Cauchy–Schwarz. Consequently,

$$\mathbf{L}_\ell(T(a+ib)) = \begin{cases} (b/\ell)K & b \neq 0, \quad \text{with domain } \operatorname{Dom} K, \\ 0 & b = 0, \quad \text{with domain } \mathscr{H}. \end{cases} \tag{307}$$

This follows by composing the real spectral functions $\log(e^{2b\lambda}) = 2b\lambda$. The zero slice has been stated separately: the everywhere-defined $\log I$ must not be replaced by a zero operator restricted to $\operatorname{Dom} K$.

K.2 The original heat multiplier and a genuine modular double

Retain the original Fourier variable v and set

$$\begin{aligned} \mathscr{H}_0 &= L^2(\mathbb{R}, dv), \qquad (C_0 f)(v) = \overline{f(-v)}, \\ (Af)(v) &= v^2 f(v), \qquad \operatorname{Dom} A = \{f : v^2 f \in L^2\}, \\ G_t &= e^{tA/4}, \\ \operatorname{Dom} G_t &= \left\{ f : \int_{\mathbb{R}} e^{(\operatorname{Re} t)v^2/2} |f(v)|^2 \, dv < \infty \right\}. \end{aligned} \tag{308}$$

The multiplication operator A is nonnegative self-adjoint, C_0 is an antiunitary involution, and $C_0 A C_0 = A$ with the displayed domain. Thus $C_0 G_t C_0 = G_{\bar{t}}$, which fixes each real heat parameter. For $\operatorname{Re} t \leq 0$, G_t is bounded with norm one. For $\operatorname{Re} t > 0$, it is densely defined and unbounded: unit vectors supported in $[n, n+1]$ have image norm at least $e^{(\operatorname{Re} t)n^2/4}$. Its domain is proper, since $f(v) = e^{-(\operatorname{Re} t)v^2/8}$ belongs to L^2 and not to that domain. Multiplication gives the exact inverse statement

$$\operatorname{Ran} G_t = \operatorname{Dom} G_{-t}, \qquad G_{-t} G_t f = f \quad (f \in \operatorname{Dom} G_t). \tag{309}$$

For the converse inclusion in the range statement, take $f = G_{-t} g$ for $g \in \operatorname{Dom} G_{-t}$; then $G_t f = g \in L^2$, proving $f \in \operatorname{Dom} G_t$. The common invariant space $\mathscr{D} = C_c^\infty(\mathbb{R})$ is a graph core for each G_t: truncate a vector in its weighted L^2 graph norm, and approximate the truncations by smooth compact functions, using equivalence of the weighted and unweighted norms on each fixed compact interval.

There is no antiunitary J_0 on this one sector satisfying $J_0 A J_0 = -A$. Such an equality would make both A and $-A$ nonnegative by preservation of real quadratic forms, forcing $\langle Af, f\rangle = 0$ for every $f \in \operatorname{Dom} A$. A nonzero smooth function supported in $(1,2)$ contradicts this. The exact construction retaining this original A is as follows.

Theorem K.2. *On $\widehat{\mathscr{H}} = \mathscr{H}_0 \oplus \mathscr{H}_0$, define*

$$\begin{aligned} \widehat{K} &= \begin{pmatrix} A & 0 \\ 0 & -A \end{pmatrix}, \quad \widehat{\Delta} = \begin{pmatrix} e^{-A} & 0 \\ 0 & e^{A} \end{pmatrix}, \\ J(f,g) &= (C_0 g, C_0 f), \quad \widehat{G}_t = e^{t\widehat{K}/4} = \begin{pmatrix} G_t & 0 \\ 0 & G_{-t} \end{pmatrix}. \end{aligned} \tag{310}$$

Here $\operatorname{Dom} \widehat{K} = \operatorname{Dom} A \oplus \operatorname{Dom} A$, $\operatorname{Dom} \widehat{\Delta} = \mathscr{H}_0 \oplus \operatorname{Dom} e^A$, *and* $\operatorname{Dom} \widehat{G}_t = \operatorname{Dom} G_t \oplus \operatorname{Dom} G_{-t}$. *Then*

$$J^2 = 1, \quad J\widehat{K}J = -\widehat{K}, \quad J\widehat{\Delta}J = \widehat{\Delta}^{-1}, \quad J\widehat{G}_t J = \widehat{G}_{-\bar{t}}. \tag{311}$$

These are the modular data of an explicitly constructed standard closed real subspace of $\widehat{\mathscr{H}}$.

Proof. The block formulas and $C_0AC_0 = A$ prove all four identities, including their domains. For the last assertion put $Q = e^{-A/2}$. This is a bounded positive injective operator, with dense range, commuting with C_0. Define the closed antilinear operator

$$S = J\widehat{\Delta}^{1/2}, \qquad \operatorname{Dom} S = \mathscr{H}_0 \oplus \operatorname{Dom} Q^{-1}, \qquad S(f,g) = (C_0Q^{-1}g, C_0Qf).$$

Its domain is invariant under S, and direct substitution gives $S^2(f,g) = (f,g)$. Its fixed real subspace is exactly

$$V = \{(f, C_0Qf) : f \in \mathscr{H}_0\}. \tag{312}$$

Both implications follow by substituting the second component in $S(f,g) = (f,g)$. The graph defining V is closed because C_0Q is bounded. If a vector in V equals i times a vector in V, its first components give $f = ig$, and its second components give $-iC_0Qg = iC_0Qg$; injectivity of Q gives $f = g = 0$. Hence $V \cap iV = \{0\}$.

Moreover $V + iV = \mathscr{H}_0 \oplus \operatorname{Ran} Q = \operatorname{Dom} S$. To prove the nontrivial inclusion, for $x \in \mathscr{H}_0$ and $y \in \operatorname{Ran} Q$ put $b = C_0Q^{-1}y$, $f = (x+b)/2$ and $g = (x-b)/(2i)$. Then

$$(f, C_0Qf) + i(g, C_0Qg) = (f + ig, C_0Q(f - ig)) = (x, y).$$

This domain is dense since $\operatorname{Ran} Q$ is dense. Thus V is standard: it is closed real, has zero intersection with iV, and has dense complex span. On that span $S(v+iw) = v - iw$ for $v, w \in V$, as follows from antilinearity and $S|_V = 1$. Its polar decomposition is exactly $S = J\widehat{\Delta}^{1/2}$. This proves the claimed modular real-subspace construction, including its canonical involution. □

For $\widehat{T}(\zeta) = e^{-i\zeta\widehat{K}}$, the Cartan logarithm is (307) with $K = \widehat{K}$, and $\widehat{G}_t = \widehat{T}(it/4)$. In particular the involution sends every real t to $-t$. For nonzero real t, the double is unbounded in one of its two sectors; its domain remains dense. The inclusion $I_+f = (f, 0)$ retains the original multiplier exactly:

$$\widehat{G}_t I_+ f = I_+ G_t f \quad (f \in \operatorname{Dom} G_t).$$

No Hilbert completion of the unnamed arithmetic topology is required for this construction.

K.3 Exact ingress and involution on the actual moving quotients

On the original entire-function space $\mathcal{H}_{\le 1}$ put $(C_E h)(s) = \overline{h(\overline{s})}$. This is an antilinear involution and preserves each previously defined growth seminorm, since $s \mapsto \overline{s}$ preserves $|s|$. For the Fourier transform already used in the test ingress,

$$(F\phi)(s) = \int_{\mathbb{R}} \phi(v) e^{-isv}\, dv,$$

the change of variable $v \mapsto -v$ proves $C_E F = FC_0$ on $\mathscr{D}$. Differentiating the integral and summing the absolutely uniformly convergent exponential series on the compact support proves

$$U_t F = F G_t, \qquad C_E U_t = U_{\overline{t}} C_E. \tag{313}$$

The second equality also follows directly from the convergent entire-space series for U_t and the real coefficient $-1/4$. Thus on $\mathcal{H}_{\le 1} \oplus \mathcal{H}_{\le 1}$ the everywhere-defined maps

$$\widehat{U}_t = (U_t, U_{-t}), \qquad \mathcal{J}(h,k) = (C_E k, C_E h)$$

satisfy $\mathcal{J}\widehat{U}_t\mathcal{J} = \widehat{U}_{-\overline{t}}$ for every complex t. The exact diagram is expressed by

$$(F \oplus F)\widehat{G}_t = \widehat{U}_t(F \oplus F), \qquad (F \oplus F)J = \mathcal{J}(F \oplus F) \quad \text{on } \mathscr{D} \oplus \mathscr{D}.$$

Here $\widehat{G}_t$ on the test space is the restriction of the closed Hilbert operator constructed above.

Keep the actual source topology τ and actual closed trace image N_{ar}. Define its conjugate presentation by $\tau^c = (C_E)_*\tau$ and $N^c = C_E N_{\mathrm{ar}}$. Since Ξ is real entire, $C_E(\mathbb{C}[s]\Xi) = \mathbb{C}[s]\Xi$, and therefore $N^c = \overline{\mathbb{C}[s]\Xi}^{\tau^c}$. This defines the conjugate topology explicitly without assuming C_E is continuous for the unnamed original τ.

Proposition K.3. *For each complex t form the topological quotient*

$$\begin{aligned} \widehat{N}_t &= U_t N_{\mathrm{ar}} \oplus U_{-t} N^c, \\ \widehat{Q}_t &= ((\mathcal{H}_{\le 1}, \tau_t) \oplus (\mathcal{H}_{\le 1}, (\tau^c)_{-t}))/\widehat{N}_t, \qquad \tau_t = (U_t)_*\tau, \quad (\tau^c)_{-t} = (U_{-t})_*\tau^c. \end{aligned} \tag{314}$$

The map $\widehat{U}_t$ induces a complex-linear homeomorphism $\overline{\widehat{U}}_t : \widehat{Q}_0 \to \widehat{Q}_t$. The involution $\mathcal{J}$ induces an antilinear homeomorphism

$$\overline{\mathcal{J}}_t : \widehat{Q}_t \longrightarrow \widehat{Q}_{-\bar{t}}, \qquad \overline{\mathcal{J}}_{-\bar{t}}\overline{\mathcal{J}}_t = 1, \qquad \overline{\mathcal{J}}_t\overline{\widehat{U}}_t = \overline{\widehat{U}}_{-\bar{t}}\overline{\mathcal{J}}_0. \tag{315}$$

For $\iota_t(\phi, \psi) = [F\phi, F\psi] \in \widehat{Q}_t$,

$$\begin{aligned} \ker \iota_t &= F^{-1}(U_t N_{\mathrm{ar}}) \oplus F^{-1}(U_{-t} N^c), \\ \iota_t \widehat{G}_t &= \overline{\widehat{U}}_t \iota_0, \qquad \overline{\mathcal{J}}_t \iota_t = \iota_{-\bar{t}} J. \end{aligned} \tag{316}$$

The inverse images here are taken inside $\mathscr{D}$.

Proof. The direct-sum heat map transports both the specified topology and the specified closed subspace exactly, proving its quotient homeomorphism. Next

$$C_E U_{-t} N^c = U_{-\bar{t}} N_{\mathrm{ar}}, \qquad C_E U_t N_{\mathrm{ar}} = U_{\bar{t}} N^c.$$

The same identities applied to the definitions of the transported topologies prove that $\mathcal{J}$ is an antilinear homeomorphism of the two ambient spaces in (314). It transports their closed relation spaces exactly, so it induces the asserted quotient map. Applying it twice is the identity on representatives. The conjugacy follows from $\mathcal{J}\widehat{U}_t = \widehat{U}_{-\bar{t}}\mathcal{J}$. The kernel formula is exactly the condition that both Fourier components lie in the two relation spaces. Finally (313) proves both equations in (316) on representatives. □

Thus the constructed Hilbert involution is antiunitary and its actual source-quotient image is a proved antilinear homeomorphism. No inner product is asserted on $\widehat{Q}_t$. For real t, this construction provides the requested negative parameter and its exact quotient target. It does not assert the distinct frozen inclusion $U_t N_{\mathrm{ar}} \subseteq N_{\mathrm{ar}}$.

K.4 The exact viscosity sign and tensor Fourier map

Let $\nu > 0$ be the original viscosity and let $\theta \ge 0$ be physical elapsed time. On $L^2(\mathbb{R}^3, dq)$ define $A_\nu = -\nu\Delta_q$ by the unitary Fourier transform

$$(\mathcal{F}_3 f)(k) = (2\pi)^{-3/2} \int_{\mathbb{R}^3} f(q) e^{-ik\cdot q}\, dq.$$

Its domain is exactly $\{f : |k|^2 \mathcal{F}_3 f \in L^2\}$, and its multiplier is $\nu|k|^2$. Integration by parts verifies this differential expression on Schwartz functions, and the spectral multiplication domain supplies the closed realization. On the frequency tensor product $L^2(\mathbb{R})^{\otimes 3} = L^2(\mathbb{R}^3)$ one has

$$\mathcal{F}_3 e^{-\theta A_\nu} \mathcal{F}_3^{-1} = e^{-\nu\theta|k|^2} = G_{-4\nu\theta} \otimes G_{-4\nu\theta} \otimes G_{-4\nu\theta}. \tag{317}$$

The equality holds first on simple tensors, where multiplication of the three factors gives the middle multiplier, and then on all of L^2 by their common boundedness and tensor density. On $\mathscr{D}^{\otimes 3}$ the original unnormalized Fourier transform $F^{\otimes 3}$ is $(2\pi)^{3/2}\mathcal{F}_3$ with the variables renamed; this scalar is retained. Each transformed test is entire in the three coordinates, and (313) gives $U^{\otimes 3}_{-4\nu\theta}$ on that entire tensor test space. This proves the sign and coefficient in the viscous bridge. The multiplier also acts componentwise on vector fields and preserves the divergence-free subspace: in Fourier coordinates $k \cdot \widehat{u}(k) = 0$ is unchanged by scalar multiplication. Its inverse at $\theta > 0$ has precisely the domain $\{f : e^{\nu\theta|k|^2}\mathcal{F}_3 f \in L^2\}$. These identities identify the linear viscous operator, including its inverse domain; they do not replace the nonlinear velocity, pressure, or forcing equations.

K.5 The public viscous source and its exact proof status

The inspected public manuscript [12] has two preserved revisions. Its earlier 165-page PDF and its later 166-page PDF have different recorded byte hashes. The later revision is the one served after 19:09:35 UTC on 8 September 2026; its first three pages were inspected directly. No equivalence of the two complete proofs is asserted. The pinned source repository commit is recorded separately in the bibliography and provenance.

Theorem 1.1 in that manuscript is an imported source claim: for every $\nu > 0$ it asserts existence of $f \in C_c^\infty(\mathbb{R}^3 \times (0,\infty); \mathbb{R}^3)$, a compact spatial set K, and smooth u, p on $\mathbb{R}^3 \times [0,1)$ satisfying

$$\partial_\theta u + (u \cdot \nabla)u - \nu\Delta u + \nabla p = f, \qquad \nabla \cdot u = 0, \qquad u(\cdot, 0) = 0,$$

with the support and norm conditions

$$\operatorname{supp} u(\cdot,\theta) \cup \operatorname{supp} p(\cdot,\theta) \subseteq K, \qquad \sup_{0\le\theta<1} \|u(\cdot,\theta)\|_2 < \infty, \qquad \limsup_{\theta\uparrow 1} \|u(\cdot,\theta)\|_\infty = \infty.$$

It also asserts nonexistence of a global smooth competitor with the same force and initial datum and uniformly bounded kinetic energy. These assertions have not been independently proved in this fragment or checked here by a Lean kernel. The identities in (317) are independently proved above and match the exact ordinary Laplacian and positive viscosity in this source equation.

The updated manuscript credits the preceding constructions of Diego Córdoba and Luis Martínez-Zoroa, and the hypodissipative work with Zheng. Separately, the primary statement [13] attributes announced smooth-forcing results for incompressible porous media, Boussinesq, and three-dimensional incompressible Euler to Levent Alpöge and Tristan Buckmaster, and credits the Córdoba–Martínez-Zoroa programme. Those are distinct source attributions. The operator calculations here do not decide any private-data or appropriation allegation.

K.6 The retained spatial pullback, exact diffusion leakage, and feedback

Keep the original Cartesian coordinates $q = (x, y, w)$ and polynomials

$$Q = 1 + xy, \qquad \sigma(q) = \tfrac{1}{2}\left[Q^3 w + y^2 Q(4 + 3xy)\right], \qquad \eta(s) = (0, 0, 2s).$$

Put $\mathscr{H} = \mathcal{O}(\mathbb{C}_s)$ and $\mathscr{G} = \mathcal{O}(\mathbb{C}^3_q)$, with uniform convergence on compact sets, and define the actual pullback and section maps

$$\mathcal{I} = \sigma^* : \mathscr{H} \longrightarrow \mathscr{G}, \qquad E_\eta = \eta^* : \mathscr{G} \longrightarrow \mathscr{H}.$$

Substitution gives $\sigma\eta(s) = s$, so $E_\eta\mathcal{I} = I$. If $K \subset \mathbb{C}^3$ is compact, then $\sigma(K)$ is compact and $\sup_K |\mathcal{I}h| = \sup_{\sigma(K)} |h|$. The corresponding identity for η proves continuity of both maps. Thus $P = \mathcal{I}E_\eta$ is a continuous projection and

$$\mathscr{G} = \mathcal{I}\mathscr{H} \oplus \ker E_\eta. \tag{318}$$

Indeed $F = \mathcal{I}E_\eta F + (F - \mathcal{I}E_\eta F)$; the second summand is in $\ker E_\eta$, and $\mathcal{I}h \in \ker E_\eta$ implies $h = 0$. These decomposition maps and their inverse, addition, are continuous. In particular $\mathcal{I}\mathscr{H} = \ker(I - P)$ is closed. This analytic splitting does not identify the source topology τ with a compact-open topology.

Retain

$$D = \partial_s, \quad \Delta = \partial_x^2 + \partial_y^2 + \partial_w^2, \quad g = \nabla\sigma, \quad N = \sum_{i=1}^{3} g_i^2, \quad L = \Delta\sigma.$$

The sum defining N is polynomial over $\mathbb{C}$ and is the Euclidean squared norm on real q. Coordinate differentiation gives

$$\partial_i(\mathcal{I}h) = g_i\mathcal{I}h', \qquad \partial_i^2(\mathcal{I}h) = g_i^2\mathcal{I}h'' + (\partial_i^2\sigma)\mathcal{I}h', \qquad \Delta\mathcal{I}h = N\mathcal{I}h'' + L\mathcal{I}h'.$$

The full coefficients before any evaluation are

$$\begin{aligned} g_1 &= \tfrac{3}{2}yQ^2w + \tfrac{1}{2}y^3(7 + 6xy), \\ g_2 &= \tfrac{3}{2}xQ^2w + \tfrac{1}{2}(8y + 21xy^2 + 12x^2y^3), \\ g_3 &= \tfrac{1}{2}Q^3, \\ L &= 3wQ(x^2 + y^2) + 3y^4 + 18x^2y^2 + 21xy + 4. \end{aligned}$$

At $\eta(s)$ these give $g = (0, 0, 1/2)$, $N = 1/4$, $L = 4$. Consequently the compressed operator and the full complementary component are

$$A = E_\eta\Delta\mathcal{I} = \tfrac{1}{4}D^2 + 4D : \mathscr{H} \longrightarrow \mathscr{H}, \tag{319}$$

$$W = (I - P)\Delta\mathcal{I}, \qquad Wh = (N - \tfrac{1}{4})\mathcal{I}h'' + (L - 4)\mathcal{I}h' : \mathscr{H} \longrightarrow \ker E_\eta, \tag{320}$$

$$\Delta\mathcal{I} = \mathcal{I}A + W.$$

Every map is continuous because composition, differentiation, and multiplication by a fixed polynomial are continuous on the specified spaces.

Proposition K.4. *The exact kernel of W is the constant functions. Equivalently,*

$$\{h \in \mathscr{H} : \Delta\mathcal{I}h \in \mathcal{I}\mathscr{H}\} = \mathbb{C}.$$

Proof. On the entire two-parameter family $(x, 0, 2s)$, direct substitution gives

$$\sigma(x, 0, 2s) = s, \quad N(x, 0, 2s) = \tfrac{1}{4} + 9x^2s^2, \quad L(x, 0, 2s) = 4 + 6x^2s.$$

Thus $Wh = 0$, evaluated with $x = 1$, implies $9s^2h'' + 6sh' = 0$ for every $s \in \mathbb{C}$. Write the convergent entire Taylor series $h(s) = \sum_{n\geq 0} a_n s^n$. The coefficient of s^n in this identity is $3n(3n - 1)a_n$. For each integer $n \geq 1$ its scalar factor is nonzero, so every such a_n vanishes. Conversely all derivatives of a constant vanish and therefore $Wh = 0$. Finally, by the splitting, $\Delta\mathcal{I}h \in \mathcal{I}\mathscr{H}$ is equivalent to $(I - P)\Delta\mathcal{I}h = 0$. □

The splitting also supplies a derivative extension with an explicit common domain and codomain:

$$B = \mathcal{I}DE_\eta : \mathscr{G} \longrightarrow \mathscr{G}, \qquad B\mathcal{I} = \mathcal{I}D.$$

The commutator with the Euclidean Laplacian is consequently well typed. For every $h \in \mathscr{H}$ it is

$$\begin{aligned} [\Delta, B]\mathcal{I}h &= \Delta\mathcal{I}h' - \mathcal{I}D(E_\eta\Delta\mathcal{I}h) \\ &= N\mathcal{I}h''' + L\mathcal{I}h'' - \mathcal{I}(\tfrac{1}{4}h''' + 4h'') \\ &= (N - \tfrac{1}{4})\mathcal{I}h''' + (L - 4)\mathcal{I}h'' = W(h'). \end{aligned} \tag{321}$$

The just-proved kernel formula gives the exact consequence $[\Delta, B]\mathcal{I}h = 0$ if and only if $h(s) = a + bs$ for some $a, b \in \mathbb{C}$. This conclusion requires no invariance of $\mathcal{I}\mathscr{H}$ under Δ.

Compression does not reproduce repeated diffusion. To compute the feedback exactly, write $H = D_q^2\sigma$. The full product rule gives

$$\begin{aligned}
\Delta(N\mathcal{I}h'') &= (\Delta N)\mathcal{I}h'' + 2(g\cdot\nabla N)\mathcal{I}h^{(3)} + NL\mathcal{I}h^{(3)} + N^2\mathcal{I}h^{(4)},\\
\Delta(L\mathcal{I}h') &= (\Delta L)\mathcal{I}h' + 2(g\cdot\nabla L)\mathcal{I}h'' + L^2\mathcal{I}h'' + LN\mathcal{I}h^{(3)},\\
\partial_j N &= 2\sum_i g_i H_{ij},\\
\Delta N &= 2\sum_{i,j} H_{ij}^2 + 2g\cdot\nabla L, \qquad g\cdot\nabla N = 2g^{\mathsf{T}}Hg.
\end{aligned}$$

Adding gives the identity on every entire h,

$$\begin{aligned}
\Delta^2\mathcal{I}h = {} & N^2\mathcal{I}h^{(4)} + (4g^{\mathsf{T}}Hg + 2NL)\mathcal{I}h^{(3)}\\
& + (2\sum_{i,j} H_{ij}^2 + 4g\cdot\nabla L + L^2)\mathcal{I}h'' + (\Delta L)\mathcal{I}h'.
\end{aligned}$$

Differentiating the original g_i, L and evaluating at the same section $\eta(s)$ gives

$$H(\eta(s)) = \begin{pmatrix} 0 & 3s & 0\\ 3s & 4 & 0\\ 0 & 0 & 0\end{pmatrix}, \qquad \sum_{i,j} H_{ij}(\eta(s))^2 = 18s^2 + 16,$$

$$(g^{\mathsf{T}}Hg)(\eta(s)) = 0, \quad (g\cdot\nabla L)(\eta(s)) = 0, \quad (\Delta L)(\eta(s)) = 24s.$$

For the last equality, the term $3w(x^2+y^2)$ in L contributes $12w = 24s$ after the two derivatives and substitution; all other terms give zero there. Therefore

$$\begin{aligned}
E_\eta\Delta^2\mathcal{I} &= \tfrac{1}{16}D^4 + 2D^3 + (36s^2+48)D^2 + 24sD,\\
E_\eta\Delta^2\mathcal{I} - A^2 &= (36s^2+32)D^2 + 24sD = E_\eta\Delta W.
\end{aligned} \tag{322}$$

The last equality also follows directly from $\Delta\mathcal{I} = \mathcal{I}A + W$. It retains the component that leaves the pullback subspace and returns at the second diffusion. It is an identity of differential operators; no Euclidean heat semigroup on all of $\mathscr{G}$ is presumed.

Finally let $h \in \mathcal{H}_{\le 1}$ and use the established entire evolution $h_t = U_t h$, $\partial_t h_t = -D^2 h_t/4$. For $\nu > 0$ put

$$f(\tau, q) = h_{-4\nu\tau}(\sigma(q)).$$

The time chain rule gives $\partial_\tau f = \nu h''_{-4\nu\tau}(\sigma(q))$. Thus the exact scalar viscous residual is

$$(\partial_\tau - \nu\Delta)f = \nu\big[(1 - N(q))h''_{-4\nu\tau}(\sigma(q)) - L(q)h'_{-4\nu\tau}(\sigma(q))\big]. \tag{323}$$

For a specified smooth spatial advection field V, its complete additional term is

$$(V\cdot\nabla\sigma)h'_{-4\nu\tau}(\sigma(q)).$$

In particular the retained material identity $V\cdot\nabla\sigma = 1$ gives coefficient $1 - \nu L$ for the first derivative. Setting $h = \Xi$ recovers $Z_{-4\nu\tau}$ with the factor -4ν and the original Cartesian metric intact. These are scalar equations with explicitly computed forcing. They do not assert the vector momentum equation, its pressure, divergence constraint, or the support, initial-value, and energy conditions of a Navier–Stokes solution.

L The published fluid input and its complete arithmetic profile map

We now apply the public fluid theorem recorded in Section K.5 through a proved map. The complete companion calculation [14] was read. The construction and every estimate used here are proved below, including the polynomial coordinates, real flow, pressure, forcing, physical metric, arithmetic Fourier constants, and inverse. The final existence assertion cites the public Theorem 1.1 as an external mathematical input; it is not an independent verification of that theorem's complete proof.

L.1 The original three coordinates and their full inverse differential

In the original real coordinates $\mathbf{x} = (x, y, w)$ put

$$\begin{aligned} F_1 &= (1+xy)^3 w + y^2(1+xy)(4+3xy), \\ F_2 &= y + 3x(1+xy)^2 w + 3xy^2(4+3xy), \\ F_3 &= 2x - 3x^2 y - x^3 w, \\ \mathbf{S} &= (F_1/2, F_2, F_3)^{\mathsf{T}}, \qquad J_S = D\mathbf{S}. \end{aligned} \tag{324}$$

Thus $S_1 = \sigma$ is exactly the earlier spatial coordinate. The bold notation distinguishes this spatial three-vector from the original quotient multiplication operator S.

Here is a complete determinant calculation. On $x \neq 0$ set $v = 1 + xy$, $h = 2 - 3xy - x^2 w$, and $A = v^2 + v - v^3 h$, $B = 4v + 2 - 3v^2 h$. Direct substitution gives $F = (A/x^2, B/x, xh)$ and

$$\det \frac{\partial(x,v,h)}{\partial(x,y,w)} = -x^3, \qquad \det \frac{\partial F}{\partial(x,v,h)} = x^{-3} \det \begin{pmatrix} -2A & 2v+1-3v^2h & -v^3 \\ -B & 4-6vh & -3v^2 \\ h & 0 & 1 \end{pmatrix} = 2x^{-3}.$$

For the last equality the determinant's numerator is

$$(4-6vh)(-2v^2-2v+3v^3h) + (2v+1-3v^2h)(4v+2-6v^2h) = 2,$$

as expansion verifies. Thus $\det DF = -2$ on $x \neq 0$ and, by polynomial identity, everywhere. Consequently $\det J_S = -1$ everywhere. Define the polynomial columns

$$\begin{aligned} W_1 = -\tfrac{1}{2}\nabla F_2 \times \nabla F_3, \quad W_2 = -\tfrac{1}{2}\nabla F_3 \times \nabla F_1, \quad W_3 = -\tfrac{1}{2}\nabla F_1 \times \nabla F_2, \\ (DF)^{-1} = [\,W_1\ W_2\ W_3\,], \qquad J_S^{-1} = [\,2W_1\ W_2\ W_3\,]. \end{aligned} \tag{325}$$

The scalar triple products with the three rows ∇F_i give $DF\, W_i = e_i$, proving both inverse identities. In particular $\ker(d\sigma)_{\mathbf{x}} = \operatorname{span}_{\mathbb{R}}\{W_2(\mathbf{x}), W_3(\mathbf{x})\}$: write any vector in the basis (W_1, W_2, W_3) and use $d\sigma(W_i) = \delta_{1i}/2$. No global inverse of F or $\mathbf{S}$ is presumed.

L.2 The complete real flow and its volume-preserving inverse

Fix $K \subset \mathbb{R}^3$ compact, $0 < T < \infty$ and $\nu > 0$. Let $\mathcal{U}_{K,T,\nu}$ consist of smooth real triples (u, p, f) on $[0, T) \times \mathbb{R}^3$ satisfying

$$\partial_t u_i + \sum_{\ell=1}^{3} u_\ell \partial_\ell u_i = \nu \Delta_{\mathbf{x}} u_i - \partial_i p + f_i, \qquad \sum_i \partial_i u_i = 0, \qquad \operatorname{supp} u(t,\cdot) \subseteq K. \tag{326}$$

There is no restriction on p, f beyond this definition for the correspondence; the public theorem supplies the stronger support and initial conditions in its stated example. For any such triple define

$$\partial_t X_t(q) = u(t, X_t(q)), \qquad X_0(q) = q, \qquad A_X(t,q) = D_q X_t(q), \qquad q \in \mathbb{R}^3.$$

For each $T_0 < T$, every spatial derivative of u is bounded on $[0, T_0] \times K$ and vanishes outside K. In particular its global spatial Lipschitz constant is uniform on this time interval. On an interval of length less than its reciprocal, the integral map $Y \mapsto q + \int u(s, Y(s))\, ds$ is a contraction on continuous curves in the uniform norm. Successive such intervals construct the solution to T_0; boundedness of u prevents finite escape. The same construction backward from each endpoint constructs its inverse. Differentiating the integral equation gives a linear integral equation for each first label derivative, and then successively for every higher label and time derivative. The same local contraction and induction prove smoothness and uniqueness of these derivatives. Thus X_t is a smooth global diffeomorphism for every $t < T$.

Differentiation and the determinant product rule give

$$\partial_t A_X = Du(t, X_t) A_X, \quad A_X(0, q) = I_3, \qquad \partial_t \det A_X = (\operatorname{div} u)(t, X_t) \det A_X = 0.$$

The determinant identity first holds where the matrix is invertible, including a neighbourhood of $t = 0$; its value one prevents any first loss of invertibility and continues the identity. Hence $\det A_X = 1$ with positive orientation. Every $q \notin K$ has its constant trajectory. An endpoint outside K also has a constant backward trajectory, so inverse uniqueness proves

$$X_t(K) = K, \qquad X_t(q) = q \ (q \notin K), \qquad X_t(L) \subseteq K \cup L \quad (L \subset \mathbb{R}^3 \text{ compact}). \tag{327}$$

All these conclusions hold on each preterminal interval without a uniform bound on $|u|$ as t approaches T.

L.3 The full complex fibre and both time derivatives

Set $a_j(t, q) = S_j(X_t(q))$ and $v_j(t, q) = (\nabla S_j \cdot u)(t, X_t(q))$. Then a_j is real and $\partial_t a_j = v_j$. At fixed t the map on the actual real and complex variables is

$$(q, \zeta) \longmapsto (q, a_j(t, q) + \zeta), \qquad (q, s) \longmapsto (q, s - a_j(t, q)).$$

These are exact inverse maps, smooth in q and holomorphic on each complex fibre. Write $\mathcal{B} = C^\infty(\mathbb{R}^3; \mathcal{O}(\mathbb{C}_\zeta))$ with seminorms consisting of suprema of finitely many label and spectral derivatives on compact products. The map

$$\mathcal{P}_t : \mathcal{O}(\mathbb{C})^3 \longrightarrow \mathcal{B}^3, \qquad \mathcal{P}_t(h_j)_j = (h_j(a_j(t, q) + \zeta))_j$$

has the left inverse, for any fixed q_*, $(\mathcal{R}_t G)_j(s) = G_j(q_*, s - a_j(t, q_*))$. Substitution proves $\mathcal{R}_t \mathcal{P}_t = I$. Images of compact products have compact spectral argument; the chain rule involves only finitely many bounded derivatives of a_j there. This proves continuity of $\mathcal{P}_t$, while evaluation and translation prove continuity of $\mathcal{R}_t$. Therefore $\mathcal{P}_t$ is a topological isomorphism onto its image with the inherited topology. At $\zeta = 0$ its first component is exactly $h_1(\sigma(X_t(q))) = X_t^* \sigma^* h_1(q)$.

Keep a smooth, explicitly chosen real Newman-time function $\vartheta : [0, T) \to \mathbb{R}$, independently of physical t. For either original profile $h_\theta = Z_\theta$ or K_θ, define

$$\Psi_j(t, q, \zeta) = h_{\vartheta(t)}(\zeta + a_j(t, q)).$$

The previously proved heat identity gives

$$\partial_t \Psi_j = -\frac{\vartheta'}{4} \partial_\zeta^2 \Psi_j + v_j \partial_\zeta \Psi_j. \tag{328}$$

Since v_j has no spectral dependence, another complete differentiation gives

$$\partial_t^2 \Psi_j = \frac{(\vartheta')^2}{16} \partial_\zeta^4 \Psi_j - \frac{\vartheta' v_j}{2} \partial_\zeta^3 \Psi_j + \left(v_j^2 - \frac{\vartheta''}{4} \right) \partial_\zeta^2 \Psi_j + b_j \partial_\zeta \Psi_j, \tag{329}$$

where, with every original momentum term,

$$b_j = \partial_t v_j = \left[\sum_i (\nu \Delta u_i - \partial_i p + f_i) \partial_i S_j + \sum_{i,\ell} u_i u_\ell \partial_i \partial_\ell S_j \right] (t, X_t(q)). \tag{330}$$

Indeed the material derivative of $\nabla S_j \cdot u$ gives the Hessian term and the full material acceleration of u; substituting (326) gives (330). In the second derivative the mixed cubic term occurs twice, each with coefficient $-\vartheta' v_j/4$, proving the factor $-1/2$.

The full Euclidean viscous operator in label coordinates is

$$G_X = A_X^{-1} A_X^{-\mathsf{T}}, \qquad \mathcal{L}_t = \sum_{\alpha,\beta} \partial_{q_\alpha} (G_X^{\alpha\beta} \partial_{q_\beta}).$$

For every smooth scalar b one has $\mathcal{L}_t X_t^* b = X_t^* \Delta b$. To prove it with all first-order coefficient derivatives, test against $\phi \in C_c^\infty(\mathbb{R}^3)$ and use the gradient chain rule and the change of variables with determinant one:

$$\begin{aligned} \int \phi \, \mathcal{L}_t X_t^* b \, dq &= - \int \nabla_q \phi \cdot A_X^{-1} (\nabla b) \circ X_t \, dq \\ &= - \int \nabla_{\mathbf{x}} (\phi \circ X_t^{-1}) \cdot \nabla b \, d\mathbf{x} = \int \phi \, X_t^* \Delta b \, dq. \end{aligned}$$

Smoothness makes this distributional identity pointwise. The ordinary spatial chain rule now proves

$$\begin{aligned} \mathcal{L}_t \Psi_j &= (|\nabla S_j|^2 \circ X_t) \partial_\zeta^2 \Psi_j + (\Delta S_j \circ X_t) \partial_\zeta \Psi_j, \\ (\partial_t - \nu \mathcal{L}_t) \Psi_j &= - \left(\frac{\vartheta'}{4} + \nu |\nabla S_j|^2 \circ X_t \right) \partial_\zeta^2 \Psi_j + (v_j - \nu \Delta S_j \circ X_t) \partial_\zeta \Psi_j. \end{aligned} \tag{331}$$

L.4 Original Fourier constants and exact velocity recovery

Define, with the original arithmetic kernel,

$$h_*(r) = \frac{\pi}{2} r^2 (2\pi r^2 - 3) e^{-\pi r^2}, \qquad k(\omega) = e^{\omega/2} \sum_{n \geq 1} h_*(n e^\omega).$$

For the Fourier convention with exponential $e^{-2\pi i r \xi}$, the second and fourth Gaussian derivative formulas are

$$\widehat{r^2 e^{-\pi r^2}} = (1/(2\pi) - \xi^2) e^{-\pi \xi^2}, \quad \widehat{r^4 e^{-\pi r^2}} = (\xi^4 - 3\xi^2/\pi + 3/(4\pi^2)) e^{-\pi \xi^2}.$$

They follow by differentiating the Gaussian transform twice and four times and dividing by $(-2\pi i)^2$ and $(-2\pi i)^4$, respectively. The coefficients π^2 and $-3\pi/2$ then give $\widehat{h_*} = h_*$ and $h_*(0) = 0$. Poisson summation gives $\sum_{n \geq 1} h_*(n\lambda) = \lambda^{-1} \sum_{n \geq 1} h_*(n/\lambda)$, and hence $k(\omega) = k(-\omega)$. Poisson summation for this Schwartz function follows directly by periodizing it: its Fourier coefficients are the displayed Fourier transform at integer frequencies; the function and coefficient series converge absolutely with derivatives, so evaluation at zero is valid.

Each fixed number of derivatives of the defining series is locally uniformly convergent. At $\omega \geq 0$ its Gaussian terms give $|k^{(r)}(\omega)| \leq C_r e^{C_r \omega} e^{-\pi e^{2\omega}}$; factor the $n = 1$ exponential and bound the remaining polynomially weighted series by $\sum n^{M_r} e^{-\pi(n^2-1)} < \infty$. Evenness gives the same bound with $|\omega|$ at the other end. Expanding the definition proves $\Phi(u) = 2k(2u)$ with every factor retained. Consequently

$$\begin{aligned} Z_\theta(s) &= 4 \int_{\mathbb{R}} k(\omega) e^{\theta \omega^2/4} e^{is\omega} \, d\omega, \\ K_\theta(s) &= 4 \int_{\mathbb{R}} m(\omega) k(\omega) e^{\theta \omega^2/4} e^{is\omega} \, d\omega, \end{aligned} \tag{332}$$

where m is exactly the three-shift, four-term multiplier in (264), including $-i\omega/336$ at shift 3. All derivatives of these amplitudes are Schwartz for real θ, locally uniformly in θ; the same is true of their Fourier transforms by integration by parts.

Set $d_h(\theta)^2 = \int_{\mathbb{R}} |h'_\theta(s)|^2\, ds$. The exact constants are

$$\begin{aligned} d_Z(\theta)^2 &= 32\pi \int_{\mathbb{R}} \omega^2 k(\omega)^2 e^{\theta\omega^2/2}\, d\omega, \\ d_K(\theta)^2 &= 32\pi \int_{\mathbb{R}} \omega^2 |m(\omega)|^2 k(\omega)^2 e^{\theta\omega^2/2}\, d\omega. \end{aligned} \tag{333}$$

For completeness, insert $e^{-\epsilon s^2}$ in the squared norm of $4\int b(\omega) e^{is\omega}\, d\omega$. Fubini and the Gaussian integral give

$$16\sqrt{\pi/\epsilon} \iint b(\omega)\overline{b(\eta)} e^{-(\omega-\eta)^2/(4\epsilon)}\, d\omega\, d\eta.$$

The Gaussian convolution has mass 2π and tends to $2\pi b$ in this integral: split at fixed distance from zero, use uniform continuity on the near part and its Gaussian tail on the far part, with Schwartz domination. On the original side dominated convergence applies because the Fourier transform is Schwartz. The result is $32\pi \int |b|^2$. Taking $b = i\omega k e^{\theta\omega^2/4}$ or $i\omega m k e^{\theta\omega^2/4}$ proves (333). Both integrals are strictly positive: $k(0) = \Phi(0)/2 > 0$ and $m(0) = 127/219024 > 0$ give an interval containing nonzero frequencies where the integrands' non-Gaussian factors are positive. The two-ended bounds prove continuity in θ by domination on compact parameter intervals. Thus on each compact $I \subset \mathbb{R}$ there are $0 < d_- \le d_h(\theta) \le d_+ < \infty$.

Define the full arithmetic temporal defect on the real spectral line by

$$\mathscr{D}_j = \partial_t \Psi_j + \frac{\vartheta'}{4} \partial_s^2 \Psi_j = v_j h'_\vartheta(s + a_j).$$

Translation by the real a_j preserves Lebesgue measure. Therefore

$$\begin{aligned} \sum_j \|\mathscr{D}_j(t,q,\cdot)\|_2^2 &= d_h(\vartheta(t))^2 |v(t,q)|^2, \\ v_j(t,q) &= \frac{\int_{\mathbb{R}} \overline{h'_\vartheta(s+a_j)}\, \mathscr{D}_j(t,q,s)\, ds}{d_h(\vartheta)^2}, \\ u(t, X_t(q)) &= 2W_1(X_t(q))v_1 + W_2(X_t(q))v_2 + W_3(X_t(q))v_3. \end{aligned} \tag{334}$$

These formulas prove an inverse correspondence $(X_t, u \circ X_t) \leftrightarrow (X_t, \mathscr{D})$ on the three real profile lines at each (t,q). The flow and its inverse sheet are retained in both sides.

For nonempty K, put $M_J = \max_K \|J_S\|_{\mathrm{op}}$ and $N_J = \max_K \|J_S^{-1}\|_{\mathrm{op}}$. They are finite and positive. Define

$$\mathcal{A}_\infty(t) = \sup_q \left(\sum_j \|\mathscr{D}_j(t,q,\cdot)\|_2^2 \right)^{1/2}, \quad \mathcal{A}_2(t) = \left(\sum_j \int_{\mathbb{R}^3} \int_{\mathbb{R}} |\mathscr{D}_j|^2\, ds\, dq \right)^{1/2}.$$

When $\vartheta([0,T)) \subseteq I$, the complete estimates are

$$\frac{d_-}{N_J} \|u(t)\|_\infty \le \mathcal{A}_\infty(t) \le d_+ M_J \|u(t)\|_\infty, \qquad \frac{d_-}{N_J} \|u(t)\|_2 \le \mathcal{A}_2(t) \le d_+ M_J \|u(t)\|_2. \tag{335}$$

Indeed $v = J_S(X_t)(u \circ X_t)$, both vanish off K, and $|J_S u| \le M_J |u|$, $|u| \le N_J |J_S u|$ on K. The supremum is preserved by the diffeomorphism X_t. For the integral, its determinant one gives the exact identity

$$\mathcal{A}_2(t)^2 = d_h(\vartheta(t))^2 \int_{\mathbb{R}^3} |J_S(\mathbf{x}) u(t,\mathbf{x})|^2\, d\mathbf{x}.$$

This proves both bounds without discarding any spatial or spectral component. For a fixed original incidence coefficient a_{ef}, the observation $a_{ef} K_\theta$ multiplies $\mathscr{D}^K$ by a_{ef} and its squared norm by $|a_{ef}|^2$; inversion uses its actual nonzero value, and the zero incidence has exactly the zero observation.

L.5 Application to the stated public singularity theorem

Theorem L.1 (Arithmetic consequence of the cited public input)**.** *Take the velocity, pressure, force and compact set supplied by Theorem 1.1 of [12], with its original $\nu > 0$. Choose the explicit Newman time $\vartheta(t) = 0$ and the actual arithmetic profile $h_0 = \Xi$, on $0 \le t < 1$. Then the constructed entire spectral fields satisfy*

$$\mathscr{D}_j = \partial_t \Psi_j, \qquad \mathscr{D}_j(0, q, s) = 0, \qquad \sup_{t<1} \mathcal{A}_2(t) < \infty, \qquad \limsup_{t\uparrow 1} \mathcal{A}_\infty(t) = \infty. \tag{336}$$

Every $\mathscr{D}_j$ is supported in $q \in K$. For every compact $L \subset \mathbb{R}^3$, compact $E_0 \subset \mathbb{C}$ and integer $r \ge 0$,

$$\sup_{t<1,\ q\in L,\ \zeta\in E_0} |\partial_\zeta^r \Psi_j(t, q, \zeta)| < \infty. \tag{337}$$

The analogous temporal-defect conclusions hold for the original full profile K_0 with its exact $d_K(0)$.

Proof. The cited public theorem provides a triple in (326) with $T = 1$, $u(0) = 0$, uniformly bounded $\|u(t)\|_2$, and unbounded terminal $\|u(t)\|_\infty$. The preceding explicit construction therefore supplies $X_t, \Psi, \mathscr{D}$. With the chosen constant Newman time the heat part of the temporal defect vanishes exactly, and $u(0) = 0$ gives $\mathscr{D}(0) = 0$. Set $d_- = d_+ = d_Z(0) > 0$ in (335). Its two inequalities give the energy bound and the unbounded terminal supremum in (336). Support follows from $v = 0$ off K. For the last bound, (327) places the argument of $\Xi^{(r)}$ in the fixed compact set $S_j(K \cup L) + E_0$. Entire continuity gives its finite maximum. All arguments apply to K_0 using the already proved positive constant $d_K(0)$ and its entire derivatives. This proof uses the external existence theorem exactly at its first step; all transfer formulas and estimates are proved within this fragment. □

L.6 The actual cutoff-summed source witness and its arithmetic image

L.6.1 The precise imported witness and its unchanged viscosity

We now use the particular field constructed in the 166-page revision of [12]. The original cutoff-summed fields, the smooth extension of the force, and their all-order estimates are the imported source input. Its actual closing equations (9.21), (10.4)–(10.14) and (10.20)–(10.23) were read directly. The complete companion specialization [15] was also read; all arithmetic, flow, energy and terminal maps used here have their proofs below. No independent verification of every lemma in the public 166-page existence proof is claimed.

Use a star to retain the source's viscosity-one fields, rather than overwriting the physical viscosity $\nu > 0$. In source equation (9.21), page 115, the actual summed Cartesian vector potential, azimuthal field, and pressure are denoted by $A_*, B_* e_\theta, p_{*,\mathrm{loc}}$. These are the cutoff-summed corrected fields of that equation, including all stages and the finite initialization. With the source's own radial-time coordinate denoted q_{src} to distinguish it from our material label q, its exact sums are

$$\begin{aligned} A_* &= A_{*,0} + \sum_{j\ge 1} \chi(a_j q_{\mathrm{src}}) A_{*,j}, \\ B_* e_\theta &= B_{*,0} e_\theta + \sum_{j\ge 1} \chi(a_j q_{\mathrm{src}}) B_{*,j}, \\ p_{*,\mathrm{loc}} &= p_{*,0} + \sum_{j\ge 1} \chi(a_j q_{\mathrm{src}}) p_{*,j}. \end{aligned}$$

The displayed a_j, χ and all finite-state representatives are the actual choices in source (9.21); the subscript star marks viscosity one and introduces no replacement of them. Source equation

(10.4), page 118, constructs

$$u_* = \operatorname{curl}(c_* A_*) + c_* B_* e_\theta, \qquad p_* = c_* p_{*,\mathrm{loc}}, \qquad c_* = \chi_x \chi_t.$$

Here the source spatial cutoff is smooth in (r^2, z), supported in $r < r_0,\ |z| < z_0$, and equals one on $r \le r_0/2,\ |z| \le z_0/2$. Its time cutoff is zero for $1 - t \ge \tau_0$ and one for $0 \le 1 - t \le \tau_0/2$, where $0 < \tau_0 < 1/2$. Retain the actual source choices of these parameters. The source force before time one is exactly

$$f_* = \partial_t u_* + (u_* \cdot \nabla) u_* - \Delta u_* + \nabla p_*.$$

Thus every cutoff derivative belongs to the force. Its smooth extension through time one is source (10.11), page 120, with all its prescribed Taylor coefficients and cutoff sequence. The source proves that its support is contained in $K_* \times [0, 2]$, where $K_* = \operatorname{supp} \chi_x$, and it vanishes near time zero. The pressure is the displayed compactly supported source pressure; no divergence-free condition is imposed on f_*.

For the physical parameter $\nu > 0$, take precisely source (10.22):

$$\begin{aligned} y &= x/\sqrt{\nu}, & x &= \sqrt{\nu}\, y, \\ u_\nu(x,t) &= \sqrt{\nu}\, u_*(x/\sqrt{\nu}, t), & p_\nu(x,t) &= \nu p_*(x/\sqrt{\nu}, t), \\ f_\nu(x,t) &= \sqrt{\nu}\, f_*(x/\sqrt{\nu}, t), & K_\nu &= \sqrt{\nu}\, K_*. \end{aligned}$$

The inverse multiplies the velocity and force by $\nu^{-1/2}$, the pressure by ν^{-1}, and substitutes $x = \sqrt{\nu}\, y$; physical time is unchanged. The chain rule gives

$$\partial_t u_\nu + (u_\nu \cdot \nabla_x) u_\nu - \nu \Delta_x u_\nu + \nabla_x p_\nu = \sqrt{\nu}\, [\partial_t u_* + (u_* \cdot \nabla_y) u_* - \Delta_y u_* + \nabla_y p_*](y,t) = f_\nu.$$

Also $\operatorname{div}_x u_\nu = \operatorname{div}_y u_* = 0$, $u_\nu(0) = 0$, and u_ν, p_ν have support in K_ν for every $t < 1$. These are exactly the inputs of Section L with $T = 1$, with all coordinate and parameter maps specified.

Let

$$F_\nu(t) = \int_0^t \|f_\nu(s)\|_{L^2(\mathbb{R}^3)}\, ds, \qquad E_\nu = F_\nu(1) = \nu^{5/4} F_*(1) < \infty.$$

The last exponent follows from the factor $\sqrt{\nu}$ in f_ν and $dx = \nu^{3/2} dy$. For completeness the energy estimate used below is proved directly for this input. Compact support and incompressibility give, by integrating the equation against u_ν,

$$\frac{1}{2} \frac{d}{dt} \|u_\nu(t)\|_2^2 + \nu \|\nabla u_\nu(t)\|_2^2 = \langle f_\nu(t), u_\nu(t) \rangle.$$

The convection integral is the integral of $\operatorname{div}(u_\nu |u_\nu|^2/2)$, and the pressure integral is minus the integral of $p_\nu \operatorname{div} u_\nu$; both vanish. Divide by $(\|u_\nu\|_2^2 + \delta^2)^{1/2}$, use Cauchy–Schwarz, integrate from zero, and let $\delta \downarrow 0$. This proves $\|u_\nu(t)\|_2 \le F_\nu(t)$. Substitution back into the integrated identity proves, with its full dissipation coefficient,

$$\|u_\nu(t)\|_2^2 + 2\nu \int_0^t \|\nabla u_\nu(s)\|_2^2 ds \le 2 \int_0^t F_\nu'(s) F_\nu(s) ds = F_\nu(t)^2.$$

L.6.2 The source's actual growing component in two arithmetic channels

The source fixes $0 < h < 1/100$, $X_{\mathrm{in}} > 0$, and $e_0 > 0$ through its construction. Its closing proof, (10.20)–(10.21), page 124, gives the actual corrected swirl along

$$x_{*,\tau} = (\sqrt{2 X_{\mathrm{in}} \tau}, 0, 0), \qquad t = 1 - \tau, \qquad (u_*)_2(x_{*,\tau}, 1 - \tau) = \tau^{-1/2-h}(e_0 + R_*(\tau)).$$

The original remainder is retained as the function $R_*(\tau) = \tau^{1/2+h}(u_*)_2(x_{*,\tau}, 1-\tau) - e_0$. The source estimate supplies constants $C_* > 0, \tau_* > 0$ with $|R_*(\tau)| \le C_*\tau^{2h}$ for $0 < \tau < \tau_*$. At these points the cylindrical angle is zero, so $e_\theta = (0,1,0)$ and the displayed Cartesian component is precisely the source swirl. The annular and higher-order corrections have already been accounted for in this remainder. The parameter h is the chosen source parameter; no claim that an arbitrary prescribed h works is needed or made.

Define

$$r_{\nu,\tau} = \sqrt{2\nu X_{\mathrm{in}}\tau}, \qquad x_{\nu,\tau} = (r_{\nu,\tau}, 0, 0).$$

The exact viscosity map proves

$$(u_\nu)_2(x_{\nu,\tau}, 1-\tau) = \sqrt{\nu}\,\tau^{-1/2-h}(e_0 + R_*(\tau)). \tag{338}$$

Use the actual material flow $X_{\nu,t}$ of u_ν proved to exist on each interval before one in Section L. Put

$$X_{\nu,t}(q) = \sqrt{\nu}\,X_{*,t}(q/\sqrt{\nu}).$$

This is an equality of the actual flows: its right side equals q at zero, and its time derivative is $\sqrt{\nu}\,u_*(t, X_{*,t}(q/\sqrt{\nu})) = u_\nu(t, \sqrt{\nu}\,X_{*,t}(q/\sqrt{\nu}))$. The previously proved uniqueness gives the equality, with inverse $X^{-1}_{\nu,t}(x) = \sqrt{\nu}\,X^{-1}_{*,t}(x/\sqrt{\nu})$. Define the labels

$$q_{\nu,\tau} = X^{-1}_{\nu,1-\tau}(x_{\nu,\tau}).$$

This is the exact inverse between the Eulerian sampling points and their material labels. It is a family of labels depending on τ; the source path has not been asserted to be a single fluid trajectory.

Write $S = \mathbf{S} = (F_1/2, F_2, F_3)$ and $J = J_S = DS$ in the physical Cartesian coordinates x, not the viscosity-one coordinate y. Direct substitution into the original polynomials gives

$$S(r,0,0) = (0,0,2r), \qquad J(r,0,0) = \begin{pmatrix} 0 & 0 & 1/2 \\ 0 & 1 & 3r \\ 2 & -3r^2 & -r^3 \end{pmatrix}.$$

Indeed the first row comes from differentiating $F_1/2 = ((1+xy)^3 w + y^2(1+xy)(4+3xy))/2$; the second comes from $F_2 = y + 3x(1+xy)^2 w + 3xy^2(4+3xy)$; the third comes from $F_3 = 2x - 3x^2 y - x^3 w$, all evaluated at $(x,y,w) = (r,0,0)$. Every entry, including the off-diagonal powers of r, is retained. Writing $v = J(r,0,0)u$ therefore gives the exact component inverse

$$u_3 = 2v_1, \qquad u_2 = v_2 - 6rv_1,$$
$$u_1 = \tfrac{1}{2}v_3 + \tfrac{3}{2}r^2 v_2 - 8r^3 v_1.$$

Substituting these expressions into all three rows proves both directions of this matrix inverse.

Fix actual arithmetic time zero. Take $H = Z_0 = \Xi$ or $H = K_0$ with the complete four-term coefficient from the preceding proof, and write

$$d_H^2 = \int_{\mathbb{R}} |H'(s)|^2 ds > 0, \qquad \Psi_{\nu,j}(t,q,\zeta) = H(S_j(X_{\nu,t}(q)) + \zeta), \qquad D_{\nu,j} = \partial_t \Psi_{\nu,j}.$$

The exact positive constants are those of (333) at $\theta = 0$, including 32π and, for K_0, the entire factor $|m|^2$. At the particular pair $(t,q) = (1-\tau, q_{\nu,\tau})$, the first two shifts are both zero. Consequently define the actual integral

$$I_{\nu,j}(\tau) = \frac{1}{d_H^2}\int_{\mathbb{R}} \overline{H'(s)} D_{\nu,j}(1-\tau, q_{\nu,\tau}, s) ds, \qquad j = 1, 2.$$

The chain rule gives $D_{\nu,j} = v_j H'(s)$ there. It follows, without an asymptotic replacement of the profiles, that

$$I_{\nu,2}(\tau) - 6r_{\nu,\tau} I_{\nu,1}(\tau) = \sqrt{\nu}\,\tau^{-1/2-h}(e_0 + R_*(\tau)). \tag{339}$$

There is also a whole-profile identity, stronger than its projection:

$$\begin{aligned} &D_{\nu,2}(1-\tau, q_{\nu,\tau}, s) - 6r_{\nu,\tau} D_{\nu,1}(1-\tau, q_{\nu,\tau}, s) \\ &\qquad = \sqrt{\nu}\,\tau^{-1/2-h}(e_0 + R_*(\tau))H'(s). \end{aligned} \tag{340}$$

Its squared $L^2(ds)$ norm is exactly $\nu\tau^{-1-2h}(e_0+R_*(\tau))^2 d_H^2$. Thus the source's actual component and remainder, not just an abstract possibility of growth, have been carried into the original arithmetic profile. The complex conjugation and integration above retain the complex values of K_0 when that channel is used.

Choose any $0 < \tau_{**} \le \tau_*$ such that $C_*\tau_{**}^{2h} \le e_0/2$ and the source cutoffs equal one on the path for $\tau < \tau_{**}$. Cauchy–Schwarz on the two component coefficients $(-6r_{\nu,\tau}, 1)$ yields

$$\left(\sum_{j=1}^{3} \|D_{\nu,j}(1-\tau, q_{\nu,\tau}, \cdot)\|_2^2\right)^{1/2} \ge \frac{d_H\sqrt{\nu}\,e_0}{2\sqrt{1+72\nu X_{\mathrm{in}}\tau}}\,\tau^{-1/2-h}. \tag{341}$$

This lower bound proves divergence in the spatial-supremum, spectral-L^2 norm for the imported constructed field. It is not a matching upper asymptotic for its global supremum. All forcing, pressure, and material diffusion terms remain those of (330) and (331), with the explicit (u_ν, p_ν, f_ν) just given.

L.6.3 A terminal arithmetic profile with a bounded energy norm

The same constructed input gives a terminal value in a precise function space. This assertion can be proved from its energy identity without controlling its unbounded spatial supremum. Let $M_\nu = \max_{K_\nu} \|J\|_{\mathrm{op}}$ and put

$$\mathcal{H}_\nu = L^2(K_\nu \times \mathbb{R}, dq\,ds; \mathbb{C}^3), \qquad B_{\nu,j}(t,q,s) = \Psi_{\nu,j}(t,q,s) - H(S_j(q)+s).$$

Subtracting the displayed initial value defines an affine coordinate with the explicit inverse $B \mapsto B + (H(S_j(q)+s))_j$. It does not discard the initial profile. Its domain and reference element are specified; outside K_ν the difference is identically zero. The full profile on K_ν is also square integrable because real translation gives norm $|K_\nu|^{1/2}\|H\|_2$ in each channel.

The already proved exact identity gives

$$\|D_\nu(t)\|_{\mathcal{H}_\nu}^2 = d_H^2 \int_{K_\nu} |J(x)u_\nu(t,x)|^2 dx \le d_H^2 M_\nu^2 E_\nu^2.$$

The identity follows by integrating the real spectral translation and then using $\det D_q X_{\nu,t} = 1$; no spatial coordinate is removed. On a closed time interval before one the functions are smooth and the Schwartz bounds justify their Hilbert-space derivatives. Integrating $\partial_t B_\nu = D_\nu$ there and using the triangle inequality for the integral yields, for $0 \le a < b < 1$,

$$\|B_\nu(b) - B_\nu(a)\|_{\mathcal{H}_\nu} \le d_H M_\nu E_\nu(b-a). \tag{342}$$

Completeness constructs a unique terminal element $B_\nu(1)$ and the exact bound $\|B_\nu(1) - B_\nu(t)\| \le d_H M_\nu E_\nu(1-t)$. This constructs a Lipschitz extension on $[0,1]$ in the specified Hilbert space, while (341) proves divergence in the stronger spatial-supremum norm of its time derivative.

We also identify the terminal element through an actual map on labels. Volume preservation gives

$$\int_{K_\nu}\int_0^1 |u_\nu(t,X_{\nu,t}(q))|dt\,dq=\int_0^1 \|u_\nu(t)\|_{L^1(K_\nu)}dt\le |K_\nu|^{1/2}E_\nu.$$

The nonnegative integral is finite, so for almost every q the full trajectory has finite total length before one. Its integral equation therefore constructs the limit $X_{\nu,1}(q)\in K_\nu$ as $t\uparrow 1$. Set its value arbitrarily in K_ν on the null exceptional set. Also the integral triangle inequality and volume preservation give

$$\|X_{\nu,b}-X_{\nu,a}\|_{L^2(K_\nu)}\le\int_a^b\|u_\nu(t)\|_2dt\le E_\nu(b-a),$$

and the same estimate with $b=1$ follows by the constructed limit and Fatou's lemma. For every continuous g on compact K_ν, dominated convergence gives

$$\int_{K_\nu} g(X_{\nu,1}(q))dq=\lim_{t\uparrow 1}\int_{K_\nu} g(X_{\nu,t}(q))dq=\int_{K_\nu} g(x)dx.$$

This identifies the pushforward of Lebesgue measure exactly. To see that continuous tests suffice, for each relatively closed set C use the continuous decreasing approximations $g_n(x)=\max(0,1-n\,\mathrm{dist}(x,C))$ to its indicator, then monotone or dominated convergence; complements give open sets and the uniqueness of finite measures on their generated Borel sigma algebra gives the stated pushforward.

For almost every q the pointwise terminal entire fibre is

$$\Psi_{\nu,j}(1,q,\zeta)=H(S_j(X_{\nu,1}(q))+\zeta).$$

It agrees with the Hilbert-space limit: real translations are continuous in $L^2(ds)$, as follows from $\|H(\cdot+a)-H(\cdot+b)\|_2\le d_H|a-b|$ by integrating H'. The shifts converge almost everywhere, remain bounded on K_ν, and their translated-profile squared differences are bounded by $4\|H\|_2^2$. Dominated convergence proves the identification in $\mathcal{H}_\nu$. Replacing H by $H^{(n)}$ in this proof proves the same assertions for every pure spectral derivative, with the retained constant $\|H^{(n+1)}\|_2M_\nu E_\nu$.

The terminal label map has a further exact operator correspondence. On $L^2(K_\nu)$ define $V_tg=g\circ X_{\nu,t}$, including $t=1$. For $t<1$ its inverse is composition with $X_{\nu,t}^{-1}$, so it is unitary. The just-proved measure identity makes V_1 an isometry. For continuous g, dominated convergence gives $V_tg\to V_1g$ in L^2. Continuous functions are dense for Lebesgue measure restricted to a compact set: approximate measurable-set indicators by continuous distance cutoffs between compact subsets and open supersets using finite-measure regularity, then approximate simple functions. The isometric bounds extend the convergence to every $g\in L^2(K_\nu)$. Thus $V_t\to V_1$ strongly. Its Hilbert adjoint satisfies $V_1^*V_1=I$ by preservation of inner products, and $V_1V_1^*$ is the orthogonal projection onto its closed range: it is self-adjoint, idempotent, acts as identity on that range, and annihilates its orthogonal complement. These are the proved terminal inverse and range statements. No pointwise inverse of $X_{\nu,1}$, or surjectivity of V_1, has been supplied by this argument.

L.6.4 The actual zero image, including the terminal fibres

For every time before one and almost every terminal label, the shift $a=S_j(X_{\nu,t}(q))$ is real. The full coordinate map is $\zeta\mapsto a+\zeta$ with inverse $s\mapsto s-a$. If $H(s)=(s-z_0)^mA_0(s)$ with $A_0(z_0)\neq 0$, the actual image is

$$H(a+\zeta)=(\zeta-(z_0-a))^mA_0(a+\zeta).$$

This retains the whole analytic unit and all its derivatives. In particular its zero has imaginary part $\operatorname{Im} z_0$ at all these times, even for the source input whose arithmetic time-derivative combination obeys (340). With $H = \Xi$ the precise completed-zeta argument is $\rho = 1/2+i(a+\zeta)$, and $\operatorname{Re}\rho = 1/2-\operatorname{Im}\zeta$. Thus the actual source witness produces the proved derivative singularity and terminal profile above; this explicit map preserves zero heights. A zero-height change is not obtained by substituting a faster-growing velocity into this same real-translation map.

M The actual source quotient of a flow-driven arithmetic profile

M.1 Translation, heat, and the original spectral generator

For $a, \theta \in \mathbb{C}$ define, with the input coordinate s and output coordinate ζ retained,

$$(T_a h)(\zeta) = h(\zeta + a), \qquad V_{\theta,a} = T_a U_\theta : \mathcal{H}_{\le 1} \longrightarrow \mathcal{H}_{\le 1}.$$

The derivative is the complex derivative in the displayed coordinate. Translation preserves the original order class. More precisely, for $1 < p < 2$ and $c > 0$ the already defined growth seminorms satisfy

$$\|T_a h\|_{p,c} \le e^{c|a|^p} \|h\|_{p,c\,2^{1-p}},$$

because $|\zeta + a|^p \le 2^{p-1}(|\zeta|^p + |a|^p)$. Thus T_a and T_{-a} are continuous for γ. Ordinary differentiation gives $DT_a = T_a D$; continuity and the convergent heat series give $T_a U_\theta = U_\theta T_a$. Consequently

$$V_{\theta,a}^{-1} = U_{-\theta} T_{-a}, \qquad V_{\theta,a} D = D V_{\theta,a}.$$

These assertions hold on all of $\mathcal{H}_{\le 1}$ and do not impose continuity of translation or differentiation for the unnamed source topology.

Retain the actual τ, N_{ar} and Q_{ar}, and define their exact transported presentations

$$\tau_{\theta,a} = (V_{\theta,a})_* \tau, \qquad N_{\theta,a} = V_{\theta,a} N_{\mathrm{ar}}, \qquad Q_{\theta,a} = (\mathcal{H}_{\le 1}, \tau_{\theta,a})/N_{\theta,a}. \tag{343}$$

Then $N_{\theta,a}$ is closed, and

$$\overline{V}_{\theta,a} : Q_{\mathrm{ar}} \longrightarrow Q_{\theta,a}, \qquad [h] \longmapsto [V_{\theta,a} h]$$

is a complex-linear homeomorphism. Its inverse is induced by the displayed inverse on representatives. The original arithmetic multiplication has the exact conjugate

$$B_{\theta,a} = V_{\theta,a} s V_{\theta,a}^{-1} = \zeta + a - \frac{\theta}{2} D_\zeta. \tag{344}$$

Indeed the previous heat commutator gives $U_\theta s U_{-\theta} = s - \theta D/2$; translation sends multiplication by s to multiplication by $\zeta + a$ and commutes with D. This full combination preserves $N_{\theta,a}$ and is continuous for $\tau_{\theta,a}$ by its conjugacy. The induced quotient operator is similar through $\overline{V}_{\theta,a}$ to the source S, so its spectrum is exactly the original s-spectrum. To verify both directions, conjugate a continuous two-sided inverse of $S - \lambda$ by $\overline{V}_{\theta,a}$, or conjugate an inverse of the transported operator back. At $\theta = 0$, multiplication by ζ itself descends, and its spectrum is $\operatorname{Spec}(S) - a$ because $B_{0,a} = \zeta + a$.

Along the retained real flow let

$$a_j(t,q) = S_j(X_t(q)), \qquad v_j(t,q) = \partial_t a_j(t,q), \qquad \Psi_j(t,q,\zeta) = V_{\vartheta(t),a_j(t,q)} \Xi(\zeta).$$

Here S_j denotes the original polynomial target component, whereas the unindexed S above denotes the arithmetic quotient operator. The source quotient is taken fibre by fibre at each fixed (t,q,j). If a topology on the whole family of quotient fibres is needed, give their disjoint

union the topology transported from $\mathbb{C}^2 \times Q_{\mathrm{ar}}$ by $((\theta,a),[h]) \mapsto ((\theta,a),[V_{\theta,a}h])$. This specifies joint continuity by an exact trivialization and does not assert joint parameter continuity for the untransported τ.

Because $\Xi \in N_{\mathrm{ar}}$, every profile represents zero in its actual transported quotient:

$$[\Psi_j]_{Q_{\vartheta,a_j}} = 0. \tag{345}$$

Its temporal defect is nevertheless a specific full representative,

$$\mathcal{D}_j = \partial_t \Psi_j + \frac{\vartheta'}{4} D_\zeta^2 \Psi_j = v_j V_{\vartheta,a_j} \Xi'.$$

This is the complete chain rule with the original heat coefficient. Write

$$\beta = [\Xi']_{Q_{\mathrm{ar}}}, \qquad \mathfrak{a}_\beta = \{c \in \mathbb{C} : c\Xi' \in N_{\mathrm{ar}}\}.$$

Then the exact quotient statement and its scalar kernel are

$$[\mathcal{D}_j]_{Q_{\vartheta,a_j}} = \overline{V}_{\vartheta,a_j}(v_j\beta), \qquad \ker(c \mapsto c\beta) = \mathfrak{a}_\beta. \tag{346}$$

The set $\mathfrak{a}_\beta$ is a closed complex linear subspace: it is the inverse image of the closed N_{ar} under the continuous scalar map $c \mapsto c\Xi'$ into $(\mathcal{H}_{\le 1}, \tau)$. Its equality with $\{0\}$ or $\mathbb{C}$ is precisely the unresolved membership question $\Xi' \notin N_{\mathrm{ar}}$ or $\Xi' \in N_{\mathrm{ar}}$. No evaluation functional on the source quotient is used to decide it.

At a fixed physical point $x = X_t(q)$, retain $v = J(x)u(t,x)$ and the full inverse $J^{-1}(x) = [2W_1(x)\ W_2(x)\ W_3(x)]$. Thus the exact kernel of the complexified velocity-to-source-class map is

$$J(x)^{-1}(\mathfrak{a}_\beta^3) \subseteq \mathbb{C}^3;$$

for real velocities take its intersection with $\mathbb{R}^3$. This follows componentwise from (346) and the invertibility of the unchanged J.

The derivative relation explains why a zero profile class does not determine its representative derivative. At a fixed (θ,a) put

$$\mathfrak{D}_{\theta,a} = \{([h],[Dh]) : h \in \mathcal{H}_{\le 1}\} \subset Q_{\theta,a}^2, \qquad \mathcal{W}_{\theta,a} = (DN_{\theta,a} + N_{\theta,a})/N_{\theta,a}.$$

It has full domain and output fibre $[Dh]+\mathcal{W}_{\theta,a}$ at $[h]$: replace h by every $h+n$ with $n \in N_{\theta,a}$. Its presenting map has kernel $\{h \in N_{\theta,a} : Dh \in N_{\theta,a}\}$. Since D commutes with $V_{\theta,a}$, conjugating both components by $\overline{V}_{\theta,a}$ gives exactly the previous source derivative relation and its ambiguity space. In particular $\overline{V}_{\theta,a}\beta$ is an actual element of its multivalued part at zero. The displayed temporal connection is an identity on the full smooth profiles; a derivative on all untransported source quotient representatives is not presumed.

M.2 An exact information-preserving refinement and its residue projection

Retain the explicit analytic ideal $I = \Xi\mathcal{H}_{\le 1}$, already proved closed for γ by the full division estimate, and put

$$H_\cap = N_{\mathrm{ar}} \cap I, \qquad C_\cap = (\mathcal{H}_{\le 1}, \tau \vee \gamma)/H_\cap, \qquad Q_I = (\mathcal{H}_{\le 1}, \gamma)/I. \tag{347}$$

The join topology is precisely the topology induced by $h \mapsto (h,h)$ into $(\mathcal{H}_{\le 1},\tau) \times (\mathcal{H}_{\le 1},\gamma)$. Both subspaces in the intersection are closed in this join, so $C_\cap$ is Hausdorff. The identity on representatives induces continuous surjections

$$p_N : C_\cap \longrightarrow Q_{\mathrm{ar}}, \qquad p_I : C_\cap \longrightarrow Q_I, \qquad \ker p_N = N_{\mathrm{ar}}/H_\cap, \quad \ker p_I = I/H_\cap. \tag{348}$$

The kernel assertions are direct membership tests; continuity follows from the quotient universal property. These kernel identifications are also homeomorphisms when N_{ar} and I have their subspace topologies from $\tau \vee \gamma$. To prove this, let $q_{\cap}$ be the open quotient map and let A denote either subspace. Because $H_{\cap} \subset A$, $q_{\cap}(A \cap O) = q_{\cap}(A) \cap q_{\cap}(O)$ for every open ambient O. Thus its restriction to A is an open quotient map onto the corresponding kernel with its subspace topology. The joint map (p_N, p_I) is continuous and injective since its kernel is zero. No inference of a topological embedding from those two properties is needed.

Let λ be any actual zero of Ξ, with its actual multiplicity $m_\lambda \geq 1$. Choose a disk containing no other zero and retain the full unit

$$\Xi(s) = (s-\lambda)^{m_\lambda} A_\lambda(s), \qquad A_\lambda(\lambda) \neq 0.$$

For $h \in \mathcal{H}_{\leq 1}$ define

$$R_\lambda(h) = \mathrm{Res}_{s=\lambda} \frac{h(s)}{\Xi(s)} = \frac{1}{2\pi i} \int_{|s-\lambda|=r} \frac{h(s)}{\Xi(s)}\, ds, \tag{349}$$

where the circle has positive orientation and lies in this disk. The integral is independent of the permitted r by Laurent expansion. It is continuous for γ: its absolute value is at most $r \sup_{|s-\lambda|=r} |h(s)| / \min_{|s-\lambda|=r} |\Xi(s)|$, and a compact supremum is bounded by a fixed growth seminorm times the maximum of its exponential weight on the circle. It annihilates I, since h/Ξ is entire there, and therefore annihilates $H_{\cap}$.

Logarithmic differentiation of the full local factor gives

$$\frac{c\Xi'(s)}{\Xi(s)} = \frac{m_\lambda c}{s-\lambda} + c\frac{A'_\lambda(s)}{A_\lambda(s)}, \qquad R_\lambda(c\Xi') = m_\lambda c. \tag{350}$$

In particular $\Xi' \notin I$: the displayed pole is nonzero. Thus $\beta_I = [\Xi']_{Q_I} \neq 0$, and $\widetilde{\beta} = [\Xi']_{C_{\cap}} \neq 0$ as well, independent of the status of β in the actual source quotient. The residue functional factors continuously through $C_{\cap}$; write its induced functional as $\widetilde{R}_\lambda$. The formula

$$\Pi_\lambda(c) = \frac{\widetilde{R}_\lambda(c)}{m_\lambda}\, \widetilde{\beta}, \qquad c \in C_{\cap} \tag{351}$$

defines a continuous projection onto the line $\mathbb{C}\widetilde{\beta}$. Indeed its square is itself by $\widetilde{R}_\lambda(\widetilde{\beta}) = m_\lambda$, and scalar multiplication on the quotient is continuous. Its inverse coordinate on that line is $\widetilde{R}_\lambda / m_\lambda$, while $c \mapsto c\widetilde{\beta}$ is its continuous inverse. Consequently this line has exactly the ordinary complex topology, and $C_{\cap} = \mathbb{C}\widetilde{\beta} \oplus \ker \widetilde{R}_\lambda$ is a continuous direct sum. This proves an information-preserving refinement of the actual source map, including a continuous recovery coordinate.

The full principal-part map $h \mapsto (\mathrm{PP}_\lambda(h/\Xi))_{\lambda \in Z(\Xi)}$ has kernel exactly I. Vanishing of every negative Laurent coefficient makes h/Ξ entire at every zero; it is already holomorphic elsewhere, and the proved entire division bound puts it in $\mathcal{H}_{\leq 1}$. The reverse inclusion is immediate. Each finite principal-part coordinate is continuous by the same circle integral with the appropriate power of $s-\lambda$. On the velocity line its entire image at every actual zero is the single term $m_\lambda c/(s-\lambda)$ from (350), with the unchanged analytic unit in the regular part.

For general (θ, a) let $I_{\theta,a} = V_{\theta,a} I$ and transport (347) by $V_{\theta,a}$. Its ambient topology is $\tau_{\theta,a} \vee \gamma$, since $V_{\theta,a}$ is a γ-homeomorphism, and its relation is $N_{\theta,a} \cap I_{\theta,a} = V_{\theta,a} H_{\cap}$. Every map, kernel and projection above is transported exactly. In particular the transported residue coordinate is

$$R_{\lambda;\theta,a}(h) = \mathrm{Res}_{s=\lambda} \frac{(U_{-\theta} T_{-a} h)(s)}{\Xi(s)}.$$

For the actual temporal defect it gives $R_{\lambda;\vartheta,a_j}(\mathcal{D}_j) = m_\lambda v_j$. The pair of source and analytic classes therefore has the explicit injective common lift $v_j[V_{\vartheta,a_j}\Xi']$ in this transported refinement. For nonzero heat time this formula retains the inverse heat operator; it does not replace $V_{\theta,a} I$ by an unproved ordinary ideal generated by $T_a Z_\theta$.

M.3 The logarithmic connection and the public blowup residue field

Take the explicit Newman-time choice $\vartheta(t) = 0$. For the original real flow set

$$\Psi_j(t, q, \zeta) = \Xi(\zeta + a_j(t, q)), \qquad \mathcal{D}_j = \partial_t \Psi_j = v_j(t, q)\Xi'(\zeta + a_j(t, q)).$$

At every actual zero λ the translated divisor is $\zeta_{\lambda,j}(t, q) = \lambda - a_j(t, q)$, and the full factorization is

$$\Psi_j = (\zeta - \zeta_{\lambda,j})^{m_\lambda} A_\lambda(\zeta + a_j).$$

Thus all multiplicities and the actual analytic unit are retained. Since a_j is real, its imaginary coordinate is exactly $\operatorname{Im}\lambda$. Differentiating its location gives $\partial_t \zeta_{\lambda,j} = -v_j$.

On the complement of the divisor in $[0, T) \times \mathbb{R}^3_q \times \mathbb{C}_\zeta$, define the complex-valued differential form, smooth in the real variables and holomorphic in ζ,

$$\omega_j = \frac{d\Psi_j}{\Psi_j} = \frac{\Xi'(\zeta + a_j)}{\Xi(\zeta + a_j)}(d\zeta + da_j), \qquad da_j = v_j\, dt + \sum_b \partial_{q_b} a_j\, dq_b. \tag{352}$$

All label and time coefficients are displayed. Put $s = \zeta + a_j$. Direct exterior differentiation gives $d\omega_j = (\Xi'/\Xi)'(s)\, ds \wedge ds + (\Xi'/\Xi)(s) d^2 s = 0$. Equivalently the connection $d - \omega_j$ on the trivial complex line has zero curvature and annihilates the section Ψ_j. This calculation is on the complement of its zeros and retains the singular divisor explicitly.

At fixed (t, q) a sufficiently small positively oriented circle around $\zeta_{\lambda,j}$ has

$$\frac{1}{2\pi i}\int \omega_j = m_\lambda,$$

where only its $d\zeta$ restriction is integrated. The pole term in (350) gives this integral; the holomorphic unit term has zero integral by Cauchy's theorem. The time component and its residue are exactly

$$\begin{aligned} \omega_j(\partial_t) &= \frac{\mathcal{D}_j}{\Psi_j} = \frac{m_\lambda v_j}{\zeta - \zeta_{\lambda,j}} + v_j \frac{A'_\lambda(\zeta + a_j)}{A_\lambda(\zeta + a_j)}, \\ r_j(t, q) &:= \operatorname{Res}_{\zeta=\zeta_{\lambda,j}} \frac{\mathcal{D}_j}{\Psi_j} = m_\lambda v_j = -m_\lambda \partial_t \zeta_{\lambda,j}. \end{aligned} \tag{353}$$

The coordinate change $s = \zeta + a_j$ has differential $ds = d\zeta$ on each spectral fibre, so this is precisely $R_{\lambda;0,a_j}(\mathcal{D}_j)$, including its orientation and multiplicity factor.

Write $r = (r_1, r_2, r_3)^{\mathsf{T}}$, and retain the full original polynomial target and its matrix inverse:

$$S = (F_1/2, F_2, F_3)^{\mathsf{T}}, \quad J = DS = \operatorname{diag}(1/2, 1, 1)DF, \quad J^{-1} = [2W_1\ W_2\ W_3].$$

The literal recovery and pointwise norm identities are

$$\begin{aligned} u(t, X_t(q)) &= \frac{2W_1(X_t(q))r_1 + W_2(X_t(q))r_2 + W_3(X_t(q))r_3}{m_\lambda}, \\ |r(t, q)|^2 &= m_\lambda^2 |J(X_t(q))u(t, X_t(q))|^2. \end{aligned} \tag{354}$$

They follow by $r = m_\lambda v$, $v = Ju$ and the stated matrix inverse, with no selection of a global inverse sheet of S. The flow X_t is retained as part of the data.

For the nonempty compact velocity support K put

$$M_K = \max_K \|J\|_{\mathrm{op}}, \qquad N_K = \max_K \|J^{-1}\|_{\mathrm{op}}.$$

They are finite and positive by compactness and $\det J = -1$. Since $X_t(K) = K$, X_t is onto, and u, v, r vanish outside the corresponding compact labels, (354) gives

$$\frac{m_\lambda}{N_K}\|u(t)\|_\infty \le \sup_q |r(t,q)| \le m_\lambda M_K \|u(t)\|_\infty. \tag{355}$$

The volume-preserving change of variables $x = X_t(q)$ gives

$$\int_{\mathbb{R}^3} |r(t,q)|^2\, dq = m_\lambda^2 \int_{\mathbb{R}^3} |J(x)u(t,x)|^2\, dx, \tag{356}$$

whose square root is between $(m_\lambda/N_K)\|u(t)\|_2$ and $m_\lambda M_K\|u(t)\|_2$. These are exact spatial estimates for the arithmetic residues. With the previously proved spectral normalization $d_Z(0)^2 = \int_{\mathbb{R}} |\Xi'(s)|^2 ds > 0$, the unchanged real translations also give

$$|r(t,q)| = \frac{m_\lambda}{d_Z(0)} \left(\sum_j \|\mathcal{D}_j(t,q,\cdot)\|^2_{L^2(\mathbb{R})} \right)^{1/2}.$$

Proposition M.1 (Application of the cited public Navier–Stokes theorem)**.** *For every $\nu > 0$, take the velocity, pressure and force supplied by the public Theorem 1.1 of [12]. The construction above, with $\vartheta = 0$, gives arithmetic logarithmic residue fields satisfying*

$$\sup_{0\le t<1} \int_{\mathbb{R}^3} |r(t,q)|^2\, dq < \infty, \qquad \limsup_{t\uparrow 1} \sup_q |r(t,q)| = \infty.$$

Every residue recovers the full physical velocity by (354)*, and each translated arithmetic divisor has the same imaginary coordinates as the original zero divisor of Ξ.*

Proof. The cited theorem supplies a smooth incompressible velocity on $[0,1)\times\mathbb{R}^3$, a fixed compact support, zero initial velocity, a uniform kinetic-energy bound, and unbounded terminal velocity supremum, with its stated smooth compactly supported force and its pressure. These are the inputs of the proved preterminal flow construction, so its volume and inverse statements apply on every $[0, T_0]$ with $T_0 < 1$. Formula (356) and the upper bound using M_K prove the uniform residue energy bound. The lower bound in (355) proves the unbounded terminal supremum. Recovery and imaginary-coordinate preservation were proved above for the actual same fields. The existence step in this corollary is exactly the cited Theorem 1.1; the correspondence and every displayed estimate are proved here. □

M.4 The published witness's exact growing residue

Take the particular corrected field of the public construction, with the source's fixed $h \in (0, 1/100)$, $X_{\mathrm{in}} > 0$, $e_0 > 0$, and remainder $R_*(\tau)$. Its closing formulas [12, equations (10.20)–(10.22), p. 124], as imported in the preceding witness construction, are

$$|R_*(\tau)| \le C_*\tau^{2h} \quad (0 < \tau < \tau_*), \qquad (u_\nu)_2(x_{\nu,\tau}, 1-\tau) = \sqrt{\nu}\,\tau^{-1/2-h}(e_0 + R_*(\tau)),$$
$$r_{\nu,\tau} = \sqrt{2\nu X_{\mathrm{in}}\tau}, \quad x_{\nu,\tau} = (r_{\nu,\tau}, 0, 0), \quad q_{\nu,\tau} = X^{-1}_{\nu,1-\tau}(x_{\nu,\tau}).$$

The remainder is the actual source remainder, including its corrected fields. The parameter h is the one chosen in that construction. The quantity $r_{\nu,\tau}$ here is the displayed physical radius; the map to the original polynomial target is retained below. The labels $q_{\nu,\tau}$ vary with τ according to the exact material inverse just displayed.

Substitution into the full original polynomials gives

$$S(r,0,0) = (0,0,2r), \qquad J(r,0,0) = \begin{pmatrix} 0 & 0 & 1/2 \\ 0 & 1 & 3r \\ 2 & -3r^2 & -r^3 \end{pmatrix}.$$

Here the same original formulas are

$$F_1 = Q^3 w + y^2 Q(4 + 3xy), \quad F_2 = y + 3xQ^2 w + 3xy^2(4 + 3xy),$$
$$F_3 = 2x - 3x^2 y - x^3 w, \quad Q = 1 + xy.$$

Differentiating $S = (F_1/2, F_2, F_3)$ and setting $(x, y, w) = (r, 0, 0)$ gives each matrix entry. Its componentwise inverse gives $u_3 = 2v_1$ and $u_2 = v_2 - 6rv_1$, retaining the off-diagonal $3r$. The first two arithmetic shifts at $(1 - \tau, q_{\nu,\tau})$ are zero. Consequently, as an identity of full entire functions of ζ,

$$\begin{aligned} \mathcal{E}_{\nu,\tau}(\zeta) &:= \mathcal{D}_{\nu,2}(1 - \tau, q_{\nu,\tau}, \zeta) - 6r_{\nu,\tau}\mathcal{D}_{\nu,1}(1 - \tau, q_{\nu,\tau}, \zeta) \\ &= \sqrt{\nu}\, \tau^{-1/2-h}(e_0 + R_*(\tau))\Xi'(\zeta). \end{aligned} \tag{357}$$

Indeed each defect is $v_j\Xi'$ because its shift is zero, so their combination is $(u_\nu)_2\Xi'$, and the stated source identity applies.

For every actual zero λ of multiplicity m_λ, its complete local logarithmic image is

$$\frac{\mathcal{E}_{\nu,\tau}(\zeta)}{\Xi(\zeta)} = \sqrt{\nu}\, \tau^{-1/2-h}(e_0 + R_*(\tau)) \left(\frac{m_\lambda}{\zeta - \lambda} + \frac{A'_\lambda(\zeta)}{A_\lambda(\zeta)} \right). \tag{358}$$

Therefore the actual unshifted arithmetic residue at the source sample is exactly

$$\operatorname{Res}_{\zeta=\lambda} \frac{\mathcal{E}_{\nu,\tau}(\zeta)}{\Xi(\zeta)} = m_\lambda \sqrt{\nu}\, \tau^{-1/2-h}(e_0 + R_*(\tau)). \tag{359}$$

All multiplicities, the whole analytic unit, the positive viscosity, the original remainder and the exact material labels remain present. Choose $\tau_{**} > 0$ within the source interval and its cutoff region with $C_*\tau_{**}^{2h} \le e_0/2$. For $0 < \tau < \tau_{**}$ the absolute value of (359) is at least $m_\lambda \sqrt{\nu}\, e_0 \tau^{-1/2-h}/2$, which diverges. This is an explicit pole-coefficient singularity of the published constructed field's arithmetic image.

For the three residue channels $r_{\nu,j}$ of (353), the same evaluation and the two-component Cauchy–Schwarz inequality give the quantitative bound

$$|r_\nu(1 - \tau, q_{\nu,\tau})| \ge \frac{m_\lambda \sqrt{\nu}\, e_0}{2\sqrt{1 + 72\nu X_{\mathrm{in}}\tau}}\, \tau^{-1/2-h}. \tag{360}$$

To verify the denominator, the coefficient vector $(-6r_{\nu,\tau}, 1)$ has squared norm $1 + 36r_{\nu,\tau}^2 = 1 + 72\nu X_{\mathrm{in}}\tau$. Its product with the first two residue coordinates is exactly the residue in (359); the third coordinate can only increase the Euclidean norm on the left. If $E_\nu = \nu^{5/4}F_*(1)$ is the full source force-integral energy bound and $M_\nu = \max_{K_\nu} \|J\|_{\mathrm{op}}$, the earlier volume identity simultaneously gives

$$\int_{\mathbb{R}^3} |r_\nu(t, q)|^2 dq \le m_\lambda^2 M_\nu^2 E_\nu^2 = m_\lambda^2 M_\nu^2 \nu^{5/2} F_*(1)^2 \quad (t < 1).$$

At these particular first two channels, their zero shifts mean the source quotient is the original Q_{ar} itself. The exact classes are

$$[\mathcal{E}_{\nu,\tau}]_{Q_{\mathrm{ar}}} = \sqrt{\nu}\, \tau^{-1/2-h}(e_0 + R_*(\tau))\beta, \qquad [\mathcal{E}_{\nu,\tau}]_{C_\cap} = \sqrt{\nu}\, \tau^{-1/2-h}(e_0 + R_*(\tau))\widetilde{\beta}.$$

The latter line has the explicitly proved continuous inverse $\widetilde{R}_\lambda/m_\lambda$, so its scalar coordinate has exactly the displayed singularity. In particular these common classes leave every bounded subset of $C_\cap$: a continuous linear functional sends bounded sets to bounded scalar sets, whereas their $\widetilde{R}_\lambda$ values diverge. Their projection to the actual source quotient and its exact kernel

remain (348); the undecided source membership is not replaced by an identification with the analytic ideal.

There is also an exact relation between the changing sample labels and the time component of the flat logarithmic connection. Let $A_\tau = D_q X_{\nu,1-\tau}(q_{\nu,\tau})$. Differentiating $X_{\nu,1-\tau}(q_{\nu,\tau}) = x_{\nu,\tau}$ gives

$$A_\tau \frac{dq_{\nu,\tau}}{d\tau} = \frac{dx_{\nu,\tau}}{d\tau} + u_\nu(1-\tau, x_{\nu,\tau}), \qquad \frac{dx_{\nu,\tau}}{d\tau} = \frac{r_{\nu,\tau}}{2\tau}(1,0,0).$$

The sign follows from $d(1-\tau)/d\tau = -1$. Since $\nabla_q a_j$ is the jth row of $J(x_{\nu,\tau})A_\tau$, and the first two rows of $J(r,0,0)$ have zero first component,

$$\nabla_q a_j \cdot \frac{dq_{\nu,\tau}}{d\tau} = v_j, \qquad j = 1,2.$$

Thus $a_j(1-\tau, q_{\nu,\tau}) = 0$ has total derivative $-v_j + \nabla_q a_j \cdot dq_{\nu,\tau}/d\tau = 0$, exactly as required. On the spectral complement of the zeros, the pullback of (352) to this curve and fixed ζ is zero in the τ direction, while its original ∂_t component has the residue (353). These statements are related by the displayed full tangent-vector identity; the sample is not silently treated as one fixed material label.

For the full four-term profile $K_\theta = U_\theta K_0$, the corresponding actual source classes are instead

$$\kappa = [K_0]_{Q_{\mathrm{ar}}}, \qquad \beta_K = [K_0']_{Q_{\mathrm{ar}}}, \qquad [T_a K_\theta] = \overline{V}_{\theta,a}\kappa, \qquad [v T_a K_\theta'] = \overline{V}_{\theta,a}(v\beta_K).$$

These follow by the same commuting translation, heat and derivative maps and retain κ without assigning it the value zero. The residue corollary above uses the actual Ξ profile. The displayed four-term quotient formulas concern the whole K profile and its derivative; they do not identify individual zeros of K with zeros of Ξ.

M.5 The translated modular double

Keep $C_E h(s) = \overline{h(\overline{s})}$, $\tau^c = (C_E)_*\tau$ and $N^c = C_E N_{\mathrm{ar}}$ from the explicit modular construction. Direct conjugation gives $C_E T_a = T_{\overline{a}} C_E$, and the established heat identity gives $C_E U_\theta = U_{\overline{\theta}} C_E$. On the two entire sectors define

$$\widehat{V}_{\theta,a} = \mathrm{diag}(T_a U_\theta, T_a U_{-\theta}), \qquad \mathcal{J}(h,k) = (C_E k, C_E h).$$

Then the exact all-parameter identity is

$$\mathcal{J}\widehat{V}_{\theta,a} = \widehat{V}_{-\overline{\theta},\overline{a}}\mathcal{J}. \tag{361}$$

Indeed its first component is $C_E T_a U_{-\theta} k = T_{\overline{a}} U_{-\overline{\theta}} C_E k$, and its second component is the same calculation with θ. For the actual real flow, $a \in \mathbb{R}$; for real Newman time it therefore sends (θ, a) to $(-\theta, a)$.

Give the two ambient sectors the topologies $(T_a U_\theta)_*\tau$ and $(T_a U_{-\theta})_*\tau^c$, and quotient by the closed relation

$$\widehat{N}_{\theta,a} = T_a U_\theta N_{\mathrm{ar}} \oplus T_a U_{-\theta} N^c.$$

Applying (361) both to these relation spaces and to the definitions of their transported topologies proves an antilinear homeomorphism from this actual doubled quotient to the one indexed by $(-\overline{\theta}, \overline{a})$. Its square is the identity, and its conjugacy with the quotient heat and translation maps follows on representatives from the same displayed equality. Thus the Cartan reversal retains the full real flow translation while reversing the radial heat parameter, with both original source and conjugate-source sectors specified.

N The exact strength of the frozen source spectral information

The primary spectrum statement in [5, Proposition 3.6(ii)] specifies the actual zero set as the spectrum of the Hadamard quotient's multiplication operator. It gives no multiplicity assertion. The cited [6, Proposition 2.24] computes the finite Hermite polynomial image; the discussion after its proof explicitly treats function-space choices. The finite multiplicity calculation in [6, Theorem 2.47 and equations (2.447)–(2.449)] belongs to the separately specified weighted Hilbert quotient. The following exact algebra records both the obstruction available from such data and the limits of transferring a spectrum set alone.

N.1 One-sided heat inclusion and the full generalized chain

The already proved commutator identity (176) and its following heat calculation show that $DN_{\mathrm{ar}} \subseteq N_{\mathrm{ar}}$ makes every $S-\lambda$ algebraically onto. The single inclusion $U_t N_{\mathrm{ar}} \subseteq N_{\mathrm{ar}}$ at any $t \neq 0$ also makes every $S-\lambda$ onto, using every derivative of $U_t\Xi$ and the exact source division map. Here is a stronger consequence that needs no descended derivative.

Lemma N.1. *Let $T : V \to V$ be a surjective endomorphism of a complex vector space and let $0 \neq v_0 \in \ker T$. Then for every integer $k \geq 1$, $\dim \ker T^k \geq k$. More precisely one can choose $Tv_j = v_{j-1}$ for every $j \geq 1$, and the vectors $v_0, \dots, v_n$ are linearly independent for every n.*

Proof. Surjectivity gives the successive preimages. It follows inductively that $T^j v_j = v_0$ and $T^{j+1} v_j = 0$. Applying T^n to a relation $\sum_{j=0}^{n} c_j v_j = 0$ gives $c_n v_0 = 0$, hence $c_n = 0$. Repeating with the remaining highest index gives every coefficient zero. The first k vectors lie in $\ker T^k$, proving the bound. □

Applied to $T = S - \lambda$, this proves that either frozen inclusion just stated is incompatible with any nonzero finite-dimensional generalized eigenspace of the actual source operator. The source-independent implication is

$$U_t N_{\mathrm{ar}} \subseteq N_{\mathrm{ar}}, \quad t \neq 0, \quad 0 \neq v_0 \in \ker(S-\lambda) \quad \Longrightarrow \quad \dim \ker(S-\lambda)^k \geq k \quad (k \geq 1). \tag{362}$$

With a descended $d[f] = [Df]$, its more specific ladder is $v_j = (-1)^j d^j v_0 / j!$; indeed $[d, S] = I$ gives $(S-\lambda) d^j v_0 = -j d^{j-1} v_0$. Thus the complete factorial and sign agree with the general preimage construction. This is a proved algebraic incompatibility; it does not assert that the unnamed source quotient has the finite generalized block needed to apply the incompatibility.

N.2 The actual finite block in the specified Sobolev source

Retain the source's real frequency t, compact character χ_0, weight $\delta > 1$, and actual zero multiplicity m. The vectors in the dual block of the specified source are

$$\eta_{t,a}(x) = \chi_0(x) |x|^{it} (\log |x|)^a, \qquad a \in \mathbb{Z}_{\geq 0}, \quad a < m, \quad a < (\delta - 1)/2.$$

The last strict bound is exactly integrability of $u^{2a}(1+u^2)^{-\delta/2}$ on $\mathbb{R}$: at infinity its power is $2a - \delta$, integrable precisely when it is less than -1; near zero it is locally integrable for all these a. Let r be the number of permitted integers. The source representation at $\lambda > 0$ acts by

$$\vartheta_m^{\mathrm{tr}}(\lambda) \eta_{t,a} = \lambda^{it} \sum_{b=0}^{a} \binom{a}{b} (\log \lambda)^b \eta_{t,a-b}.$$

Differentiating at $\lambda = e^{\epsilon}$, $\epsilon = 0$, proves

$$(D_{\chi_0}^{\mathrm{tr}} - it) \eta_{t,a} = a \eta_{t,a-1}, \qquad (D_{\chi_0}^{\mathrm{tr}} - it)^k \eta_{t,a} = \frac{a!}{(a-k)!} \eta_{t,a-k} \quad (0 \leq k \leq a). \tag{363}$$

For $k > a$ the expression is zero. These vectors are independent: divide a proposed relation by the nonzero character and use that $\log|x|$ ranges over $\mathbb{R}$, making the remaining polynomial identically zero. Thus the block has dimension r, one-dimensional ordinary eigenspace, and nilpotence index r. Its multiplicity is generalized multiplicity. The common-test morphism to the actual Hadamard source was proved earlier with its exact two kernels; this finite block is not silently identified with a block in that different quotient.

N.3 An explicit spectral model and the inverse at zero

Let $\Lambda = Z(\Xi)$ be the actual zero set. On the complex vector space

$$V_\Lambda = \bigoplus_{\lambda\in\Lambda}\bigoplus_{n\ge 0}\mathbb{C}e_{\lambda,n}$$

every vector has finite support. Give it the finest locally convex topology, defined by all seminorms on this vector space. Coordinate functionals separate points, so it is Hausdorff. Every linear map from it to any locally convex space is continuous: each pullback of a continuous seminorm is among these defining seminorms. With $e_{\lambda,-1} = 0$ set

$$S_\Lambda e_{\lambda,n} = \lambda e_{\lambda,n} + e_{\lambda,n-1}, \qquad d_\Lambda e_{\lambda,n} = -(n+1)e_{\lambda,n+1}.$$

Both are continuous, everywhere-defined, and direct evaluation on the basis proves $[d_\Lambda, S_\Lambda] = I$. For $\mu \notin \Lambda$ the full continuous inverse is

$$(S_\Lambda - \mu)^{-1}e_{\lambda,n} = \sum_{j=0}^{n}\frac{(-1)^j}{(\lambda-\mu)^{j+1}}e_{\lambda,n-j}. \tag{364}$$

Multiplication in either order cancels adjacent terms of this finite sum and leaves the basis vector. The formula preserves finite support, so it defines an inverse on the whole space, continuous by its topology. For $\mu \in \Lambda$, the μ-summand is a surjective backward shift and every other summand uses the same finite inverse formula. Thus $S_\Lambda - \mu$ is onto, its kernel is exactly $\mathbb{C}e_{\mu,0}$, and

$$\ker(S_\Lambda - \lambda)^k = \operatorname{span}\{e_{\lambda,0},\dots,e_{\lambda,k-1}\}, \quad \operatorname{Spec}_{\text{alg}} S_\Lambda = \operatorname{Spec}_{\text{end}} S_\Lambda = \operatorname{Spec}_{\text{point}} S_\Lambda = \Lambda. \tag{365}$$

Here $\operatorname{Spec}_{\text{end}}$ means failure of a continuous two-sided inverse in the algebra of continuous endomorphisms, not the Hilbert spectral subdivision called continuous spectrum. Since $\Xi(0) \neq 0$, this S_Λ has a continuous inverse at zero.

There is an exact principal-part realization of the model: send $e_{\lambda,n}$ to the class of w^{-n-1} in $\mathbb{C}[w,w^{-1}]/\mathbb{C}[w]$ in its λ-summand. The finite negative Laurent coefficients give its inverse. Multiplication by $\lambda + w$ is S_Λ and differentiation in w is d_Λ, as evaluation on each monomial proves. This complete model shows that the correct zero set as spectrum, geometric multiplicity one, and a continuous inverse at zero can coexist with a continuous Weyl pair. It does not identify this model with Q_{ar}.

For the actual quotient, injectivity of S gives exactly $N_{\text{ar}} \cap s\mathcal{H}_{\le 1} = sN_{\text{ar}}$: if $sf \in N_{\text{ar}}$, then $S[f] = 0$, so $f \in N_{\text{ar}}$; the converse is polynomial invariance. The original source inverse at zero is explicitly

$$S^{-1}[f] = \left[\frac{f(s) - f(0)\Xi(s)/\Xi(0)}{s}\right]. \tag{366}$$

The singularity is removable, and division by a degree-one polynomial preserves the original order class. Multiplication by s gives $[f]$, and injectivity gives its unique inverse. If f is changed by $n \in N_{\text{ar}}$, the numerator difference belongs to $N_{\text{ar}} \cap s\mathcal{H}_{\le 1} = sN_{\text{ar}}$, proving representative

independence directly. The usual topological reading of the source resolvent also supplies continuity of this quotient inverse, not continuity of its chosen lift. More generally,

$$S-\lambda \text{ injective} \quad\Longleftrightarrow\quad N_{\mathrm{ar}} \cap (s-\lambda)\mathcal{H}_{\le 1} = (s-\lambda)N_{\mathrm{ar}}$$

by the same calculation. Saturation at zero does not prove saturation at other points. The missing frozen-descent decision can therefore be approached through an actual finite generalized source block or a range obstruction, as well as through identification of the topology. Neither has been supplied by the primary Hadamard spectrum clauses inspected here.

O Actual angular momentum, nonlinear fluxes and arithmetic dilation

The complete companion proof [16] is retained here with its original coordinates and full corrected source fields. The source existence and correction estimates remain the explicit external input [12].

O.1 Cartesian input, angular projection and its exact section

Use the complete source witness (u_ν, p_ν, f_ν) constructed in Section L.6, for every physical time $0 \le t < 1$. Retain the viscosity $\nu > 0$ and original Cartesian coordinates (x_1, x_2, z). For $r > 0$ put $x_1 = r\cos\theta, x_2 = r\sin\theta$, $e_r = (\cos\theta, \sin\theta, 0)$ and $e_\theta = (-\sin\theta, \cos\theta, 0)$. Write $u_r = e_r \cdot u_\nu$, $u_\theta = e_\theta \cdot u_\nu$ and $u_z = (u_\nu)_3$. These components may depend on θ; in particular the complete corrected velocity is not assumed axisymmetric. Define $\langle a\rangle = (2\pi)^{-1}\int_0^{2\pi} a\, d\theta$ and retain the radial coordinate and inverse

$$\sigma = \frac{r^2}{2}, \qquad r = \sqrt{2\sigma}, \qquad r\,dr = d\sigma.$$

This radial σ is not the earlier polynomial coordinate $F_1/2$. They coexist as explicit functions on the original Cartesian space; no equality of their Laplacians is required below. Their exact common coordinate map, on $(0,\infty)\times(\mathbb{R}/2\pi\mathbb{Z})\times\mathbb{R}$, is $(\sigma,\theta,z) \mapsto (\sqrt{2\sigma}\cos\theta, \sqrt{2\sigma}\sin\theta, z)$. Its inverse is $\sigma = (x_1^2+x_2^2)/2$, the direction $(\cos\theta,\sin\theta) = (x_1,x_2)/\sqrt{x_1^2+x_2^2}$, and the unchanged z. In these coordinates the full polynomial is

$$\frac{F_1}{2} = \frac{1}{2}\big[(1+\sigma\sin(2\theta))^3 z + 2\sigma\sin^2\theta(1+\sigma\sin(2\theta))(4+3\sigma\sin(2\theta))\big].$$

Substitution of $x_1x_2 = \sigma\sin(2\theta)$ and $x_2^2 = 2\sigma\sin^2\theta$ proves this morphism; adjoining the displayed value as an extra coordinate gives its graph map with inverse projection onto (σ,θ,z). The Cartesian definition continues over the axis where the angular chart is not defined.

The actual angular-momentum density, fluxes and force torque are

$$B(\sigma,z,t) = \langle r u_\theta\rangle, \qquad P(\sigma,z,t) = \langle r^2 u_r u_\theta\rangle,$$
$$Q(\sigma,z,t) = \langle r u_z u_\theta\rangle, \qquad F(\sigma,z,t) = \langle r f_\theta\rangle.$$

These definitions include every corrected, oscillatory and cutoff term of the actual source fields before averaging. All four functions have fixed compact support in (σ,z) on the closed half-plane $\sigma \ge 0$. They are smooth there on every closed preterminal time interval, with

$$B = O(\sigma), \quad Q = O(\sigma), \quad F = O(\sigma), \qquad P = O(\sigma^2) \quad (\sigma \downarrow 0). \tag{367}$$

Here and below these estimates hold with all z, t derivatives on compact preterminal sets. To prove them without polar smoothness assumptions, observe that $ru_\theta = x_1u_2 - x_2u_1$ and

$ru_r = x_1u_1 + x_2u_2$. Angular averaging their Cartesian expressions gives smooth rotation-invariant functions of (x_1, x_2). A smooth rotation-invariant function is smooth in σ on the closed half-line: its restriction to $(r, 0)$ is even and smooth; Taylor's formula to any even order, with its differentiated remainder, expresses it as a polynomial in r^2 plus a remainder of arbitrarily high order. The chain rule on $r > 0$ then supplies every continuous right derivative in σ at zero and the fundamental theorem of calculus identifies those derivatives successively. The same Taylor proof works with the retained z, t parameters. The constants in the averages defining B, Q, F vanish at $r = 0$; odd radial powers vanish under rotation. They therefore start at r^2. For P, the quadratic Cartesian term at zero is

$$\big\langle (x_1a + x_2b)(x_1b - x_2a) \big\rangle = 0, \qquad (a, b) = (u_1, u_2)(0, 0, z, t),$$

because $\langle x_1^2 - x_2^2 \rangle = \langle x_1x_2 \rangle = 0$. Its constant term also vanishes, and rotation removes the odd powers, so P starts at r^4. This proves (367).

The projection $\Pi u = B$ has the explicit linear section

$$(\mathcal{R}B)(x_1, x_2, z) = \frac{B(\sigma, z)}{2\sigma}(-x_2, x_1, 0), \qquad \Pi\mathcal{R} = I. \tag{368}$$

The quotient at zero is the actual smooth extension: $B(\sigma, z)/\sigma = \int_0^1 \partial_\sigma B(a\sigma, z)da$. Thus the section is Cartesian smooth, compactly supported, and divergence-free. For $r > 0$ it has sole component B/r in the e_θ direction, proving the section identity. For the exact remainder $w = u_\nu - \mathcal{R}B$, one has $\Pi w = 0$ and

$$P = rB\langle w_r \rangle + r^2\langle w_r w_\theta \rangle, \qquad Q = B\langle w_z \rangle + r\langle w_z w_\theta \rangle. \tag{369}$$

These follow by substituting $u_r = w_r$, $u_z = w_z$ and $u_\theta = B/r + w_\theta$ into the original definitions. They exhibit both mean transport and the complete two quadratic correlations. The full inverse of $u \mapsto (B, w)$ is $(B, w) \mapsto \mathcal{R}B + w$ on the stated subspace $\Pi w = 0$. In particular the projection has the precise kernel $\{w : \langle rw_\theta \rangle = 0\}$; none of that kernel's contribution to the fluxes has been discarded.

O.2 The exact equation before and after angular averaging

For the unaveraged Cartesian angular momentum $L = x_1u_2 - x_2u_1 = ru_\theta$, the product rule in the full Navier–Stokes equation gives

$$\partial_t L + \operatorname{div}(u_\nu L) = \nu\Delta L - 2\nu\omega_z - \partial_\theta p_\nu + (x_1f_2 - x_2f_1), \qquad \omega_z = \partial_1u_2 - \partial_2u_1. \tag{370}$$

Indeed the material derivatives of x_1, x_2 contribute $u_1u_2 - u_2u_1 = 0$; incompressibility converts material transport to the displayed divergence. The Laplacian product rule gives $\Delta L = x_1\Delta u_2 - x_2\Delta u_1 + 2\omega_z$. Finally $x_1\partial_2 - x_2\partial_1 = \partial_\theta$ gives the negative pressure-torque term with its sign.

In polar coordinates $\omega_z = r^{-1}\partial_r(ru_\theta) - r^{-1}\partial_\theta u_r$. Average (370) over a full circle. Every angular derivative has zero integral, by the fundamental theorem of calculus and periodicity, including the actual non-axisymmetric pressure. The convection term is $r^{-1}\partial_r(r\langle u_rL \rangle) + \partial_z\langle u_zL \rangle$; the averaged viscous term is $\nu(\partial_{rr} - r^{-1}\partial_r + \partial_{zz})B$. Using $\partial_r = r\partial_\sigma$ now proves

$$\partial_t B + \partial_\sigma P + \partial_z Q = \nu(2\sigma\partial_{\sigma\sigma}B + \partial_{zz}B) + F. \tag{371}$$

The formula extends to $\sigma = 0$ by (367) and smoothness. The pressure has disappeared here by a proved projection, with its entire original torque term exhibited in (370). No scalar heat equation has replaced the actual nonlinear fluxes.

More generally a Cartesian momentum residual $f_i + \sum_j \partial_j T_{ij}$ contributes the torque $\langle x_1 \partial_j T_{2j} - x_2 \partial_j T_{1j} \rangle$. For a symmetric stress T, the product rule gives exactly

$$x_1 \partial_j T_{2j} - x_2 \partial_j T_{1j} = \partial_j (x_1 T_{2j} - x_2 T_{1j}) - (T_{21} - T_{12}) = \partial_j (x_1 T_{2j} - x_2 T_{1j}).$$

Writing its vector flux in polar components therefore turns this contribution into

$$\partial_\sigma \langle r^2 T_{\theta r} \rangle + \partial_z \langle r T_{\theta z} \rangle, \quad T_{\theta r} = e_\theta^{\mathsf{T}} T e_r, \quad T_{\theta z} = e_\theta^{\mathsf{T}} T e_z. \tag{372}$$

For a nonsymmetric tensor the retained extra torque is $-\langle T_{21} - T_{12} \rangle$. Thus every symmetric stress cancellation in the source momentum residual has this exact projected image, with its sign determined by the side of the momentum equation on which the source places $\operatorname{div} T$. The map is linear in the full Cartesian tensor; its evaluation is not an identification of the source's auxiliary-torus average with the physical angular average used here.

The source cutoff sum has this exact image before the terminal limit. In its original viscosity-one coordinates let $\mathbf{b}_j$ denote the actual Cartesian vector appearing as B_j in source (9.21). With q_* here denoting that source's concentration coordinate, retain every increment as

$$\delta u_{*,j} = \operatorname{curl}\big(c_* \chi(a_j q_*) A_j\big) + c_* \chi(a_j q_*) \mathbf{b}_j, \qquad \delta p_{*,j} = c_* \chi(a_j q_*) p_j.$$

The notation q_* in this display is the source coordinate determined by $1 - t = q_*(1 - \eta^2)$ and $z = q_*^{1/2-h} \eta$, not a material label; those two equations specify its actual dependence on (z, t). Define each viscosity-ν increment by the full scaling of Section L.6, and denote the correspondingly localized base by $(u_{\nu,0}, p_{\nu,0})$. The actual fields are $u_\nu = \sum_{j \geq 0} u_{\nu,j}$ and $p_\nu = \sum_{j \geq 0} p_{\nu,j}$, where the positive-index terms are the displayed increments with that scaling. Source local finiteness on every compact set before time one makes these sums, and every derivative of them, finite on a neighborhood of each sampled circle. Indeed the source coordinate is independent of θ, so its positive lower bound and its finite set of active stages can be chosen over the entire compact circle. Thus the exact nonlinear fluxes are

$$P = \sum_{a,b \geq 0} \langle r^2 (u_{\nu,a})_r (u_{\nu,b})_\theta \rangle, \qquad Q = \sum_{a,b \geq 0} \langle r (u_{\nu,a})_z (u_{\nu,b})_\theta \rangle.$$

Each sum contains every diagonal term and both ordered cross terms for each pair $a \neq b$. The curl in $u_{\nu,j}$ is taken after multiplying both cutoffs, so their derivative corrections belong to these same fluxes. Projection, differentiation and the sums commute by the stated local finiteness. Hence the source's exact residual identity, including its flat remainder and final smooth force, passes to (371) with all these contributions. This argument uses the actual summed fields for $t < 1$; it does not exchange an uncontrolled infinite series with the terminal limit.

O.3 The radial Mellin equation with all shifted fluxes

For each actual field $a = B, P, Q, F$ put

$$\widehat{a}(s, z, t) = \int_0^\infty a(\sigma, z, t) \sigma^{s-1} d\sigma, \qquad \sigma^{s-1} = e^{(s-1) \log \sigma}.$$

The original measure here is $d\sigma = r\, dr$, with real logarithm; angular averaging already contains its factor $1/(2\pi)$. The integral for B, Q, F is absolutely convergent and holomorphic on $\mathbb{R}es > -1$, and that for P on $\mathbb{R}es > -2$, by compact support and (367). On compact substrips every complex derivative adds an integrable power of $|\log \sigma|$, which proves holomorphy by differentiated domination. All z, t derivatives on closed preterminal sets obey the same justification.

For $\mathbb{R}es > 0$, integration by parts gives

$$\int_0^\infty \partial_\sigma P\, \sigma^{s-1} d\sigma = -(s-1)\widehat{P}(s-1),$$
$$\int_0^\infty 2\sigma \partial_{\sigma\sigma} B\, \sigma^{s-1} d\sigma = 2s(s-1)\widehat{B}(s-1).$$

All upper boundary terms vanish by compact support. The lower terms $P\sigma^{s-1}$, $B'\sigma^s$ and $B\sigma^{s-1}$ vanish by (367) and $\mathbb{R}es > 0$. Consequently the full equation is

$$\partial_t \widehat{B} - (s-1)\widehat{P}(s-1) + \partial_z \widehat{Q} = \nu\{2s(s-1)\widehat{B}(s-1) + \partial_{zz}\widehat{B}\} + \widehat{F}, \qquad \mathbb{R}es > 0. \tag{373}$$

In particular no shift $s \mapsto s-1$, polynomial factor, or force moment has been suppressed.

This is an actual injective transform with its inverse. For any real $c > 0$ and fixed (z,t) put $g(u) = e^{cu} a(e^u, z, t)$. It vanishes for sufficiently large positive u and is $O(e^{(c+1)u})$ as $u \to -\infty$, with the same bound for every derivative; for P the exponent improves to $c+2$. These facts follow from the chain rule and (367). Thus g is Schwartz and

$$\widehat{a}(c - i\eta, z, t) = \int_{\mathbb{R}} g(u) e^{-i\eta u} du, \qquad a(\sigma, z, t) = \frac{\sigma^{-c}}{2\pi} \int_{\mathbb{R}} \widehat{a}(c - i\eta, z, t) e^{i\eta \log \sigma} d\eta.$$

To justify the inverse without a distributional shortcut, twice integrate by parts in the forward integral to obtain an integrable $O((1+|\eta|)^{-2})$ bound. Insert $e^{-\epsilon\eta^2}$ in the inverse. Fubini and the Gaussian integral give convolution by $(4\pi\epsilon)^{-1/2} e^{-u^2/(4\epsilon)}$, which converges to g by continuity and its Gaussian tail. Dominated convergence in the integrable Fourier inverse then proves the displayed identity.

O.4 The same arithmetic dilation map on the actual corrected fields

Retain the source's chosen $0 < h < 1/100$. For any smooth compactly supported half-line function a with $a(0) = 0$, define

$$\begin{aligned} \phi_a(x) &= a(x) - 2^h a(2x), \\ \psi_a(x) &= \phi_a(x) - \tfrac{1}{2}\phi_a(x/2) \\ &= (1 + 2^{h-1})a(x) - 2^h a(2x) - \tfrac{1}{2}a(x/2), \\ (\mathcal{A}_h a)(x) &= \sum_{n \geq 1} \psi_a(nx), \qquad x > 0. \end{aligned}$$

These are the exact three dilation coefficients used by the parent heat-exterior calculation; here their input is the actual B, P, Q, F. Every sum is locally finite on $x > 0$, and the trace vanishes for all sufficiently large x. Substitution gives $\int_0^\infty \psi_a(x)dx = 0$ and $\psi_a(0) = 0$. Its small-x bound has a direct proof. On each interval $[nx, (n+1)x]$, two integrations by parts give the trapezoidal identity

$$\int_{nx}^{(n+1)x} \psi_a(v)dv = \frac{x}{2}\{\psi_a(nx) + \psi_a((n+1)x)\} - \frac{1}{2}\int_{nx}^{(n+1)x} (v - nx)((n+1)x - v)\psi_a''(v)dv.$$

Summing, using compact support and the two zero quantities just proved, yields

$$|(\mathcal{A}_h a)(x)| \leq \frac{x}{8}\|\psi_a''\|_1. \tag{374}$$

Thus its Mellin integral is holomorphic on $\mathbb{R}es > -1$. Parameter derivatives in z, t have the same bound because the trace is linear, preserves the zero endpoint and zero integral, and commutes with those derivatives on preterminal compact sets.

On $\mathbb{R}es > 1$ absolute Fubini gives

$$\mathcal{M}(\mathcal{A}_h a)(s) = W_h(s)\widehat{a}(s), \qquad W_h(s) = \zeta(s)(1-2^{s-1})(1-2^{h-s}). \tag{375}$$

Indeed the sum of absolute Mellin integrals is $\zeta(\mathbb{R}es)\int |\psi_a(x)|x^{\mathbb{R}es-1}dx$, and a dilation by $b>0$ multiplies its Mellin integral by b^{-s}. The only pole of ζ is its simple pole at one, and the factor $1-2^{s-1}$ cancels it. Consequently W_h is entire and the identity theorem extends (375) to $\mathbb{R}es > -1$ for these actual inputs. Its multiplier is the same arithmetic factor as in the parent exterior calculation. The gamma product there is specifically the Mellin transform of that exterior's H_h; it is not silently substituted for the present corrected-field moment $\widehat{a}$.

The complete inverse preserves the actual field and its endpoint:

$$\begin{aligned}
\psi_a(x) &= \sum_{n\geq 1} \mu(n)(\mathcal{A}_h a)(nx),\\
\phi_a(x) &= \sum_{j\geq 0} 2^{-j}\psi_a(x/2^j),\\
a(x) &= -\sum_{k\geq 1} 2^{-kh}\phi_a(x/2^k).
\end{aligned}$$

The first double sum is finite at each $x>0$, so rearranging gives $\sum_{m\geq 1}\psi_a(mx)\sum_{n|m}\mu(n) = \psi_a(x)$. The divisor sum is one for $m=1$ and zero otherwise, since the squarefree expansion gives $\prod_{p|m}(1-1)$ for $m>1$. The finite second sum through $j=J$ equals $\phi_a(x)-2^{-J-1}\phi_a(x/2^{J+1})$. The negative finite third sum through $k=K$ equals $a(x)-2^{-Kh}a(x/2^K)$. Their tails vanish by boundedness at zero and $h>0$. After any x derivative, each summand gains the additional factor $2^{-j\ell}$ or $2^{-k\ell}$; bounded derivatives on compact half-line intervals prove uniform convergence there. The arithmetic inverse is locally finite away from zero; the subsequent recovered functions extend to zero by the displayed smooth inverse. These are actual inverse maps on this stated class, including its parameter dependence, with no division by zeta values at their zeros.

Set $\mathfrak{B} = \mathcal{M}\mathcal{A}_h B$ and define $\mathfrak{P}, \mathfrak{Q}, \mathfrak{F}$ in the identical way. Multiplication of (373) by $W_h(s)$ gives the exact equation

$$\partial_t\mathfrak{B} + \partial_z\mathfrak{Q} - \nu\partial_{zz}\mathfrak{B} - \mathfrak{F} = (s-1)W_h(s)\{\widehat{P}(s-1) + 2\nu s\widehat{B}(s-1)\}.$$

Both hatted terms on the right are explicitly recovered by the proved inverses, so this equation retains the full dynamics. There is also the division-free shifted relation

$$\begin{aligned}
&W_h(s-1)\{\partial_t\mathfrak{B} + \partial_z\mathfrak{Q} - \nu\partial_{zz}\mathfrak{B} - \mathfrak{F}\} = (s-1)W_h(s)\{\mathfrak{P}(s-1) + 2\nu s\mathfrak{B}(s-1)\},\\
&\mathbb{R}es > 0.
\end{aligned} \tag{376}$$

Initially this follows on $\mathbb{R}es > 1$; every factor is holomorphic on $\mathbb{R}es > 0$ by the previous convergence proofs, so it holds throughout that half-plane. The preceding inverse form retains the information even at a zero of $W_h(s-1)$; the cross-multiplied relation alone is not being asserted to reconstruct information lost by multiplying by a vanishing scalar.

O.5 The actual concentrating circle and its recoverable arithmetic image

In the source's fixed inner similarity region all annular corrections vanish. The remaining cutoff-summed slow field is axisymmetric, so the swirl estimate on the positive x_1 axis in (10.21) holds on the full circle with the same r, z, t. The corresponding source locators are (9.21), (10.20)–(10.21), and the unchanged inner-region construction of [12]. Hence at $\sigma = \nu X_{\mathrm{in}}\tau, z = 0, t = 1-\tau$ the actual average is

$$B(\nu X_{\mathrm{in}}\tau, 0, 1-\tau) = \nu\sqrt{2X_{\mathrm{in}}}\,\tau^{-h}(e_0 + R_*(\tau)). \tag{377}$$

The factor follows from multiplying the full retained swirl in (338) by $r = \sqrt{2\nu X_{\mathrm{in}}\tau}$; the physical angular average of an axisymmetric value leaves it unchanged. In particular this actual scalar angular momentum also diverges, now at the source exponent h, with its original remainder.

Applying the proved inverse series to $\mathcal{A}_h B(\cdot, 0, 1-\tau)$ recovers exactly the value in (377); it is therefore a source-specific arithmetic image of the concentrating corrected field. The double dyadic inverse and the finite-at-each-argument arithmetic inverse retain their stated order of evaluation. No uniform bound for those inverses as the sampling point tends to zero has been inferred, so the recovery identity alone is not a claim that the un-inverted trace has the same pointwise blowup rate.

For completeness the resulting zeta factor also has a precise local zero map. On $h < \mathbb{R}es < 1$, both dilation factors are nonzero. Writing $\zeta(s) = (s - s_0)^m a_0(s)$ at a zeta zero gives

$$\mathfrak{B}(s, z, t) = (s - s_0)^m a_0(s)(1 - 2^{s-1})(1 - 2^{h-s})\widehat{B}(s, z, t).$$

This is the complete product, including the corrected-field Mellin factor. If that factor has order n at s_0, the product has order $m + n$, by multiplication of the two Taylor series; if it is nonzero, the multiplicity is exactly m. Identical statements hold for each flux and force moment. The arithmetic map and inverse are thus exact for the nonlinear witness, while no off-critical zeta zero or nonvanishing of the actual corrected-field Mellin factor is assumed.

P An exact fixed-axis singularity of the arithmetic trace

The complete companion fixed-axis proof [17] is included with a local proof of its endpoint summation formula. The angular-momentum transform can see a fixed-axis derivative of the actual source field directly, without first applying its inverse. The reason is a specific feature of the construction: every positive-order azimuthal coefficient has zero axis value. We retain the source amplitude C, the parameter h, the physical viscosity ν, all cutoffs and the original time $t = 1 - \tau$. The calculation uses the constructed field of Section L.6; it does not certify the entire existence argument of the released manuscript independently.

P.1 The axis trace of the fully corrected velocity

Here and only here let $q_{\mathrm{src}}(z, t)$ denote the scalar similarity coordinate called q in the source, not the material label in Section L. Its source coordinates are

$$\tau = 1 - t = q_{\mathrm{src}}(1 - \eta^2), \qquad z = q_{\mathrm{src}}^{D}\eta, \qquad X = \frac{r^2}{2q_{\mathrm{src}}}, \qquad D = \frac{1}{2} - h, \quad A = \frac{1}{2} + h.$$

In particular at $z = 0$ one has exactly $\eta = 0$ and $q_{\mathrm{src}} = \tau$. Source (5.1), pp. 45–46, writes each azimuthal coefficient as

$$u_{\theta,n} = \frac{r}{C} q_{\mathrm{src}}^{-1-h+2nh}\varphi_n(X, \eta).$$

Its Lemma 5.1, p. 47, prescribes $\varphi_n(0, \eta) = 0$ for every $n \geq 1$. The radial extensions and moment corrections are supported away from the unchanged inner interval. Source Proposition 5.5 and its proof, pp. 60–62, realize this series with cutoffs $\chi(c_n q_{\mathrm{src}})$; these depend on z, t, not r, and the realized angular component is

$$\frac{u_{\theta,B}}{r} = \frac{q_{\mathrm{src}}^{-1-h}}{C}\left(\varphi_0(X, \eta) + \sum_{n\geq 1} \chi(c_n q_{\mathrm{src}}) q_{\mathrm{src}}^{2nh}\varphi_n(X, \eta)\right). \tag{378}$$

For every fixed preterminal z, t, this sum is finite in a neighborhood of that point: $c_n \to \infty$, while $q_{\mathrm{src}} > 0$ and χ has compact support. Thus evaluation at $X = 0$, and the smooth extension of the quotient $u_{\theta,B}/r$, require no exchange of a terminal limit with an infinite sum.

The leading axis datum is not an unknown positive remainder. Source (B.3), p. 145, and Proposition B.2, pp. 146–147, give

$$\varphi_0(0,\eta)=\varphi_*(\eta),\qquad \varphi_*(\eta)=\exp\left(\Lambda\int_0^\eta \zeta_*(w)\,dw\right),\qquad \varphi_*(0)=1.$$

Here ζ_* is the source's auxiliary axis function, not the Riemann zeta function. Neither its value nor C is rescaled.

All annular wave and mean potentials, including the finite initialization and the later cutoff-summed stages, vanish near the axis for each $t<1$: source Proposition 9.9, (9.21), pp. 114–117. Only finitely many stages are present locally at such a time. Their Cartesian derivatives therefore vanish on a common neighborhood of the axis. The base potential is azimuthal and independent of angle, so its curl has only radial and axial components. Consequently it contributes no additional azimuthal term to (378). Finally the spatial and temporal cutoffs in source (10.4), p. 118, are identically one near the origin for all sufficiently small positive τ. Their derivatives vanish there as well. These are facts about the actual summed source field, not a replacement of that field by its leading approximation [12, Lemma 5.1; Proposition 5.5; (9.21); (10.4); (B.3)].

It follows that there is a source-determined $\tau_1>0$ such that

$$\left.\frac{u_{\theta,*}(r,0,1-\tau)}{r}\right|_{r=0}=\frac{1}{C}\tau^{-1-h}\qquad(0<\tau<\tau_1).\tag{379}$$

The quotient denotes its proved smooth extension, not division by zero. For the physical viscosity map $u_\nu(x,t)=\sqrt{\nu}\,u_*(x/\sqrt{\nu},t)$, put $r_*=r/\sqrt{\nu}$. Then, for $r>0$,

$$\frac{u_{\theta,\nu}(r,0,t)}{r}=\frac{\sqrt{\nu}\,u_{\theta,*}(r_*,0,t)}{\sqrt{\nu}\,r_*}=\frac{u_{\theta,*}(r_*,0,t)}{r_*}.$$

Passing to the smooth axis extension proves the same identity (379) for u_ν. This cancellation follows from the actual coordinate map; viscosity has not been set to one.

Use the original physical radial coordinate $\sigma=r^2/2$ and angular average $(2\pi)^{-1}\int_0^{2\pi}$. Let

$$B_\nu(\sigma,z,t)=\frac{1}{2\pi}\int_0^{2\pi} r u_{\theta,\nu}(r,\theta,z,t)\,d\theta.$$

In the axis neighborhood just established the full field is axisymmetric and $B_\nu=2\sigma(u_{\theta,\nu}/r)$. The exact consequence is

$$B_\nu(0,0,1-\tau)=0,\qquad \partial_\sigma B_\nu(0,0,1-\tau)=\frac{2}{C}\tau^{-1-h}.\tag{380}$$

No $O(\tau^{2h})$ error is present in this first axis derivative: each positive-order axis value is exactly zero. This does not remove the actual remainder at the distinct nonzero X_{in} sample of Section L.6.

P.2 The endpoint summation identity with its complete remainder

The endpoint calculation below uses an exact summation formula. We give its proof here so that the fixed-axis result has no missing local dependency. Define the Bernoulli polynomials by the convergent power series, for the auxiliary variable v near zero,

$$\frac{ve^{xv}}{e^v-1}=\sum_{n=0}^{\infty}B_n(x)\frac{v^n}{n!},\qquad B_n=B_n(0).$$

The singularity at $v = 0$ is removable. Differentiation in x gives $B_n' = nB_{n-1}$; subtracting the series at $x = 0$ from that at $x = 1$ gives $B_n(1) = B_n(0)$ for $n \neq 1$ and $B_1(1) - B_1(0) = 1$. Expansion to second degree gives $B_1(x) = x - 1/2$ and $B_2 = 1/6$. The function $v/(e^v - 1) + v/2$ is even, by direct substitution of $-v$. Therefore $B_{2k+1} = 0$ for every $k \geq 1$. Write $\widetilde{B}_n(y) = B_n(y - \lfloor y \rfloor)$ away from integers, with the common endpoint value for $n \geq 2$. For these n the function is continuous, bounded and periodic.

Let f be smooth and compactly supported on $[0, \infty)$, let $x > 0$, and let $M \geq 1$ be an integer. The exact identity is

$$\begin{aligned} \sum_{n\geq 1} f(nx) &= \frac{1}{x}\int_0^\infty f(v)\,dv - \frac{f(0)}{2} - \sum_{k=1}^{M} \frac{B_{2k}}{(2k)!} x^{2k-1} f^{(2k-1)}(0) + \mathcal{R}_M(f,x), \\ \mathcal{R}_M(f,x) &= -\frac{x^{2M-1}}{(2M)!}\int_0^\infty \widetilde{B}_{2M}(v/x) f^{(2M)}(v)\,dv, \\ |\mathcal{R}_M(f,x)| &\leq \frac{\|B_{2M}\|_{L^\infty([0,1])}}{(2M)!} x^{2M-1} \|f^{(2M)}\|_{L^1(0,\infty)}. \end{aligned} \tag{381}$$

To prove it first take mesh one on $[0, N]$, with integer N beyond the support. On every unit interval integration by parts gives

$$\int_n^{n+1} B_1(v-n) f'(v)\,dv = \frac{f(n) + f(n+1)}{2} - \int_n^{n+1} f(v)\,dv.$$

Summing produces the trapezoidal sum minus the integral. Repeated integration by parts using $B_j' = jB_{j-1}$ yields the endpoint terms $B_{2k}(f^{(2k-1)}(N) - f^{(2k-1)}(0))/(2k)!$ and the remaining integral $-\int_0^N \widetilde{B}_{2M} f^{(2M)}/(2M)!$. The internal boundary terms cancel because the endpoint values of B_j agree for $j \geq 2$; the odd endpoint coefficients above degree one vanish by the already proved parity. All upper endpoint derivatives vanish by the choice of N. Apply this identity to $v \mapsto f(xv)$ and substitute $u = xv$ in each integral. This gives every power and sign in (381), including the stated norm bound.

For completeness the value of zeta needed at the first axis pole also follows from this formula. Apply its finite-interval proof on $[1, N]$ to v^{-w} and initially let $N \to \infty$ for $\operatorname{Re} w > 1$. For $M = 2$ the result is

$$\zeta(w) = \frac{1}{w-1} + \frac{1}{2} + \frac{B_2}{2!} w + \frac{B_4}{4!} w(w+1)(w+2) - \frac{(w)_4}{4!}\int_1^\infty \widetilde{B}_4(v) v^{-w-4}\,dv.$$

Here $(w)_4 = w(w+1)(w+2)(w+3)$. Boundedness of the periodic polynomial proves that the last integral is holomorphic for $\operatorname{Re} w > -3$, with all complex derivatives justified by the integrable logarithmic weights on compact subsets. Thus the equality continues there apart from the pole at one. Substitution $w = -1$ cancels the first two terms, and the two higher products vanish. Hence $\zeta(-1) = -B_2/2 = -1/12$, with its sign fixed by the actual summation formula. These classical formulas agree with [18, §2.10(i)] and [19, §25.6(i)].

P.3 The arithmetic operator on every endpoint derivative

We prove the endpoint map on its full smooth compactly supported domain. Let $a \in C^\infty([0, \infty))$ have compact support and $a(0) = 0$. For the same fixed h define

$$\phi_a(x) = a(x) - 2^h a(2x), \qquad \psi_a(x) = \phi_a(x) - \tfrac{1}{2}\phi_a(x/2), \qquad (\mathcal{A}_h a)(x) = \sum_{n\geq 1} \psi_a(nx) \quad (x > 0).$$

This is exactly the arithmetic operator used on the nonlinear fluxes. There is no radius change hidden in its argument x. Every sum and its derivatives are locally finite for $x > 0$. Substitution

in the two integrals proves $\int_0^\infty \psi_a = 0$, and $\psi_a(0) = 0$. The full endpoint derivatives of the input are

$$\psi_a^{(j)}(0) = (1 - 2^{-j-1})(1 - 2^{h+j})a^{(j)}(0) \qquad (j \geq 0). \tag{382}$$

For clarity the passage from the sum to actual endpoint derivatives is justified here, rather than differentiating a formal asymptotic series. We use the Euler–Maclaurin identity proved with its remainder in (381); its classical provenance is [18, §2.10(i), (2.10.1)]. With $f_j(v) = v^j \psi_a^{(j)}(v)$, direct differentiation on $x > 0$ gives

$$x^j (\mathcal{A}_h a)^{(j)}(x) = \sum_{n \geq 1} f_j(nx).$$

All these f_j are smooth and compactly supported. Integration by parts j times gives $\int f_j = (-1)^j j! \int \psi_a = 0$; the factors v^j make each zero-end boundary term vanish. Their axis derivatives are

$$f_j^{(r)}(0) = 0 \quad (r < j), \qquad f_j^{(r)}(0) = \frac{r!}{(r-j)!} \psi_a^{(r)}(0) \quad (r \geq j).$$

Choose the Euler–Maclaurin order M so that $2M - 1 > j$. After division by x^j, its remainder is bounded by

$$\frac{\|B_{2M}\|_\infty}{(2M)!}\, x^{2M-1-j} \|f_j^{(2M)}\|_1,$$

and hence tends to zero. Terms below degree j vanish, so every differentiated series has a finite limit at zero. The only constant term for $j \geq 1$ is the Bernoulli term with $2k - 1 = j$; for even $j \geq 2$ it is absent. The fundamental theorem of calculus on $[\epsilon, x]$, followed by $\epsilon \downarrow 0$, now identifies these derivative limits with successive derivatives of the extended function. Boundedness of the next derivative justifies the passage under each integral. Thus $\mathcal{A}_h a$ is smooth at zero, and

$$\begin{aligned} (\mathcal{A}_h a)(0) &= 0, \\ (\mathcal{A}_h a)^{(j)}(0) &= -\frac{B_{j+1}}{j+1}(1 - 2^{-j-1})(1 - 2^{h+j})a^{(j)}(0), \qquad j \geq 1. \end{aligned} \tag{383}$$

The Bernoulli convention is $B_2 = 1/6$ and odd $B_{j+1} = 0$ for even $j \geq 2$. In particular

$$(\mathcal{A}_h a)'(0) = \frac{2^{h+1} - 1}{16}\, a'(0). \tag{384}$$

The coefficient is strictly positive for the source $h > 0$.

The preceding proofs apply also after any fixed z, t derivative of a smooth compactly supported parameter family with common support on compact preterminal parameter sets. Every bound then holds uniformly on those sets, with constants depending on the derivative order. This proves compatibility of the endpoint map with those parameter derivatives. It supplies no uniform estimate as $t \uparrow 1$.

P.4 A directly visible singularity at a fixed arithmetic coordinate

Apply this same operator to the actual compactly supported $a(\sigma) = B_\nu(\sigma, z, t)$. Write its arithmetic image as

$$\mathcal{B}_\nu(x, z, t) = \mathcal{A}_h B_\nu(x, z, t).$$

The domain and all nonlinear fluxes are those already given in the preceding angular-momentum section. Equations (380) and (384) now prove the exact formula

$$\boxed{\partial_x \mathcal{B}_\nu(0, 0, 1 - \tau) = \frac{2^{h+1} - 1}{8C}\, \tau^{-1-h}, \qquad 0 < \tau < \tau_1, \quad \nu > 0.} \tag{385}$$

Here $x = 0, z = 0$ are fixed coordinates, not a shrinking sampling path, and $\mathcal{B}_\nu(0, z, t) = 0$ for every preterminal t. Thus the value is zero while its first x-derivative diverges positively. For any fixed $\epsilon > 0$, the $C^1([0, \epsilon])$ seminorm is at least the displayed endpoint derivative. In particular no extension continuous at $t = 1$ with values in $C^1([0, \epsilon])$ can agree with this family. This assertion follows from an exact derivative of the original arithmetic sum, not merely a lower bound obtained after its inverse.

For every integer $m \geq 0$, with $(1 + h)_m = \prod_{j=0}^{m-1}(1 + h + j)$ and empty product one, differentiation of the exact equality gives

$$\partial_t^m \partial_x \mathcal{B}_\nu(0, 0, t) = \frac{2^{h+1} - 1}{8C}(1 + h)_m (1 - t)^{-1-h-m}, \qquad 1 - \tau_1 < t < 1.$$

Indeed $d_t(1 - t)^{-\alpha} = \alpha(1 - t)^{-\alpha-1}$. These are equalities on an open preterminal interval. They make no assignment of finite terminal derivatives.

The projection onto B_ν has not discarded the other velocity data in the full correspondence. Its exact smooth section and kernel are

$$RB_\nu = \frac{B_\nu(r^2/2, z, t)}{r^2}(-x_2, x_1, 0), \qquad w_\nu = u_\nu - RB_\nu, \qquad \langle r(w_\nu)_\theta \rangle = 0,$$

where B_ν/σ has its smooth integral extension at zero. The preceding section proves the inverse $(B_\nu, w_\nu) \mapsto RB_\nu + w_\nu$ and retains every covariance in both momentum fluxes. The signal (385) concerns the B_ν component of that complete map. It does not replace the full Navier–Stokes equation by a closed scalar equation for B_ν.

P.5 The exact Mellin pole carrying this axis signal

For fixed preterminal parameters, put

$$\widehat{B}_\nu(s) = \int_0^\infty B_\nu(\sigma, z, t)\sigma^{s-1}\, d\sigma, \qquad \mathfrak{B}_\nu(s) = \int_0^\infty \mathcal{B}_\nu(x, z, t)x^{s-1}\, dx.$$

Both integrals are initially holomorphic on $\Re s > -1$. The full transform calculation in the preceding section gives

$$\mathfrak{B}_\nu(s) = W_h(s)\widehat{B}_\nu(s), \qquad W_h(s) = \zeta(s)(1 - 2^{s-1})(1 - 2^{h-s}).$$

The pole of ζ at 1 is canceled by the displayed first dilation factor, making W_h entire.

Here is the continuation needed at $s = -1$. For any fixed $b > 0$, split the integral at b and subtract the linear Taylor term on $[0, b]$. Since $B_\nu(0) = 0$, the exact formula is

$$\widehat{B}_\nu(s) = \int_0^b [B_\nu(\sigma) - B_{\nu,\sigma}(0)\sigma]\sigma^{s-1} d\sigma + \int_b^\infty B_\nu(\sigma)\sigma^{s-1} d\sigma + \frac{B_{\nu,\sigma}(0)b^{s+1}}{s + 1}.$$

Taylor's integral remainder is $O(\sigma^2)$; the first integral is holomorphic for $\Re s > -2$, including after any complex s derivative on a compact subset, because the additional powers of $\log \sigma$ remain integrable. The second is entire by compact support away from zero. Hence $\operatorname{Res}_{s=-1} \widehat{B}_\nu = B_{\nu,\sigma}(0)$. The identical argument applies to the proved smooth $\mathcal{B}_\nu$, so its residue is $\partial_x \mathcal{B}_\nu(0)$.

The classical value $\zeta(-1) = -1/12$ [19, §25.6(i)] gives

$$W_h(-1) = -\frac{1}{12}\left(1 - \frac{1}{4}\right)(1 - 2^{h+1}) = \frac{2^{h+1} - 1}{16} > 0.$$

Equality on $\Re s > -1$ therefore continues to the meromorphic functions just constructed and agrees with the direct endpoint proof. At $z = 0, t = 1 - \tau$ the pole is genuine and its residue is exactly

$$\operatorname{Res}_{s=-1} \mathfrak{B}_\nu(s, 0, 1-\tau) = \frac{2^{h+1}-1}{8C}\tau^{-1-h}. \tag{386}$$

There is no zero-factor compensation at this pole, since $W_h(-1) > 0$. The sign of $\zeta(-1)$ alone is not the sign of the arithmetic residue: both dilation factors have been evaluated explicitly.

This identifies what the construction propagates in this channel: an exact axis derivative, equivalently a growing residue at the fixed Mellin point -1. The identity retains the full $\widehat{B}_\nu$, its flux equations, and its inverse maps. The nontrivial zeros of ζ remain the zero factors specified in the full product formula; no off-critical zero is inferred from this axis pole or from its positive residue.

Q Exact entrance of the angular Mellin profiles into the source quotient

Q.1 The original direct dilation and every endpoint pole

In this section $\sigma = r^2/2 \geq 0$ is the physical radial coordinate, z is the unchanged axial coordinate, t is physical time, w is a Mellin variable, and s is the original Connes spectral variable. The radial Mellin coordinate called s in Sections O and P is denoted by w here. The original Connes spectral coordinate remains s. Their exact affine relation is $w = 1/2 - is$, with inverse $s = i(w - 1/2)$. The functional equation gives $\Xi(s) = \xi(1/2 + is) = \xi(w)$. Fix (z, t) before terminal time and let

$$a \in \mathscr{A} = \{a \in C_c^\infty([0,\infty)) : a(0) = 0\}.$$

The actual inputs B, P, Q, F of the angular construction belong to this space. Dependence on (z, t) is retained as a smooth parameter on every compact preterminal set. Keep the source's chosen $0 < h < 1/100$ and its exact dilation operators

$$\phi_a(\sigma) = a(\sigma) - 2^h a(2\sigma), \quad \psi_a(\sigma) = \phi_a(\sigma) - \tfrac{1}{2}\phi_a(\sigma/2), \quad A_h a(x) = \sum_{n\geq 1} \psi_a(nx).$$

The letter $x > 0$ in this last expression is the argument of the direct arithmetic dilation; it has the same numerical value as the input radial variable and is not the Cartesian first coordinate. The source identity, with its unchanged measure, is

$$\begin{gathered} \mathcal{M}(A_h a)(w) = W_h(w)\widehat{a}(w), \\ W_h(w) = \zeta(w)(1 - 2^{w-1})(1 - 2^{h-w}), \qquad \widehat{a}(w) = \int_0^\infty a(\sigma)\sigma^{w-1} d\sigma. \end{gathered} \tag{387}$$

It initially holds on $\operatorname{Re} w > 1$ and extends to $\operatorname{Re} w > -1$ by the proved endpoint bound.

Write $a_n = a^{(n)}(0)/n!$, $n \geq 1$. The full meromorphic continuation of $\widehat{a}$ has only possible simple poles at $w = -n$, with

$$\operatorname{Res}_{w=-n} \widehat{a}(w) = a_n. \tag{388}$$

Here is a direct proof for arbitrary smooth input. Choose a smooth compact cutoff χ equal to one near zero. For any $N \geq 1$, $r_N = a - \chi \sum_{n=1}^N a_n \sigma^n$ vanishes to order $N+1$. Thus $\int r_N(\sigma)\sigma^{w-1} d\sigma$ is holomorphic on $\operatorname{Re} w > -N-1$. The cutoff Mellin transform $C_\chi(v) = \int \chi(\sigma)\sigma^{v-1} d\sigma$ satisfies $C_\chi(v) = -v^{-1} \int \chi'(\sigma)\sigma^v d\sigma$. The numerator is entire and has value -1 at $v = 0$, giving residue one. Consequently

$$\widehat{a}(w) = \int_0^\infty r_N(\sigma)\sigma^{w-1} d\sigma + \sum_{n=1}^N a_n C_\chi(w + n).$$

These expressions agree on their original common half-plane and therefore on their overlaps. They prove the continuation and every residue in (388). Euler-operator integration by parts in the remainder, with its exact endpoint vanishing order, also proves rapid decrease on every finite vertical strip away from the finitely many poles, exactly as in the previous full even-Schwartz Mellin proof.

The dilation factor $d_h(w) = (1 - 2^{w-1})(1 - 2^{h-w})$ is nonzero at every $w = -n$, $n \geq 1$. At negative even integers the simple zero of ζ cancels the possible Mellin pole. At negative odd integers $\zeta(-n) \neq 0$: in the retained completion $\xi(w) = w(w-1)\pi^{-w/2}\Gamma(w/2)\zeta(w)/2$, the other factors are finite and nonzero there, while $\xi(-n) = \xi(1+n) > 0$. The gamma recurrence makes $\Gamma(-n/2)$ finite and nonzero for odd n, starting from $\Gamma(1/2) > 0$; the positivity at $1+n$ follows from its positive Dirichlet series. At $w = 1$ the zero of $1 - 2^{w-1}$ cancels the sole zeta pole. Thus

$$W_h\widehat{a} \text{ is entire} \quad \Longleftrightarrow \quad a^{(2k+1)}(0) = 0 \text{ for every } k \geq 0. \tag{389}$$

Its uncanceled poles and residues are precisely

$$\operatorname{Res}_{w=-n} W_h(w)\widehat{a}(w) = \zeta(-n)(1 - 2^{-n-1})(1 - 2^{h+n})a_n, \qquad n \geq 1 \text{ odd}.$$

For example its first pole coefficient is $\zeta(-1)(3/4)(1 - 2^{h+1})a'(0)$; all factors multiplying $a'(0)$ are nonzero.

This condition is also the exact parity condition for the direct original trace. Taylor differentiation gives

$$\frac{\psi_a^{(n)}(0)}{n!} = (1 - 2^{-n-1})(1 - 2^{h+n})a_n.$$

The multiplier is nonzero for every integer $n \geq 1$. Therefore $\psi_a(|x|)$ is smooth and even on $\mathbb{R}$ exactly when all odd jets of a vanish. The even extension is then compactly supported, zero at zero, and has integral zero, because $\int_0^\infty \psi_a = 0$. It belongs to the precise $\mathcal{S}_0^{\mathrm{ev}}$ used in the source. Its original arithmetic trace is the entire continuation

$$\mathcal{F}_a(s) = W_h(1/2 - is)\widehat{a}(1/2 - is).$$

The identity with the source trace follows first for $\operatorname{Re} w > 1$ by $F_\psi(i(w - 1/2)) = \zeta(w)\mathcal{M}\psi(w)$, and then everywhere by analytic continuation. This proves the exact trace identity without calling a general meromorphic Mellin profile entire.

For the actual angular momentum the first obstruction is a familiar but fully specified Cartesian observable:

$$\partial_\sigma B(0, z, t) = (\partial_{x_1} u_2 - \partial_{x_2} u_1)(0, 0, z, t) = \omega_z(0, 0, z, t). \tag{390}$$

Indeed expand the smooth Cartesian u_1, u_2 to first order at the axis. In $\langle x_1 u_2 - x_2 u_1 \rangle$, constant velocity terms have zero average; the mixed quadratic terms have zero average; and $\langle x_1^2 \rangle = \langle x_2^2 \rangle = r^2/2 = \sigma$. The remaining coefficient is exactly $\partial_{x_1} u_2 - \partial_{x_2} u_1$. The higher-order remainder is $o(r^2)$ after angular averaging, and smoothness in σ identifies the displayed derivative. Thus the angular source's $B = O(\sigma)$ alone does not enforce the first, or any of the higher odd-jet entrance conditions.

Q.2 Order-one growth is an additional exact entrance condition

For either entire profile below, order at most one is tested in the original coordinate by the previously defined growth seminorms. The full direct domain is therefore the explicit linear space

$$\mathscr{A}_{\mathrm{dir}} = \Big\{ a \in \mathscr{A} : a^{(2k+1)}(0) = 0 \ (k \geq 0), \\ \sup_{w \in \mathbb{C}} |W_h(w)\widehat{a}(w)| e^{-c|i(w-1/2)|^p} < \infty \quad (1 < p < 2,\ c > 0) \Big\}. \tag{391}$$

The product in this expression uses its proved removable values. This is a necessary and sufficient condition expressed entirely through the original input's Mellin integrals and endpoint continuation above: the parity statement is exactly entireness, and the displayed bounds are exactly the proved characterization of the original set $\mathcal{H}_{\le 1}$. No derivative-size bound has been silently imposed on arbitrary smooth compact input. For $a \in \mathscr{A}_{\mathrm{dir}}$, the original Mellin homeomorphism and trace intersection theorem give

$$\mathcal{F}_a \in \mathcal{A}_{\mathrm{Sch}} \subset I = \Xi\mathcal{H}_{\le 1}, \qquad \frac{\mathcal{F}_a(i(w-1/2))}{\zeta(w)} = d_h(w)\widehat{a}(w) \in \mathfrak{M}_0^{\mathrm{ev}}. \tag{392}$$

All its Mellin strip bounds and the unique even Schwartz preimage are therefore the ones already proved, rather than an unproved new source topology.

Here is a useful sufficient entrance class with a complete growth argument. Suppose a is analytic at the endpoint, meaning that $a(\sigma) = \sum_{n\ge 1} a_n \sigma^n$ on $[0, \delta_0]$ for a convergent power series of positive radius. Choose $0 < \delta < \delta_0$ inside its convergence disk, and let $R_0 > \delta$ contain the support. Exact integration gives

$$\widehat{a}(w) = \int_\delta^{R_0} a(\sigma)\sigma^{w-1} d\sigma + \sum_{n\ge 1} \frac{a_n \delta^{w+n}}{w+n}. \tag{393}$$

The series converges normally away from its indicated poles because $\sum |a_n|\delta^n < \infty$. Outside the radius-$1/4$ disks about the negative integers it gives $|\widehat{a}(w)| \le Ce^{C|w|}$: bound the integral over its fixed positive interval and use $|w+n|^{-1} \le 4$ in the series.

For clarity the zeta factor used in the remaining growth argument also has order at most one after its pole is removed. Let $R_\Gamma(z) = 1/\Gamma(z)$. The entire product

$$R_\Gamma(z) = ze^{\gamma_{\mathrm{E}} z} \prod_{n=1}^{\infty} (1 + z/n)e^{-z/n}$$

has logarithmic modulus at most $CR\log(R+2)$ on $|z| \le R$ for $R \ge 1$. To verify both its identity and this bound, repeated integration by parts in the beta integral gives

$$\int_0^1 t^{z-1}(1-t)^n dt = \frac{n!}{z(z+1)\cdots(z+n)} \quad (\operatorname{Re} z > 0).$$

After $u = nt$, multiplication by n^z makes the integral converge to $\Gamma(z)$ by domination with $u^{\operatorname{Re} z - 1}e^{-u}$, locally uniformly in that half-plane. Rewriting its reciprocal using $H_n - \log n \to \gamma_{\mathrm{E}}$ gives the displayed product. The limit of $H_n - \log n$ exists by comparison of $1/n$ with the integral of $1/x$ over each unit interval. Its product tails converge locally uniformly because $|\log(1+v) - v| \le 2|v|^2$ for $|v| \le 1/2$. For $n > 2R$ their absolute logarithmic sum is at most $2R^2 \sum_{n>2R} n^{-2} \le CR$. For $n \le 2R$, use $(1 + R/n)e^{R/n}$ to bound each factor; the logarithmic sum is at most $2R\log(1+R) + RH_{\lfloor 2R \rfloor}$. The initial factors give the claimed bound. The finite beta-integral expression times its rewritten reciprocal is identically one; passing to the two locally uniform limits proves $\Gamma(z)R_\Gamma(z) = 1$ on the positive half-plane, without presuming nonvanishing of a limit. Analytic continuation proves the reciprocal-gamma identity everywhere, including its zeros.

The retained completion now gives

$$(w-1)\zeta(w) = 2\pi^{w/2}\xi(w)\frac{R_\Gamma(w/2)}{w}.$$

The last quotient has its removable value $1/2$ at zero. The original $\xi(w) = \Xi(i(w-1/2))$ has order at most one; the product bound just proved and multiplication by an exponential show the same for $(w-1)\zeta(w)$. Division by w is bounded outside a fixed disk, and removable

holomorphy bounds its inside. This proves the assertion with the actual completion constants. The two dyadic factors have exponential type. Combining these bounds with (393) proves the order-one bounds for $W_h\widehat{a}$ outside the fixed negative-integer disks and a fixed disk about one. When the parity conditions hold, that product is entire. The maximum-modulus principle on every omitted disk supplies the same order-one bounds throughout its interior, with radius enlarged by at most one. Hence every endpoint-analytic a with the stated odd jets zero belongs to $\mathscr{A}_{\mathrm{dir}}$. In particular every smooth compact input vanishing on a whole neighborhood of zero belongs to this domain.

Q.3 A square-root radial ingress retaining every input

There is a second exact map which enforces the source's even test-space condition and retains the original radial coordinate. Define

$$b_a(\rho) = a(\rho^2/2), \quad \chi_a(\sigma) = \phi_a(\sigma) - 2^{-1/2}\phi_a(\sigma/2), \quad \psi_a^{\mathrm{rad}}(\rho) = \chi_a(\rho^2/2).$$

Here ρ is a signed real test coordinate; $\sigma = \rho^2/2$ and $|\rho| = \sqrt{2\sigma}$. This is exactly the earlier physical radius with both signs supplied for the even test. It does not assert equality with the polynomial coordinate $F_1/2$. Each b_a and ψ_a^{rad} is smooth, compactly supported and even, with value zero at zero. Its integral satisfies

$$\int_0^\infty \psi_a^{\mathrm{rad}}(\rho)d\rho = 2^{-1/2}\int_0^\infty \chi_a(\sigma)\sigma^{-1/2}d\sigma = 0.$$

Indeed the second dilation in χ_a multiplies the last integral by $2^{1/2}$, exactly canceled by its coefficient $2^{-1/2}$. Thus $\psi_a^{\mathrm{rad}} \in \mathcal{S}_0^{\mathrm{ev}}$ for every $a \in \mathscr{A}$.

With no measure factor removed, substitution $\rho = \sqrt{2\sigma}$ gives the exact Mellin and source trace formulas

$$\begin{aligned} M_a^{\mathrm{rad}}(w) &= 2^{w/2-1}(1-2^{w/2-1/2})(1-2^{h-w/2})\widehat{a}(w/2), \\ \mathcal{F}_a^{\mathrm{rad}}(s) &= \zeta(w)M_a^{\mathrm{rad}}(w), \qquad w = 1/2 - is. \end{aligned} \tag{394}$$

Initially derive them on $\operatorname{Re} w > 2$ by absolute integration and the source trace formula. The proved even Schwartz trace is entire, so it supplies their continuation with all removable values. At $w = -2n$ the only possible pole of M_a^{rad} has the full residue

$$\operatorname{Res}_{w=-2n} M_a^{\mathrm{rad}}(w) = 2^{-n}(1-2^{-n-1/2})(1-2^{h+n})a_n, \qquad n \geq 1. \tag{395}$$

The factor 2 from the residue of $\widehat{a}(w/2)$ combines with $2^{w/2-1}$ to give 2^{-n}, as displayed. These are precisely the Taylor coefficients of $\chi_a(\rho^2/2)$ at powers ρ^{2n}; all its original radial jets have been retained.

This map is injective and has a full inverse on its image. From its entire source trace recover the meromorphic function M_a^{rad} by division by ζ, with the removable values specified by the identity, and apply the already proved even Schwartz Mellin inverse to recover ψ_a^{rad}. Then

$$\begin{aligned} \chi_a(\sigma) &= \psi_a^{\mathrm{rad}}(\sqrt{2\sigma}), \\ \phi_a(\sigma) &= \sum_{j\geq 0} 2^{-j/2}\chi_a(\sigma/2^j), \\ a(\sigma) &= -\sum_{k\geq 1} 2^{-kh}\phi_a(\sigma/2^k). \end{aligned} \tag{396}$$

The finite second sum leaves remainder $2^{-(J+1)/2}\phi_a(\sigma/2^{J+1})$ and the finite third sum leaves remainder $2^{-Kh}a(\sigma/2^K)$. Both tend to zero by boundedness at zero. After ℓ derivatives, their

terms gain $2^{-j\ell}$ and $2^{-k\ell}$; bounded derivatives on compact half-line intervals give uniform convergence for every derivative. The first recovered function has its smooth extension at zero because its data lie in this image. Thus these formulas recover the full original a, not only its zero set or finitely many moments.

In the original radial Mellin variable the exact relation between the two arithmetic transforms is the common input $\widehat{a}$:

$$\mathcal{F}_a(i(w-\tfrac{1}{2})) = \zeta(w)(1-2^{w-1})(1-2^{h-w})\widehat{a}(w),$$
$$\mathcal{F}_a^{\mathrm{rad}}(i(2w-\tfrac{1}{2})) = 2^{w-1}\zeta(2w)(1-2^{w-1/2})(1-2^{h-w})\widehat{a}(w).$$

The first line uses its meromorphic continuation for arbitrary input. The second is entire in its original spectral variable. Both retain the distinct zeta arguments, dyadic coefficients and factor 2^{w-1}. Their proved inverse constructions recover the same input even at zeros of these factors. Neither formula divides one entire trace by the other at an unexamined zero.

The full order-one entrance domain of this second map is

$$\mathscr{A}_{\mathrm{rad}} = \left\{ a \in \mathscr{A} : \sup_{w\in\mathbb{C}} |\zeta(w) M_a^{\mathrm{rad}}(w)| e^{-c|i(w-1/2)|^p} < \infty \quad (1<p<2,\ c>0) \right\}. \tag{397}$$

Entireness is already proved for every input; the displayed condition is exactly the remaining order condition. Every endpoint-analytic a belongs to this domain, without any parity restriction. To prove it, apply (393) with argument $w/2$, the just-proved zeta growth bound and the exponential dyadic factors. The only possible Mellin poles are at negative even w, and all are canceled by the simple trivial zeros. Maximum modulus fills their fixed disks as before. For every $a \in \mathscr{A}_{\mathrm{rad}}$ the trace is in $\mathcal{A}_{\mathrm{Sch}} \subset \Xi\mathcal{H}_{\le 1}$, and its precise source Schwartz input is (394) with the inverse (396).

Q.4 An exact full-input family in the original order class

Arbitrary smooth endpoint data still have an information-preserving entrance into a family of actual order-one trace functions. Fix one smooth function $\chi : [0,\infty) \to [0,1]$ equal to zero on $[0,1]$ and one on $[2,\infty)$, and set

$$\epsilon_n = 2^{-n}, \qquad a_n(\sigma) = \chi(\sigma/\epsilon_n)a(\sigma), \qquad n \ge 1.$$

These are specified functions of the full input, not replacements of it. Every a_n vanishes near zero, so both $\mathcal{F}_{a_n}$ and $\mathcal{F}_{a_n}^{\mathrm{rad}}$ lie in the original $\mathcal{H}_{\le 1}$ by the sufficient entrance theorem. Retain the whole sequence $\mathfrak{I}(a) = (\mathcal{F}_{a_n}^{\mathrm{rad}})_{n\ge 1}$. It is injective: invert each entry by (396); for every fixed $\sigma > 0$ all sufficiently large n have $a_n = a$ on a neighborhood of σ. Hence the recovered values and all derivatives there are eventually exact. At zero the full original jets are recovered by $a^{(k)}(0) = \lim_{\sigma\downarrow 0} a^{(k)}(\sigma)$, using the known smooth half-line domain. Also $a(0) = 0$ is retained. This proves the full inverse on the image, including endpoint information; no finite truncation of the sequence has been asserted sufficient.

This gives a direct actual entrance for all the corrected angular fields without an unproved endpoint analyticity assertion. For the source sample $\sigma_\tau = \nu X_{\mathrm{in}}\tau$ choose any integer $n(\tau)$ with $2\epsilon_{n(\tau)} < \sigma_\tau$. Then inversion of that specified sequence entry recovers the exact sample value

$$B(\sigma_\tau, 0, 1-\tau) = \nu\sqrt{2X_{\mathrm{in}}}\tau^{-h}(e_0 + R_*(\tau)).$$

All sample-dependent index choices recover the same full value, since the cutoff is exactly one there. This is a proved inverse evaluation of the source's actual sample, not an assertion that an uninverted trace or its source quotient class has this same pointwise rate.

Q.5 The source kernels, arithmetic refinement, and derivative defect

For $\bullet \in \{\mathrm{dir}, \mathrm{rad}\}$ give $\mathscr{A}_\bullet$ the initial topology of its inclusion in the usual compact-test space together with $a \mapsto \mathcal{F}_a^\bullet \in (\mathcal{H}_{\le 1}, \tau \vee \gamma)$. Here the compact-test topology is the inductive limit of the spaces supported in $[0, R]$, $R = 1, 2, \ldots$, with seminorms $\max_{0 \le \sigma \le R} |a^{(j)}(\sigma)|$, $j \ge 0$. This explicitly defines the topology required for the following maps; it does not assert continuity into the unnamed τ from the compact-test topology alone. The actual source quotient entrance and its exact kernel are

$$q_\bullet : \mathscr{A}_\bullet \longrightarrow Q_{\mathrm{ar}}, \quad a \longmapsto [\mathcal{F}_a^\bullet], \qquad \ker q_\bullet = \{a \in \mathscr{A}_\bullet : \mathcal{F}_a^\bullet \in N_{\mathrm{ar}}\}. \tag{398}$$

Each map is continuous by the topology just specified. Its underlying entire-trace map is injective: divide by zeta on its meromorphic image, use the proved Mellin inverse to recover the exact test, and then the convergent dilation inverses to recover a. For the direct map those inverses are the original $\phi_a = \sum_{j \ge 0} 2^{-j} \psi_a(\sigma/2^j)$ and $a = -\sum_{k \ge 1} 2^{-kh} \phi_a(\sigma/2^k)$; their telescoping proof is identical with the retained exponent 1 instead of $1/2$ in the second dilation. Thus (398) describes exactly the information lost on subsequently imposing the actual source relation, not an ambiguity of the Mellin reconstruction.

Let $H_\cap = N_{\mathrm{ar}} \cap I$ and $C_\cap = (\mathcal{H}_{\le 1}, \tau \vee \gamma)/H_\cap$ be the previously proved common refinement. The maps $a \mapsto [\mathcal{F}_a^\bullet]_{C_\cap}$ are continuous and project to (398). Their kernels are equal to those in that equation, since all the trace images already lie in I. Their images lie in the exact kernel $I/H_\cap$ of $C_\cap \to Q_I$. This proves the precise comparison with the analytic realization; it does not identify N_{ar} with I or assert $\mathcal{A}_{\mathrm{Sch}} \subseteq N_{\mathrm{ar}}$.

The raw traces alone therefore have zero analytic principal parts. Their derivative supplies a different explicitly calculated coordinate. Write its unique entire quotient as $\mathcal{F}_a^\bullet = \Xi g_a^\bullet$, where $g_a^\bullet \in \mathcal{H}_{\le 1}$ by the proved division theorem. At each actual zero λ of multiplicity m_λ,

$$\frac{D\mathcal{F}_a^\bullet}{\Xi} = g_a^{\bullet\prime} + g_a^\bullet \frac{\Xi'}{\Xi}, \qquad \mathrm{PP}_\lambda \frac{D\mathcal{F}_a^\bullet}{\Xi} = \frac{m_\lambda g_a^\bullet(\lambda)}{s - \lambda}. \tag{399}$$

To prove the complete local assertion, insert $\Xi = (s - \lambda)^{m_\lambda} A_\lambda$ into the first equality. Then $\Xi'/\Xi = m_\lambda/(s - \lambda) + A_\lambda'/A_\lambda$, and $(g_a^\bullet(s) - g_a^\bullet(\lambda))/(s - \lambda)$ is holomorphic. All remaining terms, including the actual unit, are holomorphic; the displayed simple principal part is exact.

The source class of this derivative is still its class modulo the actual N_{ar}. Its entire presenting relation has the same already calculated ambiguity $(DN_{\mathrm{ar}} + N_{\mathrm{ar}})/N_{\mathrm{ar}}$. For the specific element $\beta = [\Xi']$, equality with any angular derivative or raw entrance is exactly the respective membership test

$$\Xi' - D\mathcal{F}_a^\bullet \in N_{\mathrm{ar}} \quad \text{or} \quad \Xi' - \mathcal{F}_a^\bullet \in N_{\mathrm{ar}}.$$

Neither follows from a nonzero angular sample or a zeta factorization. At the common refinement, raw images belong to $I/H_\cap$ while $[\Xi']_{C_\cap}$ does not, because its analytic residue is $m_\lambda \neq 0$. Their relation to the source quotient remains the proved projection with kernel $N_{\mathrm{ar}}/H_\cap$, not a claimed nonrelation. Likewise the full heat domain is the inverse image of $\mathcal{A}_{\mathrm{Sch}}$ under U_θ as already computed; its zero-jet and Mellin-strip conditions apply to these exact traces. No frozen-source inclusion $U_\theta N_{\mathrm{ar}} \subseteq N_{\mathrm{ar}}$ is inferred from their arithmetic dilation or from the Navier–Stokes angular momentum sample.

Q.6 The radial and polynomial coordinates are an explicit common map

On the real angular chart retain

$$(x_1, x_2, z) = (\sqrt{2\sigma}\cos\theta, \sqrt{2\sigma}\sin\theta, z).$$

Its inverse is $\sigma = (x_1^2 + x_2^2)/2$ and the unit direction $(\cos\theta, \sin\theta) = (x_1, x_2)/\sqrt{x_1^2 + x_2^2}$, with unchanged z. The original polynomial coordinate is

$$\sigma_{\mathrm{pol}} = \frac{1}{2}\left[(1 + \sigma\sin 2\theta)^3 z + 2\sigma\sin^2\theta(1 + \sigma\sin 2\theta)(4 + 3\sigma\sin 2\theta)\right].$$

Indeed $x_1 x_2 = \sigma\sin 2\theta$ and $x_2^2 = 2\sigma\sin^2\theta$ substitute directly into the unchanged $F_1/2$. Adjoining σ_{pol} gives an injective graph map with inverse projection onto (σ, θ, z); in particular no division or inverse Jacobian is needed to relate the two observables. The square-root test ingress above uses $\sigma = \rho^2/2$, whereas the earlier fluid-profile translation uses this displayed σ_{pol}. Their exact common Cartesian chart and the graph formula retain the full coordinate dictionary.

Q.7 Analytic radial entrance for the actual corrected central field

The preceding general entrance criteria can be applied to the actual central field without an additional analyticity assumption. We retain the profiles and all cutoff choices of the cited source. The argument below proves the required radial analyticity locally at the axis; it does not claim that a general smooth cutoff is analytic on its transition region.

First consider the original viscosity-one similarity variables

$$\tau = 1 - t, \qquad \tau = q(1 - \eta^2), \qquad z = q^{1/2-h}\eta, \qquad X = \frac{r^2}{2q}.$$

For every preterminal point in the source's local domain, $q > 0$. The source's Proposition B.2 supplies a leading profile jointly analytic in X and η on a neighborhood of its initial radial rectangle. Every positive-order profile actually selected by the source is analytic in X near $X = 0$, as we now prove from its displayed recurrence.

For an explicit definition of every coefficient in the ensuing linear system, retain the complete original recurrence [12, equations (4.1)–(4.2), (5.1)–(5.6)]. In the present recurrence A, D are the source's scalar exponents, and D_X is its logarithmic radial derivative:

$$\begin{gathered} A = \tfrac{1}{2} + h, \qquad D = \tfrac{1}{2} - h, \qquad \lambda_n = 2nh, \qquad b = -A - \tfrac{1}{2}, \qquad c = -A, \\ d = 1 - \eta^2, \qquad L = 1 - 2h\eta^2, \qquad D_X = X\partial_X, \\ \tau = 1 - t = q(1 - \eta^2), \qquad z = q^D\eta, \qquad \sigma = \frac{r^2}{2}, \qquad X = \frac{\sigma}{q}. \end{gathered} \tag{400}$$

The source denotes the radial σ in this line by s; the equation explicitly identifies that source variable with our retained radial coordinate, while leaving the Connes spectral s distinct. Every other source coefficient in the recurrence is unchanged. For a real exponent a define

$$\begin{aligned} &T_a f = L^{-1}(-af + D\eta\,\partial_\eta f + D_X f), \\ &Z_a f = L^{-1}(2a\eta f + d\,\partial_\eta f - 2\eta D_X f), \\ &T_{a,n} = T_{a+\lambda_n}, \qquad Z_{a,n} = Z_{a+\lambda_n}, \qquad Z_{a,n}^{[2]} = Z_{a+\lambda_n - D} Z_{a+\lambda_n}. \end{aligned} \tag{401}$$

The rightmost operator acts first in the last expression. These are the exact physical derivative identities

$$\partial_t(q^a f) = q^{a-1} T_a f, \qquad \partial_z(q^a f) = q^{a-D} Z_a f.$$

Indeed implicit differentiation of the unchanged similarity coordinates gives

$$q_t = -L^{-1}, \quad \eta_t = \frac{D\eta}{qL}, \quad X_t = \frac{X}{qL}, \qquad q_z = \frac{2\eta q^{1-D}}{L}, \quad \eta_z = \frac{d}{q^D L}, \quad X_z = -\frac{2\eta X}{q^D L};$$

substitution in the ordinary chain rule proves both formulas. A second axial derivative changes the exponent by $-D$, which proves the stated composition order for $Z^{[2]}$.

At each nonnegative order the physical profiles are

$$\begin{aligned} u_{\theta,n} &= q^{-A+\lambda_n} E_n = r q^{-A-1/2+\lambda_n} \phi_n / C, & E_n &= \sqrt{2X}\,\phi_n / C, \\ u_{z,n} &= q^{-A+\lambda_n} U_n, & r u_{r,n} &= q^{\lambda_n} V_n, \\ p_n &= q^{-2A+\lambda_n} \Pi_n, & A_X f &= \int_0^1 f(b_0 X, \eta)\, db_0. \end{aligned} \tag{402}$$

The letter b_0 in the average is an integration variable, distinct from the exponent $b = -A - 1/2$. The leading profiles have index zero; every negative-index field is zero. Primes in the following recurrences mean ∂_X at fixed η. Incompressibility gives

$$\partial_X V_n = -Z_{-A,n} U_n, \qquad \frac{V_n}{X} = \frac{2\eta U_n - 2\eta(D+\lambda_n) A_X(U_n) - d\,\partial_\eta A_X(U_n)}{L}. \tag{403}$$

The original positive-order equations are, for $n \geq 1$,

$$2(X\phi_n'' + 2\phi_n') = T_{b,n}\phi_n + \sum_{\substack{i,j\geq 0 \\ i+j=n}} \Big\{ V_i(\phi_j' + \phi_j/X) + U_i Z_{b,j} \phi_j \Big\} - Z^{[2]}_{b,n-1}\phi_{n-1}, \tag{404}$$

$$2(XU_n'' + U_n') = T_{c,n} U_n + \sum_{\substack{i,j\geq 0 \\ i+j=n}} \Big\{ V_i U_j' + U_i Z_{c,j} U_j \Big\} + Z_{-2A,n}\Pi_n - Z^{[2]}_{c,n-1} U_{n-1}, \tag{405}$$

$$\Pi_n' = C^{-2} \sum_{\substack{i,j\geq 0 \\ i+j=n}} \phi_i \phi_j - \frac{\Omega_{n-1}}{2X}, \tag{406}$$

$$\Omega_k = T_{0,k} V_k + \sum_{\substack{i,j\geq 0 \\ i+j=k}} \Big\{ V_i(V_j' - V_j/(2X)) + U_i Z_{0,j} V_j \Big\} - 2XV_k'' - Z^{[2]}_{0,k-1} V_{k-1}. \tag{407}$$

In particular the pressure term is $-\Omega_{n-1}/(2X)$, with the factor 2 in its denominator. These formulas define each lower-order source without leaving a pressure, radial-viscous or axial-viscous contribution implicit. All ordered pairs in the sums are retained.

The explicit six-row reduction can also be stated without an unnamed source matrix. At fixed $n \geq 1$ define the three known lower-order functions

$$\begin{aligned} f_n^{\phi} &= \sum_{i=1}^{n-1} \{ V_i(\phi_{n-i}' + \phi_{n-i}/X) + U_i Z_{b,n-i}\phi_{n-i} \} - Z^{[2]}_{b,n-1}\phi_{n-1}, \\ f_n^{U} &= \sum_{i=1}^{n-1} \{ V_i U_{n-i}' + U_i Z_{c,n-i} U_{n-i} \} - Z^{[2]}_{c,n-1} U_{n-1}, \\ f_n^{\Pi} &= C^{-2} \sum_{i=1}^{n-1} \phi_i \phi_{n-i} - \Omega_{n-1}/(2X). \end{aligned}$$

Empty sums are zero. Put $\xi = \sqrt{X}$, $v_0 = V_0/X$, and write the six current unknowns as

$$W_n = (\phi, U, K, \Pi, p_\xi, q_\xi)^{\mathsf{T}}, \qquad p_\xi = \partial_\xi \phi, \quad q_\xi = \partial_\xi U, \quad K = A_X(U) - U.$$

Here p_ξ, q_ξ are labels for the last two components; they are not the physical pressure or the similarity coordinate q. The current radial-velocity coefficient is the following explicit linear expression:

$$\mathcal{V}_n[U,K] = \frac{2\eta U - 2\eta(D+\lambda_n)(U+K) - d\,\partial_\eta(U+K)}{L}.$$

Thus $V_n = X\mathcal{V}_n[U,K]$. With every known coefficient evaluated at $(X,\eta) = (\xi^2,\eta)$, the first four rows are

$$\begin{aligned} \partial_\xi \phi = p_\xi, \quad & \partial_\xi U = q_\xi, \\ \partial_\xi K + \frac{2}{\xi} K = -q_\xi, \quad & \partial_\xi \Pi = 4\xi C^{-2}\phi_0\phi + 2\xi f_n^{\Pi}. \end{aligned} \tag{408}$$

The complete last two rows are

$$\begin{aligned} \partial_\xi p_\xi + \frac{3}{\xi} p_\xi = 2\Big\{ & \frac{-(b+\lambda_n)\phi + D\eta\,\partial_\eta\phi + \xi p_\xi/2}{L} \\ & + v_0(\xi p_\xi/2 + \phi) + U_0 \frac{2(b+\lambda_n)\eta\phi + d\,\partial_\eta\phi - \eta\xi p_\xi}{L} \\ & + \mathcal{V}_n[U,K](X\phi_0' + \phi_0) + UZ_{b,0}\phi_0 + f_n^{\phi}\Big\}. \end{aligned} \tag{409}$$

$$\begin{aligned} \partial_\xi q_\xi + \frac{1}{\xi} q_\xi = 2\Big\{ & \frac{-(c+\lambda_n)U + D\eta\,\partial_\eta U + \xi q_\xi/2}{L} + v_0\xi q_\xi/2 \\ & + U_0 \frac{2(c+\lambda_n)\eta U + d\,\partial_\eta U - \eta\xi q_\xi}{L} + X\mathcal{V}_n[U,K]U_0' + UZ_{c,0}U_0 \\ & + \frac{2(-2A+\lambda_n)\eta\Pi + d\,\partial_\eta\Pi - 2\eta X(2C^{-2}\phi_0\phi + f_n^{\Pi})}{L} + f_n^{U}\Big\}. \end{aligned} \tag{410}$$

These rows follow directly from (403)–(407). For the radial conversion, $\partial_X\phi = p_\xi/(2\xi)$ and $2(X\phi'' + 2\phi') = (\partial_\xi p_\xi + 3p_\xi/\xi)/2$; the analogous axial identity has q_ξ/ξ instead of $3p_\xi/\xi$. The apparent divisions in the transport rows cancel exactly with $V_0 = Xv_0$ and $V_n = X\mathcal{V}_n[U,K]$, producing the displayed regular coefficients. In the pressure contribution to the last row, the substitution is the full equation $\Pi' = 2C^{-2}\phi_0\phi + f_n^{\Pi}$; hence its lower-order f_n^{Π} contribution is included.

Equations (408)–(410) define $A_0W_n + A_1\partial_\eta W_n + f_n$ uniquely by collecting, in this specified order of the six components, the coefficients of W_n, $\partial_\eta W_n$, and the remaining known terms. Thus the Picard argument's coefficient functions are completely specified. Extracting the ∂_η terms from these rows gives exactly the sparse A_1 displayed in that argument: the coefficient of $\partial_\eta\phi$ is $2(D\eta + dU_0)/L$, the $\partial_\eta U$ and $\partial_\eta K$ coefficients in the angular row are both $-2d(\phi_0 + X\phi_0')/L$, and the axial row gives $2(D\eta + dU_0 - dXU_0')/L$, $-2dXU_0'/L$, and $2d/L$. This supplies a direct check of every factor in the sparse matrix and preserves every lower-order forcing term.

Set $\xi = \sqrt{X}$. Source equations (5.2)–(5.7) give at each positive order n the linear regular-singular system

$$\begin{aligned} \partial_\xi W_n + \xi^{-1} C_{\mathrm{diag}} W_n &= A_0(\xi,\eta)W_n + A_1(\xi,\eta)\partial_\eta W_n + f_n(\xi,\eta), \\ C_{\mathrm{diag}} &= \mathrm{diag}(0,0,2,0,3,1), \\ W_n &= (\phi_n, U_n, K_n, \Pi_n, \partial_\xi\phi_n, \partial_\xi U_n)^{\mathsf{T}}, \\ K_n &= A_X(U_n) - U_n, \qquad W_n(0,\eta) = 0. \end{aligned} \tag{411}$$

The matrices and source here are the unchanged ones from that recurrence. In particular the only possibly nonzero entries of A_1 are exactly

$$\begin{aligned} &(A_1)_{51} = 2H_c/L, && (A_1)_{52} = (A_1)_{53} = -2d(\phi_0 + X\partial_X\phi_0)/L, \\ &(A_1)_{62} = 2(H_c - dX\partial_X U_0)/L, && (A_1)_{63} = -2dX\partial_X U_0/L, && (A_1)_{64} = 2d/L, \end{aligned}$$

where $d = 1-\eta^2$, $L = 1 - 2h\eta^2$ and $H_c = (1/2 - h)\eta + dU_0$. Thus A_1 and every parameter derivative map the first four components to the last two and annihilate the last two.

Assume inductively that all lower-order profiles are analytic near the axis. This is true at order zero by the just-cited proposition. The source's radial extensions agree with their inner profiles on one fixed initial radial interval; their moment corrections have support to the right of that interval. Thus those extensions do not change any lower-order germ used here. The equations forming A_0, A_1, f_n use finitely many sums, products, radial and parameter derivatives, and the original radial averages, with divisions only by L, nonzero leading factors on the chosen neighborhood, and the powers of X whose zero factors are explicitly canceled in the source recurrence. These operations preserve the analytic germs. For the average the exact formula is $A_X g = \int_0^1 g(bX, \eta)\, db$. For the apparently divided sources, $V_j = X v_j$ by (5.2), and every summand of Ω_j in (5.6) is divisible by X, including

$$V_i(\partial_X V_j - V_j/(2X)) = X v_i(v_j/2 + X\partial_X v_j).$$

Analytic divisibility follows from the vanishing Taylor coefficient, so no singular coefficient is discarded. Once the displayed diagonal singular part is separated, A_0, A_1, f_n consequently extend holomorphically to a product $|\xi| < R$, $\eta \in \Omega$ about the chosen real axis point. Their norms are bounded on a smaller closed product. The radii may depend on n; only positivity of each radius is used below.

For $c_i = (C_{\mathrm{diag}})_{ii}$ extend the exact source inverse to complex ξ by

$$(Gg)_i(\xi, \eta) = \xi \int_0^1 b^{c_i} g_i(b\xi, \eta)\, db, \qquad K = G(A_0 + A_1\partial_\eta).$$

It is holomorphic on the product, has value zero at $\xi = 0$, and termwise integration of a convergent Taylor series proves $(\partial_\xi + c_i/\xi)(Gg)_i = g_i$, with its holomorphic interpretation at zero. Its kernels are diagonal, independent of η, and have absolute value at most one along each straight integration segment.

For every such diagonal kernel D_0, at arbitrary radii,

$$A_1(\xi) D_0 A_1(\xi') = 0, \qquad A_1(\xi) D_0 \partial_\eta A_1(\xi') = 0.$$

Thus two adjacent derivative factors in an expansion of K^k vanish, including the term in which the left derivative strikes the right matrix. Each remaining word contains at most $p_k = \lceil k/2 \rceil$ parameter derivatives. Let a smaller parameter neighborhood have distance $\Delta > 0$ from the boundary of the larger one. Split that loss evenly among the at most p_k derivatives and use Cauchy's integral estimate at each differentiation. The ordered radial integration simplex has volume $|\xi|^{k+1}/(k+1)!$; this follows by parametrizing each nested segment by its ratio to the preceding complex radius, so the same positive real simplex appears. Summing the at most 2^k words and absorbing the bounded matrix and source norms gives

$$\|K^k G f_n\|_{|\xi| \le R', \Omega'} \le C_n^{k+1} \frac{(R')^{k+1}}{(k+1)!} \max\{1, p_k/\Delta\}^{p_k}, \qquad R' < R. \tag{412}$$

The estimate applies to complete derivative operators, so every Leibniz derivative of a coefficient is included. Its kth root tends to zero: the factorial has root comparable to k, whereas the derivative loss has root at most a constant times $k^{1/2+1/(2k)}$. Hence $\sum_{k \ge 0} K^k G f_n$ converges normally to a holomorphic solution of (411). Applying the same estimate to the iterated homogeneous integral equation proves uniqueness among bounded parameter-holomorphic solutions on the real restricted radial interval. The source's actual solution has that parameter regularity by Lemma 5.1; its restriction therefore equals this holomorphic solution.

The source equations preserve even parity in the first four components and odd parity in the last two, as stated and proved by its uniqueness argument on page 49. This also follows by reflecting the just-constructed integral equation and using the same uniqueness. The first four

convergent complex Taylor series therefore contain only powers $\xi^{2j} = X^j$. Their radius in X is the square of a positive radius in ξ, proving analytic X germs for ϕ_n, U_n, K_n, Π_n. Formula (5.2) and the analytic radial average prove the same for V_n/X. This closes the induction for the actual profiles, including their full parameter derivatives.

For completeness the locally summed slow field includes the entire meridional velocity obtained by differentiating the source's base potential, including every derivative of the summation cutoffs. The following formulas make this point explicit. Retain $A = 1/2 + h$, $D = 1/2 - h$, $\lambda_n = 2nh$, $X = \sigma/q$, $\sigma = r^2/2$, and define

$$\rho_0(q) = 1, \qquad \rho_n(q) = \chi(c_n q) \quad (n \geq 1), \qquad F_n^{\mathrm{src}}(X, \eta) = \int_0^X U_n(v, \eta)\, dv.$$

The superscript distinguishes the source's radial primitive from the angular force flux F. Source (10.1) gives exactly

$$S = \sum_{n \geq 0} \rho_n(q) q^{1-A+\lambda_n} F_n^{\mathrm{src}}(X, \eta), \qquad A_{\mathrm{base}} = \frac{S}{r} e_\theta.$$

On each sufficiently small preterminal parameter neighborhood the sum is finite. Since $F_n^{\mathrm{src}} = X A_X U_n$, put

$$\begin{aligned} \mathfrak{a}(\sigma, z, t) &= \frac{1}{2} \sum_{n \geq 0} \rho_n(q) q^{-A+\lambda_n} A_X U_n(\sigma/q, \eta), \\ \mathfrak{b}(\sigma, z, t) &= \frac{1}{C} \sum_{n \geq 0} \rho_n(q) q^{-A-1/2+\lambda_n} \phi_n(\sigma/q, \eta), \\ \mathfrak{p}(\sigma, z, t) &= \sum_{n \geq 0} \rho_n(q) q^{-2A+\lambda_n} \Pi_n(\sigma/q, \eta). \end{aligned} \tag{413}$$

Every sum here retains order zero. The radial-average formula proves that $S = 2\sigma\mathfrak{a}$, including its analytic extension to $\sigma = 0$. Consequently the original potential and its complete curl have the Cartesian representatives

$$\begin{aligned} A_{\mathrm{base}} &= \mathfrak{a}(-x_2, x_1, 0), \\ \operatorname{curl} A_{\mathrm{base}} &= (-x_1 \partial_z \mathfrak{a}, -x_2 \partial_z \mathfrak{a}, 2\mathfrak{a} + 2\sigma \partial_\sigma \mathfrak{a}). \end{aligned} \tag{414}$$

Indeed the first two curl components differentiate the two displayed potential components with respect to z, with their indicated signs, while the last is $\partial_{x_1}(x_1\mathfrak{a}) + \partial_{x_2}(x_2\mathfrak{a}) = 2\mathfrak{a} + 2\sigma\partial_\sigma\mathfrak{a}$. This proves the formula at positive radius and, because its coefficients are analytic, also at the axis.

In particular $\partial_z \mathfrak{a}$ differentiates $\rho_n(q(z,t))$ as well as the profiles. To check its cutoff contribution against the original construction, note that $q_z = 2\eta q^{1-D}/L$ and $A + D = 1$. The source's incompressibility formula yields $-Z_{D+\lambda_n} F_n^{\mathrm{src}} = V_n$, so $-\partial_z(q^{1-A+\lambda_n} F_n^{\mathrm{src}}) = q^{\lambda_n} V_n$. Differentiating the full cutoff-weighted streamfunction therefore gives, for every $n \geq 1$,

$$\begin{aligned} r u_{r,n}^{\mathrm{cut}} &= q^{\lambda_n} \left[\chi(c_n q) V_n - \frac{2\eta}{L} (c_n q) \chi'(c_n q) F_n^{\mathrm{src}} \right], \\ u_{z,n}^{\mathrm{cut}} &= q^{-A+\lambda_n} \chi(c_n q) U_n. \end{aligned} \tag{415}$$

For the second equality use $r^{-1}\partial_r S_n = \partial_\sigma S_n$ and the independence of q, η, ρ_n from σ. These are exactly source (5.45), including the negative cutoff derivative and its full factor $2\eta(c_n q)/L$.

The analytic profile germs already proved, their analytic radial averages, and local finiteness show that all three functions in (413) have analytic σ germs. This remains true after every fixed z, t derivative: the cutoffs and their derivatives are smooth functions of $q(z,t)$ independent

of σ, and all other factors are derivatives of the already established analytic profile germs. Source (9.21) and the support statement accompanying (10.1) say that every added wave and mean representative, including the finite initialization block, vanishes near the axis. The active sum is finite on the chosen preterminal neighborhood. Intersecting their vanishing neighborhoods proves that the actual local field there consists of the complete curl (414), the angular component $r\mathfrak{b}\,e_\theta$, and the pressure $\mathfrak{p}$. Thus the assertion about the actual slow field includes its full radial and axial components, rather than only its angular component.

Now apply the final localization exactly as in source (10.4). On the central slab $|z| < z_0/2$ and sufficiently near the axis, the spatial cutoff equals one on a neighborhood. Put $\kappa(t) = \chi_t(t)$. There $c(x,t) = \kappa(t)$ and $\nabla c = 0$, whence

$$u_* = \operatorname{curl}(\kappa A) + \kappa B_{\mathrm{src}} e_\theta = \kappa u_{\mathrm{slow}}, \qquad p_* = \kappa \mathfrak{p}.$$

Here B_{src} denotes the source's azimuthal velocity coefficient in (9.21), not our angular-momentum field B. The complete localized velocity has the exact form

$$\begin{aligned} u_* &= (\alpha_* x_1 - \beta_* x_2, \alpha_* x_2 + \beta_* x_1, \gamma_*), \qquad \pi_* = p_*, \\ \alpha_* &= -\kappa \partial_z \mathfrak{a}, \qquad \beta_* = \kappa \mathfrak{b}, \qquad \gamma_* = \kappa(2\mathfrak{a} + 2\sigma \partial_\sigma \mathfrak{a}), \qquad \pi_* = \kappa \mathfrak{p}. \end{aligned} \tag{416}$$

The source's support margin in (10.3) places a neighborhood of the temporal cutoff support within its local coordinate domain; outside that support the actual localized fields vanish. Thus these formulas establish the asserted germs for every $0 \le t < 1$ in the central slab, including the temporal transition.

No time derivative of this cutoff is omitted from the force. For example its exact viscosity-one residual satisfies

$$f_* = \kappa f_{\mathrm{slow}} + \kappa' u_{\mathrm{slow}} + (\kappa^2 - \kappa)(u_{\mathrm{slow}} \cdot \nabla) u_{\mathrm{slow}},$$

where f_{slow} is the full local residual of the preceding velocity and pressure. This follows by substituting $u_* = \kappa u_{\mathrm{slow}}$ and $p_* = \kappa \mathfrak{p}$ into the original momentum equation; the spatial derivatives of κ vanish, while its time derivative is retained.

The physical viscosity map gives, with $\widehat{\sigma} = \sigma/\nu$ and $\widehat{z} = z/\sqrt{\nu}$,

$$\alpha_\nu = \alpha_*(\widehat{\sigma}, \widehat{z}, t), \qquad \beta_\nu = \beta_*(\widehat{\sigma}, \widehat{z}, t), \qquad \gamma_\nu = \sqrt{\nu}\, \gamma_*(\widehat{\sigma}, \widehat{z}, t), \qquad \pi_\nu = \nu \pi_*(\widehat{\sigma}, \widehat{z}, t).$$

For an explicit check of the force, let subscripts σ, z, t denote partial derivatives of these complete localized physical coefficients and suppress their subscript ν on the right. Then

$$f_\nu = (\alpha_f x_1 - \beta_f x_2, \alpha_f x_2 + \beta_f x_1, \gamma_f),$$

where the original residual, with physical viscosity ν, gives

$$\begin{aligned} \alpha_f &= \alpha_t + 2\sigma\alpha\alpha_\sigma + \gamma\alpha_z + \alpha^2 - \beta^2 - \nu(2\sigma\alpha_{\sigma\sigma} + 4\alpha_\sigma + \alpha_{zz}) + \pi_\sigma, \\ \beta_f &= \beta_t + 2\sigma\alpha\beta_\sigma + \gamma\beta_z + 2\alpha\beta - \nu(2\sigma\beta_{\sigma\sigma} + 4\beta_\sigma + \beta_{zz}), \\ \gamma_f &= \gamma_t + 2\sigma\alpha\gamma_\sigma + \gamma\gamma_z - \nu(2\sigma\gamma_{\sigma\sigma} + 2\gamma_\sigma + \gamma_{zz}) + \pi_z. \end{aligned} \tag{417}$$

To verify every coefficient, use $u \cdot \nabla\sigma = 2\sigma\alpha$ and $u \cdot \nabla z = \gamma$; differentiating the two transverse Cartesian coordinate factors contributes $\alpha^2 - \beta^2$ and $2\alpha\beta$. For a scalar radial coefficient g,

$$\Delta g = 2\sigma g_{\sigma\sigma} + 2g_\sigma + g_{zz}, \qquad \Delta(x_j g) = x_j(2\sigma g_{\sigma\sigma} + 4g_\sigma + g_{zz}) \quad (j = 1, 2).$$

The pressure gradient is $(x_1\pi_\sigma, x_2\pi_\sigma, \pi_z)$. Substitution proves (417) with every sign and factor. All its coefficients have analytic radial germs because the formulas use the full differentiated localized coefficients from (416).

Finally the actual axisymmetric central angular projections, with the original factor $(2\pi)^{-1}$ in the angular average, are exactly

$$B_\nu = 2\sigma\beta_\nu, \qquad P_\nu = 4\sigma^2\alpha_\nu\beta_\nu, \qquad Q_\nu = 2\sigma\gamma_\nu\beta_\nu, \qquad F_\nu = 2\sigma\beta_f.$$

They prove analytic radial germs for all four actual fields on $|z| < \sqrt{\nu}\, z_0/2$, $0 \le t < 1$, and retain every meridional and cutoff contribution. The separate exact terminal-axis coefficient and its fixed spectral-value consequence require $0 < \tau < \tau_1$, when the temporal cutoff equals one; the all-preterminal analyticity assertion does not enlarge that range of the exact terminal formulas.

In the retained physical variables the domain just proved is

$$|z| < \sqrt{\nu}\, z_0/2, \qquad 0 \le t < 1. \tag{418}$$

Thus the radial entrance theorem applies to the actual untruncated four fields on (418):

$$\mathcal{F}_B^{\mathrm{rad}}, \quad \mathcal{F}_P^{\mathrm{rad}}, \quad \mathcal{F}_Q^{\mathrm{rad}}, \quad \mathcal{F}_F^{\mathrm{rad}} \in \mathcal{A}_{\mathrm{Sch}} \subset \Xi\mathcal{H}_{\le 1}.$$

The compact radial support was already proved, so the endpoint analyticity just established is all that was needed for the order-one sufficient criterion. At other axial points, the source only prescribes a smooth spatial cutoff; the previously constructed full-input packet ingress retains those points without imposing an unsupported analytic cutoff.

Q.8 The full angular equation in the actual entire radial traces

Retain the original physical variables, viscosity and source exponent, and set

$$\begin{aligned} K_h(w) &= \zeta(w)2^{w/2-1}(1-2^{w/2-1/2})(1-2^{h-w/2}), \\ \kappa_h(s) &= K_h(1/2-is), \qquad w = 1/2-is, \\ (T_h a)(s) &= \mathcal{F}_a^{\mathrm{rad}}(s) = K_h(w)\widehat{a}(w/2). \end{aligned} \tag{419}$$

All products use their previously proved entire continuations. The multiplier itself is entire and has order at most one. Indeed

$$\frac{1-2^{(w-1)/2}}{w-1} = -\frac{\log 2}{2}\int_0^1 e^{v(w-1)\log 2/2}\, dv$$

is entire of exponential type. Multiplying it by the already proved order-one entire function $(w-1)\zeta(w)$ and the two remaining exponential factors proves the assertion. Moreover

$$K_h(1) = -\frac{\log 2}{2\sqrt{2}}(1-2^{h-1/2}) \neq 0,$$

since the retained $0 < h < 1/100$ makes $h - 1/2 < 0$.

Proposition Q.1 (Actual entire angular dynamics)**.** *On the central domain* (418)*, each of the four actual untruncated profiles* $\mathcal{F}_B^{\mathrm{rad}}, \mathcal{F}_P^{\mathrm{rad}}, \mathcal{F}_Q^{\mathrm{rad}}, \mathcal{F}_F^{\mathrm{rad}}$ *is a smooth* $(\mathcal{H}_{\le 1}, \gamma)$*-valued function of* (z,t) *before terminal time. Every mixed physical parameter derivative is the radial trace of the corresponding mixed derivative of its full input. They satisfy, at every* $s \in \mathbb{C}$*,*

$$\begin{aligned} &\kappa_h(s-2i)\big(\partial_t\mathcal{F}_B^{\mathrm{rad}}(s) + \partial_z\mathcal{F}_Q^{\mathrm{rad}}(s) - \nu\partial_{zz}\mathcal{F}_B^{\mathrm{rad}}(s) - \mathcal{F}_F^{\mathrm{rad}}(s)\big) \\ &\quad = (w/2-1)\kappa_h(s)\big(\mathcal{F}_P^{\mathrm{rad}}(s-2i) + \nu w\mathcal{F}_B^{\mathrm{rad}}(s-2i)\big). \end{aligned} \tag{420}$$

On the exact radial trace image with its recovered smooth inputs and endpoint orders (367)*, this entire identity is equivalent to the full physical equation* (371)*.*

Proof. Fix a compact preterminal parameter set inside the stated central slab, and slightly enlarge it there. The actual source construction has a positive lower bound for q on this set. Its cutoff sums (9.21), (10.1) of [12] therefore have a common finite set of active profiles on a neighborhood of this compact set. The analytic germs proved in Section Q.7 give each of those finitely many profiles a holomorphic radial extension jointly with its similarity parameter on smaller product neighborhoods. A finite cover of the real parameter set and a further radial shrink give one complex σ-disk for all of them. The change $X = \sigma/(\nu q)$ preserves such a common disk because q is bounded away from zero. Its parameter changes are smooth: in viscosity-one variables the determinant of $(q, \eta) \mapsto (q(1-\eta^2), q^{1/2-h}\eta)$ is $q^{1/2-h}(1-2h\eta^2) > 0$ on the retained domain. The finite annular terms vanish on a common smaller radial interval, and all summation cutoffs depend only on the physical parameters there. Their arbitrary smooth parameter derivatives multiply the same holomorphic radial germs. The final spatial cutoff is identically one near this portion of the axis. Consequently the Cartesian products and derivatives forming the full B, P, Q, F, including the force residual, have one common holomorphic radial disk with smooth parameter dependence and mixed parameter derivatives jointly continuous and bounded on the compact parameter set times every smaller closed disk. Radial derivatives are taken on a larger disk before this last shrink. The disk can be chosen once for all derivative orders: differentiation changes the bounds, while the finite holomorphic profile domains and the parameter-only cutoff factors stay the same. The fixed compact radial support supplies the corresponding bounds away from the axis. At $t = 0$ these statements use the smooth one-sided derivatives of the source field.

Here is the resulting growth-topology estimate, including the parameter limit it controls. Choose $0 < \delta < \rho$ so that $|\sigma| \le \rho$ lies inside this common holomorphic disk, and choose $R > \delta$ containing the fixed radial support. Write $\widetilde{a}$ for the holomorphic germ and put

$$N(a) = \sup_{\delta \le \sigma \le R} |a(\sigma)| + \sup_{|\sigma| \le \rho} |\widetilde{a}(\sigma)|.$$

For every parameter derivative of any of the four fields, Cauchy's coefficient formula gives $|a_n| \le N(a)\rho^{-n}$. The exact expansion (393) at argument $w/2$ therefore gives, outside the disks $|w + 2n| < 1/2$, $n \ge 1$,

$$|\widehat{a}(w/2)| \le C_{\delta,\rho,R} N(a) e^{C_{\delta,R}|w|}.$$

In detail, the integral on $[\delta, R]$ is bounded by $(R-\delta)\delta^{-1}N(a)e^{|w|\max\{|\log\delta|,|\log R|\}/2}$; the series is bounded by $4N(a)e^{|w||\log\delta|/2}\sum_{n\ge1}(\delta/\rho)^n$. For $1 < p < 2$ choose $1 < p_0 < p$. The proved order bound on K_h gives $|K_h(w)| \le C_{p_0}e^{|w|^{p_0}}$. The entire product $K_h(w)\widehat{a}(w/2)$ then has the preceding bound multiplied by $C_{p_0}e^{|w|^{p_0}}$ off the disks. Maximum modulus on each disk extends it inside with $|w|$ replaced by $|w| + 1$. Since $|w| \le |s| + 1/2$ and $p_0 < p$, for every $c > 0$ this proves

$$p_{p,c}(T_h a) := \sup_{s\in\mathbb{C}} |T_h a(s)| e^{-c|s|^p} \le C_{\delta,\rho,R,p,c} N(a). \tag{421}$$

Every difference and parameter derivative considered here has the same support and holomorphic germ domain, so the same estimate applies to it. For a coordinate direction e_j in (z, t), the integral Taylor formula in the two sup norms gives

$$N\Big(\frac{a(y+\epsilon e_j) - a(y)}{\epsilon} - \partial_j a(y)\Big) \le \frac{|\epsilon|}{2} \sup_{0\le v\le 1} N(\partial_j^2 a(y + v\epsilon e_j)).$$

The bound tends to zero. Applying (421) proves differentiation in every defining γ seminorm. The first-order integral formula applied to each $\partial^\alpha a$ also proves continuity of its trace in those seminorms. Repeating this argument proves all mixed derivatives and the asserted smoothness.

Put $v = w/2$. On $\operatorname{Re} w > 2$ the Mellin integrations by parts already justified in (373) give

$$\partial_t \widehat{B}(v) + \partial_z \widehat{Q}(v) - \nu\partial_{zz}\widehat{B}(v) - \widehat{F}(v) = (v-1)\{\widehat{P}(v-1) + 2\nu v\widehat{B}(v-1)\}.$$

The parameter differentiations commute with T_h by the preceding proof. The exact spectral change is $1/2 - i(s-2i) = w-2$, so the two shifted traces are $K_h(w-2)\widehat{P}(v-1)$ and $K_h(w-2)\widehat{B}(v-1)$. Multiplying the last displayed equation by $K_h(w)K_h(w-2)$, and retaining $2\nu v = \nu w$, proves (420) on that half-plane. All its terms are entire, hence the identity theorem proves it at every complex s. They also lie in $\mathcal{H}_{\le 1}$: finite products, polynomial factors and a fixed translation preserve order at most one. No radial or nonlinear flux has been removed.

For the claimed exact converse retain the full inverse domain: each of the four entire functions is the radial trace of its specified smooth compact input, recovered by (396), with the common local germ and parameter regularity just proved. Define its full physical residual

$$E = \partial_t B + \partial_\sigma P + \partial_z Q - \nu(2\sigma B_{\sigma\sigma} + B_{zz}) - F.$$

The endpoint orders make $E(0) = 0$; it has the same compact support and an analytic endpoint germ. Repeating the preceding integrations without setting E to zero shows that the left side minus the right side of (420) is exactly

$$\kappa_h(s-2i)\mathcal{F}_E^{\mathrm{rad}}(s). \tag{422}$$

If the entire identity holds, this product vanishes identically. The multiplier is nonzero on an open set because $K_h(1) \neq 0$, so the identity theorem gives $\mathcal{F}_E^{\mathrm{rad}} = 0$. The proved injective radial inverse gives $E = 0$. Conversely, $E = 0$ makes the product zero. Thus even the isolated zeros of the multiplier lose no information on this specified full entire image. This conclusion uses the full functions and their inverse domain, rather than values at a single multiplier zero. □

The codomain of this proved smooth physical evolution is $(\mathcal{H}_{\le 1}, \gamma)$, with every individual trace in $\mathcal{A}_{\mathrm{Sch}} \subset \Xi\mathcal{H}_{\le 1}$. Its subsequent map to the original source quotient is the already specified $q_{\mathrm{rad}}(a) = [\mathcal{F}_a^{\mathrm{rad}}]_{Q_{\mathrm{ar}}}$, with exact kernel (398). That map is continuous for the previously defined initial topology involving $\tau \vee \gamma$. The new seminorm estimate proves no additional continuity into the unnamed source topology τ. Nor does the entire equation define its multipliers or spectral translation on source classes without invariance of the actual relation space N_{ar}. The exact relation furnished here is therefore the full smooth entire evolution, its injective radial inverse, and the specified projection and kernel to the source quotient; the frozen differential and heat descent questions remain those of the original N_{ar}.

Q.9 A fixed entire spectral value of the actual concentrating field

Throughout the exact terminal identities below, $0 < \tau < \tau_1$, with the source-determined $\tau_1 > 0$ from Section P; the final central cutoffs equal one on this range. At $z = 0$, $q = \tau$, $\eta = 0$ in the original source coordinates. The source's equation (B.3) gives $\phi_0(0,0) = 1$; Lemma 5.1 gives $\phi_n(0,0) = 0$ for every $n \ge 1$. Its locally finite sum, the vanishing of all annular corrections near the axis, and the final cutoffs equal to one near the specified terminal central point yield the exact axial identity

$$\frac{u_{\nu,\theta}}{r}\Big|_{\sigma=0,z=0,t=1-\tau} = \frac{\tau^{-1-h}}{C}, \qquad \partial_\sigma B(0,0,1-\tau) = \frac{2}{C}\tau^{-1-h}. \tag{423}$$

The first coefficient has the original amplitude $C > 1$ from the source and the original power $q^{-A-1/2} = q^{-1-h}$. The second identity follows from $B = ru_\theta = 2\sigma(u_\theta/r)$. Under viscosity scaling $u_\nu = \sqrt{\nu}\, u_*(x/\sqrt{\nu}, t)$, both the numerator u_θ and radius r have the same factor $\sqrt{\nu}$, which cancels in this ratio. Thus no viscosity factor is absent from the displayed coefficient.

For the direct arithmetic Mellin profile this gives the genuine uncanceled pole

$$\mathrm{Res}_{w=-1} W_h(w)\widehat{B}(w,0,1-\tau) = \frac{2^{h+1}-1}{8C}\tau^{-1-h}.$$

Here $W_h(-1) = (2^{h+1}-1)/16$ follows from $\zeta(-1) = -1/12$ and the two retained dyadic factors. The pole is compatible with the entire radial ingress because the two precise maps have different Mellin arguments and different moment-canceling dilation coefficients.

For the latter, set the fixed original spectral point $s_0 = -5i/2$, so that $w = 1/2 - is_0 = -2$. Formula (395) gives

$$\operatorname{Res}_{w=-2} M_B^{\mathrm{rad}}(w, 0, 1-\tau) = -\frac{(1-2^{-3/2})(2^{h+1}-1)}{C}\,\tau^{-1-h}.$$

The completed zeta function determines the exact canceled zero coefficient:

$$\xi(-2) = \xi(3) = \frac{3\zeta(3)}{2\pi}, \qquad \Gamma(w/2) = -\frac{2}{w+2} + O(1), \qquad \zeta'(-2) = -\frac{\zeta(3)}{4\pi^2}.$$

For the first equality evaluate the retained completion at 3, using $\Gamma(3/2) = \sqrt{\pi}/2$, and use the functional equation $\xi(-2) = \xi(3)$. For the second, the gamma recurrence gives the residue -1 at its argument -1, and the change of argument $w/2$ supplies the factor two. Multiplication by $w(w-1)\pi^{-w/2}/2$, whose value at -2 is 3π, shows $\xi(-2) = -6\pi\zeta'(-2)$, proving the third equality.

Multiplying the simple zeta zero by the exact Mellin pole therefore gives the actual entire trace value

$$\mathcal{F}_B^{\mathrm{rad}}(s_0, 0, 1-\tau) = \frac{\zeta(3)}{4\pi^2 C}(1-2^{-3/2})(2^{h+1}-1)\tau^{-1-h}. \tag{424}$$

The function here is the untruncated actual radial source trace already proved to lie in $\Xi\mathcal{H}_{\le 1}$ for each preterminal time. Writing its unique entire quotient as $\mathcal{F}_B^{\mathrm{rad}} = \Xi g_B^{\mathrm{rad}}$ and retaining $\Xi(s_0) = \xi(3) = 3\zeta(3)/(2\pi)$ gives

$$g_B^{\mathrm{rad}}(s_0, 0, 1-\tau) = \frac{(1-2^{-3/2})(2^{h+1}-1)}{6\pi C}\tau^{-1-h}. \tag{425}$$

All factors are strictly positive, so both displayed values diverge with exactly the source exponent. This is a fixed-spectral-point consequence for actual order-one entire functions, not an uninverted trace claim at a moving sample.

Its precise relation to the source quotient retains the original relation space. Since $\Xi(s_0) \neq 0$, the function $e(s) = \Xi(s)/\Xi(s_0)$ belongs to $\mathbb{C}[s]\Xi \subseteq N_{\mathrm{ar}}$ and satisfies $e(s_0) = 1$. For any representative f and any prescribed scalar b, $f + (b - f(s_0))e$ has the same source quotient class and value b at s_0. Thus the representative relation $\{([f]_{Q_{\mathrm{ar}}}, f(s_0)) : f \in \mathcal{H}_{\le 1}\}$ has full scalar fibres. This proves the exact information lost by imposing the source quotient on the now-constructed growing trace value. The full trace itself and its inverse radial input remain specified by the preceding injective maps and kernels; the calculation does not decide the distinct membership of Ξ' in N_{ar}.

R Original-space Fréchet rigidity and the ring obstruction

We retain the original ring $\mathcal{H}_{\le 1} = \mathcal{H}_{\le 1}$, ordinary pointwise multiplication, its coordinate s, and $Mf = sf$. The source relation remains $N_{\mathrm{ar}} = \overline{\mathbb{C}[s]\Xi}^{\tau}$ and $Q_{\mathrm{ar}} = \mathcal{H}_{\le 1}/N_{\mathrm{ar}}$. The word "topological ring" in [5, §3.6, p.19, typeset line 552 and Proposition 3.6] does not itself assert completeness or metrizability. The cited [6, Proposition 2.24 and the following discussion, pp.379–381] computes the finite Hermite image and discusses the choice of function spaces. Those inspected clauses do not state that the Hadamard topology is Fréchet or that polynomials are dense in it. The theorem below classifies a precisely specified class of topologies on the original space; it does not insert Fréchet structure into the primary source.

R.1 The exact division map before an open-mapping argument

Lemma R.1 (Original division and closed evaluation hyperplanes)**.** *For each $\lambda \in \mathbb{C}$, the map*

$$\mathsf{L}_\lambda : \mathcal{H}_{\le 1} \times \mathbb{C} \longrightarrow \mathcal{H}_{\le 1}, \qquad \mathsf{L}_\lambda(g, c) = (s - \lambda)g + c \tag{426}$$

is a linear bijection, with inverse

$$\mathsf{L}_\lambda^{-1} f = (\mathsf{A}_\lambda f, f(\lambda)), \qquad \mathsf{A}_\lambda f(s) = \frac{f(s) - f(\lambda)}{s - \lambda}. \tag{427}$$

In any complex topological-vector-space topology τ with continuous M, L_λ is continuous. For a Hausdorff such topology, writing $\mathfrak{m}_\lambda = (s - \lambda)\mathcal{H}_{\le 1}$, one has

$$\mathrm{ev}_\lambda \text{ is continuous} \quad \Longleftrightarrow \quad \mathfrak{m}_\lambda \text{ is closed.} \tag{428}$$

Proof. The singularity in (427) is removable by the Taylor expansion at the retained point λ. For $|s| \ge 2(|\lambda| + 1)$ one has $|s - \lambda| \ge |s|/2 \ge 1$, so $|\mathsf{A}_\lambda f(s)| \le |f(s)| + |f(\lambda)|$. The remaining compact disk contributes a finite maximum. The filled quotient therefore belongs to the same order class $\mathcal{H}_{\le 1}$. Evaluation of $\mathsf{L}_\lambda(g, c) = 0$ at λ gives $c = 0$; then $(s - \lambda)g = 0$ gives $g = 0$. Substitution verifies both inverse identities. Continuity of L_λ follows from continuity of M, scalar multiplication, addition, and the constant embedding $c \mapsto c1$.

The kernel of evaluation is exactly $\mathfrak{m}_\lambda$, which proves the forward implication of (428). If this hyperplane is closed, $\mathcal{H}_{\le 1}/\mathfrak{m}_\lambda$ is a one-dimensional Hausdorff complex topological vector space. The map $\mathbb{C} \to \mathcal{H}_{\le 1}/\mathfrak{m}_\lambda$, $c \mapsto [c1]$, is a continuous linear bijection. Its inverse is continuous: the image of the unit circle is compact and excludes zero; choose a balanced zero neighbourhood avoiding that compact set. It contains no $c[1]$ with $|c| \ge 1$, since balance would put $(c/|c|)[1]$ in it. This proves inverse continuity at zero, and scalar rescaling and translation prove it everywhere. Composing with the quotient map gives continuous evaluation. □

The lemma does not infer continuity of the inverse of (426) from its bijectivity. Even when evaluation is continuous, the continuous bijection $\mathcal{H}_{\le 1} \to \mathfrak{m}_\lambda$, $g \mapsto (s - \lambda)g$, has a further inverse-continuity question. The next argument settles both inverse components in its stated class.

R.2 The original growth space and the open-mapping step

Lemma R.2 (Complete growth topology on the unchanged order class)**.** *Retain every original seminorm*

$$p_{p,a}(f) = \|f\|_{p,a} = \sup_{s \in \mathbb{C}} |f(s)| e^{-a|s|^p}, \qquad 1 < p < 2, \quad a > 0.$$

These seminorms define a Hausdorff Fréchet topology γ on exactly $\mathcal{H}_{\le 1}$. The countable subfamily

$$q_n = p_{1+1/n,1}, \qquad n = 2, 3, \ldots, \tag{429}$$

is cofinal for this topology. Ordinary multiplication and M are continuous.

Proof. For $f \in \mathcal{H}_{\le 1}$ and $1 < r < p$, the order bound gives $|f(s)| \le C \exp(|s|^r)$ for a finite C. Since $\sup_{t \ge 0} \exp(t^r - at^p) < \infty$, every displayed seminorm is finite. Conversely, finiteness of all these seminorms gives $\log M_f(R) \le \log p_{p,a}(f) + aR^p$ for every $p > 1$ in the displayed range, hence order at most one. The zero function and constants satisfy the same membership conclusion.

For a given p, a, choose n with $r_n = 1 + 1/n < p$. Then

$$\begin{aligned} p_{p,a}(f) &\le C_{p,a,n} q_n(f), \\ C_{p,a,n} &= \exp\left(\sup_{t \ge 0} (t^{r_n} - at^p) \right) < \infty. \end{aligned} \tag{430}$$

Every q_n is already one of the original seminorms, proving cofinality without replacing their definition. They separate entire functions, giving a Hausdorff metrizable locally convex space.

Let (f_k) be Cauchy in all q_n. On each disk, a fixed q_n bounds the compact supremum by a finite factor, so f_k converges uniformly on compact disks to an entire f. For fixed n and $\epsilon > 0$, choose k_0 such that $q_n(f_k - f_l) \le \epsilon$ for $k, l \ge k_0$. Sending $l \to \infty$ pointwise and taking the supremum gives $q_n(f_k - f) \le \epsilon$. Thus all $q_n(f)$ are finite; (430) and the order calculation put f in the original $\mathcal{H}_{\le 1}$. This proves completeness and convergence in the entire original seminorm family. Finally

$$p_{p,a}(fg) \le p_{p,a/2}(f)p_{p,a/2}(g),$$
$$p_{p,a}(sf) \le \left(\sup_{t\ge 0} te^{-at^p/2}\right) p_{p,a/2}(f)$$

prove the two continuity assertions. □

Lemma R.3 (The open-mapping step used here)**.** *A continuous linear surjection between Fréchet spaces is open. In particular a continuous linear bijection between such spaces is a homeomorphism.*

Proof. Let $T : F \to G$ be the surjection and let U be an arbitrary zero neighbourhood in F. Using countably many defining seminorms, choose balanced convex zero neighbourhoods U_n such that every series $x_n \in U_n$ converges in F and has sum in U. Explicitly, bound the first n seminorms by 2^{-n}, and at every stage impose the finitely many bounds of a smaller neighbourhood inside U with summable coefficients. Each seminorm then has summable tails, and completeness gives convergence; the finite bounds keep the sum inside U.

Each U_n is absorbing, so $G = \bigcup_{k\ge 1} kT(U_n)$. The complete-metric Baire theorem implies that $\overline{T(U_n)}$ has nonempty interior. Balance and convexity put a zero neighbourhood inside it: average a neighbourhood about an interior point with its reflected neighbourhood. Choose zero neighbourhoods $V_n \subseteq \overline{T(U_n)}$ whose diameters tend to zero in a compatible metric of G. Given $y \in V_1$, choose $x_1 \in U_1$ with $y - Tx_1 \in V_2$. Recursively choose $x_n \in U_n$ so that the remainder after n terms lies in V_{n+1}; membership in the preceding closure gives each choice. The series converges to $x \in U$, its image converges to y, and continuity gives $Tx = y$. Thus $V_1 \subseteq T(U)$, proving openness. The Baire property used here follows by the nested-closed-ball proof in a complete compatible metric. □

Theorem R.4 (Original-space Fréchet rigidity)**.** *On the original vector space $\mathcal{H}_{\le 1}$, there is exactly one Hausdorff Fréchet complex-vector-space topology in which $Mf = sf$ is continuous: it is γ. In particular the same uniqueness holds for Hausdorff Fréchet topological-algebra structures with the original ordinary multiplication and usual complex scalars.*

Proof. Let τ belong to the stated class. The map L_λ in (426) is a continuous linear bijection between the Fréchet spaces $(\mathcal{H}_{\le 1}, \tau) \times \mathbb{C}$ and $(\mathcal{H}_{\le 1}, \tau)$. Lemma R.3 makes it a homeomorphism. Thus evaluation at every λ, and the divided difference A_λ, are τ-continuous.

Fix p, a from the unchanged growth family. Its seminorm $p_{p,a}$ is a supremum of the continuous functions $f \mapsto |f(s)|e^{-a|s|^p}$, hence is lower semicontinuous for τ. It is finite everywhere by Lemma R.2. The closed sets $F_m = \{f : p_{p,a}(f) \le m\}$, $m \ge 1$, cover $\mathcal{H}_{\le 1}$. Baire supplies m, f_0 and a zero neighbourhood V with $f_0 + V \subseteq F_m$. Taking $0 \in V$ gives $f_0 \in F_m$, so for $v \in V$,

$$p_{p,a}(v) \le p_{p,a}(f_0 + v) + p_{p,a}(f_0) \le 2m.$$

Scaling V proves continuity of $p_{p,a}$. All growth seminorms are therefore τ-continuous, and $\mathrm{id} : (\mathcal{H}_{\le 1}, \tau) \to (\mathcal{H}_{\le 1}, \gamma)$ is a continuous linear bijection. Both spaces are Fréchet, the latter by Lemma R.2; the open-mapping lemma proves inverse continuity. Hence $\tau = \gamma$. Conversely γ belongs to the stated class by that same lemma, completing existence and uniqueness. □

No family of seminorms for τ, evaluation continuity, polynomial density, or principal-ideal description was assumed in this theorem. They are consequences within its explicit Fréchet domain. The inspected primary topology has not yet been proved to lie in that domain.

R.3 The complete closure and blocks in the classified domain

Proposition R.5 (Closure, generalized blocks, and frozen failure)**.** *For every topology classified in Theorem R.4, its original source-form closure is*

$$\begin{aligned} \overline{\mathbb{C}[s]\Xi}^{\tau} = \overline{\mathbb{C}[s]\Xi}^{\gamma} = I := \Xi\mathcal{H}_{\le 1}, \\ I = \left\{ f \in \mathcal{H}_{\le 1} : \begin{matrix} f^{(j)}(\lambda) = 0 \text{ for every } \lambda \in \mathbb{C} \text{ with } \Xi(\lambda) = 0, \\ \text{and every } j \in \mathbb{Z}_{\ge 0} \text{ with } 0 \le j < m_\lambda \end{matrix} \right\}. \end{aligned} \tag{431}$$

At every actual zero λ of multiplicity m, put

$$v_{\lambda,j} = \left[\frac{\Xi(s)}{(s-\lambda)^j} \right]_I, \qquad 1 \le j \le m.$$

For the original multiplication S on $\mathcal{H}_{\le 1}/I$ and every integer $k \ge 1$,

$$\begin{aligned} \ker(S-\lambda)^k &= \operatorname{span}\{v_{\lambda,1}, \dots, v_{\lambda,\min(k,m)}\}, \\ \dim \ker(S-\lambda)^k &= \min(k,m). \end{aligned} \tag{432}$$

In this classified domain neither D nor D^2 descends to the frozen quotient by all representatives, and $U_t I \not\subseteq I$ for every $t \neq 0$.

Proof. The topology equality has just been proved. Lemma C.2 proves Taylor-polynomial density in γ, and Lemma F.1 proves entire division with the retained estimate

$$\|f/g\|_{p,a} \le \left(\frac{\max(1, \|g\|_{p,b})}{|g(0)|} \right)^3 \|f\|_{p,b}, \qquad b = \frac{a}{6 \cdot 2^p}.$$

The original center applies since $\Xi(0) > 0$. The complete closure calculation in Proposition F.2 therefore gives (431); it uses closed vanishing jets for one containment, entire division for the quotient's order, and Taylor approximation followed by multiplication for the reverse containment. The principal-ideal equality is thus a calculated closure.

All $v_{\lambda,j}$ belong to the original order class. Writing $T = S - \lambda$, one has $Tv_{\lambda,1} = 0$ and $Tv_{\lambda,j} = v_{\lambda,j-1}$ for $j > 1$. A relation among them would make a finite sum of the principal parts $(s-\lambda)^{-j}$ entire after division by Ξ, forcing every coefficient to vanish. If $T^k[f] = 0$, then $(s-\lambda)^k f = \Xi h$ for some $h \in \mathcal{H}_{\le 1}$. For $k \le m$, subtract the Taylor polynomial of h through degree $k-1$ at λ. Division of the remainder by $(s-\lambda)^k$ stays in $\mathcal{H}_{\le 1}$ by iterating Lemma R.1. Modulo I, this expresses $[f]$ in the span of the first k displayed vectors. For $k > m$, entire regularity of f forces h to vanish to order at least $k-m$ at λ. Divide by that exact factor and reduce to $k = m$. This proves the full equality and dimension in (432).

Finally Theorem F.5 proves the direct frozen failures for this calculated I. Its witnesses retain every multiplicity: Ξ' loses one order at an actual zero; for D^2 use Ξ'' when $m \ge 2$, and $((s-\lambda)\Xi)''(\lambda) = 2\Xi'(\lambda) \neq 0$ when $m = 1$. For heat the complete identity (204) gives $U_t(e^{as}\Xi)(s) = e^{as - ta^2/4} Z_t(s - ta/2)$. Evaluation at an actual zero, under a proposed $U_t I \subseteq I$, would make nonzero Z_t vanish at every $\lambda - ta/2$, $a \in \mathbb{C}$. For $t \neq 0$ these fill $\mathbb{C}$, proving the contradiction. The full representative-ambiguity calculation in (172) gives the asserted all-representative domain failure. □

R.4 The exact multiplier and annihilator maps without completeness

Return now to a general Hausdorff complex topological vector algebra $(\mathcal{H}_{\le 1}, \tau)$ with the original multiplication, and keep its actual closed N_{ar}. Define

$$\begin{aligned} R_\tau &= \overline{\mathbb{C}[s]}^{\,\tau}, \\ \mathrm{Mult}(N_{\mathrm{ar}}) &= \{a \in \mathcal{H}_{\le 1} : aN_{\mathrm{ar}} \subseteq N_{\mathrm{ar}}\}, \\ J_\tau &= \{a \in \mathcal{H}_{\le 1} : a\mathcal{H}_{\le 1} \subseteq N_{\mathrm{ar}}\}. \end{aligned} \tag{433}$$

Proposition R.6 (The original-ring representation)**.** *R_τ and $\mathrm{Mult}(N_{\mathrm{ar}})$ are closed unital subalgebras, and $R_\tau \subseteq \mathrm{Mult}(N_{\mathrm{ar}})$. The space J_τ is the largest $\mathcal{H}_{\le 1}$-ideal contained in N_{ar}, and it is closed. There is an exact faithful algebra representation*

$$\mathrm{Mult}(N_{\mathrm{ar}})/J_\tau \longrightarrow \mathrm{End}_{\mathrm{cont}}(Q_{\mathrm{ar}}), \qquad [a] \longmapsto \rho(a), \quad \rho(a)[f] = [af], \qquad S = \rho(s). \tag{434}$$

If $R_\tau = \mathcal{H}_{\le 1}$, then N_{ar} is an $\mathcal{H}_{\le 1}$-ideal and $N_{\mathrm{ar}} = \overline{\Xi\mathcal{H}_{\le 1}}^{\,\tau}$.

Proof. For every polynomial p, continuous multiplication by p preserves $\mathbb{C}[s]\Xi$, and hence its closure N_{ar}. This proves the original polynomial-module property. Joint multiplication of convergent polynomial nets proves that R_τ is a subalgebra; the corresponding addition and scalar maps prove the remaining algebra operations, and its definition gives closedness.

The set $\mathrm{Mult}(N_{\mathrm{ar}})$ is a unital subalgebra by its definition. It is closed because

$$\mathrm{Mult}(N_{\mathrm{ar}}) = \bigcap_{n \in N_{\mathrm{ar}}} \{a : an \in N_{\mathrm{ar}}\},$$

an intersection of continuous inverse images of the closed space N_{ar}. It contains all polynomials and therefore their closure R_τ. Similarly J_τ is closed by its intersection-of-inverse-images definition. If $a \in J_\tau$ and $b, f \in \mathcal{H}_{\le 1}$, then $(ba)f = a(bf) \in N_{\mathrm{ar}}$, so it is an ideal. Setting $f = 1$ shows $J_\tau \subseteq N_{\mathrm{ar}}$. Every ideal contained in N_{ar} has this multiplication property and is therefore contained in J_τ.

For $a \in \mathrm{Mult}(N_{\mathrm{ar}})$ the map $[f] \mapsto [af]$ is representative-independent. Its composite with the quotient map is continuous, so the quotient universal property proves its continuity. Its vanishing on all classes is exactly $a\mathcal{H}_{\le 1} \subseteq N_{\mathrm{ar}}$. Thus its kernel is J_τ, giving the faithful representation (434). No topology on the algebra of endomorphisms is needed for this algebra representation.

If $R_\tau = \mathcal{H}_{\le 1}$, take polynomials $p_i \to a$. For $n \in N_{\mathrm{ar}}$, $p_i n \in N_{\mathrm{ar}}$ and continuity gives $an \in N_{\mathrm{ar}}$, so N_{ar} is an ideal. In particular $\Xi\mathcal{H}_{\le 1} \subseteq N_{\mathrm{ar}}$. The other containment after closure follows from $\mathbb{C}[s]\Xi \subseteq \Xi\mathcal{H}_{\le 1}$. □

Thus ordinary topological-ring structure already supplies (433) and (434). It does not justify replacing R_τ by $\mathcal{H}_{\le 1}$ without a density proof. The abstract Weyl model of Section N was a model of vector-space and operator data. It did not construct the original ring, this closure, or these multiplier maps; it cannot refute additional deductions from them.

R.5 The closed-ideal spectral obstruction

Lemma R.7 (Surjectivity gives an explicit continuous ring inverse)**.** *Let N_{ar} be an actual closed $\mathcal{H}_{\le 1}$-ideal in the preceding topological algebra. If $S - \lambda$ on $Q_{\mathrm{ar}} = \mathcal{H}_{\le 1}/N_{\mathrm{ar}}$ is algebraically onto, it has a continuous inverse. More precisely, a preimage $b \in Q_{\mathrm{ar}}$ of $[1]$ gives $(S-\lambda)^{-1}[f] = b[f]$ on all of Q_{ar}.*

Proof. The quotient map is open: the preimage of the image of an open set U is $U + N_{\mathrm{ar}}$, a union of translates of U. Its product with itself is therefore an open quotient map, so ordinary multiplication induces continuous multiplication on the quotient algebra. Surjectivity gives b with $([s] - \lambda[1])b = [1]$. Multiplication by b is continuous and, by commutativity, is both a left and right inverse of $S - \lambda$. No operator-series inverse or open-unit-group hypothesis enters this proof. □

Proposition R.8 (Nonempty source spectrum excludes frozen ideal descent)**.** *Among the original source-form closed $\mathcal{H}_{\leq 1}$-ideals whose multiplication operator has the nonempty spectrum stated in [5, Proposition 3.6(ii)], one has*

$$DN_{\mathrm{ar}} \not\subseteq N_{\mathrm{ar}}, \qquad D^2 N_{\mathrm{ar}} \not\subseteq N_{\mathrm{ar}}, \qquad U_t N_{\mathrm{ar}} \not\subseteq N_{\mathrm{ar}} \quad (t \neq 0). \tag{435}$$

No finite-multiplicity assertion is required for this conclusion.

Proof. The original calculations (176)–(179) prove that either derivative invariance or a single nonzero one-sided heat inclusion makes every $S - \lambda$ algebraically onto. Their exact mechanism is retained: derivative invariance puts every $D^j \Xi$ in N_{ar}; the heat commutator polynomials have recurrence

$$P_0 = 1, \qquad P_{k+1}(X) = \frac{t}{2} X P_k(X) - P_k'(X),$$

with leading coefficient $(t/2)^k$, and put every $D^k U_t N_{\mathrm{ar}}$ in N_{ar}. At each λ at least one derivative of the relevant nonzero entire function Ξ or Z_t is nonzero. The actual division formula (174) then proves surjectivity.

Lemma R.7 would make every $S - \lambda$ continuously invertible on this quotient algebra, giving empty spectrum and contradicting the stated nonempty arithmetic spectrum. This excludes derivative and heat inclusion. The exact identity $D^2(sn) - sD^2 n = 2Dn$, with $sN_{\mathrm{ar}} \subseteq N_{\mathrm{ar}}$, gives the equivalence of the first and second derivative invariance questions and excludes the second inclusion. □

There is a direct value form of the same ring obstruction. If $n \in N_{\mathrm{ar}}$ has $n(\lambda) \neq 0$ and N_{ar} is an ideal, write

$$1 = \frac{n(s)}{n(\lambda)} + (s - \lambda) b(s), \qquad b(s) = \frac{1 - n(s)/n(\lambda)}{s - \lambda} \in \mathcal{H}_{\leq 1}.$$

Then multiplication by $[b]$ is the continuous inverse of $S - \lambda$. Thus every spectral point forces every relation element to vanish there. This conclusion is algebraic and uses no assumed evaluation continuity. The actual ideal property of the unnamed primary closure has not been established merely by proving this obstruction.

R.6 Why unit openness cannot be added to the original ring

Proposition R.9 (Nonopenness of the original unit set)**.** *For any complex topological-vector-algebra topology on the original $\mathcal{H}_{\leq 1}$, its group of units $\mathcal{H}_{\leq 1}^{\times}$ is not open. In fact*

$$1 \in \overline{\mathcal{H}_{\leq 1} \setminus \mathcal{H}_{\leq 1}^{\times}}^{\tau}. \tag{436}$$

Proof. The map $\epsilon \mapsto 1 - \epsilon s$ is continuous and tends to 1 as $\epsilon \to 0$ by the usual complex scalar multiplication. For every $\epsilon \neq 0$ it vanishes at $s = 1/\epsilon$, so it has no entire reciprocal and is not a unit of $\mathcal{H}_{\leq 1}$. This proves (436), and in particular no neighbourhood of the unit 1 is contained in the unit set. □

Thus unit openness is impossible on this original ring, not a technical condition that can be supplied by a choice of its complex-vector topology. The explicit quotient inverse above avoids that premise. The bounded primary-source reading recorded at the start of this section has supplied neither Fréchet structure for τ nor the actual ideal property of N_{ar}. These remain specific unresolved source assertions. The original-space classification, complete consequences within its stated class, exact ring representation, closed-ideal spectrum obstruction, and unit-set nonopenness have all been proved without identifying that source topology by assumption.

S Proof scope and reproducibility

The new source-topology assertions proved here are the complete transported closure, homeomorphic quotient, conjugated spectral operator, source-domain differential correspondence, exact coordinate kernels and cokernels, correlated jet extension, finite free cover, compact-test heat ingress with its weighted-moment inverse domains, and exact comparison diagrams. The compact and growth relation subspaces both equal $\Xi\mathcal{H}_{\leq 1}$ by the complete division proof. Their quotient topologies are computed: the compact quotient has its exact jet-image topology and product completion, whereas the growth quotient is complete and the comparison inverse is not continuous. All derivative relation domains and finite ambiguity projections retain the full zero multiplicities and analytic units. The arithmetic principal-part realization has the exact kernel $\Xi\mathcal{H}_{\leq 1}$, every finite image, the same product completion, and explicit original-coordinate spectral blocks. The full Schwartz trace factorization and its order-class intersection are proved without assigning every Schwartz trace to that order class. Its exact Mellin image has a proved inverse and complete finite-strip estimates, including every Taylor residue at the input origin. The exponential trace orbit has its exact open Schwartz boundary. The meromorphic arithmetic heat action retains the full zeta connection, its pole cancellations and the convergent numerator jets determining its Schwartz domain. The full persistent heat and even-derivative cores of $\Xi\mathcal{H}_{\leq 1}$ are zero. Every nonzero input returns to that analytic relation, or to the original Schwartz trace intersection, only at a closed discrete set of times; the complete proof retains every actual zero multiplicity and uses the original Hadamard product. The actual four-term ξ^4 coefficient is proved nonzero from its complete Fourier multiplier, commutes exactly with heat, and has those same discrete arithmetic return restrictions, with its original incidence factor retained. The retained Cartesian bilaplacian has its complete four coefficients and exact scaled residue. That original residue has full complex ambiguity on every actual moving source quotient, including time zero. Its finite trace sections and the precise quotient topology give an explicit homeomorphic residue-zero presentation. Every finite origin jet has a reciprocal-Ξ polynomial section, and at every nonzero time every one-point finite jet is realized by transported original polynomial traces. The complete normal-order correction, nonzero-polynomial obstruction, finite section construction and both quotient maps are proved. The separately transported residue has its full convergent derivative series and the exact heat-conjugate quotient. The Cartan construction now has a proved standard-real-subspace modular double for the original heat multiplier, full spectral domains at both signs of the parameter, and an exact conjugate moving-source quotient diagram. Its involution reverses the radial parameter. The threefold Fourier map identifies the positive Euclidean viscous generator with the original heat coefficient. For the retained nonlinear spatial pullback, the complete splitting, compressed diffusion, constant leakage kernel, affine commutator kernel, and second-diffusion feedback are computed explicitly. The separate public Navier–Stokes theorem is recorded as an imported source claim with distinct revision receipts; its full proof has not been independently verified in this fragment. Its actual cutoff-summed witness is now an explicit imported input to the full arithmetic flow map. The original polynomial target, inverse differential, pressure, forcing, viscosity scaling, material metric, both time

generators, and the two exact Fourier norm constants are retained. The source's particular growing swirl produces a complete arithmetic derivative channel and an unbounded residue with its whole source remainder. The joint profile and residue energies stay bounded, and the full profile has a proved terminal limit in its stated Hilbert space. The actual moving source class and its exact scalar kernel are calculated. A common quotient by $N_{\mathrm{ar}} \cap \Xi\mathcal{H}_{\leq 1}$ maps to both source and analytic quotients with proved kernels and a continuous residue recovery projection. The full logarithmic connection, varying source sample labels, and translated Cartan reversal are proved. A separate exact spectral model and generalized-chain argument identify what additional actual source spectral data would decide frozen descent without imposing a replacement topology. The specified source Sobolev cokernel has an exact dense compact-test ingress and a common-test comparison with Q_{ar}, with both kernels and all generator signs proved. Both explicit realizations have proved frozen differential and heat noninvariance and the original-coordinate divisor spectrum. Each heated finite Hermite trace also has its proved unique Möbius inverse, precise Mellin domain and full nontrivial-zero pole orders, with actual failures of boundedness at the origin for suitable original seeds at every nonzero time. The missing identification of the paper's unnamed topology is recorded explicitly and is not replaced by a conditional arithmetic theorem. Nothing in these results establishes an off-critical zero, a negative Weil value, or an arithmetic endpoint by analogy with the inverse-fibre collision. The exact transport through that collision cover and the derivative information it requires have been calculated above.

The angular-momentum map retains the exact Cartesian torque and symmetric-stress divergence. Its full nonlinear fluxes, half-line endpoint jets and inverse arithmetic dilation are proved. The source's actual fixed-axis coefficient gives a genuine Mellin pole at -1. The separate radial lift has a proved inverse and entire-function growth criterion. The full source recurrence proves radial analyticity of the actual corrected central field, including its pressure and force. Thus its untruncated radial arithmetic trace belongs to $\Xi\mathcal{H}_{\leq 1}$ on the stated central slab, and its value at the fixed original spectral point $-5i/2$ grows with exactly the source power τ^{-1-h}. Evaluation on the actual source quotient has full scalar ambiguity, with an explicit trace section. Whole input sequences provide the exact ingress elsewhere without presuming analyticity of arbitrary smooth spatial cutoffs.

The central meridional streamfunction and every summation-cutoff derivative are written explicitly, as are all three physical-force coefficients. The full radial entire evolution has a proved growth-topology smoothness estimate and an exact shift by $-2i$ in the original spectral coordinate. Its equation defect is the nonzero entire multiplier times the radial trace of the complete physical momentum residual. Injectivity proves equivalence of the two full equations on the stated inverse domain, including at the isolated multiplier zeros.

The original entire-function ring has a unique Hausdorff Fréchet vector-space topology with continuous coordinate multiplication, proved here to be the complete growth topology. The actual primary source has not been proved to be in that class. Without that identification, its exact closed multiplier algebra and largest annihilator ideal still give a faithful quotient representation. The full closed-ideal spectral obstruction and nonopenness of the original unit set are proved; the actual source ideal property, and therefore its remaining frozen-descent decision, remain open in the inspected source data.

The companion checker uses rational polynomial arithmetic to verify the heat conjugation on finite polynomial inputs, the retained product terms, the matrix commutator, the iterated branch-domain coefficients, the original-coordinate relation, and sample finite jet multiplication, the multiplicity-sensitive derivative domains, the source heat commutator recurrence, exact Möbius divisor sums, and Mellin principal-part coordinate inverses. The continuation checks the exponential Gaussian input and its generator, inverse Mellin differentiation and arithmetic residues, the Sobolev cutoff and Fourier moments, and the meromorphic heat conjugacy with all local pole terms. It also checks the unchanged reflection composition, time-jet coefficients

and positive Hadamard counting integrand used to compute the persistent core. The final spatial checks retain all Cartesian derivatives, all four bilaplacian coefficients, the residue limit and original four-term coefficient, the reciprocal-jet sections, and every tested normal-order and heat-conjugation correction. The Cartesian replay also checks the full spatial compression, leakage and its returning component, and the original scalar viscous residual. The modular domains and quotient diagram have an independent written proof audit. The new original three-coordinate replay checks its full determinant and cofactor inverse, the actual source sample matrix and inverse, the swirl channel, both time derivatives, all three material accelerations, the translated spectral conjugacy, and every tested logarithmic unit and residue coefficient. The varying material-label tangent signs and the quantitative source lower-bound denominator are retained. The angular replay checks the full Cartesian torque identities, physical radial diffusion and graph coordinate, dyadic endpoint coefficients, exact radial Mellin residues, similarity derivative maps, sparse recurrence products and the fixed entire trace value. These are reproducible checks of the displayed identities, not a substitute for their written proofs. Source hashes and exact reading locators are in the companion provenance receipt; no numerical zero search or Lean run is part of this work.

Using the released Navier–Stokes construction as the arithmetic input

The precise imported witness and its unchanged viscosity

We now use the particular field constructed in the released manuscript [8], not merely an arbitrary member of the input class of Section 16. The complete 166-page source proof is retained in this repository as `sources/ns_public/navier_stokes_166.pdf`; its SHA-256 is recorded in the source provenance. The source existence and all-order residual construction are imported from that manuscript. Our independent reading of its witness construction and closing proof is recorded in `logbook/ns_witness_source_read.md`; it is not a claim of independent verification of every source lemma. The formulas below specify the imported object and prove its arithmetic image.

Use a star to retain the source's viscosity-one fields, rather than overwriting the physical viscosity $\nu > 0$. In source equation (9.21), page 115, the actual summed Cartesian vector potential, azimuthal field, and pressure are denoted by $A_*, B_* e_\theta, p_{*,\mathrm{loc}}$. These are the cutoff-summed corrected fields of that equation, including all stages and the finite initialization. Source equation (10.4), page 118, constructs

$$u_* = \operatorname{curl}(c_* A_*) + c_* B_* e_\theta, \qquad p_* = c_* p_{*,\mathrm{loc}}, \qquad c_* = \chi_x \chi_t.$$

Here the source spatial cutoff is smooth in (r^2, z), supported in $r < r_0,\ |z| < z_0$, and equals one on $r \le r_0/2,\ |z| \le z_0/2$. Its time cutoff is zero for $1 - t \ge \tau_0$ and one for $0 \le 1 - t \le \tau_0/2$, where $0 < \tau_0 < 1/2$. Retain the actual source choices of these parameters. The source force before time one is exactly

$$f_* = \partial_t u_* + (u_* \cdot \nabla) u_* - \Delta u_* + \nabla p_*.$$

Thus every cutoff derivative belongs to the force. Its smooth extension through time one is source (10.11), page 120, with all its prescribed Taylor coefficients and cutoff sequence. The source proves that its support is contained in $K_* \times [0, 2]$, where $K_* = \operatorname{supp} \chi_x$, and it vanishes near time zero. The pressure is the displayed compactly supported source pressure; no divergence-free condition is imposed on f_*.

For the physical parameter $\nu > 0$, take precisely source (10.22):

$$\begin{aligned} y &= x/\sqrt{\nu}, & x &= \sqrt{\nu}\, y, \\ u_\nu(x,t) &= \sqrt{\nu}\, u_*(x/\sqrt{\nu}, t), & p_\nu(x,t) &= \nu p_*(x/\sqrt{\nu}, t), \\ f_\nu(x,t) &= \sqrt{\nu}\, f_*(x/\sqrt{\nu}, t), & K_\nu &= \sqrt{\nu}\, K_*. \end{aligned}$$

The inverse multiplies the velocity and force by $\nu^{-1/2}$, the pressure by ν^{-1}, and substitutes $x = \sqrt{\nu}\, y$; physical time is unchanged. The chain rule gives

$$\partial_t u_\nu + (u_\nu \cdot \nabla_x) u_\nu - \nu \Delta_x u_\nu + \nabla_x p_\nu = \sqrt{\nu}\, [\partial_t u_* + (u_* \cdot \nabla_y) u_* - \Delta_y u_* + \nabla_y p_*](y,t) = f_\nu.$$

Also $\operatorname{div}_x u_\nu = \operatorname{div}_y u_* = 0$, $u_\nu(0) = 0$, and u_ν, p_ν have support in K_ν for every $t < 1$. These are exactly the inputs of Section 16 with $T = 1$, with all coordinate and parameter maps specified.

Let

$$F_\nu(t) = \int_0^t \|f_\nu(s)\|_{L^2(\mathbb{R}^3)}\, ds, \qquad E_\nu = F_\nu(1) = \nu^{5/4} F_*(1) < \infty.$$

The last exponent follows from the factor $\sqrt{\nu}$ in f_ν and $dx = \nu^{3/2} dy$. For completeness the energy estimate used below is proved directly for this input. Compact support and incompressibility give, by integrating the equation against u_ν,

$$\frac{1}{2} \frac{d}{dt} \|u_\nu(t)\|_2^2 + \nu \|\nabla u_\nu(t)\|_2^2 = \langle f_\nu(t), u_\nu(t) \rangle.$$

The convection integral is the integral of $\operatorname{div}(u_\nu|u_\nu|^2/2)$, and the pressure integral is minus the integral of $p_\nu \operatorname{div} u_\nu$; both vanish. Divide by $(\|u_\nu\|_2^2+\delta^2)^{1/2}$, use Cauchy–Schwarz, integrate from zero, and let $\delta \downarrow 0$. This proves $\|u_\nu(t)\|_2 \le F_\nu(t)$. Substitution back into the integrated identity proves, with its full dissipation coefficient,

$$\|u_\nu(t)\|_2^2 + 2\nu \int_0^t \|\nabla u_\nu(s)\|_2^2 ds \le 2\int_0^t F_\nu'(s)F_\nu(s)ds = F_\nu(t)^2.$$

The source's actual growing component in two arithmetic channels

The source fixes $0 < h < 1/100$, $X_{\text{in}} > 0$, and $e_0 > 0$ through its construction. Its closing proof, (10.20)–(10.21), page 124, gives the actual corrected swirl along

$$x_{*,\tau} = (\sqrt{2X_{\text{in}}\tau}, 0, 0), \qquad t = 1-\tau, \qquad (u_*)_2(x_{*,\tau}, 1-\tau) = \tau^{-1/2-h}(e_0 + R_*(\tau)).$$

The original remainder is retained as the function $R_*(\tau) = \tau^{1/2+h}(u_*)_2(x_{*,\tau}, 1-\tau) - e_0$. The source estimate supplies constants $C_* > 0, \tau_* > 0$ with $|R_*(\tau)| \le C_*\tau^{2h}$ for $0 < \tau < \tau_*$. At these points the cylindrical angle is zero, so $e_\theta = (0,1,0)$ and the displayed Cartesian component is precisely the source swirl. The annular and higher-order corrections have already been accounted for in this remainder. The parameter h is the chosen source parameter; no claim that an arbitrary prescribed h works is needed or made.

Define

$$r_{\nu,\tau} = \sqrt{2\nu X_{\text{in}}\tau}, \qquad x_{\nu,\tau} = (r_{\nu,\tau}, 0, 0).$$

The exact viscosity map proves

$$(u_\nu)_2(x_{\nu,\tau}, 1-\tau) = \sqrt{\nu}\,\tau^{-1/2-h}(e_0 + R_*(\tau)). \tag{437}$$

Use the actual material flow $X_{\nu,t}$ of u_ν proved to exist on each interval before one in Section 16. Put

$$X_{\nu,t}(q) = \sqrt{\nu}\, X_{*,t}(q/\sqrt{\nu}).$$

This is an equality of the actual flows: its right side equals q at zero, and its time derivative is $\sqrt{\nu}\, u_*(t, X_{*,t}(q/\sqrt{\nu})) = u_\nu(t, \sqrt{\nu}\, X_{*,t}(q/\sqrt{\nu}))$. The previously proved uniqueness gives the equality, with inverse $X_{\nu,t}^{-1}(x) = \sqrt{\nu}\, X_{*,t}^{-1}(x/\sqrt{\nu})$. Define the labels

$$q_{\nu,\tau} = X_{\nu,1-\tau}^{-1}(x_{\nu,\tau}).$$

This is the exact inverse between the Eulerian sampling points and their material labels. It is a family of labels depending on τ; the source path has not been asserted to be a single fluid trajectory.

Retain $S = (F_1/2, F_2, F_3)$ and $J = DS$ in the physical Cartesian coordinates x, not the viscosity-one coordinate y. Direct substitution into the original polynomials gives

$$S(r,0,0) = (0,0,2r), \qquad J(r,0,0) = \begin{pmatrix} 0 & 0 & 1/2 \\ 0 & 1 & 3r \\ 2 & -3r^2 & -r^3 \end{pmatrix}.$$

Indeed the first row comes from differentiating $F_1/2 = ((1+xy)^3 w + y^2(1+xy)(4+3xy))/2$; the second comes from $F_2 = y + 3x(1+xy)^2 w + 3xy^2(4+3xy)$; the third comes from $F_3 = 2x - 3x^2 y - x^3 w$, all evaluated at $(x,y,w) = (r,0,0)$. Every entry, including the off-diagonal powers of r, is retained. Writing $v = J(r,0,0)u$ therefore gives the exact component inverse

$$u_3 = 2v_1, \qquad u_2 = v_2 - 6rv_1,$$
$$u_1 = \tfrac{1}{2}v_3 + \tfrac{3}{2}r^2 v_2 - 8r^3 v_1.$$

Substituting these expressions into all three rows proves both directions of this matrix inverse.

Fix actual arithmetic time zero. Take $H = Z_0 = \Xi$ or $H = K_0$ with the complete four-term coefficient from the preceding proof, and write

$$d_H^2 = \int_{\mathbb{R}} |H'(s)|^2 ds > 0, \qquad \Psi_{\nu,j}(t,q,\zeta) = H(S_j(X_{\nu,t}(q)) + \zeta), \qquad D_{\nu,j} = \partial_t \Psi_{\nu,j}.$$

The exact positive constants are those of (152) at $\theta = 0$, including 32π and, for K_0, the entire factor $|m|^2$. At the particular pair $(t,q) = (1-\tau, q_{\nu,\tau})$, the first two shifts are both zero. Consequently define the actual integral

$$I_{\nu,j}(\tau) = \frac{1}{d_H^2} \int_{\mathbb{R}} \overline{H'(s)} D_{\nu,j}(1-\tau, q_{\nu,\tau}, s) ds, \qquad j = 1, 2.$$

The chain rule gives $D_{\nu,j} = v_j H'(s)$ there. It follows, without an asymptotic replacement of the profiles, that

$$I_{\nu,2}(\tau) - 6 r_{\nu,\tau} I_{\nu,1}(\tau) = \sqrt{\nu}\, \tau^{-1/2-h}(e_0 + R_*(\tau)). \tag{438}$$

There is also a whole-profile identity, stronger than its projection:

$$\begin{aligned} D_{\nu,2}(1-\tau, q_{\nu,\tau}, s) - 6 r_{\nu,\tau} D_{\nu,1}(1-\tau, q_{\nu,\tau}, s) \\ = \sqrt{\nu}\, \tau^{-1/2-h}(e_0 + R_*(\tau)) H'(s). \end{aligned} \tag{439}$$

Its squared $L^2(ds)$ norm is exactly $\nu \tau^{-1-2h}(e_0 + R_*(\tau))^2 d_H^2$. Thus the source's actual component and remainder, not just an abstract possibility of growth, have been carried into the original arithmetic profile. The complex conjugation and integration above retain the complex values of K_0 when that channel is used.

Choose any $0 < \tau_{**} \le \tau_*$ such that $C_* \tau_{**}^{2h} \le e_0/2$ and the source cutoffs equal one on the path for $\tau < \tau_{**}$. Cauchy–Schwarz on the two component coefficients $(-6 r_{\nu,\tau}, 1)$ yields

$$\left(\sum_{j=1}^{3} \| D_{\nu,j}(1-\tau, q_{\nu,\tau}, \cdot) \|_2^2 \right)^{1/2} \ge \frac{d_H \sqrt{\nu}\, e_0}{2\sqrt{1 + 72 \nu X_{\mathrm{in}} \tau}}\, \tau^{-1/2-h}. \tag{440}$$

This lower bound proves divergence in the spatial-supremum, spectral-L^2 norm for the imported constructed field. It is not a matching upper asymptotic for its global supremum. All forcing, pressure, and material diffusion terms remain those of (149) and (150), with the explicit (u_ν, p_ν, f_ν) just given.

A terminal arithmetic profile with a bounded energy norm

The same constructed input gives a terminal value in a precise function space. This assertion can be proved from its energy identity without controlling its unbounded spatial supremum. Let $M_\nu = \max_{K_\nu} \|J\|_{\mathrm{op}}$ and put

$$\mathcal{H}_\nu = L^2(K_\nu \times \mathbb{R}, dq\, ds; \mathbb{C}^3), \qquad B_{\nu,j}(t,q,s) = \Psi_{\nu,j}(t,q,s) - H(S_j(q) + s).$$

Subtracting the displayed initial value defines an affine coordinate with the explicit inverse $B \mapsto B + (H(S_j(q) + s))_j$. It does not discard the initial profile. Its domain and reference element are specified; outside K_ν the difference is identically zero. The full profile on K_ν is also square integrable because real translation gives norm $|K_\nu|^{1/2} \|H\|_2$ in each channel.

The already proved exact identity gives

$$\|D_\nu(t)\|_{\mathcal{H}_\nu}^2 = d_H^2 \int_{K_\nu} |J(x) u_\nu(t,x)|^2 dx \le d_H^2 M_\nu^2 E_\nu^2.$$

The identity follows by integrating the real spectral translation and then using $\det D_q X_{\nu,t} = 1$; no spatial coordinate is removed. On a closed time interval before one the functions are smooth and the Schwartz bounds justify their Hilbert-space derivatives. Integrating $\partial_t B_\nu = D_\nu$ there and using the triangle inequality for the integral yields, for $0 \le a < b < 1$,

$$\|B_\nu(b) - B_\nu(a)\|_{\mathcal{H}_\nu} \le d_H M_\nu E_\nu(b-a). \tag{441}$$

Completeness constructs a unique terminal element $B_\nu(1)$ and the exact bound $\|B_\nu(1) - B_\nu(t)\| \le d_H M_\nu E_\nu(1-t)$. This constructs a Lipschitz extension on $[0,1]$ in the specified Hilbert space, while (440) proves divergence in the stronger spatial-supremum norm of its time derivative.

We also identify the terminal element through an actual map on labels. Volume preservation gives

$$\int_{K_\nu} \int_0^1 |u_\nu(t, X_{\nu,t}(q))| dt\, dq = \int_0^1 \|u_\nu(t)\|_{L^1(K_\nu)} dt \le |K_\nu|^{1/2} E_\nu.$$

The nonnegative integral is finite, so for almost every q the full trajectory has finite total length before one. Its integral equation therefore constructs the limit $X_{\nu,1}(q) \in K_\nu$ as $t \uparrow 1$. Set its value arbitrarily in K_ν on the null exceptional set. Also the integral triangle inequality and volume preservation give

$$\|X_{\nu,b} - X_{\nu,a}\|_{L^2(K_\nu)} \le \int_a^b \|u_\nu(t)\|_2 dt \le E_\nu(b-a),$$

and the same estimate with $b = 1$ follows by the constructed limit and Fatou's lemma. For every continuous g on compact K_ν, dominated convergence gives

$$\int_{K_\nu} g(X_{\nu,1}(q)) dq = \lim_{t \uparrow 1} \int_{K_\nu} g(X_{\nu,t}(q)) dq = \int_{K_\nu} g(x) dx.$$

This identifies the pushforward of Lebesgue measure exactly. To see that continuous tests suffice, for each relatively closed set C use the continuous decreasing approximations $g_n(x) = \max(0, 1 - n \operatorname{dist}(x, C))$ to its indicator, then monotone or dominated convergence; complements give open sets and the uniqueness of finite measures on their generated Borel sigma algebra gives the stated pushforward.

For almost every q the pointwise terminal entire fibre is

$$\Psi_{\nu,j}(1, q, \zeta) = H(S_j(X_{\nu,1}(q)) + \zeta).$$

It agrees with the Hilbert-space limit: real translations are continuous in $L^2(ds)$, as follows from $\|H(\cdot + a) - H(\cdot + b)\|_2 \le d_H |a - b|$ by integrating H'. The shifts converge almost everywhere, remain bounded on K_ν, and their translated-profile squared differences are bounded by $4\|H\|_2^2$. Dominated convergence proves the identification in $\mathcal{H}_\nu$. Replacing H by $H^{(n)}$ in this proof proves the same assertions for every pure spectral derivative, with the retained constant $\|H^{(n+1)}\|_2 M_\nu E_\nu$.

The terminal label map has a further exact operator correspondence. On $L^2(K_\nu)$ define $V_t g = g \circ X_{\nu,t}$, including $t = 1$. For $t < 1$ its inverse is composition with $X_{\nu,t}^{-1}$, so it is unitary. The just-proved measure identity makes V_1 an isometry. For continuous g, dominated convergence gives $V_t g \to V_1 g$ in L^2. Continuous functions are dense for Lebesgue measure restricted to a compact set: approximate measurable-set indicators by continuous distance cutoffs between compact subsets and open supersets using finite-measure regularity, then approximate simple functions. The isometric bounds extend the convergence to every $g \in L^2(K_\nu)$. Thus $V_t \to V_1$ strongly. Its Hilbert adjoint satisfies $V_1^* V_1 = I$ by preservation of inner products, and $V_1 V_1^*$ is the orthogonal projection onto its closed range: it is self-adjoint, idempotent, acts as identity on that range, and annihilates its orthogonal complement. These are the proved terminal inverse and range statements. No pointwise inverse of $X_{\nu,1}$, or surjectivity of V_1, has been supplied by this argument.

The actual zero image, including the terminal fibres

For every time before one and almost every terminal label, the shift $a = S_j(X_{\nu,t}(q))$ is real. The full coordinate map is $\zeta \mapsto a + \zeta$ with inverse $s \mapsto s - a$. If $H(s) = (s - z_0)^m A_0(s)$ with $A_0(z_0) \neq 0$, the actual image is

$$H(a + \zeta) = (\zeta - (z_0 - a))^m A_0(a + \zeta).$$

This retains the whole analytic unit and all its derivatives. In particular its zero has imaginary part $\operatorname{Im} z_0$ at all these times, even for the source input whose arithmetic time-derivative combination obeys (439). With $H = \Xi$ the precise completed-zeta argument is $\rho = 1/2 + i(a+\zeta)$, and $\operatorname{Re} \rho = 1/2 - \operatorname{Im} \zeta$. Thus the actual source witness produces the proved derivative singularity and terminal profile above; this explicit map preserves zero heights. A zero-height change is not obtained by substituting a faster-growing velocity into this same real-translation map.

Actual angular momentum, nonlinear fluxes and arithmetic dilation

Cartesian input, angular projection and its exact section

Use the complete source witness (u_ν, p_ν, f_ν) constructed in the preceding appendix, for every physical time $0 \le t < 1$. Retain the viscosity $\nu > 0$ and original Cartesian coordinates (x_1, x_2, z). For $r > 0$ put $x_1 = r\cos\theta, x_2 = r\sin\theta$, $e_r = (\cos\theta, \sin\theta, 0)$ and $e_\theta = (-\sin\theta, \cos\theta, 0)$. Write $u_r = e_r \cdot u_\nu$, $u_\theta = e_\theta \cdot u_\nu$ and $u_z = (u_\nu)_3$. These components may depend on θ; in particular the complete corrected velocity is not assumed axisymmetric. Define $\langle a\rangle = (2\pi)^{-1}\int_0^{2\pi} a\, d\theta$ and retain the radial coordinate and inverse

$$\sigma = \frac{r^2}{2}, \qquad r = \sqrt{2\sigma}, \qquad r\, dr = d\sigma.$$

This radial σ is not the earlier polynomial coordinate $F_1/2$. They coexist as explicit functions on the original Cartesian space; no equality of their Laplacians is required below. Their exact common coordinate map, on $(0,\infty)\times(\mathbb{R}/2\pi\mathbb{Z})\times\mathbb{R}$, is $(\sigma,\theta,z) \mapsto (\sqrt{2\sigma}\cos\theta, \sqrt{2\sigma}\sin\theta, z)$. Its inverse is $\sigma = (x_1^2 + x_2^2)/2$, the direction $(\cos\theta, \sin\theta) = (x_1, x_2)/\sqrt{x_1^2 + x_2^2}$, and the unchanged z. In these coordinates the full polynomial is

$$\frac{F_1}{2} = \frac{1}{2}\big[(1 + \sigma\sin(2\theta))^3 z + 2\sigma\sin^2\theta(1 + \sigma\sin(2\theta))(4 + 3\sigma\sin(2\theta))\big].$$

Substitution of $x_1x_2 = \sigma\sin(2\theta)$ and $x_2^2 = 2\sigma\sin^2\theta$ proves this morphism; adjoining the displayed value as an extra coordinate gives its graph map with inverse projection onto (σ,θ,z). The Cartesian definition continues over the axis where the angular chart is not defined.

The actual angular-momentum density, fluxes and force torque are

$$B(\sigma,z,t) = \langle r u_\theta\rangle, \qquad P(\sigma,z,t) = \langle r^2 u_r u_\theta\rangle,$$
$$Q(\sigma,z,t) = \langle r u_z u_\theta\rangle, \qquad F(\sigma,z,t) = \langle r f_\theta\rangle.$$

These definitions include every corrected, oscillatory and cutoff term of the actual source fields before averaging. All four functions have fixed compact support in (σ, z) on the closed half-plane $\sigma \ge 0$. They are smooth there on every closed preterminal time interval, with

$$B = O(\sigma), \quad Q = O(\sigma), \quad F = O(\sigma), \qquad P = O(\sigma^2) \quad (\sigma \downarrow 0). \tag{442}$$

Here and below these estimates hold with all z, t derivatives on compact preterminal sets. To prove them without polar smoothness assumptions, observe that $ru_\theta = x_1u_2 - x_2u_1$ and $ru_r = x_1u_1 + x_2u_2$. Angular averaging their Cartesian expressions gives smooth rotation-invariant functions of (x_1, x_2). A smooth rotation-invariant function is smooth in σ on the closed half-line: its restriction to $(r, 0)$ is even and smooth; Taylor's formula to any even order, with its differentiated remainder, expresses it as a polynomial in r^2 plus a remainder of arbitrarily high order. The chain rule on $r > 0$ then supplies every continuous right derivative in σ at zero and the fundamental theorem of calculus identifies those derivatives successively. The same Taylor proof works with the retained z, t parameters. The constants in the averages defining B, Q, F vanish at $r = 0$; odd radial powers vanish under rotation. They therefore start at r^2. For P, the quadratic Cartesian term at zero is

$$\big\langle (x_1a + x_2b)(x_1b - x_2a)\big\rangle = 0, \qquad (a,b) = (u_1,u_2)(0,0,z,t),$$

because $\langle x_1^2 - x_2^2\rangle = \langle x_1x_2\rangle = 0$. Its constant term also vanishes, and rotation removes the odd powers, so P starts at r^4. This proves (442).

The projection $\Pi u = B$ has the explicit linear section

$$(\mathcal{R}B)(x_1, x_2, z) = \frac{B(\sigma, z)}{2\sigma}(-x_2, x_1, 0), \qquad \Pi\mathcal{R} = I. \tag{443}$$

The quotient at zero is the actual smooth extension: $B(\sigma, z)/\sigma = \int_0^1 \partial_\sigma B(a\sigma, z)da$. Thus the section is Cartesian smooth, compactly supported, and divergence-free. For $r > 0$ it has sole component B/r in the e_θ direction, proving the section identity. For the exact remainder $w = u_\nu - \mathcal{R}B$, one has $\Pi w = 0$ and

$$P = rB\langle w_r\rangle + r^2\langle w_r w_\theta\rangle, \qquad Q = B\langle w_z\rangle + r\langle w_z w_\theta\rangle. \tag{444}$$

These follow by substituting $u_r = w_r$, $u_z = w_z$ and $u_\theta = B/r + w_\theta$ into the original definitions. They exhibit both mean transport and the complete two quadratic correlations. The full inverse of $u \mapsto (B, w)$ is $(B, w) \mapsto \mathcal{R}B + w$ on the stated subspace $\Pi w = 0$. In particular the projection has the precise kernel $\{w : \langle rw_\theta\rangle = 0\}$; none of that kernel's contribution to the fluxes has been discarded.

The exact equation before and after angular averaging

For the unaveraged Cartesian angular momentum $L = x_1u_2 - x_2u_1 = ru_\theta$, the product rule in the full Navier–Stokes equation gives

$$\partial_t L + \operatorname{div}(u_\nu L) = \nu\Delta L - 2\nu\omega_z - \partial_\theta p_\nu + (x_1f_2 - x_2f_1), \qquad \omega_z = \partial_1u_2 - \partial_2u_1. \tag{445}$$

Indeed the material derivatives of x_1, x_2 contribute $u_1u_2 - u_2u_1 = 0$; incompressibility converts material transport to the displayed divergence. The Laplacian product rule gives $\Delta L = x_1\Delta u_2 - x_2\Delta u_1 + 2\omega_z$. Finally $x_1\partial_2 - x_2\partial_1 = \partial_\theta$ gives the negative pressure-torque term with its sign.

In polar coordinates $\omega_z = r^{-1}\partial_r(ru_\theta) - r^{-1}\partial_\theta u_r$. Average (445) over a full circle. Every angular derivative has zero integral, by the fundamental theorem of calculus and periodicity, including the actual non-axisymmetric pressure. The convection term is $r^{-1}\partial_r(r\langle u_rL\rangle) + \partial_z\langle u_zL\rangle$; the averaged viscous term is $\nu(\partial_{rr} - r^{-1}\partial_r + \partial_{zz})B$. Using $\partial_r = r\partial_\sigma$ now proves

$$\partial_t B + \partial_\sigma P + \partial_z Q = \nu(2\sigma\partial_{\sigma\sigma}B + \partial_{zz}B) + F. \tag{446}$$

The formula extends to $\sigma = 0$ by (442) and smoothness. The pressure has disappeared here by a proved projection, with its entire original torque term exhibited in (445). No scalar heat equation has replaced the actual nonlinear fluxes.

More generally a Cartesian momentum residual $f_i + \sum_j \partial_j T_{ij}$ contributes the torque $\langle x_1\partial_j T_{2j} - x_2\partial_j T_{1j}\rangle$. For a symmetric stress T, the product rule gives exactly

$$x_1\partial_j T_{2j} - x_2\partial_j T_{1j} = \partial_j(x_1T_{2j} - x_2T_{1j}) - (T_{21} - T_{12}) = \partial_j(x_1T_{2j} - x_2T_{1j}).$$

Writing its vector flux in polar components therefore turns this contribution into

$$\partial_\sigma\langle r^2T_{\theta r}\rangle + \partial_z\langle rT_{\theta z}\rangle, \quad T_{\theta r} = e_\theta^{\mathsf{T}}Te_r, \quad T_{\theta z} = e_\theta^{\mathsf{T}}Te_z. \tag{447}$$

For a nonsymmetric tensor the retained extra torque is $-\langle T_{21} - T_{12}\rangle$. Thus every symmetric stress cancellation in the source momentum residual has this exact projected image, with its sign determined by the side of the momentum equation on which the source places $\operatorname{div} T$. The map is linear in the full Cartesian tensor; its evaluation is not an identification of the source's auxiliary-torus average with the physical angular average used here.

The source cutoff sum has this exact image before the terminal limit. In its original viscosity-one coordinates let $\mathbf{b}_j$ denote the actual Cartesian vector appearing as B_j in source (9.21). With q_* here denoting that source's concentration coordinate, retain every increment as

$$\delta u_{*,j} = \operatorname{curl}\bigl(c_*\chi(a_jq_*)A_j\bigr) + c_*\chi(a_jq_*)\mathbf{b}_j, \qquad \delta p_{*,j} = c_*\chi(a_jq_*)p_j.$$

The notation q_* in this display is the source coordinate determined by $1-t = q_*(1-\eta^2)$ and $z = q_*^{1/2-h}\eta$, not a material label; those two equations specify its actual dependence on (z,t). Define each viscosity-ν increment by the full scaling of the preceding appendix, and denote the correspondingly localized base by $(u_{\nu,0}, p_{\nu,0})$. The actual fields are $u_\nu = \sum_{j\ge0} u_{\nu,j}$ and $p_\nu = \sum_{j\ge0} p_{\nu,j}$, where the positive-index terms are the displayed increments with that scaling. Source local finiteness on every compact set before time one makes these sums, and every derivative of them, finite on a neighborhood of each sampled circle. Indeed the source coordinate is independent of θ, so its positive lower bound and its finite set of active stages can be chosen over the entire compact circle. Thus the exact nonlinear fluxes are

$$P = \sum_{a,b\ge0} \langle r^2 (u_{\nu,a})_r (u_{\nu,b})_\theta \rangle, \qquad Q = \sum_{a,b\ge0} \langle r (u_{\nu,a})_z (u_{\nu,b})_\theta \rangle.$$

Each sum contains every diagonal term and both ordered cross terms for each pair $a \neq b$. The curl in $u_{\nu,j}$ is taken after multiplying both cutoffs, so their derivative corrections belong to these same fluxes. Projection, differentiation and the sums commute by the stated local finiteness. Hence the source's exact residual identity, including its flat remainder and final smooth force, passes to (446) with all these contributions. This argument uses the actual summed fields for $t<1$; it does not exchange an uncontrolled infinite series with the terminal limit.

The radial Mellin equation with all shifted fluxes

For each actual field $a = B, P, Q, F$ put

$$\widehat{a}(s,z,t) = \int_0^\infty a(\sigma,z,t)\sigma^{s-1}d\sigma, \qquad \sigma^{s-1} = e^{(s-1)\log\sigma}.$$

The original measure here is $d\sigma = r\,dr$, with real logarithm; angular averaging already contains its factor $1/(2\pi)$. The integral for B, Q, F is absolutely convergent and holomorphic on $\Re s > -1$, and that for P on $\Re s > -2$, by compact support and (442). On compact substrips every complex derivative adds an integrable power of $|\log\sigma|$, which proves holomorphy by differentiated domination. All z,t derivatives on closed preterminal sets obey the same justification.

For $\Re s > 0$, integration by parts gives

$$\int_0^\infty \partial_\sigma P\,\sigma^{s-1}d\sigma = -(s-1)\widehat{P}(s-1),$$
$$\int_0^\infty 2\sigma\partial_{\sigma\sigma}B\,\sigma^{s-1}d\sigma = 2s(s-1)\widehat{B}(s-1).$$

All upper boundary terms vanish by compact support. The lower terms $P\sigma^{s-1}$, $B'\sigma^s$ and $B\sigma^{s-1}$ vanish by (442) and $\Re s > 0$. Consequently the full equation is

$$\partial_t\widehat{B} - (s-1)\widehat{P}(s-1) + \partial_z\widehat{Q} = \nu\{2s(s-1)\widehat{B}(s-1) + \partial_{zz}\widehat{B}\} + \widehat{F}, \qquad \Re s > 0. \tag{448}$$

In particular no shift $s \mapsto s-1$, polynomial factor, or force moment has been suppressed.

This is an actual injective transform with its inverse. For any real $c>0$ and fixed (z,t) put $g(u) = e^{cu}a(e^u,z,t)$. It vanishes for sufficiently large positive u and is $O(e^{(c+1)u})$ as $u\to-\infty$,

with the same bound for every derivative; for P the exponent improves to $c+2$. These facts follow from the chain rule and (442). Thus g is Schwartz and

$$\widehat{a}(c-i\eta,z,t)=\int_{\mathbb{R}} g(u)e^{-i\eta u}du, \qquad a(\sigma,z,t)=\frac{\sigma^{-c}}{2\pi}\int_{\mathbb{R}}\widehat{a}(c-i\eta,z,t)e^{i\eta\log\sigma}d\eta.$$

To justify the inverse without a distributional shortcut, twice integrate by parts in the forward integral to obtain an integrable $O((1+|\eta|)^{-2})$ bound. Insert $e^{-\epsilon\eta^2}$ in the inverse. Fubini and the Gaussian integral give convolution by $(4\pi\epsilon)^{-1/2}e^{-u^2/(4\epsilon)}$, which converges to g by continuity and its Gaussian tail. Dominated convergence in the integrable Fourier inverse then proves the displayed identity.

The same arithmetic dilation map on the actual corrected fields

Retain the source's chosen $0<h<1/100$. For any smooth compactly supported half-line function a with $a(0)=0$, define

$$\begin{aligned}\phi_a(x)&=a(x)-2^h a(2x),\\ \psi_a(x)&=\phi_a(x)-\tfrac{1}{2}\phi_a(x/2)\\ &=(1+2^{h-1})a(x)-2^h a(2x)-\tfrac{1}{2}a(x/2),\\ (\mathcal{A}_h a)(x)&=\sum_{n\ge1}\psi_a(nx), \qquad x>0.\end{aligned}$$

These are the exact three dilation coefficients used by the parent heat-exterior calculation; here their input is the actual B,P,Q,F. Every sum is locally finite on $x>0$, and the trace vanishes for all sufficiently large x. Substitution gives $\int_0^\infty\psi_a(x)dx=0$ and $\psi_a(0)=0$. Its small-x bound has a direct proof. On each interval $[nx,(n+1)x]$, two integrations by parts give the trapezoidal identity

$$\int_{nx}^{(n+1)x}\psi_a(v)dv=\frac{x}{2}\{\psi_a(nx)+\psi_a((n+1)x)\}-\frac{1}{2}\int_{nx}^{(n+1)x}(v-nx)((n+1)x-v)\psi_a''(v)dv.$$

Summing, using compact support and the two zero quantities just proved, yields

$$|(\mathcal{A}_h a)(x)|\le\frac{x}{8}\|\psi_a''\|_1. \tag{449}$$

Thus its Mellin integral is holomorphic on $\Re s>-1$. Parameter derivatives in z,t have the same bound because the trace is linear, preserves the zero endpoint and zero integral, and commutes with those derivatives on preterminal compact sets.

On $\Re s>1$ absolute Fubini gives

$$\mathcal{M}(\mathcal{A}_h a)(s)=W_h(s)\widehat{a}(s), \qquad W_h(s)=\zeta(s)(1-2^{s-1})(1-2^{h-s}). \tag{450}$$

Indeed the sum of absolute Mellin integrals is $\zeta(\Re s)\int|\psi_a(x)|x^{\Re s-1}dx$, and a dilation by $b>0$ multiplies its Mellin integral by b^{-s}. The only pole of ζ is its simple pole at one, and the factor $1-2^{s-1}$ cancels it. Consequently W_h is entire and the identity theorem extends (450) to $\Re s>-1$ for these actual inputs. Its multiplier is the same arithmetic factor as in the parent exterior calculation. The gamma product there is specifically the Mellin transform of that exterior's H_h; it is not silently substituted for the present corrected-field moment $\widehat{a}$.

The complete inverse preserves the actual field and its endpoint:

$$\begin{aligned}\psi_a(x)&=\sum_{n\ge1}\mu(n)(\mathcal{A}_h a)(nx),\\ \phi_a(x)&=\sum_{j\ge0}2^{-j}\psi_a(x/2^j),\\ a(x)&=-\sum_{k\ge1}2^{-kh}\phi_a(x/2^k).\end{aligned}$$

The first double sum is finite at each $x > 0$, so rearranging gives $\sum_{m\ge1} \psi_a(mx) \sum_{n|m} \mu(n) = \psi_a(x)$. The divisor sum is one for $m = 1$ and zero otherwise, since the squarefree expansion gives $\prod_{p|m}(1-1)$ for $m > 1$. The finite second sum through $j = J$ equals $\phi_a(x) - 2^{-J-1}\phi_a(x/2^{J+1})$. The negative finite third sum through $k = K$ equals $a(x) - 2^{-Kh}a(x/2^K)$. Their tails vanish by boundedness at zero and $h > 0$. After any x derivative, each summand gains the additional factor $2^{-j\ell}$ or $2^{-k\ell}$; bounded derivatives on compact half-line intervals prove uniform convergence there. The arithmetic inverse is locally finite away from zero; the subsequent recovered functions extend to zero by the displayed smooth inverse. These are actual inverse maps on this stated class, including its parameter dependence, with no division by zeta values at their zeros.

Set $\mathfrak{B} = \mathcal{M}\mathcal{A}_h B$ and define $\mathfrak{P}, \mathfrak{Q}, \mathfrak{F}$ in the identical way. Multiplication of (448) by $W_h(s)$ gives the exact equation

$$\partial_t \mathfrak{B} + \partial_z \mathfrak{Q} - \nu \partial_{zz} \mathfrak{B} - \mathfrak{F} = (s-1)W_h(s)\{\widehat{P}(s-1) + 2\nu s \widehat{B}(s-1)\}.$$

Both hatted terms on the right are explicitly recovered by the proved inverses, so this equation retains the full dynamics. There is also the division-free shifted relation

$$W_h(s-1)\{\partial_t \mathfrak{B} + \partial_z \mathfrak{Q} - \nu \partial_{zz} \mathfrak{B} - \mathfrak{F}\} = (s-1)W_h(s)\{\mathfrak{P}(s-1) + 2\nu s \mathfrak{B}(s-1)\}, \quad \Re s > 0. \tag{451}$$

Initially this follows on $\Re s > 1$; every factor is holomorphic on $\Re s > 0$ by the previous convergence proofs, so it holds throughout that half-plane. The preceding inverse form retains the information even at a zero of $W_h(s-1)$; the cross-multiplied relation alone is not being asserted to reconstruct information lost by multiplying by a vanishing scalar.

The actual concentrating circle and its recoverable arithmetic image

In the source's fixed inner similarity region all annular corrections vanish. The remaining cutoff-summed slow field is axisymmetric, so the swirl estimate on the positive x_1 axis in (10.21) holds on the full circle with the same r, z, t. The source-read receipt records the corresponding construction locators. Hence at $\sigma = \nu X_{\mathrm{in}}\tau, z = 0, t = 1-\tau$ the actual average is

$$B(\nu X_{\mathrm{in}}\tau, 0, 1-\tau) = \nu\sqrt{2X_{\mathrm{in}}}\,\tau^{-h}(e_0 + R_*(\tau)). \tag{452}$$

The factor follows from multiplying the full retained swirl in (437) by $r = \sqrt{2\nu X_{\mathrm{in}}\tau}$; the physical angular average of an axisymmetric value leaves it unchanged. In particular this actual scalar angular momentum also diverges, now at the source exponent h, with its original remainder.

Applying the proved inverse series to $\mathcal{A}_h B(\cdot, 0, 1-\tau)$ recovers exactly the value in (452); it is therefore a source-specific arithmetic image of the concentrating corrected field. The double dyadic inverse and the finite-at-each-argument arithmetic inverse retain their stated order of evaluation. No uniform bound for those inverses as the sampling point tends to zero has been inferred, so the recovery identity alone is not a claim that the un-inverted trace has the same pointwise blowup rate.

For completeness the resulting zeta factor also has a precise local zero map. On $h < \Re s < 1$, both dilation factors are nonzero. Writing $\zeta(s) = (s-s_0)^m a_0(s)$ at a zeta zero gives

$$\mathfrak{B}(s, z, t) = (s-s_0)^m a_0(s)(1-2^{s-1})(1-2^{h-s})\widehat{B}(s, z, t).$$

This is the complete product, including the corrected-field Mellin factor. If that factor has order n at s_0, the product has order $m+n$, by multiplication of the two Taylor series; if it is nonzero, the multiplicity is exactly m. Identical statements hold for each flux and force moment. The arithmetic map and inverse are thus exact for the nonlinear witness, while no off-critical zeta zero or nonvanishing of the actual corrected-field Mellin factor is assumed.

An exact divergence-free arithmetic heat correspondence

The subsequent full proof *Exact angular-momentum composition and growth in the retained complement*, whose complete received source is retained with this edition proves that this note's six-channel image has zero physical averaged swirl. Applied to the same actual source witness, it locates the reported swirl supremum growth in the retained complement. The entire coordinate construction below is preserved; the later proof strengthens the component-specific conclusion and supplies an additional exact toroidal heat channel.

This note extends the explicit two-angular-channel correspondence received from the RH arithmetic task. The received note reproduces the complete underlying coordinate map below; the immutable source snapshot and its full independent review are retained under `sources/solenoidal_correspondence_009/`. Its original map, Fourier convention and positive viscosity are retained below. The additional construction supplies a divergence-free range, a full inverse on that range, the complementary velocity, and the complete nonlinear equations in those coordinates. It proves operator and solution-coordinate identities; it does not construct a singular Navier–Stokes solution or establish a statement about zeros of a zeta function.

1. Spaces, coordinates and the original map

Fix the original viscosity $\nu > 0$, and put

$$\lambda_\nu = 2\sqrt{\nu}, \qquad a_\nu = \sqrt{2}\,\nu^{1/4}. \tag{1}$$

Let $H = L^2(\mathbb{R}, ds; \mathbb{C}^3)$, $V = L^2(\mathbb{R}^3, dx; \mathbb{C}^3)$, and let $\widehat{V}$ denote the distinct Fourier-coordinate copy with measure $d\xi$. Inner products are linear in the first argument. The Fourier convention is

$$(\mathcal{F}_3 v)(\xi) = (2\pi)^{-3/2} \int_{\mathbb{R}^3} e^{-ix\cdot\xi} v(x)\, dx. \tag{2}$$

We use its Plancherel extension. Thus $\mathcal{F}_3 : V \to \widehat{V}$ is unitary, and $\mathcal{F}_3(\partial_{x_j} v) = i\xi_j \mathcal{F}_3 v$ on the corresponding derivative domain. The Fourier foundation is the same one used in the incoming note; no assertion of a new proof of Plancherel is made here.

For $\xi = \rho\omega$, $\rho > 0$, $\omega \in S^2$, define

$$Y_0(\omega) = (4\pi)^{-1/2}, \qquad Y_1(\omega) = \sqrt{3/(4\pi)}\,\omega_3. \tag{3}$$

The incoming map, now applied to vectors component by component, is

$$(J_\nu h)(\rho\omega) = \frac{a_\nu}{\rho}\left[Y_0(\omega) h(\lambda_\nu \rho) + Y_1(\omega) h(-\lambda_\nu \rho)\right]. \tag{4}$$

The value at $\xi = 0$ is immaterial. Since Y_0, Y_1 are orthonormal on the sphere and $a_\nu^2 = \lambda_\nu$, polar integration gives

$$\|J_\nu h\|_{\widehat{V}}^2 = a_\nu^2 \int_0^\infty \left(|h(\lambda_\nu\rho)|^2 + |h(-\lambda_\nu\rho)|^2\right) d\rho = \|h\|_H^2. \tag{5}$$

Its adjoint, for vector angular coefficients $g_j(\rho) = \int_{S^2} Y_j(\omega) g(\rho\omega)\, d\omega$, is

$$(J_\nu^* g)(s) = \begin{cases} \rho g_0(\rho)/a_\nu, & s > 0, \quad \rho = s/\lambda_\nu, \\ \rho g_1(\rho)/a_\nu, & s < 0, \quad \rho = -s/\lambda_\nu. \end{cases} \tag{6}$$

Indeed the inner product in (4) has factor $a_\nu \rho\, d\rho$, whereas substituting (6) into the H inner product has factor $\lambda_\nu \rho / a_\nu\, d\rho$; these factors agree by (1). This proves the adjoint formula for the full vector spaces.

The physical divergence-free space and its Fourier image are

$$V_\sigma = \{v \in V : \operatorname{div}_x v = 0 \text{ in distributions}\}, \qquad \widehat{V}_\sigma = \{g \in \widehat{V} : \xi \cdot g(\xi) = 0 \text{ a.e.}\}. \tag{7}$$

These are mapped onto each other by $\mathcal{F}_3$: testing the distributional divergence against Schwartz functions and applying (2) gives the multiplier $i\xi \cdot g$. Its vanishing as a distribution is equivalent to its almost-everywhere vanishing, since $\xi \cdot g$ is locally integrable. In particular, $\widehat{V}_\sigma$ is not the kernel of differentiation with respect to ξ.

2. The original range and incompressibility

There is an exact statement

$$\operatorname{ran}(J_\nu) \cap \widehat{V}_\sigma = \{0\}. \tag{8}$$

To prove it, fix a radius for which the transversality equation holds for almost every angular coordinate. Multiplying the angular coefficients by their nonzero fixed constants reduces the equation, without changing the field, to

$$\omega \cdot b + \omega_3\, \omega \cdot c = 0, \qquad b = (4\pi)^{-1/2} h(\lambda_\nu \rho), \quad c = \sqrt{3/(4\pi)}\, h(-\lambda_\nu \rho). \tag{9}$$

The left side is continuous on the sphere; an almost-everywhere zero continuous function is zero everywhere. Evaluating at ω and $-\omega$ gives separately $\omega \cdot b = 0$ and $\omega_3 \omega \cdot c = 0$. Evaluation at the three coordinate unit vectors proves $b = 0$. For c, evaluation at e_3, $(e_1 + e_3)/\sqrt{2}$ and $(e_2 + e_3)/\sqrt{2}$ proves $c_3 = c_1 = c_2 = 0$. Hence both half-line values of h vanish almost everywhere, proving (8).

Equation (8) describes this particular componentwise range. The following explicit map places the same arithmetic space isometrically inside the divergence-free space.

3. Leray projection and its complete angular Gram matrix

For $\omega \in S^2$, set

$$L(\omega) = I - \omega\omega^{\mathsf{T}}, \qquad (Lg)(\rho\omega) = L(\omega) g(\rho\omega). \tag{10}$$

The coordinate identities $L^{\mathsf{T}} = L$, $L^2 = L$, and $\omega^{\mathsf{T}} L = 0$ prove that L is the orthogonal projection $\widehat{V} \to \widehat{V}_\sigma$. The physical projection is $\mathbb{P} = \mathcal{F}_3^{-1} L \mathcal{F}_3$. Values assigned at the origin do not affect these operators.

The sphere integrals needed for the entire calculation are

$$\begin{aligned} \int_{S^2} \omega_i \omega_j \, d\omega &= \frac{4\pi}{3} \delta_{ij}, \\ \int_{S^2} \omega_i \omega_j \omega_3^2 \, d\omega &= \frac{4\pi}{15} (\delta_{ij} + 2\delta_{i3}\delta_{j3}). \end{aligned} \tag{11}$$

Off-diagonal integrals vanish by reflection in one coordinate. Equality of the three second moments and their sum 4π gives the first line. In polar coordinates about the third axis, $d\omega = d\varphi\, dz$, $-1 < z < 1$, so $\int \omega_3^4 = 2\pi \int_{-1}^{1} z^4\, dz = 4\pi/5$. The other two fourth moments in (11) are equal; their sum is $\int \omega_3^2 (1 - \omega_3^2) = 4\pi/3 - 4\pi/5$. Each is therefore $4\pi/15$, as displayed.

For arbitrary $b, c \in \mathbb{C}^3$, equations (3), (10), and (11) give

$$\begin{aligned} \int_{S^2} |L(\omega)[Y_0 b + Y_1 c]|^2 \, d\omega = \frac{2}{3}(|b_1|^2 + |b_2|^2 + |b_3|^2) \\ + \frac{4}{5}(|c_1|^2 + |c_2|^2) + \frac{2}{5}|c_3|^2. \end{aligned} \tag{12}$$

For completeness, $|Lz|^2 = |z|^2 - |\omega \cdot z|^2$. The Y_0 contribution to the removed term is $|b|^2/3$; the Y_1 contribution is $(|c_1|^2 + |c_2|^2 + 3|c_3|^2)/5$. The cross term integrates to zero under $\omega \mapsto -\omega$.

Subtracting these displayed quantities from $|b|^2 + |c|^2$ proves (12), including all complex cross terms. Equivalently, the complete operator $G = J_\nu^* L J_\nu$ is

$$(Gh)(s) = \begin{cases} \frac{2}{3}h(s), & s > 0, \\ \operatorname{diag}(\frac{4}{5}, \frac{4}{5}, \frac{2}{5})h(s), & s < 0. \end{cases} \tag{13}$$

The off-diagonal entries vanish by the same reflection calculations; thus (13) follows either entry by entry or by the polarization identity applied to (12) and (5).

Define the explicit, bounded, invertible, self-adjoint arithmetic-coordinate operator

$$(Dh)(s) = \begin{cases} \sqrt{3/2}\, h(s), & s > 0, \\ \operatorname{diag}(\sqrt{5}/2, \sqrt{5}/2, \sqrt{5/2})h(s), & s < 0. \end{cases} \tag{14}$$

Its inverse has positive branch $\sqrt{2/3}I$ and negative branch $\operatorname{diag}(2/\sqrt{5}, 2/\sqrt{5}, \sqrt{2/5})$. No viscosity, spatial coordinate, time, or source field has been replaced. Equations (13)–(14) give $DGD = I$ and $G = D^{-2}$ with their literal factors.

4. The isometry, its range, inverse and reality

Set

$$K_\nu = \mathcal{F}_3^{-1} L J_\nu D : H \longrightarrow V_\sigma. \tag{15}$$

Its adjoint as an operator into the full space V is

$$K_\nu^* = D J_\nu^* L \mathcal{F}_3 : V \longrightarrow H. \tag{16}$$

Every factor is bounded, so the adjoint of the product follows directly by successive inner-product identities. Equations (10) and (13) prove

$$K_\nu^* K_\nu = D J_\nu^* L^2 J_\nu D = DGD = I, \qquad \|K_\nu h\|_V = \|h\|_H. \tag{17}$$

Consequently the range $W_\nu = \operatorname{ran}(K_\nu)$ is closed, its inverse is the restriction of (16), and $P_\nu = K_\nu K_\nu^*$ is its orthogonal projection on all of V. Its kernel is exactly $W_\nu^\perp$.

Here is an explicit coordinate description of the range and inverse. Define six real vector functions on the sphere by

$$\begin{aligned} U_j^+(\omega) &= \sqrt{3/2}\, Y_0(\omega) L(\omega) e_j, \quad j = 1, 2, 3, \\ U_j^-(\omega) &= d_j Y_1(\omega) L(\omega) e_j, \quad (d_1, d_2, d_3) = (\sqrt{5}/2, \sqrt{5}/2, \sqrt{5/2}). \end{aligned} \tag{18}$$

Their pairwise inner products are δ_{ij} within each sign and zero between signs by (12)–(14). All six satisfy $\omega \cdot U_j^\pm = 0$. Formula (15) becomes

$$\widehat{K_\nu h}(\rho\omega) = \frac{a_\nu}{\rho} \sum_{j=1}^{3} \left[U_j^+(\omega) h_j(\lambda_\nu \rho) + U_j^-(\omega) h_j(-\lambda_\nu \rho) \right]. \tag{19}$$

Thus W_ν consists exactly of physical fields whose Fourier angular functions belong to the span of these six functions at almost every radius. Arbitrary square-integrable radial coefficients are allowed. The inverse radial substitution in (19) proves this asserted surjectivity onto the described range, using $a_\nu^2 = \lambda_\nu$ to verify integrability. In particular this subspace, though its parametrization depends on ν, is the same angular subspace for every positive ν.

For every $v \in V$, (16) has the component formula

$$(K_\nu^* v)_j(s) = \begin{cases} \dfrac{\rho}{a_\nu} \displaystyle\int_{S^2} U_j^+(\omega) \cdot \widehat{v}(\rho\omega)\, d\omega, & s > 0, \\ \dfrac{\rho}{a_\nu} \displaystyle\int_{S^2} U_j^-(\omega) \cdot \widehat{v}(\rho\omega)\, d\omega, & s < 0, \end{cases} \qquad \rho = |s|/\lambda_\nu. \tag{20}$$

The six functions are real, so no conjugates have been suppressed in (20). The kernel consists exactly of fields whose six displayed angular coefficients vanish almost everywhere. On the Fourier side P_ν is the orthogonal projection onto these six angular functions at each radius.

The reality condition is also exact. Put

$$(C_H h)(s) = \begin{cases} \overline{h(s)}, & s > 0, \\ -\overline{h(s)}, & s < 0. \end{cases} \tag{21}$$

Since $U_j^+(-\omega) = U_j^+(\omega)$ and $U_j^-(-\omega) = -U_j^-(\omega)$, (19) and the identity $\widehat{\overline{v}}(\xi) = \overline{\widehat{v}(-\xi)}$ prove

$$K_\nu C_H h = \overline{K_\nu h}. \tag{22}$$

Injectivity of K_ν then proves that its physical velocity is real if and only if $h(s)$ is real for almost every $s > 0$ and purely imaginary for almost every $s < 0$. Thus the real physical state space is retained rather than silently identified with all complex arithmetic coordinates.

5. The full heat-generator domains

Define

$$\begin{aligned} (Ah)(s) = s^2 h(s)/4, \quad & \mathrm{Dom}(A) = \{h \in H : \int_{\mathbb{R}} s^4 |h(s)|^2/16 \, ds < \infty\}, \\ B_\nu v = -\nu \Delta v, \quad & \mathrm{Dom}(B_\nu) = \{v \in V : \int_{\mathbb{R}^3} \nu^2 |\xi|^4 |\widehat{v}(\xi)|^2 \, d\xi < \infty\}. \end{aligned} \tag{23}$$

Both are multiplication operators in the indicated coordinates. The pointwise equality

$$\nu\rho^2 = (\lambda_\nu \rho)^2/4 = (-\lambda_\nu \rho)^2/4 \tag{24}$$

commutes with every angular factor in (19). The polar norm calculation with an additional factor $(\nu\rho^2)^2$ proves, without an unbounded-operator interchange,

$$K_\nu h \in \mathrm{Dom}(B_\nu) \iff h \in \mathrm{Dom}(A), \quad B_\nu K_\nu h = K_\nu A h, \quad \|B_\nu K_\nu h\|_V = \|Ah\|_H. \tag{25}$$

Since the six-channel projection acts at each radius, P_ν commutes with the radial multipliers and preserves their domains. For $v \in \mathrm{Dom}(B_\nu)$, (20) or the projected norm bound therefore gives $K_\nu^* v \in \mathrm{Dom}(A)$ and $K_\nu^* B_\nu v = A K_\nu^* v$. The identical argument proves these statements for every nonnegative integer power, retaining the factors $(\nu\rho^2)^m$ and $s^{2m}/4^m$.

For all $t \geq 0$, pointwise multiplication further gives

$$e^{-tB_\nu} K_\nu = K_\nu e^{-tA}, \qquad K_\nu^* e^{-tB_\nu} = e^{-tA} K_\nu^*. \tag{26}$$

Here the exponentials are respectively the explicit bounded multipliers $e^{-t\nu|\xi|^2}$ and $e^{-ts^2/4}$; dominated convergence proves their strong continuity. In the incoming arithmetic convention $W_z = e^{zA}$, physical forward time remains $z = -t$. No positive-time backward heat operator is asserted bounded on H.

6. An invertible decomposition of every divergence-free velocity

Let $V_{\sigma,c} = V_\sigma \cap W_\nu^\perp$. The map

$$\mathcal{T}_\nu : V_\sigma \longrightarrow H \oplus V_{\sigma,c}, \qquad \mathcal{T}_\nu u = (h, w) = (K_\nu^* u, (I - P_\nu)u) \tag{27}$$

is onto and one-to-one, with

$$\mathcal{T}_\nu^{-1}(h, w) = K_\nu h + w, \qquad \|u\|_V^2 = \|h\|_H^2 + \|w\|_V^2. \tag{28}$$

Indeed $P_\nu u \in V_\sigma$, so $w \in V_\sigma$ and is perpendicular to W_ν. Conversely $K_\nu^* w = 0$, $P_\nu K_\nu h = K_\nu h$, and $P_\nu w = 0$; substitution proves both inverse identities and the orthogonal norm equality. Complex conjugation preserves both subspaces by (22) and the reality of the angular projection. Therefore the real version of (27)–(28) is obtained exactly by imposing (21) on h and ordinary physical reality on w.

The operators are nonlocal in physical space. Formula (28) retains any compact support of the original u as a property of the sum; it does not assert compact support of its two separate summands. Their smoothness on a compact time interval follows from (25) for every power whenever u and its time derivatives belong to every corresponding Sobolev domain. One can verify the required continuous derivative representatives directly: if $v \in \mathrm{Dom}(B_\nu^m)$ for every m, Cauchy–Schwarz makes $|\xi|^k \widehat{v}$ integrable by choosing $2m > k + 3/2$; its inverse Fourier integral is then a bounded continuous derivative of order k. Thus no spatial support or smoothness has been silently transferred to a different norm.

7. All nonlinear, pressure and forcing terms

We retain the physical variables (x, t, u, p, f, ν) of the incoming source. On a fixed compact time interval before its endpoint, smooth fields with a fixed compact spatial support have all the derivative and integrability properties used below. Write the convection components in full as

$$N_i(u) = \sum_{j=1}^{3} u_j \partial_{x_j} u_i, \qquad i = 1, 2, 3. \tag{29}$$

For the actual divergence-free velocity, define (h, w) by (27). The complete component expansion is

$$\begin{aligned} N_i(K_\nu h + w) = \sum_{j=1}^{3} \big[&(K_\nu h)_j \partial_{x_j} (K_\nu h)_i + (K_\nu h)_j \partial_{x_j} w_i \\ &+ w_j \partial_{x_j} (K_\nu h)_i + w_j \partial_{x_j} w_i \big]. \end{aligned} \tag{30}$$

Set $Q_\nu = I - P_\nu$, and define the physical solenoidal source

$$F_\sigma = \mathbb{P}[f - N(K_\nu h + w)]. \tag{31}$$

Application of $\mathbb{P}$ to the physical equation, followed by (25) and (27), gives exactly

$$\begin{aligned} \partial_t h + Ah &= K_\nu^* F_\sigma, \\ \partial_t w + B_\nu w &= Q_\nu F_\sigma, \\ h(0) = K_\nu^* u(0), &\qquad w(0) = Q_\nu u(0). \end{aligned} \tag{32}$$

The pressure is retained by the additional exact equation

$$\nabla p = (I - \mathbb{P})[f - N(K_\nu h + w)]. \tag{33}$$

To justify the pressure step explicitly, Fourier transformation of $\nabla p \in L^2$ gives a vector parallel to ξ, so $L\widehat{\nabla p} = 0$. The vectors $\partial_t u$ and $B_\nu u$ remain solenoidal by differentiation of (7). Consequently projecting the original equation onto the orthogonal complement of V_σ gives (33). Thus no condition on the divergence of the original force has been added.

These calculations are reversible. For fields with the displayed smoothness and domain properties, the initial conditions in (32), the constraint $w(t) \in V_{\sigma,c}$, and equation (33), apply K_ν to the first equation of (32) and add the second. By (25) and $P_\nu + Q_\nu = I$, the result is

$$\partial_t (K_\nu h + w) + B_\nu (K_\nu h + w) = \mathbb{P}[f - N(K_\nu h + w)]. \tag{34}$$

Adding (33), with its actual gradient and sign, proves

$$\partial_t u + N(u) - \nu\Delta u + \nabla p = f, \qquad \operatorname{div}_x u = 0, \quad u = K_\nu h + w. \tag{35}$$

Equations (27) and (28) prove exact inverse maps on this solution data. The additive spatially constant value of p, if allowed by the chosen physical pressure class, is retained as part of the data; no inverse of the gradient that would discard it has been claimed.

The associated integral equations retain all the same terms:

$$\begin{aligned} h(t) &= e^{-tA}h(0) + \int_0^t e^{-(t-s)A}K_\nu^* F_\sigma(s)\,ds, \\ w(t) &= e^{-tB_\nu}w(0) + \int_0^t e^{-(t-s)B_\nu}Q_\nu F_\sigma(s)\,ds. \end{aligned} \tag{36}$$

On each stated compact time interval, F_σ is continuous in the Hilbert space. The contractions in (26) make these Bochner integrals well defined. Taking their Fourier multipliers gives the scalar integrating-factor identity at almost every frequency. Differentiation on the actual smooth domain, or integration of (32) with this scalar identity, proves both directions between (32) and (36). Equations (30)–(36) show explicitly where the angular complement re-enters the arithmetic equation. Dropping w or any term of (30) would change the dynamics; no such deletion is made.

8. Exact energy and a concrete supremum-norm test

Equations (27)–(28) prove for every time at which the physical velocity is in L^2

$$\frac{1}{2}\int_{\mathbb{R}^3} |u(x,t)|^2\,dx = \frac{1}{2}\int_{\mathbb{R}} |h(s,t)|^2\,ds + \frac{1}{2}\int_{\mathbb{R}^3} |w(x,t)|^2\,dx. \tag{37}$$

For each derivative order, the analogous radial generator identity is

$$\|B_\nu^m u\|_V^2 = \|A^m h\|_H^2 + \|B_\nu^m w\|_V^2, \tag{38}$$

on exactly the domains in (25). This follows from commutation of the angular projection with the radial multiplier and the same orthogonality.

Even the six-channel divergence-free range has no bounded map from its L^2 norm to its physical supremum norm. Here is a full explicit calculation. Choose a real nonnegative nonzero $\varphi \in C_c^\infty((1,2))$. For $R > 0$, put

$$h_R(s) = \frac{R^{-1/2}}{\|\varphi\|_{L^2(\mathbb{R})}}\varphi(s/R)e_3, \qquad v_R = K_\nu h_R. \tag{39}$$

Its negative half-line values vanish, and substitution $s = Rr$ proves $\|h_R\|_H = \|v_R\|_V = 1$. Formula (22) proves that v_R is real. Its Fourier transform is smooth and compactly supported in the annulus $R/\lambda_\nu < \rho < 2R/\lambda_\nu$, hence v_R is a smooth rapidly decreasing divergence-free field. Repeated integration by parts in the inverse Fourier integral proves the asserted rapid decrease; the frequency support is away from the origin and all derivatives vanish at its boundary.

The exact angular integral is

$$\int_{S^2} U_3^+(\omega)\,d\omega = \sqrt{3/2}\,(4\pi)^{-1/2}\frac{8\pi}{3}e_3, \tag{40}$$

because $\int L(\omega)e_3 = 4\pi e_3 - (4\pi/3)e_3$. Evaluating the absolutely convergent inverse Fourier integral at $x = 0$, including its polar factor ρ^2, now gives

$$\begin{aligned} v_R(0) &= C_{\nu,\varphi} R^{3/2} e_3, \\ C_{\nu,\varphi} &= (2\pi)^{-3/2} \frac{a_\nu}{\lambda_\nu^2} \sqrt{3/2}\, (4\pi)^{-1/2} \frac{8\pi}{3} \frac{\int_1^2 r\varphi(r)\, dr}{\|\varphi\|_{L^2(\mathbb{R})}} > 0. \end{aligned} \tag{41}$$

Indeed $\int_0^\infty \rho\varphi(\lambda_\nu \rho/R)\, d\rho = (R^2/\lambda_\nu^2) \int_1^2 r\varphi(r)\, dr$, which accounts for every factor in (41). Therefore $\|v_R\|_\infty \geq C_{\nu,\varphi} R^{3/2}$ with $\|v_R\|_2 = 1$.

This is a family of states with an exact inverse and norm calculation. The parameter R is not physical time, and no trajectory through this family solving (35) has been asserted. It proves precisely that bounded arithmetic Hilbert norm does not control physical velocity supremum, including inside this divergence-free image. The full dynamics relevant to a particular trajectory remain (30)–(36).

9. Application to the actual source witness, with its existence dependency retained

The frozen 166-page manuscript is `sources/ns_public/navier_stokes_166.pdf`, SHA256 `0e779481c4da40bd28d1e642e1d8ca57447d129610df28dfa5a11e9af8ae228f`. Its equations (9.21), (10.4), (10.5) and (10.11) define the cutoff-summed, localized velocity, pressure and smoothly extended force. Denote that viscosity-one triple by (u_*, p_*, f_*). Its stated existence and preceding increment estimates are imported source results here. The correspondence calculation in this section does not replace that construction's proof.

For the given physical viscosity $\nu > 0$, retain the source's spatial map and time:

$$u_\nu(x,t) = \sqrt{\nu}\, u_*(x/\sqrt{\nu}, t), \qquad p_\nu(x,t) = \nu p_*(x/\sqrt{\nu}, t), \qquad f_\nu(x,t) = \sqrt{\nu}\, f_*(x/\sqrt{\nu}, t). \tag{42}$$

Every term in $\partial_t u_\nu + (u_\nu \cdot \nabla_x)u_\nu - \nu\Delta_x u_\nu + \nabla_x p_\nu$ equals $\sqrt{\nu}$ times the corresponding viscosity-one term evaluated at $x/\sqrt{\nu}$. For example $\partial_{x_j}(u_\nu)_i = (\partial_{y_j} u_{*,i})(y,t)$, $\Delta_x(u_\nu)_i = \nu^{-1/2}\Delta_y u_{*,i}$, and $\partial_{x_i} p_\nu = \sqrt{\nu}\partial_{y_i} p_*$, where $y = x/\sqrt{\nu}$. This proves the exact residual and force relation, including zero initial data, divergence, unchanged endpoint time, support $\sqrt{\nu}K_*$, and the inverse obtained by $x = \sqrt{\nu}y$. The volume factor is $dx = \nu^{3/2}dy$, giving

$$\|u_\nu(t)\|_2^2 = \nu^{5/2}\|u_*(t)\|_2^2, \qquad \|u_\nu(t)\|_\infty = \sqrt{\nu}\|u_*(t)\|_\infty. \tag{43}$$

Apply the proved inverse pair (27)–(28) to these actual source fields, at each $0 \leq t < 1$:

$$h_\nu(t) = K_\nu^* u_\nu(t), \qquad w_\nu(t) = (I - P_\nu)u_\nu(t), \qquad u_\nu(t) = K_\nu h_\nu(t) + w_\nu(t). \tag{44}$$

The complete projected equations are (30)–(36) with $(h, w, p, f) = (h_\nu, w_\nu, p_\nu, f_\nu)$, and both initial coordinates are zero. Thus their forcing and nonlinear terms are the actual ones defined by (42), rather than an independently chosen input. Formula (37) and the source's bounded energy give

$$\|h_\nu(t)\|_H^2 + \|w_\nu(t)\|_V^2 = \nu^{5/2}\|u_*(t)\|_2^2. \tag{45}$$

The source's (10.20)–(10.21) retain $A = 1/2 + h$, $0 < h < 1/100$, the original $X_{\mathrm{in}} \in (0, X_a)$, $e_0 = E_0(X_{\mathrm{in}}, 0) > 0$, and an actual remainder $R_*(\tau) = O(\tau^{2h})$. Its positive inner-circle swirl has Cartesian second component at

$$x_{\nu,\tau} = (\sqrt{2\nu X_{\mathrm{in}}\tau}, 0, 0), \qquad t = 1 - \tau. \tag{46}$$

Here is the full source interval on which the localized-field identity is used. In (10.4), the original spatial cutoff is one for $r \leq r_0/2$, $|z| \leq z_0/2$, and the original time cutoff is one for

$0 \le 1 - t \le \tau_0/2$, with $r_0, z_0, \tau_0 > 0$. The source local domain is $q < q_*$. Choose the actual remainder constants $\tau_{\mathrm{rem}} > 0$, $C_* \ge 0$ supplied by (3.6), so $|R_*(\tau)| \le C_* \tau^{2h}$ on $0 < \tau < \tau_{\mathrm{rem}}$, and retain

$$\tau_* := \min\left\{\tau_{\mathrm{rem}}, \frac{\tau_0}{2}, \frac{r_0^2}{8X_{\mathrm{in}}}, q_*\right\} > 0. \tag{46a}$$

For $0 < \tau < \tau_*$, the unscaled point $x_{\nu,\tau}/\sqrt{\nu}$ has $r = \sqrt{2X_{\mathrm{in}}\tau} < r_0/2$, $z = 0$, $q = \tau < q_*$, and time remaining $\tau < \tau_0/2$. Both cutoffs equal one in a neighborhood of this point. Thus their spatial derivatives vanish there, and the full curl localization $\mathrm{curl}(cA) + cBe_\theta$ equals the local velocity, including the fact that $\nabla c \times A = 0$ there. The original annular corrections vanish because $X = X_{\mathrm{in}} < X_a$. The positive-order base cutoff sum remains in $R_*(\tau)$. At angular coordinate zero the cylindrical vector e_θ is the positive Cartesian second basis vector. These facts prove the exact point, component, and time-domain correspondence without imposing a viscosity-dependent restriction on τ_*.

Substitution in the exact inverse (44), retaining the whole remainder, gives the concrete arithmetic-and-complement value

$$\big[K_\nu h_\nu(1-\tau) + w_\nu(1-\tau)\big]_2(x_{\nu,\tau}) = \sqrt{\nu}\,\tau^{-1/2-h}\big(e_0 + R_*(\tau)\big). \tag{47}$$

The pointwise evaluation is justified on every preterminal time by the actual smooth source and the all-order domain argument following (28); the specific source equality (47) is asserted on $0 < \tau < \tau_*$ from (46a). For $C_* > 0$ and $0 < \tau < \min\{\tau_*, (e_0/(2C_*))^{1/(2h)}\}$, equation (47) is at least $\sqrt{\nu} e_0 \tau^{-1/2-h}/2$; if $C_* = 0$, the same inequality holds throughout $(0, \tau_*)$. The triangle inequality therefore yields

$$\|K_\nu h_\nu(1-\tau)\|_\infty + \|w_\nu(1-\tau)\|_\infty \ge \frac{\sqrt{\nu} e_0}{2}\tau^{-1/2-h}. \tag{48}$$

Equations (45), (47) and (48) specify the actual transported bounded-energy and supremum-growth statements. These equations alone do not assign the growth to either angular component separately. The subsequent complete angular-momentum composition proof linked above establishes zero averaged swirl of this six-channel image and proves that the source's reported swirl supremum growth is carried by the retained complement. No zeta-zero assertion follows from these coordinate identities. The source existence dependency and the retained angular complement are explicit throughout the application.

10. Provenance and verification scope

The incoming RH note supplies (1)–(6), the multiplier $s^2/4$, and its scalar forced-heat interpretation. Sections 2–9 above add the typed transversality result, the Leray angular Gram calculation, six-channel divergence-free isometry, exact complement, reality, full nonlinear and pressure equations, the explicit fixed-energy supremum-norm family, and the application to the actual source witness. The projection and Fourier foundations are stated in their actual spaces; the finite-dimensional, integral, domain, and dynamical calculations are displayed in full.

An independent bounded reviewer checked the Gram factors, operator adjoint, isometry and reality calculation. The reviewer required one type clarification before writing: the spectral constraint is $\xi \cdot g(\xi) = 0$, not differentiation of g with respect to ξ; (7) implements that correction. The review does not certify the incoming manuscript's Navier–Stokes construction or all prior arithmetic results. No Lean, Lake, Elan, source-download code, or cumulative publication build was executed for this note.

Final independent review of the solenoidal heat correspondence, Sections 6–9

Review completed on 2026-09-09. This is the previously unfinished second independent mathematical review. It covers the complete equations (27)–(48), including the new (46a), their required definitions and generator domains in Sections 1–5, and their composition with the complete later swirl note. The corrected source passes this bounded review. The correction made during review is the explicit original-source localization interval in (46a); its proof is checked below. No equation, forcing term, pressure term, viscosity factor, inverse, or energy factor in the previous Sections 6–9 required correction.

The reviewed original live-file hashes are retained below. The exact supplied snapshots and version mapping accompany this edition under `sources/solenoidal_correspondence_009/`:

- `solenoidal_arithmetic_heat_correspondence.md`.
  SHA-256 `0b727b10d97ccd2efbf7a34102784a103da4c44fa53950a4297c884a6a110fe3`.
- `solenoidal_swirl_composition.md`.
  SHA-256 `8e4856e1264735342ff14a6ad9833eb48285f5dfd2979c36e0dde75fc7b67dbf`.

The earlier heat-note snapshot read at the start of this review had SHA-256 `f9dcd233 7680cf586a8f3c03dd527d71f4bc9cb2284e490f616d04c36240b5cb`. The programme lane's author added (46a) and its surrounding domain proof after receiving the source-domain finding. The reviewer reread the resulting complete Section 9 and checked the resulting hash above. The existing proofs were not edited by this reviewer.

The primary source is the retained 166-page `sources/ns_public/navier_stokes_166.pdf`, SHA-256 `0e779481c4da40bd28d1e642e1d8ca57447d129610df28dfa5a11e9af8ae228f`. Its extracted text is the originally reviewed extraction, SHA-256 `f83414ea543ffd89910b9b8a 35d1680a2e3295246c91cbedf54e719a7bc316d0`. Direct source reading for this review covered Theorem 3.1, Proposition 9.9 Step 5, Proposition 10.1 and (10.4)–(10.5), Lemma 10.4 and (10.13)–(10.14), and (10.20)–(10.21). A separately delegated source-match check also inspected the source's original PDF pages. The bounded coordinate calculations below use the source witness with its existence explicitly imported; this report supplies no independent proof of all preceding increment estimates, no complete Navier–Stokes counterexample certificate, and no assertion about zeta zeros. Plancherel and the distributional Fourier foundation are the stated shared foundations of the notes.

1. Required operator identities and their types

Retain the original positive viscosity, measures, complex inner-product convention and Fourier convention:

$$\nu > 0, \quad a_\nu = \sqrt{2}\,\nu^{1/4}, \quad \lambda_\nu = 2\sqrt{\nu}, \quad a_\nu^2 = \lambda_\nu,$$

$$H = L^2(\mathbb{R}, ds; \mathbb{C}^3), \qquad V = L^2(\mathbb{R}^3, dx; \mathbb{C}^3), \qquad \widehat{v}(\xi) = (2\pi)^{-3/2} \int e^{-ix\cdot\xi} v(x)\, dx.$$

The Fourier variable is $\xi = \rho\omega$, with $\rho > 0$, $\omega \in S^2$, and measure $\rho^2 d\rho\, d\omega$. Inner products are linear in the first argument. The solenoidal projection has the literal multiplier

$$L(\omega) = I - \omega\omega^{\mathsf{T}}, \qquad \mathbb{P} = \mathcal{F}_3^{-1} L \mathcal{F}_3.$$

The equation defining V_σ is distributional divergence in physical coordinates, equivalent to $\xi \cdot \widehat{v}(\xi) = 0$ almost everywhere. It is not differentiation of $\widehat{v}$ with respect to ξ. The entries of L are bounded and real; direct multiplication gives $L^2 = L = L^*$.

The six original angular functions are

$$U_j^+ = \sqrt{3/2}\,(4\pi)^{-1/2} L e_j, \qquad U_j^- = d_j \sqrt{3/(4\pi)}\,\omega_3 L e_j, \quad (d_1, d_2, d_3) = (\sqrt{5}/2, \sqrt{5}/2, \sqrt{5/2}).$$

The underlying Gram matrices before these displayed diagonal factors are $(2/3)I$ on the positive branch and $\mathrm{diag}(4/5, 4/5, 2/5)$ on the negative branch; the cross-branch matrix is zero by antipodal parity. Multiplication by the displayed factors on both sides gives the identity matrices. Consequently all six functions are orthonormal in $L^2(S^2; \mathbb{C}^3)$, and $\omega \cdot U_j^{\pm} = 0$ entry by entry.

The resulting map and adjoint, with their full coefficients, are

$$\widehat{K_\nu h}(\rho\omega) = \frac{a_\nu}{\rho} \sum_{j=1}^{3} \left(U_j^+(\omega) h_j(\lambda_\nu \rho) + U_j^-(\omega) h_j(-\lambda_\nu \rho) \right), \tag{R1}$$

$$(K_\nu^* v)_j(s) = \frac{\rho}{a_\nu} \int_{S^2} U_j^{\mathrm{sgn}(s)}(\omega) \cdot \widehat{v}(\rho\omega)\, d\omega, \quad \rho = |s|/\lambda_\nu. \tag{R2}$$

For each half-line the adjoint computation compares the factors $a_\nu \rho\, d\rho$ and $\lambda_\nu \rho / a_\nu\, d\rho$; they agree. Orthonormality then gives

$$K_\nu^* K_\nu = I_H, \qquad \|K_\nu h\|_V^2 = a_\nu^2 \int_0^\infty (|h(\lambda_\nu \rho)|^2 + |h(-\lambda_\nu \rho)|^2) d\rho = \|h\|_H^2. \tag{R3}$$

Thus $W_\nu = K_\nu H$ is closed, $P_\nu = K_\nu K_\nu^*$ is its orthogonal projection on the full V, and $Q_\nu = I - P_\nu$. At each radius P_ν projects onto exactly these six angular functions. This description also proves that P_ν is the same operator for all positive ν, although the parametrization K_ν retains its viscosity dependence. Since the six angular functions are tangent,

$$\mathbb{P} P_\nu = P_\nu \mathbb{P} = P_\nu, \qquad K_\nu^* \mathbb{P} = K_\nu^*. \tag{R4}$$

The second product identity in the first pair follows either by taking adjoints of the first one or by the same angular calculation.

Now $V_{\sigma,c} = V_\sigma \cap W_\nu^\perp$ has the original Hilbert norm. For every $u \in V_\sigma$, set $h = K_\nu^* u$, $w = Q_\nu u$. Equation (R4) proves $w \in V_\sigma$; projection orthogonality proves $w \perp W_\nu$. Conversely, for each $(h, w) \in H \oplus V_{\sigma,c}$,

$$K_\nu^*(K_\nu h + w) = h, \qquad Q_\nu(K_\nu h + w) = w, \qquad K_\nu K_\nu^* u + Q_\nu u = u. \tag{R5}$$

These are both inverse identities claimed in (27)–(28), on the stated whole spaces.

Antipodal parity is $U_j^+(-\omega) = U_j^+(\omega)$ and $U_j^-(-\omega) = -U_j^-(\omega)$. Substituting into (R1), together with $\widehat{\overline{v}}(\xi) = \overline{\widehat{v}(-\xi)}$, gives $K_\nu C_H h = \overline{K_\nu h}$, with C_H exactly as in (21). The real coordinate subspace is precisely $C_H h = h$, namely real values on the positive half-line and purely imaginary values on the negative half-line; w must be real in physical space. Injectivity in (R3) proves necessity as well as sufficiency. Therefore the real source is preserved by the full inverse pair.

2. Generator domains and the regularity actually used

Write the two original positive generators as

$$Ah = s^2 h/4, \qquad B_\nu v = -\nu \Delta v,$$

with their weighted multiplication domains in (23). For every integer $m \geq 0$, direct substitution into the polar integral gives the exact equality

$$\int_{\mathbb{R}^3} (\nu|\xi|^2)^{2m} |\widehat{K_\nu h}(\xi)|^2 d\xi = \int_{\mathbb{R}} (s^2/4)^{2m} |h(s)|^2 ds. \tag{R6}$$

Indeed $\nu\rho^2 = (\lambda_\nu\rho)^2/4 = (-\lambda_\nu\rho)^2/4$ at each radius, and every angular factor remains unchanged. Equality of these nonnegative integrals proves equivalence of the domains, including when the integrals are infinite. On the finite domains it gives

$$B_\nu^m K_\nu h = K_\nu A^m h, \qquad K_\nu^* B_\nu^m v = A^m K_\nu^* v. \tag{R7}$$

For the second identity, angular projection commutes with the scalar radial multiplier, and its norm at each radius is at most one. This also proves that $P_\nu, Q_\nu, \mathbb{P}$ preserve all these domains and commute with the generators there. No commutation of P_ν with an individual Cartesian derivative is needed or asserted: (38) is the radial generator identity for integer m.

Here is an explicit regularity estimate behind the post-(28) paragraph. For an integer derivative order $k \geq 0$, choose an integer m with $4m > 2k + 3$. Cauchy–Schwarz with the original measure gives

$$\int_{\mathbb{R}^3} |\xi|^k |\widehat{v}(\xi)|\, d\xi \leq I_{\nu,k,m}^{1/2} \big(\|v\|_V^2 + \|B_\nu^m v\|_V^2\big)^{1/2}, \tag{R8}$$

$$I_{\nu,k,m} = 4\pi \int_0^\infty \frac{\rho^{2k+2}}{1 + \nu^{2m}\rho^{4m}} d\rho \leq 4\pi\nu^{-(2k+3)/2} \left(\frac{1}{2k+3} + \frac{1}{4m-2k-3}\right) < \infty. \tag{R9}$$

The inequality is obtained by splitting the original integral at $\rho = \nu^{-1/2}$. On the first part the denominator is at least one; on the second it is at least $\nu^{2m}\rho^{4m}$. Integrating both displayed powers gives exactly (R9). For every multi-index α with $|\alpha| = k$, the inverse Fourier integral with multiplier $(i\xi)^\alpha$ is absolutely convergent. Difference quotients, dominated by the next integrable power, show that it is the distributional derivative $\partial_x^\alpha v$; dominated convergence gives continuity. In particular

$$\|\partial_x^\alpha v\|_\infty \leq (2\pi)^{-3/2} I_{\nu,k,m}^{1/2} \big(\|v\|_V^2 + \|B_\nu^m v\|_V^2\big)^{1/2}. \tag{R10}$$

This proves the required smooth bounded representatives for every field in all the generator domains, with all viscosity and Fourier factors retained.

The time assertion uses continuity in these graph norms, not merely pointwise set membership without time control. The actual class in Section 7 has exactly that continuity. On a fixed interval $[0, T]$, $T < 1$, its original fields and all their time derivatives are smooth with fixed compact spatial support. Each Cartesian derivative is uniformly continuous and bounded on that compact support times $[0, T]$. Integration over the fixed compact set therefore makes every such time-dependent derivative continuous in L^2. The same argument for one further time derivative proves differentiation in L^2. Equation (R7) transports these graph-norm statements to $K_\nu h$ and w; (R10) then transports continuity and differentiation to the continuous spatial representatives. The two projected components need not be compactly supported. Their smoothness and the estimates used below follow from this argument, without such a support claim.

3. Complete convection, pressure, force and inverse equations

For two vector fields define the original ordered bilinear expression

$$N_i(a, b) = \sum_{j=1}^3 a_j \partial_{x_j} b_i, \qquad N(u) = N(u, u). \tag{R11}$$

For each spatial point, Cauchy–Schwarz in the three derivative indices yields

$$|N(a,b)|^2 \leq |a|^2 \sum_{i,j=1}^3 |\partial_{x_j} b_i|^2, \quad \|N(a,b)\|_2 \leq \|a\|_\infty \left(\sum_{i,j=1}^3 \|\partial_{x_j} b_i\|_2^2\right)^{1/2}. \tag{R12}$$

All four terms with a, b chosen from $K_\nu h, w$ consequently belong to L^2. The full differentiated expression is

$$\partial_x^\alpha N_i(a,b) = \sum_{j=1}^{3} \sum_{\beta \le \alpha} \binom{\alpha}{\beta} (\partial_x^\beta a_j)(\partial_x^{\alpha-\beta+e_j} b_i). \tag{R13}$$

Each term is an L^∞ factor times an L^2 factor by Section 2, with the literal binomial coefficient retained. The same finite product rule in time proves all required time and graph-norm continuity. This supplies the full nonlinear regularity used in (31) and (36), even though the summands are nonlocal fields.

Bilinearity gives, for each i, exactly

$$N_i(K_\nu h + w) = \sum_{j=1}^{3} \left[(K_\nu h)_j \partial_{x_j} (K_\nu h)_i + (K_\nu h)_j \partial_{x_j} w_i + w_j \partial_{x_j} (K_\nu h)_i + w_j \partial_{x_j} w_i \right]. \tag{R14}$$

There are four ordered contributions. None is removed by the fact that $w \perp W_\nu$, which is a linear Hilbert-space statement.

Let $g = f - N(K_\nu h + w)$. The original equation reads

$$\partial_t u + B_\nu u + \nabla p = g. \tag{R15}$$

Both $\partial_t u$ and $B_\nu u$ lie in V_σ: the former follows by differentiating the distributional divergence with the proved time regularity, and the latter follows from the multiplier $\nu|\xi|^2$. The actual pressure has $\nabla p \in L^2$. Its Fourier vector is longitudinal. This last fact can also be justified without assigning a Fourier transform to a pressure outside a tempered class: the L^2 vector $b = \nabla p$ satisfies $\partial_i b_j - \partial_j b_i = 0$; Fourier transformation of this identity gives $\xi_i \widehat{b}_j - \xi_j \widehat{b}_i = 0$. These products are locally integrable, so the identities hold almost everywhere. At each nonzero ξ, choose an index with $\xi_i \neq 0$; the equations imply $\widehat{b}_j = \xi_j \widehat{b}_i / \xi_i$ for every j, hence $L\widehat{b} = 0$. Thus $\mathbb{P}\nabla p = 0$ on the stated class.

Applying $\mathbb{P}$ and $I - \mathbb{P}$ to (R15) proves both equations

$$\partial_t u + B_\nu u = F_\sigma, \qquad F_\sigma = \mathbb{P} g, \qquad \nabla p = (I - \mathbb{P}) g. \tag{R16}$$

The force has no additional divergence condition. In Fourier coordinates the last equation retains its full components:

$$\widehat{\partial_{x_i} p}(\xi) = \sum_{j=1}^{3} \frac{\xi_i \xi_j}{|\xi|^2} \widehat{g}_j(\xi) \quad (\xi \neq 0). \tag{R17}$$

The original pressure itself remains part of the data. Equal gradients determine pressures up to a spatially constant function of time on connected $\mathbb{R}^3$; the notes do not choose or discard that function. Equation (R17) confirms the sign in (33), including the original force-divergence term in the associated pressure equation.

Applying K_ν^* and Q_ν to the first equation of (R16), using (R7), gives

$$\partial_t h + A h = K_\nu^* F_\sigma, \qquad \partial_t w + B_\nu w = Q_\nu F_\sigma, \tag{R18}$$

and the original initial data become $h(0) = K_\nu^* u(0)$, $w(0) = Q_\nu u(0)$. Conversely apply K_ν to the first equation of (R18) and add the second. The forcing is

$$K_\nu K_\nu^* F_\sigma + Q_\nu F_\sigma = P_\nu F_\sigma + Q_\nu F_\sigma = F_\sigma.$$

The left side is $\partial_t (K_\nu h + w) + B_\nu (K_\nu h + w)$. Adding the retained pressure equation in (R16) and then the full convection (R14) proves (35) with $-\nu\Delta$, $+\nabla p$ and $+N(u)$ in precisely their

original positions. The constraint $w(t) \in V_{\sigma,c}$, together with (R1), proves divergence zero. Equation (R5) proves both inverse identities on this solution data, including force, pressure and initial data left unchanged. This is an exact equivalence of the stated solution classes, not an equation on a truncated angular field.

For completeness, the mild equations in (36) hold in the same strong class. The semigroups are the explicit contraction multipliers $e^{-ts^2/4}$ and $e^{-t\nu|\xi|^2}$. Each commutes with its generator on its domain. A forcing G continuous in the generator graph norm has a graph-norm Bochner integral

$$J(t) = \int_0^t e^{-(t-s)D} G(s)\, ds, \qquad DJ(t) = \int_0^t e^{-(t-s)D} DG(s)\, ds, \tag{R19}$$

where $D = A$ or B_ν. The two displayed Hilbert-space integrals show membership in the closed graph of D, by approximation by Riemann sums. For a positive increment δ,

$$J(t+\delta) = e^{-\delta D} J(t) + \int_t^{t+\delta} e^{-(t+\delta-s)D} G(s)\, ds.$$

Divide by δ, subtract $J(t)/\delta$, and use the multiplier limit $(e^{-\delta d} - 1)/\delta \to -d$, dominated by d on the domain. The last integral quotient tends to $G(t)$ by continuity and contraction. The same argument with a negative increment gives $J' = G - DJ$. The homogeneous term has derivative $-De^{-tD}v_0$ when $v_0 \in \operatorname{Dom} D$. This proves mild-to-strong. For strong-to-mild, the Fourier coordinate satisfies the scalar integrating-factor formula; Fubini is justified by the continuous Hilbert norm on a finite interval. Alternatively, subtract the constructed mild solution: the difference solves $z' + Dz = 0$, $z(0) = 0$, and $d\|z\|^2/dt = -2\langle Dz, z\rangle \le 0$, because the multiplier is nonnegative. It vanishes identically. Section 2 and (R13) give the graph continuity required by (R19) for the actual source. Mere continuity of a generic Hilbert-space forcing, without the additional stated strong-domain class, is not being used to claim automatic strong differentiability.

4. Energy identities and the exact supremum test

The two summands in (R5) are perpendicular. Expanding the Hilbert norm, retaining the two conjugate cross terms and using $K_\nu^* w = 0$, gives

$$\|K_\nu h + w\|_V^2 = \|K_\nu h\|_V^2 + \|w\|_V^2 + \langle K_\nu h, w\rangle + \langle w, K_\nu h\rangle = \|h\|_H^2 + \|w\|_V^2. \tag{R20}$$

Multiplication by $1/2$ proves (37) exactly. For every integer $m \ge 0$ on the domain of B_ν^m, (R7) and commutation of Q_ν with the generator give $B_\nu^m u = K_\nu A^m h + B_\nu^m w$, with $B_\nu^m w \in W_\nu^\perp$. Applying (R20) gives (38). This does not claim that a single nonsummed Cartesian derivative has an orthogonal splitting.

In the explicit supremum test retain the original nonnegative nonzero real $\varphi \in C_c^\infty((1,2))$, the original parameter $R > 0$, and

$$h_R(s) = R^{-1/2}\varphi(s/R)e_3/\|\varphi\|_{L^2(\mathbb{R})}.$$

The change of integration variable $s = Rr$ gives $\|h_R\|_H^2 = R^{-1}R\int|\varphi(r)|^2 dr/\|\varphi\|_2^2 = 1$; (R3) gives $\|v_R\|_V = 1$ for $v_R = K_\nu h_R$. Negative-half-line data vanish, so the precise reality constraint holds. Formula (R1) has smooth frequency support bounded away from $\rho = 0$; its radial factors vanish with all derivatives at the ends of their compact support. Thus the full Cartesian Fourier field belongs to $C_c^\infty(\mathbb{R}^3)$. Repeated integration by parts in each frequency variable, keeping the factors $(2\pi)^{-3/2}$, proves rapid decrease of every derivative of its inverse Fourier transform.

The angular integral has every coefficient shown in the note:

$$\int_{S^2} U_3^+(\omega)\, d\omega = \sqrt{3/2}(4\pi)^{-1/2}\left(4\pi e_3 - \frac{4\pi}{3}e_3\right) = \sqrt{3/2}(4\pi)^{-1/2}\frac{8\pi}{3}e_3.$$

Evaluating the absolutely convergent inverse Fourier integral at zero supplies one factor ρ^2 from measure and one factor ρ^{-1} from (R1). Therefore

$$v_R(0) = (2\pi)^{-3/2}\frac{a_\nu R^{-1/2}}{\|\varphi\|_2}\left(\int_0^\infty \rho\varphi(\lambda_\nu\rho/R)d\rho\right)\sqrt{3/2}(4\pi)^{-1/2}\frac{8\pi}{3}e_3.$$

The remaining substitution $r = \lambda_\nu\rho/R$ gives $R^2\lambda_\nu^{-2}\int_1^2 r\varphi(r)dr$. Combining the displayed factors yields precisely

$$v_R(0) = R^{3/2}(2\pi)^{-3/2}\frac{a_\nu}{\lambda_\nu^2}\sqrt{3/2}(4\pi)^{-1/2}\frac{8\pi}{3}\frac{\int_1^2 r\varphi(r)dr}{\|\varphi\|_2}e_3. \tag{R21}$$

Every factor multiplying $R^{3/2}$ is positive. Since v_R is continuous, its point value is bounded by its essential supremum, proving the stated lower bound in (41). This proves failure of an L^2-to-L^∞ bound even on this six-channel range. It proves no trajectory statement because R is the freely chosen frequency parameter, and the notes correctly keep it distinct from physical time in (32).

5. The actual viscosity map, source, pressure and energy

Let (u_*, p_*, f_*) be exactly the viscosity-one localized, cutoff-summed, force-extended triple cited in Section 9. For $y = x/\sqrt{\nu}$, the fields are

$$u_\nu(x,t) = \sqrt{\nu}\,u_*(y,t), \qquad p_\nu(x,t) = \nu p_*(y,t), \qquad f_\nu(x,t) = \sqrt{\nu}\,f_*(y,t). \tag{R22}$$

The complete mixed derivative formulas are, for each spatial multi-index α and time order $b \geq 0$,

$$\partial_x^\alpha\partial_t^b u_\nu = \nu^{(1-|\alpha|)/2}(\partial_y^\alpha\partial_t^b u_*)\circ(y,t),$$
$$\partial_x^\alpha\partial_t^b p_\nu = \nu^{1-|\alpha|/2}(\partial_y^\alpha\partial_t^b p_*)\circ(y,t), \qquad \partial_x^\alpha\partial_t^b f_\nu = \nu^{(1-|\alpha|)/2}(\partial_y^\alpha\partial_t^b f_*)\circ(y,t). \tag{R23}$$

These follow by one chain-rule factor $\nu^{-1/2}$ for each spatial derivative and no time factor. In particular, for each component i,

$$\partial_t u_{\nu,i} = \sqrt{\nu}(\partial_t u_{*,i})\circ(y,t), \quad \sum_j u_{\nu,j}\partial_{x_j}u_{\nu,i} = \sqrt{\nu}\sum_j(u_{*,j}\partial_{y_j}u_{*,i})\circ(y,t),$$
$$-\nu\Delta_x u_{\nu,i} = -\sqrt{\nu}(\Delta_y u_{*,i})\circ(y,t), \quad \partial_{x_i}p_\nu = \sqrt{\nu}(\partial_{y_i}p_*)\circ(y,t). \tag{R24}$$

Adding these four complete terms gives $\sqrt{\nu}$ times the full original residual and hence exactly f_ν. No component of the original force is replaced. Divergence transforms with factor one: $\sum_i \partial_{x_i}u_{\nu,i} = (\sum_i \partial_{y_i}u_{*,i})\circ(y,t) = 0$. The inverse is

$$u_*(y,t) = \nu^{-1/2}u_\nu(\sqrt{\nu}y,t), \quad p_*(y,t) = \nu^{-1}p_\nu(\sqrt{\nu}y,t), \quad f_*(y,t) = \nu^{-1/2}f_\nu(\sqrt{\nu}y,t). \tag{R25}$$

Thus the force, pressure, zero initial values, endpoint time and full source smoothness all have an exact inverse correspondence. Spatial support K_* becomes $\sqrt{\nu}K_*$; the time supports are unchanged. The positive spatial Jacobian is $\nu^{3/2}$, so

$$\|u_\nu(t)\|_2^2 = \nu\,\nu^{3/2}\|u_*(t)\|_2^2 = \nu^{5/2}\|u_*(t)\|_2^2, \quad \|u_\nu(t)\|_\infty = \sqrt{\nu}\|u_*(t)\|_\infty. \tag{R26}$$

The supremum equality follows because the spatial map is onto, and the amplitude is positive. The corresponding dissipation factor, if the source energy estimate (10.13) is retained, is exactly

$$\nu\|\nabla_x u_\nu(t)\|_2^2 = \nu\,\nu^{3/2}\|\nabla_y u_*(t)\|_2^2 = \nu^{5/2}\|\nabla_y u_*(t)\|_2^2.$$

Likewise $\|f_\nu(t)\|_2 = \nu^{5/4}\|f_*(t)\|_2$. Thus the entire original energy-and-dissipation estimate scales with $\nu^{5/2}$, not only its velocity term. This checks the energy factor used in (43) and (45).

The actual arithmetic coordinates in (44) are now the uniquely determined $K_\nu^* u_\nu$ and $Q_\nu u_\nu$. Section 3 applies to them with the actual p_ν, f_ν, and the initial coordinates are zero. Applying (R20) to precisely this velocity proves (45). Its boundedness uses the source's bounded energy as an imported result; none is inferred from the coordinate decomposition alone.

6. Exact source-domain correction and the inner-circle value

The required original source constants and their roles are explicit. Theorem 3.1, printed pages 15–16, supplies $0 < h < 1/100$, $q_* > 0$, $0 < X_a < X_{\text{ext}} < \infty$, and the local domain $\tau > 0, q < q_*$. The original exponent is $A = 1/2 + h$. Theorem 3.1(iv), equation (3.6), and Proposition 9.9 Step 5, printed page 116, choose a fixed

$$X_{\text{in}} \in (0, X_a), \qquad e_0 = E_0(X_{\text{in}}, 0) > 0. \tag{R27}$$

All annular corrections vanish there, and the positive-order background cutoff sum remains in the actual relative remainder $R_*(\tau) = O(\tau^{2h})$. This statement does not remove the cutoff sum from the remainder. At $z = 0$, the original coordinates have $\eta = 0$ and $q = \tau = 1 - t$.

In Proposition 10.1, printed page 118, the source chooses $0 < \tau_0 < 1/2$ and $z_0 > 0$ such that $C_0(\tau_0 + z_0^{1/D}) < q_*/2$, with the actual comparison constant C_0 in (10.3) and $D = 1/2 - h$. The original $r_0 > 0$ is arbitrary and fixed. The spatial cutoff equals one for $r \le r_0/2, |z| \le z_0/2$, while its compact support lies in $r < r_0, |z| < z_0$. The time cutoff equals one when $0 \le 1 - t \le \tau_0/2$ and zero when $1 - t \ge \tau_0$. The localized fields in (10.4) are

$$u_* = \operatorname{curl}(c\mathcal{A}) + cBe_\theta, \qquad p_* = cp_{\text{loc}}, \qquad c = \chi_x\chi_t. \tag{R28}$$

Here $\mathcal{A}$ denotes the source vector potential called A in (10.4); this report writes $\mathcal{A}$ only to prevent collision with the arithmetic generator and the already retained scalar exponent, without changing the source object. The original product identity is

$$\operatorname{curl}(c\mathcal{A}) = c\operatorname{curl}\mathcal{A} + \nabla c \times \mathcal{A}. \tag{R29}$$

Choose actual $\tau_{\text{rem}} > 0$, $C_* \ge 0$ realizing the source remainder bound $|R_*(\tau)| \le C_*\tau^{2h}$ for $0 < \tau < \tau_{\text{rem}}$. The added equation (46a) retains

$$\tau_* = \min\left\{\tau_{\text{rem}}, \frac{\tau_0}{2}, \frac{r_0^2}{8X_{\text{in}}}, q_*\right\} > 0. \tag{R30}$$

At the original, unscaled point $y_\tau = (\sqrt{2X_{\text{in}}\tau}, 0, 0)$, every $0 < \tau < \tau_*$ gives

$$r = \sqrt{2X_{\text{in}}\tau} < r_0/2, \quad z = 0, \quad |z| < z_0/2, \quad q = \tau < q_*, \quad 1 - t = \tau < \tau_0/2. \tag{R31}$$

Strict inequalities ensure that both cutoff functions equal one in a neighborhood. Therefore $c = 1$ and all its spatial derivatives are zero there; in particular the full second term of (R29) is zero. The localized velocity is exactly the original local velocity at the point. The same argument holds at every angle of this circle because both cutoffs are axisymmetric. This proves the needed localization identity directly, with no omitted curl term and no dependence of τ_* on viscosity.

The original cylindrical basis is $e_\theta = (-\sin\theta, \cos\theta, 0)$. At angle $\theta = 0$ it is the positive Cartesian second unit vector. Thus the source (3.6), combined with (R27)–(R31), is precisely

$$(u_*)_2(y_\tau, 1 - \tau) = \tau^{-1/2-h}(e_0 + R_*(\tau)), \quad 0 < \tau < \tau_*. \tag{R32}$$

Multiplication by $\sqrt{\nu}$ and the spatial map in (R22) give $x_{\nu,\tau} = \sqrt{\nu} y_\tau = (\sqrt{2\nu X_{\mathrm{in}}\tau}, 0, 0)$. Inserting the exact inverse pair (R5) proves (47) with its whole remainder and original positive component. The all-order graph argument gives pointwise representatives at every fixed preterminal time; the particular equality (47) uses exactly the smaller interval (R30).

For $C_* > 0$, throughout

$$0 < \tau < \min\{\tau_*, (e_0/(2C_*))^{1/(2h)}\},$$

the remainder satisfies $|R_*(\tau)| < e_0/2$. The value in (47) is therefore at least $\sqrt{\nu} e_0 \tau^{-1/2-h}/2$. For $C_* = 0$, the remainder vanishes on its original interval and the same bound holds on $(0, \tau_*)$. Pointwise triangle inequality and the continuous representatives give exactly (48). The new (46a) resolves the only source-domain precision issue found by this review. It preserves the original constants, source equality, positive-order remainder, signs and support construction.

7. Compatibility with the complete later swirl result

The later note has the original scalar space

$$\mathcal{B} = L^2((0,\infty) \times \mathbb{R}, 2\pi r^{-1} dr\, dz),$$

and the maps

$$\Pi v(r,z) = \frac{1}{2\pi} \int_0^{2\pi} r\, v(r\cos\theta, r\sin\theta, z) \cdot e_\theta \, d\theta, \qquad (\mathcal{R}B)(x) = B(r,z) r^{-1} e_\theta. \tag{R33}$$

The later proof's circle Cauchy–Schwarz calculation gives $\|\Pi v\|_{\mathcal{B}} \le \|v\|_V$; polar integration gives $\mathcal{R}^* = \Pi$, $\Pi\mathcal{R} = I$ and $\|\mathcal{R}B\|_V = \|B\|_{\mathcal{B}}$. These retain the original 2π, radial weight and division by r.

Its composition proof is valid on the full six-channel range. Rotation about the third axis leaves Y_0, Y_1 unchanged and commutes with the negative-branch diagonal coefficient because $d_1 = d_2$. Averaging the rotated coefficient vectors in (R1) leaves their third components alone. Reflection $S = \mathrm{diag}(1,-1,1)$ satisfies $U_3^{\pm}(S\omega) = S U_3^{\pm}(\omega)$. The original Fourier change of coordinates intertwines the polar-vector actions, with absolute Jacobian one for this reflection. On the fixed physical plane the second component of the reflected-invariant rotational mean is therefore zero. Its value times r equals the circle integral in (R33). This proves $\Pi K_\nu h = 0$ first on smooth frequency data with compact support away from zero. Such data are dense in H, so the boundedness just established proves

$$\Pi K_\nu = 0 : H \to \mathcal{B}, \qquad \Pi Q_\nu = \Pi : V \to \mathcal{B}, \qquad K_\nu^* \mathcal{R} = 0 : \mathcal{B} \to H. \tag{R34}$$

The last equation follows by taking the adjoint of the first. Thus this is an exact operator composition, not a conclusion from the nonidentity of two angular presentations.

For the actual source on the circle in (R31), every annular correction vanishes; the remaining background and its cutoff sum are axisymmetric, as used by the later source application. Localization is one on the entire circle by (R31). Therefore its azimuthal component is constant around that circle and equals (R32) after the viscosity map. Applying (R33) and (R34) to the exact original velocity gives

$$\Pi w_\nu(r_{\nu,\tau}, 0, 1-\tau) = \Pi u_\nu(r_{\nu,\tau}, 0, 1-\tau) = \nu\sqrt{2X_{\mathrm{in}}}\, \tau^{-h}(e_0 + R_*(\tau)), \tag{R35}$$

where $r_{\nu,\tau} = \sqrt{2\nu X_{\mathrm{in}}\tau} > 0$. The original integral triangle inequality is

$$|\Pi w(r,z)| \le r \sup_\theta |w(r\cos\theta, r\sin\theta, z)| \le r\|w\|_\infty.$$

Dividing (R35) by the actual positive radius proves exactly

$$\|w_\nu(1-\tau)\|_\infty \geq \sqrt{\nu}\,\tau^{-1/2-h}|e_0 + R_*(\tau)|. \tag{R36}$$

Its valid interval is (R30), intersected with the displayed positivity interval when the positive half-leading lower bound is used. The later note's asymptotic interval may always be reduced to this intersection. Equation (R36) assigns the growing supremum to the retained complement; it does not assign growth of the complement at the particular angle zero, nor a bound on the separate six-channel field. The earlier heat note now states exactly this stronger subsequent conclusion and no contradictory component claim.

The later additional toroidal heat channel has its two original functions $\sqrt{15/(8\pi)}\omega_3(\omega\times e_3)$ and $\sqrt{3/(8\pi)}(\omega\times e_3)$, with $\omega\times e_3 = (\omega_2, -\omega_1, 0)$. Its orthogonality to all six functions follows from $(Le_j)\cdot(\omega\times e_3) = (\omega\times e_3)_j$ and oddness in ω_1 or ω_2. Its source application never replaces the full complement by these two coefficients. Consequently it introduces no dropped component into (R14)–(R18) or the source value (R32). The full pressure, force and viscosity statements of the reviewed note remain compatible with this later construction.

8. Exact completed coverage and limitation

This report completes the requested independent review of Sections 6–9, including all earlier definitions needed to type their maps, all original nonlinear and pressure signs, full forcing, both solution inverse directions, the actual strong regularity class, generator and mild-equation domains, energy factors, the explicit supremum family, source viscosity transport, the actual source remainder, the corrected localization interval and the subsequent swirl assignment. The written calculations above are the review evidence; no finite symbolic test is being substituted for them.

The reviewed source existence is imported from the frozen manuscript. This bounded report does not validate its complete local construction, every finite-stage estimate, a formal kernel closure, the comparison theorem excluding all global competitors, or any RH conclusion. No Lean, Lake, Elan, parent proof edit, PDF build, publication or new user-owned task was executed by this reviewer. Root remains responsible for integrating the corrected full heat proof and this completed review into the cumulative artifact.

Exact mean-swirl component of the solenoidal arithmetic correspondence

The retained six-channel map and the physical angular projection

The preceding complete imported note defines $K_\nu = \mathcal{F}_3^{-1} L J_\nu D$, its adjoint, range projection $P_\nu = K_\nu K_\nu^*$, generator $A = s^2/4$ and all their domains. Its J_ν is the displayed radial Fourier isometry; the original polynomial differential $J = DS$ remains the different explicitly defined matrix in the earlier appendices. Composition here uses the physical velocity as common data, through the exact inverses $u = K_\nu h + w$ and $(X, D) \mapsto u \circ X$ already proved; neither map's coordinates are silently substituted for the other's. All six angular functions, their Gram factors and viscosity constants are retained exactly as in the imported equations (1)–(25).

Use the physical angular-momentum projection $\Pi u = \langle r u_\theta \rangle$ of the preceding Mellin appendix, with $\sigma = r^2/2$. Its full Hilbert target is

$$\mathcal{B} = L^2\left((0,\infty)\times\mathbb{R}, \frac{2\pi}{2\sigma}\, d\sigma\, dz; \mathbb{C}\right), \qquad \|B\|_{\mathcal{B}}^2 = 2\pi \int_{\mathbb{R}} \int_0^\infty \frac{|B(\sigma,z)|^2}{2\sigma} d\sigma dz.$$

Jensen's inequality on each circle and $d\sigma = r\, dr$ prove

$$\|\Pi u\|_{\mathcal{B}}^2 = 2\pi \int |\langle u_\theta \rangle|^2 r\, dr\, dz \le \int |u_\theta|^2 r\, d\theta\, dr\, dz \le \|u\|_V^2.$$

Thus the formula defines a bounded map on the full physical L^2 space, by Fubini, without a pointwise trace assumption at the axis. The exact section of (443) extends to an isometry $\mathcal{R} : \mathcal{B} \to V$ because

$$(\mathcal{R}B)(x) = \frac{B(\sigma,z)}{r} e_\theta, \qquad \|\mathcal{R}B\|_V^2 = \|B\|_{\mathcal{B}}^2, \qquad \Pi\mathcal{R}B = B.$$

Its divergence is zero in distributions: its pairing with a gradient has the angular factor $\partial_\theta \phi$, whose integral over the circle is zero; absolute integrability follows from the L^2 bound and the compactly supported smooth test gradient. For the inner products linear in the first argument, polar integration also gives $\langle u, \mathcal{R}B\rangle_V = \langle \Pi u, B\rangle_{\mathcal{B}}$. Consequently $\mathcal{R}^* = \Pi$ and

$$C = \mathcal{R}\Pi = C^* = C^2$$

is the orthogonal projection onto all axisymmetric azimuthal fields. This gives both its full range and its kernel, and the norm retains the complete $2\pi/(2\sigma)$ weight.

Rotation and reflection prove that the six channels have zero mean swirl

Let R_α be rotation by α about e_3, and let $S = \mathrm{diag}(1,-1,1)$. For any orthogonal matrix O define the unitary polar-vector action

$$(T_O u)(x) = O u(O^{-1}x).$$

On the Fourier-coordinate copy $\widehat{V} = L^2(d\xi; \mathbb{C}^3)$ define $(\widehat{T}_O g)(\xi) = O g(O^{-1}\xi)$. Substitution $x = Oy$, with the absolute Jacobian one, in the original Fourier integral proves $\mathcal{F}_3 T_O = \widehat{T}_O \mathcal{F}_3$, first on integrable square integrable fields and then on L^2 by density and unitarity. For $O = R_\alpha$ or $O = S$, both Y_0 and Y_1 are invariant, $L(O\omega) = OL(\omega)O^{-1}$, and both branch matrices in D commute with O. Substituting these three identities into the full formula for K_ν proves

$$T_O K_\nu h = K_\nu(Oh), \qquad (Oh)(s) = Oh(s). \tag{453}$$

The exact mean rotation operator is the bounded integral $\mathcal{Z} = (2\pi)^{-1}\int_0^{2\pi} T_{R_\alpha}\,d\alpha$. Equation (453) gives

$$\mathcal{Z}K_\nu h = K_\nu(h_3 e_3), \qquad T_S\mathcal{Z}K_\nu h = \mathcal{Z}K_\nu h.$$

For a smooth vector field fixed by all rotations, its radial, azimuthal and axial components are functions of (r,z). At $x=(r,0,z)$ reflection invariance under S makes its second component zero. Rotation then gives zero azimuthal component everywhere. Equivalently, for every smooth u,

$$Cu = \tfrac{1}{2}(I - T_S)\mathcal{Z}u. \tag{454}$$

Indeed at $(r,0,z)$ the right side retains precisely the azimuthal component of $\mathcal{Z}u$, and its value is $\langle u_\theta\rangle e_\theta$. Rotation transports the identity over each circle. The formula extends to all L^2 fields by density and boundedness. Applying it to the preceding reflection identity proves

$$CK_\nu = 0, \qquad \Pi K_\nu = 0, \qquad CP_\nu = P_\nu C = 0. \tag{455}$$

For the second identity use $\Pi C = \Pi$; the last follows by taking adjoints as well. This is an exact operator computation, not a nonidentity assertion about two coordinate presentations. It does not say that $K_\nu h$ has zero azimuthal component at each angle: nonconstant angular components can remain before averaging.

An exact three-component state space, energy and heat domains

Set $V_{\mathrm{rem}} = V_\sigma \cap \ker K_\nu^* \cap \ker\Pi$ and $Q_\nu^{\mathrm{rem}} = I - P_\nu - C$ on V_σ. Orthogonality in (455) makes this the projection onto V_{rem}. The complete correspondence is

$$u \longmapsto (h, B, v) = (K_\nu^* u, \Pi u, Q_\nu^{\mathrm{rem}} u), \qquad u = K_\nu h + \mathcal{R}B + v, \tag{456}$$

$$\|u\|_V^2 = \|h\|_H^2 + \|B\|_{\mathcal{B}}^2 + \|v\|_V^2. \tag{457}$$

The domain is V_σ and the target is exactly $H \oplus \mathcal{B} \oplus V_{\mathrm{rem}}$. Applying K_ν^*, Π and Q_ν^{rem} to the displayed inverse proves surjectivity and both inverse identities. Pairwise orthogonality proves the energy equality. On real physical velocities the first component obeys the imported branch conjugation (21), while B, v are real; the projections commute with the corresponding conjugations. The original two-component remainder is exactly

$$w = (I - P_\nu)u = \mathcal{R}B + v. \tag{458}$$

Its inverse coordinates are $B = \Pi w$ and $v = (I - C)w$.

Let $B_\nu = -\nu\Delta$ with the full Fourier domain in imported (23). Orthogonal-coordinate actions T_O commute with its radial Fourier multiplier and preserve its domain. Hence (454) proves that C commutes with B_ν, all its nonnegative integer powers, and the heat multipliers, with preservation of their domains. One may also see domain preservation by applying the angular projection at each Fourier radius before integrating its weighted squared norm. Define the exact swirl generator and domain by

$$G_\nu = \Pi B_\nu \mathcal{R},$$

$$\mathrm{Dom}(G_\nu) = \left\{ B \in \mathcal{B} : \int_{\mathbb{R}^3} \nu^2 |\xi|^4 |\mathcal{F}_3(\mathcal{R}B)(\xi)|^2 d\xi < \infty \right\}.$$

This preserves the actual spatial and Fourier domain, including the axis; no unrestricted formal half-line boundary condition replaces it. On this domain $B_\nu\mathcal{R} = \mathcal{R}G_\nu$ and $\|G_\nu B\|_{\mathcal{B}} = \|B_\nu \mathcal{R}B\|_V$, by invariance of the range and the isometry. For smooth axis-regular compactly supported B, the original cylindrical vector Laplacian gives explicitly

$$G_\nu B = -\nu(2\sigma B_{\sigma\sigma} + B_{zz}).$$

Indeed its azimuthal component is $(\partial_{rr} + r^{-1}\partial_r - r^{-2} + \partial_{zz})(B/r)$; multiplying this by r gives the displayed expression using $\partial_r = r\partial_\sigma$. The exact form integral is

$$\begin{aligned}\nu\|\nabla\mathcal{R}B\|_V^2 &= 2\pi\nu\int\left[\left|B_\sigma - \frac{B}{2\sigma}\right|^2 + \frac{|B|^2}{4\sigma^2} + \frac{|B_z|^2}{2\sigma}\right]d\sigma dz\\ &= 2\pi\nu\int\left[|B_\sigma|^2 + \frac{|B_z|^2}{2\sigma}\right]d\sigma dz.\end{aligned}$$

For the second equality the retained cross term is $-\operatorname{Re}(B_\sigma\overline{B})/\sigma$. Integration of $(|B|^2)'/\sigma$ cancels precisely the displayed $|B|^2/(2\sigma^2)$ terms; the boundary value $|B|^2/\sigma$ vanishes at zero by $B = O(\sigma)$ and at infinity by compact support. Thus both expressions account for the same full spatial derivatives.

The corresponding heat action on $\mathcal{B}$ is explicitly $E_\nu(t) = \Pi e^{-tB_\nu}\mathcal{R}$, $t \geq 0$. Range invariance and $\Pi\mathcal{R} = I$ prove the semigroup law, strong continuity and the intertwiner $e^{-tB_\nu}\mathcal{R} = \mathcal{R}E_\nu(t)$. Its generator is $-G_\nu$ on exactly the stated domain, by conjugating the Fourier difference quotient through the isometry. This establishes the heat correspondence at the axis as well as on $r > 0$.

The full nonlinear equations in the three retained components

For the actual source set $u^{(1)} = K_\nu h$, $u^{(2)} = \mathcal{R}B$, and $u^{(3)} = v$. Retain all nine terms

$$N_i(u) = \sum_{a,b=1}^{3}\sum_{j=1}^{3}(u^{(a)})_j\partial_j(u^{(b)})_i, \qquad \mathcal{F}_\sigma = \mathbb{P}(f_\nu - N(u)).$$

The source pressure and initial conditions give exactly

$$\begin{aligned}\partial_t h + Ah &= K_\nu^*\mathcal{F}_\sigma, & h(0) &= 0,\\ \partial_t B + G_\nu B &= \Pi\mathcal{F}_\sigma, & B(0) &= 0,\\ \partial_t v + B_\nu v &= Q_\nu^{\mathrm{rem}}\mathcal{F}_\sigma, & v(0) &= 0,\\ \nabla p_\nu &= (I - \mathbb{P})(f_\nu - N(u)).\end{aligned}$$

All expressions have the original physical x, t, ν, and the force is the complete smooth extension of the source residual. The identities follow by applying the three projections to the full physical equation and the just-proved generator intertwiners. Conversely apply K_ν, $\mathcal{R}$ and the inclusion to the three equations and add. The projections sum to identity on the solenoidal space, so adding the retained pressure equation recovers every term of the original Navier–Stokes equation. The pressure's additive constant, when part of the source pressure class, remains part of the data exactly as in the imported note.

The scalar equation here is precisely (446): $\Pi\mathcal{F}_\sigma = F - \partial_\sigma P - \partial_z Q$. To prove that the Leray projection introduces no missing torque, use $\mathcal{R}\mathcal{B} \subset V_\sigma$ and $\Pi = \mathcal{R}^*$ to get $\Pi\mathbb{P} = \Pi$ by adjoints. Then apply the already computed Cartesian angular-momentum projection of the full nonlinear term. Hence the Mellin equation (448), its arithmetic multiplier and all inverse series apply to this exact orthogonal swirl component. The full momentum fluxes still contain all nine pairs of components above.

The actual blowup belongs to an explicit orthogonal component

For every time before one, the actual compact smooth source is in every domain needed above. Its projected components have smooth representatives by the imported Fourier-domain

argument. The operator equality $\Pi K_\nu = 0$ therefore holds on every circle, and $X_{\mathrm{in}} \in (0, X_a)$ is the original inner-circle label. Retain $e_0 = E_0(X_{\mathrm{in}}, 0) > 0$, $|R_*(\tau)| \le C_* \tau^{2h}$ and

$$\tau_* = \min\left\{\tau_{\mathrm{rem}}, \frac{\tau_0}{2}, \frac{r_0^2}{8X_{\mathrm{in}}}, q_*\right\} > 0.$$

For $0 < \tau < \tau_*$, the unscaled circle satisfies $r^2 = 2X_{\mathrm{in}}\tau < r_0^2/4$, $z = 0$, $q = \tau < q_*$ and $1 - t = \tau < \tau_0/2$. The original spatial and temporal cutoffs are one in a neighborhood of that circle. In the complete product $\operatorname{curl}(cA) = c \operatorname{curl} A + \nabla c \times A$, the latter term therefore vanishes there. All annular corrections vanish because $X_{\mathrm{in}} < X_a$, while every positive-order background term remains in R_*. The viscosity map evaluates these cutoffs at $x/\sqrt{\nu}$, retaining this same time interval. Thus (452) gives, for $0 < \tau < \tau_*$,

$$(\mathcal{R}B)_\theta(r_{\nu,\tau}, \theta, 0, 1 - \tau) = \sqrt{\nu}\, \tau^{-1/2-h}(e_0 + R_*(\tau)), \qquad r_{\nu,\tau} = \sqrt{2\nu X_{\mathrm{in}}\tau}.$$

This value is independent of θ by the definition of $\mathcal{R}$. Its full source remainder and viscosity are unchanged. Define the full positivity interval by

$$\tau_+ = \begin{cases} \min\{\tau_*, (e_0/(2C_*))^{1/(2h)}\}, & C_* > 0, \\ \tau_*, & C_* = 0. \end{cases}$$

For $0 < \tau < \tau_+$, the original remainder bound gives $|R_*(\tau)| \le e_0/2$, proving

$$\|\mathcal{R}B(1 - \tau)\|_\infty \ge \frac{\sqrt{\nu} e_0}{2} \tau^{-1/2-h}, \qquad \|w(1 - \tau)\|_\infty \ge \frac{\sqrt{\nu} e_0}{2} \tau^{-1/2-h}. \tag{459}$$

For the second inequality, $\Pi w = B$ gives $|B|/r = |\langle w_\theta \rangle| \le \sup_\theta |w(r, \theta, z)|$. It assigns the circle-average and supremum growth to the old orthogonal remainder, not to w at an arbitrarily fixed angle. No growth assertion about $K_\nu h$ alone is necessary or inferred. At the same time (457) gives the complete bound

$$\|h(t)\|_H^2 + \|B(t)\|_{\mathcal{B}}^2 + \|v(t)\|_V^2 = \|u_\nu(t)\|_2^2 \le E_\nu^2.$$

Thus a specified orthogonal component has the actual source's supremum singularity while its exact weighted energy stays bounded. The arithmetic transform of this same B is the recoverable one proved above, with its full corrected-field Mellin factor retained. This strengthens the imported sum-of-suprema conclusion without asserting an off-critical zeta zero.

The original source files remain in `sources/solenoidal_correspondence/`; the corrected complete source and full independent review are retained in `sources/solenoidal_correspondence_009/`. The complete review is included immediately before this proof; its equations (R6)–(R19) supply the full generator-domain, regularity, all-pairs nonlinear, pressure and strong/mild arguments. The rotation/reflection annihilation, three-component domain proof, and source lower bound have also received independent review. These are reviews of the stated coordinate correspondence and its imported source applications. No complete independent proof of the entire 166-page source construction or new Lean run is claimed.

An exact fixed-axis singularity of the arithmetic trace

The angular-momentum transform can see a fixed-axis derivative of the actual source field directly, without first applying its inverse. The reason is a specific feature of the construction: every positive-order azimuthal coefficient has zero axis value. We retain the source amplitude C, the parameter h, the physical viscosity ν, all cutoffs and the original time $t = 1 - \tau$. The calculation uses the constructed field of the preceding actual-source appendix; it does not certify the entire existence argument of the released manuscript independently.

The source amplitude C in this appendix is a positive scalar. The orthogonal operator called C in the preceding composition remains exactly $\mathcal{R}\Pi$; these two typed uses are stated explicitly without altering either object. The scalar source parameter h likewise retains its original value $0 < h < 1/100$; when the solenoidal coefficient function is needed below it is denoted $h_{\mathrm{sol}} = K_\nu^* u_\nu$.

The axis trace of the fully corrected velocity

Here and only here let $q_{\mathrm{src}}(z,t)$ denote the scalar similarity coordinate called q in the source, not the material label in the preceding material-flow appendix. Its source coordinates are

$$\tau = 1 - t = q_{\mathrm{src}}(1 - \eta^2), \qquad z = q_{\mathrm{src}}^D \eta, \qquad X = \frac{r^2}{2q_{\mathrm{src}}}, \qquad D = \frac{1}{2} - h, \quad A = \frac{1}{2} + h.$$

In particular at $z = 0$ one has exactly $\eta = 0$ and $q_{\mathrm{src}} = \tau$. Source (5.1), pp. 45–46, writes each azimuthal coefficient as

$$u_{\theta,n} = \frac{r}{C} q_{\mathrm{src}}^{-1-h+2nh} \varphi_n(X, \eta).$$

Its Lemma 5.1, p. 47, prescribes $\varphi_n(0,\eta) = 0$ for every $n \geq 1$. The radial extensions and moment corrections are supported away from the unchanged inner interval. Source Proposition 5.5 and its proof, pp. 60–62, realize this series with cutoffs $\chi(c_n q_{\mathrm{src}})$; these depend on z, t, not r, and the realized angular component is

$$\frac{u_{\theta,B}}{r} = \frac{q_{\mathrm{src}}^{-1-h}}{C} \left(\varphi_0(X,\eta) + \sum_{n \geq 1} \chi(c_n q_{\mathrm{src}}) q_{\mathrm{src}}^{2nh} \varphi_n(X,\eta) \right). \tag{460}$$

For every fixed preterminal z, t, this sum is finite in a neighborhood of that point: $c_n \to \infty$, while $q_{\mathrm{src}} > 0$ and χ has compact support. Thus evaluation at $X = 0$, and the smooth extension of the quotient $u_{\theta,B}/r$, require no exchange of a terminal limit with an infinite sum.

The leading axis datum is not an unknown positive remainder. Source (B.3), p. 145, and Proposition B.2, pp. 146–147, give

$$\varphi_0(0,\eta) = \varphi_*(\eta), \qquad \varphi_*(\eta) = \exp\left(\Lambda \int_0^\eta \zeta_*(w)\, dw \right), \qquad \varphi_*(0) = 1.$$

Here ζ_* is the source's auxiliary axis function, not the Riemann zeta function. Neither its value nor C is rescaled.

All annular wave and mean potentials, including the finite initialization and the later cutoff-summed stages, vanish near the axis for each $t < 1$: source Proposition 9.9, (9.21), pp. 114–117. Only finitely many stages are present locally at such a time. Their Cartesian derivatives therefore vanish on a common neighborhood of the axis. The base potential is azimuthal and independent of angle, so its curl has only radial and axial components. Consequently it contributes no additional azimuthal term to (460). Finally the spatial and temporal cutoffs in source (10.4), p. 118, are identically one near the origin for all sufficiently small positive τ. Their derivatives vanish there as well. These are facts about the actual summed source field,

not a replacement of that field by its leading approximation [8, Lemma 5.1; Proposition 5.5; (9.21); (10.4); (B.3)].

It follows that there is a source-determined $\tau_1 > 0$ such that

$$\left.\frac{u_{\theta,*}(r,0,1-\tau)}{r}\right|_{r=0} = \frac{1}{C}\tau^{-1-h} \qquad (0<\tau<\tau_1). \tag{461}$$

The quotient denotes its proved smooth extension, not division by zero. For the physical viscosity map $u_\nu(x,t) = \sqrt{\nu}\, u_*(x/\sqrt{\nu}, t)$, put $r_* = r/\sqrt{\nu}$. Then, for $r > 0$,

$$\frac{u_{\theta,\nu}(r,0,t)}{r} = \frac{\sqrt{\nu}\, u_{\theta,*}(r_*,0,t)}{\sqrt{\nu}\, r_*} = \frac{u_{\theta,*}(r_*,0,t)}{r_*}.$$

Passing to the smooth axis extension proves the same identity (461) for u_ν. This cancellation follows from the actual coordinate map; viscosity has not been set to one.

Use the original physical radial coordinate $\sigma = r^2/2$ and angular average $(2\pi)^{-1}\int_0^{2\pi}$. Let

$$B_\nu(\sigma,z,t) = \frac{1}{2\pi}\int_0^{2\pi} r u_{\theta,\nu}(r,\theta,z,t)\, d\theta.$$

In the axis neighborhood just established the full field is axisymmetric and $B_\nu = 2\sigma(u_{\theta,\nu}/r)$. The exact consequence is

$$B_\nu(0,0,1-\tau) = 0, \qquad \partial_\sigma B_\nu(0,0,1-\tau) = \frac{2}{C}\tau^{-1-h}. \tag{462}$$

No $O(\tau^{2h})$ error is present in this first axis derivative: each positive-order axis value is exactly zero. This does not remove the actual remainder at the distinct nonzero X_{in} sample of the preceding actual-source appendix.

Complete endpoint summation formula and remainder

We prove the needed endpoint summation formula explicitly. Let $B_m(v)$ be the Bernoulli polynomials defined by $B_0 = 1$, $B_m' = mB_{m-1}$, and $\int_0^1 B_m(v)\, dv = 0$ for $m \geq 1$; write $B_m = B_m(0)$. In particular $B_1(v) = v - 1/2$. For $m \geq 2$ integration of B_m' shows $B_m(1) = B_m(0)$. Uniqueness of this defining recursion proves $B_m(1-v) = (-1)^m B_m(v)$, so $B_m(0) = 0$ for odd $m \geq 3$. Let $\widetilde{B}_m(v) = B_m(\{v\})$; the values at the integers are immaterial inside integrals.

Suppose $f \in C^{2M}([0,\infty))$ and its derivatives through order $2M$ are integrable, have the indicated one-sided values at zero, and tend to zero at infinity through order $2M-1$. Summing integration by parts over the intervals $[nx,(n+1)x]$ gives

$$\sum_{n\geq 1} f(nx) = x^{-1}\int_0^\infty f(v)\, dv - \frac{f(0)}{2} + \int_0^\infty \widetilde{B}_1(v/x) f'(v)\, dv.$$

For example, the integral on one interval equals the half-sum of its endpoint values minus x^{-1} times its integral. For every subsequent integration the identity $\frac{d}{dv}\widetilde{B}_m(v/x) = mx^{-1}\widetilde{B}_{m-1}(v/x)$ holds between endpoints. The endpoint values match for $m \geq 2$, and the odd endpoint constants vanish as just proved. Repeating these integrations gives

$$\sum_{n\geq 1} f(nx) = x^{-1}\int_0^\infty f(v)\, dv - \frac{f(0)}{2} - \sum_{k=1}^{M} \frac{B_{2k}}{(2k)!} f^{(2k-1)}(0) x^{2k-1} + R_M(f,x), \tag{463}$$

$$|R_M(f,x)| \leq \frac{\|B_{2M}\|_{L^\infty[0,1]}}{(2M)!} x^{2M-1}\int_0^\infty |f^{(2M)}(v)|\, dv.$$

The remainder is the negative of $x^{2M-1}(2M)!^{-1}\int \widetilde{B}_{2M}(v/x) f^{(2M)}(v)\, dv$; its displayed bound justifies the infinite endpoint limit as well.

The recursion gives $B_2(v) = v^2 - v + 1/6$, so $B_2 = 1/6$. For the compact smooth inputs below all integrability and boundary hypotheses of this formula hold at every finite order.

The arithmetic operator on every endpoint derivative

We prove the endpoint map on its full smooth compactly supported domain. Let $a \in C^\infty([0,\infty))$ have compact support and $a(0) = 0$. For the same fixed h define

$$\phi_a(x) = a(x) - 2^h a(2x), \qquad \psi_a(x) = \phi_a(x) - \tfrac{1}{2}\phi_a(x/2), \qquad (\mathcal{A}_h a)(x) = \sum_{n\geq 1} \psi_a(nx) \quad (x > 0).$$

This is exactly the arithmetic operator used on the nonlinear fluxes. There is no radius change hidden in its argument x. Every sum and its derivatives are locally finite for $x > 0$. Substitution in the two integrals proves $\int_0^\infty \psi_a = 0$, and $\psi_a(0) = 0$. The full endpoint derivatives of the input are

$$\psi_a^{(j)}(0) = (1 - 2^{-j-1})(1 - 2^{h+j})a^{(j)}(0) \qquad (j \geq 0). \tag{464}$$

For clarity the passage from the sum to actual endpoint derivatives is justified here, rather than differentiating a formal asymptotic series. We use the Euler–Maclaurin identity proved with its remainder in (463); its classical provenance is DLMF, §2.10(i), (2.10.1). With $f_j(v) = v^j \psi_a^{(j)}(v)$, direct differentiation on $x > 0$ gives

$$x^j (\mathcal{A}_h a)^{(j)}(x) = \sum_{n\geq 1} f_j(nx).$$

All these f_j are smooth and compactly supported. Integration by parts j times gives $\int f_j = (-1)^j j! \int \psi_a = 0$; the factors v^j make each zero-end boundary term vanish. Their axis derivatives are

$$f_j^{(r)}(0) = 0 \quad (r < j), \qquad f_j^{(r)}(0) = \frac{r!}{(r-j)!}\psi_a^{(r)}(0) \quad (r \geq j).$$

Choose the Euler–Maclaurin order M so that $2M - 1 > j$. After division by x^j, its remainder is bounded by

$$\frac{\|B_{2M}\|_\infty}{(2M)!}\, x^{2M-1-j} \|f_j^{(2M)}\|_1,$$

and hence tends to zero. Terms below degree j vanish, so every differentiated series has a finite limit at zero. The only constant term for $j \geq 1$ is the Bernoulli term with $2k - 1 = j$; for even $j \geq 2$ it is absent. The fundamental theorem of calculus on $[\epsilon, x]$, followed by $\epsilon \downarrow 0$, now identifies these derivative limits with successive derivatives of the extended function. Boundedness of the next derivative justifies the passage under each integral. Thus $\mathcal{A}_h a$ is smooth at zero, and

$$\begin{aligned} (\mathcal{A}_h a)(0) &= 0, \\ (\mathcal{A}_h a)^{(j)}(0) &= -\frac{B_{j+1}}{j+1}(1 - 2^{-j-1})(1 - 2^{h+j})a^{(j)}(0), \qquad j \geq 1. \end{aligned} \tag{465}$$

The Bernoulli convention is $B_2 = 1/6$ and odd $B_{j+1} = 0$ for even $j \geq 2$. In particular

$$(\mathcal{A}_h a)'(0) = \frac{2^{h+1} - 1}{16}\, a'(0). \tag{466}$$

The coefficient is strictly positive for the source $h > 0$.

The preceding proofs apply also after any fixed z, t derivative of a smooth compactly supported parameter family with common support on compact preterminal parameter sets. Every bound then holds uniformly on those sets, with constants depending on the derivative order. This proves compatibility of the endpoint map with those parameter derivatives. It supplies no uniform estimate as $t \uparrow 1$.

A directly visible singularity at a fixed arithmetic coordinate

Apply this same operator to the actual compactly supported $a(\sigma) = B_\nu(\sigma, z, t)$. Write its arithmetic image as

$$\mathcal{B}_\nu(x, z, t) = \mathcal{A}_h B_\nu(x, z, t).$$

The domain and all nonlinear fluxes are those already given in the preceding angular-momentum section. Equations (462) and (466) now prove the exact formula

$$\boxed{\partial_x \mathcal{B}_\nu(0, 0, 1-\tau) = \frac{2^{h+1}-1}{8C}\,\tau^{-1-h}, \qquad 0 < \tau < \tau_1, \quad \nu > 0.} \tag{467}$$

Here $x = 0, z = 0$ are fixed coordinates, not a shrinking sampling path, and $\mathcal{B}_\nu(0, z, t) = 0$ for every preterminal t. Thus the value is zero while its first x-derivative diverges positively. For any fixed $\epsilon > 0$, the $C^1([0, \epsilon])$ seminorm is at least the displayed endpoint derivative. In particular no extension continuous at $t = 1$ with values in $C^1([0, \epsilon])$ can agree with this family. This assertion follows from an exact derivative of the original arithmetic sum, not merely a lower bound obtained after its inverse.

For every integer $m \geq 0$, differentiation of the exact equality gives

$$\partial_t^m \partial_x \mathcal{B}_\nu(0, 0, t) = \frac{2^{h+1}-1}{8C}(1+h)_m (1-t)^{-1-h-m}, \qquad 1 - \tau_1 < t < 1.$$

Indeed $d_t(1-t)^{-\alpha} = \alpha(1-t)^{-\alpha-1}$. These are equalities on an open preterminal interval. They make no assignment of finite terminal derivatives.

The projection onto B_ν has not discarded the other velocity data in the full correspondence. Its exact smooth section and kernel are

$$RB_\nu = \frac{B_\nu(r^2/2, z, t)}{r^2}(-x_2, x_1, 0), \qquad w_\nu = u_\nu - RB_\nu, \qquad \langle r(w_\nu)_\theta \rangle = 0,$$

where B_ν/σ has its smooth integral extension at zero. The preceding section proves the inverse $(B_\nu, w_\nu) \mapsto RB_\nu + w_\nu$ and retains every covariance in both momentum fluxes. The signal (467) concerns the B_ν component of that complete map. It does not replace the full Navier–Stokes equation by a closed scalar equation for B_ν.

The exact Mellin pole carrying this axis signal

For fixed preterminal parameters, put

$$\widehat{B}_\nu(s) = \int_0^\infty B_\nu(\sigma, z, t)\sigma^{s-1}\,d\sigma, \qquad \mathfrak{B}_\nu(s) = \int_0^\infty \mathcal{B}_\nu(x, z, t)x^{s-1}\,dx.$$

Both integrals are initially holomorphic on $\Re s > -1$. The full transform calculation in the preceding section gives

$$\mathfrak{B}_\nu(s) = W_h(s)\widehat{B}_\nu(s), \qquad W_h(s) = \zeta(s)(1 - 2^{s-1})(1 - 2^{h-s}).$$

The pole of ζ at 1 is canceled by the displayed first dilation factor, making W_h entire.

Here is the continuation needed at $s = -1$. For any fixed $b > 0$, split the integral at b and subtract the linear Taylor term on $[0, b]$. Since $B_\nu(0) = 0$, the exact formula is

$$\widehat{B}_\nu(s) = \int_0^b [B_\nu(\sigma) - B_{\nu,\sigma}(0)\sigma]\sigma^{s-1}d\sigma + \int_b^\infty B_\nu(\sigma)\sigma^{s-1}d\sigma + \frac{B_{\nu,\sigma}(0)b^{s+1}}{s+1}.$$

Taylor's integral remainder is $O(\sigma^2)$; the first integral is holomorphic for $\Re s > -2$, including after any complex s derivative on a compact subset, because the additional powers of

$\log\sigma$ remain integrable. The second is entire by compact support away from zero. Hence $\operatorname{Res}_{s=-1}\widehat{B}_\nu = B_{\nu,\sigma}(0)$. The identical argument applies to the proved smooth $\mathcal{B}_\nu$, so its residue is $\partial_x\mathcal{B}_\nu(0)$.

We also derive the required zeta value from the same finite-endpoint calculation, so that this value does not stand in for a proof. For $\Re s > 1$, apply the cell integrations on $[1, N]$ to $f(v) = v^{-s}$, include the first endpoint, and let $N \to \infty$. Since $f^{(k)}(v) = (-1)^k(s)_k v^{-s-k}$, the exact result is

$$\zeta(s) = \frac{1}{s-1} + \frac{1}{2} + \sum_{k=1}^{M} \frac{B_{2k}}{(2k)!}(s)_{2k-1} \\ - \frac{(s)_{2M}}{(2M)!}\int_1^\infty \widetilde{B}_{2M}(v)v^{-s-2M}\,dv, \qquad (s)_k = \prod_{j=0}^{k-1}(s+j), \quad (s)_0 = 1.$$

The integral is holomorphic on $\Re s > 1 - 2M$: on every compact subset, its integrand and every complex derivative are bounded by a constant times $v^{-1-\delta}(1 + |\log v|^j)$ for some $\delta > 0$, an integrable function. This follows from boundedness of the periodic Bernoulli polynomial. Identity on the initial half-plane identifies the meromorphic continuation. Increasing M covers the whole plane; the formulas agree on their overlaps by the identity theorem, and the only pole is the displayed pole at $s = 1$ with residue one. At $s = -1$, use $M = 2$. The products $(s)_3$ and $(s)_4$ vanish there, while the remainder integral is holomorphic there. Thus

$$\zeta(-1) = \frac{1}{-1-1} + \frac{1}{2} + \frac{B_2}{2!}(-1) = -\frac{1}{2} + \frac{1}{2} - \frac{1}{12} = -\frac{1}{12}.$$

This also proves directly the claimed cancellation of the sole zeta pole by $1 - 2^{s-1}$, since its derivative at one is $-\log 2$.

The classical value $\zeta(-1) = -1/12$ DLMF, §25.6(i) gives

$$W_h(-1) = -\frac{1}{12}\left(1 - \frac{1}{4}\right)(1 - 2^{h+1}) = \frac{2^{h+1}-1}{16} > 0.$$

Equality on $\Re s > -1$ therefore continues to the meromorphic functions just constructed and agrees with the direct endpoint proof. At $z = 0, t = 1 - \tau$ the pole is genuine and its residue is exactly

$$\operatorname{Res}_{s=-1}\mathfrak{B}_\nu(s, 0, 1-\tau) = \frac{2^{h+1}-1}{8C}\tau^{-1-h}. \tag{468}$$

There is no zero-factor compensation at this pole, since $W_h(-1) > 0$. The sign of $\zeta(-1)$ alone is not the sign of the arithmetic residue: both dilation factors have been evaluated explicitly.

This identifies what the construction propagates in this channel: an exact axis derivative, equivalently a growing residue at the fixed Mellin point -1. The identity retains the full $\widehat{B}_\nu$, its flux equations, and its inverse maps. The nontrivial zeros of ζ remain the zero factors specified in the full product formula; no off-critical zero is inferred from this axis pole or from its positive residue.

The fixed-axis arithmetic jet in the full orthogonal correspondence

The precise conversion between the retained pair and triple

Use the complete Hilbert spaces, domains, reality conditions and operators of the preceding solenoidal and swirl appendices. In particular $P_\nu = K_\nu K_\nu^*$, $\Pi\mathcal{R} = I$, $\mathcal{R}^* = \Pi$, and $\Pi K_\nu = 0$. Let w_{axis} denote exactly the $w_\nu = u_\nu - \mathcal{R}B_\nu$ in the preceding axis appendix. It has the typed domain

$$w_{\mathrm{axis}} \in V_\sigma \cap \ker \Pi, \qquad B_\nu \in \mathcal{B}.$$

The complete conversion from that pair to the solenoidal triple is

$$\begin{aligned} (B_\nu, w_{\mathrm{axis}}) &\longmapsto (h_{\mathrm{sol}}, B_\nu, v), \\ h_{\mathrm{sol}} &= K_\nu^* w_{\mathrm{axis}}, & v &= (I - P_\nu) w_{\mathrm{axis}}, \\ (h_{\mathrm{sol}}, B_\nu, v) &\longmapsto (B_\nu, K_\nu h_{\mathrm{sol}} + v), & v &\in V_\sigma \cap \ker K_\nu^* \cap \ker \Pi. \end{aligned}$$

Indeed $K_\nu^* \mathcal{R} = (\Pi K_\nu)^* = 0$, so $K_\nu^* w_{\mathrm{axis}} = K_\nu^* u_\nu$. Also $\Pi P_\nu = 0$, and hence v is in the displayed kernel intersection; $K_\nu^*(I - P_\nu) = 0$ supplies its other kernel condition. Adding $P_\nu w_{\mathrm{axis}}$ and $(I - P_\nu) w_{\mathrm{axis}}$ proves the first inverse identity. Conversely, applying K_ν^* and $I - P_\nu$ to $K_\nu h_{\mathrm{sol}} + v$ returns h_{sol} and v, respectively. Thus both maps are onto their displayed targets and are mutual inverses. The full reconstruction and energy are still exactly

$$u_\nu = K_\nu h_{\mathrm{sol}} + \mathcal{R}B_\nu + v, \qquad \|u_\nu\|_V^2 = \|h_{\mathrm{sol}}\|_H^2 + \|B_\nu\|_{\mathcal{B}}^2 + \|v\|_V^2.$$

The original solenoidal pair remainder is instead $w_{\mathrm{sol}} = (I - P_\nu) u_\nu = \mathcal{R}B_\nu + v$. Its exact conversion is $B_\nu = \Pi w_{\mathrm{sol}}$ and $v = (I - \mathcal{R}\Pi) w_{\mathrm{sol}}$. This specifies the relation between both remainder coordinates without discarding either one. Applying the arithmetic operator to the same B_ν therefore composes the two correspondences literally, with the original $\sigma = r^2/2$, physical z, t, ν, all nine nonlinear ordered pairs, and the previously displayed pressure and force.

An exact on-axis vorticity functional and its arithmetic image

The endpoint functional is defined on smooth physical fields, not on arbitrary L^2 equivalence classes. Fix z, orient the (x_1, x_2)-disk by $+e_3$, and orient its circle by increasing θ. For a smooth vector field u, write $\omega_3 = \partial_{x_1} u_2 - \partial_{x_2} u_1$. The original angular-momentum projection has the exact circulation form

$$\begin{aligned} (\Pi u)(r^2/2, z) &= \frac{1}{2\pi} \oint_{|x_\perp| = r} u \cdot d\ell \\ &= \frac{1}{2\pi} \int_{|x_\perp| \le r} \omega_3(x_\perp, z)\, dx_1 dx_2 = \int_0^r \rho \, \langle \omega_3(\rho, \theta, z) \rangle \, d\rho. \end{aligned}$$

The first equality uses $d\ell = r e_\theta \, d\theta$; the second is Stokes' formula with the stated orientation. For $r > 0$, differentiating and using $d\sigma/dr = r$ gives

$$\partial_\sigma (\Pi u)(r^2/2, z) = \langle \omega_3(r, \theta, z) \rangle.$$

Continuity at the axis gives the endpoint value: the difference from $\omega_3(0, 0, z)$ is bounded in absolute value by $\sup_{|x_\perp| \le r} |\omega_3(x_\perp, z) - \omega_3(0, 0, z)|$, which tends to zero. The integral formula also proves the one-sided derivative at $\sigma = 0$ directly. Consequently the typed functional

$$\mathcal{J}_z(u) := \partial_\sigma (\Pi u)(0, z) = \omega_3(0, 0, z)$$

annihilates every smooth field in $\ker \Pi$. In particular the smooth preterminal components satisfy

$$\mathcal{J}_z(K_\nu h_{\mathrm{sol}}) = 0, \qquad \mathcal{J}_z(v) = 0, \qquad \mathcal{J}_z(u_\nu) = \mathcal{J}_z(\mathcal{R}B_\nu).$$

Smoothness here follows from the full Fourier-domain preservation proved earlier: the actual source lies in every spatial Sobolev domain at each preterminal time, and each of the three orthogonal projections preserves those domains. Sobolev embedding at successively larger orders gives smooth representatives for all three components.

For the compact smooth source field, the complete endpoint summation proof above now gives the exact linear-functional identity

$$\partial_x \mathcal{A}_h(\Pi u_\nu)(x,z,t)|_{x=0} = \frac{2^{h+1}-1}{16}\,\mathcal{J}_z(u_\nu) = \frac{2^{h+1}-1}{16}\,\mathcal{J}_z(\mathcal{R}B_\nu).$$

Thus the arithmetic endpoint functional acts on the explicit orthogonal swirl component. For the actual source at $z = 0, t = 1-\tau$, the proved axis identity supplies, on its original interval $0 < \tau < \tau_1$,

$$\omega_{3,\nu}(0,0,0,1-\tau) = \frac{2}{C}\tau^{-1-h},$$
$$\partial_x \mathcal{A}_h B_\nu(0,0,1-\tau) = \mathrm{Res}_{s=-1}\,\mathfrak{B}_\nu(s,0,1-\tau) = \frac{2^{h+1}-1}{8C}\tau^{-1-h}.$$

The source amplitude $C > 0$ and exponent h retain their original values; the complete viscosity map proves why no additional ν factor remains in this fixed-axis derivative. The exact inner-circle velocity estimate, which has its own $\sqrt{\nu}$ and original remainder, remains the different evaluation already proved in the preceding appendix.

The fixed-axis result and every endpoint derivative were received as a complete source proof, preserved in `sources/ns_axis_jet/parent_29l.tex`. The accompanying review records its precise reviewed hash; the provenance file records the frozen source hash and reader adaptations separately. The complete Euler–Maclaurin argument needed for the adaptation is included above, together with a direct derivation of $\zeta(-1)$. The present pair/triple conversion and circulation calculation supply the exact composition with the full solenoidal state. The entire preceding solenoidal-swirl extension also received a complete independent incoming owner review with no mathematical correction; its final physical/Fourier operator notation clarification was checked locally.

The actual entire radial trace of the orthogonal swirl component

A full state map retaining the spatial complement and every radial input

Retain the physical Hilbert spaces and operators of the solenoidal correspondence: $H = L^2(\mathbb{R}, ds; \mathbb{C}^3)$, $V = L^2(\mathbb{R}^3, dx; \mathbb{C}^3)$, its solenoidal subspace V_σ, K_ν, $P_\nu = K_\nu K_\nu^*$, $\Pi u = \langle r u_\theta \rangle$, and the section $\mathcal{R}$. Here $\sigma = r^2/2$ is unchanged. Write $C_{\mathrm{sw}} = \mathcal{R}\Pi$ for the orthogonal operator previously called C; the source amplitude $C > 1$ remains a positive scalar. This typed renaming leaves both objects unchanged. Put

$$H_\infty = \bigcap_{m \geq 0} \operatorname{Dom} A^m, \qquad V_\infty = \bigcap_{m \geq 0} \operatorname{Dom} B_\nu^m, \qquad V_{\mathrm{rem},\infty} = V_\infty \cap V_\sigma \cap \ker K_\nu^* \cap \ker \Pi,$$

where $Ah = s^2 h/4$ and $B_\nu = -\nu\Delta$ have their full Fourier multiplication domains. Let

$$\mathscr{B}_c = \{B \in C_c^\infty([0,\infty) \times \mathbb{R}) : B(0,z) = 0 \text{ for every } z\}, \qquad \mathscr{U} = \{u \in V_\infty \cap V_\sigma : \Pi u \in \mathscr{B}_c\}.$$

The actual complete corrected source belongs to $\mathscr{U}$ at each preterminal time, by its compact support and the proved angular endpoint orders. For every $B \in \mathscr{B}_c$, $B/\sigma = \int_0^1 B_\sigma(a\sigma, z)\, da$ has its smooth extension at zero. Thus $\mathcal{R}B$ is a compactly supported smooth solenoidal vector field and lies in every displayed domain. In particular its weighted energy is finite, with the unchanged measure $2\pi(2\sigma)^{-1} d\sigma\, dz$.

The complete orthogonal correspondence already proved restricts to the following bijection with the displayed full target:

$$\mathscr{U} \longleftrightarrow H_\infty \oplus \mathscr{B}_c \oplus V_{\mathrm{rem},\infty}.$$

$$u \longmapsto (h_{\mathrm{sol}}, B, v) = (K_\nu^* u, \Pi u, (I - P_\nu - C_{\mathrm{sw}})u),$$
$$u = K_\nu h_{\mathrm{sol}} + \mathcal{R}B + v.$$

Indeed preservation of every generator domain proves the forward inclusions. Conversely each displayed triple reconstructs a field in $V_\infty \cap V_\sigma$; applying Π returns B, because $\Pi K_\nu = 0$, $\Pi\mathcal{R} = I$, and $\Pi v = 0$. The identities $K_\nu^* K_\nu = I$, $K_\nu^* \mathcal{R} = 0$ and $K_\nu^* v = 0$ return h_{sol}; subtraction returns v. These prove both inverse identities and surjectivity. Real fields retain the original positive-real/negative-pure-imaginary branch condition in h_{sol}, with real B, v.

For arbitrary $B \in \mathscr{B}_c$, use the complete sequence ingress $\mathfrak{I}$ of Section Q, with its one fixed cutoff and $\epsilon_n = 2^{-n}$. Define

$$\mathscr{P}(B) = \big(\mathfrak{I}(B(\cdot, z))\big)_{z \in \mathbb{R}}.$$

Every entry is an actual order-one entire radial trace. Its precise target is the set of such families whose individually recovered inputs fit a function in $\mathscr{B}_c$, with the fixed cutoff sequence specified there. This is a checked image condition on the explicit inverse, rather than a claim about arbitrary families of entire functions. For $\sigma > 0$, inversion of the n-th entry gives exactly $\chi(\sigma/\epsilon_n)B(\sigma, z)$. Once $2\epsilon_n < \sigma$, it equals B on a neighborhood of that point. The smooth recovered function supplies every endpoint derivative; its value at zero is the retained value zero. Thus $\mathscr{P}$ is injective and its stated inverse recovers the full joint input. Consequently the full state has the equally exact encoding

$$u \longleftrightarrow (h_{\mathrm{sol}}, \mathscr{P}(B), v), \qquad u = K_\nu h_{\mathrm{sol}} + \mathcal{R}\mathscr{P}^{-1}(\mathscr{P}(B)) + v.$$

No finite sequence entry or axial restriction replaces the full input. Its energy remains

$$\|u\|_V^2 = \|h_{\mathrm{sol}}\|_H^2 + 2\pi \int_{\mathbb{R}} \int_0^\infty \frac{|B(\sigma, z)|^2}{2\sigma}\, d\sigma\, dz + \|v\|_V^2.$$

The central entire observable and its exact axis functional

On the actual central slab $|z| < \sqrt{\nu} z_0/2$, the complete coefficient recurrence, all six complex Picard rows, every cutoff term, and the full physical force in Section Q.7 prove endpoint analyticity of the untruncated B, P, Q, F at every $0 \le t < 1$. The single entire observable there is exactly

$$\mathcal{T}_{h,z}(u)(s) = T_h(\Pi u)(s,z) = K_h(w)\widehat{\Pi u}(w/2, z), \qquad w = \tfrac{1}{2} - is,$$

on the subspace of $\mathscr{U}$ whose $\Pi u(\cdot, z)$ has an analytic endpoint germ. Its target lies in $(\mathcal{H}_{\le 1}, \gamma)$, and the inverse of T_h is precisely the full radial Mellin and two dyadic inverse series already proved. In particular

$$\mathcal{T}_{h,z}(u) = \mathcal{T}_{h,z}(\mathcal{R}\Pi u), \qquad \ker \mathcal{T}_{h,z} = \{u \text{ in this domain} : \Pi u(\cdot, z) = 0\}.$$

The first equality follows by $\Pi\mathcal{R}\Pi = \Pi$; the second follows by the injectivity of T_h. Both $K_\nu h_{\mathrm{sol}}$ and v belong to that kernel whenever they arise from the full state above. This specifies exactly what this one axial observable sees; the full encoding $(h_{\mathrm{sol}}, \mathscr{P}(B), v)$ still recovers all of u.

There is an exact numerical correspondence between the earlier direct Mellin pole and the value of this entire function. For any endpoint analytic $a \in C_c^\infty([0,\infty))$ with $a(0) = 0$, define $s_0 = -5i/2$. The full radial residue formula and the proved value $\zeta'(-2) = -\zeta(3)/(4\pi^2)$ give

$$(T_h a)(s_0) = \frac{\zeta(3)}{8\pi^2}(1 - 2^{-3/2})(2^{h+1} - 1)a'(0),$$
$$\frac{(T_h a)(s_0)}{\Xi(s_0)} = \frac{(1 - 2^{-3/2})(2^{h+1} - 1)}{12\pi} a'(0), \qquad \Xi(s_0) = \frac{3\zeta(3)}{2\pi}.$$

Indeed the residue of M_a^{rad} at $w = -2$ is $2^{-1}(1 - 2^{-3/2})(1 - 2^{h+1})a'(0)$. Multiplication by the displayed negative derivative of zeta gives the first formula with all signs and factors; division by the nonzero displayed $\Xi(s_0)$ gives the second. The direct arithmetic endpoint calculation gives $(\mathcal{A}_h a)'(0) = (2^{h+1} - 1)a'(0)/16$, hence

$$(T_h a)(s_0) = \frac{2\zeta(3)}{\pi^2}(1 - 2^{-3/2})(\mathcal{A}_h a)'(0), \qquad \frac{(T_h a)(s_0)}{\Xi(s_0)} = \frac{4}{3\pi}(1 - 2^{-3/2})(\mathcal{A}_h a)'(0).$$

All scalar coefficients are nonzero. Thus these are invertible scalar maps between the endpoint derivative, the direct Mellin residue, and the fixed entire value on this input class. They do not identify the two full transforms; their distinct arguments and full inverses remain those proved in the imported radial correspondence.

For the actual field, the independently proved circulation identity $\partial_\sigma \Pi u(0, z) = \omega_3(0, 0, z)$ now yields

$$\mathcal{T}_{h,z}(u)(s_0) = \frac{\zeta(3)}{8\pi^2}(1 - 2^{-3/2})(2^{h+1} - 1)\omega_3(0, 0, z).$$

Both other orthogonal summands have zero value of this functional. At $z = 0, t = 1 - \tau$, the actual coefficient $\omega_{3,\nu}(0, 0, 0, 1 - \tau) = 2\tau^{-1-h}/C$ therefore proves

$$\mathcal{T}_{h,0}(u_\nu)(s_0, 1 - \tau) = \mathcal{T}_{h,0}(\mathcal{R}B_\nu)(s_0, 1 - \tau) = \frac{\zeta(3)(1 - 2^{-3/2})(2^{h+1} - 1)}{4\pi^2 C}\tau^{-1-h}.$$

The interval is exactly $0 < \tau < \tau_1$ for this axis identity; the source amplitude, exponent and viscosity transformation are unchanged.

The retained nonlinear evolution and source quotient

The actual four central traces obey the entire angular equation (420), with its exact residual $\kappa_h(s-2i)\mathcal{F}_E^{\mathrm{rad}}(s)$. In the present full state, write $u^{(1)} = K_\nu h_{\mathrm{sol}}$, $u^{(2)} = \mathcal{R}B$, and $u^{(3)} = v$. Its actual fluxes are still precisely

$$P = \sum_{a,b=1}^{3} \langle r^2 u_r^{(a)} u_\theta^{(b)} \rangle, \qquad Q = \sum_{a,b=1}^{3} \langle r u_z^{(a)} u_\theta^{(b)} \rangle.$$

These follow by expanding both factors in the original definitions. All nine ordered terms are retained, even though $\Pi u^{(1)}$ and $\Pi u^{(3)}$ vanish. Only their complete sums are used as the compact endpoint-analytic inputs to T_h; no compact-support or analyticity assertion is imposed on an individual nonlocal summand. The original pressure equation remains the longitudinal projection of $f - \sum_{a,b} N(u^{(a)}, u^{(b)})$, with its spatial constant part retained. The source force has every cutoff and physical viscosity term already displayed. The full inverse state map and the injective residual calculation consequently recover the same physical angular equation, together with the other two component equations from the preceding orthogonal correspondence.

Finally write $q : \mathcal{H}_{\le 1} \to Q_{\mathrm{ar}} = \mathcal{H}_{\le 1}/N_{\mathrm{ar}}$ for the original source projection. Its composite with this actual trace has the exact kernel

$$\ker(q\mathcal{T}_{h,z}) = \{u \text{ in the stated domain} : T_h(\Pi u(\cdot, z)) \in N_{\mathrm{ar}}\}.$$

The inverse before this projection remains the radial inverse above. At the fixed point s_0, put $e = \Xi/\Xi(s_0)$. The original polynomial trace subspace gives $e \in N_{\mathrm{ar}}$ and $e(s_0) = 1$. The representative relation has the exact image

$$\{(q(f), f(s_0)) : f \in \mathcal{H}_{\le 1}\} = Q_{\mathrm{ar}} \times \mathbb{C}.$$

To prove surjectivity, choose any representative f of the prescribed class and replace it by $f + (b - f(s_0))e$; its class is unchanged and its value is exactly the prescribed b. Its kernel as a linear map from $\mathcal{H}_{\le 1}$ to that product is exactly $N_{\mathrm{ar}} \cap \{f : f(s_0) = 0\}$, by the two definitions. This gives the complete algebraic correspondence for the loss of this representative value. No continuity into the unnamed source topology, identification of that topology as Fréchet, ideal property of the actual N_{ar}, or membership of Ξ' is inferred.